CHAPTER 11

Feedback

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IN THIS CHAPTER YOU WILL LEARN

1. The general structure of the negative-feedback amplifier and the basic principle that underlies its operation.
2. The advantages of negative feedback, how these come about, and at what cost.
3. The appropriate feedback topology to employ with amplifiers of each of the four types: voltage, current, transconductance, and transresistance.
4. An intuitive and insightful approach for the analysis of practical feedback-amplifier circuits.
5. Why and how negative-feedback amplifiers can become unstable (i.e., oscillate) and how to design the circuit to ensure stable performance.

Introduction

Most physical systems incorporate some form of feedback. It is interesting to note, though, that the theory of negative feedback has been developed by electronics engineers. In his search for methods for the design of amplifiers with stable gain for use in transatlantic telephone repeaters, Harold Black, an electronics engineer with the Western Electric Company, invented the feedback amplifier in 1928. Since then, the technique has been so widely used that it is almost impossible to think of electronic circuits without some form of feedback, either implicit or explicit. Furthermore, the concept of feedback and its associated theory are currently used in areas other than engineering, such as in the modeling of biological systems.

Feedback can be either **negative** or **positive**. In amplifier design, negative feedback is applied to effect one or more of the following goals:

1. *Desensitize the gain:* that is, make the value of the gain less sensitive to variations in the values of circuit components, such as might be caused by changes in temperature.
2. *Reduce nonlinear distortion:* that is, make the output proportional to the input (in other words, make the gain constant, independent of signal level).
3. *Reduce the effect of noise:* that is, minimize the contribution to the output of unwanted electric signals generated, either by the circuit components themselves or by extraneous interference.
4. *Control the input and output resistances:* that is, raise or lower the input and output resistances by the selection of an appropriate feedback topology.
5. *Extend the bandwidth* of the amplifier.

在本章中，您将了解到

1. 负反馈放大器的一般结构及其工作的基本原理。

2. 负面反馈的优势、如何产生以及代价是什么。

3. 电压、电流、跨导和跨阻四种类型的放大器各自应采用的适当反馈拓扑结构。

4. 分析实用反馈放大器电路的直观而深刻的方法。

5. 负反馈放大器为何以及如何变得不稳定（即振荡），以及如何设计电路以确保性能稳定。

导言

大多数物理系统都包含某种形式的反馈。但有趣的是，负反馈理论是由电子工程师提出的。1928 年，美国西部电气公司的电子工程师哈罗德-布莱克（Harold Black）在寻找用于跨大西洋电话中继器的具有稳定增益的放大器设计方法时，发明了反馈放大器。从那时起，这项技术就得到了广泛应用，以至于几乎不可能想到没有某种形式的隐式或显式反馈的电子电路。此外，反馈概念及其相关理论目前还用于工程学以外的领域，如生物系统建模。

反馈可以是负反馈，也可以是正反馈。在放大器设计中，负反馈用于实现以下一个或多个目标：

1. 使增益脱敏：即降低增益值对电路元件值变化的敏感性，如温度变化可能引起的变化。
2. 减少非线性失真：即使输出与输入成正比（换句话说，使增益恒定，与信号电平无关）。
3. 减少噪音影响：即尽量减少电路元件本身或外来干扰产生的不需要的电信号对输出的影响。
4. 控制输入和输出电阻：即通过选择适当的反馈拓扑结构来提高或降低输入和输出电阻。
5. 扩展放大器的带宽。

All of the desirable properties above are obtained at the expense of a reduction in gain. It will be shown that the gain-reduction factor, called the **amount of feedback**, is the factor by which the circuit is desensitized, by which the input resistance of a voltage amplifier is increased, by which the bandwidth is extended, and so on. In short, *the basic idea of negative feedback is to trade off gain for other desirable properties*. This chapter is devoted to the study of negative-feedback amplifiers: their analysis, design, and characteristics.

Under certain conditions, the negative feedback in an amplifier can become positive and of such a magnitude as to cause oscillation. In fact, in Chapter 18 we will study the use of positive feedback in the design of oscillators and bistable circuits. Here, in this chapter, however, we are interested in the design of stable amplifiers. We shall therefore study the stability problem of negative-feedback amplifiers and their potential for oscillation.

It should not be implied, however, that positive feedback always leads to instability. In fact, positive feedback is quite useful in a number of nonregenerative applications, such as the design of active filters, which are studied in Chapter 17.

Before we begin our study of negative feedback, we wish to remind the reader that we have already encountered negative feedback in a number of applications. Almost all op-amp circuits (Chapter 2) employ negative feedback. Another popular application of negative feedback is the use of the emitter resistance R_E to stabilize the bias point of bipolar transistors and to increase the input resistance, bandwidth, and linearity of a BJT amplifier. In addition, the source follower and the emitter follower both employ a large amount of negative feedback. The question then arises about the need for a formal study of negative feedback. As will be appreciated by the end of this chapter, the formal study of feedback provides an invaluable tool for the analysis and design of electronic circuits. Also, the insight gained by thinking in terms of feedback can be extremely profitable.

11.1 The General Feedback Structure

11.1.1 Signal-Flow Diagram

Figure 11.1 shows the basic structure of a feedback amplifier. Rather than showing voltages and currents, Fig. 11.1 is a signal-flow diagram, where each of the quantities x can represent either a voltage or a current signal. The basic amplifier is unilateral and has a gain A, known as the **open-loop gain**; thus its output x_o is related to the input x_i by

$$x_o = Ax_i \tag{11.1}$$

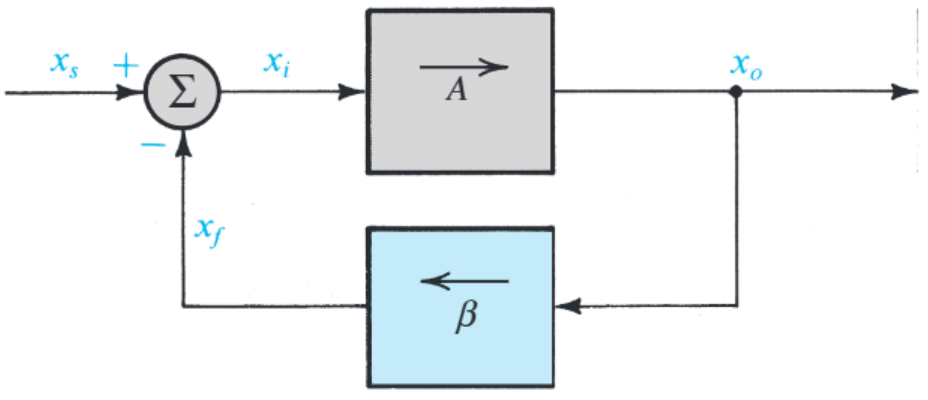

Figure 11.1 General structure of the feedback amplifier. This is a signal-flow diagram, and the quantities x represent either voltage or current signals.

上述所有理想特性都是以降低增益为代价的。我们将看到，增益降低系数，即反馈量，是电路脱敏、电压放大器输入电阻增大、带宽扩展等的系数。简而言之，负反馈的基本思想是以增益换取其他理想特性。本章专门研究负反馈放大器：它们的分析、设计和特性。

在某些条件下，放大器中的负反馈可以变为正反馈，其大小可以导致振荡。事实上，在第 18 章中，我们将研究正反馈在振荡器和双稳态电路设计中的应用。不过，在本章中，我们感兴趣的是设计稳定的放大器。因此，我们将研究负反馈放大器的稳定性问题及其产生振荡的可能性。

但这并不意味着正反馈总是导致不稳定。事实上，正反馈在许多非再生应用中都非常有用，例如有源滤波器的设计，第 17 章将对此进行研究。

在开始研究负反馈之前，我们希望提醒读者，我们已经在许多应用中遇到过负反馈。几乎所有运算放大器电路（第 2 章）都采用了负反馈。负反馈的另一个常见应用是使用发射极电阻 R 来稳定双极晶体管的偏置点，并增加 BJT 放大器的输入电阻、带宽和线性度。此外，源极跟随器和发射极跟随器都采用了大量负反馈。这就产生了是否需要对负反馈进行正式研究的问题。本章结束时，我们将了解到，对反馈的正式研究为分析和设计电子电路提供了宝贵的工具。此外，从反馈角度思考问题所获得的洞察力也会让人受益匪浅。

11.1 一般反馈结构

11.1.1 信号流程图

图 11.1 显示了反馈放大器的基本结构。图 11.1 显示的不是电压和电流，而是信号流图，其中每个量 x 都可以代表电压或电流信号。基本放大器是单边的，其增益为 A，即开环增益；因此其输出 x 与输入 x 的关系为

$$x = Ax \tag{11.1}$$

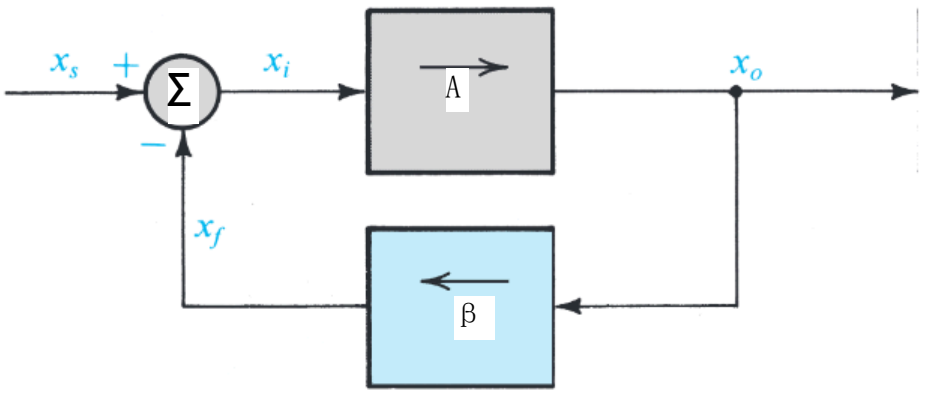

图 11.1 反馈放大器的一般结构。这是一个信号流图，量 x 代表电压或电流信号。

The *feedback network* measures or samples the output signal x_o and provides a *feedback signal* x_f that is related to x_o by the **feedback factor** β,

$$x_f = \beta x_o \tag{11.2}$$

It is assumed that connecting the feedback network to the amplifier output does not change the gain A or the value of x_o; that is, *the feedback network does not load the amplifier output*. Also, the feedback network is unilateral.

The feedback signal x_f is *subtracted* from the source signal x_s, which is the input to the complete feedback amplifier,[1] to produce the signal x_i, which is the input to the basic amplifier,

$$x_i = x_s - x_f \tag{11.3}$$

Here we note that it is this subtraction that makes the feedback negative. In essence, *negative feedback reduces the signal that appears at the input of the basic amplifier*. Here, too, we assume that connecting the output of the feedback network to the amplifier input, through the subtractor or differencing circuit, does not change the gain A; that is, *the feedback network does not load the amplifier input.*

11.1.2 The Closed-Loop Gain

The gain of the feedback amplifier, known as the closed-loop gain or the **gain-with-feedback** and denoted A_f, is defined as

$$A_f \equiv \frac{x_o}{x_s}$$

Combining Eqs. (11.1) through (11.3) provides the following expression for A_f:

$$A_f = \frac{A}{1 + A\beta} \tag{11.4}$$

The quantity $A\beta$ is called the **loop gain**, a name that follows from Fig. 11.1. For the feedback to be negative, the loop gain $A\beta$ must be positive; that is, the feedback signal x_f should have the same sign as x_s, thus resulting in a smaller difference signal x_i. Equation (11.4) indicates that for positive $A\beta$ the gain with feedback A_f will be smaller than the open-loop gain A by a factor equal to $1 + A\beta$, which is called the **amount of feedback**.

If, as is the case in many circuits, the loop gain $A\beta$ is large, $A\beta \gg 1$, then from Eq. (11.4) it follows that

$$A_f \simeq \frac{1}{\beta} \tag{11.5}$$

which is a very interesting result: *When the loop gain is large, the gain of the feedback amplifier is almost entirely determined by the feedback network*. Since the feedback network usually consists of passive components, which usually can be chosen to be as accurate as one wishes, the advantage of negative feedback in obtaining accurate, predictable, and stable gain

[1] In earlier chapters, we used the subscript "sig" for quantities associated with the signal source (e.g., v_{sig} and R_{sig}). We did that to avoid confusion with the subscript "s," which is usually used with FETs to denote quantities associated with the source terminal of the transistor. At this point, however, it is expected that readers have become sufficiently familiar with the subject that the possibility of confusion is minimal. Therefore, we will revert to using the simpler subscript s for signal-source quantities.

反馈网络测量或采样输出信号 x，并提供反馈信号 x，该信号与 x 的反馈系数 β 有关、

$$x = \beta x \tag{11.2}$$

假设将反馈网络连接到放大器输出端不会改变增益 A 或 x 的值；也就是说，反馈网络不会对放大器输出端产生负载。

此外，反馈网络是单方面的。

反馈信号 x 从源信号 x 中减去，源信号 x 是完整反馈放大器的输入，而反馈信号 x 则是基本放大器的输入、

$$x = x - x \tag{11.3}$$

这里我们要注意的是，正是这种减法使反馈变成了负反馈。实质上，负反馈减少了基本放大器输入端的信号。在此，我们还假定，通过减法器或差分电路将反馈网络的输出连接到放大器输入端不会改变增益 A；也就是说，反馈网络不会对放大器输入端造成负载。

11.1.2 闭环增益

反馈放大器的增益称为闭环增益或带反馈增益，用 A 表示，其定义为

$$A \equiv \frac{x}{x}$$

将公式（11.1）至（11.3）结合起来，可以得到 A 的如下表达式 ：

$$A = \frac{A}{1 + A\beta} \tag{11.4}$$

根据图 11.1，Aβ 称为环路增益。要使反馈为负，环路增益 Aβ 必须为正；也就是说，反馈信号 x 的符号应与 x 相同，从而使差分信号 x 变小。 公式（11.4）表明，对于正 Aβ，反馈增益 A 将小于开环增益 A，其系数等于 1 + Aβ，这就是反馈量。

如果在许多电路中，环路增益 Aβ 都很大，即 Aβ 为 1，那么根据公式（11.4）可以得出

$$A \approx \frac{1}{\beta} \tag{11.5}$$

这是一个非常有趣的结果：当环路增益较大时，反馈放大器的增益几乎完全由反馈网络决定。由于反馈网络通常由无源元件组成，而无源元件的选择通常可以随心所欲，因此负反馈在获得精确、可预测和稳定的增益方面具有优势。

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1 在前面几章中，我们用下标 "sig "表示与信号源相关的量（如 v 和 R）。我们这样做是为了避免与下标 "s "混淆，后者通常用于场效应晶体管，表示与晶体管源极相关的量。不过，在这一点上，预计读者对这一主题已经足够熟悉，混淆的可能性微乎其微。因此，我们将恢复使用更简单的下标 s 来表示信号源量。

should be apparent. In other words, the overall gain will have very little dependence on the gain of the basic amplifier, A, a desirable property because the gain A is usually a function of many manufacturing and application parameters, some of which might have wide tolerances. We have seen a dramatic illustration of all of these effects in op-amp circuits in Chapter 2, where the closed-loop gain is almost entirely determined by the feedback elements. Generally, we will consider $(1/\beta)$ to be the ideal value of A_f.

Equations (11.1) through (11.3) can be combined to obtain the following expression for the feedback signal x_f:

$$x_f = \frac{A\beta}{1+A\beta}x_s \tag{11.6}$$

Thus for $A\beta \gg 1$ we see that $x_f \simeq x_s$, which implies that the signal x_i at the input of the basic amplifier is reduced to almost zero. Thus if a large amount of negative feedback is employed, the feedback signal x_f becomes an almost identical replica of the input signal x_s. The difference between x_s and x_f, which is x_i, is sometimes referred to as the **error signal**.[2] Accordingly, the **input differencing circuit** is often also called a **comparison circuit**. (It is also known as a **mixer**.) An expression for x_i can be easily determined as

$$x_i = \frac{1}{1+A\beta}x_s \tag{11.7}$$

from which we can verify that for $A\beta \gg 1$, x_i becomes very small. An outcome of this property is the tracking of the two input terminals of an op amp. Observe that negative feedback reduces the signal that appears at the input terminals of the basic amplifier by the amount of feedback $(1 + A\beta)$. As will be seen later, it is this reduction of input signal that results in the increased linearity of the feedback amplifier.[3]

11.1.3 The Loop Gain

From the discussion above we see that the loop gain $A\beta$ is a very important—in fact, the most important—characteristic parameter of a feedback amplifier:

1. The sign of $A\beta$ determines the polarity of the feedback; the loop gain $A\beta$ must be positive for the feedback to be negative.
2. The magnitude of $A\beta$ determines how close the closed-loop gain A_f is to the ideal value of $1/\beta$.
3. The magnitude of $A\beta$ determines the amount of feedback $(1 + A\beta)$ and hence, as we shall see in the next section, the magnitude of the various improvements in amplifier performance resulting from the negative feedback.
4. As we shall see in later sections, the inevitable variation of $A\beta$ with frequency can cause $A\beta$ to become negative, which in turn can cause the feedback amplifier to become unstable. It follows that the design of a stable feedback amplifier may involve modifying the frequency behaviors of its loop gain $A\beta$ appropriately (Section 11.10).

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[2] This terminology is more common in feedback control systems than in feedback amplifiers.

[3] We have in fact already seen examples of this: adding a resistance R_e in the emitter of a CE amplifier (or a resistance R_s in the source of a CS amplifier) increases the linearity of these amplifiers because for the same input signal as before, v_{be} and v_{gs} are now smaller (by the amount of feedback).

应该是显而易见的。换句话说，整体增益对基本放大器增益 A 的依赖性很小，这是一个理想的特性，因为增益 A 通常是许多制造和应用参数的函数，其中一些参数的公差可能很大。我们在第 2 章中看到了运算放大器电路中所有这些影响的生动说明，其中闭环增益几乎完全由反馈元件决定。一般来说，我们将（1/β）视为 A 的理想值。

将公式（11.1）至（11.3）结合起来，可以得到反馈信号 x 的如下表达式：

$$x = \frac{A\beta}{1 + A\beta}x \tag{11.6}$$

因此，对于 Aβ 1，我们可以看到 x x，这意味着基本放大器输入端的信号 x 几乎为零。因此，如果采用大量负反馈，反馈信号 x 就会成为输入信号 x 的几乎完全相同的复制品。 x 与 x 之间的差值，即 x ，有时称为误差信号。因此，输入差分电路通常也被称为比较电路。(x 的表达式很容易确定为

$$x = \frac{1}{1 + A\beta}x \tag{11.7}$$

由此我们可以验证，当 Aβ 为 1 时，x 会变得非常小。这一特性的一个结果就是运算放大器两个输入终端的跟踪。请注意，负反馈会通过反馈量（1 + Aβ）减少出现在基本放大器输入端的信号。正如稍后将看到的，正是输入信号的减少导致了反馈放大器线性度的提高。

11.1.3 回路增益

从上面的讨论中我们可以看出，环路增益 Aβ 是反馈放大器的一个非常重要的特征参数，实际上也是最重要的特征参数：

1. Aβ 的符号决定了反馈的极性；环路增益 Aβ 必须为正，反馈才会为负。

2. Aβ 的大小决定了闭环增益 A 与理想值 1/β 的接近程度。

3. Aβ 的大小决定了反馈量（1 + Aβ），因此，正如我们将在下一节看到的，也决定了负反馈对放大器性能的各种改进的大小。

4. 我们将在后面的章节中看到，Aβ 随频率的变化不可避免地会导致 Aβ 变为负值，这反过来又会导致反馈放大器变得不稳定。因此，设计稳定的反馈放大器可能需要适当修改其环路增益 Aβ 的频率行为（第 11.10 节）。

2 这一术语在反馈控制系统中比在反馈放大器中更为常见。

3 事实上，我们已经看到过这样的例子：在 CE 放大器的发射极上增加一个电阻 R（或在 CS 放大器的源极上增加一个电阻 R）会提高这些放大器的线性度，因为对于与之前相同的输入信号，v 和 v 现在会变小（由反馈量决定）。

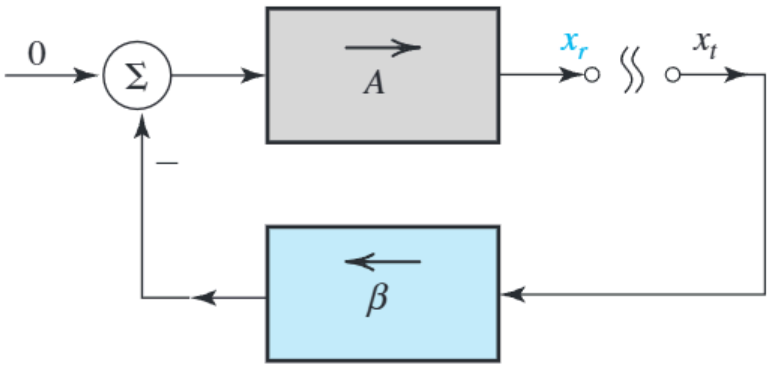


Figure 11.2 Determining the loop gain by breaking the feedback loop at the output of the basic amplifier, applying a test signal x_t, and measuring the returned signal x_r : $A\beta \equiv -x_r/x_t$.

The significance of the loop gain requires us to consider its determination. Reference to Fig. 11.1 indicates that the value of the loop gain $A\beta$ can be determined as follows:

1. Set $x_s = 0$.
2. Break the feedback loop at a convenient location, ensuring that the values of A and β *do not change*. Since we assumed that the feedback network does not load the amplifier output, we can break the loop at the amplifier output (see Fig. 11.2) without causing A to change.
3. Apply a test signal x_t to the input of the loop (where the break has been made) and determine the *returned signal* x_r at the loop output (i.e., at the other side of the break). From Fig. 11.2 we see that

$$x_r = -A\beta x_t$$

and the loop gain $A\beta$ is obtained as

$$A\beta = -\frac{x_r}{x_t} \tag{11.8}$$

We observe that since $A\beta$ is positive, the returned signal x_r will be out of phase with the test signal x_t, verifying that the feedback is indeed negative. In fact, this approach is used qualitatively to ascertain the polarity of the feedback. We will have a lot more to say about the loop gain in subsequent sections.

Example 11.1

The noninverting op-amp configuration shown in Fig. 11.3(a) provides a direct implementation of the feedback loop of Fig. 11.1.

(a) Assume that the op amp has infinite input resistance and zero output resistance. Find an expression for the feedback factor β.
(b) Find the condition the open-loop gain A must satisfy so that the closed-loop gain A_f is almost entirely determined by the feedback network. Also, give the value of A_f in this case.

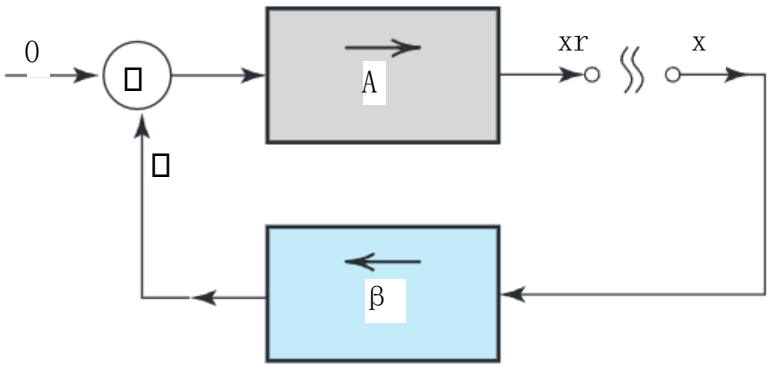


图 11.2 通过断开基本放大器输出端的反馈回路、施加测试信号 x 并测量返回信号 x 来确定环路增益：Aβ ≡ -x/x。

环路增益的重要性要求我们考虑其确定方法。参考图 11.1 可以得出环路增益 Aβ 的值如下：

1. 设 x = 0。
2. 在方便的位置断开反馈回路，确保 A 和 β 的值不发生变化。由于我们假定反馈网络不负载放大器输出，因此可以在放大器输出端断开回路（见图 11.2），而不会导致 A 发生变化。
3. 在环路输入端（断路处）施加测试信号 x，并确定环路输出端（即断路的另一端）的返回信号 x。从图 11.2 可以看出

$$x = -A\beta x$$

环路增益 Aβ 的计算公式为

$$A\beta = -\frac{x}{x} \tag{11.8}$$

我们观察到，由于 Aβ 为正，返回信号 x 将与测试信号 x 反相，从而验证反馈确实为负。事实上，这种方法可用于定性地确定反馈的极性。关于环路增益，我们将在后续章节中详细介绍。

例 11.1

图 11.3(a) 所示的非反相运算放大器配置直接实现了图 11.1 的反馈回路。(a) 假设运算放大器的输入电阻为无穷大，输出电阻为零。(b) 找出开环增益 A 必须满足的条件，以便闭环增益 A 几乎完全由反馈网络决定。并给出这种情况下 Ain 的值。

Example 11.1 *continued*

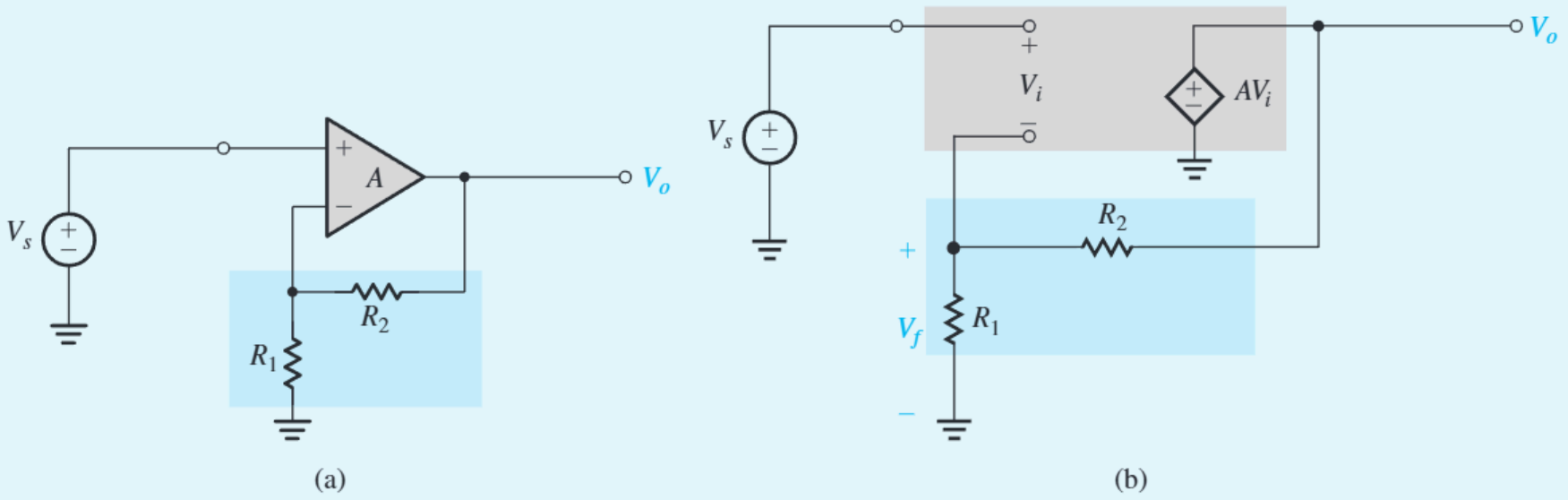


Figure 11.3 (**a**) A noninverting op-amp circuit for Example 11.1. (**b**) The circuit in (**a**) with the op amp replaced with its equivalent circuit.

(c) If the open-loop gain $A = 10^4$ V/V, find R_2/R_1 to obtain a closed-loop gain A_f of 10 V/V.
(d) What is the amount of feedback in decibels?
(e) If $V_s = 1$ V, find V_o, V_f, and V_i.
(f) If A decreases by 20%, what is the corresponding decrease in A_f?

Solution

(a) To be able to see more clearly the direct correspondence between the circuit in Fig. 11.3(a) and the block diagram in Fig. 11.1, we replace the op amp with its equivalent-circuit model, as shown in Fig. 11.3(b). Since the op amp is assumed to have infinite input resistance and zero output resistance, its model is simply an ideal voltage-controlled voltage source of gain A. From Fig. 11.3(b) we observe that the feedback network, consisting of the voltage divider (R_1, R_2), is connected directly to the output and feeds a signal V_f to the inverting input terminal of the op amp. It is important at this point to note that the zero output resistance of the op amp causes the output voltage to be AV_i irrespective of the values of R_1 and R_2. That is what we meant by the statement that in the block diagram of Fig. 11.1, the feedback network is assumed to not load the basic amplifier. Now we can easily determine the feedback factor β from

$$\beta \equiv \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

Let's next examine how V_f is subtracted from V_s at the input side. The subtraction is effectively performed by the differential action of the op amp; by its very nature, a differential-input amplifier takes the difference between the signals at its two input terminals. Observe also that because the input resistance of the op amp is assumed to be infinite, no current flows into the negative input terminal of the op amp and that the feedback network does not load the amplifier at the input side.

(b) The closed-loop gain A_f is given by

$$A_f = \frac{A}{1 + A\beta}$$

例 11.1 续

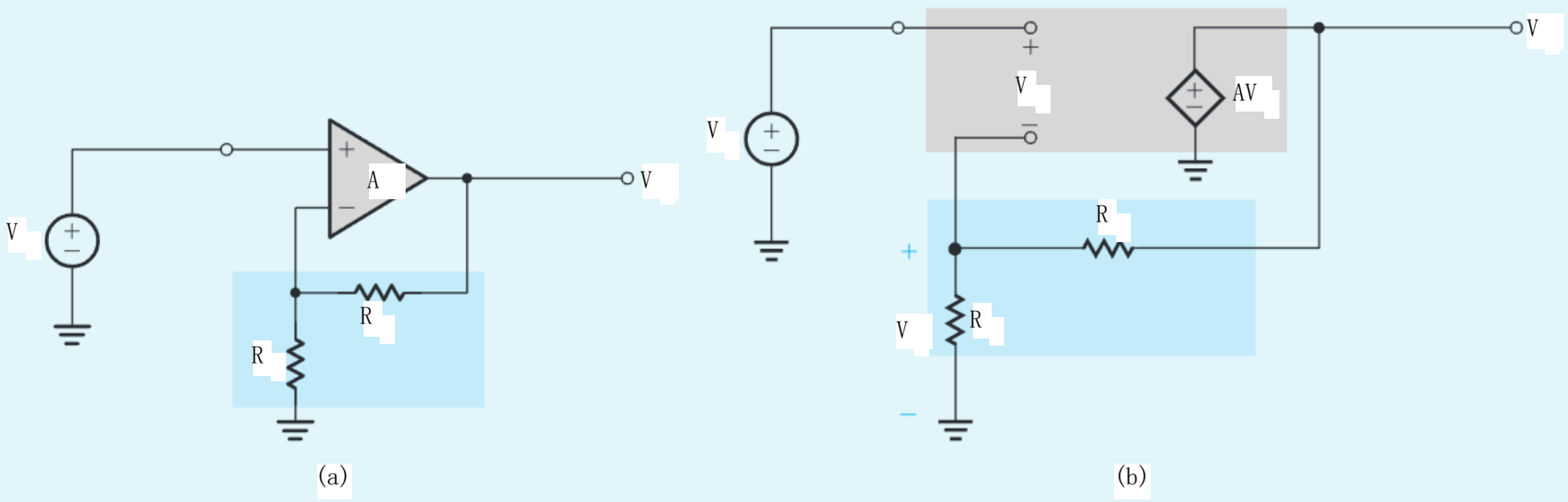


图 11.3 (a) 例 11.1 中的非反相运算放大器电路。 (b) (a) 中的电路，用等效电路替换了运算放大器。

(c) 如果开环增益 A = 10 V/V，求 R/R，以获得 10 V/V 的闭环增益 A。
(d) 以分贝为单位的反馈量是多少？(e) 如果 V=
1 V，求 V、V 和 V。
(f) 如果 A 减少 20%，A 的相应减少量是多少？

解决方案

(a) 为了更清楚地看到图 11.3(a)中电路与图 11.1 框图之间的直接对应关系，我们用图 11.3(b) 中的等效电路模型替换运算放大器。

从图 11.3(b)中我们可以看到，由分压器 (R, R) 组成的反馈网络直接连接到输出端，并将信号 V 送入运算放大器的反相输入端。我们在图 11.1 的方框图中假定反馈网络不给基本放大器加载负载，就是这个意思。现在，我们可以根据以下公式轻松确定反馈系数 β

$$\beta \equiv \frac{V}{V} = \frac{R}{R+R}$$

接下来让我们看看 Vis 如何从输入端 Vat 中减去。减法实际上是由运算放大器的差分作用完成的；就其本质而言，差分输入放大器取其两个输入端信号之差。还要注意的是，由于假定运算放大器的输入电阻为无穷大，因此没有电流流入运算放大器的负输入端，反馈网络也不会在输入端给放大器加载负载。

(b) 闭环增益 A 由以下公式给出

$$A = \frac{A}{1 + A\beta}$$

To make A_f nearly independent of A, we must ensure that the loop gain $A\beta$ is much larger than unity

$$A\beta \gg 1$$

in which case

$$A_f \simeq 1/\beta$$

Thus,

$$A/A_f \gg 1$$

or equivalently,

$$A \gg A_f$$

and

$$A_f \simeq \frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

(c) For $A = 10^4$ V/V and $A_f = 10$ V/V, we see that $A \gg A_f$, thus we can select R_1 and R_2 to obtain

$$\beta \simeq \frac{1}{A_f} = 0.1$$

Thus,

$$\frac{1}{\beta} = 1 + \frac{R_2}{R_1} = A_f = 10$$

which yields

$$R_2/R_1 = 9$$

A more exact value for the required ratio R_2/R_1 can be obtained from

$$A_f = \frac{A}{1 + A\beta}$$

$$10 = \frac{10^4}{1 + 10^4\beta}$$

which results in

$$\beta = 0.0999$$

and,

$$\frac{R_2}{R_1} = 9.01$$

为了使 A 接近独立于 A，我们必须确保环路增益 Aβ 远远大于统一值

$$A\beta \quad 1$$

在这种情况下

$$A \quad 1/\beta$$

因此、

$$A/A \quad 1$$

或等同于

$$A \quad A$$

and

$$A \quad \frac{1}{\beta} = 1 + \frac{R}{R}$$

(c) 对于 A = 10 V/V 和 A = 10 V/V，我们可以看到 A A，因此我们可以选择 Rand R，从而得到

$$\beta \quad \frac{1}{A} = 0.1$$

因此、

$$\frac{1}{\beta} = 1 + \frac{R}{R} = A = 10$$

得出

$$R/R = 9$$

所需比率 R/R 的更精确值可从以下公式中得出

$$A = \frac{A}{1 + A\beta}$$

$$10 = \frac{10}{1 + 10\,\beta}$$

导致

$$\beta = 0.0999$$

and,

$$\frac{R}{R} = 9.01$$

Example 11.1 *continued*

(d) The amount of feedback is

$$1+A\beta=\frac{A}{A_f}=\frac{10^4}{10}=1000$$

which is 60 dB.
(e) For $V_s=1$ V,

$$V_o=A_fV_s=10\times 1=10\text{ V}$$
$$V_f=\beta V_o=0.0999\times 10=0.999\text{ V}$$
$$V_i=\frac{V_o}{A}=\frac{10}{10^4}=0.001\text{ V}$$

Note that if we had used the approximate value of $\beta=0.1$, we would have obtained $V_f=1$ V and $V_i=0$ V.
(f) If A decreases by 20%, thus becoming

$$A=0.8\times 10^4\text{ V/V}$$

the value of A_f becomes

$$A_f=\frac{0.8\times 10^4}{1+0.8\times 10^4\times 0.0999}=9.9975\text{ V/V}$$

that is, it decreases by 0.025%, which is less than the percentage change in A by approximately a factor $(1+A\beta)$.

EXERCISES

11.1 Repeat Example 11.1 (c) to (f) for $A=100$ V/V.
Ans. (c) 10.11; (d) 20 dB; (e) 10 V, 0.9 V, 0.1 V; (f) 2.44%

11.2 Repeat Example 11.1 (c) to (f) for $A_f=10^3$ V/V. For (e) use $V_s=0.01$ V.
Ans. (c) 1110.1; (d) 20 dB; (e) 10 V, 0.009 V, 0.001 V; (f) 2.44%

11.1.4 Summary

We conclude this section by presenting in Table 11.1 a summary of the important parameters and formulas that characterize the ideal negative-feedback amplifier structure of Fig. 11.1.

例 11.1 续

(d) 反馈量为

$$1 + A\beta = \frac{A}{A} = \frac{10}{10} = 1000$$

即 60 分贝。

(e) 对于 V= 1 V、

$$V = AV = 10 \times 1 = 10 \text{ V} \quad V = \beta V = 0.0999 \times 10 = 0.999 \text{ V}$$

$$V = \frac{V}{A} = \frac{10}{10} = 0.001 \text{ V}$$

请注意，如果我们使用近似值 β = 0.1，我们将得到 V= 1 V 和 V= 0 V。

$$A = 0.8 \times 10 \text{ V/V}$$

的价值

$$A = \frac{0.8 \times 10}{1 + 0.8 \times 10 \times 0.0999} = 9.9975 \text{ V/V}$$

也就是说，它减少了 0.025%，比 A 的百分比变化小大约一个系数（1 + Aβ）。

练习

11.1 重复例 11.1(c)至(f)，A = 100 V/V。答案：(c) 10.11；(d) 20 dB；(e) 10 V、0.9 V、0.1 V；(f) 2.44

11.2 重复例 11.1 (c) 至 (f)，A= 10 V/V。(e) 使用 V= 0.01 V。答案：(c) 1110.1；(d) 20 dB；(e) 10 V、0.009 V、0.001 V；(f) 2.44%。

11.1.4 小结

在本节的最后，我们在表 11.1 中列出了表征图 11.1 中理想负反馈放大器结构的重要参数和公式。

Table 11.1 Summary of the Parameters and Formulas for the Ideal Feedback-Amplifier Structure of Fig. 11.1

- Open-loop gain $\equiv A$
- Feedback factor $\equiv \beta$
- Loop gain $\equiv A\beta$ (positive number)
- Amount of feedback $\equiv 1 + A\beta$
- Closed-loop gain $\equiv A_f = \dfrac{x_o}{x_s} = \dfrac{A}{1 + A\beta}$
- Feedback signal $\equiv x_f = \dfrac{A\beta}{1 + A\beta} x_s$
- Input signal to basic amplifier $\equiv x_i = \dfrac{1}{1 + A\beta} x_s$
- Closed-loop gain as a function of the ideal value $\dfrac{1}{\beta}$: $A_f = \left(\dfrac{1}{\beta}\right) \dfrac{1}{1 + 1/A\beta}$
- For large loop gain, $A\beta \gg 1$,

$$A_f \simeq \frac{1}{\beta} \qquad x_f \simeq x_s \qquad x_i \simeq 0$$

11.2 Some Properties of Negative Feedback

The properties of negative feedback were mentioned in the introduction. In the following, we shall consider some of these properties in more detail.

11.2.1 Gain Desensitivity

The effect of negative feedback on desensitizing the closed-loop gain was demonstrated in Example 11.1, where we saw that a 20% reduction in the gain of the basic amplifier gave rise to only a 0.025% reduction in the gain of the closed-loop amplifier. This sensitivity-reduction property can be analytically established as follows.

Assume that β is constant. Taking differentials of both sides of Eq. (11.4) results in

$$dA_f = \frac{dA}{(1 + A\beta)^2} \tag{11.9}$$

Dividing Eq. (11.9) by Eq. (11.4) yields

$$\frac{dA_f}{A_f} = \frac{1}{(1 + A\beta)} \frac{dA}{A} \tag{11.10}$$

which says that the percentage change in A_f (due to variations in some circuit parameter) is smaller than the percentage change in A by a factor equal to the amount of feedback. For this reason, the amount of feedback, $1 + A\beta$, is also known as the **desensitivity factor**.

表 11.1 图 11.1 理想反馈放大器结构的参数和公式摘要

- 开环增益 ≡ A
- 反馈系数 ≡ β
- 环路增益 ≡ Aβ（正数）
- 反馈量 ≡ 1 + Aβ
- 闭环增益 ≡ A= $\frac{x}{x} = \frac{A}{1 + A\beta}$
- 反馈信号 ≡ x= $\frac{A\beta}{1 + A\beta} x$
- 基本放大器的输入信号 ≡ x = $\frac{1}{1 + A\beta} x$
- 闭环增益与理想值的函数关系 $\frac{1}{\beta}$: A= $\frac{(\quad)1}{\frac{1}{\beta}\quad 1 + 1/A\beta}$
- 对于大环路增益，Aβ 为 1、

A□ $\frac{1}{\beta}$ x x x 0

11.2 负反馈的一些特性

导言中提到了负反馈的特性。下面，我们将更详细地讨论其中的一些特性。

11.2.1 增益灵敏度降低

例 11.1 演示了负反馈对闭环增益脱敏的作用，我们看到，基本放大器的增益降低 20%，闭环放大器的增益仅降低 0.025%。这一灵敏度降低特性可通过分析确定如下。

假设 β 为常数。对式（11.4）两边进行微分，结果是

$$dA = \frac{dA}{(1 + A\beta)} \tag{11.9}$$

将公式（11.9）除以公式（11.4）得出

$$\frac{dA}{A} = \frac{1}{(1 + A\beta)} \frac{dA}{A} \tag{11.10}$$

即 A 的百分比变化（由某些电路参数的变化引起）小于 A 的百分比变化，其系数等于反馈量。因此，反馈量（1 + Aβ）也被称为脱敏系数。

EXERCISE

11.3 An amplifier with a nominal gain $A = 1000$ V/V exhibits a gain change of 10% as the operating temperature changes from 25°C to 75°C. If it is required to constrain the change to 0.1% by applying negative feedback, what is the largest closed-loop gain possible? If three of these feedback amplifiers are placed in cascade, what overall gain and gain stability are achieved?

Ans. 10 V/V; 1000 V/V, with a maximum variability of 0.3% over the specified temperature range.

11.2.2 Bandwidth Extension

Consider an amplifier whose high-frequency response is characterized by a single pole. Its gain at mid and high frequencies can be expressed as

$$A(s) = \frac{A_M}{1 + s/\omega_H} \tag{11.11}$$

where A_M denotes the midband gain and ω_H is the upper 3-dB frequency. Application of negative feedback, with a frequency-independent factor β, around this amplifier results in a closed-loop gain $A_f(s)$ given by

$$A_f(s) = \frac{A(s)}{1 + \beta A(s)}$$

Substituting for $A(s)$ from Eq. (11.11) results, after a little manipulation, in

$$A_f(s) = \frac{A_M/(1 + A_M\beta)}{1 + s/\omega_H(1 + A_M\beta)} \tag{11.12}$$

Thus the feedback amplifier will have a midband gain of $A_M/(1 + A_M\beta)$ and an upper 3-dB frequency ω_{Hf} given by

$$\omega_{Hf} = \omega_H(1 + A_M\beta) \tag{11.13}$$

It follows that the upper 3-dB frequency is increased by a factor equal to the amount of feedback.

Similarly, it can be shown that if the open-loop gain is characterized by a dominant low-frequency pole giving rise to a lower 3-dB frequency ω_L, then the feedback amplifier will have a lower 3-dB frequency ω_{Lf},

$$\omega_{Lf} = \frac{\omega_L}{1 + A_M\beta} \tag{11.14}$$

Note that the amplifier bandwidth is increased by the same factor by which its midband gain is decreased, *maintaining the gain–bandwidth product at a constant value*. This point is further illustrated by the Bode plot in Fig. 11.4.

Finally, note that the action of negative feedback in extending the amplifier bandwidth should not be surprising: Negative feedback works to minimize the change in gain magnitude, including its change with frequency.

运动

11.3 一个放大器的标称增益 A = 1000 V/V，当工作温度从 25° C 变为 75° C 时，增益变化为 10%。如果要求通过负反馈将增益变化控制在 0.1%，那么最大闭环增益可能是多少？如果将三个这样的反馈放大器级联，可以获得多大的整体增益和增益稳定性？

答案：10 V/V；1000 V/V，在指定温度范围内最大变化率为 0.3%。

11.2.2 扩展带宽

考虑一个放大器，其高频响应以单极为特征。它在中频和高频的增益可表示为

$$A(s) = \frac{A}{1 + s/\omega} \tag{11.11}$$

其中 Aden 表示中频增益，ω 表示 3 分贝上限频率。在该放大器周围应用与频率无关的系数 β 的负反馈，可获得闭环增益 A(s)，其值为

$$A(s) = \frac{A(s)}{1 + \beta A(s)}$$

从公式（11.11）中代入 A(s)，稍加运算后得出

$$A(s) = \frac{A/(1 + A\beta)}{1 + s/\omega(1 + A\beta)} \tag{11.12}$$

因此，反馈放大器的中频增益为 A /（1 + Aβ），上 3 分贝频率 ω 的计算公式为

$$\omega = \omega(1 + A\beta) \tag{11.13}$$

因此，上限 3 分贝频率的增加系数等于反馈量。

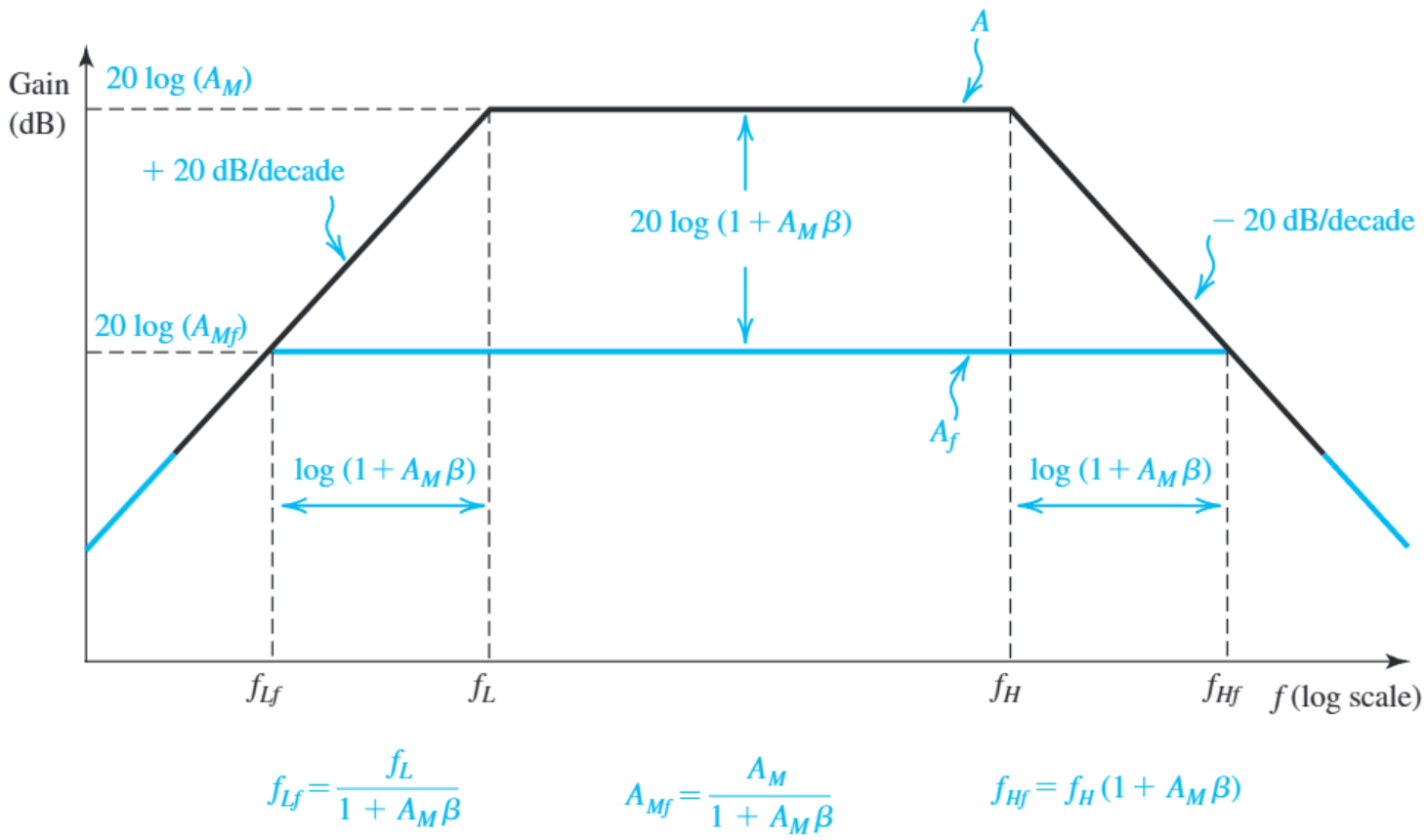
同样，可以证明，如果开环增益的特点是一个主要的低频极点会导致较低的 3 分贝频率 ω，那么反馈放大器也会有较低的 3 分贝频率 ω、

$$\omega = \frac{\omega}{1 + A\beta} \tag{11.14}$$

请注意，放大器带宽的增加系数与其中频带宽的增加系数相同。

增益降低，增益-带宽乘积保持不变。图 11.4 中的 Bode 图进一步说明了这一点。

最后，请注意负反馈在扩展放大器带宽方面的作用并不令人惊讶：负反馈的作用是最大限度地减小增益幅度的变化，包括其随频率的变化。



$$f_{Lf} = \frac{f_L}{1 + A_M\beta} \qquad A_{Mf} = \frac{A_M}{1 + A_M\beta} \qquad f_{Hf} = f_H(1 + A_M\beta)$$

Figure 11.4 Application of negative feedback reduces the midband gain, increases f_H, and reduces f_L, all by the same factor, $(1 + A_M\beta)$, which is equal to the amount of feedback.

EXERCISE

11.4 Consider the noninverting op-amp circuit of Example 11.1. Let the open-loop gain A have a low-frequency value of 10^4 and a uniform –6-dB/octave rolloff at high frequencies with a 3-dB frequency of 100 Hz. Find the low-frequency gain and the upper 3-dB frequency of a closed-loop amplifier with $R_1 = 1$ kΩ and $R_2 = 9$ kΩ.
Ans. 9.99 V/V; 100.1 kHz

11.2.3 Interference Reduction

Negative feedback can be employed to reduce the interference in an amplifier or, more precisely, to increase the ratio of signal to interference. However, as we shall now explain, this interference-reduction process is possible only under certain conditions. Consider the situation illustrated in Fig. 11.5. Figure 11.5(a) shows an amplifier with gain A_1, an input signal V_s, and interference, V_n. It is assumed that for some reason this amplifier suffers from interference and that the interference can be assumed to be introduced at the input of the amplifier. The **signal-to-interference ratio** for this amplifier is

$$S/I = V_s/V_n \tag{11.15}$$

Consider next the circuit in Fig. 11.5(b). Here we assume that it is possible to build another amplifier stage with gain A_2 that does not suffer from the interference problem. If this is the

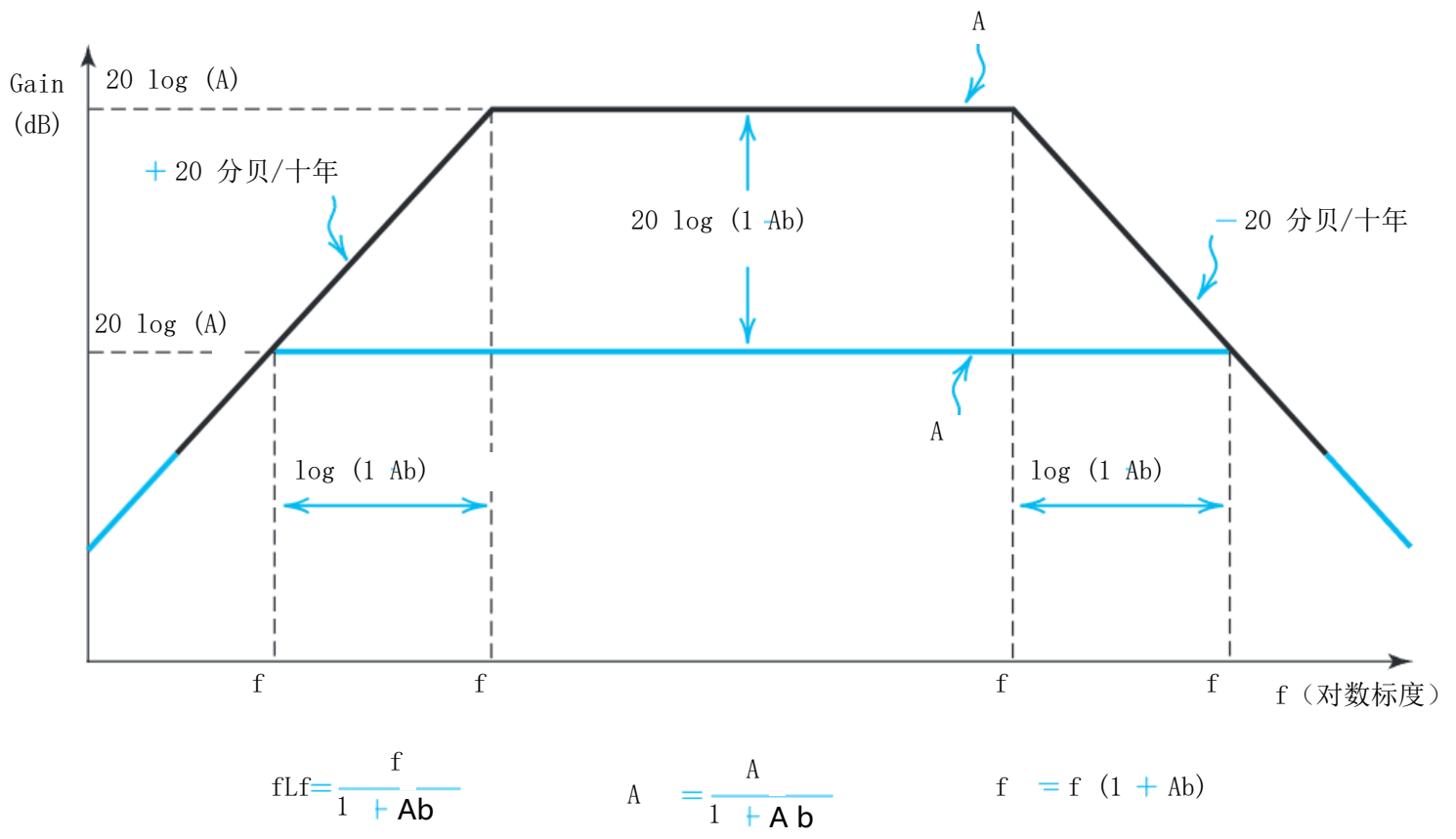


图 11.4 负反馈的应用会降低中频增益、增大 f 和减小 f，其系数均为 (1 + Aβ)，等于反馈量。

运动

11.4 考虑例 11.1 中的非反相运算放大器电路。假设开环增益 A 的低频值为 10，高频均匀为 -6 分贝/倍频程滚降，3 分贝频率为 100 赫兹。求 R= 1 k 和 R= 9 k 的闭环放大器的低频增益和上限 3 分贝频率。

答案：9.99 V/V；100.1 kHz

11.2.3 减少干扰

负反馈可以用来减少放大器中的干扰，或者更准确地说，增加信号与干扰的比率。然而，正如我们现在要解释的那样，这种降低干扰的过程只有在特定条件下才有可能实现。请看图 11.5 所示的情况。图 11.5(a) 显示了一个增益为 A 的放大器、输入信号 V 和干扰 V。假定由于某种原因，该放大器受到干扰，并且可以假定干扰是从放大器的输入端引入的。该放大器的信号干扰比为

$$S/I = V/V \tag{11.15}$$

再看图 11.5(b)中的电路。在此，我们假设可以建立另一个增益为 A 的放大器级，它不会受到干扰问题的影响。如果这是

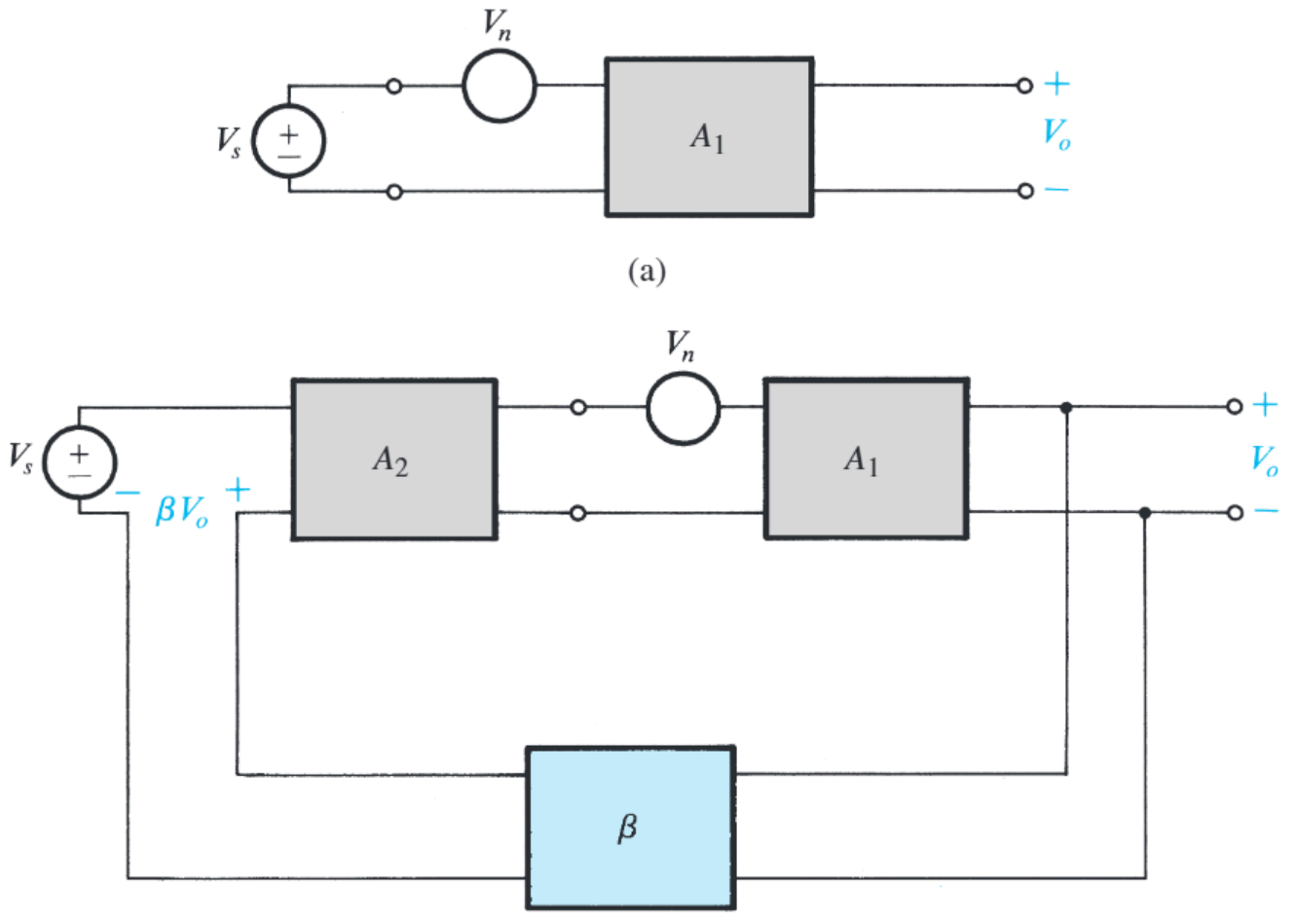


Figure 11.5 Illustrating the application of negative feedback to improve the signal-to-interference ratio in amplifiers.

case, then we may precede our original amplifier A_1 by the *clean* amplifier A_2 and apply negative feedback around the overall cascade of such an amount as to keep the overall gain unchanged. The output voltage of the circuit in Fig. 11.5(b) can be found by superposition:

$$V_o = V_s \frac{A_1 A_2}{1 + A_1 A_2 \beta} + V_n \frac{A_1}{1 + A_1 A_2 \beta} \tag{11.16}$$

Thus the signal-to-interference ratio at the output becomes

$$\frac{S}{I} = \frac{V_s}{V_n} A_2 \tag{11.17}$$

which is A_2 times higher than in the original case.

We emphasize once more that the improvement in signal-to-interference ratio by the application of feedback is possible only if one can precede the interference-prone stage by a (relatively) interference-free stage. This situation, however, is not uncommon in practice. The best example is found in the output power-amplifier stage of an audio amplifier. Such a stage usually suffers from a problem known as **power-supply hum**. The problem arises because of the large currents that this stage draws from the power supply and because it is difficult to provide adequate power-supply filtering inexpensively. The power-output stage is required to provide large power gain but little or no voltage gain. We may therefore precede the power-output stage by a small-signal amplifier that provides large voltage gain and then apply a large amount of negative feedback, thus restoring the voltage gain to its original value. Since the small-signal amplifier can be fed from another, less hefty (and hence better regulated) power supply, it will not suffer from the hum problem. The

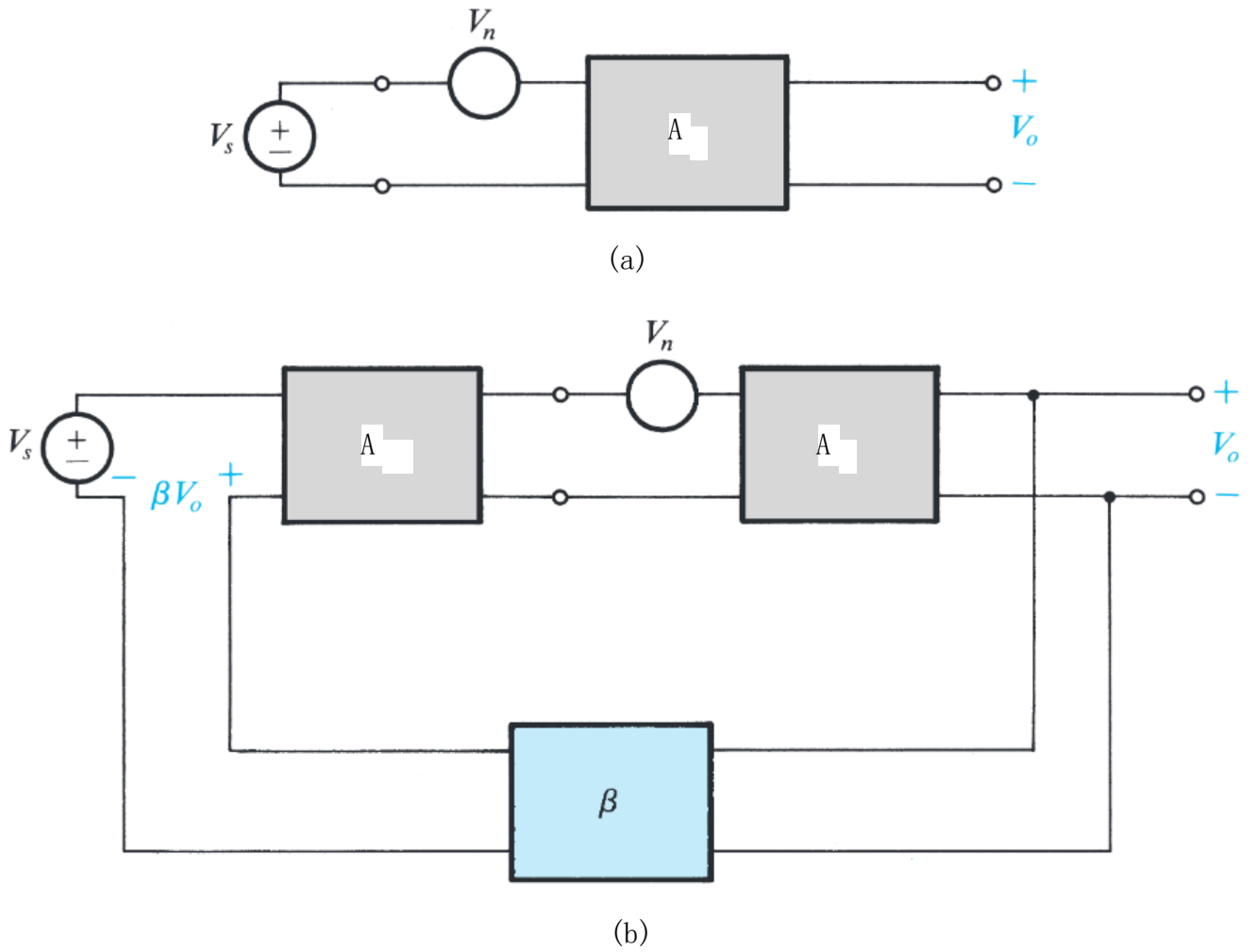


图 11.5 说明负反馈在放大器中的应用，以提高信噪比。

在这种情况下，我们可以在原始放大器 A 之前添加干净的放大器 A，并在整个级联电路周围施加一定量的负反馈，以保持整体增益不变。图 11.5(b) 中电路的输出电压可通过叠加求得：

$$V = V \frac{AA}{1 + AA\beta} + V \frac{A}{1 + AA\beta} \tag{11.16}$$

因此，输出端的信噪比变为

$$\frac{S}{I} = \frac{V}{V} A \tag{11.17}$$

比原来高出 Atimes。

我们再次强调，只有在易受干扰阶段之前有一个（相对）无干扰阶段时，才有可能通过应用反馈来提高信噪比。然而，这种情况在实践中并不少见。音频放大器的输出功率放大器级就是最好的例子。这种级通常会出现电源嗡嗡声的问题。出现这一问题的原因是该级从电源中获取的电流较大，而且很难以低廉的价格提供足够的电源滤波。电源输出级需要提供较大的功率增益，但电压增益很小或没有。因此，我们可以在功率输出级之前安装一个小信号放大器，提供较大的电压增益，然后再应用大量的负反馈，从而将电压增益恢复到原来的值。由于小信号放大器可以从另一个较小功率（因此稳压效果更好）的电源供电，因此不会出现嗡嗡声问题。嗡嗡声

hum at the output will then be reduced by the amount of the voltage gain of this added **preamplifier**.

EXERCISE

11.5 Consider a power-output stage with voltage gain $A_1 = 1$, an input signal $V_s = 1$ V, and a hum V_n of 1 V. Assume that this power stage is preceded by a small-signal stage with gain $A_2 = 100$ V/V and that overall feedback with $\beta = 1$ is applied. If V_s and V_n remain unchanged, find the signal and interference voltages at the output and hence the improvement in S/I.
Ans. $\simeq$1 V; $\simeq$0.01 V; 100 (40 dB)

11.2.4 Reduction in Nonlinear Distortion

Curve (a) in Fig. 11.6 shows the transfer characteristic v_O versus v_I of an amplifier. As indicated, the characteristic is piecewise linear, with the voltage gain changing from 1000 to 100 and then to 0. This nonlinear transfer characteristic will result in this amplifier generating a large amount of nonlinear distortion.

The amplifier transfer characteristic can be considerably **linearized** (i.e., made less nonlinear) through the application of negative feedback. That this is possible should not be too surprising, since we have already seen that negative feedback reduces the dependence of the overall closed-loop amplifier gain on the open-loop gain of the basic amplifier. Thus large

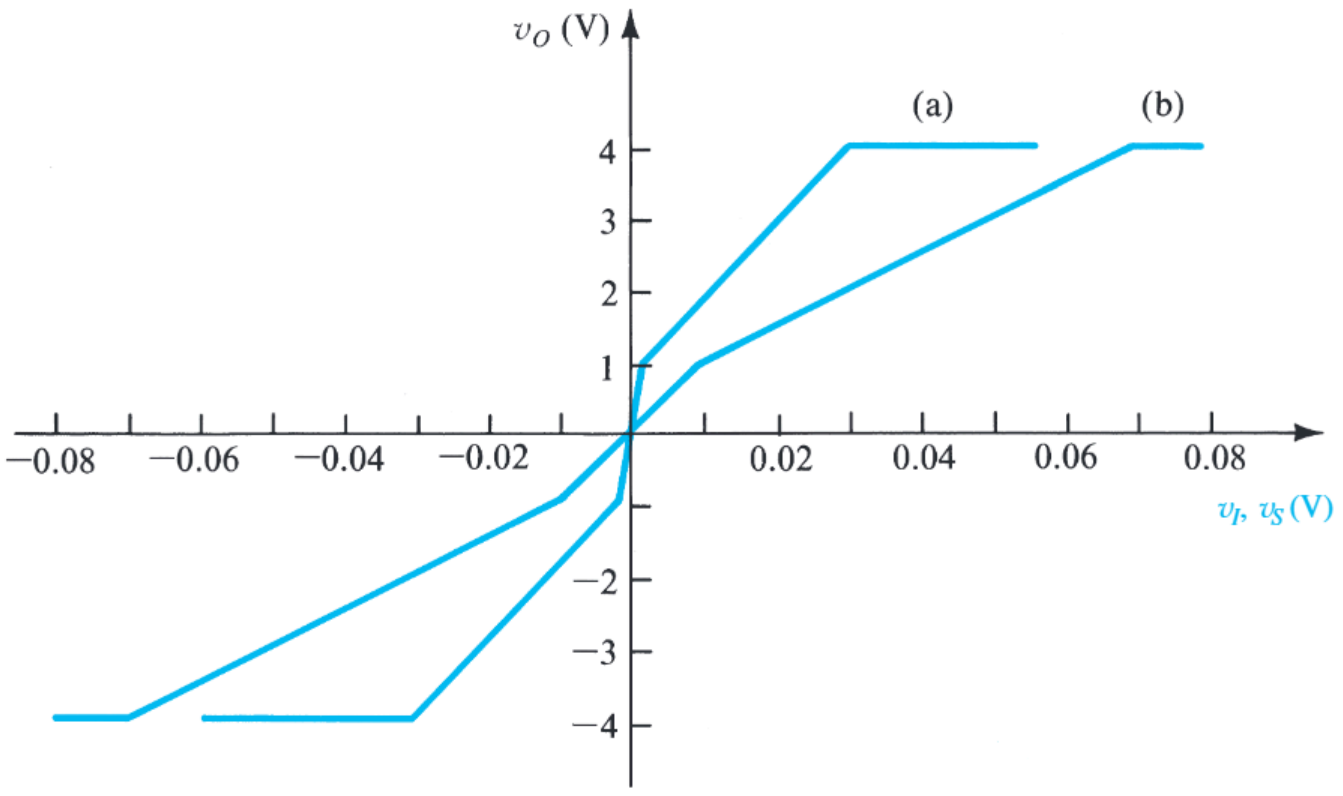


Figure 11.6 Illustrating the application of negative feedback to reduce the nonlinear distortion in amplifiers. Curve (a) shows the amplifier transfer characteristic (v_o versus v_I) without feedback. Curve (b) shows the characteristic (v_o versus v_S) with negative feedback ($\beta = 0.01$) applied.

这样，输出端的嗡嗡声就会减小，减小的幅度与添加的前置放大器的电压增益相同。

运动

11.5 考虑一个功率输出级，其电压增益 A= 1，输入信号 V= 1 V，哼声 V 为 1 V。假设该功率级之前有一个增益 A= 100 V/V 的小信号级，并应用了 β = 1 的整体反馈。如果 V 和 V 保持不变，请求解输出端的信号电压和干扰电压，从而求解 S/I 的改进。

答案：1 V；0.01 V；100 (40 dB)

11.2.4 减少非线性失真

图 11.6 中的曲线 (a) 显示了放大器 vvers 与 v 的传递特性。如图所示，该特性是片断线性的，电压增益从 1000 变为 100，然后又变为 0。这种非线性传输特性将导致放大器产生大量非线性失真。

通过负反馈，放大器的传输特性可以大大线性化（即降低非线性）。这一点不足为奇，因为我们已经看到，负反馈降低了整体闭环放大器增益对基本放大器开环增益的依赖性。因此，大

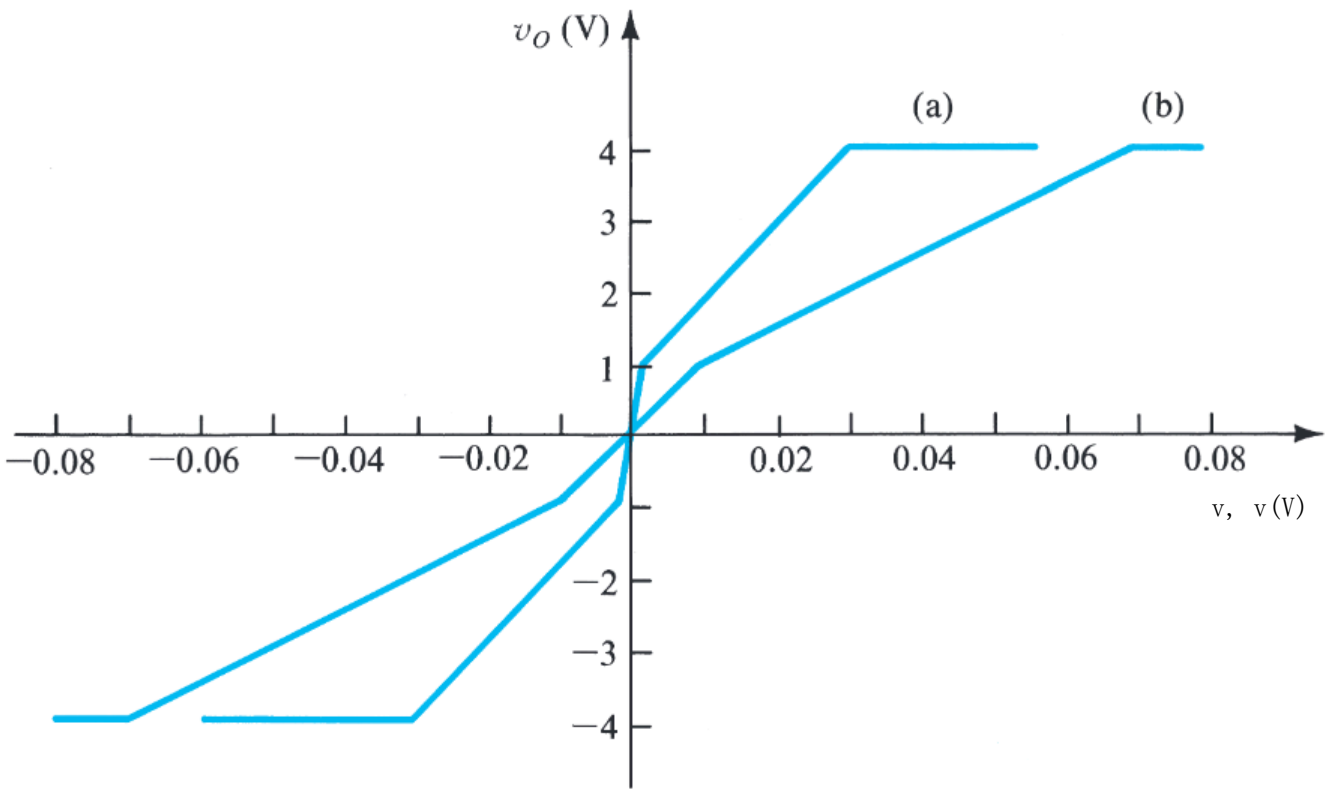


图 11.6 说明负反馈在减少放大器非线性失真的应用。曲线 (a) 显示的是无反馈时的放大器传输特性（v 对 v）。曲线 (b) 显示了应用负反馈 (β = 0.01) 时的特性（vversus v）。

changes in open-loop gain (1000 to 100 in this case) give rise to much smaller corresponding changes in the closed-loop gain.

To illustrate, let us apply negative feedback with $\beta = 0.01$ to the amplifier whose open-loop voltage transfer characteristic is depicted in Fig. 11.6. The resulting transfer characteristic of the closed-loop amplifier, v_o versus v_s, is shown in Fig. 11.6 as curve (b). Here the slope of the steepest segment is given by

$$A_{f1} = \frac{1000}{1 + 1000 \times 0.01} = 90.9$$

and the slope of the next segment is given by

$$A_{f2} = \frac{100}{1 + 100 \times 0.01} = 50$$

Thus the order-of-magnitude change in slope has been considerably reduced. The price paid, of course, is a reduction in voltage gain. Thus if the overall gain has to be restored, a preamplifier should be added. This preamplifier should not present a severe nonlinear-distortion problem, since it will be dealing with smaller signals.

Finally, it should be noted that negative feedback can do nothing at all about amplifier saturation, since in saturation the gain is very small (almost zero) and hence the amount of feedback is almost unity.

11.3 The Feedback Voltage Amplifier

Based on the quantity to be amplified (voltage or current) and on the desired form of output (voltage or current), amplifiers can be classified into four categories. These categories were discussed in Chapter 1. In this section we study the most common amplifier type: the voltage amplifier. We begin by identifying the appropriate configuration for applying negative feedback to a voltage amplifier. Then, we present a simple method for the analysis of the feedback voltage amplifier. The method makes use of the loop gain $A\beta$, whose determination was discussed in Section 11.1.3.

11.3.1 The Series–Shunt Feedback Topology

Voltage amplifiers are intended to amplify an input voltage signal and provide an output voltage signal. The voltage amplifier is essentially a voltage-controlled voltage source. The input resistance is required to be high, and the output resistance is required to be low. Since the signal source is essentially a voltage source, it is appropriately represented in terms of a Thévenin equivalent circuit. As the output quantity of interest is the output voltage, the feedback network should *sample* the output *voltage*, just as a voltmeter measures a voltage. Also, because of the Thévenin representation of the source, the feedback signal x_f should be a *voltage* that can be *mixed* with the source voltage in *series*.

From the discussion above, it follows that the most suitable feedback topology for the voltage amplifier is the **voltage-mixing**, **voltage-sampling** one shown in Fig. 11.7. Because of the series connection at the input and the parallel or shunt connection at the output, this feedback topology is also known as **series–shunt feedback**. As will be shown, this topology not only stabilizes the voltage gain V_o/V_s but also results in a higher input resistance R_{in} (intuitively, a result of the series connection at the input) and a lower output resistance R_{out}

开环增益的变化（本例中为 1000 到 100）导致闭环增益的相应变化要小得多。

为了说明这一点，让我们将 β = 0.01 的负反馈应用于放大器，其开环电压传输特性如图 11.6 所示。由此产生的闭环放大器传输特性 vversus v 如图 11.6 中的曲线 (b) 所示。其中最陡段的斜率为

$$A = \frac{1000}{1 + 1000 \times 0.01} = 90.9$$

而下一段的斜率则为

$$A = \frac{100}{1 + 100 \times 0.01} = 50$$

这样，斜率的数量级变化就大大降低了。当然，代价是电压增益的降低。因此，如果需要恢复整体增益，则应增加一个前置放大器。由于前置放大器处理的信号较小，因此不会产生严重的非线性失真问题。

最后需要指出的是，负反馈对放大器的饱和完全不起作用，因为在饱和状态下，增益非常小（几乎为零），因此反馈量几乎为零。

11.3 反馈电压放大器

根据需要放大的量（电压或电流）和所需的输出形式（电压或电流），放大器可分为四类。这些类别已在第 1 章中讨论过。在本节中，我们将研究最常见的放大器类型：电压放大器。首先，我们要确定将负反馈应用于电压放大器的适当配置。然后，我们介绍一种分析反馈电压放大器的简单方法。该方法利用环路增益 Aβ，其确定方法已在第 11.1.3 节中讨论过。

11.3.1 串并联反馈拓扑结构

电压放大器用于放大输入电压信号并提供输出电压信号。电压放大器本质上是一个压控电压源。输入电阻要求高，输出电阻要求低。由于信号源本质上是一个电压源，因此可以用等效电路来表示。由于所关注的输出量是输出电压，因此反馈网络应采样输出电压，就像电压表测量电压一样。此外，由于信号源采用 Th′evenin 表示法，反馈信号 x 应该是可以与串联的信号源电压混合的电压。

综上所述，最适合电压放大器的反馈拓扑结构是图 11.7 所示的电压混合、电压采样拓扑结构。由于输入端采用串联连接，输出端采用并联或并联连接，这种反馈拓扑结构也被称为串并联反馈。如图所示，这种拓扑结构不仅能稳定电压增益 V /V，还能使输入电阻 R 变大（直观地说，这是输入端串联的结果），输出电阻 R 变小（直观地说，这是输出端并联的结果）。

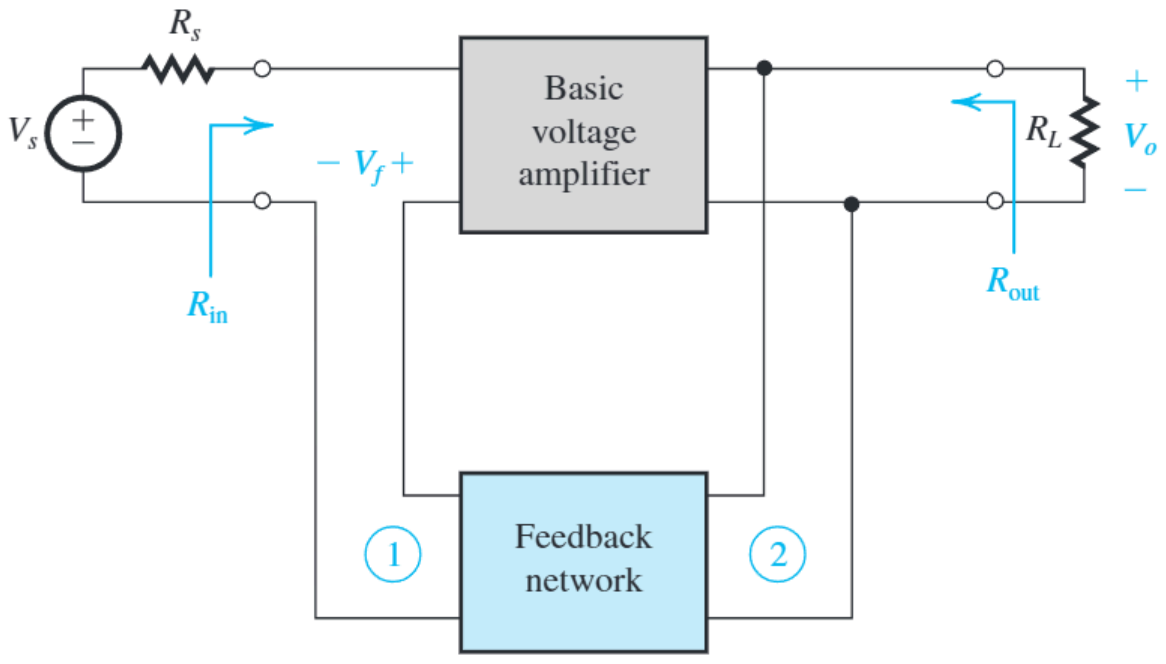


Figure 11.7 Block diagram of a feedback voltage amplifier. Here the appropriate feedback topology is series–shunt.

(intuitively, a result of the parallel connection at the output), which are desirable properties for a voltage amplifier.

The increased input resistance results because V_f subtracts from V_s, resulting in a smaller signal V_i at the input of the basic amplifier. The lower V_i, in turn, causes the input current to be smaller, with the result that the resistance seen by V_s will be larger. We shall derive a formula for the input resistance of the feedback voltage amplifier in the next section.

The decreased output resistance results because the feedback works to keep V_o as constant as possible. Thus if the current drawn from the amplifier output changes by ΔI_o, the change ΔV_o in V_o will be lower than it would have been if feedback were not present. Thus the output resistance $\Delta V_o/\Delta I_o$ will be lower than that of the open-loop amplifier. In the following section we shall derive an expression for the output resistance of the feedback voltage amplifier.

11.3.2 Examples of Series–Shunt Feedback Amplifiers

Three examples of series–shunt feedback amplifiers are shown in Fig. 11.8. The amplifier in Fig. 11.8(a) is the familiar noninverting op-amp configuration. The feedback network, composed of the voltage divider (R_1, R_2), develops a voltage V_f that is applied to the negative input terminal of the op amp. The subtraction of V_f from V_s is achieved by utilizing the differencing action of the op-amp differential input. For the feedback to be negative, V_f must be of the same polarity as V_s, thus resulting in a smaller signal at the input of the basic amplifier. To ascertain that this is the case, we follow the signal around the loop, as follows: As V_s increases, V_o increases and the voltage divider causes V_f to increase. Thus the change in V_f is of the same polarity as the change in V_s, and the feedback is negative.

The second feedback voltage amplifier, shown in Fig. 11.8(b), utilizes two MOSFET amplifier stages in cascade. The output voltage V_o is sampled by the feedback network composed of the voltage divider (R_1, R_2), and the feedback signal V_f is fed to the source terminal of Q_1. The subtraction is implemented by applying V_s to the gate of Q_1 and V_f to its source, with the result that the signal at this amplifier input $V_i = V_{gs} = V_s - V_f$. To ascertain that the feedback is negative, let V_s increase. The drain voltage of Q_1 will decrease, and since this is applied to the gate of Q_2, its drain voltage V_o will increase. The feedback network will

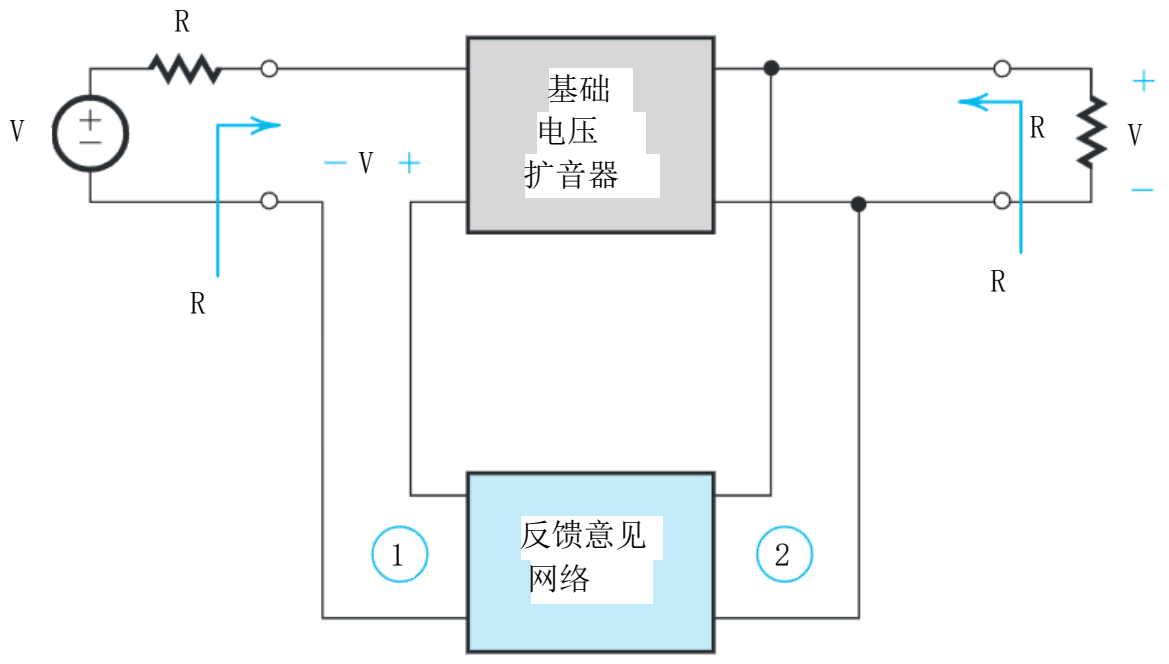


图 11.7 反馈电压放大器方框图。此处的反馈拓扑结构为串并联。

（直观地说，这是输出端并联的结果），这是电压放大器的理想特性。

输入电阻增大的原因是 Vs 从 V 中减去，导致基本放大器输入端的信号 V 变小。反过来，V 越小，输入电流就越小，V 所见的电阻也就越大。我们将在下一节推导出反馈电压放大器的输入电阻公式。

输出电阻减小的原因是反馈作用使 Vas 尽可能保持恒定。因此，如果放大器输出电流发生 I 变化，Vin Vw 的变化将低于没有反馈时的变化。因此，输出电阻 V/I 将低于开环放大器的输出电阻。在下一节中，我们将推导出反馈电压放大器输出电阻的表达式。

11.3.2 串并联反馈放大器示例

图 11.8 显示了串并联反馈放大器的三个示例。图 11.8(a) 中的放大器是我们熟悉的非反相运算放大器结构。由分压器 (R, R) 组成的反馈网络产生一个电压 V，该电压被施加到运算放大器的负输入端。利用运算放大器差分输入端的差分作用，实现 Vf 的减法。要使反馈为负，Vm 的极性必须与 V 相同，从而使基本放大器输入端的信号变小。为了确定情况是否如此，我们可在环路上跟踪信号，如下所示：随着 Vin 的增加，Vin 也会增加，分压器会导致 Vto 增加。因此，Vis 的变化与 V 的变化具有相同的极性，反馈为负。

第二个反馈电压放大器如图 11.8(b)所示，采用两个级联的 MOSFET 放大器级。输出电压由分压器（R，R）组成的反馈网络采样，反馈信号输入 Q 的源极。Q 的漏极电压将减小，而由于该电压施加在 Q 的栅极上，其漏极电压 Vw 将增大。反馈网络将

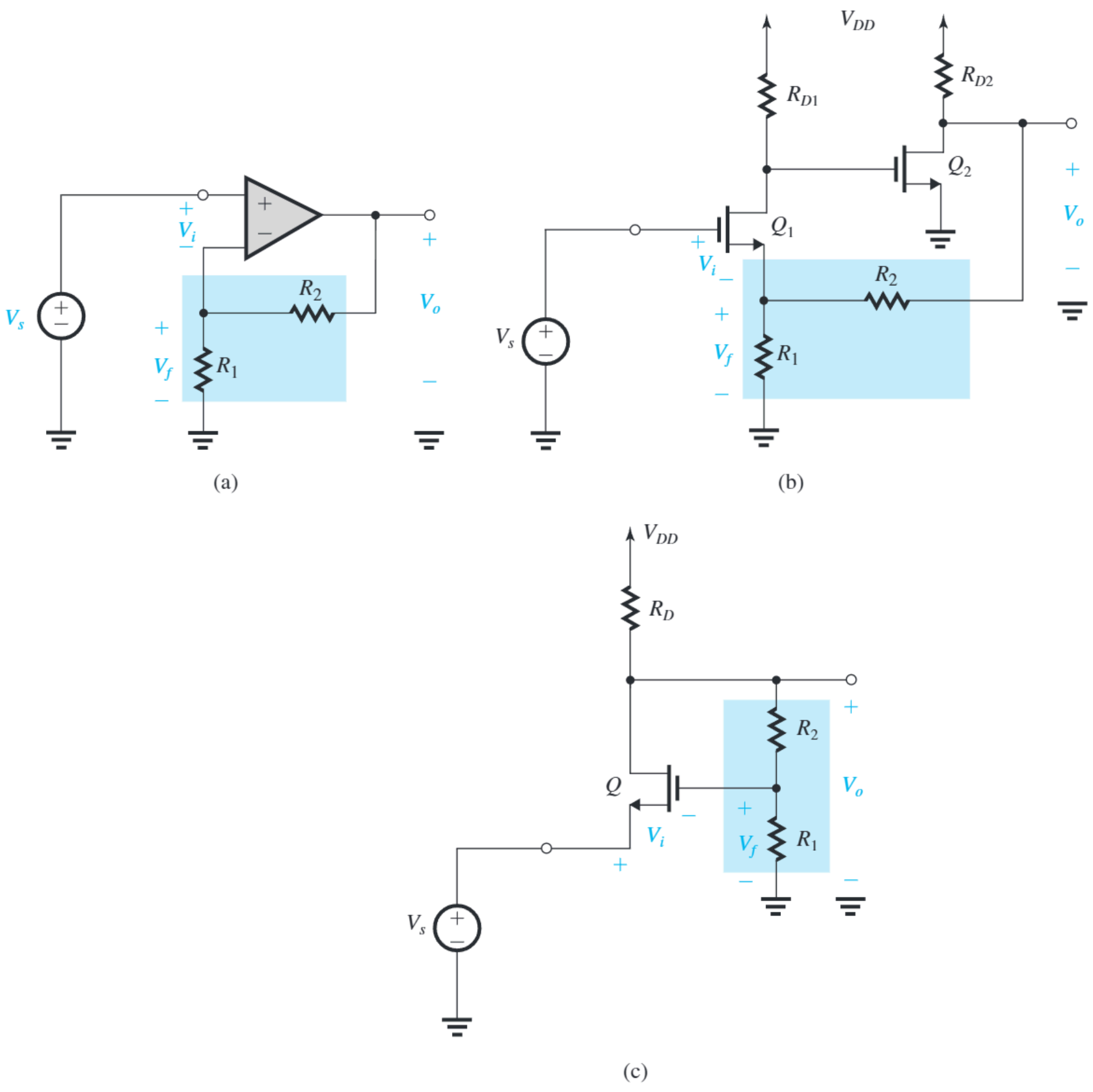


Figure 11.8 Examples of a feedback voltage amplifier. All these circuits employ series–shunt feedback. Note that the dc bias circuits are only partially shown.

then cause V_f to increase, which is the same polarity initially assumed for the change in V_s. Thus the feedback is indeed negative.

The third example of series–shunt feedback, shown in Fig. 11.8(c), utilizes a CG transistor Q with a fraction V_f of the output voltage V_o fed back to the gate through a voltage divider (R_1, R_2). Observe that the subtraction of V_f from V_s is effected by applying V_s to the source, thus the input V_i to the CG amplifier is obtained as $V_s - V_f$. As usual, however, we must check the polarity of the feedback: If V_s increases, V_d (which is V_o) will increase and V_f will correspondingly increase. Thus V_f and V_s change in the same direction, verifying that the feedback is negative.

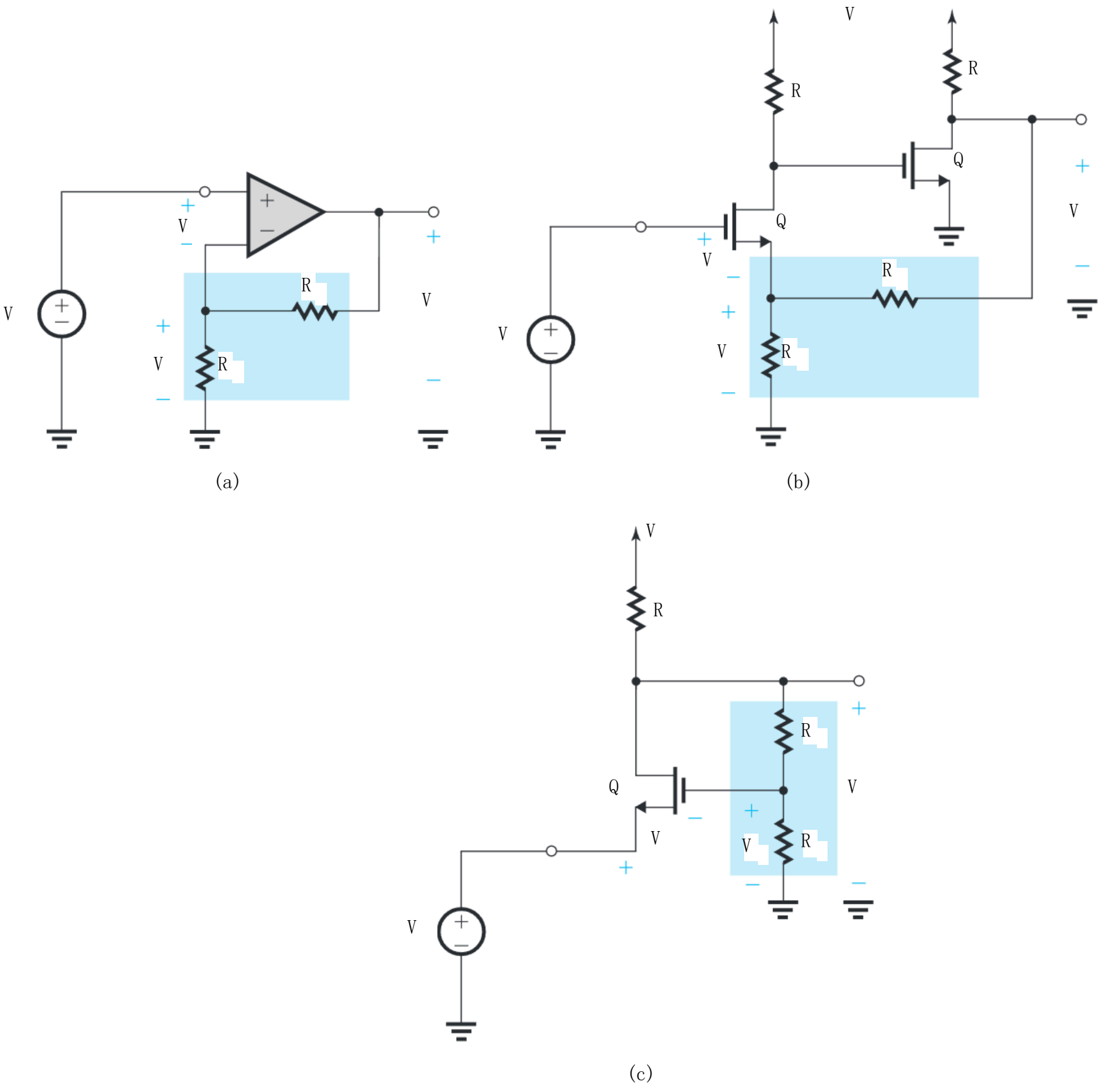


图 11.8 反馈电压放大器示例。所有这些电路都采用串并联反馈。
请注意，直流偏置电路只显示了一部分。

然后导致 Vto 增加，这与最初假设的 V 变化的极性相同。
因此，反馈确实是负面的。
第三个串并联反馈示例如图 11.8(c)所示，利用 CG 晶体管 Q 将输出电压 Vf 的一部分 V 通过分压器 (R, R) 传回栅极。请注意，Vf 与 Vis 的减法是通过向源极施加 V 来实现的，因此 CG 放大器的输入 V 为 V-V：如果 Vin 增加，V（即 V）将增加，Vw 也将相应增加。因此，V 和 V 的变化方向相同，证明反馈为负。

FEEDBACK—HISTORICAL NOTE:

The idea of feedback as an element of self-regulating behavior dates back to the eighteenth century, but the term itself did not appear in the context of a discussion on economics until the 1860s. Still later, in 1909, Karl Ferdinand Braun, a German physicist working at the University of Strasbourg, referred publicly to feedback as an undesired coupling between components of a vacuum-tube electronic system. The occasion was the lecture Braun delivered as a recipient of the Nobel Prize in Physics, shared with Guglielmo Marconi (often solely credited as the inventor of radio).

In 1927, Harold Black, at Bell Labs, invented the negative-feedback amplifier, which he described in detail in a seminal paper, "Stabilized Feedback Amplifiers," published in 1934. This invention was motivated by the need to provide low-distortion amplifiers that could be concatenated in long-distance transcontinental telephone circuits.

11.3.3 Analysis of the Feedback Voltage Amplifier Utilizing the Loop Gain

The feedback analysis method studied in Section 11.1 *cannot* be directly applied to a practical feedback voltage amplifier such as those in Fig. 11.8. This is because the analysis method of Section 11.1 is predicated on the assumption that the feedback network does not load the basic amplifier. Unfortunately, this assumption does not hold in most practical amplifier circuits. As shown in the circuits of Fig. 11.8, the feedback network is a simple resistive circuit that obviously loads the basic amplifier. As an example, in the circuit of Fig. 11.8(b), the values of the resistances R_2 and R_1, which comprise the feedback network, affect the gain of the common-source stage Q_2, which is part of the basic amplifier. Also, the value of the feedback-network resistance R_1 affects the gain of the Q_1 amplifier stage, which is part of the basic amplifier. It follows that we cannot easily disassemble a practical amplifier circuit to determine A and β and thus be able to use the feedback formulas of Sections 11.1 and 11.2.

While it is not easy to determine A and β, their product, the loop gain $A\beta$, can always be determined using the method presented in Section 11.1.3. Also, we can easily obtain the value of β by identifying and isolating the feedback network (e.g., the resistive divider (R_1, R_2) in each of the circuits in Fig. 11.8). We can then use the values of $A\beta$ and β to determine A and A_f. This loop-gain method is simple, and we shall use it in this section to perform the analysis of the feedback voltage amplifier. The method, however, has limitations that will be mentioned later. A more accurate and systematic approach for the analysis of feedback voltage amplifiers will be presented in the next section.

The loop-gain analysis method comprises four steps:

1. Identify the feedback network and use it to determine the value of β.
2. Determine the ideal value of the closed-loop gain A_f as $1/\beta$. This value of A_f is approached when $A\beta \gg 1$. The ideal or upper-bound value of A_f can be used in the initial design of the feedback amplifier. It also serves as a check on the actual value of A_f calculated below.
3. Use the method described in Section 11.1.3 to determine the loop gain $A\beta$. Recall that in breaking the loop, care should be taken to not change the conditions in the loop. Thus, if we break a feedback loop at XX′, as shown in Fig. 11.9(a), and apply a test voltage V_t to the terminals thus created to the left of XX′, the terminals to the right of XX′ must be connected to an impedance Z_t. The value of Z_t is equal

反馈历史注意：

反馈作为自我调节行为的一个要素，其概念可追溯到十八世纪，但该术语本身直到十九世纪六十年代才出现在经济学的讨论中。再晚一些，1909 年，在斯特拉斯堡大学工作的德国物理学家卡尔-费迪南德-布劳恩（Karl Ferdinand Braun）公开将反馈称为真空管电子系统元件之间不希望出现的耦合。当时，布劳恩作为诺贝尔物理学奖获得者发表了演讲，他与古列尔莫-马可尼（通常被认为是无线电的唯一发明者）共同获得了诺贝尔物理学奖。

1927 年，贝尔实验室的哈罗德-布莱克发明了负反馈放大器，他在 1934 年发表的开创性论文《稳定反馈放大器》中对此进行了详细描述。这项发明的动机是需要提供低失真放大器，以便在横贯大陆的长途电话电路中进行串联。

11.3.3 利用环路增益分析反馈电压放大器

第 11.1 节中研究的反馈分析方法不能直接应用于实际反馈电压放大器，如图 11.8 中的放大器。这是因为第 11.1 节中的分析方法是基于反馈网络不会对基本放大器造成负载的假设。遗憾的是，这一假设在大多数实际放大器电路中并不成立。如图 11.8 的电路所示，反馈网络是一个简单的电阻电路，显然会对基本放大器造成负载。例如，在图 11.8(b) 的电路中，构成反馈网络的电阻 Rand R 的值会影响共源级 Q 的增益，而共源级 Q 是基本放大器的一部分。此外，反馈网络电阻 Raff 的值也会影响作为基本放大器一部分的 Q 放大级的增益。因此，我们无法轻易拆解一个实用的放大器电路来确定 A 和 β，从而使用第 11.1 和 11.2 节中的反馈公式。

虽然确定 A 和 β 并不容易，但它们的乘积，即环路增益 Aβ，总是可以用第 11.1.3 节中介绍的方法来确定。此外，我们还可以通过识别和隔离反馈网络（例如图 11.8 中各电路中的电阻分压器 (R，R)）来轻松获得 β 值。这种环路增益方法非常简单，我们将在本节中使用它来分析反馈电压放大器。不过，这种方法也有局限性，稍后会提到。下一节将介绍一种更精确、更系统的反馈电压放大器分析方法。

环路增益分析方法包括四个步骤：

1. 确定反馈网络，并利用它来确定 β 值。
2. 确定闭环增益 A 的理想值为 1/β。当 Aβ 为 1 时，A 值就接近这个值。Ac 的理想值或上限值可用于反馈放大器的初始设计。它还可用于检验下面计算的 Ac 实际值。
3. 使用第 11.1.3 节所述方法确定环路增益 Aβ。回想一下，在断开回路时，应注意不要改变回路中的条件。因此，如果我们在 XX 处断开反馈回路（如图 11.9(a)所示），并在 XX 左侧的端子上施加测试电压 V，则 XX 右侧的端子必须与阻抗 Z 相连。Z 的值等于

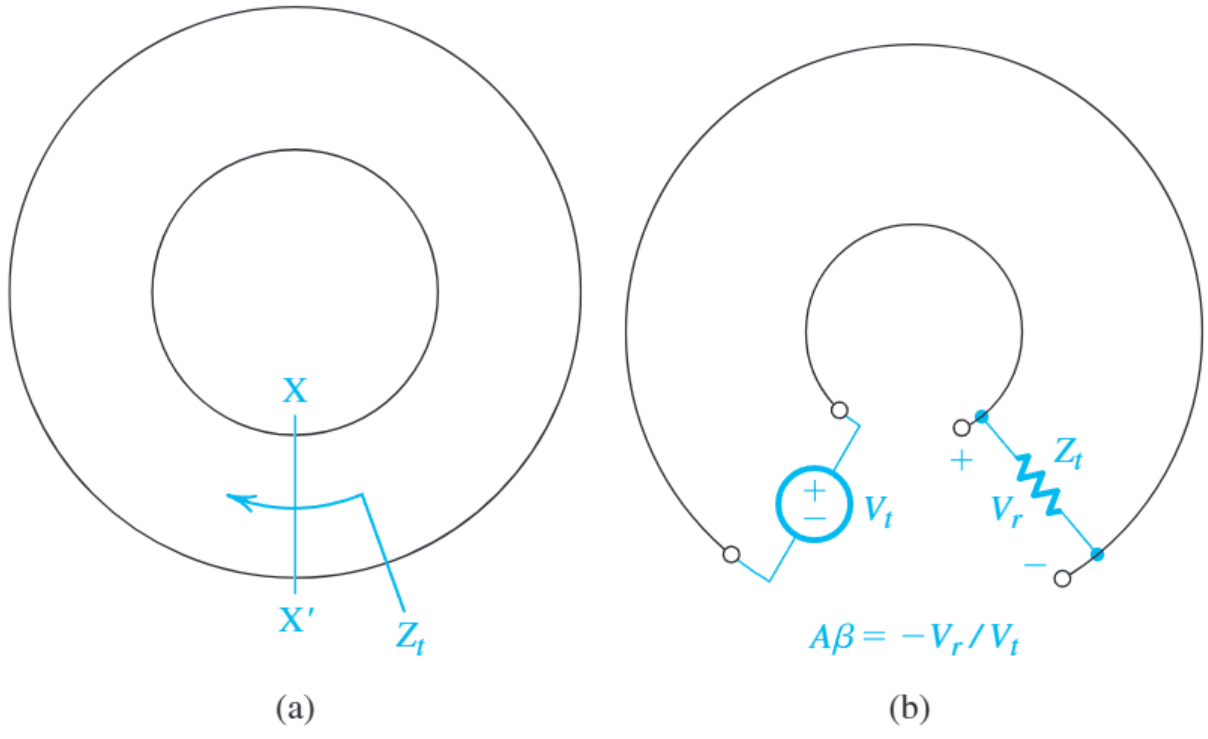


Figure 11.9 Breaking the conceptual feedback loop in (**a**) to determine the loop gain requires the termination of the loop as shown in (**b**), to ensure that the loop conditions do not change.

to the impedance previously seen looking to the left of XX′. The loop gain is then determined from

$$A\beta = -\frac{V_r}{V_t}$$

Whenever possible, we should break the loop at a location where Z_t is infinite.

4. Use the value of $A\beta$ together with that of β to determine the open-loop gain A. Then, determine the closed-loop gain A_f from

$$A_f = \frac{A}{1 + A\beta}$$

We shall illustrate the application of the loop-gain method via two examples.

Example 11.2

For the series–shunt feedback amplifier of Fig. 11.8(b), neglect the MOSFETs' r_o and

(a) give the feedback network (β circuit) and an expression for β. Also give an expression for the ideal or upper-bound value of the closed-loop gain A_f.
(b) find the ratio R_2/R_1 that results in an ideal closed-loop gain of 10 V/V. If $R_1 = 1$ kΩ, what value must R_2 have?
(c) find an expression for the loop gain $A\beta$.
(d) if $g_{m1} = g_{m2} = 4$ mA/V and $R_{D1} = R_{D2} = 10$ kΩ, determine the values of $A\beta$, A, and A_f.

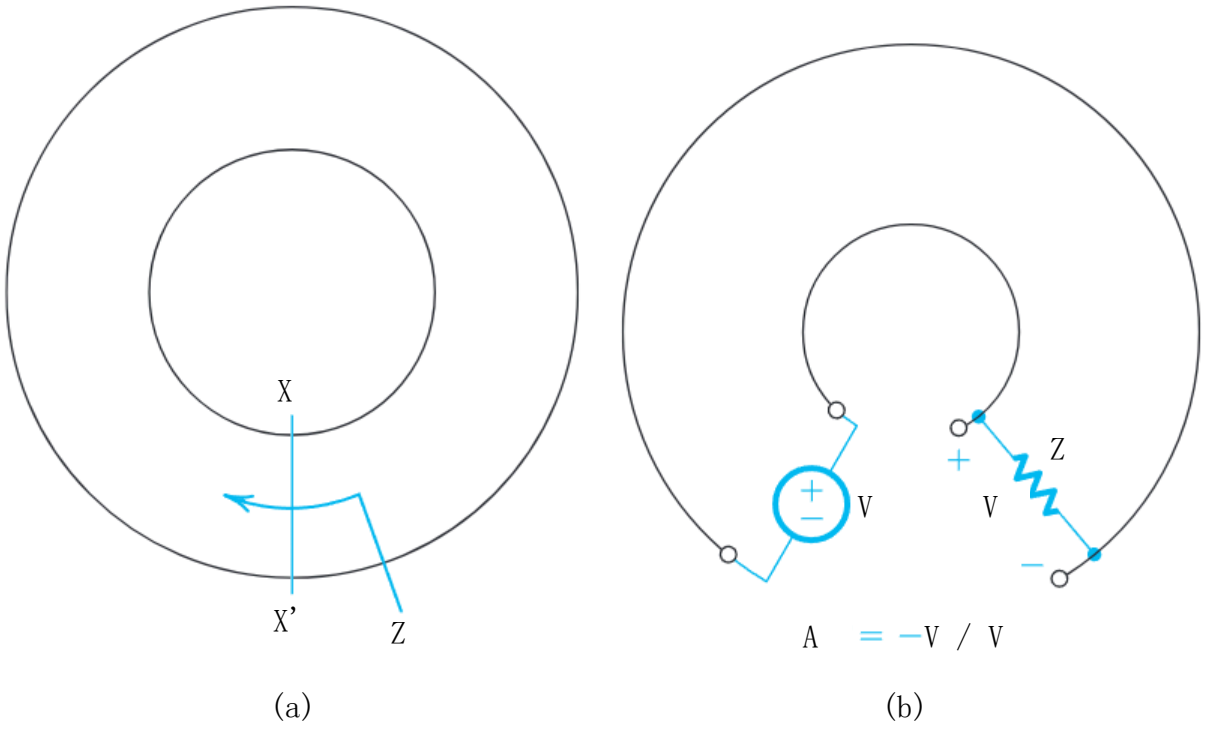


图 11.9 利用（a）中的概念反馈回路来确定环路增益，需要终止（b）中所示的环路，以确保环路条件不发生变化。

的阻抗。环路增益由以下公式确定

$$A\beta = -\frac{V}{V^-}$$

只要有可能，我们就应该在 Z 无穷大的地方打破循环。

4. 利用 Aβ 值和 β 值确定开环增益 A，然后确定闭环增益 Afrom

$$A = \frac{A}{1 + A\beta}$$

我们将通过两个例子来说明环路增益法的应用。

例 11.2

对于图 11.8(b)中的串联-并联反馈放大器，忽略 MOSFET 的 r 和 r。

(a) 请给出反馈网络（β 电路）和 β 的表达式，并给出闭环增益 A 的理想值或上限值的表达式。如果 R= 1 k，Rh 的值必须是多少？(c) 求环路增益 Aβ 的表达式。(d) 如果 g = g= 4 mA/V，R= R= 10 k，请确定 Aβ、A 和 A 的值。

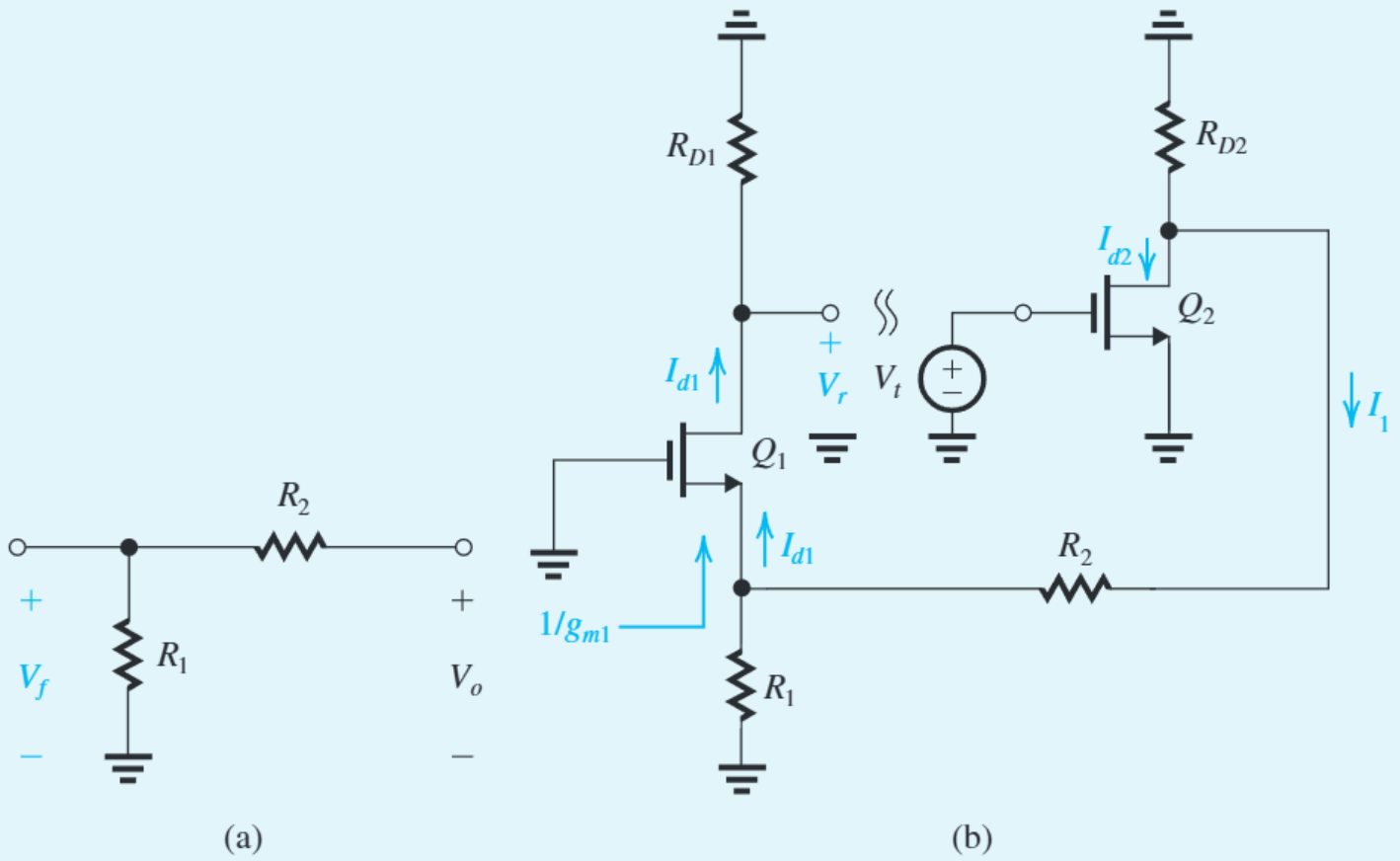


Figure 11.10 Determining: (**a**) the feedback factor β; and (**b**) the loop gain $A\beta$ for the feedback voltage amplifier of Fig. 11.8(b).

Solution

(a) The feedback network is highlighted in Fig. 11.8(b) and is redrawn in Fig. 11.10(a). It is a simple resistive voltage divider. Thus,

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

For $A\beta \gg 1$, the closed-loop gain $A_f = V_o/V_s$ is given by

$$A_f \simeq \frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

This is the ideal or upper-bound value of A_f.

(b) For A_f to have an ideal value of 10,

$$10 = 1 + \frac{R_2}{R_1}$$

Thus,

$$\frac{R_2}{R_1} = 9$$

For $R_1 = 1\ \text{k}\Omega$, $R_2 = 9\ \text{k}\Omega$.

(c) To determine the loop gain, we set $V_s = 0$. Examining the feedback loop reveals that it is most convenient to break the loop at the connection between the drain of Q_1 and the gate of Q_2. This is

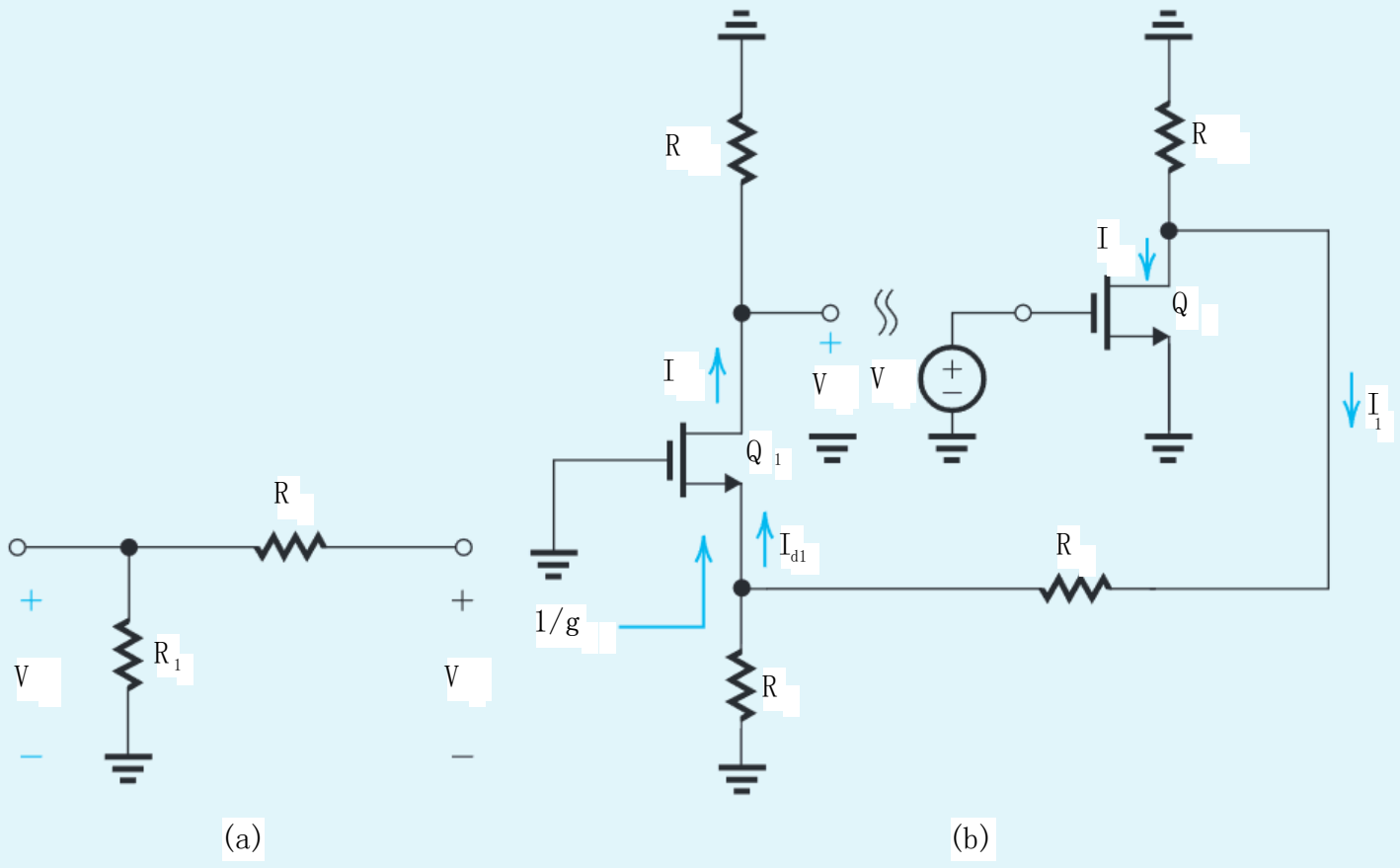


图 11.10 确定：(a) 图 11.8(b)中反馈电压放大器的反馈系数 β 和 (b) 环路增益 Aβ。

解决方案

(a) 图 11.8(b)突出显示了反馈网络，图 11.10(a)重新绘制了该网络。这是一个简单的电阻分压器。因此

$$\beta = \frac{V}{V} = \frac{R}{R+R}$$

对于 Aβ 1，闭环增益 A= V/V 的计算公式为

$$A \approx \frac{1}{\beta} = 1 + \frac{R}{R}$$

这就是 A 的理想值或上限值。 (b) A 的理想值为 10、

$$10 = 1 + \frac{R}{R}$$

因此、

$$\frac{R}{R} = 9$$

对于 R= 1 k，R= 9 k。

(c) 为了确定环路增益，我们设置 V= 0。检查反馈环路发现，在 Q 的漏极和 Q 的栅极之间的连接处断开环路最为方便。

Example 11.2 *continued*

because of the infinite input impedance at the gate of Q_2. The resulting circuit is shown in Fig. 11.10(b), for which the analysis to determine $A\beta \equiv -V_r/V_t$ proceeds as follows:

$$I_{d2} = g_{m2} V_t$$

$$I_1 = -I_{d2} \frac{R_{D2}}{R_{D2} + R_2 + \left(R_1 \| \frac{1}{g_{m1}}\right)}$$

$$I_{d1} = I_1 \frac{R_1}{R_1 + \frac{1}{g_{m1}}}$$

$$V_r = I_{d1} R_{D1}$$

Combining these four equations results in

$$A\beta \equiv -\frac{V_r}{V_t} = (g_{m1}R_{D1})(g_{m2}R_{D2}) \frac{1}{1 + g_{m1}R_1} \frac{R_1}{R_{D2} + R_2 + \left(R_1 \| \frac{1}{g_{m1}}\right)}$$

(d)

$$A\beta = 4 \times 10 \times 4 \times 10 \times \frac{1}{1 + 4 \times 1} \times \frac{1}{10 + 9 + (1 \| \frac{1}{4})}$$

$$= 16.67$$

Since

$$\beta = \frac{R_1}{R_1 + R_2} = \frac{1}{1 + 9} = 0.1$$

$$A = \frac{A\beta}{\beta} = \frac{16.67}{0.1} = 166.7 \text{ V/V}$$

and

$$A_f = \frac{A}{1 + A\beta} = \frac{166.7}{1 + 16.67} = \frac{166.7}{17.67} = 9.43 \text{ V/V}$$

which, given that the loop gain (16.67) is only moderately high, is reasonably close to the ideal value of 10 V/V.

Example 11.3

In the series–shunt feedback amplifier of Fig. 11.11(a), the op amp has an input resistance R_{id}, an open-circuit voltage gain μ, and an output resistance r_o. Find expressions for β, the ideal value of $A_f \equiv V_o/V_s$, and the loop gain $A\beta$.

例 11.2 续

结果电路如图 11.10(b)所示，确定 $A\beta \equiv -V/V$ 的分析过程如下：

$$I = g\,V$$

$$I = -I\,\frac{R}{R + R + \left(R \parallel \frac{1}{g}\right)}$$

$$I = I\,\frac{R}{R + \frac{1}{g}}$$

$$V = I\,R$$

将这四个等式合并得出

$$A\beta \equiv -\frac{V}{V} = (gR)(g\,R)\,\frac{1}{1 + gR}\,\frac{R}{R + R + \left(R \parallel \frac{1}{g}\right)}$$

(d)

$$A\beta = 4 \times 10 \times 4 \times 10 \times \frac{1}{1 + 4 \times 1} \times \frac{1}{10 + 9 + (1 \parallel)}$$

$$= 16.67$$

因为

$$\beta = \frac{R}{R + R} = \frac{1}{1 + 9} = 0.1$$

$$A = \frac{A\beta}{\beta} = \frac{16.67}{0.1} = 166.7\ V/V$$

and

$$A = \frac{A}{1 + A\beta} = \frac{166.7}{1 + 16.67} = \frac{166.7}{17.67} = 9.43\ V/V$$

考虑到环路增益（16.67）仅为中等偏高，该值与 10 V/V 的理想值相当接近。

例 11.3

在图 11.11(a)的串并联反馈放大器中，运算放大器有一个输入电阻 R、一个开路电压增益 μ 和一个输出电阻 r。求 β、$A \equiv V/V$ 的理想值和环路增益 Aβ 的表达式。

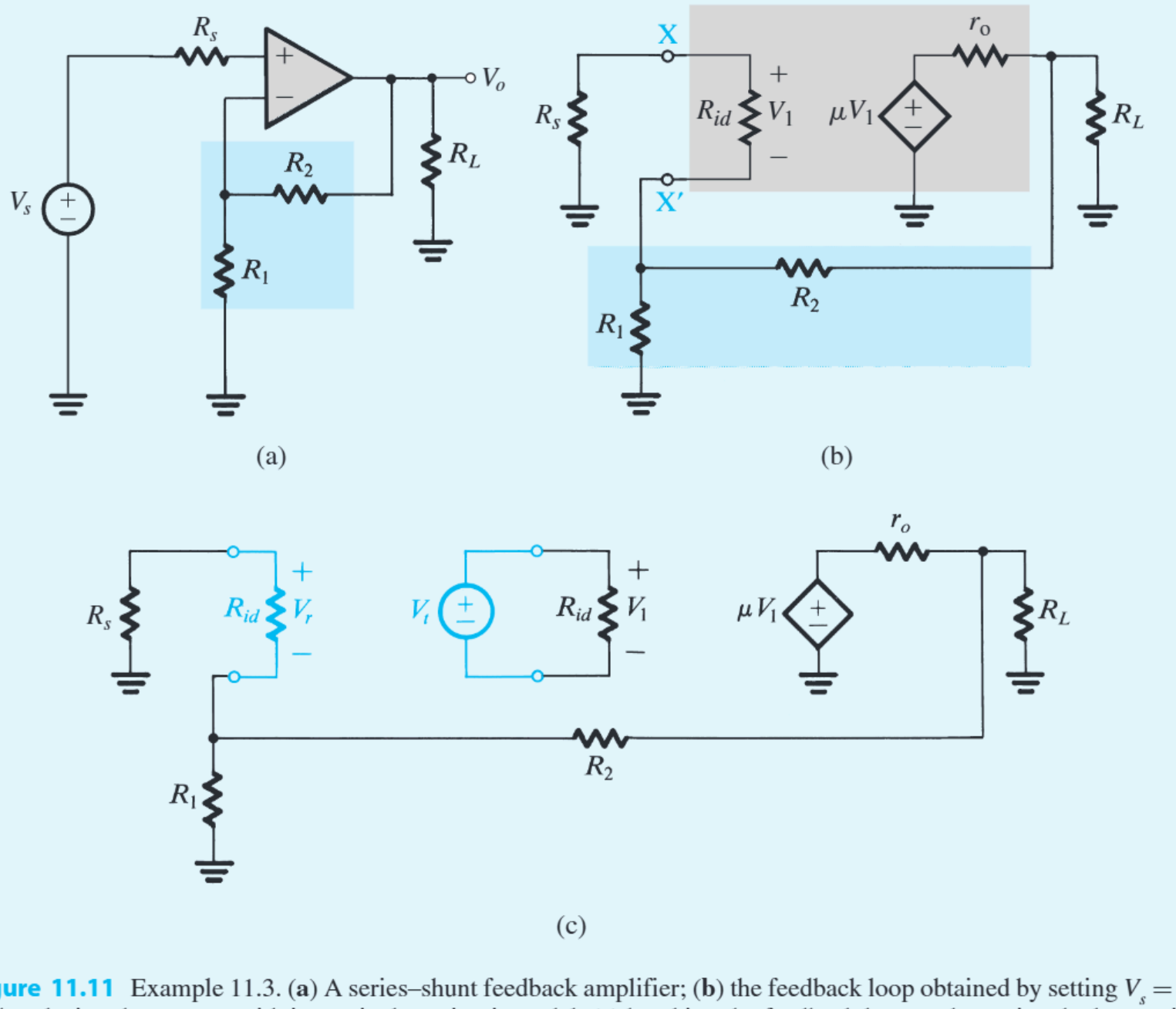


Figure 11.11 Example 11.3. (**a**) A series–shunt feedback amplifier; (**b**) the feedback loop obtained by setting $V_s = 0$ and replacing the op amp with its equivalent-circuit model; (**c**) breaking the feedback loop to determine the loop gain $A\beta = -V_r/V_t$.

Solution

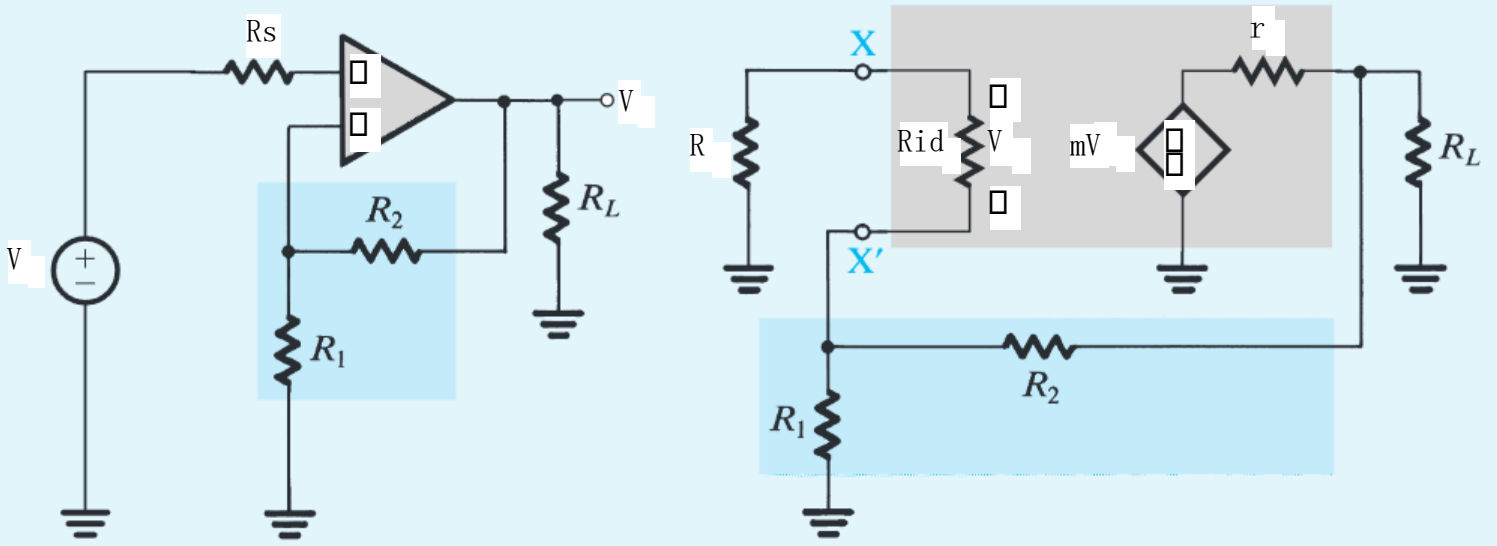
The feedback network consists of the voltage divider (R_1, R_2), thus

$$\beta = \frac{R_1}{R_1 + R_2}$$

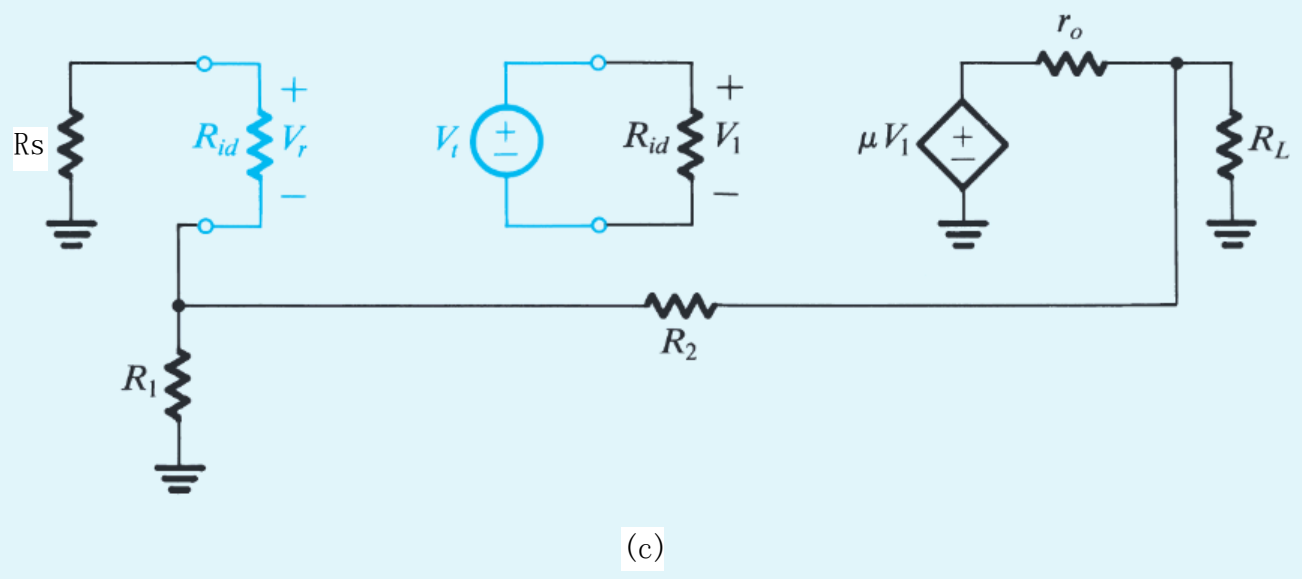
and the ideal value of A_f is

$$A_f = \frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

To determine the loop gain, we set $V_s = 0$ and replace the op amp with its equivalent-circuit model. The resulting circuit is shown in Fig. 11.11(b). Next, we break the loop to apply a test voltage V_t while terminating the loop at the break in an impedance equal to that seen prior to breaking the loop. The resulting circuit is shown in Fig. 11.11(c), where the loop has been broken at the input terminals of the op amp and the left-hand-side terminals are connected to a resistance equal to R_{id}. Analysis of the circuit to determine



(a) (b)



(c)

图 11.11 例 11.3 (a) 串并联反馈放大器；(b) 设置 V = 0 并用等效电路模型替换运算放大器后得到的反馈环路；(c) 打破反馈环路以确定环路增益 Aβ = -V/V。

解决方案

反馈网络由分压器 (R, R) 组成，因此

$$\beta = \frac{R}{R+R}$$

而 A 的理想值为

$$A = \frac{1}{\beta} = 1 + \frac{R}{R}$$

为了确定环路增益，我们设置 V= 0，并用等效电路模型替换运算放大器。由此产生的电路如图 11.11(b)所示。接下来，我们断开环路，施加测试电压 V，同时在断路处终止环路，阻抗等于断路前的阻抗。由此产生的电路如图 11.11(c)所示，环路在运算放大器的输入端断开，左侧端子连接到等于 R 的电阻上。分析电路以确定

Example 11.3 *continued*

$A\beta \equiv -V_r/V_t$ involves repeated application of the voltage divider rule, resulting in

$$A\beta = \mu \frac{\{R_L \| [R_2 + R_1 \| (R_{id} + R_s)]\}}{\{R_L \| [R_2 + R_1 \| (R_{id} + R_s)]\} + r_o} \times \frac{[R_1 \| (R_{id} + R_s)]}{[R_1 \| (R_{id} + R_s)] + R_2} \times \frac{R_{id}}{R_{id} + R_s}$$

EXERCISE

11.6 For the feedback voltage amplifier of Fig. 11.8(c):

(a) Find an expression for β.

(b) Neglecting the MOSFET r_o, find an expression for the loop gain $A\beta$. (*Hint*: Break the loop at the gate of Q.)

(c) Find an expression for the open-loop gain A.

(d) For $g_m = 4$ mA/V, $R_D = 10$ kΩ, $R_1 = 20$ kΩ, and $R_2 = 80$ kΩ, find the values of β, $A\beta$, A, and the closed-loop gain A_f. What would A_f be if $A\beta$ were much greater than unity?

Ans. (a) $\frac{R_1}{R_1 + R_2}$; (b) $g_m \frac{R_D R_1}{R_D + R_1 + R_2}$; (c) $g_m \frac{R_D(R_1 + R_2)}{R_D + R_1 + R_2}$; (d) 0.2, 7.27, 36.36 V/V, 4.4 V/V, 5 V/V.

11.3.4 A Final Remark

The loop-gain analysis method, though simple, is not complete: It does not enable us to find the input and output resistances of the feedback amplifier. This shortcoming is remedied in the next section, where we present a systematic approach to the analysis of feedback voltage amplifiers.

11.4 Systematic Analysis of Feedback Voltage Amplifiers

In this section we provide a systematic procedure for the analysis of feedback voltage amplifiers. The procedure essentially disassembles a given feedback voltage-amplifier circuit so as to obtain the "A circuit," from which we can determine the open-loop gain A and other parameters of the open-loop amplifier, such as the input and output resistances, and the "β circuit" from which the value of the feedback factor β can be found. We can then use the

例 11.3 续

$A\beta \equiv -V/V$ 涉及重复应用分压规则，结果是

$$A\beta = \mu \frac{\{r_o \| [r + r_o \| (r + r_o)]\}}{\{r_o \| [r + r_o \| (r + r_o)]\} + r_o \| [r + r_o \| (r + r_o)]} \times \frac{[g_m r_o \| (g_m r_o + g_m r_o)]}{[r_o' \| (r + r_o)] + r} \times \frac{R}{R + R}$$

运动

11.6 对于图 11.8(c)中的反馈电压放大器：

分析电路以确定 (a) 求出 β 的表达式。

(b) 忽略 MOSFET r_o，求环路增益 $A\beta$ 的表达式。(提示：在 Q 的栅极处断开回路。) (c) 求开环增益 A 的表达式。如果 $A\beta$ 远大于统一值，A 会是多少？

答案：(a) $\frac{R}{R + R}$ (b) $g_m \frac{R \| R}{R + R + R}$ (c) $g_m \frac{R \| (R + R)}{R + R + R}$ (d) 0.2、7.27、36.36 V/V、4.4 V/V、5 V/V。

11.3.4 结束语

环路增益分析方法虽然简单，但并不全面：它无法让我们找到反馈放大器的输入和输出电阻。下一节将弥补这一缺陷，我们将介绍一种分析反馈电压放大器的系统方法。

11.4 反馈电压放大器的系统分析

在本节中，我们将提供一个分析反馈电压放大器的系统程序。该程序主要是对给定的反馈电压放大器电路进行拆解，从而得到 "A 电路"，从中我们可以确定开环增益 A 和开环放大器的其他参数，如输入和输出电阻，以及 "β 电路"，从中我们可以找到反馈系数 β 的值。然后，我们可以使用

feedback formulas to determine the characteristic parameters of the feedback amplifier, such as the closed-loop gain A_f and the input and output resistances with feedback.

Our approach will be to first consider the *ideal* case in which the feedback network does *not* load the basic amplifier. Then, we consider the practical case in which not only does the feedback network load the basic amplifier, but also there is a finite source resistance R_s and a finite load resistance R_L.

11.4.1 The Ideal Case

As mentioned before, series–shunt is the appropriate feedback topology for a voltage amplifier.The ideal structure of the series–shunt feedback amplifier is shown in Fig. 11.12(a). It consists of a *unilateral* open-loop amplifier (the A circuit) and an ideal voltage-sampling, voltage-mixing feedback network (the β circuit). The A circuit has an input resistance R_i, an open-circuit voltage gain A, and an output resistance R_o. It is assumed that the source is ideal with a zero resistance and that there is no load resistance. Furthermore, note that the β circuit

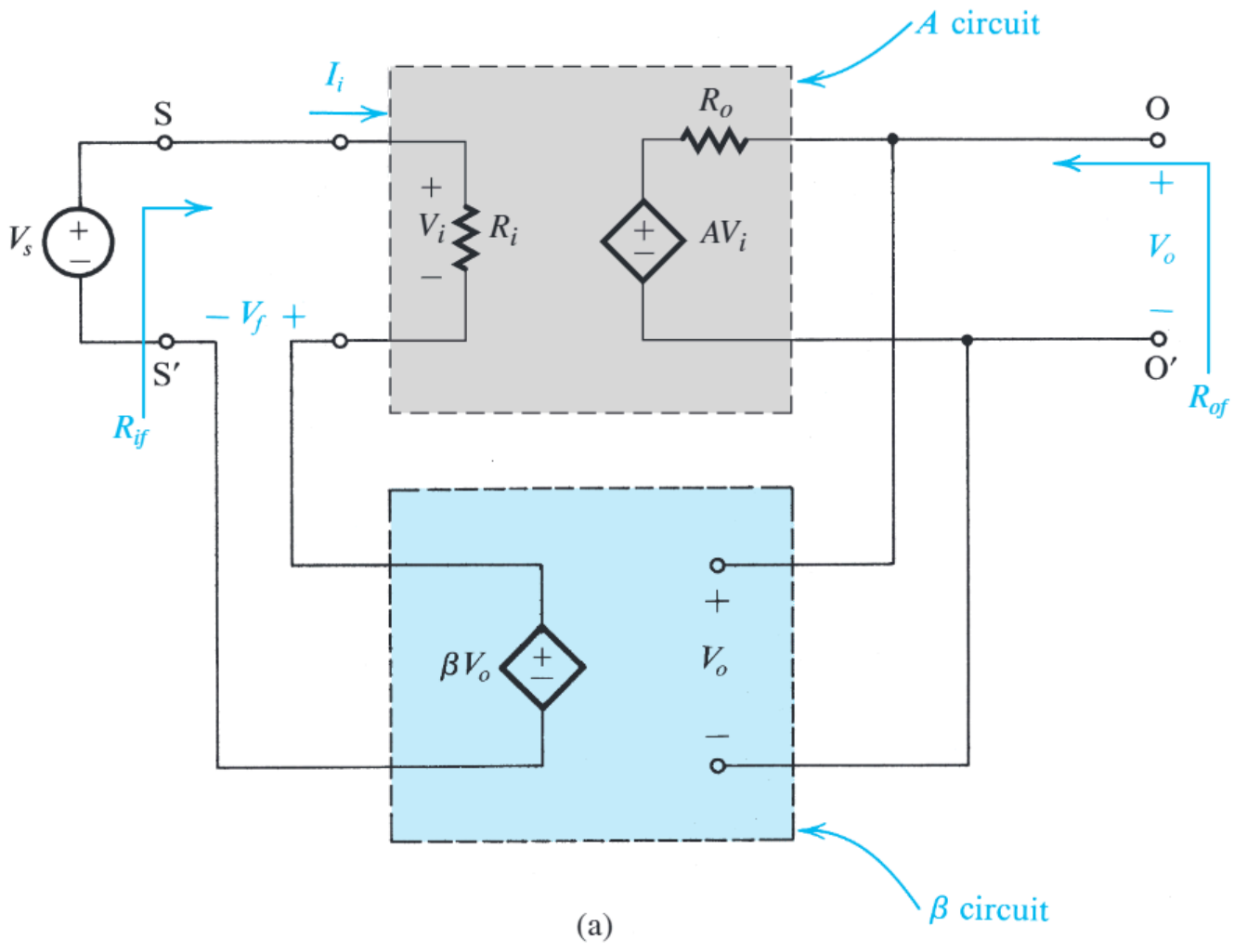


Figure 11.12 The series–shunt feedback amplifier: (**a**) ideal structure; (**b**) equivalent circuit.

然后，我们就可以利用反馈公式来确定反馈放大器的特性参数，如闭环增益 A 以及带反馈的输入和输出电阻。

我们将首先考虑反馈网络不给基本放大器加载的理想情况。然后，我们考虑实际情况，即反馈网络不仅给基本放大器加载，而且还有一个有限的源电阻 Rand 和一个有限的负载电阻 R。

11.4.1 理想情况

如前所述，串并联是电压放大器的适当反馈拓扑结构。串并联反馈放大器的理想结构如图 11.12(a)所示。它由一个单边开环放大器（A 电路）和一个理想的电压采样、电压混合反馈网络（β 电路）组成。A 电路有一个输入电阻 R、一个开路电压增益 A 和一个输出电阻 R。此外，请注意 β 电路

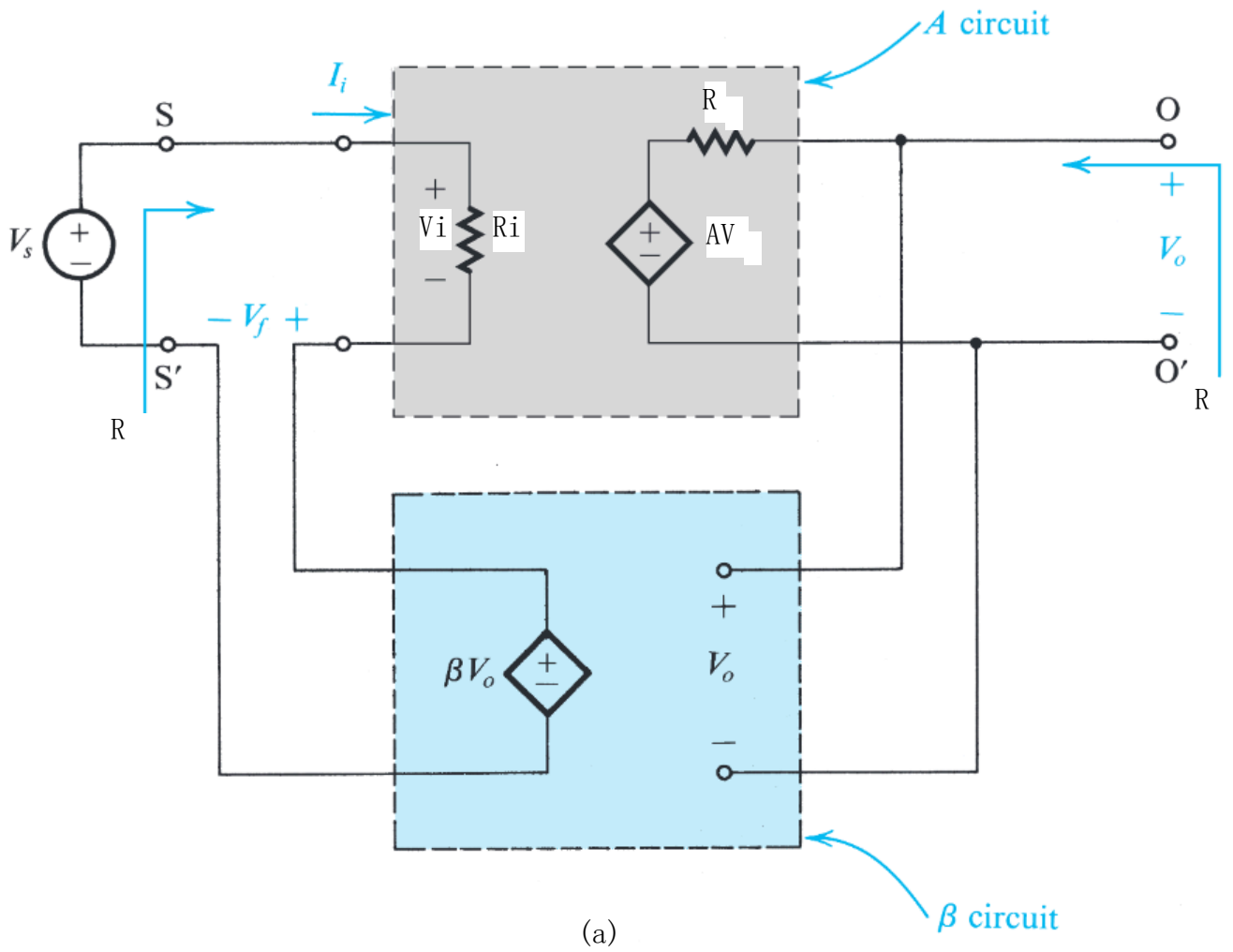


图 11.12 串并联反馈放大器：（a）理想结构；（b）等效电路。

does *not* load the A circuit; that is, connecting the β circuit does not change the value of A (defined as $A \equiv V_o/V_i$).

The circuit of Fig. 11.12(a) exactly follows the ideal feedback model of Fig. 11.1. Therefore the closed-loop voltage gain A_f is given by

$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{1+A\beta} \tag{11.18}$$

The equivalent-circuit model of the series–shunt feedback amplifier is shown in Fig. 11.12(b). Observe that A_f is the open-circuit voltage gain of the feedback amplifier, R_{if} is its input resistance, and R_{of} is its output resistance. Expressions for R_{if} and R_{of} can be derived as follows.

For R_{if}, refer to the input loop of the circuit in Fig. 11.12(a). The series mixing subtracts V_f from V_s and thus reduces V_i by a factor equal to the amount of feedback (Eq. 11.7),

$$V_i = \frac{V_s}{1+A\beta}$$

Thus the input current I_i becomes

$$I_i = \frac{V_i}{R_i} = \frac{V_s}{(1+A\beta)R_i} \tag{11.19}$$

Since I_i is the current drawn from V_s, the input resistance R_{if} can be expressed as

$$R_{if} \equiv \frac{V_s}{I_i}$$

and using Eq. (11.19) is found to be

$$R_{if} = (1+A\beta)R_i \tag{11.20}$$

Thus, as expected, the series-mixing feedback results in an increase in the amplifier input resistance by a factor equal to the amount of feedback, $(1+A\beta)$, a highly desirable property for a voltage amplifier.

To determine the output resistance R_{of} of the feedback amplifier in Fig. 11.12(a), we set $V_s = 0$ and apply a test voltage V_x between the output terminals, as shown in Fig. 11.13. If the current drawn from V_x is I_x, the output resistance R_{of} is

$$R_{of} \equiv \frac{V_x}{I_x} \tag{11.21}$$

An equation for the output loop yields

$$I_x = \frac{V_x - AV_i}{R_o} \tag{11.22}$$

From the input loop we see that

$$V_i = -V_f$$

Now $V_f = \beta V_o = \beta V_x$; thus,

$$V_i = -\beta V_x$$

不给 A 电路加载，也就是说，连接 β 电路不会改变 A 的值（定义为 A ≡ V/V）。
图 11.12(a) 的电路完全符合图 11.1 的理想反馈模型。因此，闭环电压增益 A 为

$$A\equiv \frac{V}{V} = \frac{A}{1 + A\beta} \tag{11.18}$$

串并联反馈放大器的等效电路模型如图 11.12(b)所示。请注意，A 是反馈放大器的开路电压增益，Ris 是其输入电阻，R 是其输出电阻。R 和 R 的表达式可推导如下。
关于 R，请参见图 11.12(a)中的电路输入回路。串联混合将 V 从 V 中减去，从而使 V 减小，减幅等于反馈量（公式 11.7）、

$$V= \frac{V}{1 + A\beta}$$

因此，输入电流 Ibec 成为

$$I = \frac{V}{R} = \frac{V}{(1 + A\beta)R} \tag{11.19}$$

由于 I 是 V 的电流，输入电阻 R 可以表示为

$$R\equiv \frac{V}{I}$$

并利用公式（11.19）求得

$$R = (1 + A\beta)R \tag{11.20}$$

因此，正如预期的那样，串联混合反馈会导致放大器输入电阻增加，其系数等于反馈量（1 + Aβ），这是电压放大器非常理想的特性。
如图 11.13 所示，为了确定图 11.12(a)中反馈放大器的输出电阻 R，我们设置 V= 0，并在输出端之间施加测试电压 V。如果从 Vis 吸入的电流为 I，则输出电阻 R 为

$$R \equiv \frac{V}{I} \tag{11.21}$$

输出回路方程得出

$$I= \frac{V- AV}{R} \tag{11.22}$$

从输入回路中我们可以看到

$$V= -V$$

现在，V= βV= βV；因此、

$$V= -\beta V$$

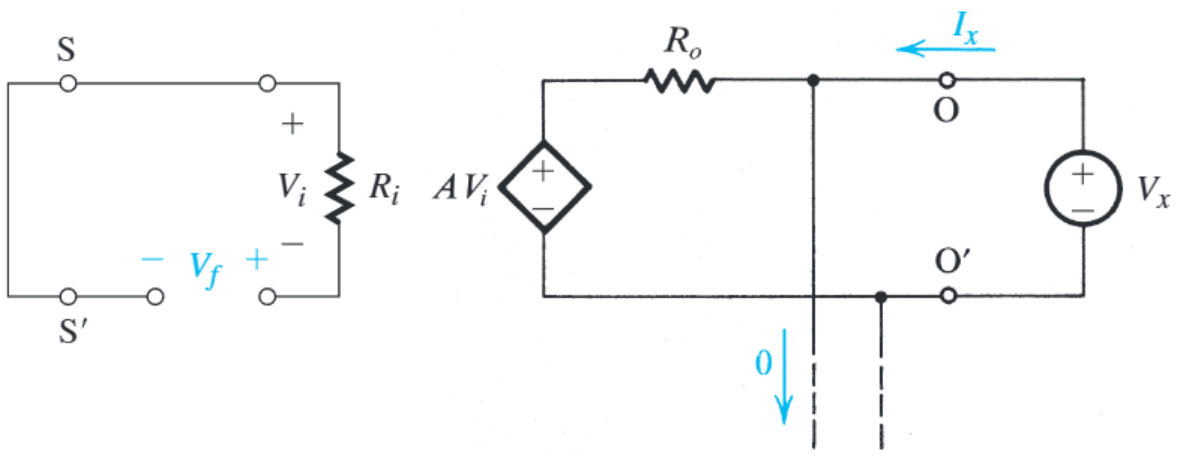


Figure 11.13 Determining the output resistance of the feedback amplifier of Fig. 11.12(a): $R_{of} = V_x/I_x$.

which when substituted in Eq. (11.22) yields

$$I_x = \frac{V_x(1+A\beta)}{R_o}$$

Substituting this value of I_x into Eq. (11.21) provides the following expression for R_{of},

$$R_{of} = \frac{R_o}{1+A\beta} \tag{11.23}$$

Thus, as expected, the shunt sampling (or voltage sampling) at the output results in a decrease in the amplifier output resistance by a factor equal to the amount of negative feedback, $(1+A\beta)$, another highly desirable property for a voltage amplifier.

Although perhaps not entirely obvious, the reduction of the output resistance is a result only of the method of sampling the output and does not depend on the method of mixing. Thus, the transistance amplifier, which is the other amplifier type in which shunt (or voltage) sampling is employed, will also exhibit a reduced output resistance.

11.4.2 The Practical Case

In a practical series–shunt feedback amplifier, the feedback network will not be an ideal voltage-controlled voltage source. Rather, as in the circuits of Fig. 11.8, the feedback network is usually resistive and hence will load the basic amplifier and thus affect the values of A, R_i, and R_o. In addition, there will be finite source and load resistances, which in turn will affect these three parameters. Thus the problem we have is as follows: Given a series–shunt feedback amplifier represented by the block diagram of Fig. 11.14(a), find the A circuit and the β circuit.

The problem in essence is to represent the general feedback voltage amplifier of Fig. 11.14(a) with the ideal structure of Fig. 11.12(a). The solution is presented, without derivation, in Fig. 11.14(b). We make the following observations.

1. The A circuit is obtained by augmenting the basic amplifier at the input with the source resistance R_s and a resistance R_{11}, and at the output with the load resistance R_L and a resistance R_{22}. Resistances R_{11} and R_{22} represent the loading effect of the feedback network on the basic amplifier at the input and the output, respectively.
2. Resistance R_{11} is the resistance looking into port 1 of the feedback network while port 2 is short-circuited. Resistance R_{22} is the resistance looking into port 2 of the

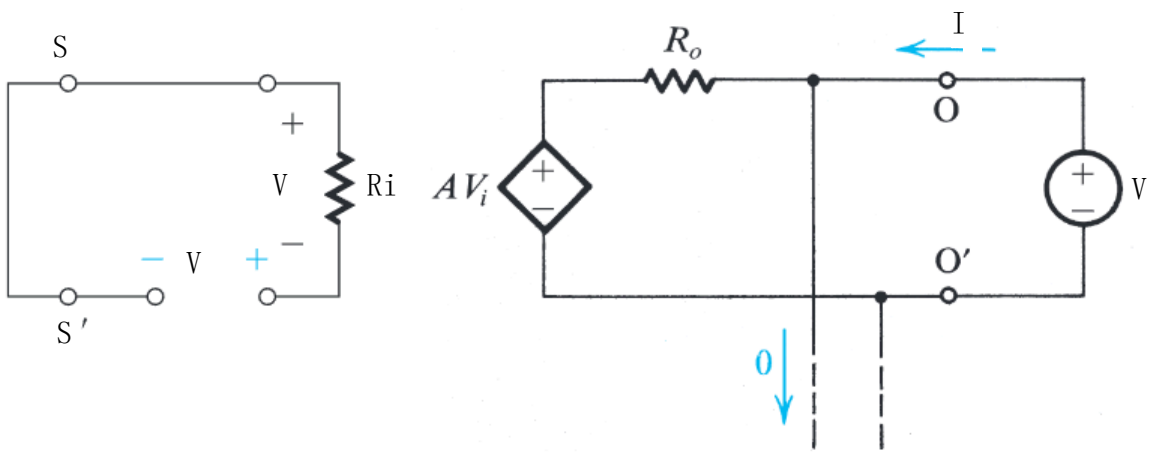


图 11.13 确定图 11.12(a) 中反馈放大器的输出电阻：R = V /I 。

代入公式 (11.22) 即可得出

$$I = \frac{V(1 + A\beta)}{R}$$

将 I 的这一数值代入公式 (11.21) 即可得到 R 的如下表达式、

$$R = \frac{R}{1 + A\beta} \qquad (11.23)$$

因此，正如预期的那样，输出端的并联取样（或电压取样）会使放大器的输出电阻减小，减小的系数等于负反馈量 (1 + Aβ)，这是电压放大器的另一个非常理想的特性。

输出电阻的减小只是输出取样方法的结果，并不取决于混合方法，尽管这一点可能并不十分明显。因此，采用分流（或电压）取样的另一种放大器--跨阻放大器，也会显示输出电阻减小。

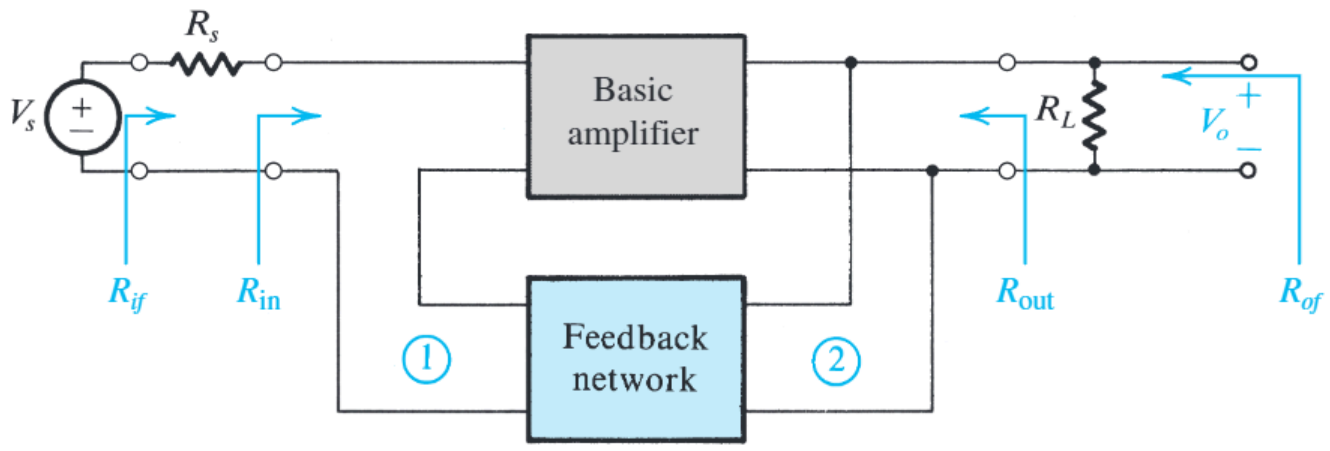
11.4.2 实际案例

在实际的串并联反馈放大器中，反馈网络并非理想的压控电压源。相反，如图 11.8 中的电路，反馈网络通常是电阻性的，因此会对基本放大器产生负载，从而影响 A、R 和 R 的值。因此，我们面临的问题如下：给出以图 11.14(a) 方框图表示的串并联反馈放大器，求 A 电路和 β 电路。

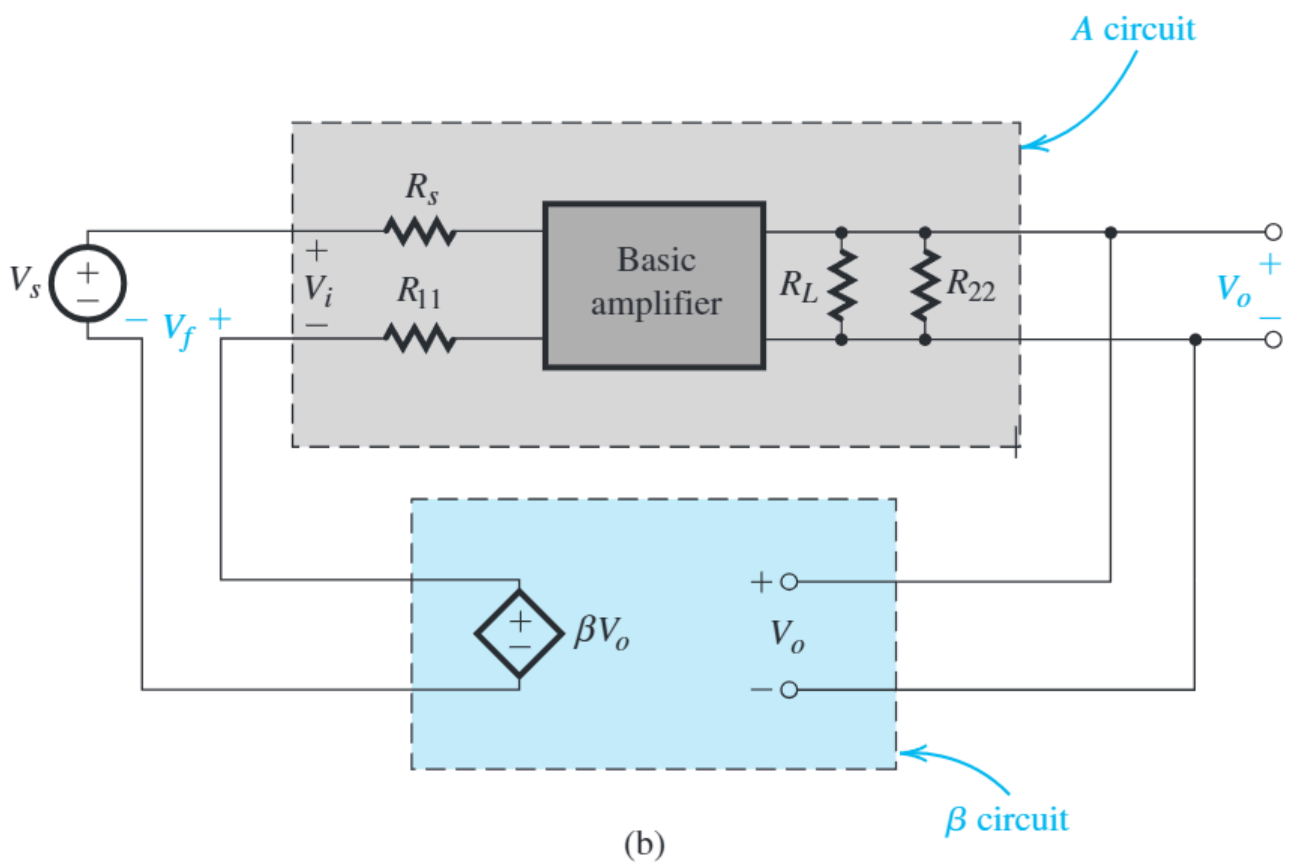
问题的实质是用图 11.12(a)的理想结构来表示图 11.14(a)的一般反馈电压放大器。图 11.14(b)给出了解决方案，但未作推导。我们得出以下结论。

1. 电阻 Rand R 分别表示反馈网络在输入和输出端对基本放大器的负载效应。

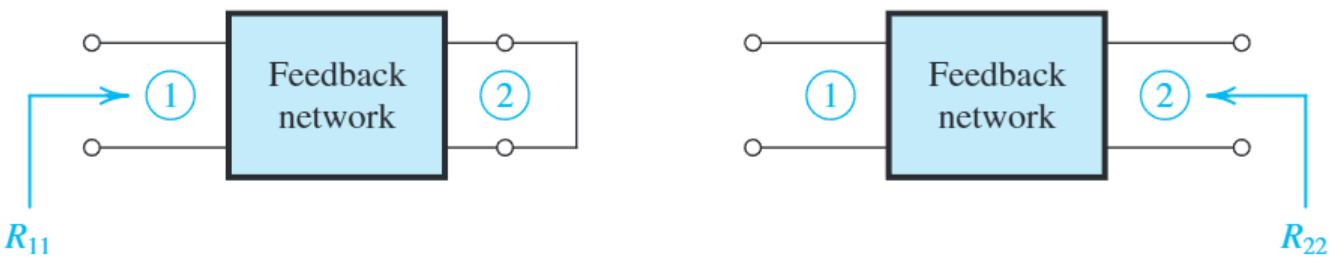
2. 当端口 2 短路时，进入反馈网络端口 1 的电阻值 Ris。电阻 Ris 是反馈网络端口 2 的电阻值。



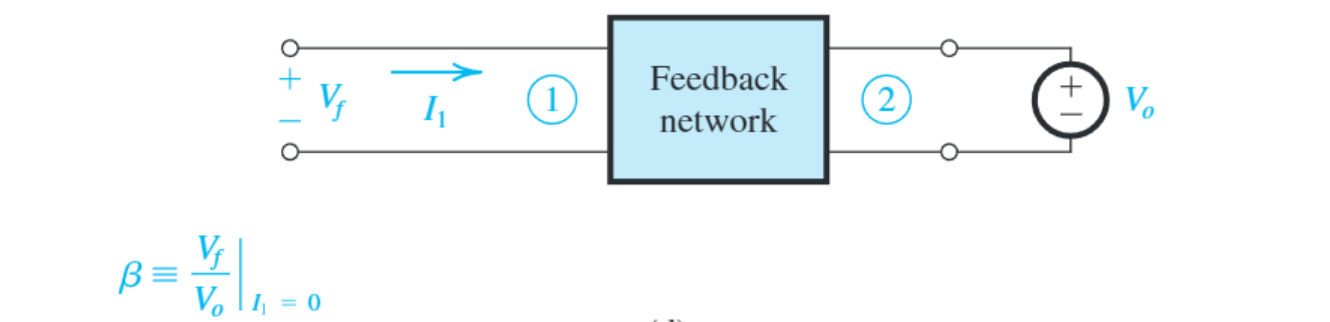
(a)



(b)

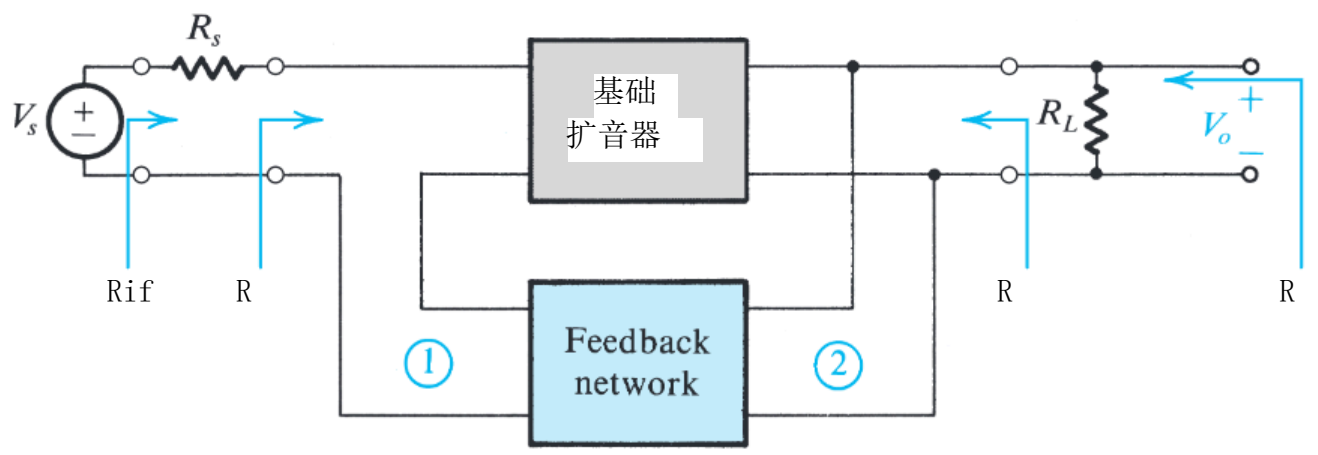


(c)

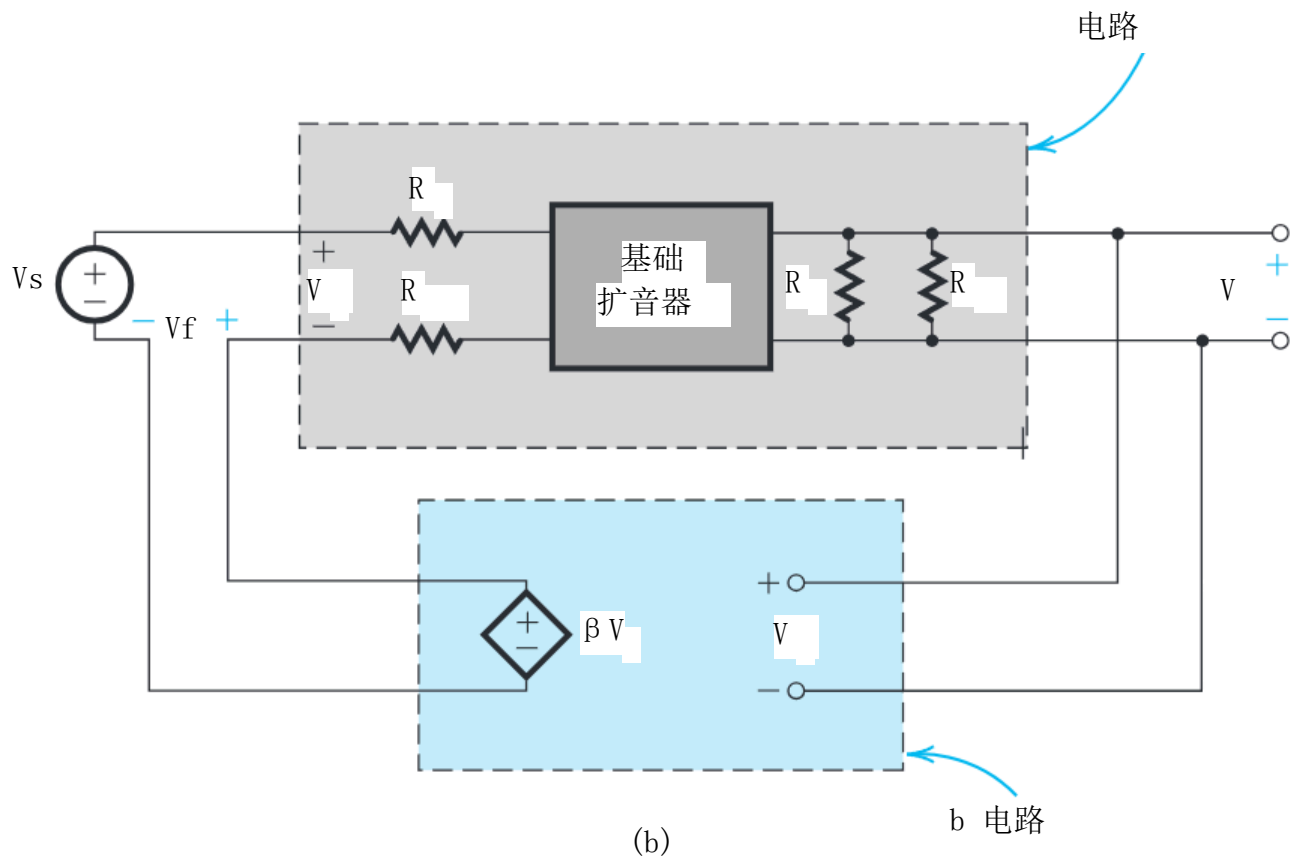


(d)

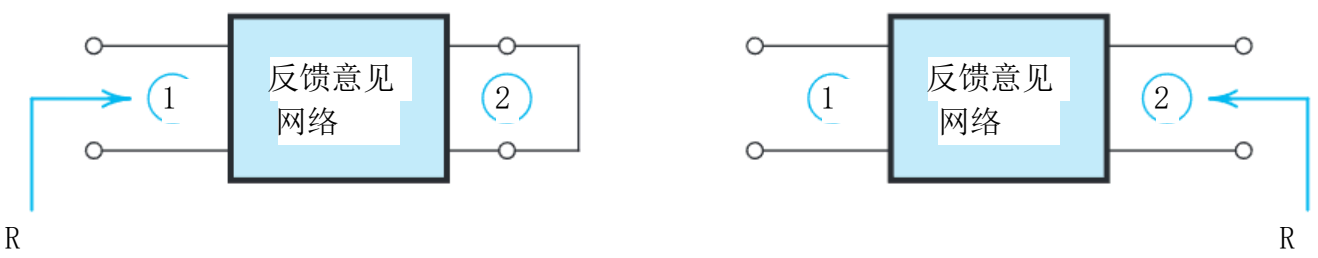
Figure 11.14 **(a)** Block diagram of a practical series–shunt feedback amplifier. **(b)** The circuit in **(a)** represented by the ideal structure of Fig. 11.12(a). **(c)** Definition of R_{11} and R_{22}. **(d)** Determination of the feedback factor β. **(e)** The A circuit, showing the open-loop resistances R_i and R_o.



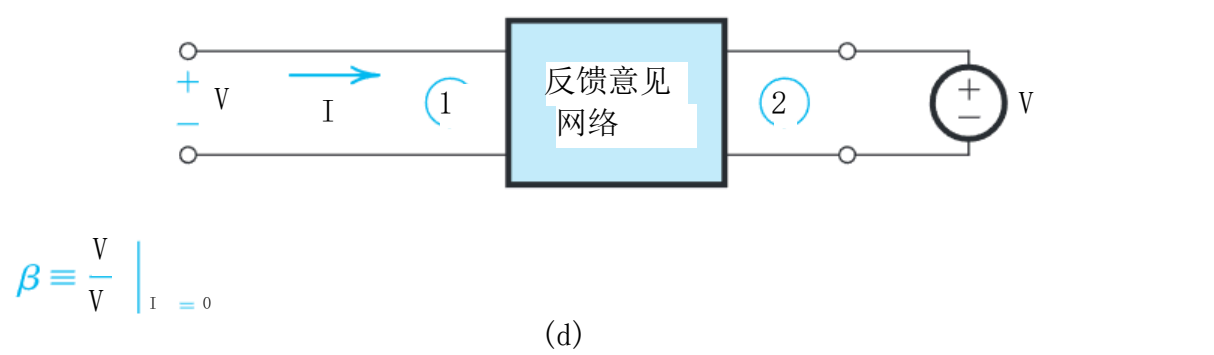
(a)



(b)

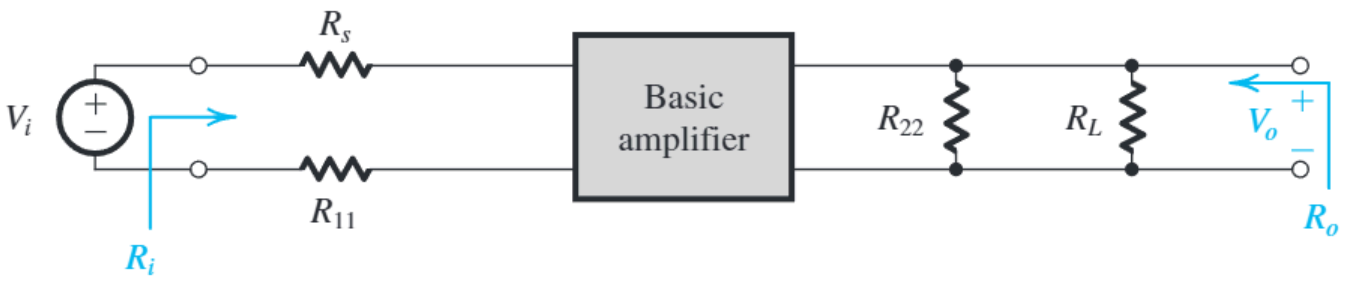


(c)



(d)

图 11.14 (a) 实用串并联反馈放大器方框图。(b) 图 11.12(a)中的理想结构表示 (a) 中的电路。(c) Rand R 的定义 (d) 反馈系数 β 的确定 (e) A 电路，显示开环电阻 Rand R。



(e)

Figure 11.14 *continued*

feedback network while port 1 is open-circuited. These definitions are illustrated in Fig. 11.14(c). Since the feedback network is connected in shunt with the output, shorting its port 2 destroys the feedback. Similarly, because the feedback network is connected in series with the input, opening its port 1 destroys the feedback. It follows that the loading effect of the feedback network is obtained by looking into its appropriate port while the other port is open-circuited or short-circuited so as to destroy the feedback.[4]

3. The feedback factor β is the transmission from port 2 to port 1 of the feedback network, with port 1 open-circuited (which destroys the feedback). Reference to Fig. 11.14(c) shows that

$$\beta \equiv \left.\frac{V_f}{V_o}\right|_{I_1=0} \tag{11.24}$$

4. The open-loop gain A can be obtained from the A circuit in Fig. 11.14(e) as $A = \dfrac{V_o}{V_i}$

5. The values of A and β can be used to determine the closed-loop gain A_f,

$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$

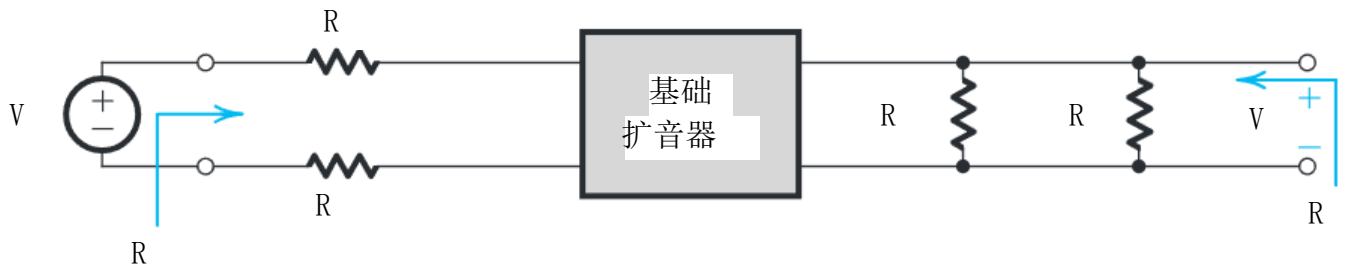
6. The open-loop input resistance R_i and output resistance R_o can be determined from the A circuit [see Fig. 11.14(e)]. These values can be used to determine the input and output resistances with feedback,

$$R_{if} = R_i(1 + A\beta)$$
$$R_{of} = R_o/(1 + A\beta)$$

From Fig. 11.14(a) we see that R_{if} is the resistance seen by the ideal signal source V_s. The actual input resistance of the feedback amplifier R_{in} excludes R_s and is found from R_{if},

$$R_{\text{in}} = R_{if} - R_s \tag{11.25}$$

[4] A simple rule to remember: If the connection is *sh*unt, *sh*ort it; if *se*ries, *se*ver it.



(e)

图 11.14 续

反馈网络，而端口 1 处于开路状态。这些定义如图 11.14(c)所示。由于反馈网络与输出并联，端口 2 短路会破坏反馈。同样，由于反馈网络与输入端串联，打开其端口 1 会破坏反馈。由此可见，反馈网络的负载效应是通过查看其相应端口，同时将其他端口开路或短路以破坏反馈来实现的。

3. 反馈系数 β 是在端口 1 开路的情况下，从端口 2 到反馈网络端口 1 的传输（这会破坏反馈）。参考图 11.14(c)可以看出

$$\beta \equiv \left.\frac{V}{V}\right|_{1=0} \tag{11.24}$$

4. 从图 11.14(e)中的 A 电路可以得到开环增益 A，即 A = $\frac{V}{V}$

5. A 和 β 的值可用来确定闭环增益 A、

$$A\equiv \frac{V}{V} = \frac{A}{1 + A\beta}$$

6. 开环输入电阻和输出电阻 R 可通过 A 电路确定［见图 11.14(e)］。这些值可用来确定带反馈的输入和输出电阻、

$$R = R(1 + A\beta)$$

$$R = R/(1 + A\beta)$$

从图 11.14(a)可以看出，R 是理想信号源 V 看到的电阻、

$$R = R - R \tag{11.25}$$

[4]记住一条简单的规则：如果是并联，则短接；如果是串联，则断开。

Similarly, R_{of} is the output resistance of the feedback amplifier including R_L. The actual output resistance excludes R_L (see Fig. 11.14(a)) and is found from R_{of},

$$R_{\text{out}} = 1 \Big/ \left(\frac{1}{R_{of}} - \frac{1}{R_L} \right) \tag{11.26}$$

A final and important note: The representation in Fig. 11.14(b) is only *approximately* equivalent to the original circuit in Fig. 11.14(a). The approximation is a result of neglecting the small forward transmission in the feedback network relative to the much larger forward transmission in the basic amplifier. Also, recall that we continue to assume that the basic amplifier is unilateral—that is, it does not have internal feedback; all the feedback occurs in the feedback network and is represented by the feedback factor β.

Example 11.4

Figure 11.15(a) shows an op amp connected in the noninverting configuration. The op amp has an open-circuit voltage gain μ, a differential input resistance R_{id}, and an output resistance r_o. Recall that in our analysis of op-amp circuits in Chapter 2, we neglected the effects of R_{id} (assumed it to be infinite) and of r_o (assumed it to be zero). Here we wish to use the feedback method to analyze the circuit taking both R_{id} and r_o into account. Find expressions for A, β, the closed-loop gain V_o/V_s, the input resistance R_{in} [see Fig. 11.15(a)], and the output resistance R_{out}. Also find numerical values, given $\mu = 10^4$, $R_{id} = 100\ \text{k}\Omega$, $r_o = 1\ \text{k}\Omega$, $R_L = 2\ \text{k}\Omega$, $R_1 = 1\ \text{k}\Omega$, $R_2 = 1\ \text{M}\Omega$, and $R_s = 10\ \text{k}\Omega$. Note that this circuit was analyzed in Example 11.3 using the loop-gain method and, where appropriate, compare results.

Solution

We observe that the feedback network consists of R_2 and R_1. This network samples the output voltage V_o and provides a voltage signal (across R_1) that is mixed in series with the input source V_s.

The A circuit can be easily obtained following the rules of Fig. 11.14, and is shown in Fig. 11.15(b). Observe that the loading effect of the feedback network at the input side is obtained by short-circuiting port 2 of the feedback network (because it is connected in shunt) and looking into port 1, with the result that $R_{11} = R_1 \parallel R_2$. The loading effect of the feedback network at the output side is found by open-circuiting port 1 (because it is connected in series) and looking into port 2, with the result that $R_{22} = R_2 + R_1$. For the resulting A circuit in Fig. 11.15(b), we can write by inspection:

$$A \equiv \frac{V_o}{V_i} = \mu \frac{R_L \parallel (R_1 + R_2)}{\left[R_L \parallel (R_1 + R_2)\right] + r_o} \frac{R_{id}}{R_{id} + R_s + (R_1 \parallel R_2)}$$

For the values given, we find that $A \simeq 6000$ V/V.

The circuit for determining β is shown in Fig. 11.15(c), from which we obtain

$$\beta \equiv \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2} \simeq 10^{-3}\ \text{V/V}$$

Thus,

$$A\beta = 6000 \times 10^{-3} = 6$$

同样，计算包括 R 在内的反馈放大器的输出电阻。实际输出电阻不包括 R（见图 11.14(a)），由 R 、

$$R = 1 \quad /(1 \quad \frac{}{R} - \frac{1}{R}) \tag{11.26}$$

最后一个重要说明：图 11.14(b)中的表示法仅近似等效于图 11.14(a)中的原始电路。之所以近似，是因为忽略了反馈网络中较小的正向传输，而基本放大器中的正向传输要大得多。另外，请注意我们仍然假设基本放大器是单边的，即它没有内部反馈；所有反馈都发生在反馈网络中，并用反馈系数 β 表示。

例 11.4

图 11.15(a) 显示了一个以非反相配置连接的运算放大器。运算放大器具有开路电压增益 μ、差分输入电阻 R 和输出电阻 r。回想一下，在第 2 章对运算放大器电路的分析中，我们忽略了 R 的影响（假设其为无限大）和 r 的影响（假设其为零）。在这里，我们希望使用反馈法分析电路，同时考虑 R 和 r 的影响。给定 μ = 10、R= 100 k、r=1 k、R=2 k、R=1 k、R=1 M 和 R=10 k，求出 A、β、闭环增益 V/V、输入电阻 R [见图 11.15(a)]和输出电阻 R 的表达式。请注意，在例 11.3 中使用环路增益法分析了该电路，并酌情比较结果。

解决方案

我们注意到，反馈网络由 Rand R 组成。该网络对输出电压 V 进行采样，并提供与输入源 V 串联混合的电压信号（跨 R）。

按照图 11.14 的规则可以很容易地得到 A 电路，如图 11.15(b)所示。

观察输入侧反馈网络的负载效应，可通过短路反馈网络的端口 2（因为它是并联连接的），并观察端口 1，得出 R= R ‖ R；输出侧反馈网络的负载效应，可通过开路端口 1（因为它是串联连接的），并观察端口 2，得出 R= R+ R：

$$A \equiv \frac{V}{V} = \mu \left[\frac{R \,\|}{R \,\|} \frac{(R+R)}{(R+R)} \right] \frac{}{+ r} \frac{R}{R + R + (R \| R)} \Bigg|$$

根据给出的数值，我们发现 A 为 6000 V/V。

确定 β 的电路如图 11.15(c)所示，由此可得

$$\beta \equiv \frac{V}{V} = \frac{R}{R+R} \quad 10 V/V$$

因此、

$$A\beta = 6000 \times 10 = 6$$

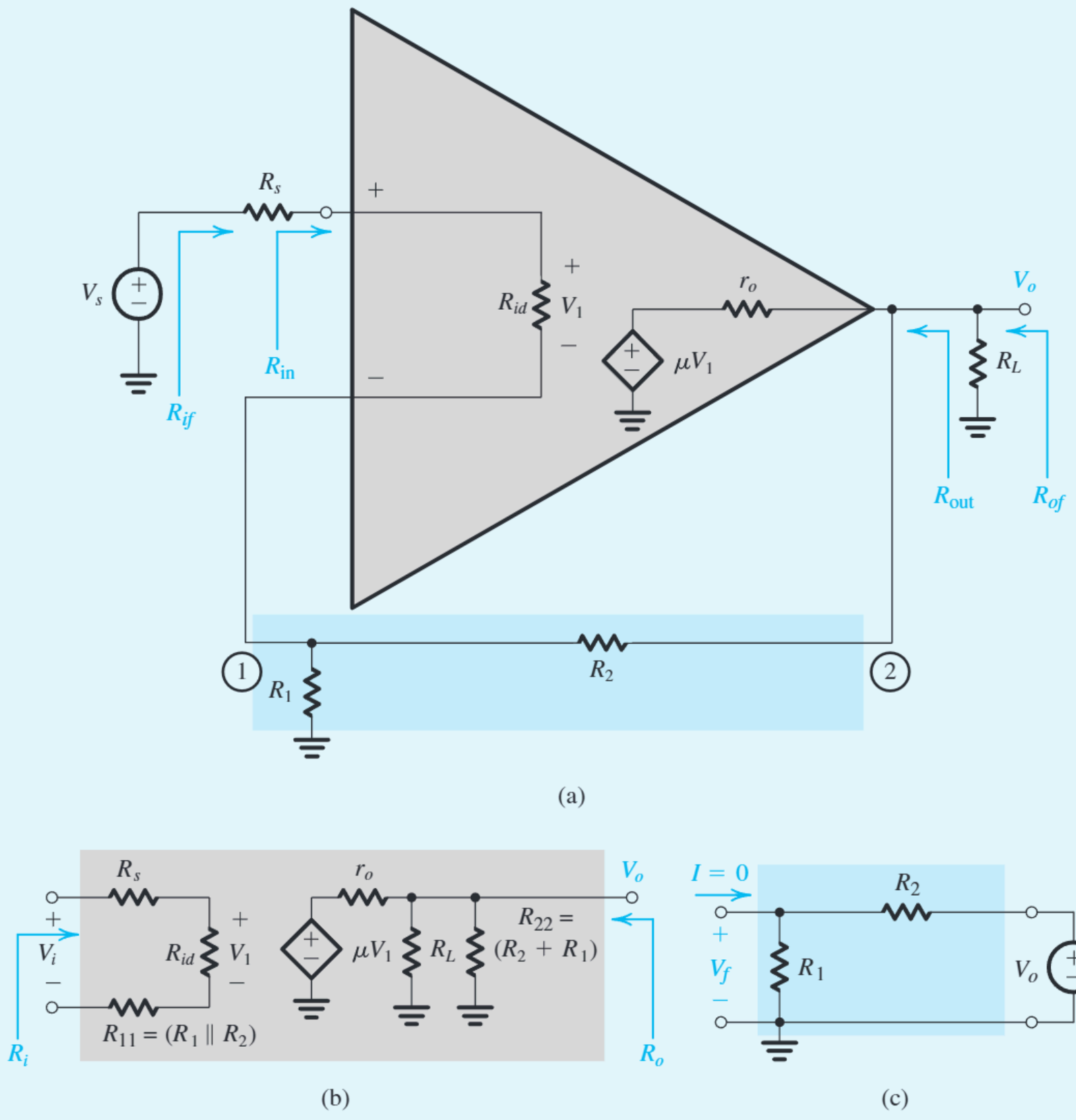


Figure 11.15 Circuits for Example 11.4.

and

$$1 + A\beta = 7$$

Identical results are obtained by substituting the given numerical values into the expression for $A\beta$ derived in Example 11.3.

The voltage gain with feedback can now be obtained as

$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{1 + A\beta} = \frac{6000}{7} = 857 \text{ V/V}$$

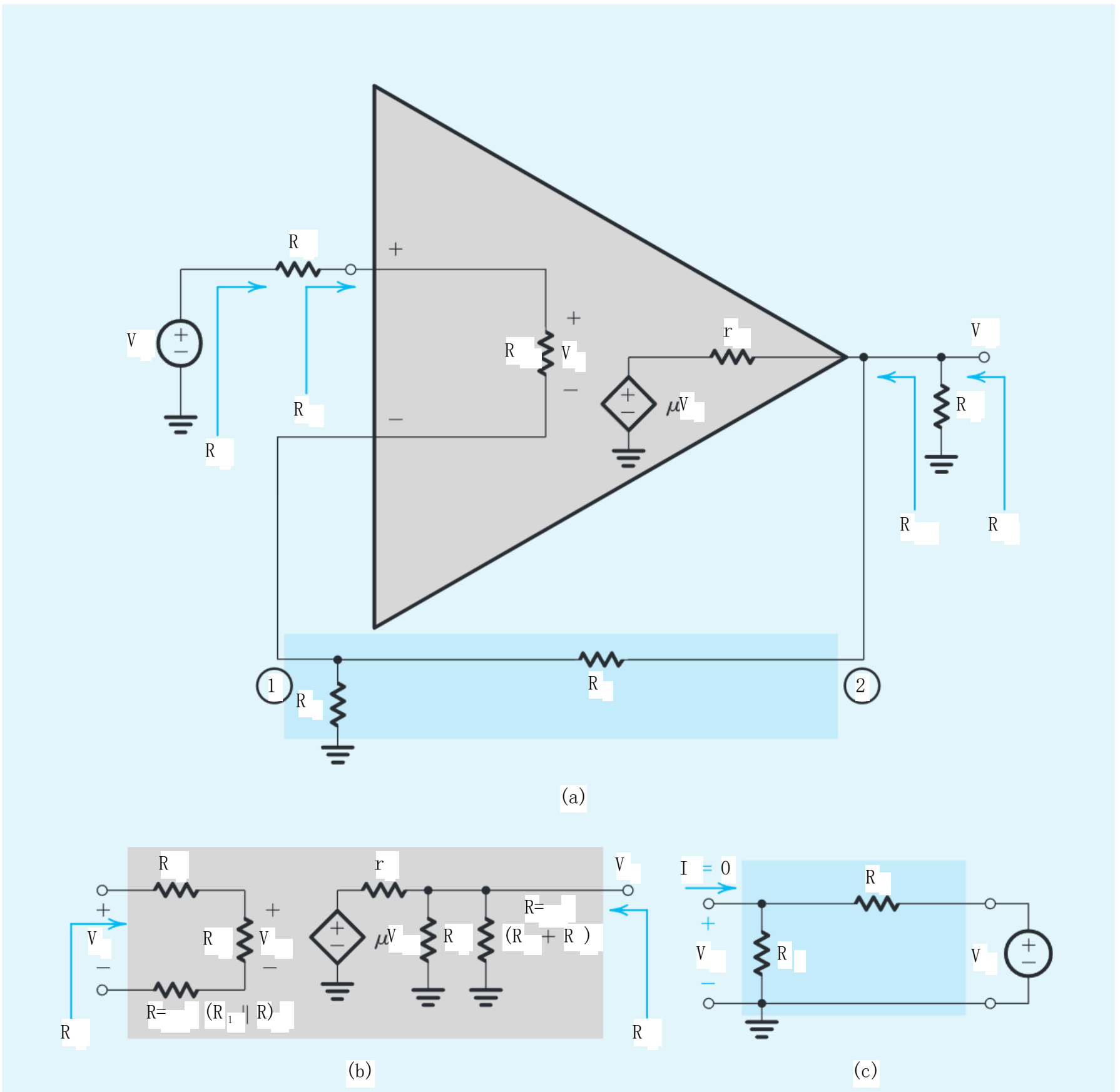


图 11.15 例 11.4 的电路

and

$$1 + A\beta = 7$$

将给定的数值代入例 11.3 中得出的 $A\beta$ 表达式，可以得到相同的结果。

带反馈的电压增益现在可以求得

$$A \equiv \frac{V}{V} = \frac{A}{1 + A\beta} = \frac{6000}{7} = 857 \text{ V/V}$$

Example 11.4 *continued*

The input resistance R_{if} determined by the feedback equations is the resistance seen by the external source (see Fig. 11.15a), and is given by

$$R_{if} = R_i(1 + A\beta)$$

where R_i is the input resistance of the A circuit in Fig. 11.15(b):

$$R_i = R_s + R_{id} + (R_1 \parallel R_2)$$

For the values given, $R_i \simeq 111$ kΩ, resulting in

$$R_{if} = 111 \times 7 = 777 \text{ k}\Omega$$

This, however, is not the resistance asked for. What is required is R_{in}, indicated in Fig. 11.15(a). To obtain R_{in} we subtract R_s from R_{if}:

$$R_{\text{in}} = R_{if} - R_s = 767 \text{ k}\Omega$$

The resistance R_{of} given by the feedback equations is the output resistance of the feedback amplifier, including the load resistance R_L, as indicated in Fig. 11.15(a). R_{of} is given by

$$R_{of} = \frac{R_o}{1 + A\beta}$$

where R_o is the output resistance of the A circuit. R_o can be obtained by inspection of Fig.11.15(b) with V_i set to zero, as

$$R_o = r_o \parallel R_L \parallel (R_2 + R_1)$$

For the values given, $R_o \simeq 666\,\Omega$, and

$$R_{of} = \frac{666}{7} = 95.2\ \Omega$$

The resistance asked for, R_{out}, is the output resistance of the feedback amplifier excluding R_L. From Fig. 11.15(a) we see that

$$R_{of} = R_{\text{out}} \parallel R_L$$

Thus

$$R_{\text{out}} \simeq 100\ \Omega$$

例 11.4 续

由反馈方程确定的输入电阻 Rd 是外部信号源看到的电阻（见图 11.15a），其值为

$$R = R(1 + A\beta)$$

其中 Ris 为图 11.15（b）中 A 电路的输入电阻：

$$R = R + R + (R \| R)$$

对于给出的值，R 111 k，结果是

$$R = 111 \times 7 = 777\ k$$

然而，这并不是所需的电阻。需要的是 R，如图 11.15(a)所示。为了得到 R，我们从 R 中减去 R ：

$$R = R - R = 767\ k$$

反馈方程给出的电阻 R 是反馈放大器的输出电阻，包括负载电阻 R，如图 11.15（a）所示。Ris 由

$$R = \frac{R}{1 + A\beta}$$

其中 Ris 为 A 电路的输出电阻。通过观察图 11.15(b)（V 设为零），可以得出 R 为

$$R = r \| R \| (R + R)$$

对于给出的数值，R 666 和

$$R = \frac{666}{7} = 95.2$$

从图 11.15(a)可以看出

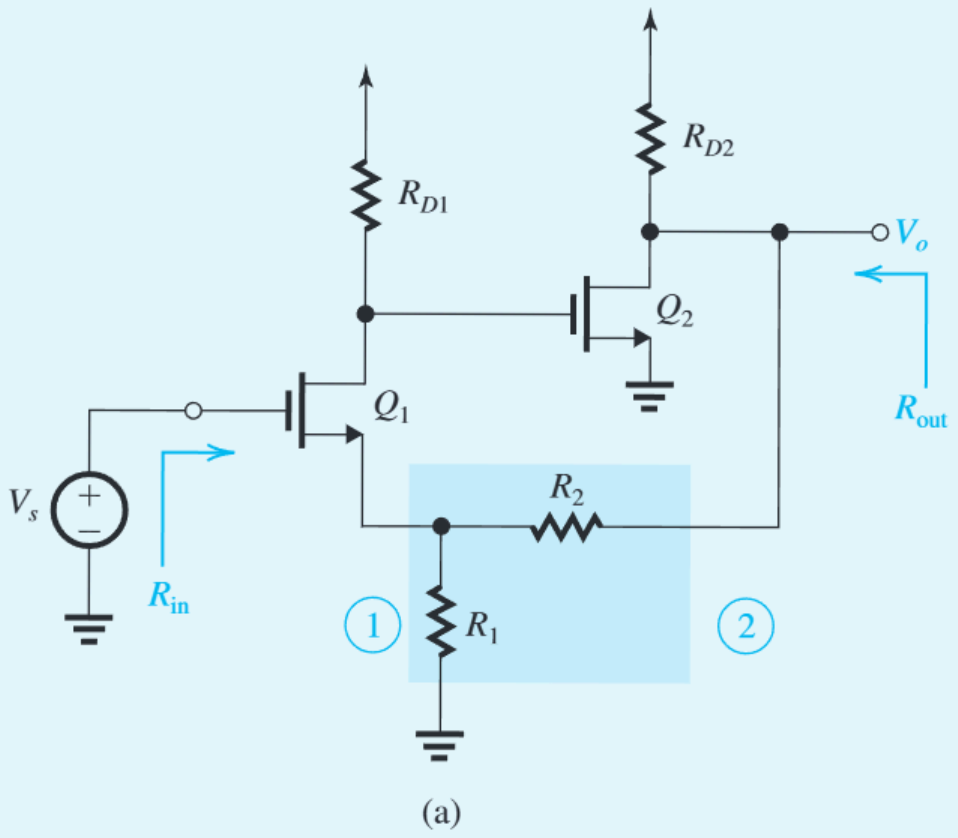
$$R = R \| R$$

Thus

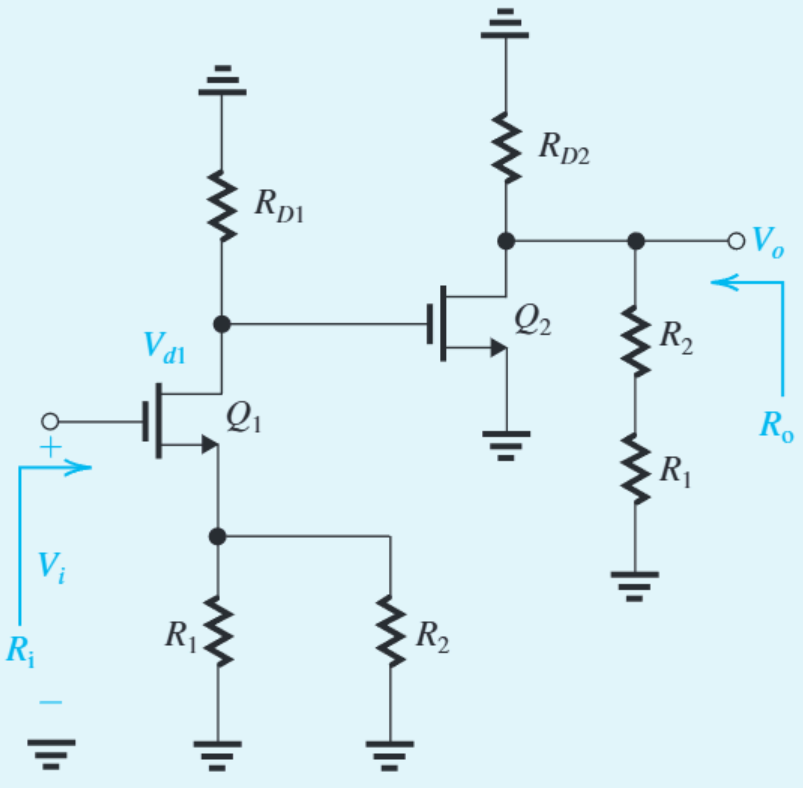
$$R\ 100$$

Example 11.5

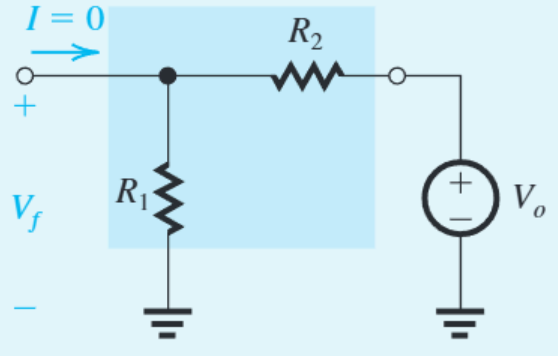
As another example of a series–shunt feedback amplifier, consider the circuit shown in Fig. 11.8(b), which we analyzed in Example 11.2 by determining the loop gain $A\beta$. In this example we wish to first analyze the circuit using our systematic procedure and then compare the results to those obtained in Example 11.2. For convenience, the circuit is repeated in Fig. 11.16(a). It is required to obtain the voltage gain V_o/V_s, input resistance R_{in}, and output resistance R_{out}. Find numerical values for the case $g_{m1} = g_{m2} = 4$ mA/V, $R_{D1} = R_{D2} = 10$ kΩ, $R_1 = 1$ kΩ, and $R_2 = 9$ kΩ. For simplicity, neglect r_o of each of Q_1 and Q_2.



(a)



(b)

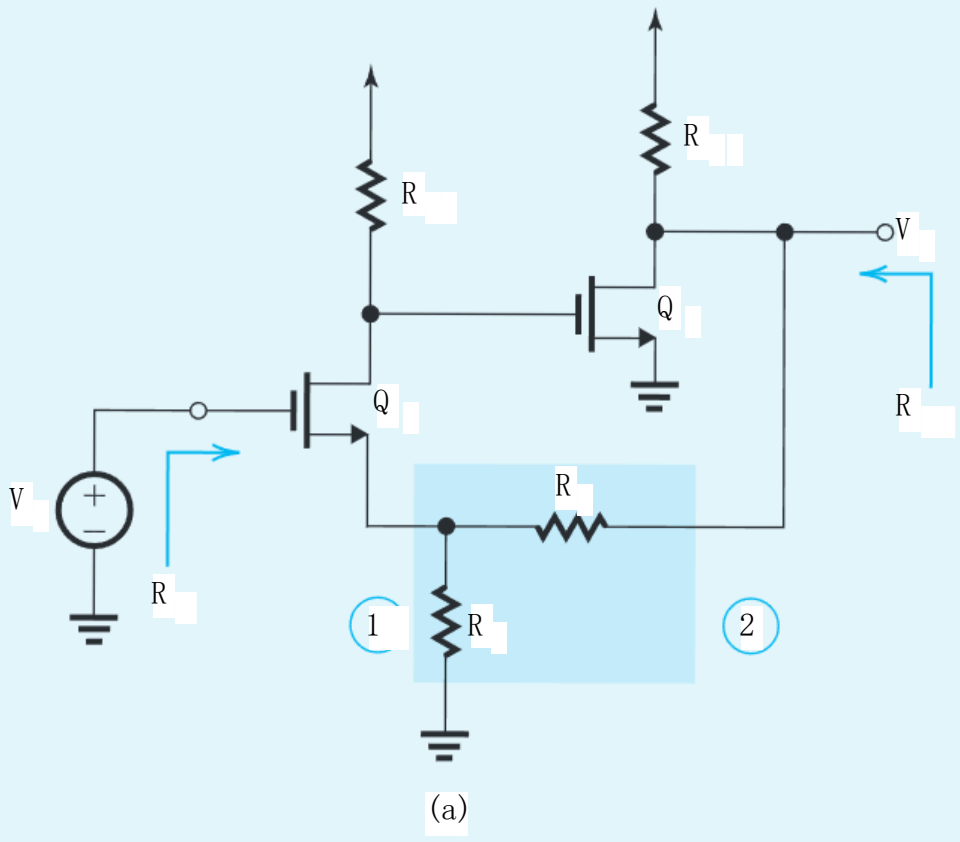


(c)

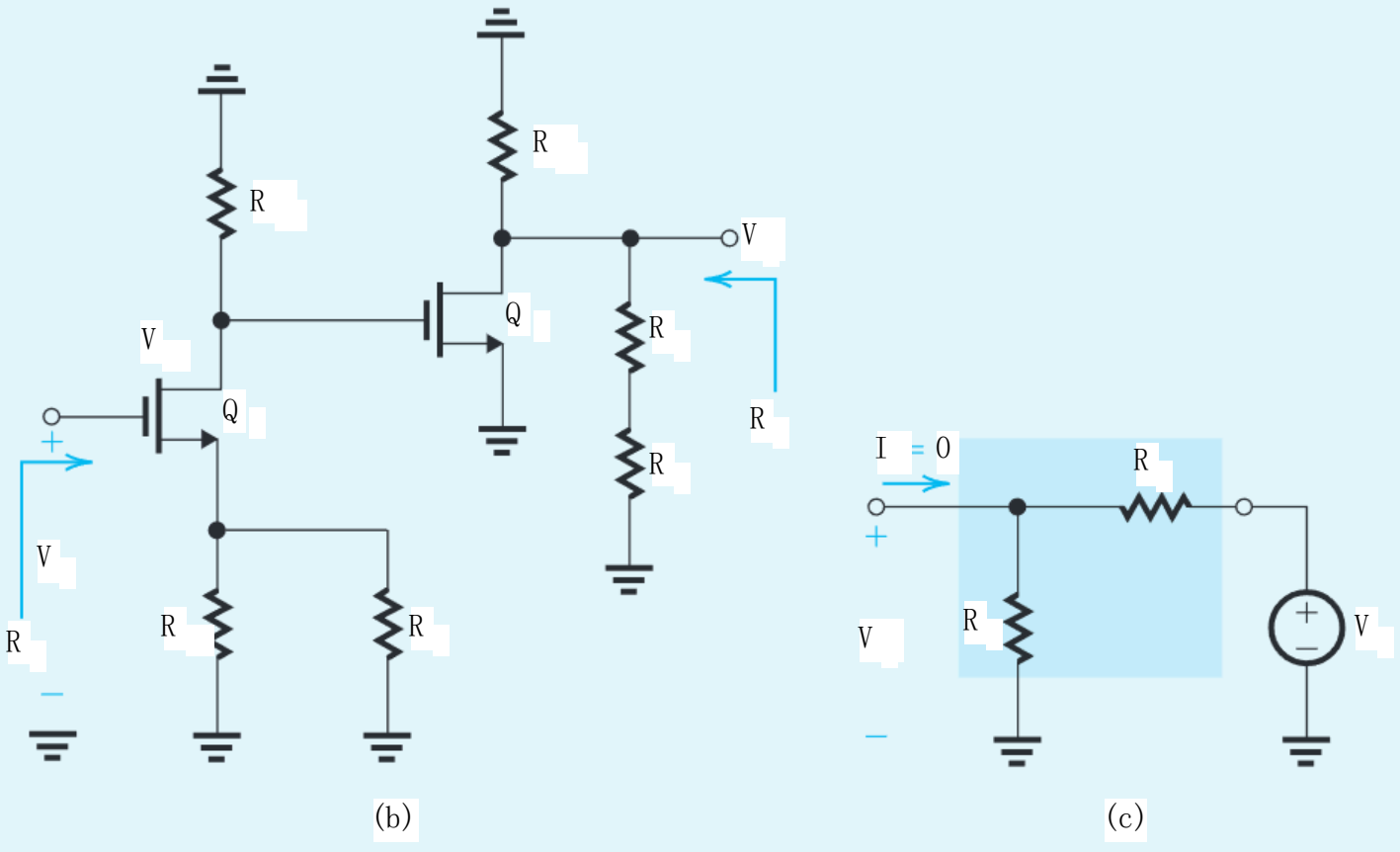
Figure 11.16 (**a**) Series–shunt feedback amplifier for Example 11.5; (**b**) the A circuit; (**c**) the β circuit.

例 11.5

作为串并联反馈放大器的另一个例子，请看图 11.8(b)所示电路，我们在例 11.2 中通过确定环路增益 Aβ 分析了该电路。在本例中，我们希望首先使用我们的系统程序对电路进行分析，然后将分析结果与例 11.2 中得到的结果进行比较。为方便起见，在图 11.16(a)中重复了该电路。在 g = g= 4 mA/V、R = R= 10 k、R= 1 k 和 R= 9 k 的情况下，求得电压增益 V/V、输入电阻 R 和输出电阻 R 的数值。为简单起见，忽略 Q 和 Q 中各电阻的 r。



(a)



(b) (c)

图 11.16 (a) 例 11.5 中的串并联反馈放大器；(b) A 电路；(c) β 电路。

Example 11.5 *continued*

Solution

We identify the feedback network as the voltage divider (R_1, R_2). Its loading effect at the input is obtained by short-circuiting its port 2 (because it is connected in shunt with the output). Then, looking into its port 1, we see $R_1 \| R_2$. The loading effect at the output is obtained by open-circuiting port 1 of the feedback network (because it is connected in series with the input). Then, looking into port 2, we see R_2 in series with R_1. The A circuit will therefore be as shown in Fig. 11.16(b). The gain A is determined as the product of the gain of Q_1 and the gain of Q_2 as follows:

$$A_1 = \frac{V_{d1}}{V_i} = -\frac{R_{D1}}{1/g_{m1} + (R_1 \| R_2)} = -\frac{g_{m1}R_{D1}}{1 + g_{m1}(R_1 \| R_2)}$$

$$A_2 = \frac{V_o}{V_{d1}} = -g_{m2}\left[R_{D2} \| (R_1 + R_2)\right]$$

$$A = \frac{V_o}{V_i} = A_1 A_2 = \frac{g_{m1}R_{D1}g_{m2}\left[R_{D2} \| (R_1 + R_2)\right]}{1 + g_{m1}(R_1 \| R_2)}$$

For the numerical values given,

$$A = \frac{4 \times 10 \times 4[10 \| (1+9)]}{1 + 4(1 \| 9)} = 173.9 \text{ V/V}$$

which is reasonably close to the value of 166.7 V/V obtained in Example 11.2.

The value of β is determined from the β circuit in Fig. 11.16(c),

$$\beta \equiv \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

For the numerical values given,

$$\beta = \frac{1}{1+9} = 0.1$$

The closed-loop gain V_o/V_s can now be found as

$$\frac{V_o}{V_s} = A_f = \frac{A}{1 + A\beta} = \frac{173.9}{1 + 173.9 \times 0.1} = 9.46 \text{ V/V}$$

which is very close to the value of 9.43 V/V obtained in Example 11.2.

The input resistance is obviously infinite because of the infinite input resistance of the MOSFET. The output resistance R_{out} is obtained as follows,

$$R_{\text{out}} = R_{of} = \frac{R_o}{1 + A\beta}$$

例 11.5 续

解决方案

我们将反馈网络确定为分压器 (R，R)。短路端口 2（因为它与输出端并联）可获得输入端的负载效应。打开反馈网络的 1 号端口（因为它与输入端串联），就能获得输出端的负载效果。因此，A 电路如图 11.16(b)所示。增益 A 取决于 Q 的增益与 Q 的增益的乘积，如下所示：

$$A = \frac{V}{V} = -\frac{R}{1/g + (R \| R)} = -\frac{g R}{1 + g}\left(\frac{}{R \| R}\right)$$

$$A = \frac{V}{V} = -g\,[R \| (R + R)]$$

$$A = \frac{V}{V} = AA = \frac{gR\ g}{1 + g}\left(\frac{[R \| (R + R)]}{R \| R}\right)$$

对于给出的数值、

$$A = \frac{4 \times 10 \times 4[10 \| (1 + 9)]}{1 + 4(1 \| 9)} = 173.9\ \text{V/V}$$

这与例 11.2 中得到的 166.7 V/V 值相当接近。

根据图 11.16（c）中的 β 电路确定 β 值、

$$\beta \equiv \frac{V}{V} = \frac{R}{R + R}$$

对于给出的数值、

$$\beta = \frac{1}{1 + 9} = 0.1$$

闭环增益 V/V 现在可求得

$$\frac{V}{V} = A = \frac{A}{1 + A\beta} = \frac{173.9}{1 + 173.9 \times 0.1} = 9.46\ \text{V/V}$$

这与例 11.2 中得到的 9.43 V/V 值非常接近。

由于 MOSFET 的输入电阻为无限，因此输入电阻显然是无限的。输出电阻的计算公式如下

$$R = R = \frac{R}{1 + A\beta}$$

where R_o is the output resistance of the A circuit. From Fig. 11.16(b),

$$R_o = R_{D2} \parallel (R_1 + R_2)$$
$$= 10 \parallel 10 = 5\,\text{k}\Omega$$

The amount of feedback is

$$1 + A\beta = 1 + (173.9 \times 0.1) = 18.39$$

Thus,

$$R_{\text{out}} = \frac{5000}{18.39} = 272\ \Omega$$

which is relatively low given that the open-loop amplifier has $R_o = 5000\ \Omega$. We finally note that the loop-gain method utilized in Example 11.2 cannot provide the input and output resistances because the A circuit is not determined.

EXERCISES

11.7 If the op amp of Example 11.4 has a uniform –6-dB/octave high-frequency rolloff with $f_{3\text{dB}} = 1$ kHz, find the 3-dB frequency of the closed-loop gain V_o/V_s.
Ans. 7 kHz

11.8 The circuit shown in Fig. E11.18 consists of a differential stage followed by an emitter follower, with series–shunt feedback supplied by the resistors R_1 and R_2. Assuming that the dc component of V_s is zero, and that β of the BJTs is very high, find the dc operating current of each of the three transistors and show that the dc voltage at the output is approximately zero. Then find the values of A, β, $A_f \equiv V_o/V_s$, R_{in}, and R_{out}. Assume that the transistors have $\beta = 100$.
Ans. 0.5 mA, 0.5 mA, 5 mA; 85.7 V/V; 0.1 V/V; 8.96 V/V; 191 kΩ; 19.1 Ω.

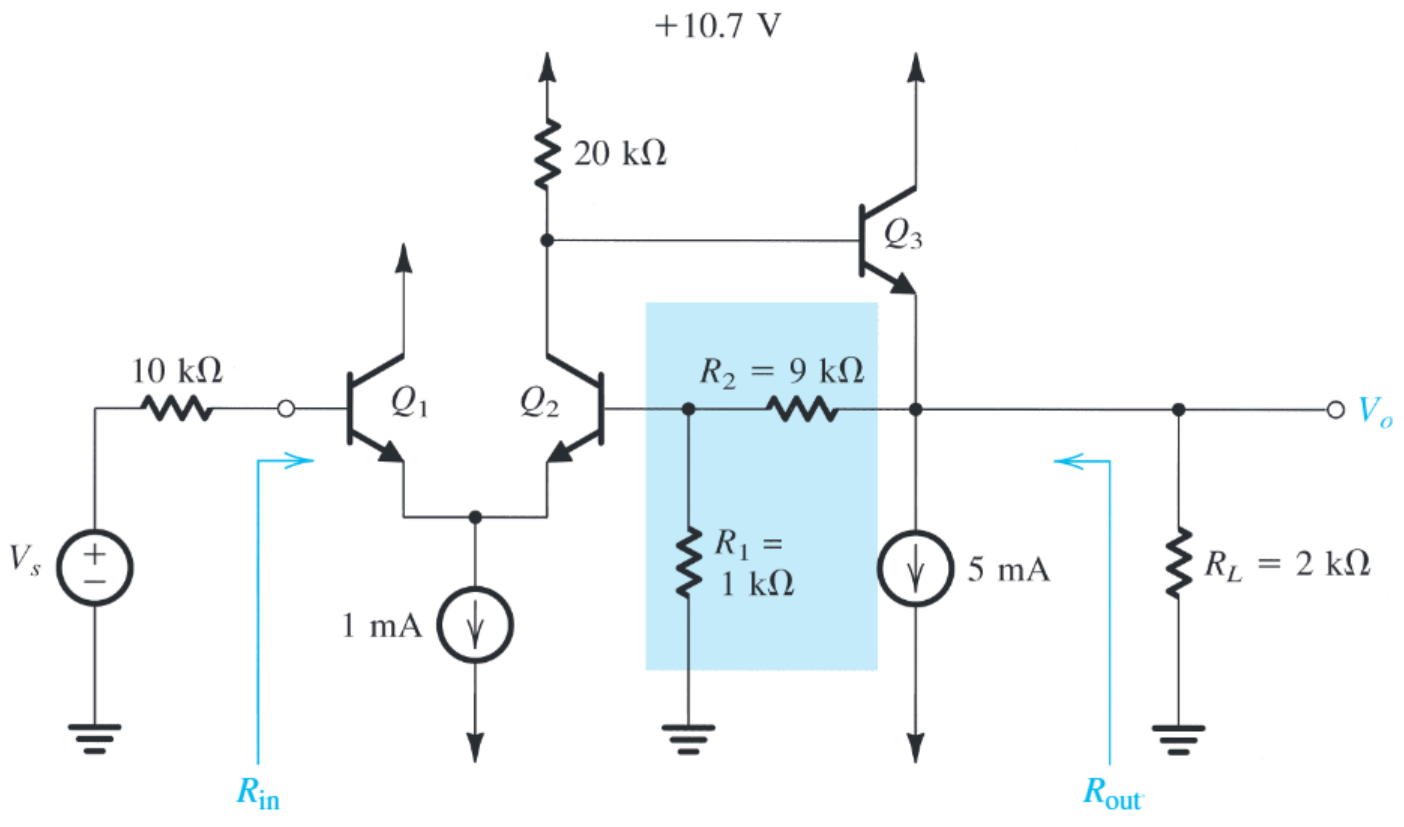


Figure E11.8

其中 Ris 为 A 电路的输出电阻。根据图 11.16(b)、

$$R = R_{is} \| (R_2 + R_1)$$
$$= 10 \| 10 = 5\ \text{k}\Omega$$

反馈量是

$$1 + A\beta = 1 + (173.9 \times 0.1) = 18.39$$

因此、

$$R_{if} = \frac{5000}{18.39} = 272\ \Omega$$

鉴于开环放大器的 R_{if}= 5000，这个值相对较低。最后我们注意到，例 11.2 中使用的环路增益法无法提供输入和输出电阻，因为 A 电路尚未确定。

练习

11.7 如果例 11.4 中的运算放大器在 f= 1 kHz 时具有均匀的 -6-dB/octave 高频滚降，求闭环增益 V/V 的 3-dB 频率。答案 7 kHz

11.8 假设 Vis 的直流分量为零，且 BJT 的 β 非常高，请分别求出三个晶体管的直流工作电流，并证明输出端的直流电压近似为零。然后求出 A、β、A ≡ V/V、R_{if} 和 R_{of} 的值。

答案：0.5 mA、0.5 mA、5 mA；85.7 V/V；0.1 V/V；8.96 V/V；191 k；19.1 .

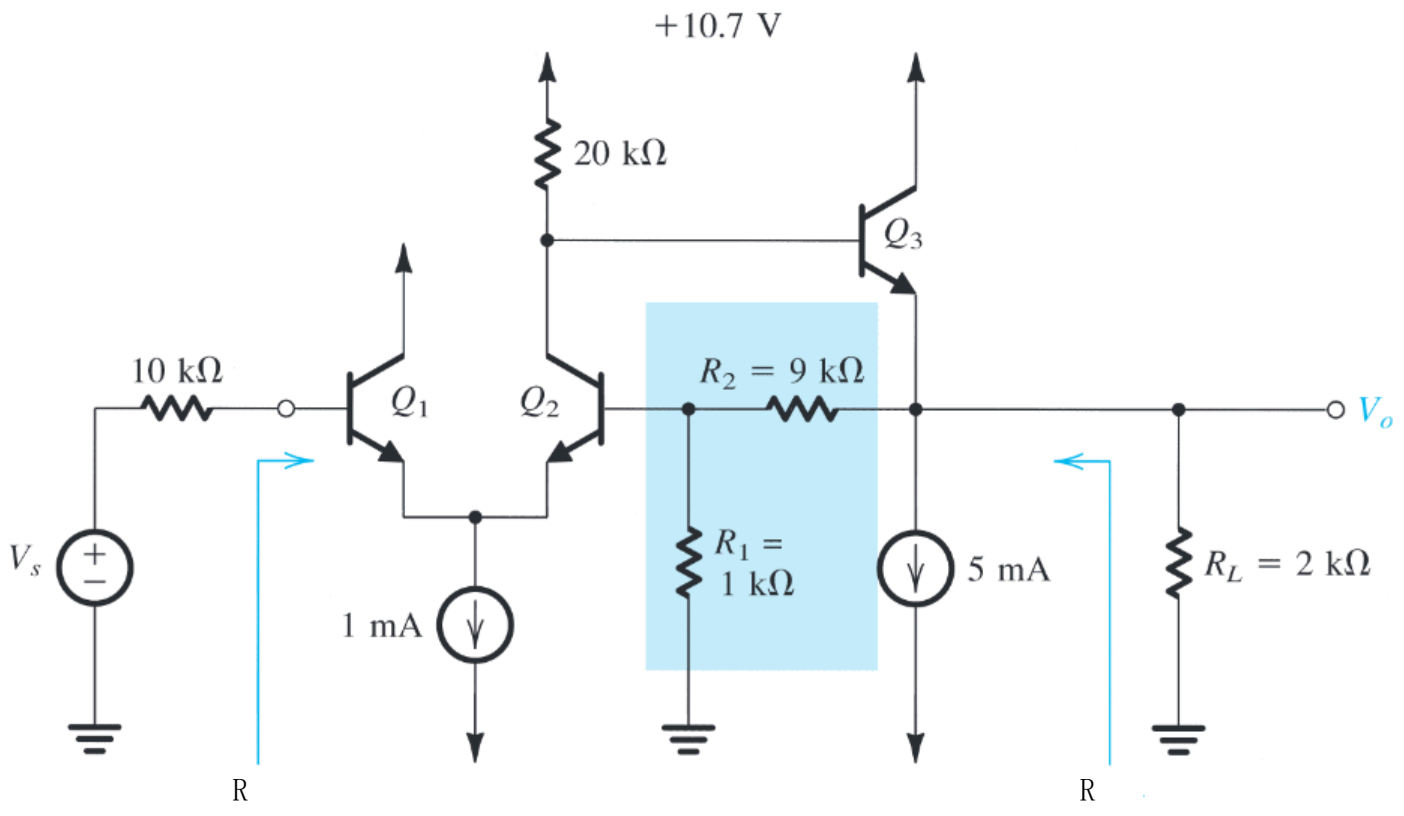


图 E11.8

11.9 For the series–shunt amplifier in Fig. 11.8(c), which was considered in Exercise 11.6, find A, β, A_f, R_{in}, and R_{out}. Neglect r_o of Q. Compare results to those obtained in Exercise 11.6.

Ans. $A = g_m[R_D \| (R_1 + R_2)]$; $\beta = R_1/(R_1 + R_2)$;
$A_f = A/(1 + A\beta)$; $R_{in} = (1/g_m)(1 + A\beta)$;
$R_{out} = [R_D \| (R_1 + R_2)]/(1 + A\beta)$

Comparison: A and β are identical to the corresponding expressions found in Exercise 11.6. However, R_{in} and R_{out} cannot be determined using the method of Exercise 11.6.

11.5 Other Feedback-Amplifier Types

Having studied in detail the most common feedback-amplifier type, the feedback voltage amplifier, we now consider the three other types of feedback amplifier: the feedback transconductance amplifier, the feedback current amplifier, and the feedback transresistance amplifier. The presentation builds on that of the feedback voltage amplifier, and the results will be given without derivation. The analysis method will be illustrated with a large number of worked-out examples dealing with practical and widely utilized circuits.

11.5.1 Basic Principles

1. *Sensing:* The feedback network must sample the output signal of interest. Thus if V_o is the output signal of interest, as in the case of voltage and transresistance amplifiers, the feedback network is connected in parallel (or *shunt*) with the amplifier output node, just as a voltmeter is connected to measure a voltage. On the other hand, if I_o is the output signal of interest, as in the case of transconductance and current amplifiers, the feedback network is connected in *series* with the output loop, just as a current meter is connected to measure a current.
2. *Mixing:* If the input signal to be amplified is a voltage, as in the case of voltage and transconductance amplifiers, the signal source is represented by its Thévenin equivalent and the feedback voltage signal V_f is connected in *series* with the input signal source V_s. On the other hand, if the signal to be amplified is a current, as in the case of current and transresistance amplifiers, the Norton form is used to represent the signal source, and the feedback current signal I_f is connected in parallel (*shunt*) with the input signal source I_s.
3. *Feedback topology:* From the above, it follows that for each of the four amplifier types there is a uniquely appropriate feedback topology:

Amplifier Type	Appropriate Feedback Topology
Voltage	Series–Shunt
Transconductance	Series–Series
Current	Shunt–Series
Transresistance	Shunt–Shunt

11.9 对于练习 11.6 中的图 11.8(c)中的串并联放大器，求 A、β、A、R 和 R。

答案 A = g [R || (R+ R)] ; β = R/R+ R ( );
A= A/(1 + Aβ); R= 1/g (1 + Aβ);
R= [R|| (R+ R)]/(1 + Aβ)

比较：A 和 β 与练习 11.6 中的相应表达式相同。然而，用练习 11.6 的方法无法确定 Rand Rc。

11.5 其他反馈放大器类型

在详细研究了最常见的反馈放大器类型--反馈电压放大器之后，我们现在来研究其他三种类型的反馈放大器：反馈跨导放大器、反馈电流放大器和反馈跨阻放大器。我们将以反馈电压放大器的分析为基础，给出结果而不作推导。分析方法将通过大量实际应用广泛的电路实例加以说明。

11.5.1 基本原则

1. 感应：反馈网络必须对相关输出信号进行采样。因此，如果 V 是所需的输出信号，如电压放大器和跨阻放大器，则反馈网络与放大器输出节点并联（或并联），就像电压表测量电压一样。另一方面，如果 I 是所需的输出信号，就像跨导放大器和电流放大器一样，反馈网络与输出回路串联，就像电流表测量电流一样。

2. 混合：如果要放大的输入信号是电压，如电压放大器和跨导放大器，则信号源用其Th´evenin等效形式表示，反馈电压信号V与输入信号源V串联。另一方面，如果要放大的信号是电流，如电流放大器和跨阻放大器，则使用诺顿形式表示信号源，反馈电流信号 I 与输入信号源 I 平行（并联）连接。

3. 反馈拓扑结构：由此可见，四种放大器类型中的每一种都有独特的合适反馈拓扑结构：

放大器类型	适当的反馈拓扑结构
电压 跨导串联电流分流串联	串联-并联 跨阻分流并联

The appropriate feedback topology not only stabilizes the gain of interest (e.g., the transconductance $A_f \equiv I_o/V_s$ in a transconductance amplifier), but also makes the input and output resistances more ideal (e.g., the shunt–series topology decreases the input resistance and increases the output resistance of a current amplifier).

4. *Input and output resistance:* The increase or decrease of the input or output resistance depends *solely* on the type of connection: Series connection always increases the resistance; parallel (shunt) connection always decreases the resistance. Furthermore, the increase or decrease is always by the amount of feedback, $(1 + A\beta)$. Thus, as an example, for the feedback current amplifier, the shunt connection at the input decreases the input resistance; $R_{if} = R_i/(1 + A\beta)$, and the series connection at the output increases the output resistance; $R_{of} = (1 + A\beta)R_o$, where R_i and R_o are the input and output resistances of the open-loop amplifier (A circuit).
5. *Dimensions of $A, \beta, A\beta$, and A_f:* Depending on the amplifier type, A, β, and A_f have the dimensions of V/V, A/A, V/A, or A/V. However, $A\beta$ is always dimensionless. For a feedback transconductance amplifier, for example, $A \equiv I_o/V_i$ (A/V), $\beta \equiv V_f/I_o$ (V/A), $A_f \equiv I_o/V_s$ (A/V), and $A\beta$ is in V/V or essentially dimensionless.
6. *Analysis using the loop gain:* For any feedback-amplifier type, the loop gain $A\beta$ can be determined using the method described in Section 11.3.3. The loop gain can then be used together with the feedback factor β to determine the open-loop gain A and hence the closed-loop gain A_f. This approach, however, does not enable the determination of the input and output resistances of the feedback amplifier. For these, we need to obtain the A circuit using the systematic analysis approach described below.

Example 11.6

Figure 11.17(a) shows a feedback transconductance amplifier utilizing an op amp together with an NMOS transistor. The feedback network consists of a resistor R_F that senses the output current I_o (recall that the drain and source currents of the MOSFET are equal) and provides a feedback voltage that is subtracted from V_s by means of the differencing action of the op-amp input. Observe that the feedback topology is series–series, which is uniquely appropriate for transconductance amplifiers.

(a) Find β and hence the closed-loop gain $A_f \equiv I_o/V_s$ obtained when $A\beta \gg 1$.
(b) Replace the op amp with its equivalent-circuit model characterized by an open-circuit voltage gain μ, an input differential resistance R_{id}, and an output resistance r_{o1}. Also, replace the MOSFET with its hybrid-π model characterized by a transconductance g_m and an output resistance r_{o2}. With the complete equivalent circuit in hand, set $V_s = 0$ and break the loop to determine $A\beta$.
(c) Use $A\beta$ together with β to find A.
(d) For $\mu = 1000$, $R_{id} = 100\ \text{k}\Omega$, $g_m = 2\ \text{mA/V}$, $r_{o2} = 20\ \text{k}\Omega$, and $R_F = 1\ \text{k}\Omega$, find the values of β, $A\beta$, A, and A_f. Compare A_f to the ideal value obtained when $A\beta \gg 1$.

适当的反馈拓扑结构不仅能稳定相关增益（例如，跨导放大器中的跨导 A ≡ I/V），还能使输入和输出电阻更加理想（例如，并联串联拓扑结构可降低输入电阻，增加电流放大器的输出电阻）。

4. 输入和输出阻抗：输入或输出电阻的增减完全取决于连接方式：串联总是增加电阻，并联（并联）总是减少电阻。此外，电阻的增大或减小总是取决于反馈量 (1 + Aβ)。因此，以反馈电流放大器为例，输入端的并联会减小输入电阻；R = R/（1 + Aβ），而输出端的串联会增大输出电阻；R= (1 + Aβ)R ，其中 Rand 是开环放大器（A 电路）的输入和输出电阻。

5. A、β、Aβ 和 A 的尺寸：根据放大器类型的不同，A、β 和 A 的尺寸分别为 V/V、A/A、V/A 或 A/V。不过，Aβ 始终是无量纲的。例如，对于反馈跨导放大器，A ≡ I /V (A/V)， β ≡ V /I (V/A)，A ≡ I/V (A/V)，而 Aβ 的单位是 V/V，或基本上是无量纲。

6. 利用环路增益进行分析对于任何类型的反馈放大器，均可使用第 11.3.3 节所述方法确定环路增益 Aβ。环路增益与反馈系数 β 可用于确定开环增益 A，进而确定闭环增益 A。然而，这种方法无法确定反馈放大器的输入和输出电阻。为此，我们需要使用下文所述的系统分析方法来获取 A 电路。

例 11.6

图 11.17(a) 显示了一个利用运算放大器和 NMOS 晶体管组成的反馈跨导放大器。反馈网络由一个电阻器 R 组成，该电阻器可检测输出电流 I（请注意 MOSFET 的漏极和源极电流相等），并提供一个反馈电压，通过运算放大器输入端的差分作用从 V 中减去该电压。请注意，反馈拓扑结构是串联式的，这对于跨导放大器来说是独一无二的。

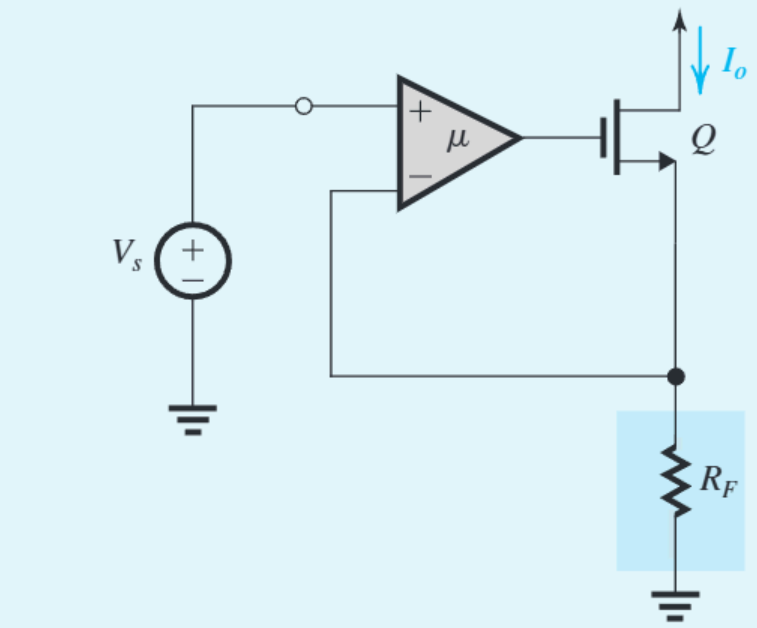
(a) 求出 β，进而求出 Aβ 1 时的闭环增益 A≡ I/V。

(b) 用运算放大器的等效电路模型代替运算放大器，其特征是开路电压增益 μ、输入差分电阻 R 和输出电阻 r。同时，将 MOSFET 换成混合-π 模型，其特征为跨导 g 和输出电阻 r。有了完整的等效电路后，设置 V = 0 并断开回路以确定 Aβ。

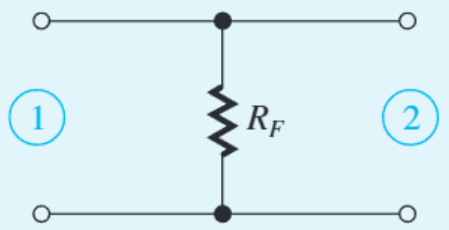
(c) 利用 Aβ 和 β 求出 A。

(d) 对于 μ = 1000、R = 100 k、g = 2 mA/V、r = 20 k 和 R = 1 k，找出 β、Aβ、A 和 A 的值。 将 A 与 Aβ 1 时得到的理想值进行比较。

Example 11.6 *continued*



(a)



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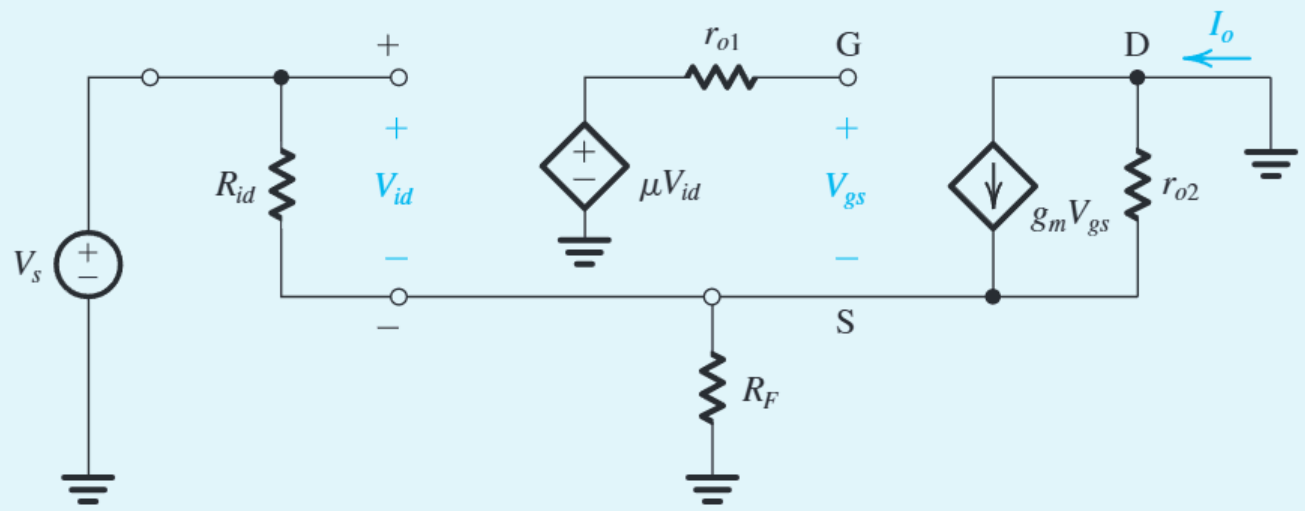
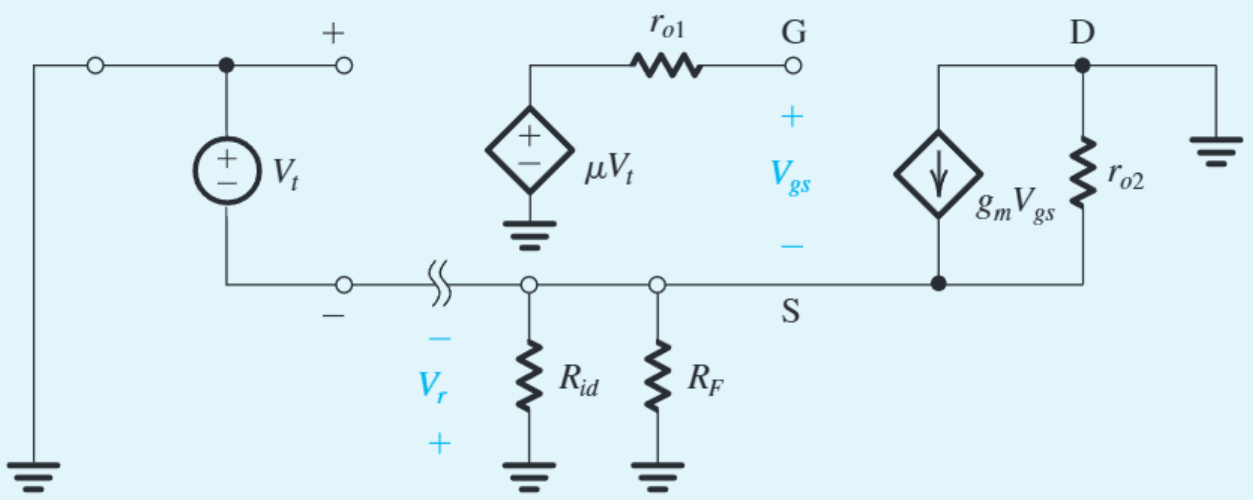


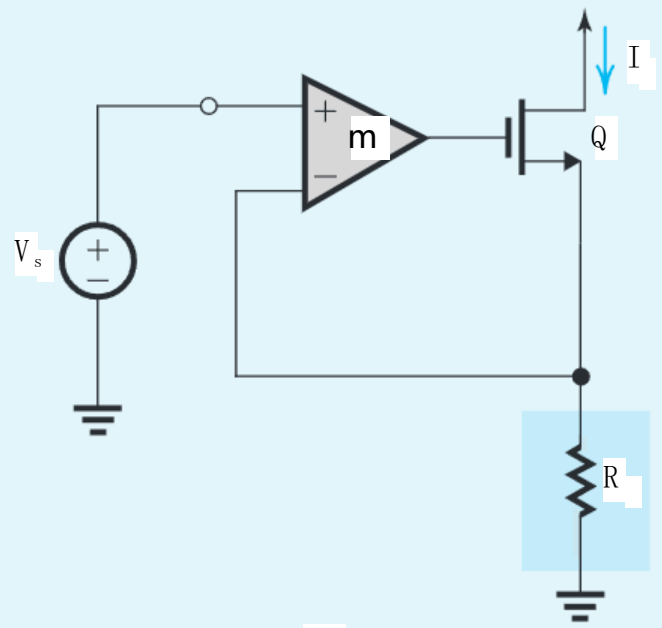
(c)



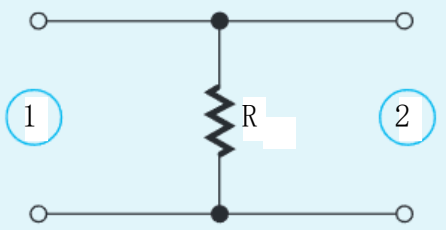
(d)

Figure 11.17 Example 11.6.

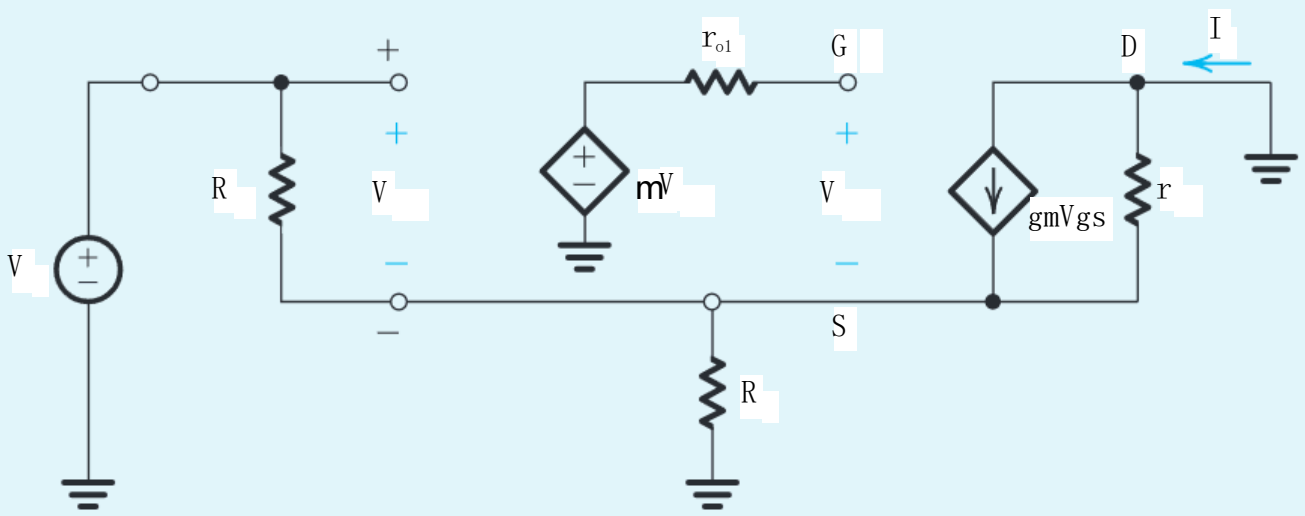
例 11.6 续



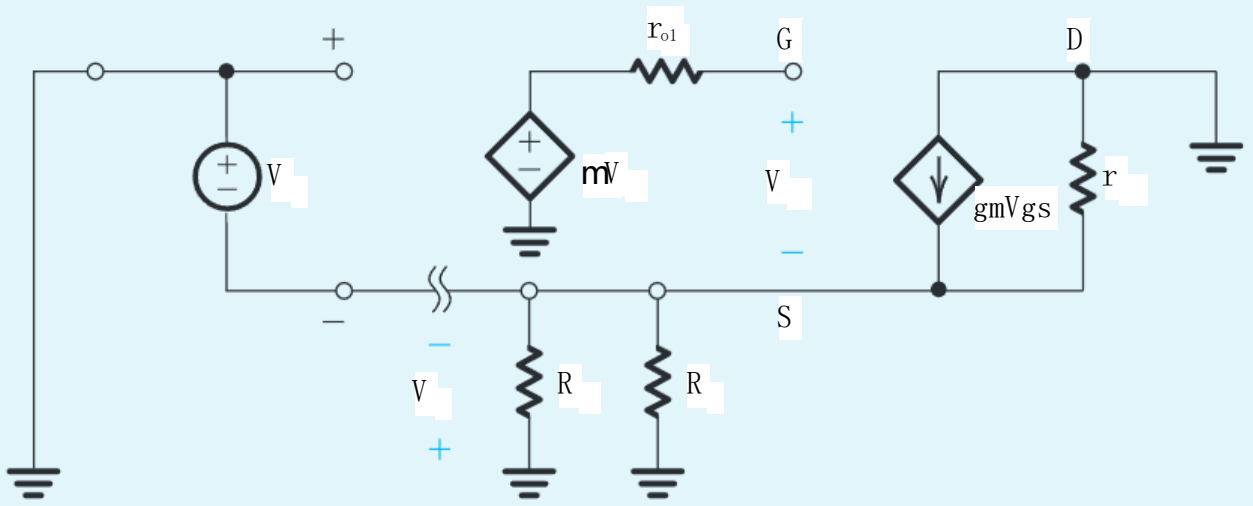
(a)



(b)



(c)



(d)

图 11.17 例 11.6.

Solution

(a) The two-port feedback network is shown in Fig. 11.17(b). When port 2 is fed with a current I_o the voltage V_f appearing across port 1 is $I_o R_F$, thus

$$\beta = R_F$$

When $A\beta \gg 1$,

$$A_f \equiv \frac{I_o}{V_s} \simeq \frac{1}{\beta} = \frac{1}{R_F}$$

(b) Figure 11.7(c) shows the equivalent-circuit model of the feedback transconductance amplifier. To determine $A\beta$, we set $V_s = 0$ and break the loop at the input terminals of the op amp as shown in Fig. 11.17(d). Note that we have applied an input voltage V_t and terminated the loop at the break in the resistance R_{id} to prevent any change in the loop conditions. This circuit can be easily analyzed to determine $A\beta \equiv -V_r/V_t$ as follows:

$$V_{gs} = \mu V_t - (-V_r) = \mu V_t + V_r$$
$$-V_r = g_m V_{gs}(R_F \parallel R_{id} \parallel r_{o2})$$

Combining these two equations yields

$$A\beta \equiv -\frac{V_r}{V_t} = \mu \frac{g_m(R_F \parallel R_{id} \parallel r_{o2})}{1 + g_m(R_F \parallel R_{id} \parallel r_{o2})}$$

(c) Substituting $\beta = R_F$ provides

$$A = \frac{\mu}{R_F} \frac{g_m(R_F \parallel R_{id} \parallel r_{o2})}{1 + g_m(R_F \parallel R_{id} \parallel r_{o2})}$$

(d) For the given numerical values,

$$\beta = 1\ \text{k}\Omega$$
$$A\beta = 1000 \times \frac{2(1 \parallel 100 \parallel 20)}{1 + 2(1 \parallel 100 \parallel 20)}$$
$$= 653.6$$
$$A = 653.6\ \text{mA/V}$$
$$A_f \equiv \frac{I_o}{V_s} = \frac{653.6}{1 + 653.6} = 0.9985\ \text{mA/V}$$

Since the ideal value of A_f is 1 mA/V, the actual A_f is only 0.15% lower than ideal.

解决方案

(a) 双端口反馈网络如图 11.17(b)所示。当端口 2 通入电流 I 时，端口 1 上出现的电压 V 为 I R，因此

$$\beta = R$$

当 Aβ 1、

$$A \equiv \frac{I}{V} \cdot \frac{1}{\beta} = \frac{1}{R}$$

(b) 图 11.7(c) 显示反馈跨导放大器的等效电路模型。如图 11.17(d)所示，为确定 Aβ，我们设置 V = 0 并在运算放大器的输入端断开环路。请注意，我们施加了输入电压 V，并在电阻 R 的断开处终止了环路，以防止环路条件发生任何变化。可以很容易地分析出该电路

如下确定 Aβ ≡ -V /V：

$$V = \mu V - (-V) = \mu V + V$$

$$-V = g V (R || R || r)$$

将这两个等式合并得出

$$A\beta \equiv -\frac{V}{V} = \mu \frac{g ("R "R "r)}{1 + g(R || R || r)}$$

(c) 将 β = R 代入可得

$$A = \frac{\mu}{R} \frac{g ("R "R "r)}{1 + g(R || R || r)}$$

(d) 对于给定的数值、

$$\beta = 1 \text{ k}$$

$$A\beta = 1000 \times \frac{2(1 || 100 || 20)}{1 + 2(1 || 100 || 20)}$$

$$= 653.6$$

$$A = 653.6 \text{ mA/V}$$

$$A \equiv \frac{I}{V} = \frac{653.6}{1 + 653.6} = 0.9985 \text{ mA/V}$$

由于 A 的理想值为 1 mA/V，因此实际 A 仅比理想值低 0.15%。

EXERCISES

11.10 For the circuit in Example 11.6, let the op-amp gain decrease by 10%. What is the corresponding percentage change in $A_f \equiv I_o/V_s$?
Ans. -0.02%

D11.11 Redesign the circuit in Example 11.6 to obtain a nominal closed-loop transconductance of 2 mA/V. What is the required value of R_F, and what is the actual transconductance A_f realized?
Ans. $R_F = 500\ \Omega$; $A_f = 1.996$ mA/V

11.5.2 The Feedback Transconductance Amplifier (Series–Series)

Figure 11.18(a) shows the ideal structure of the feedback transconductance amplifier. The open-loop amplifier (A circuit) is unilateral, has an input resistance R_i, a short-circuit transconductance gain A in A/V, and an output resistance R_o. The short-circuit output current $I_o = AV_i$ is sensed by the feedback network. Note that since the resistance looking into port 2 of the feedback network is zero, the feedback network does not load the amplifier output. The feedback network provides at port 1 a voltage signal $V_f = \beta I_o$, where the feedback factor β has the dimensions of V/A. The feedback signal is connected in series with the input signal source V_s, and the feedback network does not load the amplifier input. Finally, note the definitions of the input resistance with feedback, R_{if}, and the output resistance with feedback, R_{of}. The latter is the resistance found by looking into the output loop between any two nodes such as O and O'.

The equivalent circuit of the feedback transconductance amplifier is shown in Fig. 11.18(b). Note that the closed-loop gain A_f is the ratio of the short-circuit output current I_o and the input voltage V_s, thus it is the short-circuit transconductance of the feedback amplifier. Also, R_{of} is the resistance seen between any two nodes in the output loop, such as O and O', while V_s is set to zero. Finally, Fig. 11.18(c) provides the formulas for determining A_f, R_{if}, and R_{of}.

With the ideal case in hand, we now consider the general or practical case of a feedback transconductance amplifier, shown in Fig. 11.19(a). To be able to apply feedback analysis to this circuit, we have to find the A circuit and β. These are shown in Fig. 11.19(b). The A circuit is obtained by augmenting the basic amplifier with R_s and R_L and the two resistances R_{11} and R_{22}, which represent the loading effect of the feedback network on the basic amplifier at the input and output, respectively. Figure 11.19(b) shows how R_{11} and R_{22} are determined. Here, the series connection at both the input and the output means that the other port of the feedback network must be left open-circuited. This is also the case when β is determined.

练习

11.10 对于例 11.6 中的电路，让运算放大器增益降低 10%。A≡ I/V 的相应变化百分比是多少？

答案-0.02%

D11.11 重新设计例 11.6 中的电路，以获得 2 mA/V 的额定闭环跨导。
所需的 R 值是多少，实际实现的跨导 A 又是多少？答R = 500 ；A= 1.996 mA/V

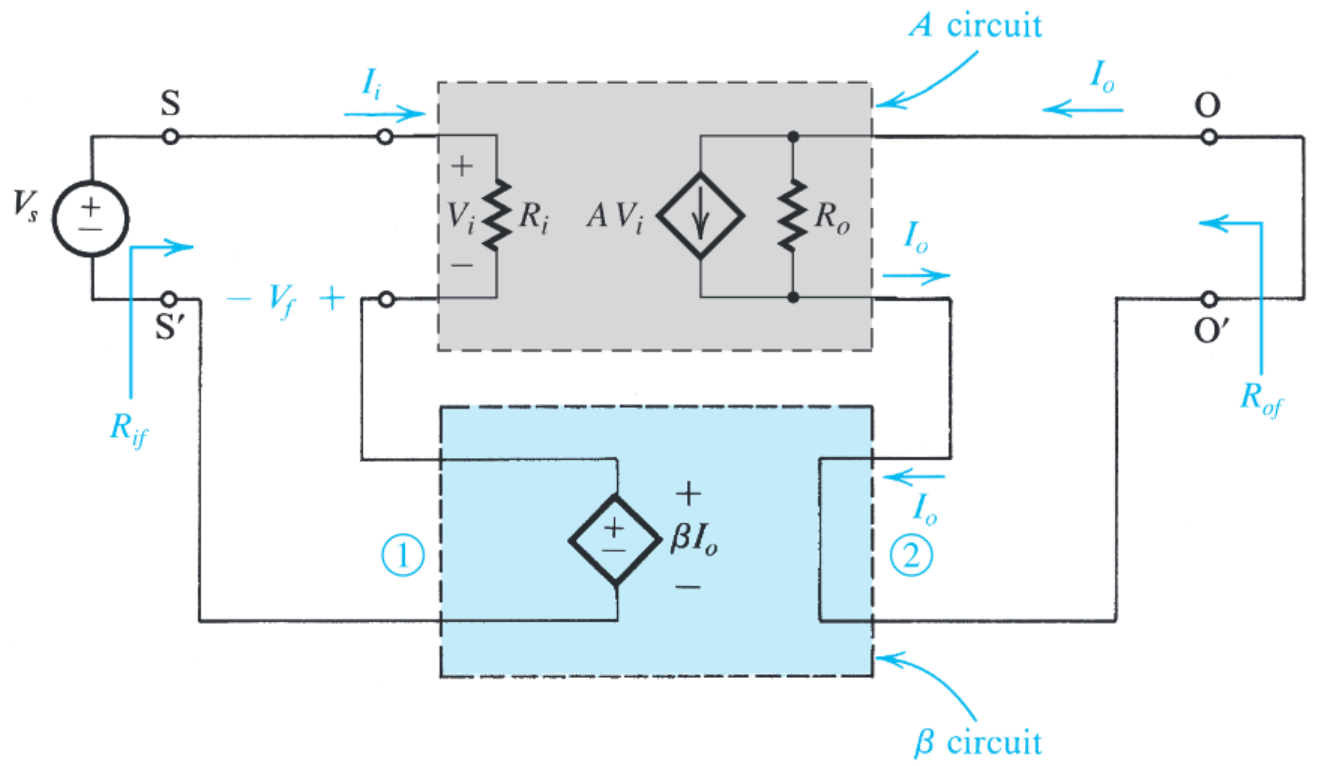
11.5.2 反馈跨导放大器（串联式）

图 11.18(a) 显示了反馈跨导放大器的理想结构。开环放大器（A 电路）是单边的，具有输入电阻 R、以 A/V 为单位的短路跨导增益 A 和输出电阻 R。请注意，由于反馈网络端口 2 的电阻为零，因此反馈网络不会对放大器的输出造成负载。反馈网络在端口 1 提供电压信号 V = βI ，其中反馈系数 β 的大小为 V/A。反馈信号与输入信号源 V 串联，反馈网络不会对放大器输入端造成负载。最后，请注意带反馈的输入电阻 R 和带反馈的输出电阻 R 的定义。后者是通过观察任意两个节点（如 O 和 O）之间的输出环路发现的电阻。

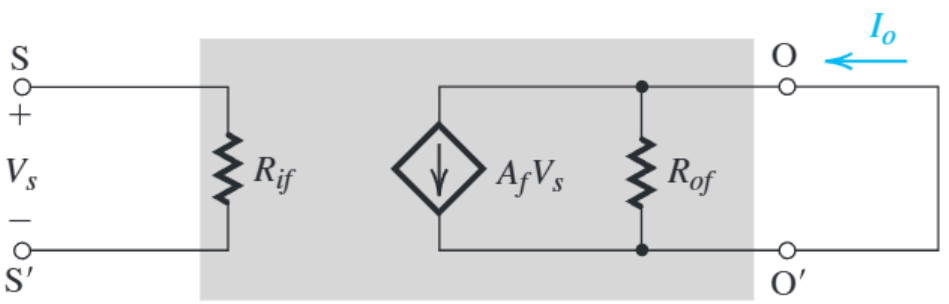
反馈跨导放大器的等效电路如图 11.18(b)所示。请注意，闭环增益 A 是短路输出电流 I 与输入电压 V 之比，因此它是反馈放大器的短路跨导。另外，将输出环路中任意两个节点（如 O 和 O）之间的电阻设为 Ris，而 Vis 设为零。最后，图 11.18(c) 提供了确定 A、R 和 R 的公式。

有了理想情况，我们现在来考虑一般或实际情况下的反馈跨导放大器，如图 11.19(a)所示。要对该电路进行反馈分析，我们必须找到 A 电路和 β。A 电路是用 R 和 R 以及两个电阻 Rand R 增强基本放大器后得到的，这两个电阻分别表示反馈网络在输入和输出端对基本放大器的负载效应。图 11.19(b) 显示了如何确定 Rand R。在这里，输入和输出端的串联意味着反馈网络的另一个端口必须保持开路状态。在确定 β 时也是如此。

(a) Ideal Structure



(b) Equivalent Circuit



(c) Formulas

$$A_f = \frac{I_o}{V_s} = \frac{A}{1+A\beta}$$

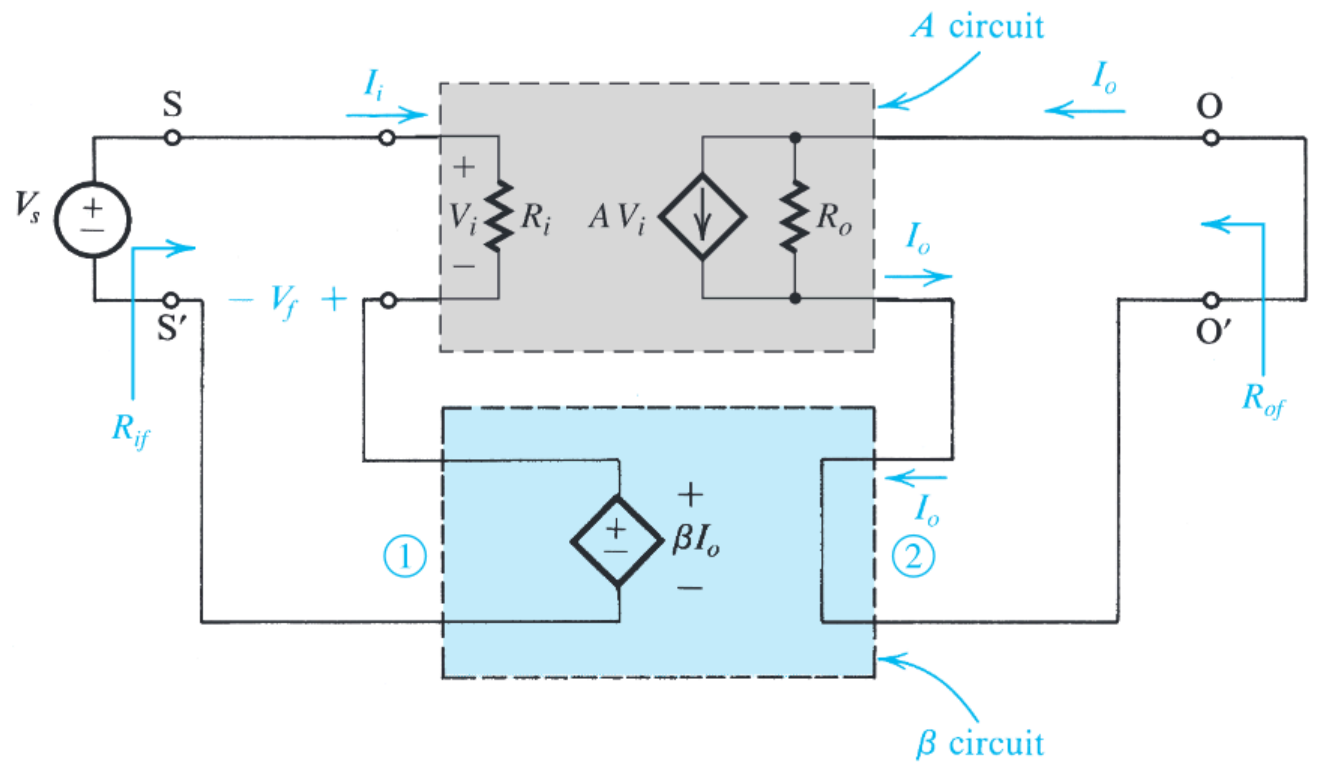
$$R_{if} = (1+A\beta)R_i$$

$$R_{of} = (1+A\beta)R_o$$

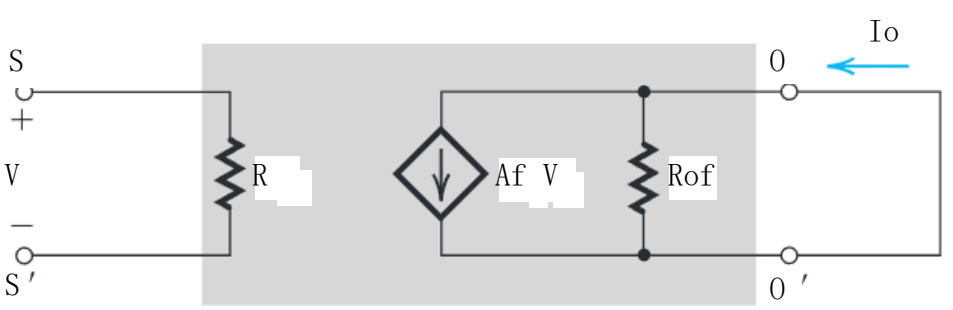
Figure 11.18 The feedback transconductance amplifier (series–series).

Finally Fig. 11.19(c) gives the formulas for determining the actual values of the input and output resistances, R_{in} and R_{out}, of the feedback amplifier from R_{if} and R_{of}. To see how these formulas come about, note from Fig. 11.19(a) that unlike R_{if}, R_{in} does not include R_s, and unlike R_{of}, R_{out} does not include R_L.

(a) Ideal Structure



(b) 等效电路



(c) 公式

$$A = \frac{I}{V} = \frac{A}{1 + A\beta}$$

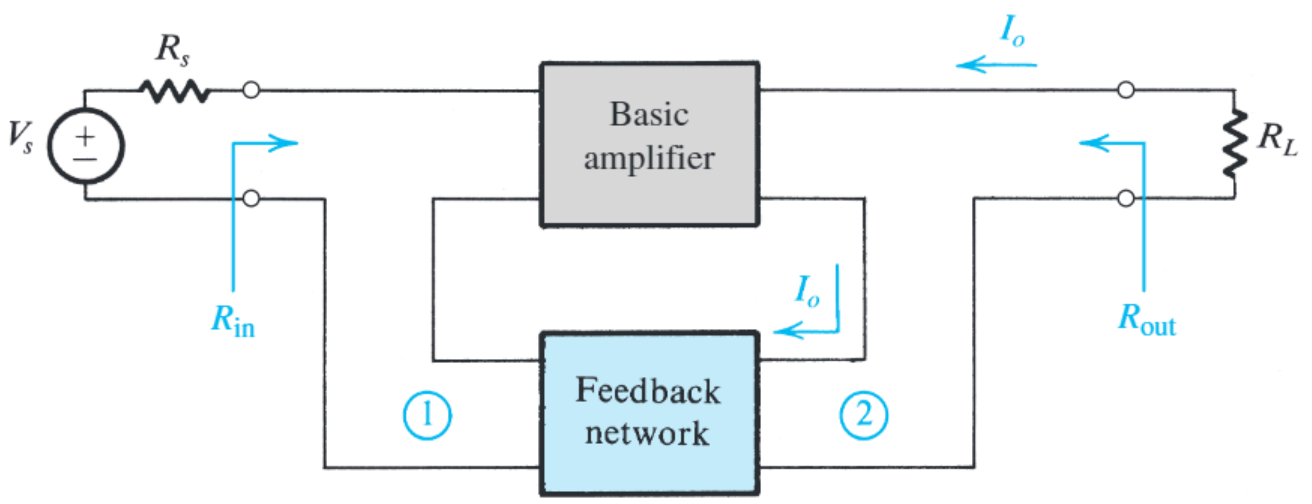
$$R = (1 + A\beta)R$$

$$R = (1 + A\beta)R$$

图 11.18 反馈跨导放大器（串联）。

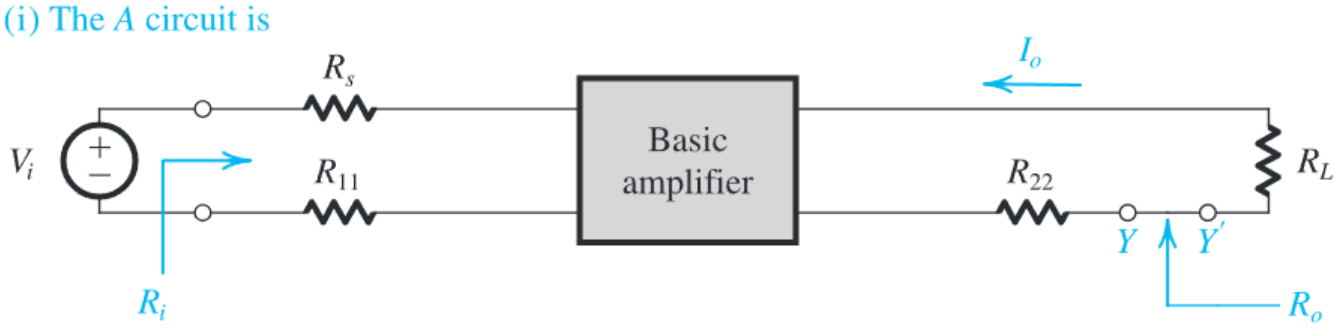
最后，图 11.19(c)给出了根据 R 和 R 确定反馈放大器输入和输出电阻 Rand R 实际值的公式。要了解这些公式是如何得出的，请注意图 11.19(a)，与 R 不同，Rd 不包括 R ，与 R 不同，Rd 不包括 R 。

(a) General Structure

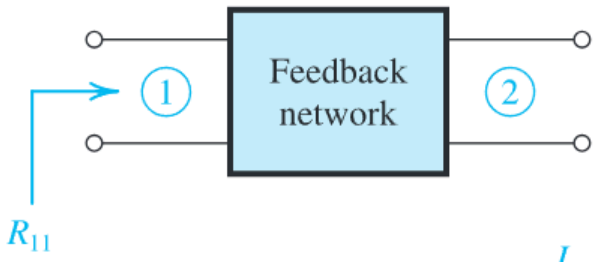


(b) Finding the A Circuit and β

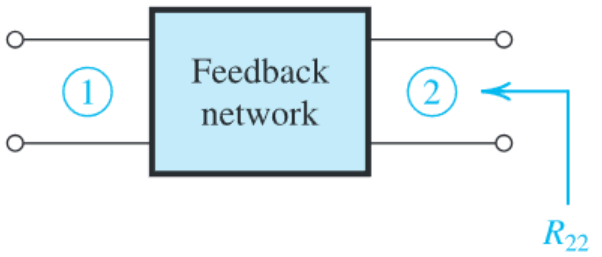
(i) The A circuit is



where R_{11} is obtained from

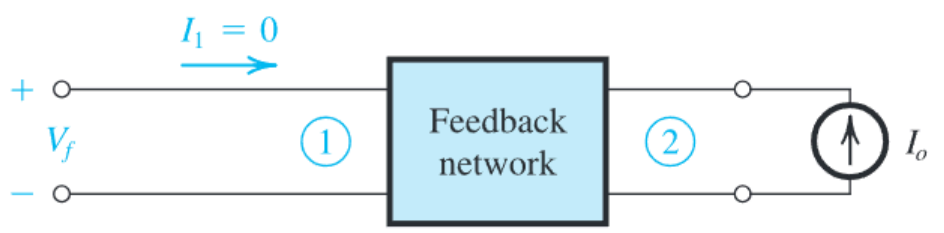


and R_{22} is obtained from



and the gain A is defined $A \equiv \dfrac{I_o}{V_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{V_f}{I_o}\right|_{I_1 = 0}$$

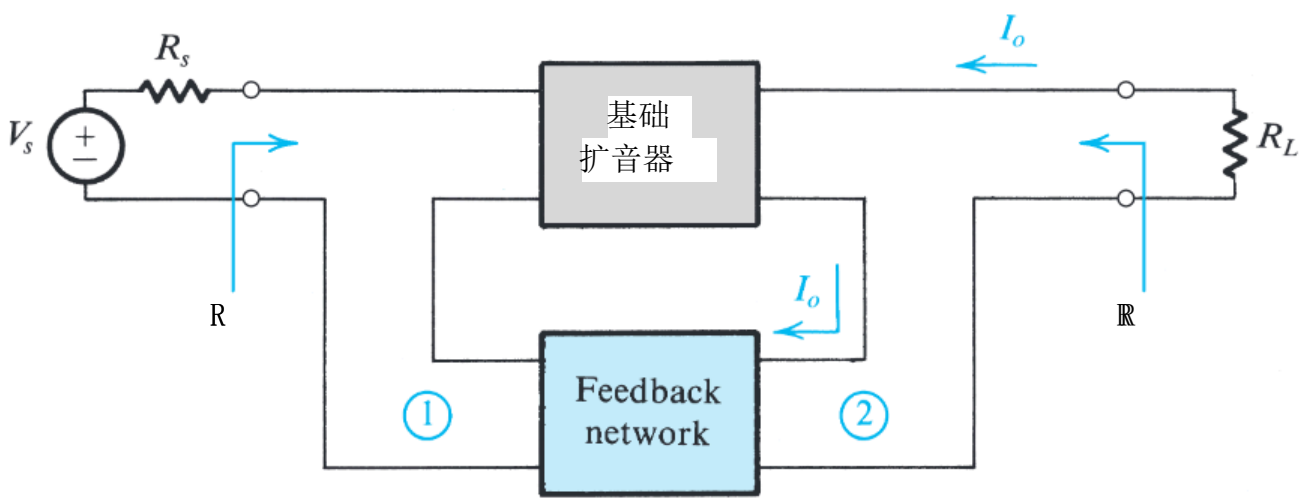
(c) Gain, Input, and Output Resistance

- Use the formulas in Fig. 11.18 to find A_f, R_{if}, and R_{of}.
- R_{in} and R_{out} can then be found from

$$R_{in} = R_{if} - R_s$$
$$R_{out} = R_{of} - R_L$$

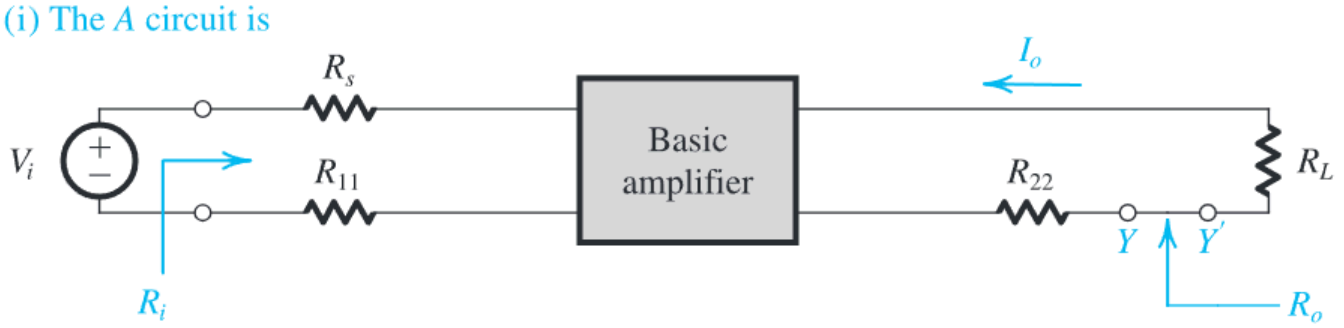
Figure 11.19 The feedback transconductance amplifier (series–series).

(a) 总体结构

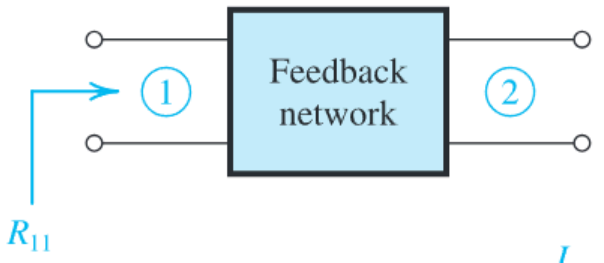


(b) Finding the A Circuit and β

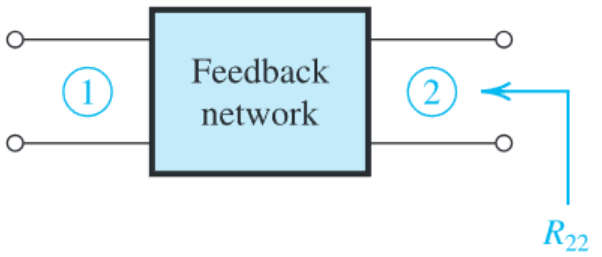
(i) The A circuit is



where R_{11} is obtained from

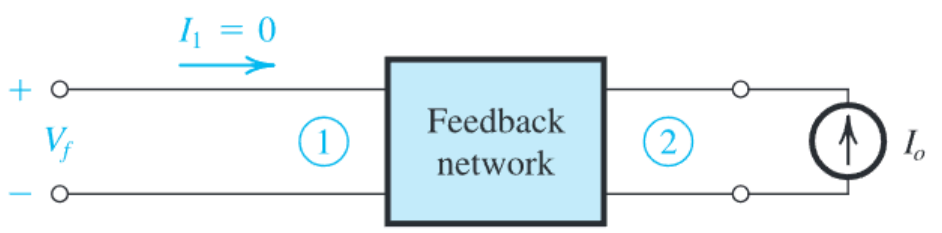


and R_{22} is obtained from



and the gain A is defined $A \equiv \dfrac{I_o}{V_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{V_f}{I_o}\right|_{I_1 = 0}$$

(c) 增益、输入和输出电阻

- 使用图 11.18 中的公式求出 A、R 和 R。
- Rand R 的计算公式为

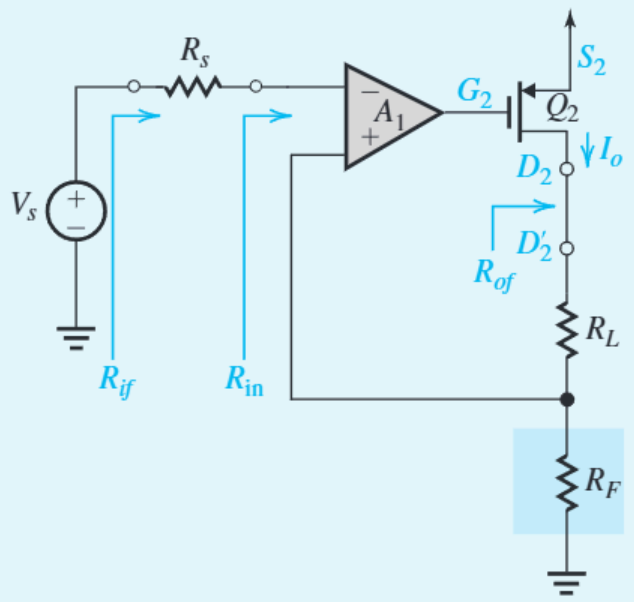
R= R − R

R= R − R

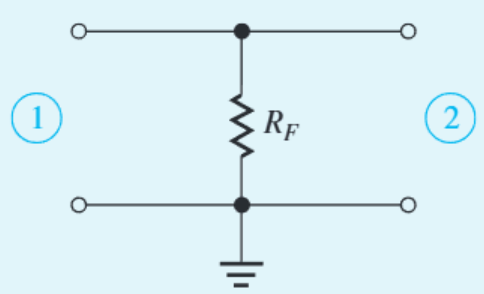
图 11.19 反馈跨导放大器（串联）。

Example 11.7

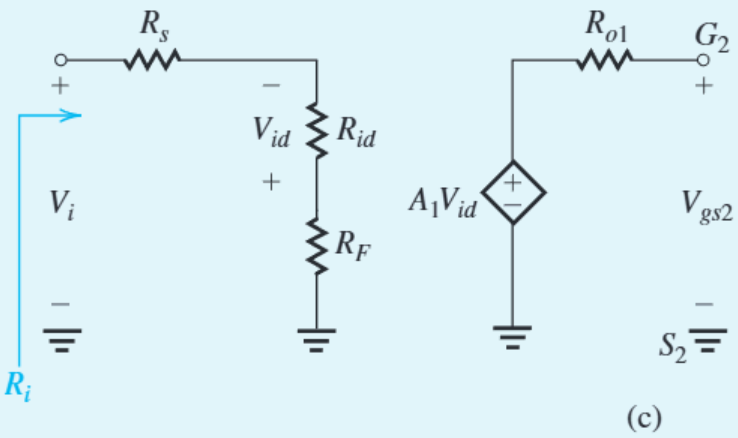
Figure 11.20(a) shows a feedback transconductance amplifier composed of a differential amplifier A_1 with an input differential resistance R_{id}, an open-circuit voltage gain A_1, and an output resistance R_{o1}, connected in cascade with a common source MOSFET Q_2 having a transconductance g_{m2} and an output resistance r_{o2}. Use the feedback-analysis method to determine the closed-loop transconductance $A_f \equiv I_o/V_s$, the input resistance R_{in}, and the output resistance R_{out}. The latter is the resistance seen between the terminals of R_L, looking back into the output loop.

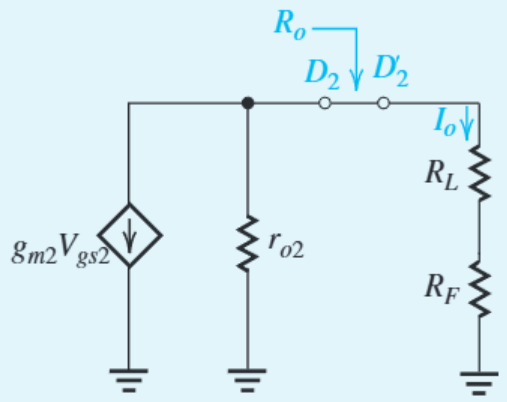


(a)

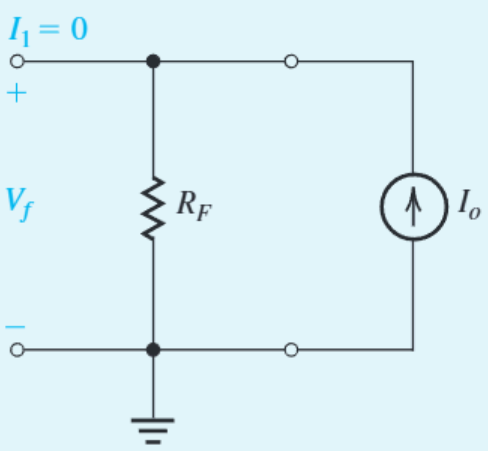


(b)





(c)

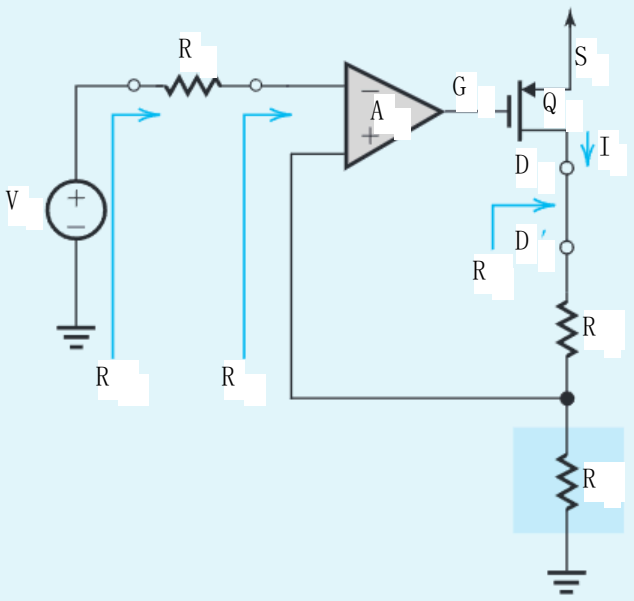


(d)

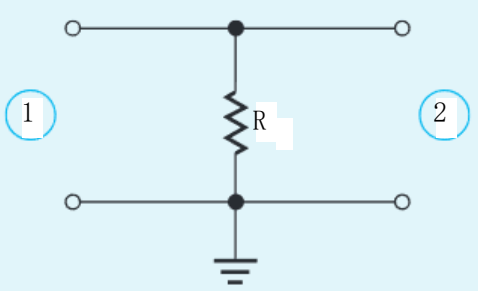
Figure 11.20 Circuits for Example 11.7.

例 11.7

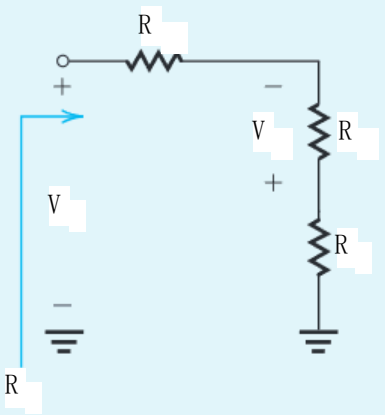
图 11.20(a) 显示了一个反馈跨导放大器，它由一个差分放大器 Aw（具有输入差分电阻 R、开路电压增益 A 和输出电阻 R）与一个共源 MOSFET Q（具有跨导率 g 和输出电阻 r）级联组成。使用反馈分析法确定闭环跨导 A≡ I/V、输入电阻 R 和输出电阻 R。

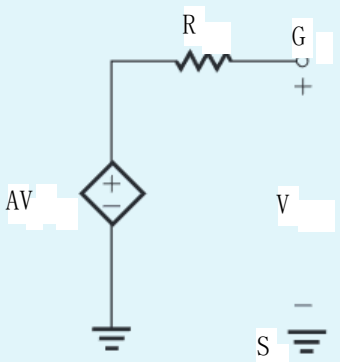


(a)

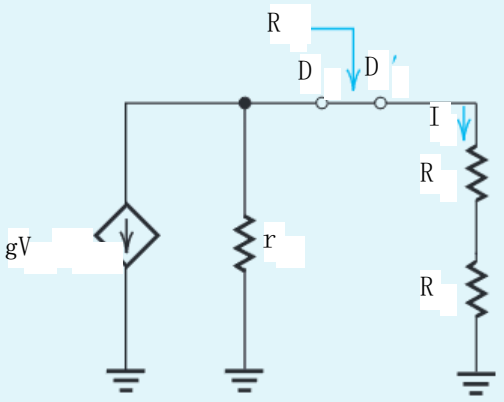


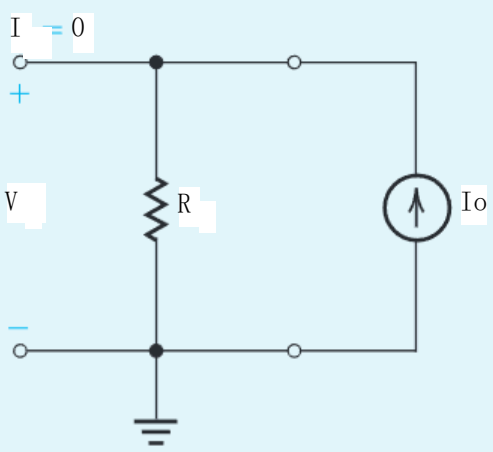
(b)





(c)





(d)

图 11.20 例 11.7 的电路

Example 11.7 *continued*

Solution

First we identify the basic amplifier and the feedback circuit. The basic amplifier consists of the differential amplifier A_1 cascaded with the CS PMOS transistor Q_2. The output current I_o is sensed by the series resistance R_F. The latter is the feedback network (Fig. 11.20b). It develops a voltage V_f that is mixed in series with the input loop.

The second step is to ascertain that the feedback is negative. This can be done by assuming an increase in V_s and following the resulting change around the loop. An increase in V_s will cause the voltage at the inverting input terminal of A_1 to increase. This in turn causes a decrease in the output voltage of A_1, which is the voltage at the gate of Q_2. Thus, transistor Q_2 will have a larger V_{SG}, which will cause I_o to increase. This in turn causes an increase in the voltage across R_F, which is the same polarity as the change in V_s. Thus, the feedback is indeed negative.

Next, we determine an approximate value for $A_f \equiv I_o/V_s$ under the assumption that the loop gain $A\beta$ is much greater than unity. This value, found before any analysis is undertaken, will help us determine at the end whether our analysis is correct: If the loop gain is found to be much greater than unity, then the final A_f should be close to the value initially determined. From the circuit of Fig. 11.20(d),

$$\beta = R_F$$

and thus for large $A\beta$,

$$A_f \simeq \frac{1}{\beta} = \frac{1}{R_F}$$

Next, we determine the A circuit. Since the feedback network (Fig. 11.20b) is connected in series with both the input and output loops, we include a resistance R_F in each of these loops (which is equivalent to saying we include, at the input, the input resistance of the feedback circuit with port 2 open and, at the output, the input resistance of the feedback circuit with port 1 open). Doing this, including R_s and R_L in the A circuit, and replacing A_1 and Q_2 with their small-signal models, results in the A circuit shown in Fig. 11.20(c). Analysis of this circuit is straightforward:

$$V_{id} = -V_i \frac{R_{id}}{R_{id} + R_s + R_F} \tag{11.27}$$

$$V_{gs2} = A_1 V_{id} \tag{11.28}$$

$$I_o = -g_{m2} V_{gs2} \frac{r_{o2}}{r_{o2} + R_L + R_F} \tag{11.29}$$

Combining these three equations results in

$$A \equiv \frac{I_o}{V_i} = (A_1 g_{m2}) \left(\frac{R_{id}}{R_{id} + R_s + R_F} \right) \left(\frac{r_{o2}}{r_{o2} + R_L + R_F} \right) \tag{11.30}$$

Usually $R_{id} \gg (R_s + R_F)$, $r_{o2} \gg (R_L + R_F)$, resulting in the approximate expression for A:

$$A \simeq A_1 g_{m2} \tag{11.31}$$

例 11.7 续

解决方案

首先，我们确定基本放大器和反馈电路。基本放大器由差分放大器 Acascaded 和 CS PMOS 晶体管 Q 组成。后者是反馈网络（图 11.20b）。它产生与输入回路串联的混合电压 V。

第二步是确定反馈为负。要做到这一点，可以假定 V 值增加，并跟踪环路上的变化。V 的增加将导致 At 反相输入端的电压增加。这样，晶体管 Q 的 V 值将增大，从而导致 I 值增大。这反过来又会导致 R 两端的电压升高，其极性与 V 的变化相同。

因此，反馈确实是负面的。

接下来，我们假设环路增益 Aβ 远大于统一值，从而确定 A ≡ I /V 的近似值。这个值是在进行任何分析之前确定的，它将帮助我们在最后确定分析是否正确：如果发现环路增益远大于 1，那么最终的 A 值应该接近最初确定的值。从图 11.20(d)的电路中可以看出

$$\beta = R$$

因此对于大 Aβ、

$$A \approx \frac{1}{\beta} = \frac{1}{R}$$

接下来，我们确定 A 电路。由于反馈网络（图 11.20b）与输入和输出回路串联，因此我们在每个回路中都加入一个电阻 R（相当于在输入端加入端口 2 开路时反馈回路的输入电阻，在输出端加入端口 1 开路时反馈回路的输入电阻）。这样，在 A 电路中加入 R 和 R，并用它们的小信号模型代替 A 和 Q，就得到了图 11.20(c)所示的 A 电路。对该电路的分析非常简单：

$$V = -V \frac{R}{R + R + R} \tag{11.27}$$

$$V = AV \tag{11.28}$$

$$I = -g V \frac{r}{r + R + R} \tag{11.29}$$

将这三个等式合并得出

$$A \equiv \frac{I}{V} = (Ag)\left(\frac{R}{R + R + R}\right)\left(\frac{r}{r + R + R}\right) \tag{11.30}$$

通常为 R (R+R)、r (R+R)，从而得到 A 的近似表达式：

$$A \approx Ag \tag{11.31}$$

The input resistance R_i can be found by inspection as

$$R_i = R_s + R_{id} + R_F \tag{11.32}$$

The output resistance R_o is found by setting $V_i = 0$, and breaking the output loop at any location, say between D_2 and D_2'. Thus,

$$R_o = r_{o2} + R_L + R_F \tag{11.33}$$

The loop gain $A\beta$ is thus

$$A\beta = (A_1 g_{m2} R_F)\left(\frac{R_{id}}{R_{id} + R_s + R_F}\right)\left(\frac{r_{o2}}{r_{o2} + R_L + R_F}\right) \tag{11.34}$$

$$\simeq A_1 g_{m2} R_F \tag{11.35}$$

With numerical values, one can now obtain the value of $A\beta$ and determine whether it is indeed much greater than unity. We next determine the closed-loop gain

$$A_f = \frac{A}{1 + A\beta}$$

Substituting for A from Eq. (11.31) and for $A\beta$ from Eq. (11.35), we have

$$A_f \simeq \frac{A_1 g_{m2}}{1 + A_1 g_{m2} R_F} \tag{11.36}$$

For $A_1 g_{m2} R_F \gg 1$,

$$A_f \simeq \frac{1}{R_F}$$

which is the value we found at the outset.

The series mixing raises the input resistance with feedback,

$$R_{if} = R_i(1 + A\beta)$$

and R_{in} can be obtained by subtracting R_s from R_{if}.

To obtain R_{of}, we note that the series connection at the output raises the output resistance, thus,

$$R_{of} = R_o(1 + A\beta)$$

and R_{out}, which is the resistance seen by R_L, can be obtained by subtracting R_L from R_{of}.

31）输入电阻 R 的计算公式为 R = R+ R+ R

(11.32)

通过设置 V= 0，并在任意位置（如 D 和 D 之间）断开输出回路，即可求得输出电阻、

R= r+ R+ R (11.33)

因此，环路增益 Aβ 为

$$A\beta = (\quad)\left(\frac{R}{R + R+ R}\right)\left(\frac{r}{r + R+ R}\right) \qquad (11.34)$$

Ag R (11.35)

有了数值，我们就可以得到 Aβ 的值，并确定它是否确实远大于整数。接下来，我们将确定闭环增益

$$A= \frac{A}{1 + A\beta}$$

代入公式（11.31）中的 A 和公式（11.35）中的 Aβ，我们得出

$$A \square \frac{Ag}{1+ Ag R} \qquad (11.36)$$

对于 Ag R 1、

$$A \square \frac{1}{R}$$

这就是我们一开始发现的值。

串联混合可通过反馈提高输入电阻、

$$R = R(1 + A\beta)$$

从 R 中减去 R 即可得到 R。

为了得到 R，我们注意到输出端的串联会提高输出电阻，因此、

$$R= R(1 + A\beta)$$

而 R 是 R 看到的电阻，可以通过从 R 中减去 R 得到。

EXERCISES

D11.12 For the circuit analyzed in Example 11.7, select a value for R_F that will result in $A_f \simeq 5$ mA/V. Now, for $A_1 = 200$ V/V, $g_{m2} = 2$ mA/V, $R_{id} = 100$ kΩ, $r_{o2} = 20$ kΩ, and assuming that $R_s \ll R_{id}$ and $R_L \ll r_{o2}$, find the value of A_f realized and the input and output resistances of the feedback transconductance amplifier. If for some reason g_{m2} drops in value by 50%, what is the corresponding percentage change in A_f?
Ans. 200 Ω; 4.94 mA/V; 8.1 MΩ; 1.64 MΩ; −1.25%

11.13 Determine the loop gain of the feedback amplifier of Fig. 11.20(a) by setting $V_s = 0$, breaking the feedback loop at G_2, applying a voltage V_{g2}, and determining the voltage V_{o1} that appears at the output of A_1; $A\beta \equiv -V_{o1}/V_{g2}$. Assume $R_F \ll (R_{id} + R_s)$ and $r_{o2} \gg R_L + R_F$.
Ans. $A\beta \simeq A_1 g_{m2} R_F$

11.14 Utilizing the full expression for $A\beta$ in Eq. (11.34) but assuming that $r_{o2} \gg R_L + R_F$ and $R_F \ll R_{id}$, show that

$$R_{\text{in}} \simeq R_{id}(1 + A_1 g_{m2} R_F)$$

11.15 Utilizing the full expression for $A\beta$ in Eq. (11.34) but assuming that $R_{id} \gg R_s + R_F$ and $R_F \ll r_{o2}$ show that

$$R_{\text{out}} \simeq r_{o2}(1 + A_1 g_{m2} R_F)$$

Example 11.8

Because negative feedback extends the amplifier bandwidth, it is commonly used in the design of broadband amplifiers. One such amplifier is the MC1553. Part of the circuit of the MC1553 is shown in Fig. 11.21(a). The circuit shown (called a **feedback triple**) is composed of three gain stages with series–series feedback provided by the network composed of R_{E1}, R_F, and R_{E2}.

Observe that the feedback network samples the emitter current I_o of Q_3, and thus I_o is the output quantity of the feedback amplifier. However, practically speaking, I_o is rather difficult to utilize. Thus it is usual to take as the output I_c, the collector current of Q_3. This current is of course almost equal to I_o; $I_c = \alpha I_o$. Thus, as a transconductance amplifier with I_c as the output current, the output resistance of interest is that labeled R_{out} in Fig. 11.21(a). In some applications, I_c is passed through a load resistance, such as R_{C3}, and the voltage V_o is taken as the output. Assume that the bias circuit, which is not shown, establishes $I_{C1} = 0.6$ mA, $I_{C2} = 1$ mA, and $I_{C3} = 4$ mA. Also assume that for all three transistors,[5] $h_{fe} = 100$ and $r_o = \infty$.

[5]To avoid possible confusion of the BJT current gain β and the feedback factor β, we sometimes use h_{fe} to denote the transistor β.

练习

D11.12 对于例 11.7 中分析的电路，请选择能使电流 A 达到 5 mA/V 的 R 值。

现在，对于 A= 200 V/V，g= 2 mA/V，R= 100 k，r= 20 k，并假设 R R 和 R r ，求反馈跨导放大器的阿雷化值以及输入和输出电阻。如果由于某种原因 g 值下降了 50%，A 的相应变化百分比是多少？

答案：200；4.94 mA/V；8.1 M；1.64 M；-1.25

11.13 通过设置 V = 0、断开 G 处的反馈回路、施加电压 V 并确定 A 输出端的电压 V，来确定图 11.20(a)中反馈放大器的环路增益；Aβ ≡ -V/V 。假设 R (R+ R) 和 r R+ R。

答案：Aβ Ag R

11.14 利用公式 (11.34) 中 Aβ 的完整表达式，但假设 r R+ R 和 R R，证明

$$R_R(1 + A_g R)$$

11.15 利用公式 (11.34) 中 Aβ 的完整表达式，并假设 R R+ R 和 R r，可以得出

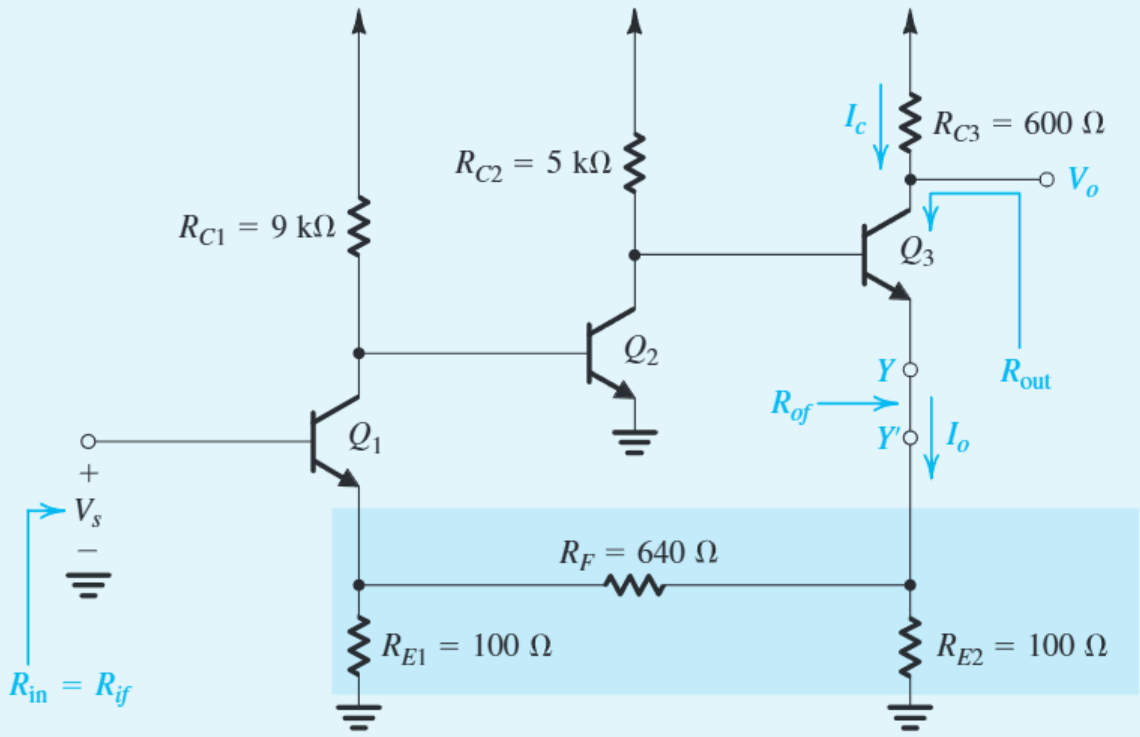
$$R_r(1 + A_g R)$$

例 11.8

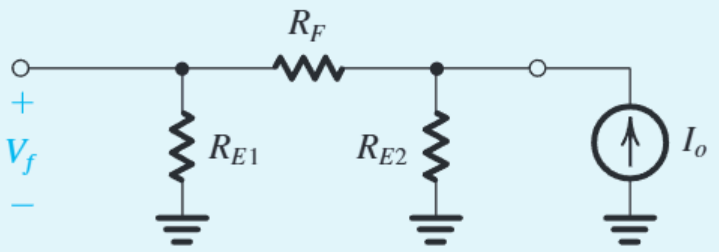
由于负反馈可扩展放大器带宽，因此常用于设计宽带放大器。MC1553 就是这样一款放大器。MC1553 的部分电路如图 11.21(a)所示。图中电路（称为三反馈）由三个增益级组成，由 R 、R 和 R 组成的网络提供串联反馈。

请注意，反馈网络会对 Q 的发射极电流 I 进行采样，因此 I 就是反馈放大器的输出量。但实际上，I 很难利用。因此，通常将 Q 的集电极电流 I 作为输出量。因此，作为以 I 作为输出电流的跨导放大器，图 11.21(a)中标注的输出电阻是值得关注的。在某些应用中，I 会通过一个负载电阻（如 R），并将电压 Vis 作为输出。假定偏置电路（图中未显示）确定 I= 0.6 mA、I= 1 mA 和 I= 4 mA。还假设所有三个晶体管的 h = 100 和 r = ∞。

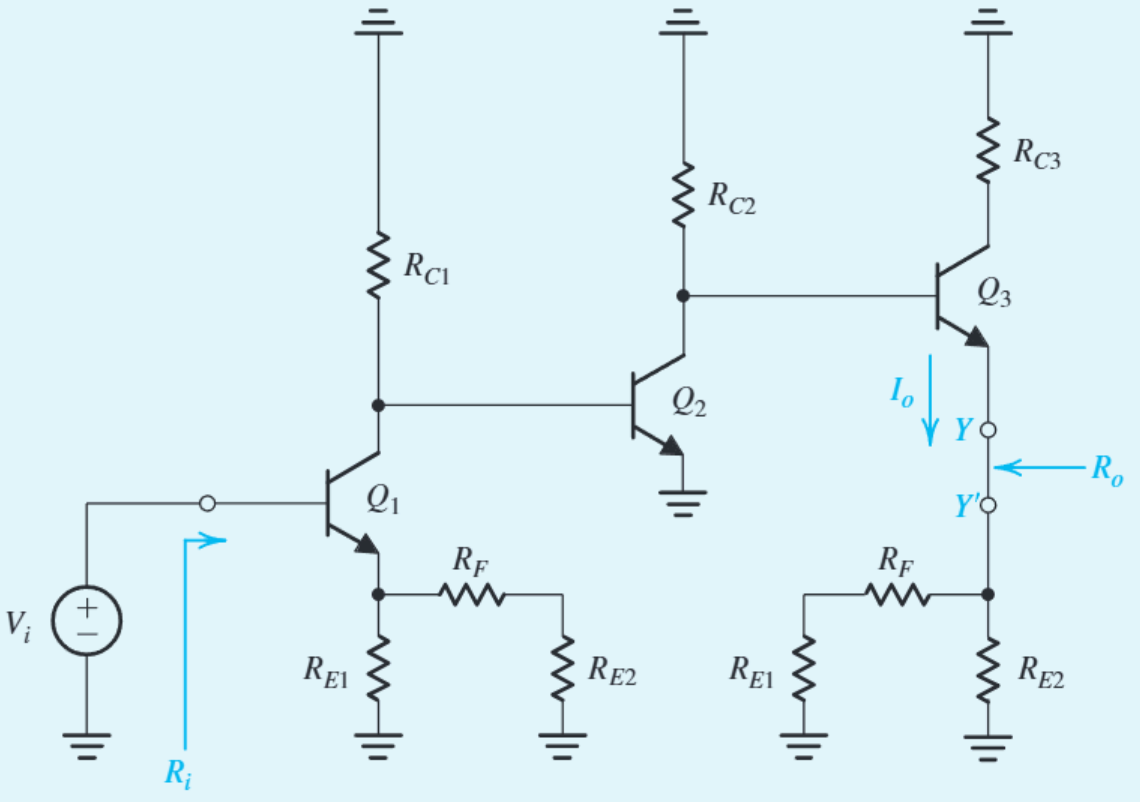
5 为了避免混淆 BJT 电流增益 β 和反馈系数 β，我们有时使用 h 来表示晶体管 β。



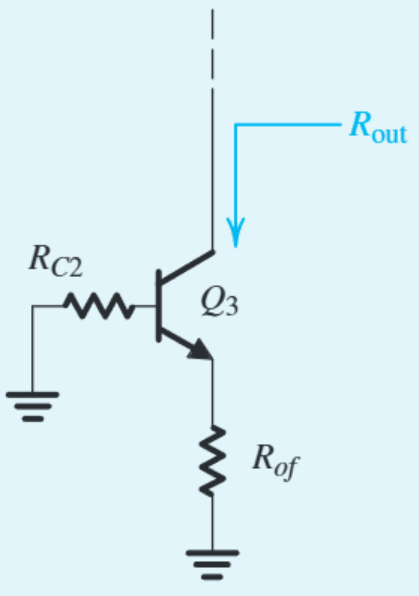
(a)



(b)

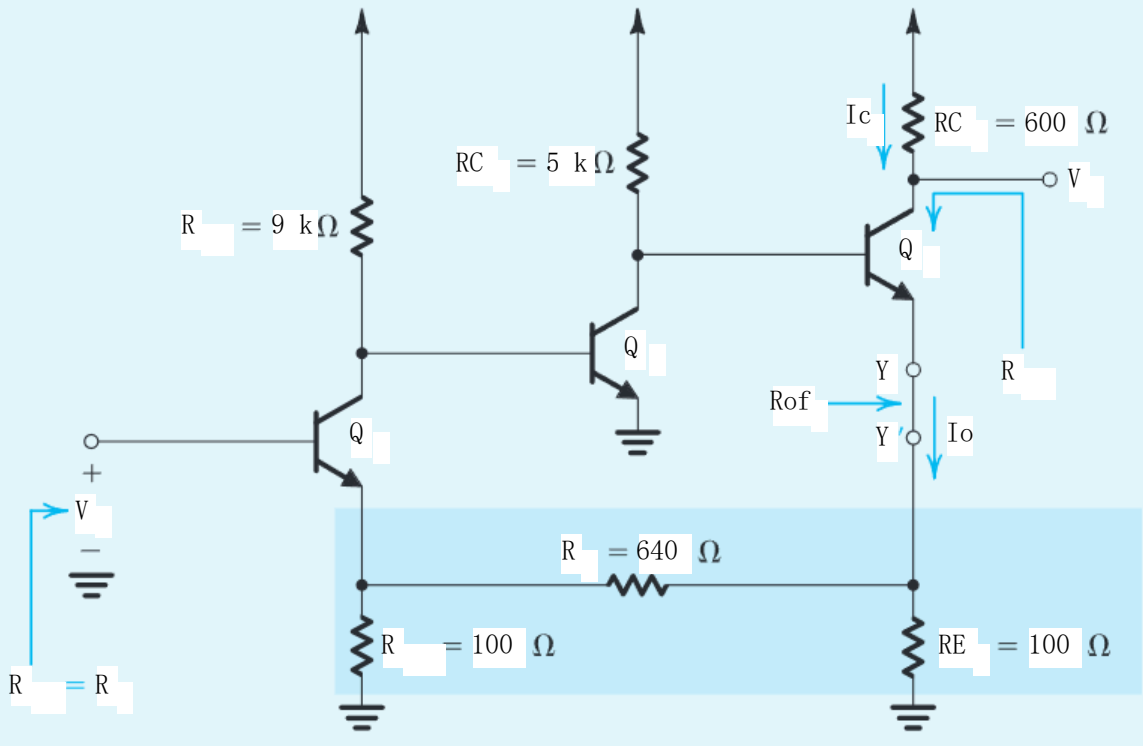


(c)

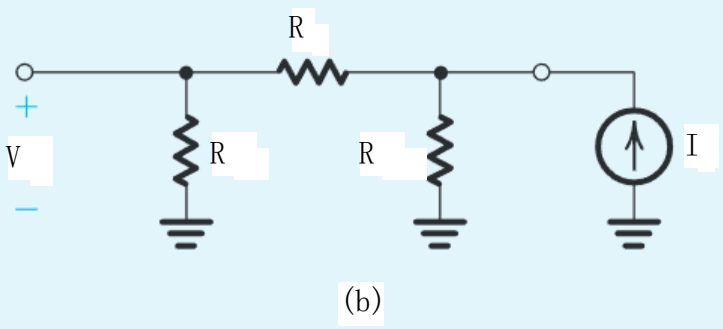


(d)

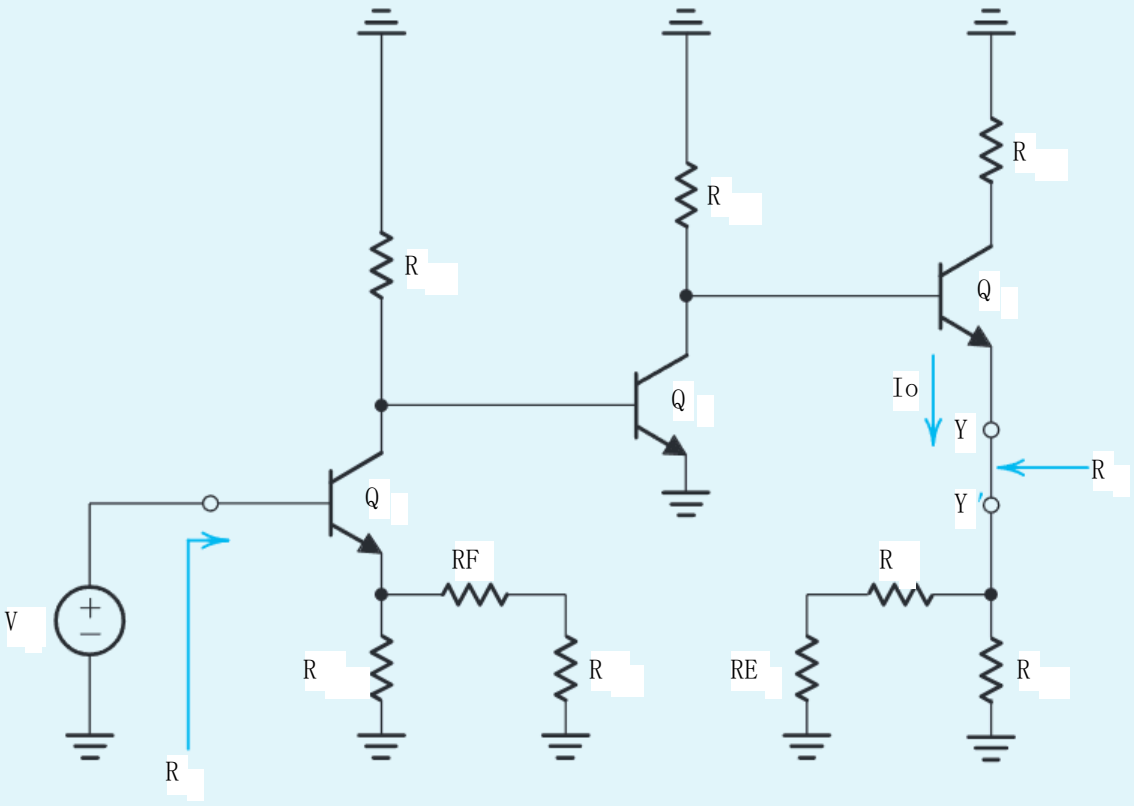
Figure 11.21 Circuits for Example 11.8.



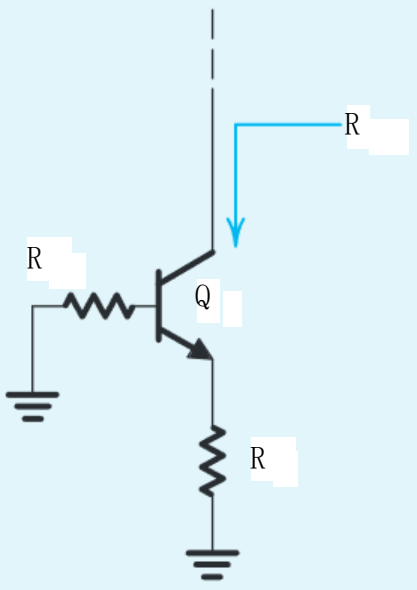
(a)



(b)



(c)



(d)

图 11.21 例 11.8 的电路

Example 11.8 *continued*

(a) Anticipating that the loop gain will be large, find an approximate expression and value for the closed-loop gain $A_f \equiv I_o/V_s$ and hence for I_c/V_s. Also find V_o/V_s.
(b) Use feedback analysis to find A, β, A_f, V_o/V_s, R_{in}, and R_{out}. For the calculation of R_{out}, assume that r_o of Q_3 is 25 kΩ.

Solution

(a) When $A\beta \gg 1$,

$$A_f \equiv \frac{I_o}{V_s} \simeq \frac{1}{\beta}$$

where the feedback factor β can be found from the feedback network. The feedback network is highlighted in Fig. 11.21(a), and the determination of the value of β is illustrated in Fig. 11.21(b), from which we find

$$\beta \equiv \frac{V_f}{I_o} = \frac{R_{E2}}{R_{E2}+R_F+R_{E1}} \times R_{E1}$$
$$= \frac{100}{100+640+100} \times 100 = 11.9\ \Omega$$

Thus,

$$A_f \simeq \frac{1}{\beta}$$
$$= \frac{1}{R_{E2}}\left(1+\frac{R_{E2}+R_F}{R_{E1}}\right)$$
$$= \frac{1}{11.9} = 84\ \text{mA/V}$$
$$\frac{I_c}{V_s} \simeq \frac{I_o}{V_s} = 84\ \text{mA/V}$$
$$\frac{V_o}{V_s} = \frac{-I_c R_{C3}}{V_s} = -84 \times 0.6 = -50.4\ \text{V/V}$$

(b) Employing the loading rules given in Fig. 11.19, we obtain the A circuit shown in Fig. 11.21(c). To find $A \equiv I_o/V_i$ we first determine the gain of the first stage. This can be written by inspection as

$$\frac{V_{c1}}{V_i} = \frac{-\alpha_1\left(R_{C1} \parallel r_{\pi 2}\right)}{r_{e1}+\left[R_{E1} \parallel \left(R_F+R_{E2}\right)\right]}$$

Since Q_1 is biased at 0.6 mA, $r_{e1} = 41.7\ \Omega$. Transistor Q_2 is biased as 1 mA; thus $r_{\pi 2} = h_{fe}/g_{m2} = 100/40 = 2.5$ kΩ. Substituting these values together with $\alpha_1 = 0.99$, $R_{C1} = 9$ kΩ, $R_{E1} = 100\ \Omega$, $R_F = 640\ \Omega$, and $R_{E2} = 100\ \Omega$, results in

$$\frac{V_{c1}}{V_i} = -14.92\ \text{V/V}$$

例 11.8 续

(a) 预计环路增益会很大，求闭环增益 A ≡ I/V 的近似表达式和数值，以及 I /V 的近似表达式和数值。同时求出 V/V。(b) 利用反馈分析法求出 A、β、A、V/V、R 和 R。

解决方案

(a) 当 Aβ 1、

$$A \equiv \frac{I}{V} \approx \frac{1}{\beta}$$

其中反馈系数 β 可通过反馈网络求得。图 11.21(a)突出显示了反馈网络，图 11.21(b)说明了 β 值的确定。

$$\beta \equiv \frac{V}{I} = \frac{R}{R + R + R} \times R$$

$$= \frac{100}{100 + 640 + 100} \times 100 = 11.9$$

因此、

$$A \approx \frac{1}{\beta}$$

$$= \frac{1}{R}\left(1 + \frac{R + R}{R}\right)$$

$$= \frac{1}{11.9} = 84 \text{ mA/V}$$

$$\frac{I}{V} \approx \frac{I}{V} = 84 \text{ mA/V}$$

$$\frac{V}{V} = \frac{-I R}{V} = -84 \times 0.6 = -50.4 \text{ v/v}$$

(b) 利用图 11.19 中给出的加载规则，我们可以得到图 11.21(c)所示的 A 电路。为了找到 A ≡ I/V，我们首先要确定第一级的增益。可以写成

$$\frac{V}{V} = \frac{-\alpha}{r + \left[R \| \left(\frac{R}{R + R}\right)\right]}$$

由于 Q 的偏压为 0.6 mA，因此 r = 41.7。晶体管 Q 的偏压为 1 mA，因此 r= h /g = 100/40 = 2.5 k。将这些数值与 α= 0.99、R = 9 k、R = 100 , R = 640 , 和 R = 100 一起代入，结果是

$$\frac{V}{V} = -14.92 \text{ V/V}$$

Next, we determine the gain of the second stage, which can be written by inspection (noting that $V_{b2} = V_{c1}$) as

$$\frac{V_{c2}}{V_{c1}} = -g_{m2}\{R_{C2} \| (h_{fe}+1)[r_{e3} + (R_{E2} \| (R_F + R_{E1}))]\}$$

Substituting $g_{m2} = 40\,\text{mA/V}$, $R_{C2} = 5\,\text{k}\Omega$, $h_{fe} = 100$, $r_{e3} = 25/4 = 6.25\ \Omega$, $R_{E2} = 100\,\Omega$, $R_F = 640\,\Omega$, and $R_{E1} = 100\,\Omega$ results in

$$\frac{V_{c2}}{V_{c1}} = -131.2\ \text{V/V}$$

Finally, for the third stage we can write by inspection

$$\frac{I_o}{V_{c2}} = \frac{I_{e3}}{V_{b3}} = \frac{1}{r_{e3} + (R_{E2} \| (R_F + R_{E1}))}$$

$$= \frac{1}{6.25 + (100 \| 740)} = 10.6\ \text{mA/V}$$

Combining the gains of the three stages results in

$$A \equiv \frac{I_o}{V_i} = -14.92 \times -131.2 \times 10.6 \times 10^{-3}$$

$$= 20.7\ \text{A/V}$$

The closed-loop gain A_f can now be found from

$$A_f \equiv \frac{I_o}{V_s} = \frac{A}{1 + A\beta}$$

$$= \frac{20.7}{1 + 20.7 \times 11.9} = 83.7\ \text{mA/V}$$

which we note is very close to the ideal value found in (a) above. This is not surprising, since the loop gain $A\beta = 20.7 \times 11.9 = 246.3$ is large ($\gg 1$).

The voltage gain is found from

$$\frac{V_o}{V_s} = \frac{-I_c R_{C3}}{V_s} \simeq \frac{-I_o R_{C3}}{V_s} = -A_f R_{C3}$$

$$= -83.7 \times 10^{-3} \times 600 = -50.2\ \text{V/V}$$

which is also very close to the approximate value found in (a) above.

接下来，我们确定第二级的增益，通过观察（注意 V = V）可以写成

$$\frac{V}{V} = -g\{R \| (h+1)[r + (R \| (R+R))]\}$$

将 g = 40 mA/V、R= 5 k、h = 100、r= 25/4 = 6.25 ， R = 100 ， R = 640 和 R = 100 代入，结果是

$$\frac{V}{V} = -131.2 \text{ V/V}$$

最后，对于第三阶段，我们可以通过检查写出

$$\frac{I}{V} = \frac{I}{V} = \frac{1}{r + (R \| (R+R))}$$

$$= \frac{1}{6.25 + (100 \| 740)} = 10.6 \text{ mA/V}$$

综合三个阶段的收益，结果是

$$A \equiv \frac{I}{V} = -14.92 \times -131.2 \times 10.6 \times 10$$

$$= 20.7 \text{ A/V}$$

闭环增益 Ac 现在可以通过以下公式求得

$$A \equiv \frac{I}{V} = \frac{A}{1 + A\beta}$$

$$= \frac{20.7}{1 + 20.7 \times 11.9} = 83.7 \text{ mA/V}$$

我们注意到这非常接近上文（a）中的理想值。这并不奇怪，因为环路增益 Aβ = 20.7 × 11.9 = 246.3 很大（ 1)。

电压增益的计算公式为

$$\frac{V}{V} = \frac{-I R}{V} \square \frac{-I R}{V} = -A R$$

$$= -83.7 \times 10 \times 600 = -50.2 \text{ v/v}$$

这也非常接近上文（a）中的近似值。

Example 11.8 *continued*

The input resistance of the feedback amplifier is given by

$$R_{\text{in}} = R_{if} = R_i(1 + A\beta)$$

where R_i is the input resistance of the A circuit. The value of R_i can be found from the circuit in Fig. 11.21(c) as follows:

$$\begin{aligned} R_i &= (h_{fe} + 1)\left[r_{e1} + \left(R_{E1} \parallel (R_F + R_{E2})\right)\right] \\ &= 13.11\ \text{k}\Omega \end{aligned}$$

Thus,

$$R_{if} = 13.11(1 + 20.7 \times 11.9) = 3.24\,\text{M}\Omega$$

To determine the output resistance R_{out}, which is the resistance looking into the collector of Q_3, we face a dilemma. The feedback does not sample I_c and thus we cannot employ the feedback formulas directly.[6] Nevertheless, we present a somewhat indirect solution to this problem below. Here we note parenthetically that had Q_1 been a MOSFET, this problem would not have existed, since $I_d = I_s$.

Since the feedback senses the emitter current I_o, the output resistance given by the feedback analysis will be the resistance seen in the emitter circuit, say between Y and Y',

$$R_{of} = R_o(1 + A\beta)$$

where R_o can be determined from the A circuit in Fig. 11.21(c) by breaking the circuit between Y and Y'. The resistance looking between these two nodes can be found to be

$$R_o = \left[R_{E2} \parallel (R_F + R_{E1})\right] + r_{e3} + \frac{R_{C2}}{h_{fe} + 1}$$

which, for the values given, yields $R_o = 143.9\ \Omega$. The output resistance R_{of} of the feedback amplifier can now be found as

$$R_{of} = R_o(1 + A\beta) = 143.9(1 + 20.7 \times 11.9) = 35.6\ \text{k}\Omega$$

[6] This important point was first brought to the authors' attention by Gordon Roberts (see Roberts and Sedra, 1992).

例 11.8 续

反馈放大器的输入电阻为

$$R = R = R(1 + A\beta)$$

其中 Ris 为 A 电路的输入电阻。根据图 11.21(c)中的电路，可求得 R 的值如下：

$$R = (h + 1)[r + (R \,||\, (R + R))]$$

$$= 13.11 \text{ k}$$

因此、

$$r = 13.11(1 + 20.7 \times 11.9) = 3.24 \text{ 米}$$

要确定输出电阻 R（即 Q 集电极的电阻），我们面临着一个难题。反馈并不对 I 进行采样，因此我们无法直接使用反馈公式。 尽管如此，我们还是在下文中提出了解决这一问题的间接方法。在此，我们顺便指出，如果 Q 是 MOSFET，这个问题就不会存在，因为 I = I。

由于反馈感应的是发射极电流 I，因此反馈分析得出的输出电阻将是发射极电路中的电阻，例如 Y 和 Y 之间的电阻、

$$R = R(1 + A\beta)$$

其中，R 可以通过断开 Y 和 Y 之间的电路，从图 11.21(c)中的 A 电路中确定。这两个节点之间的电阻值为

$$R = [R \,||\, (R + R)] + r + \frac{R}{h + 1}$$

根据给出的数值，得出 R= 143.9。现在可以求出反馈放大器的输出电阻 R 为

$$R = R(1 + A\beta) = 143.9(1 + 20.7 \times 11.9) = 35.6 \text{ k}$$

6 戈登-罗伯茨首先提请作者注意这一重要问题（见罗伯茨和塞德拉、1992）.

We can now use the value of R_{of} to obtain an approximate value for R_{out}. To do this, we assume that the effect of the feedback is to place a resistance R_{of} (35.6 kΩ) in the emitter of Q_3, and find the output resistance from the equivalent circuit shown in Fig. 11.21(d). This is the output resistance of a BJT with a resistance R_{of} in its emitter and a resistance R_{C2} in its base. The formula we have for this (Eq. 8.66) does not unfortunately account for a resistance in the base. The formula, however, can be modified (see Problem 11.54) to obtain

$$R_{\text{out}} = r_{o3} + \left[R_{of} \parallel \left(r_{\pi 3} + R_{C2}\right)\right]\left[1 + g_{m3} r_{o3} \frac{r_{\pi 3}}{r_{\pi 3} + R_{C2}}\right]$$

$$= 25 + [35.6 \parallel (0.625 + 5)]\left[1 + 160 \times 25 \times \frac{0.625}{0.625 + 5}\right]$$

$$= 2.19\,\text{M}\Omega$$

Thus R_{out} is increased (from r_{o3}) but not by $(1 + A\beta)$.

EXERCISES

D11.16 For the feedback triple in Fig. 11.21(a), analyzed in Example 11.8, modify the value of R_F to obtain a closed-loop transconductance I_o/V_s of approximately 100 mA/V. Assume that the loop gain remains large. What is the new value of R_F? For this value, what is the approximate value of the voltage gain if the output voltage is taken at the collector of Q_3?

Ans. 800 Ω; –60 V/V

11.17 Determine the loop gain of the feedback amplifier of Fig. 11.21(a). Set $V_s = 0$, break the loop between the collector of Q_1 and the base of Q_2, apply a voltage V_t to the base of Q_2, and connect a resistance equal to $r_{\pi 2}$ between the collector of Q_1 and ground. Find $A\beta$ as $(-V_r/V_t)$ where $V_r = V_{c1}$.

Ans. $A\beta = 248.9$ (slightly different from the value found in Example 11.8 because of the approximations inherent in the systematic feedback-analysis method).

11.5.3 The Feedback Transresistance Amplifier (Shunt–Shunt)

Figure 11.22(a) shows the ideal feedback transresistance amplifier, which, as expected, utilizes the shunt–shunt topology. The amplifier equivalent circuit is shown in Fig. 11.22(b), with the formulas for determining A_f, R_{if}, and R_{of} given in Fig. 11.22(c).

为此，我们假设反馈的作用是在 Q 的发射极放置一个电阻 R (35.6 k)，并从图 11.21(d)所示的等效电路中找出输出电阻。这就是一个 BJT 的输出电阻，它的发射极有一个电阻 R，基极有一个电阻 R。不幸的是，我们的计算公式（公式 8.66）没有考虑基极电阻。不过，该公式可以修改（见问题 11.54），从而得到

$$R = r + [R \| (r + R)][1 + g_m r \frac{r}{r + R}]$$

$$= 25 + [35.6 \| (0.625 + 5)] + 160 \times 25 \times \frac{0.625}{0.625 + 5}$$

$$= 2.19\ \text{M}\Omega$$

因此，Ris 增加了（从 r 开始），但并没有增加（1 + Aβ）。

练习

D11.16 对于在例 11.8 中分析的图 11.21（a）中的反馈三重电路，将 R 值修改为获得约为 100 mA/V 的闭环跨导 I /V。假设环路增益仍然很大。R 的新值是多少？对于该值，如果在 Q 的集电极上测量输出电压，电压增益的近似值是多少？

答案：800；-60 V/V

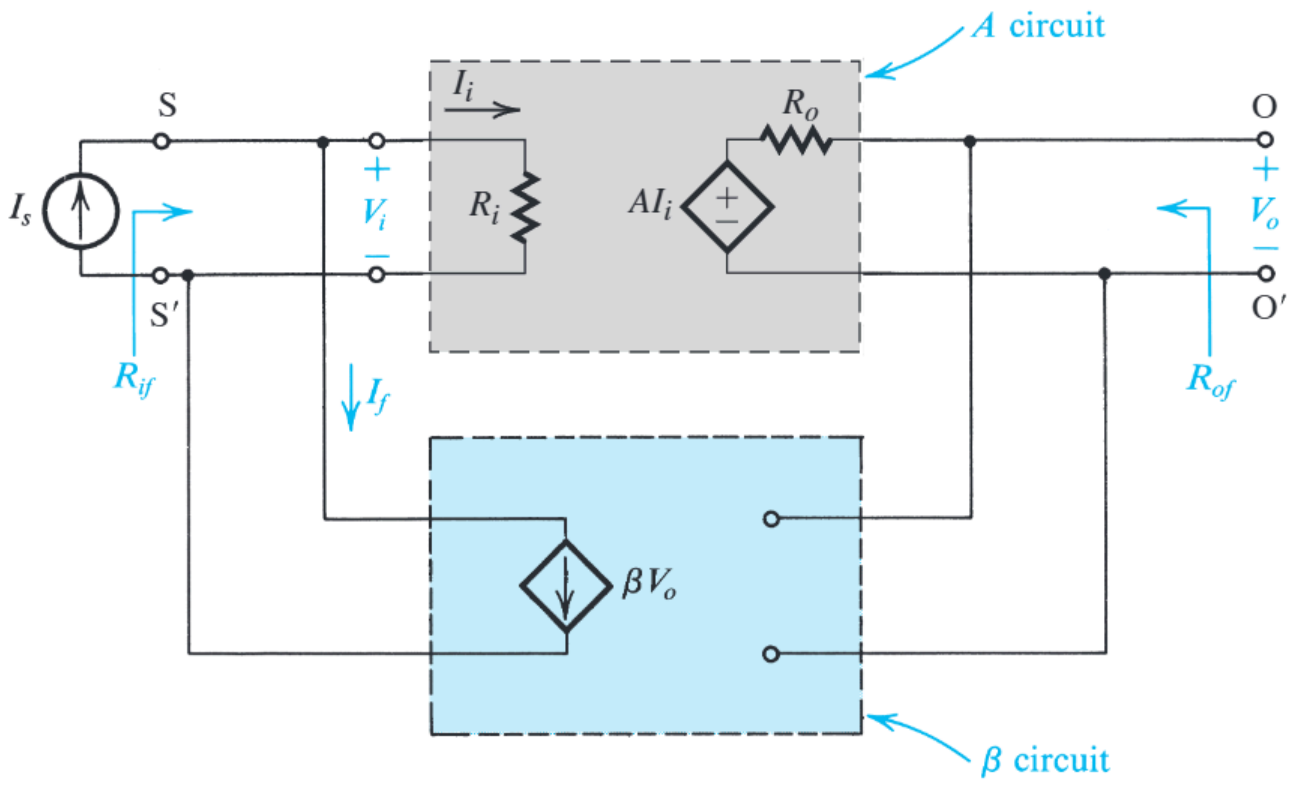
11.17 确定图 11.21(a)中反馈放大器的环路增益。设 V = 0，断开 Q 的集电极和 Q 的基极之间的环路，给 Q 的基极施加电压 V，并在 Q 的集电极和地之间连接一个等于 r 的电阻。求 Aβ 为（- V/V ），其中 V= V。

答案：Aβ = 248.9（与例 11.8 中求得的值略有不同，因为系统反馈分析法中固有的近似值）。

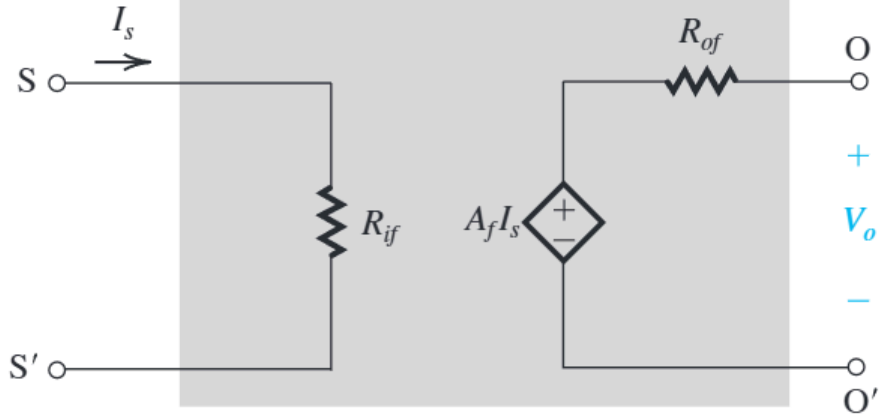
11.5.3 反馈跨阻放大器（并联式）

图 11.22(a)显示了理想的反馈跨阻放大器，正如预期的那样，它采用了并联拓扑结构。放大器等效电路见图 11.22(b)，A、R 和 R 的计算公式见图 11.22(c)。

(a) Ideal Structure



(b) Equivalent Circuit



(c) Formulas

$$A_f \equiv \frac{V_o}{I_s} = \frac{A}{1+A\beta}$$

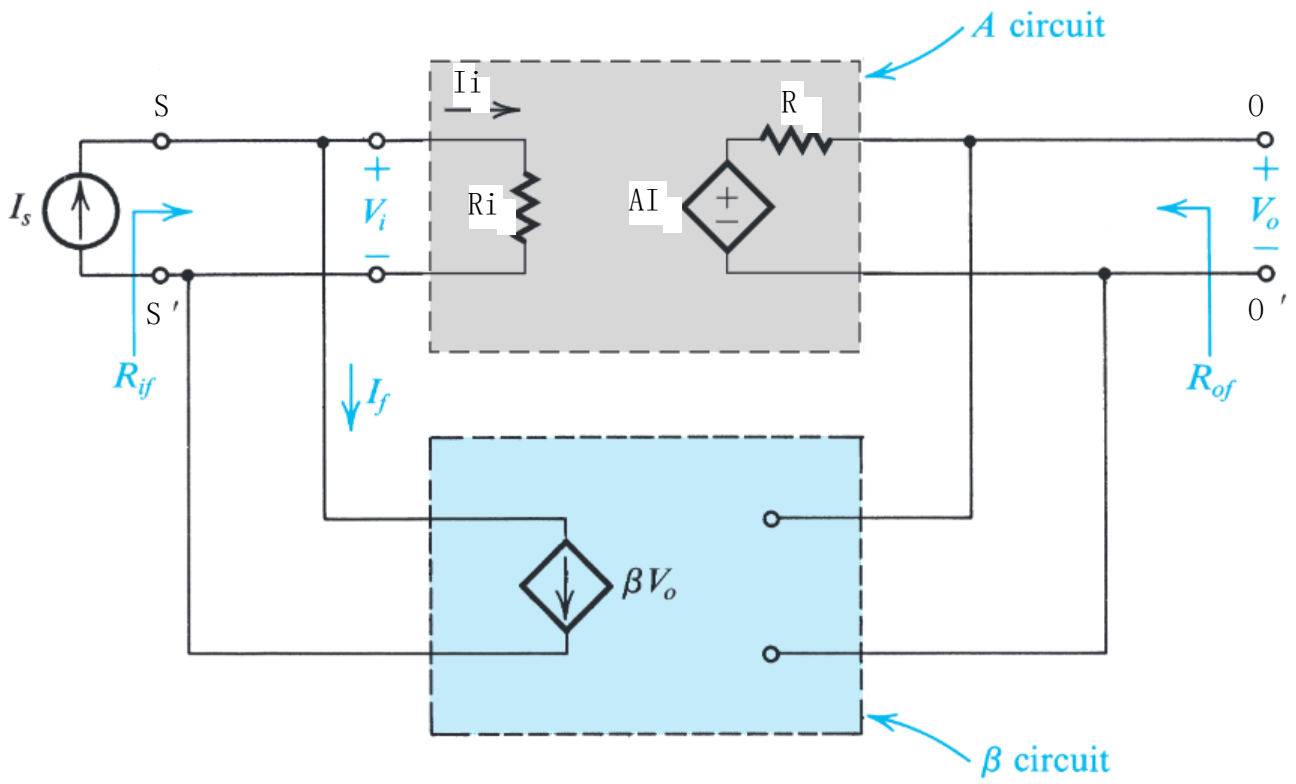
$$R_{if} = R_i/(1+A\beta)$$

$$R_{of} = R_o/(1+A\beta)$$

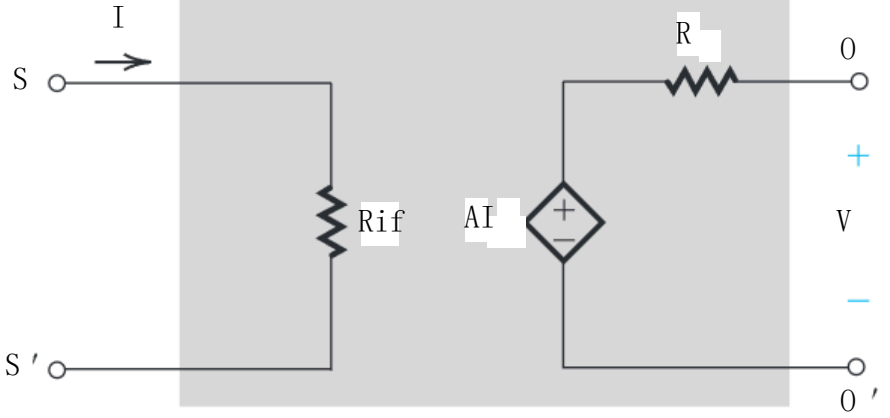
Figure 11.22 The feedback transresistance amplifier (shunt–shunt).

Application of the feedback-analysis method to a general transresistance amplifier is shown in Fig. 11.23, which presents all the steps and formulas needed. The method is now illustrated by a detailed example.

(a) 理想结构



(b) 等效电路



(c) 公式

$$A \equiv \frac{V}{I} = \frac{A}{1 + A\beta}$$

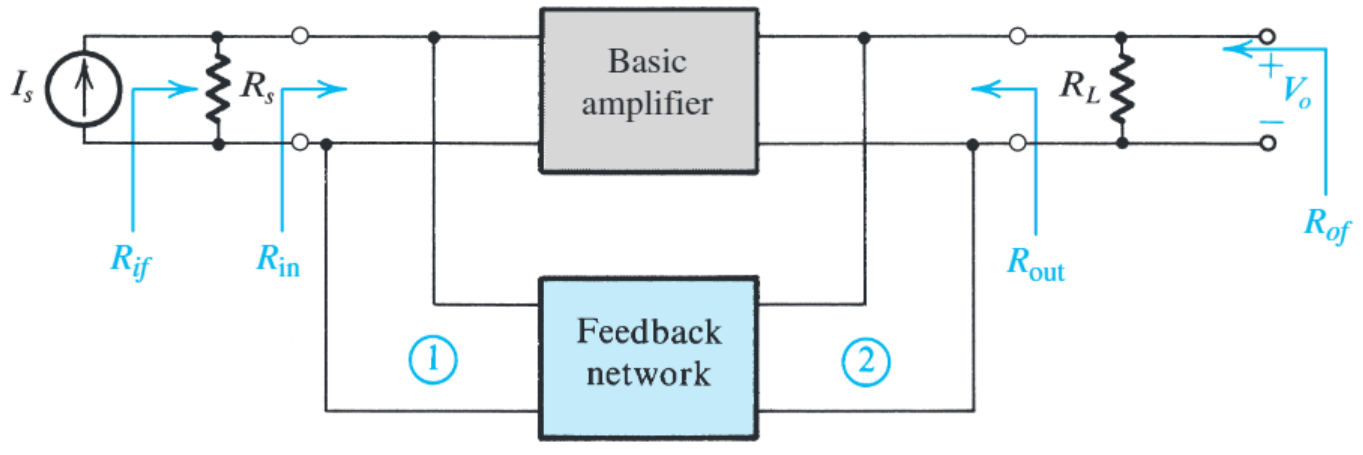
$$R = R/(1 + A\beta)$$

$$R = R/(1 + A\beta)$$

图 11.22 反馈跨阻放大器（并联）。

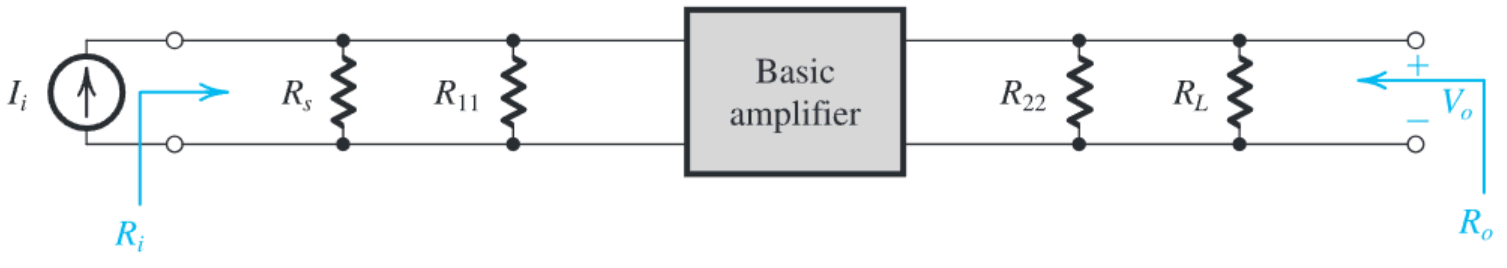
图 11.23 显示了反馈分析法在普通跨阻放大器中的应用，并给出了所需的所有步骤和公式。现在通过一个详细的例子来说明该方法。

(a) General Structure

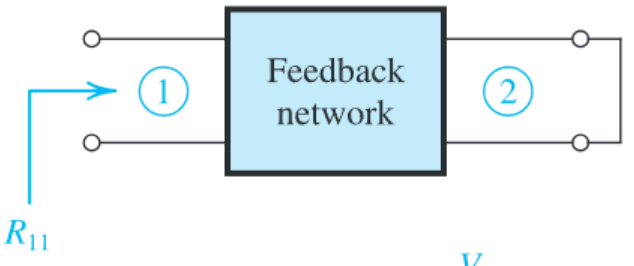


(b) Finding the A Circuit and β

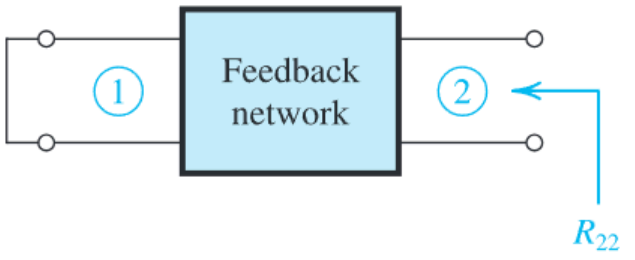
(i) The A circuit is



where R_{11} is obtained from

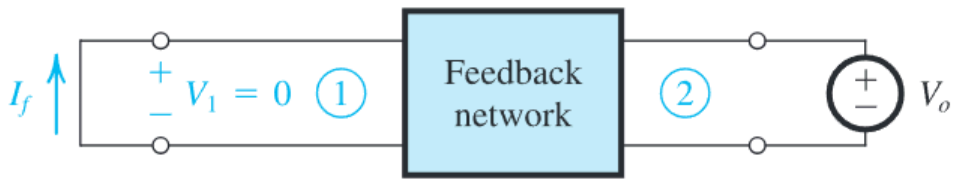


and R_{22} is obtained from



and the gain A is defined $A \equiv \dfrac{V_o}{I_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{I_f}{V_o}\right|_{V_1 = 0}$$

(c) Gain, Input, and Output Resistance

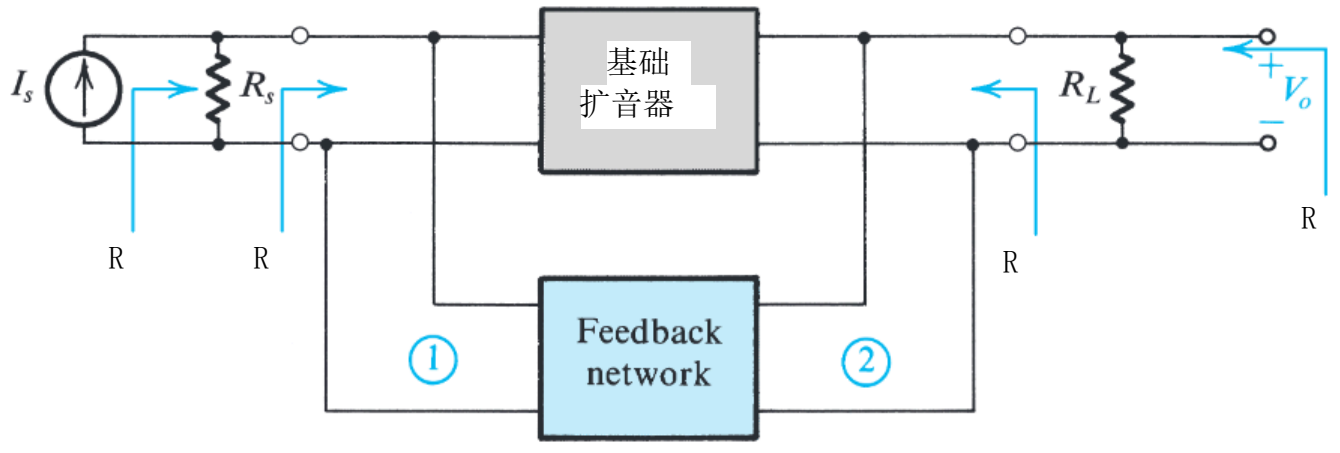
- Use the formulas in Fig. 11.21 to find A_f, R_{if}, and R_{of}.
- R_{in} and R_{out} can then be found from

$$R_{\text{in}} = 1 \Big/ \left(\frac{1}{R_{if}} - \frac{1}{R_s}\right)$$

$$R_{\text{out}} = 1 \Big/ \left(\frac{1}{R_{of}} - \frac{1}{R_L}\right)$$

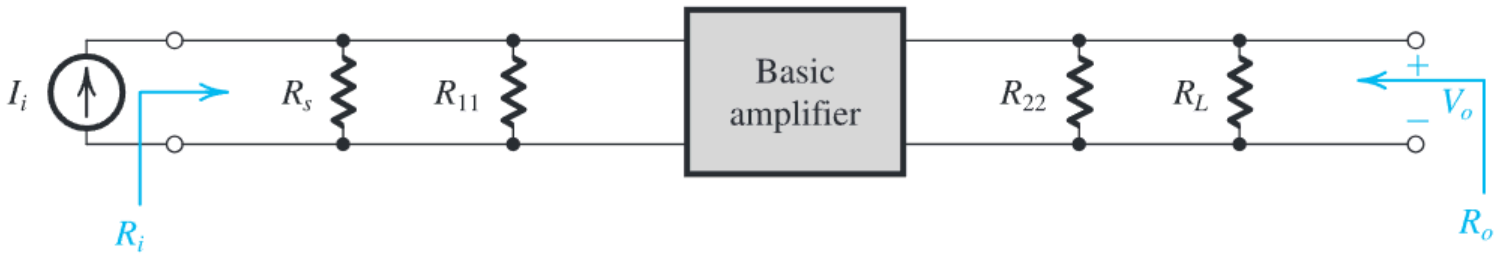
Figure 11.23 The feedback transresistance amplifier (shunt–shunt).

(a) 总体结构

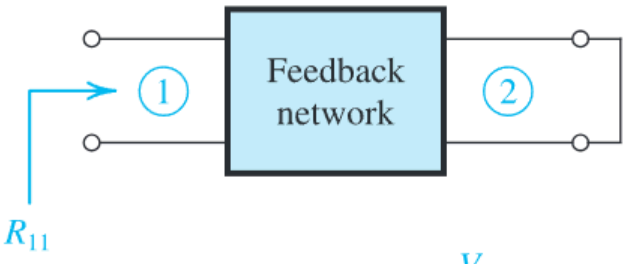


(b) Finding the A Circuit and β

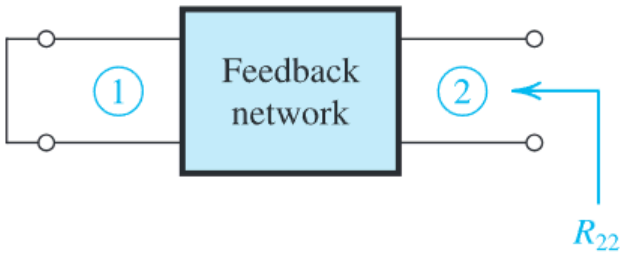
(i) The A circuit is



where R_{11} is obtained from

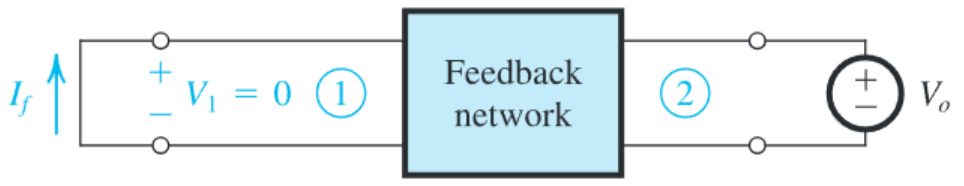


and R_{22} is obtained from



and the gain A is defined $A \equiv \dfrac{V_o}{I_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{I_f}{V_o}\right|_{V_1 = 0}$$

(c) 增益、输入和输出电阻

- 使用图 11.21 中的公式求出 A、R 和 R。
- Rand R 的计算公式为

$$R = 1 \Big/ \left(\frac{1}{R} - \frac{1}{R} \right)$$

$$R = 1 \Big/ \left(\frac{1}{R} - \frac{1}{R} \right)$$

图 11.23 反馈跨阻放大器（并联）。

Example 11.9

Figure 11.24(a) shows a feedback transresistance amplifier. It is formed by connecting a resistance R_F in the negative-feedback path of a voltage amplifier with gain μ, an input resistance R_{id}, and an output resistance r_o. The amplifier μ can be implemented with an op amp, a simple differential amplifier, a single-ended inverting amplifier, or, in the limit, a single-transistor CE or CS amplifier. The latter case will be considered in Exercise 11.18. Of course, the higher the gain μ, the more ideal the characteristics of the feedback transresistance amplifier will be, simply because of the concomitant increase in loop gain.

(a) If the loop gain is large, find an approximate expression for the closed-loop transresistance V_o/I_s of the feedback amplifier.
(b) Find the A circuit and expressions for A, R_i, and R_o.
(c) Find expressions for the loop gain, A_f, R_{if}, R_{in}, R_{of}, and R_{out}.
(d) Find the values of R_i, R_o, A, β, $A\beta$, A_f, R_{if}, R_{in}, R_{of}, and R_{out} for the case $\mu = 10^4$ V/V, $R_{id} = \infty$, $r_o = 100\ \Omega$, $R_F = 10\ \text{k}\Omega$, and $R_s = R_L = 1\ \text{k}\Omega$.
(e) If instead of a current source I_s having a source resistance $R_s = 1\ \text{k}\Omega$, the amplifier is fed from a voltage source V_s having a source resistance $R_s = 1\ \text{k}\Omega$, find an expression for and the value of the voltage gain V_o/V_s.

Solution

(a) If the loop gain $A\beta$ is large,

$$A_f \equiv \frac{V_o}{I_s} \simeq \frac{1}{\beta}$$

where β can be found from the β circuit in Fig. 11.24(b) as shown in Fig. 11.24(c),

$$\beta \equiv \frac{I_f}{V_o} = -\frac{1}{R_F} \tag{11.37}$$

Thus,

$$\frac{V_o}{I_s} \simeq -R_F$$

Note that in this case the voltage at the input node (the inverting input terminal of μ) will be very close to ground and thus very little, if any, current flows into the input terminal of the amplifier. Nearly all of I_s will flow through R_F, resulting in $V_o \simeq 0 - I_s R_F = -I_s R_F$. This should be reminiscent of the inverting op-amp configuration studied in Section 2.2.

(b) From the feedback network in Fig. 11.24(b), we see that the loading effect at the amplifier input and output will simply be R_F. This is indicated in the A circuit shown in Fig. 11.24(c), where we have replaced the amplifier μ with its equivalent-circuit model. The open-loop transresistance A can be obtained as follows:

$$V_{id} = I_i R_i \tag{11.38}$$

例 11.9

图 11.24(a) 显示了一个反馈跨阻放大器。它是通过在增益为 μ 的电压放大器的负反馈路径上连接电阻 R、输入电阻 R 和输出电阻 r 而形成的。放大器 μ 可以用运算放大器、简单差分放大器、单端反相器放大器来实现，或者在极限情况下，用单晶体管 CE 或 CS 放大器来实现。练习 11.18 将考虑后一种情况。当然，增益 μ 越高，反馈晶体管放大器的特性就越理想，原因很简单，环路增益也会随之增加。

(a) 如果环路增益很大，求反馈放大器闭环跨阻 V /I 的近似表达式。

(b) 找出 A 电路以及 A、R 和 R 的表达式。
(c) 找出环路增益、A、R ， R、R ， 和 R 的表达式。
(d) 在 μ = 10 V/V、R= ∞、r= 100、R= 10 k 和 R= 1 k 的情况下，求 R、R、A、β、Aβ、A、R、R、R 和 R 的值。(e) 如果放大器不是由源电阻 R=1 k 的电流源 I 供电，而是由源电阻 R=1 k 的电压源 V 供电，求电压增益 V/V 的表达式和值。

解决方案

(a) 如果环路增益 Aβ 较大、

$$A \equiv \frac{V}{I} \approx \frac{1}{\beta}$$

其中，β 可以从图 11.24（b）中的 β 电路中求得，如图 11.24（c）所示、

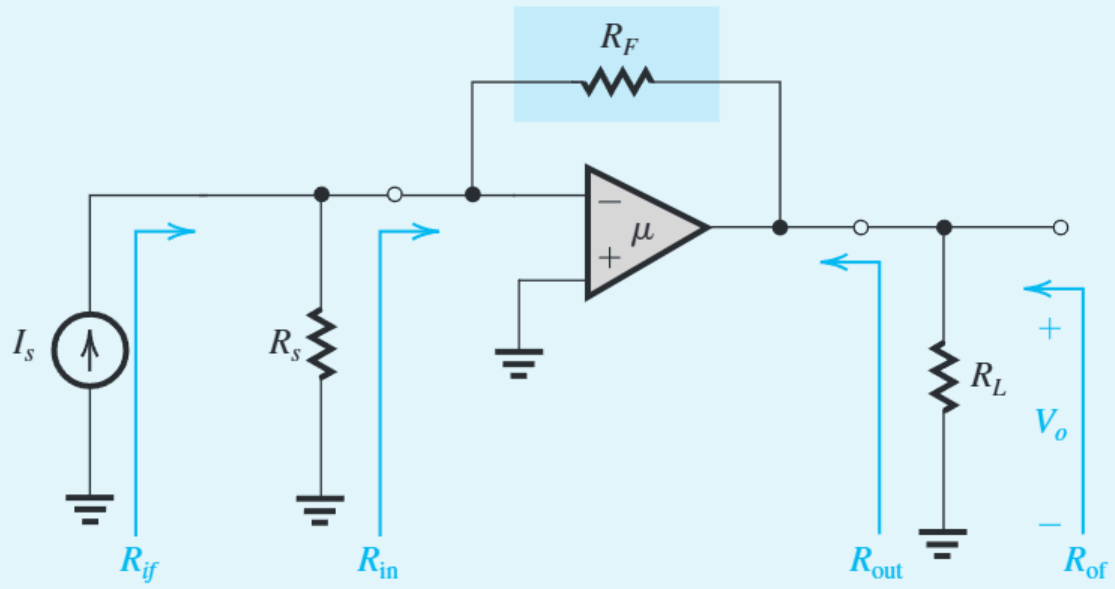
$$\beta \equiv \frac{I}{V} = -\frac{1}{R} \tag{11.37}$$

因此、

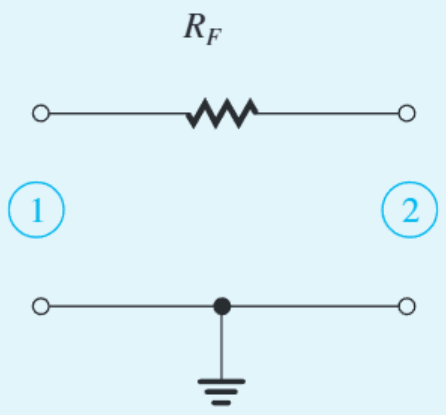
$$\frac{V}{I} \approx -R$$

请注意，在这种情况下，输入节点（ μ 的反相输入端）的电压将非常接近接地，因此流入放大器输入端的电流非常小（如果有的话）。几乎所有的 I 都将流经 R，结果是 V 0 - IR = -I R。这让人想起第 2.2 节中研究的反相运算放大器配置。
(b) 从图 11.24(b)中的反馈网络可以看出，放大器输入和输出端的负载效应为 R。图 11.24(c)所示的 A 电路表明了这一点，我们用等效电路模型替换了放大器 μ。开环跨阻 A 的计算公式如下：

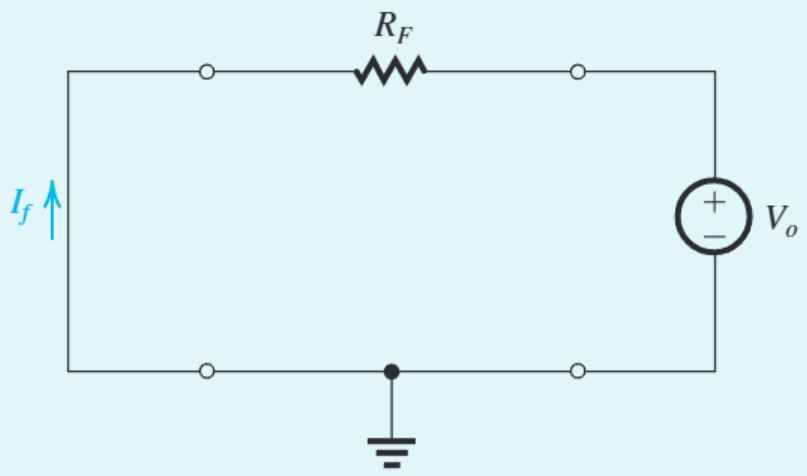
$$V = IR \tag{11.38}$$



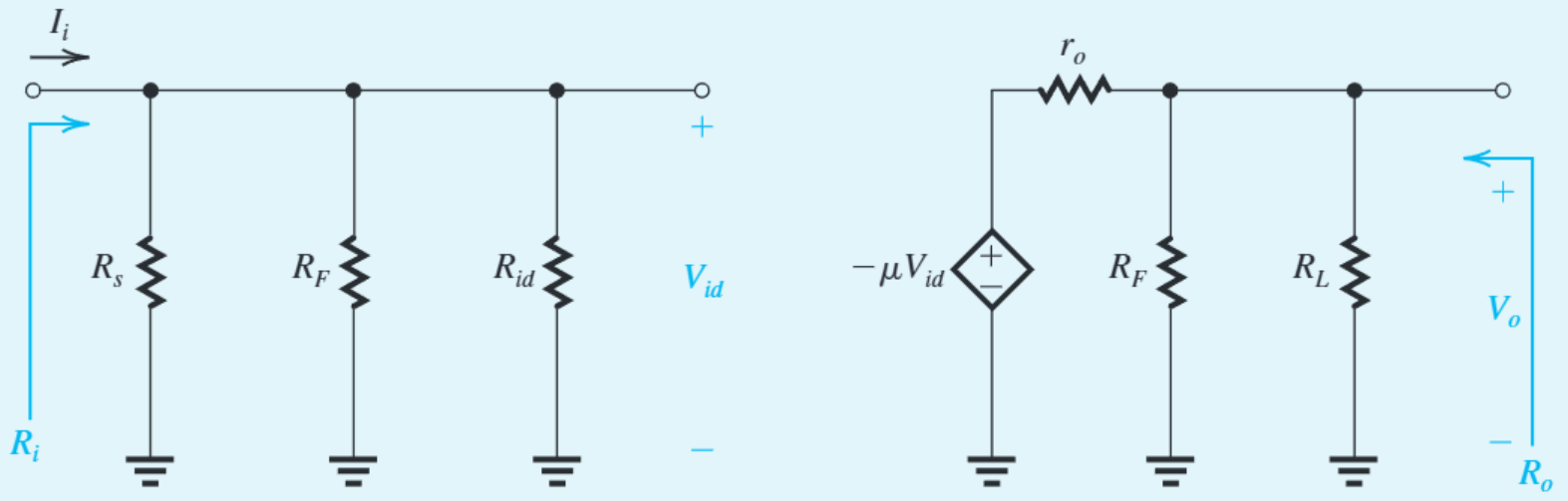
(a)



(b)

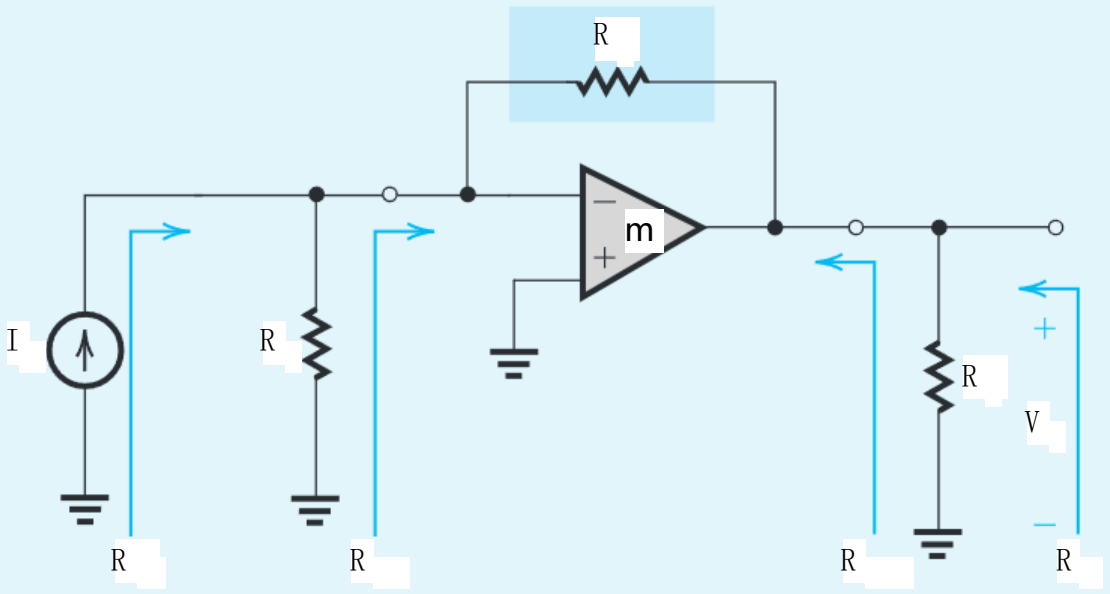


(c)

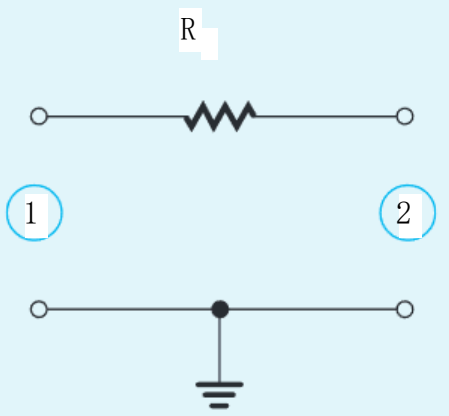


(d)

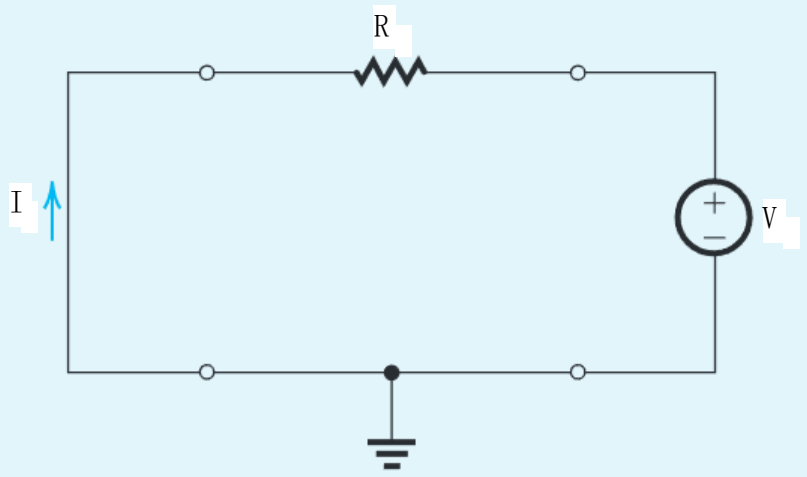
Figure 11.24 (**a**) A feedback transresistance amplifier; (**b**) the β circuit; (**c**) determining β; (**d**) the A circuit.



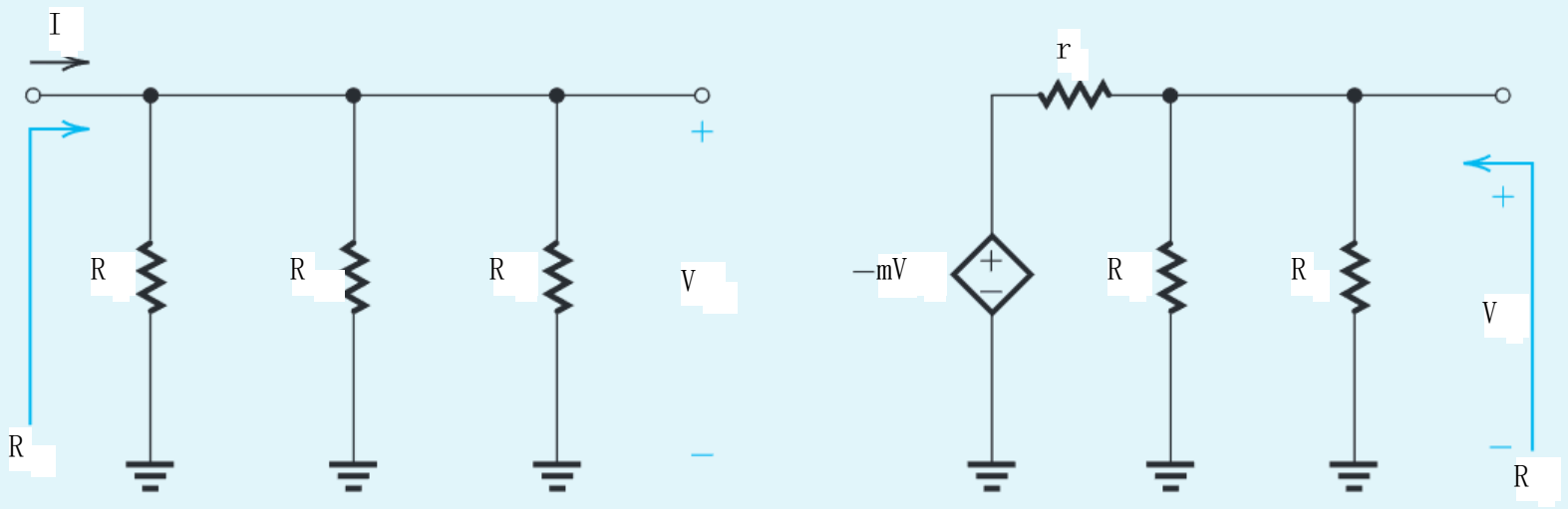
(a)



(b)



(c)



(d)

图 11.24 (a) 反馈跨阻放大器；(b) β 电路；(c) 确定 β；(d) A 电路。

Example 11.9 *continued*

where

$$R_i = R_{id} \| R_F \| R_s \tag{11.39}$$

$$V_o = -\mu V_{id} \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \tag{11.40}$$

Combining Eqs. (11.38) and (11.40) gives

$$A \equiv \frac{V_o}{I_i} = -\mu R_i \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \tag{11.41}$$

The open-loop output resistance can be obtained by inspection of the A circuit with I_i set to 0. We see that $V_{id} = 0$, and

$$R_o = r_o \| R_F \| R_L \tag{11.42}$$

(c) The loop gain $A\beta$ can be obtained by combining Eqs. (11.37) and (11.41),

$$A\beta = \mu \left(\frac{R_i}{R_F} \right) \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \tag{11.43}$$

Observe that although both A and β are negative, $A\beta$ is positive, a comforting fact confirming that the feedback is negative. Also note that $A\beta$ is dimensionless, as it must always be.

The closed-loop gain A_f can now be found as

$$A_f \equiv \frac{V_o}{I_s} = \frac{A}{1 + A\beta}$$

Thus

$$A_f = \frac{-\mu R_i \dfrac{(R_F \| R_L)}{r_o + (R_F \| R_L)}}{1 + \mu \dfrac{R_i}{R_F} \dfrac{(R_F \| R_L)}{r_o + (R_F \| R_L)}} \tag{11.44}$$

Note that the condition of $A\beta \gg 1$ that results in $A_f \simeq -R_F$ corresponds to

$$\mu \left(\frac{R_i}{R_F} \right) \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \gg 1 \tag{11.45}$$

例 11.9 续

其中

$$R = R \| R \| R \tag{11.39}$$

$$V = -\mu V \frac{\left(R \| R\right)}{r + \left(R \| R\right)} \tag{11.40}$$

结合公式 (11.38) 和 (11.40) 可得出

$$A \equiv \frac{V}{I} = -\mu R \frac{\left(R \| R\right)}{r + \left(R \| R\right)} \tag{11.41}$$

通过检查 Iset 为 0 的 A 电路，可以得到开环输出电阻。

$$R = r \| R \| R \tag{11.42}$$

(c) 结合公式 (11.37) 和 (11.41) 可以得到环路增益 Aβ、

$$A\beta = \mu \left(\frac{R}{R}\right) \frac{\left(R \| R\right)}{r + \left(R \| R\right)} \tag{11.43}$$

请注意，虽然 A 和 β 都是负值，但 Aβ 是正值，这是一个令人欣慰的事实，证明反馈是负的。另外请注意，Aβ 是无量纲的，因为它必须始终是无量纲的。

闭环增益 Ac 现在可求得

$$A \equiv \frac{V}{I} = \frac{A}{1 + A\beta}$$

Thus

$$A = \frac{-\mu R \frac{\left(R \| R\right)}{r + \left(R \| R\right)}}{1 + \mu \frac{R}{R} \frac{\left(R \| R\right)}{r + \left(R \| R\right)}} \tag{11.44}$$

注意，导致 A -R 的 Aβ 1 条件对应于

$$\mu \left(\frac{R}{R}\right) \frac{\left(R \| R\right)}{r + \left(R \| R\right)} \quad 1 \tag{11.45}$$

The input resistance with feedback, R_{if}, is obtained by dividing R_i by $(1+A\beta)$ with the result

$$R_{if} = \frac{R_i}{1+A\beta}$$

or

$$\frac{1}{R_{if}} = \frac{1}{R_i} + \frac{A\beta}{R_i} = \frac{1}{R_i} + \frac{\mu}{R_F}\frac{(R_F \parallel R_L)}{r_o + (R_F \parallel R_L)}$$

Substituting for R_i from Eq. (11.39) and replacing $\mu(R_F \parallel R_L)/[r_o + (R_F \parallel R_L)]$ by μ', where μ' is lower than but usually close to the value of μ, results in

$$R_{if} = R_{id} \parallel R_F \parallel R_s \parallel (R_F/\mu')$$

The two terms containing R_F can be combined,

$$R_{if} = R_s \parallel R_{id} \parallel [R_F/(\mu'+1)] \tag{11.46}$$

Since $R_{if} = R_s \parallel R_{\text{in}}$, we see that

$$R_{\text{in}} = R_{id} \parallel [R_F/(\mu'+1)]$$

Usually R_{id} is large and thus

$$R_{\text{in}} \simeq \frac{R_F}{\mu'+1} \simeq \frac{R_F}{\mu'} \tag{11.47}$$

from which we observe that for large amplifier gain μ, the input resistance will be low.

The output resistance with feedback R_{of} can be found by dividing R_o by $(1+A\beta)$:

$$R_{of} = \frac{R_o}{1+A\beta}$$

Thus,

$$\frac{1}{R_{of}} = \frac{1}{R_o} + \frac{A\beta}{R_o}$$

$$= \frac{1}{R_o} + \mu\frac{R_i}{R_F}\frac{(R_F \parallel R_L)}{r_o + (R_F \parallel R_L)}\frac{1}{R_o}$$

45）将 R 除以（1 + Aβ），得出带反馈的输入电阻 R=

$$\frac{R}{1 + A\beta}$$

or

$$\frac{1}{R} = \frac{1}{R} + \frac{A\beta}{R} = \frac{1}{R} + \frac{\mu}{R}\frac{(R \parallel R)}{r + (R \parallel R)}$$

代入公式 (11.39) 中的 R 并替换 μ $(R \parallel R)/[r + (R \parallel R)]$，其中 μ 是较低的通常接近 μ 值，结果是

$$R = R \parallel R \parallel R \parallel (R/\mu)$$

包含 R 的两个术语可以合并、

$$R = R \parallel R \parallel [R/(\mu + 1)]\ R/(\mu + 1) \qquad (11.46)$$

由于 R= R ‖ R，我们可以看到

$$R = R \parallel [R/(\mu + 1)]\ R/(\mu + 1)$$

通常是 Ris 大，因此

$$R \approx \frac{R}{\mu + 1} \approx \frac{R}{\mu} \qquad (11.47)$$

从中我们可以看出，放大器增益 μ 大时，输入电阻会很低。

将 R 除以（1 + Aβ），即可求出带反馈 R 的输出电阻：

$$R = \frac{R}{1 + A\beta}$$

因此、

$$\frac{1}{R} = \frac{1}{R} + \frac{A\beta}{R}$$

$$= \frac{1}{R} + \mu\frac{R}{R}\frac{(R \parallel R)}{r + (R \parallel R)}\frac{1}{R}$$

Example 11.9 *continued*

Substituting for R_o from Eq. (11.42),

$$\frac{1}{R_{of}} = \frac{1}{R_L} + \frac{1}{R_F} + \frac{1}{r_o} + \mu\frac{R_i}{R_F}\frac{1}{r_o}$$
$$= \frac{1}{R_L} + \frac{1}{R_F} + \frac{1}{r_o}\left(1 + \mu\frac{R_i}{R_F}\right)$$

Thus,

$$R_{of} = R_L \| R_F \| \frac{r_o}{1 + \mu\dfrac{R_i}{R_F}}$$

Since, moreover,

$$R_{of} = R_L \| R_{\text{out}}$$

we obtain for R_{out}

$$R_{\text{out}} = R_F \| \frac{r_o}{1 + \mu\dfrac{R_i}{R_F}}$$

Usually $R_F \gg r_o/\left[1 + \mu\left(R_i/R_F\right)\right]$; thus,

$$R_{\text{out}} \simeq \frac{r_o}{1 + \mu\dfrac{R_i}{R_F}} \simeq \left(\frac{R_F}{R_i}\right)\left(\frac{r_o}{\mu}\right)$$

from which we see that for large μ, the output resistance will be considerably reduced.
(d) For the numerical values given:

$$R_i = R_{id} \| R_F \| R_s$$
$$= \infty \| 10 \| 1 = 0.91 \text{ k}\Omega$$
$$R_o = r_o \| R_F \| R_s$$
$$= 0.1 \| 10 \| 1 = 90 \ \Omega$$
$$A = -\mu R_i \frac{\left(R_F \| R_L\right)}{r_o + \left(R_F \| R_L\right)}$$

例 11.9 续

代入公式（11.42）中的 R、

$$\frac{1}{R} = \frac{1}{R} + \frac{1}{R} + \frac{1}{r} + \mu\frac{R}{R}\frac{1}{r}$$

$$= \frac{1}{R} + \frac{1}{R} + \frac{1}{r}\left(1 + \mu\frac{R}{R}\right)$$

因此、

$$R = R \| R \| \frac{r}{1 + \mu\frac{R}{R}}$$

此外

$$R = R \| R$$

我们得到 R

$$R = R \| \frac{r}{1 + \mu\frac{R}{R}}$$

通常 R r /[1 + μ(R/R)] 因此、

$$R \approx \frac{r}{1 + \mu\frac{R}{R}} \approx \left(\frac{R}{R}\right)\left(\frac{r}{\mu}\right)$$

由此可见，当 μ 较大时，输出电阻将大大减小。(d) 对于给出的数值：

$$R = R \| R \| R$$

$$= \infty \| 10 \| 1 = 0.91\ k$$

$$R = r \| R \| R$$

$$= 0.1 \| 10 \| 1 = 90$$

$$A = -\mu R \frac{\left(R \| R\right)}{r + \left(R \| R\right)}$$

$$= -10^4 \times 0.91 \times \frac{(10 \| 1)}{0.1 + (10 \| 1)} = -8198 \text{ k}\Omega$$

$$\beta = -\frac{1}{R_F} = -\frac{1}{10} = -0.1 \text{ mA/V}$$

$$A\beta = 819.8$$

$$1 + A\beta = 820.8$$

$$A_f = \frac{A}{1 + A\beta} = -\frac{8198}{820.8} = -9.99 \text{ k}\Omega$$

which is very close to the ideal value of $-R_F = -10$ kΩ.

$$R_{if} = \frac{R_i}{1 + A\beta} = \frac{910}{820.8} = 1.11 \ \Omega$$

$$R_{\text{in}} = \frac{1}{\frac{1}{R_{if}} - \frac{1}{R_i}} - \frac{1}{\frac{1}{1.11} - \frac{1}{1000}} \simeq 1.11 \ \Omega$$

which is very low, a highly desirable property. We also have

$$R_{of} = \frac{R_o}{1 + A\beta} = \frac{90}{820.8} = 0.11 \ \Omega$$

$$R_{\text{out}} = \frac{1}{\frac{1}{R_{of}} - \frac{1}{R_L}} = \frac{1}{\frac{1}{0.11} - \frac{1}{1000}} \simeq 0.11 \ \Omega$$

which as well is very low, another highly desirable property.

(e) If the amplifier is fed with a voltage source V_s having a resistance $R_s = 1$ kΩ, the output voltage can be found from

$$V_o = A_f I_s = A_f \frac{V_s}{R_s}$$

Thus,

$$\frac{V_o}{V_s} = \frac{A_f}{R_s} = \frac{9.99 \text{ k}\Omega}{1 \text{ k}\Omega} = -9.99 \text{ V/V}$$

$$= -10 \times 0.91 \times \frac{(10 \| 1)}{0.1 + (10 \| 1)} = -8198 \text{ k}$$

$$\beta = -\frac{1}{R} = -\frac{1}{10} = -0.1 \text{ mA/V}$$

$$A\beta = 819.8$$

$$1 + A\beta = 820.8$$

$$A = \frac{A}{1 + A\beta} = -\frac{8198}{820.8} = -9.99 \text{ k}$$

非常接近理想值 $-R = -10$ k。

$$R = \frac{R}{1 + A\beta} = \frac{910}{820.8} = 1.11$$

$$R = \frac{1}{\frac{1}{R} - \frac{1}{R}} - \frac{1}{\frac{1}{1.11} - \frac{1}{1000}} 1.11$$

这是很低的价格，是非常理想的房产。我们还有

$$R = \frac{R}{1 + A\beta} = \frac{90}{820.8} = 0.11$$

$$R = \frac{1}{\frac{1}{R} - \frac{1}{R}} = \frac{1}{\frac{1}{0.11} - \frac{1}{1000}} 0.11$$

这也是另一处非常理想的房产。

(e) 如果给放大器输入一个电阻 $R = 1$ k 的电压源 V，则输出电压可由以下公式求得

$$V = AI = A \frac{V}{R}$$

因此、

$$\frac{V}{V} = \frac{A}{R} = \frac{9.99 \text{ k}}{1 \text{ k}\Omega} = -9.99 \text{ V/V}$$

EXERCISES

11.18 It is required to determine the loop gain of the amplifier of Fig. 11.24(a) directly. Set $I_s = 0$, replace the amplifier μ with its equivalent circuit, and break the loop at the amplifier input, ensuring that a resistance equal to R_{id} is connected across R_s. Show that

$$A\beta = \frac{\mu R_L (R_{id} \| R_s)}{r_o[R_L + R_F + (R_{id} \| R_s)] + R_L[R_F + (R_{id} \| R_s)]}$$

Evaluate $A\beta$ using the numerical values given in Example 11.9.

Ans. $A\beta = 819.7$

11.19 For the transresistance amplifier in Fig. E11.19, replace the MOSFET with its small-signal equivalent-circuit model and use feedback analysis to show the following:

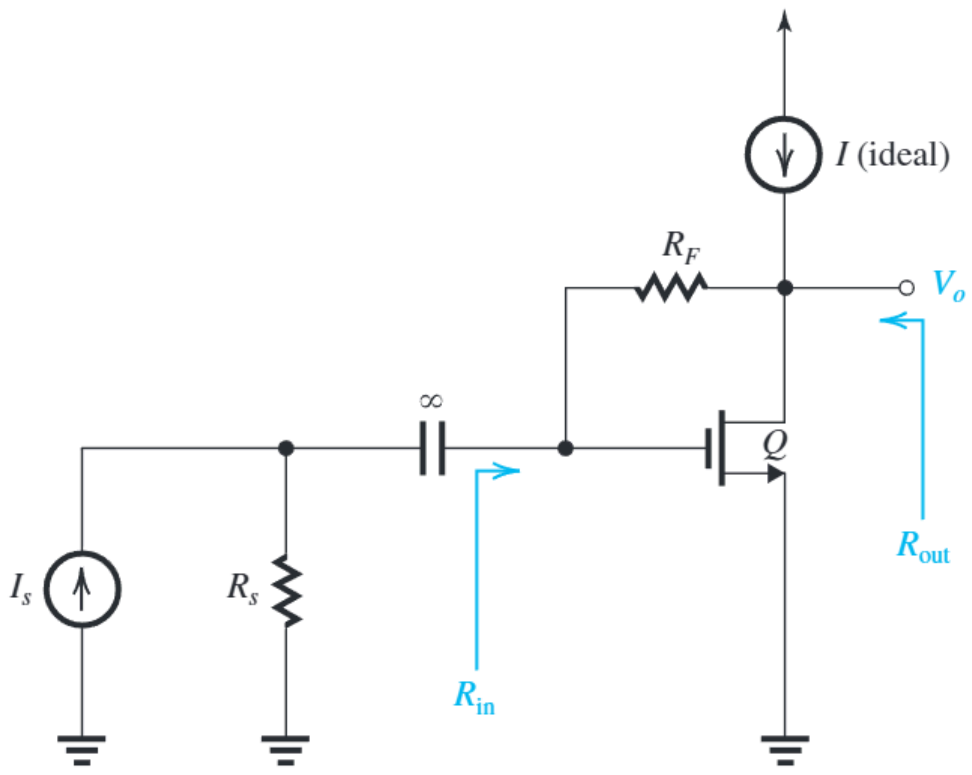


Figure E11.19

(a) For large loop gain (which cannot be achieved here), $A_f \equiv V_o/I_s \simeq -R_F$.

(b) $A_f = \dfrac{-(R_s \| R_F) g_m (r_o \| R_F)}{1 + (R_s \| R_F) g_m (r_o \| R_F)/R_F}$

(c) $R_{\text{in}} = \dfrac{R_F}{[1 + g_m (r_o \| R_F)]}$

(d) $R_{\text{out}} = r_o \| \dfrac{R_F}{1 + g_m (R_s \| R_F)}$

(e) For $g_m = 5$ mA/V, $r_o = 20$ kΩ, $R_F = 10$ kΩ, and $R_s = 1$ kΩ, find A, β, $A\beta$, A_f R_i, R_o, R_{if}, R_{in}, R_{of}, and R_{out}.

Ans. (e) −30.3 kΩ; −0.1 mA/V; 3.03; −7.52 kΩ (compare to the ideal value of −10 kΩ); 909 Ω; 6.67 kΩ; 226 Ω; 291 Ω; 1.66 kΩ; 1.66 kΩ

练习

11.18 需要直接确定图 11.24(a)中放大器的环路增益。设 I = 0，用等效电路代替放大器 μ，并在放大器输入端断开环路，确保在 R 之间连接一个等于 Ris 的电阻。

$$A\beta = \frac{\mu R \quad (R \,||\, R) \quad R\,[R + R + \quad (R}{||\, R) \;] + R\,[R + \quad (R \,||\, R]}$$

用例 11.9 中给出的数值计算 Aβ。

答案：Aβ = 819.7

11.19 对于图 E11.19 中的跨阻放大器，用其小信号等效电路模型替换 MOSFET，并使用反馈分析表明如下：

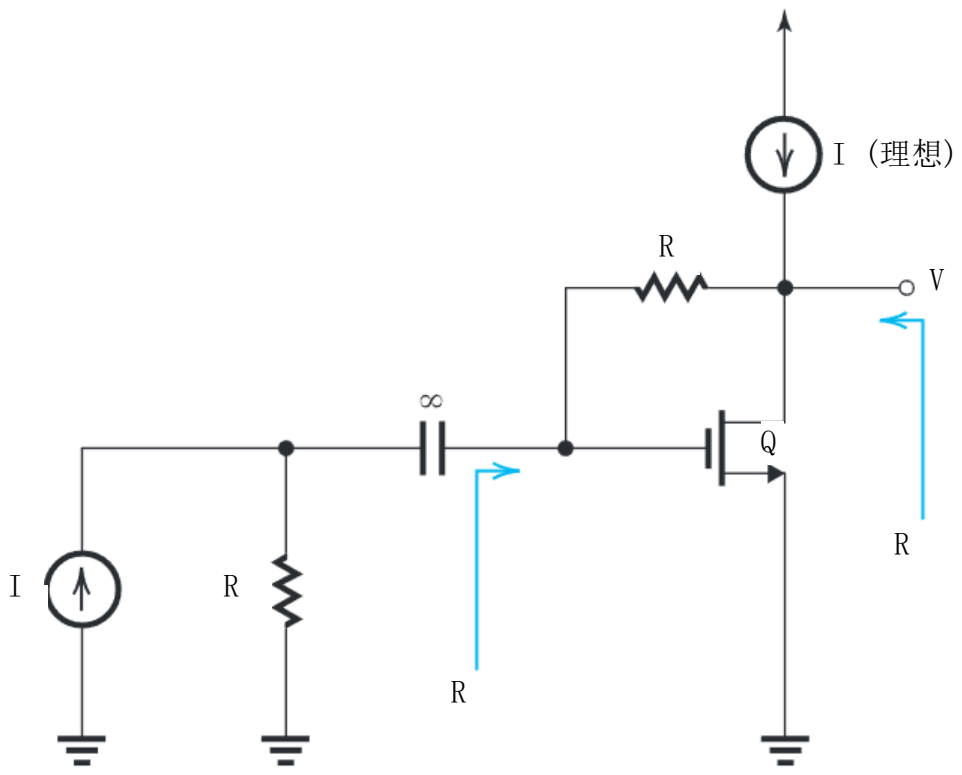


图 E11.19

(a) 对于大环路增益（此处无法实现），A ≡ V/I -R 。

(b) $A = \dfrac{-\left(\dfrac{R \,\|\, R}{R \,\|\, R}\right) g \left(\dfrac{r \,\|\, R}{r \,\|\, R/R}\right)}{1 +}$

(c) $R = \left[\dfrac{R}{1 + g\left(\dfrac{\quad}{r \,\|\, R}\right)}\right]$

(d) $R = r \| \dfrac{R}{1 + g\left(\dfrac{\quad}{R \,\|\, R}\right)}$

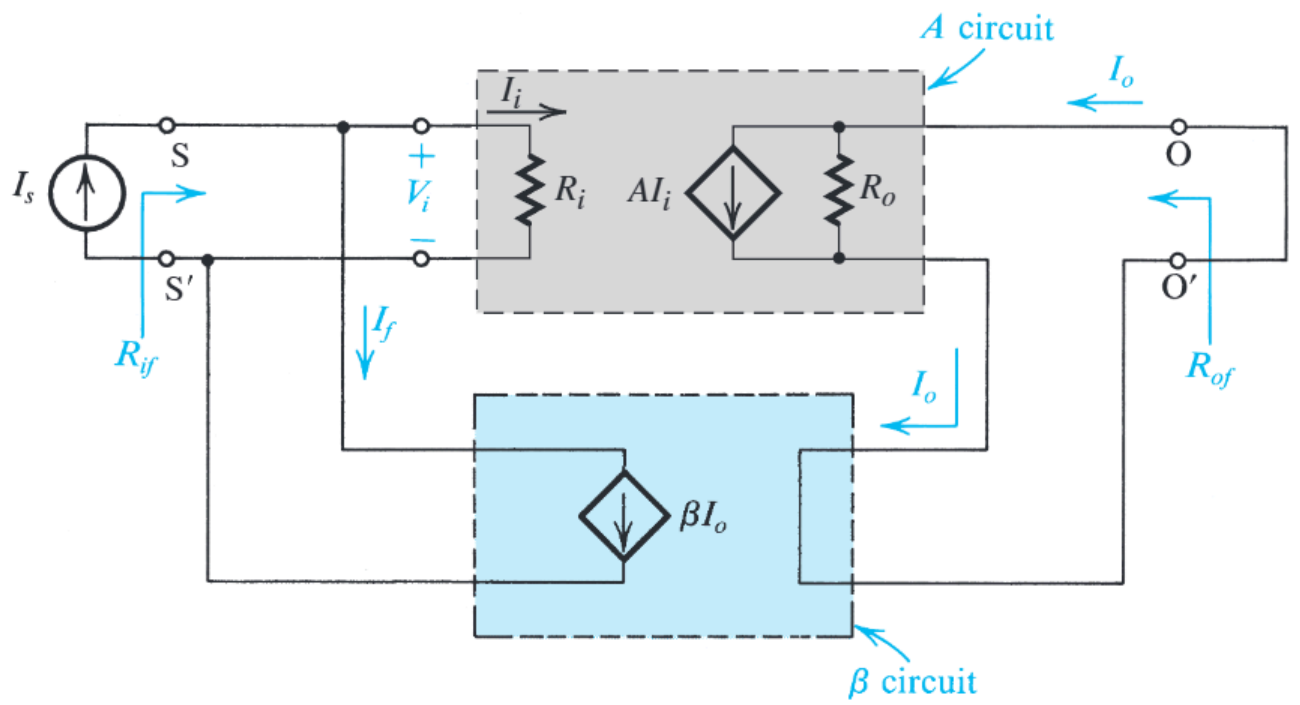
(e) 对于 g = 5 mA/V、r = 20 k、R = 10 k 和 R = 1 k，求 A、β、A β、A R ， R ， R ， R， R 和 R。 答案 (e) -30.3 k; -0.1 mA/V; 3.03; -7.52 k（与理想值 -10 k 比较）; 909 ; 6.67 k; 226 ; 291 ; 1.66 k; 1.66 k

11.5.4 The Feedback Current Amplifier (Shunt–Series)

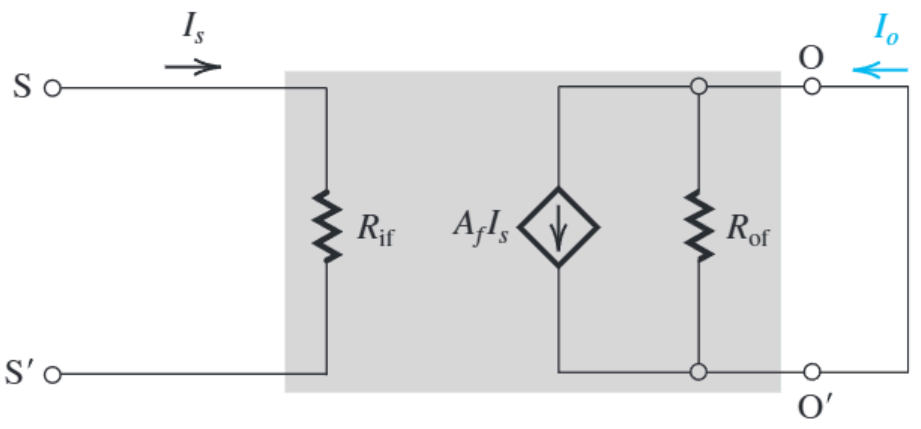
Figure 11.25(a) shows the ideal feedback current amplifier, which, as expected, utilizes the shunt–series topology. The amplifier equivalent circuit is shown in Fig. 11.25(b), with the formulas for determining A_f, R_{if}, and R_{of} given in Fig. 11.25(c).

Figure 11.26 shows how the feedback-analysis method can be applied to a general feedback current amplifier. All the necessary analysis steps and the formulas are given. The method is now illustrated by a detailed example.

(a) Ideal Structure



(b) Equivalent Circuit



(c) Formulas

$$A_f \equiv \frac{I_o}{I_s} = \frac{A}{1+A\beta}$$

$$R_{if} = R_i/(1+A\beta)$$

$$R_{of} = (1+A\beta)R_o$$

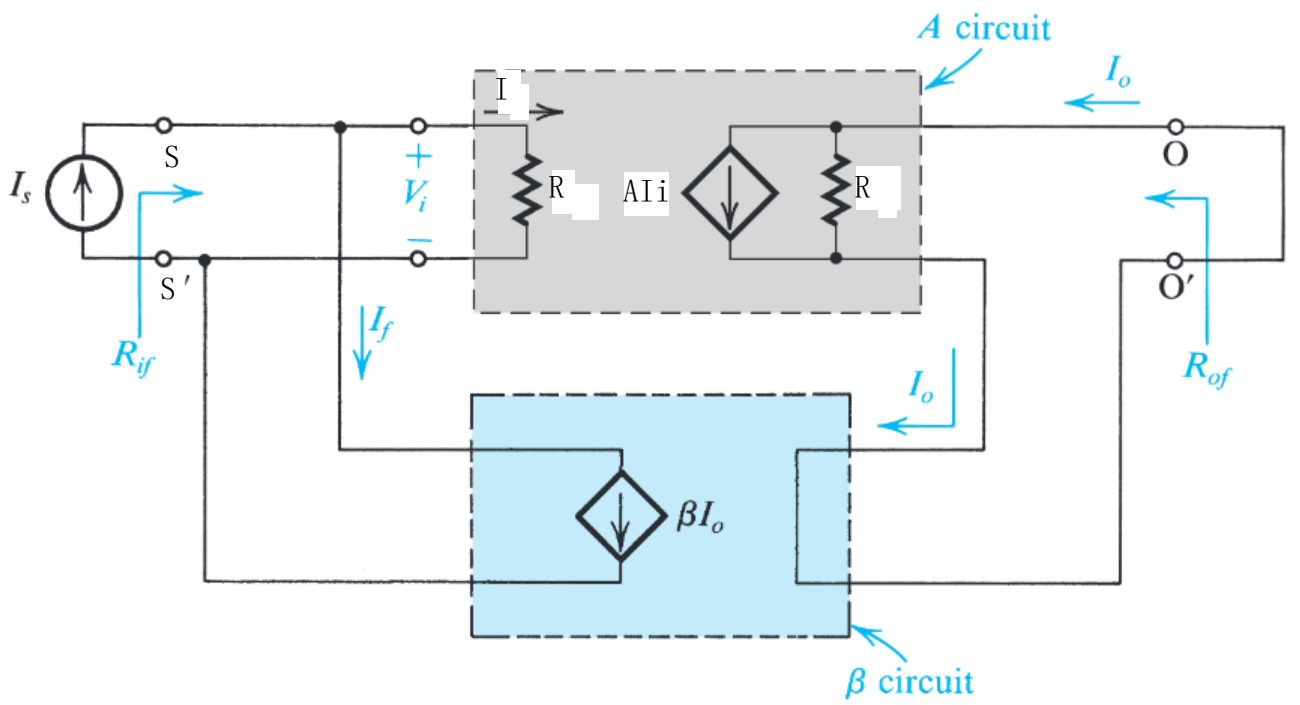
Figure 11.25 The feedback current amplifier (shunt–series).

11.5.4 反馈电流放大器（顺特系列）

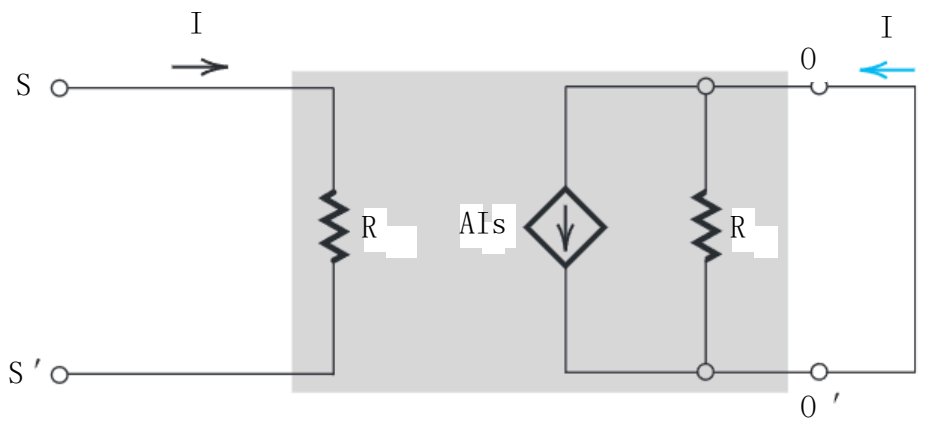
图 11.25(a) 显示了理想的反馈电流放大器，正如预期的那样，它采用了并联-串联拓扑结构。放大器等效电路见图 11.25(b)，A、R 和 R 的计算公式见图 11.25(c)。

图 11.26 展示了如何将反馈分析方法应用于一般反馈电流放大器。图中给出了所有必要的分析步骤和公式。现在通过一个详细的示例来说明该方法。

(a) 理想结构



(b) 等效电路



(c) 公式

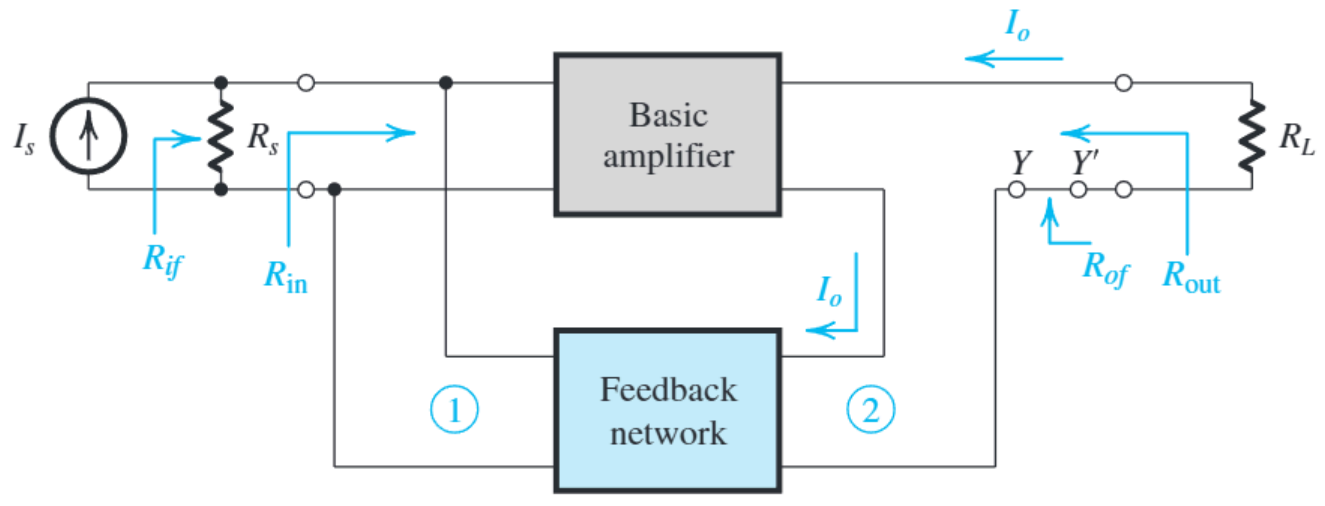
$$A \equiv \frac{I}{I} = \frac{A}{1 + A\beta}$$

$$R = R/(1 + A\beta)$$

$$R = (1 + A\beta)R$$

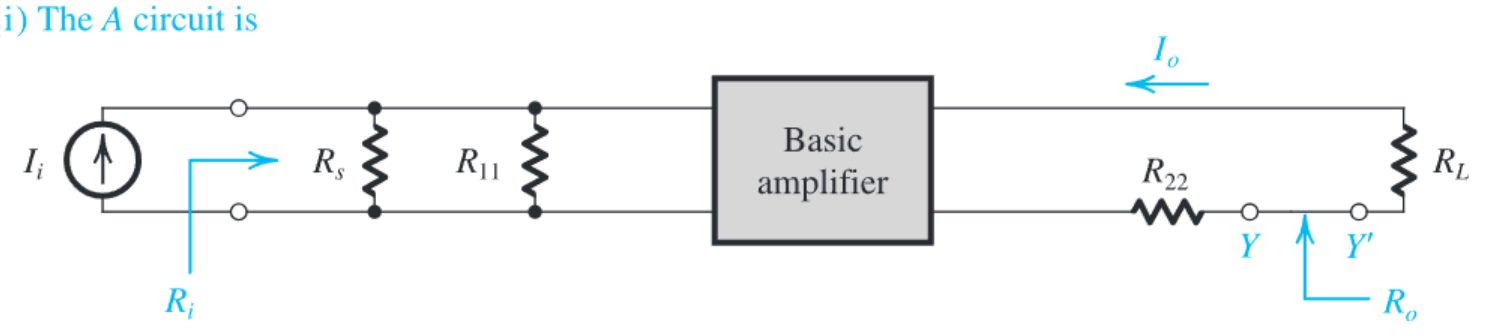
图 11.25 反馈电流放大器（并联串联）。

(a) General Structure

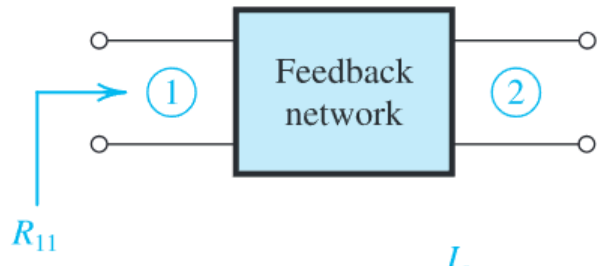


(b) Finding the A Circuit and β

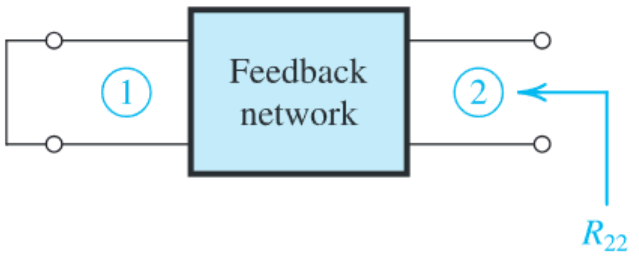
(i) The A circuit is



where R_{11} is obtained from

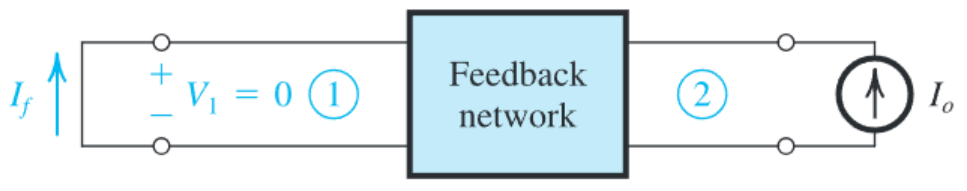


and R_{22} is obtained from



and the gain A is defined as $A \equiv \dfrac{I_o}{I_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{I_f}{I_o}\right|_{V_1 = 0}$$

(c) Gain, Input, and Output Resistance

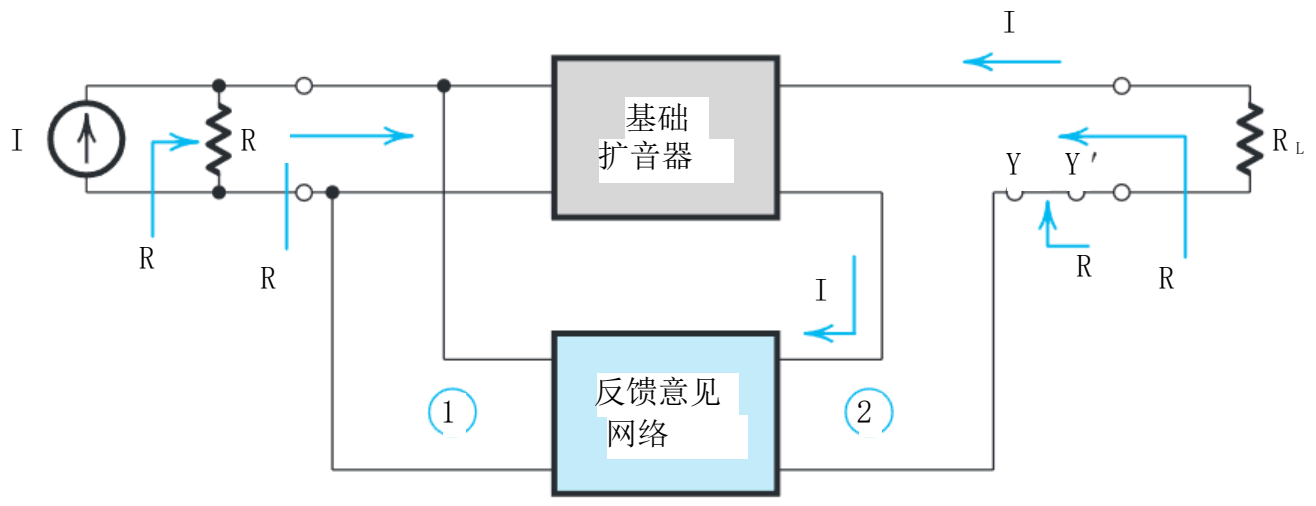
- Use the formulas in Fig. 11.25 to find A_f, R_{if}, and R_{of}.
- R_{in} and R_{out} can then be found from

$$R_{\text{in}} = 1 \Big/ \left(\frac{1}{R_{if}} - \frac{1}{R_s}\right)$$

$$R_{\text{out}} = R_{of} - R_L$$

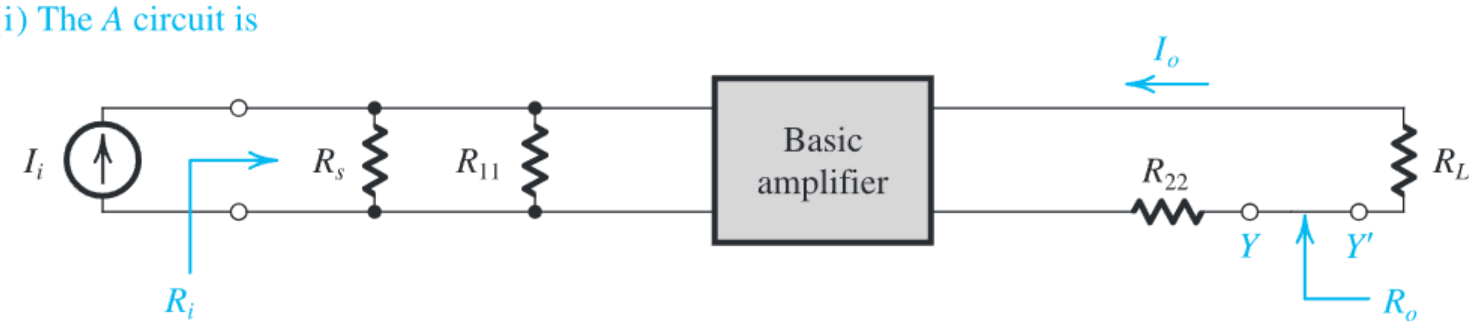
Figure 11.26 The feedback current amplifier (shunt–series).

(a) 总体结构

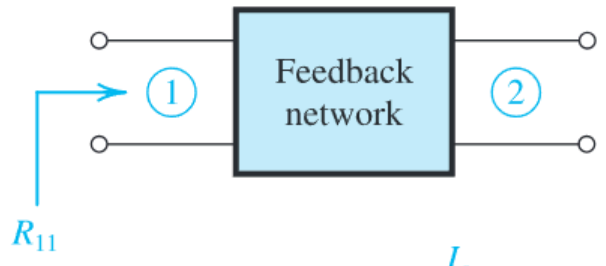


(b) Finding the *A* Circuit and β

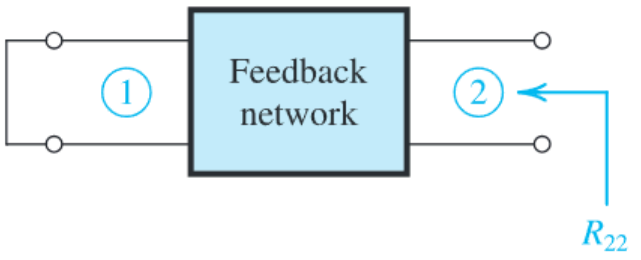
(i) The *A* circuit is



where R_{11} is obtained from

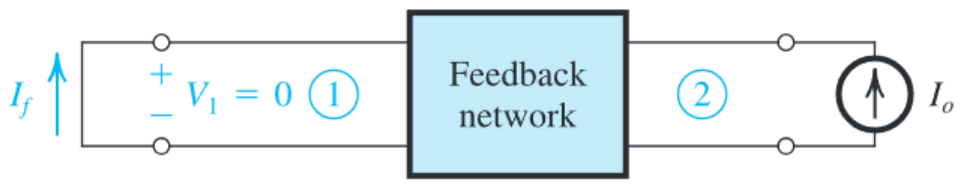


and R_{22} is obtained from



and the gain *A* is defined as $A \equiv \frac{I_o}{I_i}$

(ii) β is obtained from



$$\beta \equiv \left.\frac{I_f}{I_o}\right|_{V_1 = 0}$$

(c) 增益、输入和输出电阻

- 使用图 11.25 中的公式求出 A、R 和 R。
- Rand R 的计算公式为

$$R = 1 \Big/ \left(\frac{1}{R} - \frac{1}{R}\right)$$

$$R = R - R$$

图 11.26 反馈电流放大器（并联串联）。

Example 11.10

Figure 11.27 shows a feedback current amplifier formed by cascading an inverting voltage amplifier μ with a MOSFET Q. The output current I_o is the drain current of Q. The feedback network, consisting of resistors R_1 and R_2, senses an exactly equal current, namely, the source current of Q, and provides a feedback current signal that is mixed with I_s at the input node. Note that the bias arrangement is *not* shown.

The amplifier μ can be implemented in a variety of ways, including by means of an op amp, a differential amplifier, or a single-ended inverting amplifier. The simplest approach is to implement μ with a CS

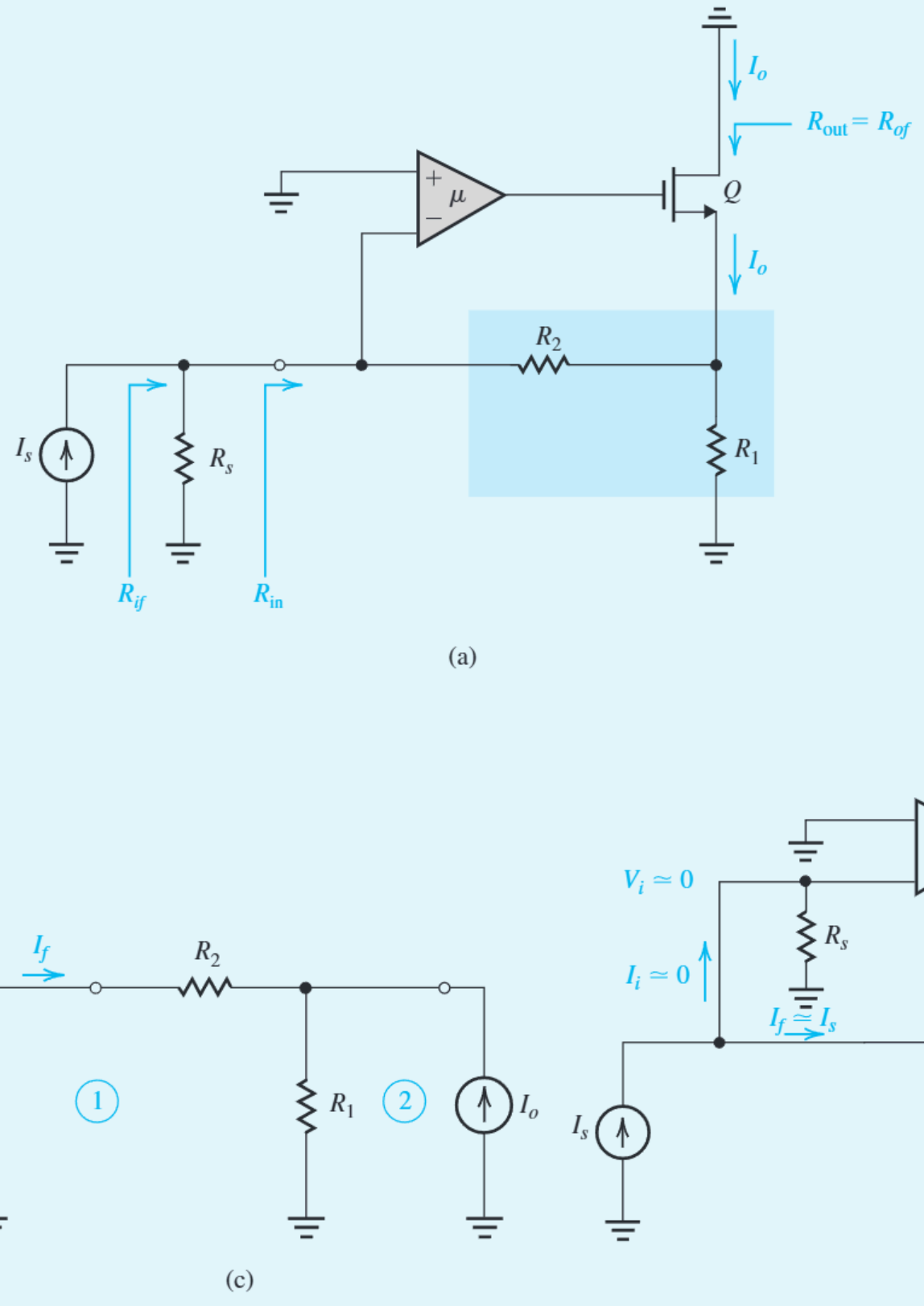


Figure 11.27 Circuit for Example 11.10.

例 11.10

输出电流 I 是 Q 的漏极电流。由电阻 Rand R 组成的反馈网络感应到一个完全相等的电流，即 Q 的源极电流，并提供一个反馈电流信号，在输入节点与 I 混合。请注意，偏置安排并未显示。

放大器 μ 可以通过多种方式实现，包括运算放大器、差分放大器或单端反相放大器。最简单的方法是用一个 CS

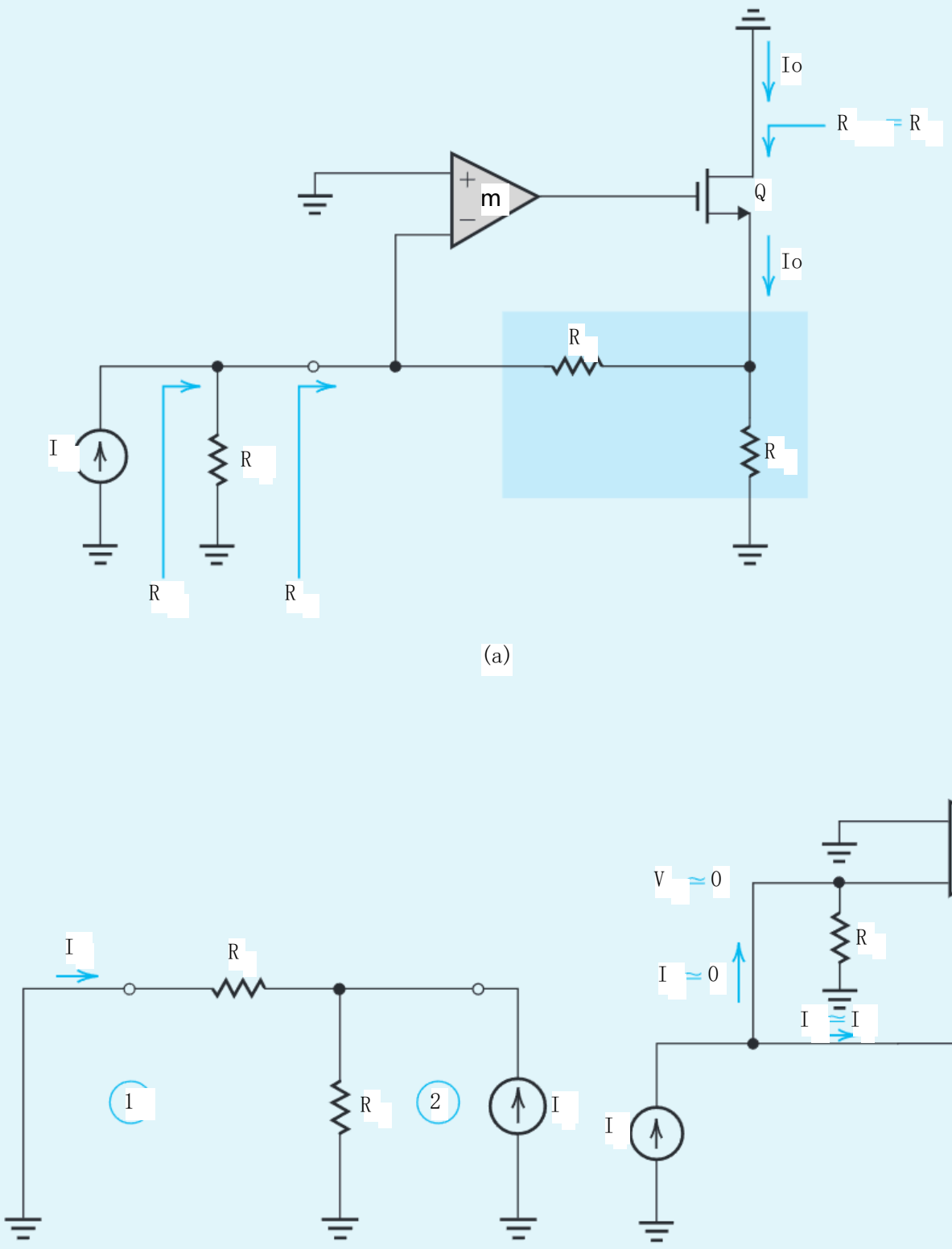


图 11.27 例 11.10 的电路。

Example 11.10 *continued*

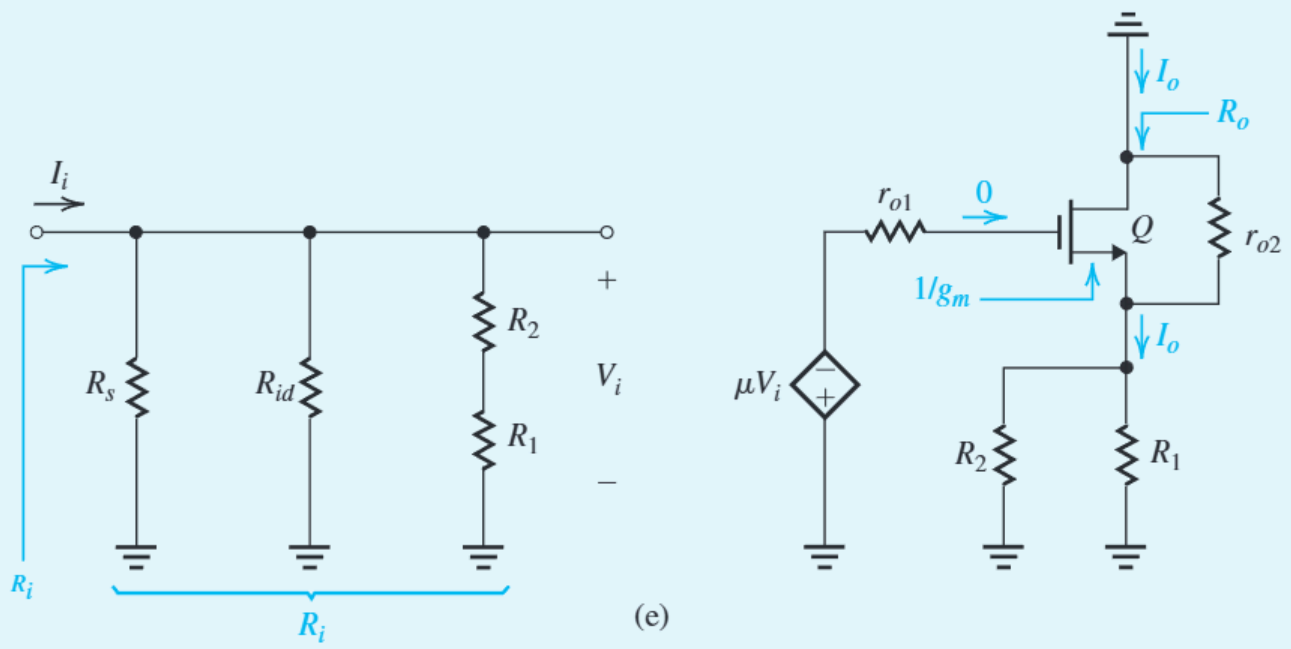


Figure 11.27 *continued*

MOSFET amplifier. However, in such a case the loop gain will be very limited. Assume that the amplifier μ has an input resistance R_{id}, an open-circuit voltage gain μ, and an output resistance r_{o1}.

(a) If the loop gain is large, find an approximate expression for the closed-loop gain $A_f \equiv I_o/I_s$.
(b) Find the A circuit and derive expressions for A, R_i, and R_o.
(c) Give expressions for $A\beta$, A_f, R_{if}, R_{in}, R_{of}, and R_{out}.
(d) Find numerical values for A, β, $A\beta$, A_f, R_i, R_{if}, R_{in}, R_o, R_{of}, and R_{out} for the following case: $\mu = 1000$ V/V, $R_s = \infty$, $R_{id} = \infty$, $r_{o1} = 1$ kΩ, $R_1 = 10$ kΩ, $R_2 = 90$ kΩ; for Q: $g_m = 5$ mA/V and $r_o = 20$ kΩ.

Solution

(a) When the loop gain $A\beta \gg 1$, $A_f \simeq 1/\beta$. The β circuit, shown in Fig. 11.27(b), is used to determine β as shown in Fig. 11.27(c),

$$\beta \equiv \frac{I_f}{I_o} = -\frac{R_1}{R_1 + R_2} \tag{11.48}$$

Thus,

$$A_f \simeq \frac{1}{\beta} = -\left(1 + \frac{R_2}{R_1}\right) \tag{11.49}$$

To see what happens in this case more clearly, refer to Fig. 11.27(d). Here we have assumed the loop gain to be large, so that $I_i \simeq 0$ and thus $I_f \simeq I_s$. Also note that because $I_i \simeq 0$, V_i will be close to zero. Thus, we can easily determine the voltage at the source of Q as $-I_f R_2 \simeq -I_s R_2$. The current through R_1 will then be $I_s R_2/R_1$. The source current of Q will be $-\left(I_s + I_s R_2/R_1\right)$, which means that the output current I_o will be

$$I_o = -I_s\left(1 + \frac{R_2}{R_1}\right)$$

which confirms the expression for A_f obtained above (Eq. 11.49).

例 11.10 续

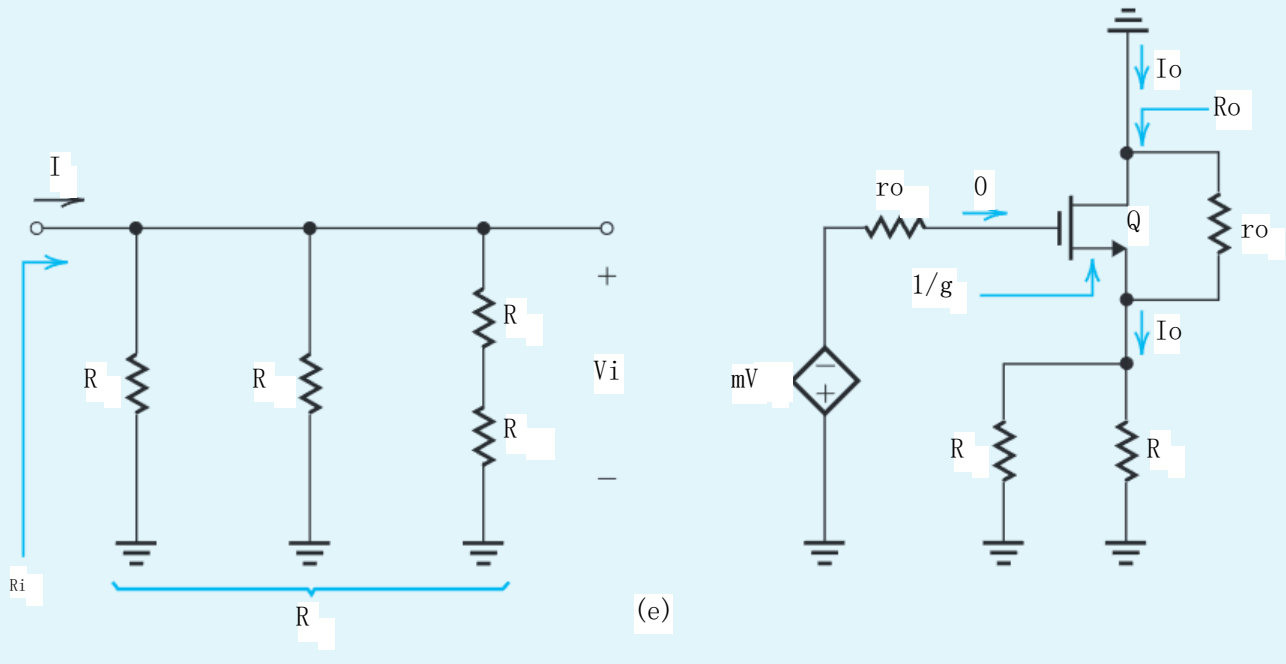


图 11.27 续

MOSFET 放大器。然而，在这种情况下，环路增益将非常有限。假设放大器 μ 具有输入电阻 R、开路电压增益 μ 和输出电阻 r。

(a) 如果环路增益很大，求闭环增益 A≡ I/I 的近似表达式。
(b) 找出 A 电路，并推导出 A、R 和 R 的表达式。
(c) 给出 Aβ、A、R ， R、R ， 和 R 的表达式。
(d) 为下列情况找出 A、β、Aβ、A、R , R , R、R , R , R , 和 R 的数值：μ = 1000 V/V，R= ∞，R= ∞，r= 1 k，R= 10 k，R= 90 k；Q：g = 5 mA/V，r= 20 k。

解决方案

(a) 当环路增益 Aβ 为 1 时，A 1/β。图 11.27(b)所示的 β 电路用于确定 β，如图 11.27(c)所示、

$$\beta \equiv \frac{I}{I} = -\frac{R}{R+R} \tag{11.48}$$

因此、

$$A \approx \frac{1}{\beta} = -\left(1+\frac{R}{R}\right) \tag{11.49}$$

要更清楚地了解这种情况，请参阅图 11.27(d)。在这里，我们假设环路增益很大，因此 I 0，从而 I I。还需要注意的是，由于 I 0，Vw 将接近于零。因此，我们可以很容易地确定 Q 的源极电压为 -I R -I R。Q 的源电流为

$(I + IR/R)$ 这意味着输出电流 I 将为

$$I = -I\left(1+\frac{R}{R}\right)$$

这证实了上文得到的 A 表达式（公式 11.49）。

(b) To obtain the A circuit we load the input side of the basic amplifier with R_s and R_{11}. The latter in this case is simply $R_1 + R_2$ (because port 2 of the feedback network is opened). We also load the output of the basic amplifier with R_{22}, which in this case is $R_1 \| R_2$ (because port 1 of the feedback network is shorted). The resulting A circuit is shown in Fig. 11.27(e), where we have replaced the amplifier μ with its equivalent circuit. Analysis of the A circuit is straightforward and proceeds as follows:

$$R_i = R_s \| R_{id} \| \left(R_1 + R_2\right) \tag{11.50}$$

$$V_i = I_i R_i \tag{11.51}$$

$$I_o = -\mu V_i \frac{1}{1/g_m + \left(R_1 \| R_2 \| r_{o2}\right)} \frac{r_{o2}}{r_{o2} + \left(R_1 \| R_2\right)} \tag{11.52}$$

Combining Eqs. (11.51) and (11.52) results in A:

$$A \equiv \frac{I_o}{I_i} = -\mu \frac{R_i}{1/g_m + \left(R_1 \| R_2 \| r_{o2}\right)} \frac{r_{o2}}{r_{o2} + \left(R_1 \| R_2\right)} \tag{11.53}$$

which with some manipulation can be expressed in the form

$$A = -\mu \frac{R_i}{\left(R_1 \| R_2\right)} \frac{\left(R_1 \| R_2 \| r_{o2}\right)}{1/g_m + \left(R_1 \| R_2 \| r_{o2}\right)} \tag{11.54}$$

Noting that R_o is the output resistance of Q, which has a resistance $(R_1 \| R_2)$ in its source lead, we can write

$$R_o = r_{o2} + \left(R_1 \| R_2\right) + \left(g_m r_{o2}\right)\left(R_1 \| R_2\right) \tag{11.55}$$

(c) The loop gain is obtained by combining Eqs. (11.48) and (11.53),

$$A\beta = \mu \frac{R_i}{R_2} \frac{\left(R_1 \| R_2 \| r_{o2}\right)}{1/g_m + \left(R_1 \| R_2 \| r_{o2}\right)} \tag{11.56}$$

The closed-loop gain A_f can be obtained by substituting the expressions of A and $A\beta$ (Eqs. 11.54 and 11.56) in $A_f = A/(1 + A\beta)$.

The input resistance R_{if} is found as

$$R_{if} = R_i/(1 + A\beta)$$

Since by definition,

$$R_{if} = R_s \| R_{\text{in}}$$

we can easily find R_{in}.

For the output resistance, we have

$$R_{\text{out}} = R_{of} = R_o(1 + A\beta)$$

(d) For the numerical values given,

$$R_i = \infty \| \infty \| (10 + 90) = 100 \text{ k}\Omega$$

(b) 为了得到 A 电路，我们在基本放大器的输入端加载 R 和 R。我们还在基本放大器的输出端加载 R，在本例中为 R ‖ R（因为反馈网络的端口 1 短路）。由此产生的 A 电路如图 11.27(e)所示，我们用等效电路替换了放大器 μ。对 A 电路的分析简单明了，步骤如下：

$$R = R|| R|| (R+ R) \tag{11.50}$$

$$V= IR \tag{11.51}$$

$$I= -\mu V \frac{1}{1/g + (R \| R \| r)} \frac{r}{r + (R|| R)} \tag{11.52}$$

将公式 (11.51) 和 (11.52) 相结合，得出 A：

$$A \equiv \frac{I}{I} = -\mu \frac{R}{1/g + (R \| R \| r)} \frac{r}{r + (R|| R)} \tag{11.53}$$

经过一些处理，可以表示为

$$A = -\mu \frac{R}{(R||R)} \frac{(R||R||R)}{1/g + (R|| R|| r)} \tag{11.54}$$

注意到 Q 的输出电阻 Ris 的源引线上有一个电阻 (R ‖ R)，我们可以写出

$$R = r+ (R|| R) + (gr) (R|| R) \tag{11.55}$$

(c) 结合公式 (11.48) 和 (11.53) 可得到环路增益、

$$A\beta = \mu \frac{R}{R} \frac{(R||R||R)}{1/g + (R|| R|| r)} \tag{11.56}$$

将 A 和 Aβ（公式 11.54 和 11.56）代入 A= A/(1 + Aβ)，即可得到闭环增益 Ac。输入电阻为

$$R = R/(1 + A\beta)$$

因为根据定义

$$R = R \| R$$

我们可以很容易地找到 R。

对于输出电阻，我们有

$$R= R= R(1 + A\beta)$$

(d) 对于给出的数值、

$$R= \infty|| \infty|| (10 + 90) = 100 \text{ k}$$

Example 11.10 *continued*

The open-loop gain A can be determined by using Eq. (11.54),

$$A = -10,764 \text{ A/A}$$

and the feedback factor β can be found by using Eq. (11.48),

$$\beta = -\frac{R_1}{R_1 + R_2} = -\frac{10}{10 + 90} = -0.1 \text{ A/A}$$

Thus,

$$A\beta = 1076.4$$

and

$$A_f = -\frac{10,764}{1 + 1076.4} = -9.99 \text{ A/A}$$

which is very close to the ideal value of

$$A_f \simeq -\left(1 + \frac{R_2}{R_1}\right) = -\left(1 + \frac{90}{10}\right) = -10 \text{ A/A}$$

$$R_{if} = \frac{R_i}{1 + A\beta} = \frac{100 \text{ k}\Omega}{1 + 1076.4} = 92.8 \ \Omega$$

since $R_s = \infty$,

$$R_{\text{in}} = R_{if} = 92.8 \ \Omega$$

$$R_o = r_{o2} + (R_1 \parallel R_2) + g_m r_{o2}(R_1 \parallel R_2) = 929 \text{ k}\Omega$$

$$R_{\text{out}} = R_{of} = (1 + A\beta)R_o = 1077.4 \times 929 = 1001 \text{ M}\Omega$$

EXERCISES

11.20 For the amplifier in Example 11.10, find the values of A_f, R_{in}, and R_{out} when the value of μ is 10 times lower, that is, when $\mu = 100$.

Ans. -9.91 A/A; 920 Ω; 101 MΩ

11.21 If in the circuit in Fig. 11.27(a), R_2 is short-circuited, find the ideal value of A_f. For the case $R_s = R_{id} = \infty$, give expressions for R_i, R_o, A, β, A_f, R_{in}, and R_{out}.

Ans. $A_f = -1$ A/A; $R_i = R_1$; $R_o = r_{o2}$, $A = -\mu g_m R_1$; $\beta = -1$; $A_f = -\mu g_m R_1/(1 + \mu g_m R_1)$; $R_{\text{in}} \simeq 1/\mu g_m$; $R_{\text{out}} \simeq \mu(g_m r_{o2})R_1$.

例 11.10 续
开环增益 A 可通过公式 (11.54) 确定、

$$a = -10,\ 764\ a/a$$

和反馈系数 β 可通过公式 (11.48) 求得、

$$\beta = -\frac{R}{R+R} = -\frac{10}{10+90} = -0.1\ A/A$$

因此、

$$A\beta = 1076.4$$

and

$$A = -\frac{10,\ 764}{1+1076.4} = 9.99\ A/A$$

非常接近理想值

$$A \approx -\left(1+\frac{R}{R}\right) = -\left(1+\frac{90}{10}\right) = -10\ A/A$$

$$R = \frac{R}{1+A\beta} = \frac{100\ k}{1+1076.4} = 92.8$$

因为 R = ∞、

R= R= 92.8

$$R = r + (R \parallel R) + gr\left(R \parallel R\right) = 929\ k$$

R= R= (1 + Aβ)R= 1077.4 × 929 = 1001 M

练习

11.20 对于例 11.10 中的放大器，当 μ 值降低 10 倍（即 μ = 100）时，求 A、R 和 Rw 的值。答案-9.91 a/a; 920 ; 101 m

11.21 对于 R = R= ∞ 的情况，请给出 R、R , A、β、A , R 和 R 的表达式。

答案 A = -1 A/A; R = R; R = r, A = -μgR; β = -1; A = -μg R/(1 + μgR); R 1/μg ; R≈ μ (gr) R.

11.6 Summary of the Feedback-Analysis Method

Table 11.2 provides a summary of the rules and relationships employed in the analysis and design of the four types of feedback amplifier. In addition to the wealth of information in Table 11.2, we offer the following important analysis tips.

1. Always begin the analysis by determining an approximate value for the closed-loop gain A_f, assuming that the loop gain $A\beta$ is large and thus

$$A_f \simeq 1/\beta$$

   This value should serve as a check on the final value you find for A_f. How close the actual A_f is to this ideal value will depend on how large $A\beta$ is compared to unity.
2. The shunt connection at input or output always results in reducing the corresponding resistance (input or output). The series connection at input or output always results in increasing the corresponding resistance (input or output).
3. In utilizing negative feedback to improve the properties of an amplifier under design, the starting point in the design is the selection of the feedback topology appropriate for the application at hand. Then the required amount of negative feedback $(1 + A\beta)$ can be ascertained utilizing the fact that it is this quantity that determines the magnitude of improvement in the various amplifier parameters. Also, the feedback factor β can be determined from the required closed-loop gain A_f,

$$\beta \simeq 1/A_f$$

11.7 The Stability Problem

11.7.1 Transfer Function of the Feedback Amplifier

In a feedback amplifier such as that represented by the general structure of Fig. 11.1, the open-loop gain A is generally a function of frequency, and it should therefore be more accurately called the **open-loop transfer function**, $A(s)$. Also, we have been assuming for the most part that the feedback network is resistive and hence that the feedback factor β is constant, but this need not be always the case. We shall therefore assume that in the general case the **feedback transfer function** is $\beta(s)$. It follows that the **closed-loop transfer function** $A_f(s)$ is given by

$$A_f(s) = \frac{A(s)}{1 + A(s)\beta(s)} \tag{11.57}$$

To focus attention on the points central to our discussion in this section, we shall assume that the amplifier is direct coupled with constant dc gain A_0 and with poles and zeros occurring in the high-frequency band. Also, for the time being let us assume that at low frequencies $\beta(s)$ reduces to a constant value. Thus at low frequencies the loop gain $A(s)\beta(s)$ becomes a constant, which should be a positive number; otherwise the feedback would not be negative. The question then is: What happens at higher frequencies?

11.6 反馈分析法总结

表 11.2 概述了分析和设计四种反馈放大器时采用的规则和关系。除了表 11.2 中的大量信息外，我们还提供了以下重要的分析提示。

1. 在开始分析时，一定要先确定闭环增益 A 的近似值，假设环路增益 Aβ 较大，因此

$$A \approx 1/\beta$$

实际 A 与理想值的接近程度取决于 Aβ 与统一值相比有多大。

2. 输入或输出端的并联总是会减小相应的电阻（输入或输出）。输入或输出端的串联总是会增加相应的电阻（输入或输出）。

3. 在利用负反馈改善设计中放大器的性能时，设计的出发点是选择适合当前应用的反馈拓扑结构。负反馈量（1 + Aβ）决定了放大器各种参数的改善程度，因此可以利用这一事实来确定所需的负反馈量。此外，反馈系数 β 也可根据所需的闭环增益 A 确定、

$$\beta \approx 1/A$$

11.7 稳定性问题

11.7.1 反馈放大器的传递函数

在反馈放大器（如图 11.1 所示的一般结构）中，开环增益 A 通常是频率的函数，因此应更准确地称为开环传递函数 A(s)。此外，我们在大多数情况下都假设反馈网络是电阻性的，因此反馈系数 β 是恒定的，但情况并非总是如此。因此，我们假设在一般情况下，反馈传递函数为 β(s)。由此可见，闭环传递函数 A(s) 取值为

$$A(s) = \frac{A(s)}{1 + A(s)\beta(s)} \tag{11.57}$$

为了将注意力集中在本节讨论的中心点上，我们假设放大器是直接耦合的，直流增益 A 保持不变，极点和零点出现在高频段。另外，我们暂且假设在低频时，β(s) 降为恒定值。因此，在低频时，环路增益 A(s)β(s) 变成了一个常数，应该是一个正数，否则反馈就不会是负数。那么问题来了：在较高频率下会发生什么？

Table 11.2 Summary of Relationships for the Four Feedback-Amplifier Topologies

Feedback Amplifier	Feedback Topology	x_i	x_o	x_f	x_s	A	β	A_f	Source Form	Loading of Feedback Network Is Obtained: At Input	Loading of Feedback Network Is Obtained: At Output	To Find β, Apply to Port 2 of Feedback Network	R_{if}	R_{of}	Refer to Figs.
Voltage	Series–shunt	V_i	V_o	V_f	V_s	$\frac{V_o}{V_i}$	$\frac{V_f}{V_o}$	$\frac{V_o}{V_s}$	Thévenin	By short-circuiting port 2 of feedback network	By open-circuiting port 1 of feedback network	a voltage, and find the open-circuit voltage at port 1	$R_i(1+A\beta)$	$\frac{R_o}{1+A\beta}$	11.12 11.14
Current	Shunt–series	I_i	I_o	I_f	I_s	$\frac{I_o}{I_i}$	$\frac{I_f}{I_o}$	$\frac{I_o}{I_s}$	Norton	By open-circuiting port 2 of feedback network	By short-circuiting port 1 of feedback network	a current, and find the short-circuit current at port 1	$\frac{R_i}{1+A\beta}$	$R_o(1+A\beta)$	11.25 11.26
Transconductance	Series–series	V_i	I_o	V_f	V_s	$\frac{I_o}{V_i}$	$\frac{V_f}{I_o}$	$\frac{I_o}{V_s}$	Thévenin	By open-circuiting port 2 of feedback network	By open-circuiting port 1 of feedback network	a current, and find the open-circuit voltage at port 1	$R_i(1+A\beta)$	$R_o(1+A\beta)$	11.18 11.19
Transresistance	Shunt–shunt	I_i	V_o	I_f	I_s	$\frac{V_o}{I_i}$	$\frac{I_f}{V_o}$	$\frac{V_o}{I_s}$	Norton	By short-circuiting port 2 of feedback network	By short-circuiting port 1 of feedback network	a voltage, and find the short-circuit current at port 1	$\frac{R_i}{1+A\beta}$	$\frac{R_o}{1+A\beta}$	11.22 11.23

表 11.2 四种反馈放大器拓扑结构的关系概述

反馈意见放大器	反馈意见拓扑结构	x	x_o	x_f	x_s	A	β	A_f	资料来源 Form	加载反馈获得网络：在输入端	加载反馈获得网络：输出时	查找 β，适用于反馈意见网络	R_{if}	R_{of}	参考至图
电压	串联分流	V_i	V_o	V_f	V_s	$\frac{V_o}{V_i}$	$\frac{V_f}{V_o}$	$\frac{V_o}{V_s}$	Th'晚上	按短期 反馈网络	通过 open- 循环 第 1 端口 反馈网络	a 电压、 并找到 开路 的电压 端口 1	$R_i(1+A\beta)$	$\frac{R_o}{1+A\beta}$	11.12 电路 11.14 端口 2 的
当前	并联系列	I	I	I	I	$\frac{I}{I}$	$\frac{I}{I}$	$\frac{I}{I}$	诺顿	通过 open- 循环 第 1 端口 反馈网络	按短期 反馈网络	a 电流、 并找到 短路 电流 端口 1	$\frac{R_i}{1+A\beta}$	$R_o(1+A\beta)$	11.25 电路 11.26 端口 2 的
跨导	系列-系列	V_i	I	V_f	V_s	$\frac{IV}{i}$	$\frac{V_f}{I}$	$\frac{IV}{s}$	Th'晚上	通过 open- 循环 第 1 端口 反馈网络	通过 open- 反馈网络	a 电流、 并找到 开路 的电压 端口 1	$R_i(1+A\beta)$	$R_o(1+A\beta)$	11.18 电路 11.19 端口 2 的
跨阻	分流-分流	I	V_o	I	I	$\frac{V_o}{I}$	$\frac{IV}{o}$	$\frac{V_o}{I}$	诺顿	按短期 循环 第 1 端口 反馈网络	按短期 反馈网络	a 电压、 并找到 短路 电流 端口 1	$\frac{R_i}{1+A\beta}$	$\frac{R_o}{1+A\beta}$	11.22 电路 11.23 端口 2 的

For physical frequencies $s = j\omega$, Eq. (11.57) becomes

$$A_f(j\omega) = \frac{A(j\omega)}{1 + A(j\omega)\beta(j\omega)} \tag{11.58}$$

Thus the loop gain $A(j\omega)\beta(j\omega)$ is a complex number that can be represented by its magnitude and phase,

$$\begin{aligned} L(j\omega) &\equiv A(j\omega)\beta(j\omega) \\ &= |A(j\omega)\beta(j\omega)|e^{j\phi(\omega)} \end{aligned} \tag{11.59}$$

It is the manner in which the loop gain varies with frequency that determines the stability or instability of the feedback amplifier. To appreciate this fact, consider the frequency at which the phase angle $\phi(\omega)$ becomes 180°. At this frequency, ω_{180}, the loop gain $A(j\omega)\beta(j\omega)$ will be a real number with a negative sign. Thus at this frequency the feedback will become positive. If at $\omega = \omega_{180}$ the magnitude of the loop gain is less than unity, then from Eq. (11.58) we see that the closed-loop gain $A_f(j\omega)$ will be greater than the open-loop gain $A(j\omega)$, since the denominator of Eq. (11.58) will be smaller than unity. Nevertheless, the feedback amplifier will be stable.

On the other hand, if at the frequency ω_{180} the magnitude of the loop gain is equal to unity, it follows from Eq. (11.58) that $A_f(j\omega)$ will be infinite. This means that the amplifier will have an output for zero input; this is by definition an **oscillator**. To visualize how this feedback loop may oscillate, consider the general loop of Fig. 11.1 with the external input x_s set to zero. Any disturbance in the circuit, such as the closure of the power-supply switch, will generate a signal $x_i(t)$ at the input to the amplifier. Such a noise signal usually contains a wide range of frequencies, and we shall now concentrate on the component with frequency $\omega = \omega_{180}$, that is, the signal $X_i \sin(\omega_{180}t)$. This input signal will result in a feedback signal given by

$$X_f = A(j\omega_{180})\beta(j\omega_{180})X_i = -X_i$$

Since X_f is further multiplied by –1 in the summer block at the input, we see that the feedback causes the signal X_i at the amplifier input to be *sustained*. That is, from this point on, there will be sinusoidal signals at the amplifier input and output of frequency ω_{180}. Thus the amplifier is said to oscillate at the frequency ω_{180}.

The question now is: What happens if at ω_{180} the magnitude of the loop gain is greater than unity? We shall answer this question, not in general, but for the restricted yet very important class of circuits in which we are interested here. The answer, which is not obvious from Eq. (11.58), is that the circuit will oscillate, and the oscillations will grow in amplitude until some nonlinearity (which is always present in some form) reduces the magnitude of the loop gain to exactly unity, at which point sustained oscillations will be obtained. This mechanism for starting oscillations by using positive feedback with a loop gain greater than unity, and then using a nonlinearity to reduce the loop gain to unity at the desired amplitude, will be exploited in the design of sinusoidal oscillators in Chapter 18. Our objective here is just the opposite: Now that we know how oscillations could occur in a negative-feedback amplifier, we wish to find methods to prevent their occurrence.

11.7.2 The Nyquist Plot

The Nyquist plot is a formalized approach for testing for stability based on the discussion above. It is simply a polar plot of loop gain, with frequency used as a parameter. Figure 11.28

对于物理频率 s = jω，公式（11.57）变为

$$A(j\omega) = \frac{A(j\omega)}{1 + A(j\omega)\beta(j\omega)} \tag{11.58}$$

因此，环路增益 A(jω)β(jω) 是一个复数，可用其幅值和相位表示、

$$\begin{aligned} L(j\omega) &\equiv A(j\omega)\beta(j\omega) \\ &= |A(j\omega)\beta(j\omega)|e \end{aligned} \tag{11.59}$$

环路增益随频率变化的方式决定了反馈放大器的稳定性或不稳定性。为了理解这一事实，请考虑相位角 φ(ω) 变为 180° 的频率。在此频率 ω 下，环路增益 A(jω)β(jω) 将是一个带负号的实数。因此，在此频率下，反馈将变为正值。如果在 ω = ω 时，环路增益的大小小于 1，那么从公式（11.58）可以看出，闭环增益 A(jω) 将大于开环增益 A(jω)，因为公式（11.58）的分母将小于 1。尽管如此，反馈放大器仍将保持稳定。

另一方面，如果在频率 ω 下，环路增益的大小等于 1，则根据公式（11.58）可知，A(jω) 将为无限大。这意味着输入为零时，放大器将有输出；顾名思义，这就是振荡器。为直观了解这一反馈回路如何产生振荡，请考虑图 11.1 中的一般回路，将外部输入 x 设为零。电路中的任何干扰，如电源开关的闭合，都会在放大器的输入端产生一个信号 x(t)。这种噪声信号通常包含多种频率，我们现在将集中讨论频率为 ω = ω 的分量，即信号 Xsin(ωt)。该输入信号将产生一个反馈信号，其值为

$$X = A(j\omega)\beta(j\omega)X = -X$$

由于 X 在输入端的夏季块中进一步乘以-1，我们可以看到，反馈使放大器输入端的信号 X 保持不变。也就是说，从这时起，放大器输入和输出端将出现频率为 ω 的正弦信号。因此，可以说放大器在频率 ω 下振荡。

现在的问题是如果在 ω 处的环路增益大于 1，会发生什么情况？我们将回答这个问题，但不是泛泛而谈，而是针对我们在此感兴趣的这一类有限但非常重要的电路。答案从式（11.58）中并不明显，那就是电路会发生振荡，而且振荡幅度会越来越大，直到某个非线性因素（总是以某种形式存在）将环路增益的幅度减小到正好为 1，此时才会出现持续振荡。在第 18 章的正弦振荡器设计中，我们将利用这种机制，通过使用环路增益大于 1 的正反馈来启动振荡，然后利用非线性将环路增益降低到所需振幅的 1。我们的目的恰恰相反：既然我们已经知道负反馈放大器如何产生振荡，我们就希望找到防止振荡发生的方法。

11.7.2 奈奎斯特图

根据上述讨论，奈奎斯特图是一种正式的稳定性测试方法。它只是以频率为参数的环路增益极坐标图。图 11.28

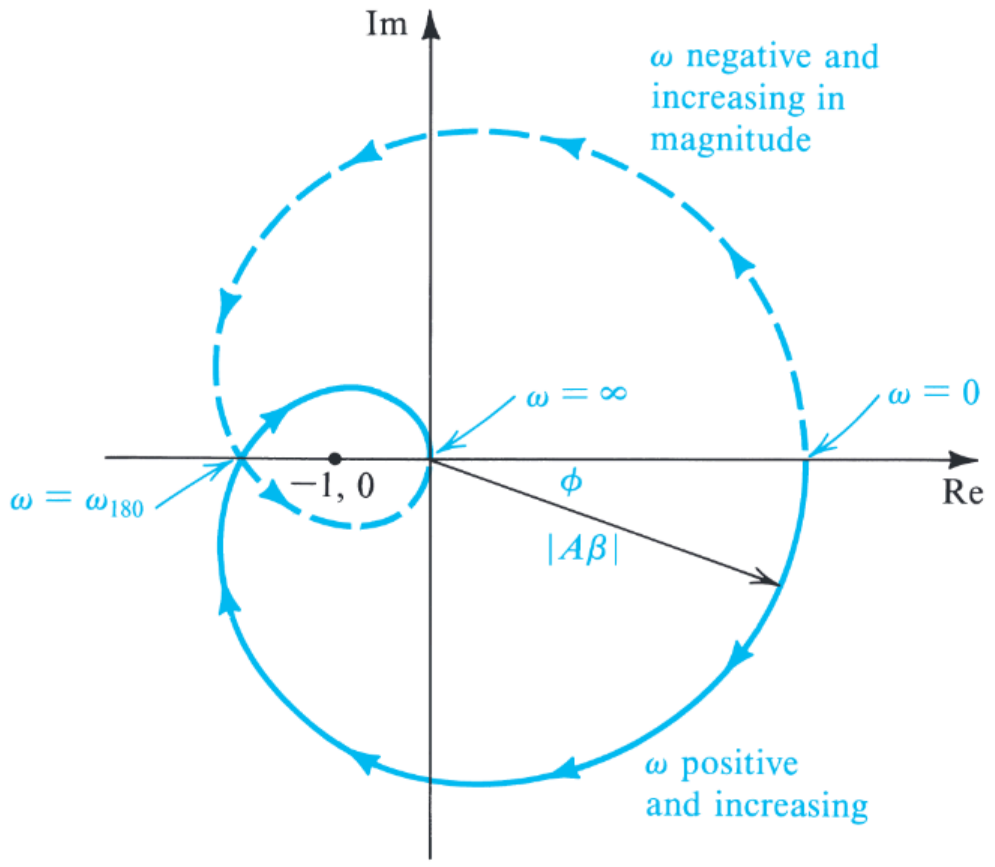


Figure 11.28 The Nyquist plot of an unstable amplifier.

shows such a plot. Note that the radial distance is $|A\beta|$ and the angle is the phase angle ϕ. The solid-line plot is for positive frequencies. Since the loop gain—and for that matter any gain function of a physical network—has a magnitude that is an even function of frequency and a phase that is an odd function of frequency, the $A\beta$ plot for negative frequencies (shown in Fig. 11.28 as a broken line) can be drawn as a mirror image through the Re axis.

The Nyquist plot intersects the negative real axis at the frequency ω_{180}. Thus, if this intersection occurs to the left of the point (–1, 0), we know that the magnitude of loop gain at this frequency is greater than unity and the amplifier will be unstable. On the other hand, if the intersection occurs to the right of the point (–1, 0) the amplifier will be stable. It follows that if the Nyquist plot *encircles* the point (–1, 0) then the amplifier will be unstable. It should be mentioned, however, that this statement is a simplified version of the **Nyquist criterion**; nevertheless, it applies to all the circuits in which we are interested. For the full theory behind the Nyquist method and for details of its application, consult Haykin (1970).

EXERCISE

11.22 Consider a feedback amplifier for which the open-loop transfer function $A(s)$ is given by

$$A(s) = \left(\frac{10}{1 + s/10^4}\right)^3$$

Let the feedback factor β be a constant independent of frequency. Find the frequency ω_{180} at which the phase shift is 180°. Then, show that the feedback amplifier will be stable if the feedback

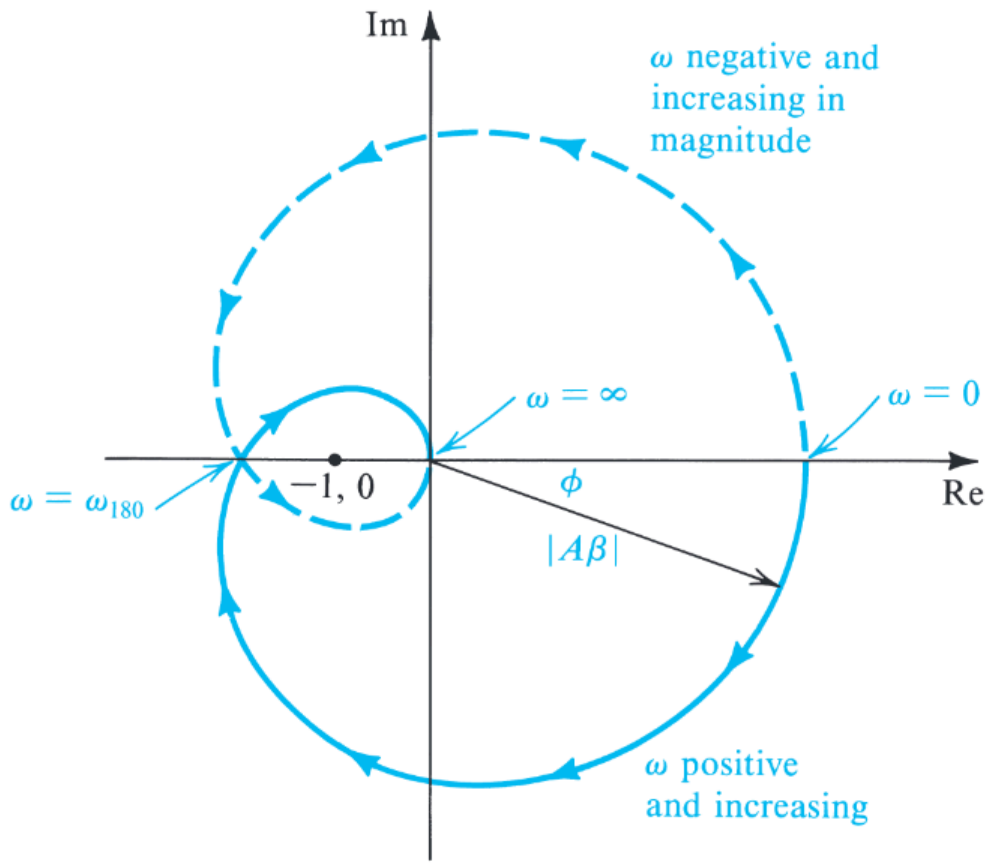


图 11.28 不稳定放大器的奈奎斯特图。

显示了这样一幅图。请注意，径向距离为 |Aβ|，角度为相位角 ϕ。实线图表示的是正频率。由于环路增益以及物理网络的任何增益函数的幅度都是频率的偶函数，而相位则是频率的奇函数，因此负频率的 Aβ 图（图 11.28 中以折线表示）可以通过 Re 轴绘制成镜像。奈奎斯特图与负实数轴相交于频率 ω。因此，如果该交点位于点（-1，0）的左侧，我们就知道该频率下的环路增益大于 1，放大器将不稳定。另一方面，如果交点出现在点（-1，0）的右侧，放大器将保持稳定。由此可见，如果奈奎斯特图环绕点（-1，0），那么放大器将是不稳定的。不过，应该指出的是，这种说法只是奈奎斯特准则的简化版本；不过，它适用于我们感兴趣的所有电路。有关奈奎斯特方法的完整理论及其应用细节，请查阅 Haykin (1970)。

运动

11.22 考虑一个反馈放大器，其开环传递函数 A（s）由以下公式给出

$$A(s) = \left(\frac{10}{1 + s/10} \right)$$

假设反馈系数 β 是一个与频率无关的常数。求相移为 180° 的频率 ω。然后，证明如果反馈

factor β is less than a critical value β_{cr} and unstable if $\beta \geq \beta_{cr}$, and find the value of β_{cr}. Hence, find the minimum value of the closed-loop gain for which the amplifier is stable.

Ans. $\omega_{180} = \sqrt{3} \times 10^4$ rad/s; $\beta_{cr} = 0.008$; $A_{f\,min} = 111.1$

HARRY NYQUIST—A DIVERSE ELECTRONICS FUNDAMENTALIST:

Harry Nyquist, a Swedish-born electrical engineer working for Bell Labs and its predecessor, was responsible for developments in communications electronics involving thermal noise, feedback-amplifier stability, telegraphy, facsimile, television, and many other areas.

In *The Idea Factory*, an excellent book on the history of Bell Labs, Jon Gertner notes that

> [S]ome lawyers in the patent office of the Bell Labs decided to study whether there was an organizing principle that could explain why certain individuals were more productive than others. They discerned only one common thread: Workers with the most patents often shared lunch or breakfast with a Bell Labs electrical engineer named Harry Nyquist. It wasn't the case that Nyquist gave them specific ideas. Rather, as one scientist recalled, "he drew people out, got them thinking." More than anything, Nyquist asked good questions.

11.8 Effect of Feedback on the Amplifier Poles

The amplifier frequency response and stability are determined directly by its poles. Therefore we shall investigate the effect of feedback on the poles of the amplifier.[7]

11.8.1 Stability and Pole Location

We shall begin by considering the relationship between stability and pole location. For an amplifier or any other system to be stable, its poles should lie in the left half of the s plane. A pair of complex-conjugate poles on the $j\omega$ axis gives rise to sustained sinusoidal oscillations. Poles in the right half of the s plane give rise to growing oscillations.

To verify the statement above, consider an amplifier with a pole pair at $s = \sigma_0 \pm j\omega_n$. If this amplifier is subjected to a disturbance, such as that caused by closure of the power-supply switch, its transient response will contain terms of the form

$$v(t) = e^{\sigma_0 t}\left[e^{+j\omega_n t} + e^{-j\omega_n t}\right] = 2e^{\sigma_0 t}\cos(\omega_n t) \tag{11.60}$$

This is a sinusoidal signal with an envelope $e^{\sigma_0 t}$. Now if the poles are in the left half of the s plane, then σ_0 will be negative and the oscillations will decay exponentially toward zero, as shown in Fig. 11.29(a), indicating that the system is stable. If, on the other hand, the poles are in

[7] For a brief review of poles and zeros and related concepts, refer to Appendix F.

因此，请找出放大器稳定的最小闭环增益值。

答案：ω= $\sqrt{3} \times 10$ 弧度/秒； $\beta = 0.008$； $A = 111.1$

哈里-尼奎斯特多样化电子原教旨主义者：

哈里-奈奎斯特（Harry Nyquist）是一位出生于瑞典的电气工程师，曾在贝尔实验室及其前身工作，负责通信电子技术的开发，涉及热噪声、反馈放大器稳定性、电报、传真、电视和许多其他领域。

在《创意工厂》这本介绍贝尔实验室历史的优秀著作中，乔恩-格特纳指出

> [贝尔实验室专利办公室的一些律师决定研究是否有一种组织原则可以解释为什么某些人比其他人更有生产力。他们只发现了一个共同点：获得专利最多的工人经常与贝尔实验室一位名叫哈里-奈奎斯特（Harry Nyquist）的电气工程师共进午餐或早餐。这并不是说奈奎斯特给了他们具体的想法。相反，正如一位科学家所回忆的，"他把人们吸引出来，让他们思考"。更重要的是，奈奎斯特提出了很多好问题。

11.8 反馈对放大器极点的影响

放大器的频率响应和稳定性直接取决于其极点。因此，我们将研究反馈对放大器极点的影响。

11.8.1 稳定性和电线杆位置

我们将首先考虑稳定性与极点位置之间的关系。要使放大器或任何其他系统保持稳定，其极点应位于 s 平面的左半部分。一对位于 $j\omega$ 轴上的复共轭极点会产生持续的正弦振荡。

位于 s 平面右半部的极点会产生越来越大的振荡。
为了验证上述说法，请考虑一个在 $s = \sigma \pm j\omega$ 处具有极对的放大器。如果该放大器受到干扰，例如电源开关闭合造成的干扰，其瞬态响应将包含以下形式的项

$$v(t) = e^{[\quad e+\ e\quad\quad\quad]} = 2e\cos(\omega t) \tag{11.60}$$

现在，如果极点位于 s 平面的左半部分，那么 σ 将为负值，振荡将以指数形式向零衰减，如图 11.29(a)所示，表明系统是稳定的。另一方面，如果极点位于

[7] 有关极点和零点及相关概念的简要回顾，请参阅附录 F。

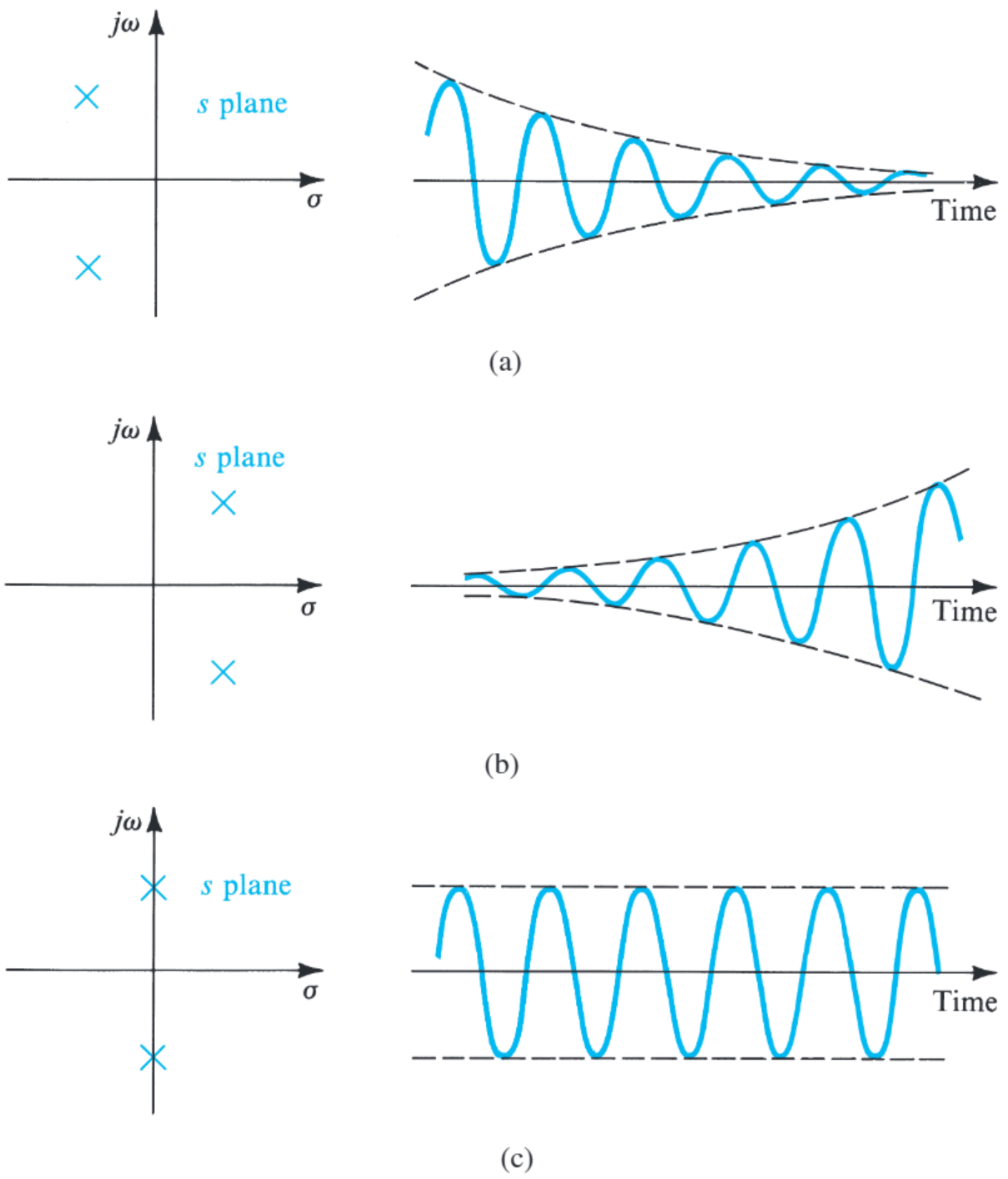


Figure 11.29 Relationship between pole location and transient response.

the right half-plane, then σ_0 will be positive, and the oscillations will grow exponentially (until some nonlinearity limits their growth), as shown in Fig. 11.29(b). Finally, if the poles are on the $j\omega$ axis, then σ_0 will be zero and the oscillations will be sustained, as shown in Fig. 11.29(c).

Although the discussion above is in terms of complex-conjugate poles, it can be shown that the existence of any right-half-plane poles results in instability.

11.8.2 Poles of the Feedback Amplifier

From the closed-loop transfer function in Eq. (11.57), we see that the poles of the feedback amplifier are the zeros of $1 + A(s)\beta(s)$. That is, the feedback amplifier poles are obtained by solving the equation

$$1 + A(s)\beta(s) = 0 \tag{11.61}$$

which is called the **characteristic equation** of the feedback loop. It should therefore be apparent that applying feedback to an amplifier changes its poles.

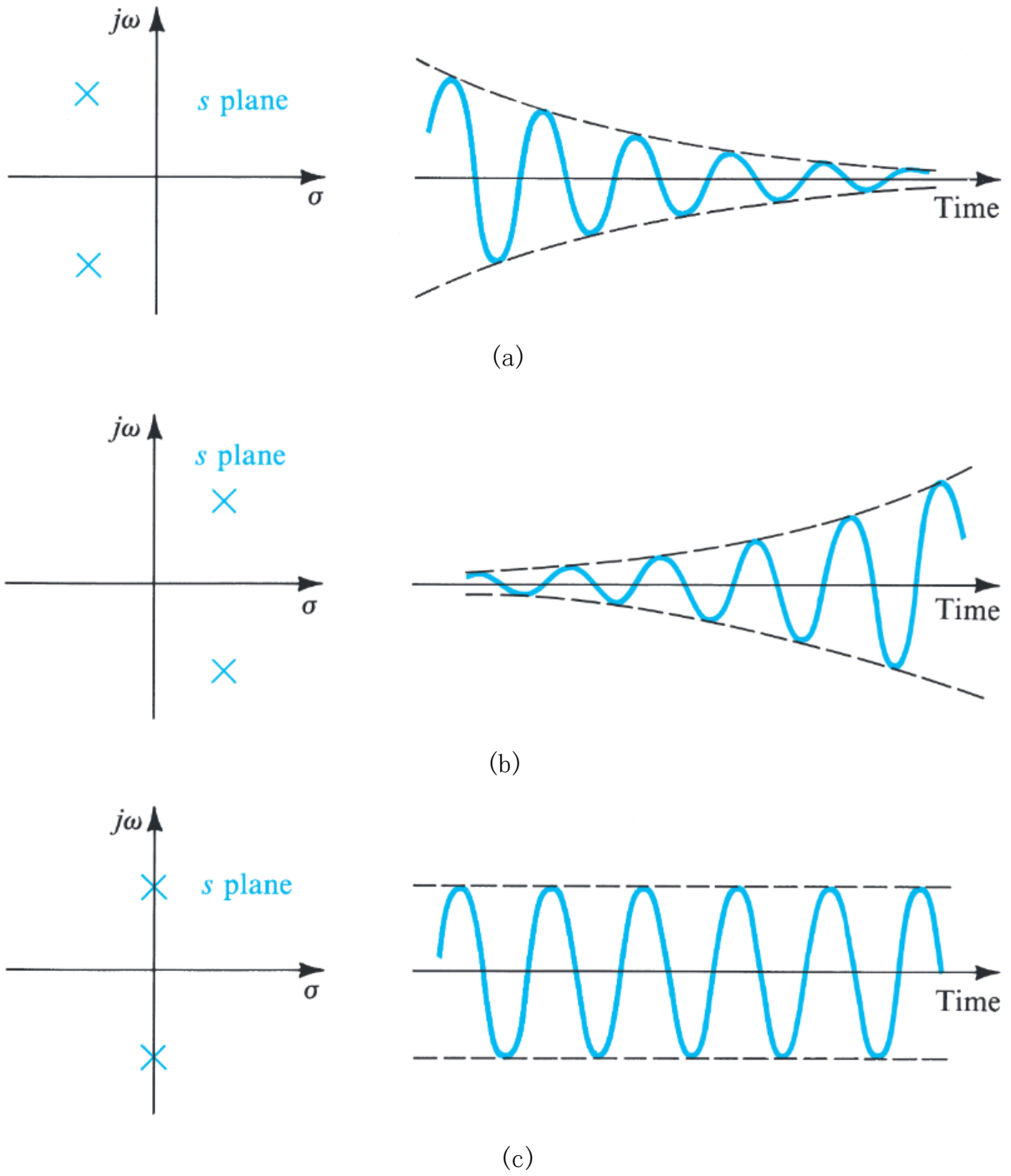


图 11.29 极点位置与瞬态响应之间的关系

如图 11.29(b)所示，如果极点在右半平面上，则 σ 为正值，振荡将呈指数增长（直到某些非线性限制了振荡的增长）。最后，如果极点位于 jω 轴上，则 σ 为零，振荡将持续，如图 11.29(c)所示。

尽管上述讨论是以复共轭极点为基础的，但可以证明任何右半平面极点的存在都会导致不稳定。

11.8.2 反馈放大器的极点

从公式 (11.57) 中的闭环传递函数可以看出，反馈放大器的极点是 1 + A(s) β (s) 的零点。也就是说，反馈放大器的极点可以通过解方程得到

$$1 + A(s)\,\beta(s) = 0 \tag{11.61}$$

这就是反馈回路的特性方程。由此可见，对放大器施加反馈会改变其极点。

In the following, we shall consider how feedback affects the amplifier poles. For this purpose we shall assume that the open-loop amplifier has real poles and no finite zeros (i.e., all the zeros are at $s = \infty$). This will simplify the analysis and enable us to focus our attention on the fundamental concepts involved. We shall also assume that the feedback factor β is independent of frequency.

11.8.3 Amplifier with a Single-Pole Response

Consider first the case of an amplifier whose open-loop transfer function is characterized by a single pole:

$$A(s) = \frac{A_0}{1 + s/\omega_P} \tag{11.62}$$

The closed-loop transfer function is given by

$$A_f(s) = \frac{A_0/(1 + A_0\beta)}{1 + s/\omega_P(1 + A_0\beta)} \tag{11.63}$$

Thus the feedback moves the pole along the negative real axis to a frequency ω_{Pf},

$$\omega_{Pf} = \omega_P(1 + A_0\beta) \tag{11.64}$$

This process is illustrated in Fig. 11.30(a). Figure 11.30(b) shows Bode plots for $|A|$ and $|A_f|$. Note that while at low frequencies the difference between the two plots is $20 \log(1 + A_0\beta)$, the two curves coincide at high frequencies. One can show that this indeed is the case by approximating Eq. (11.63) for frequencies $\omega \gg \omega_P(1 + A_0\beta)$:

$$A_f(s) \simeq \frac{A_0\omega_P}{s} \simeq A(s) \tag{11.65}$$

Physically speaking, at such high frequencies the loop gain is much smaller than unity and the feedback is ineffective.

Figure 11.30(b) clearly illustrates the fact that applying negative feedback to an amplifier results in extending its bandwidth at the expense of a reduction in gain. Since the pole of the

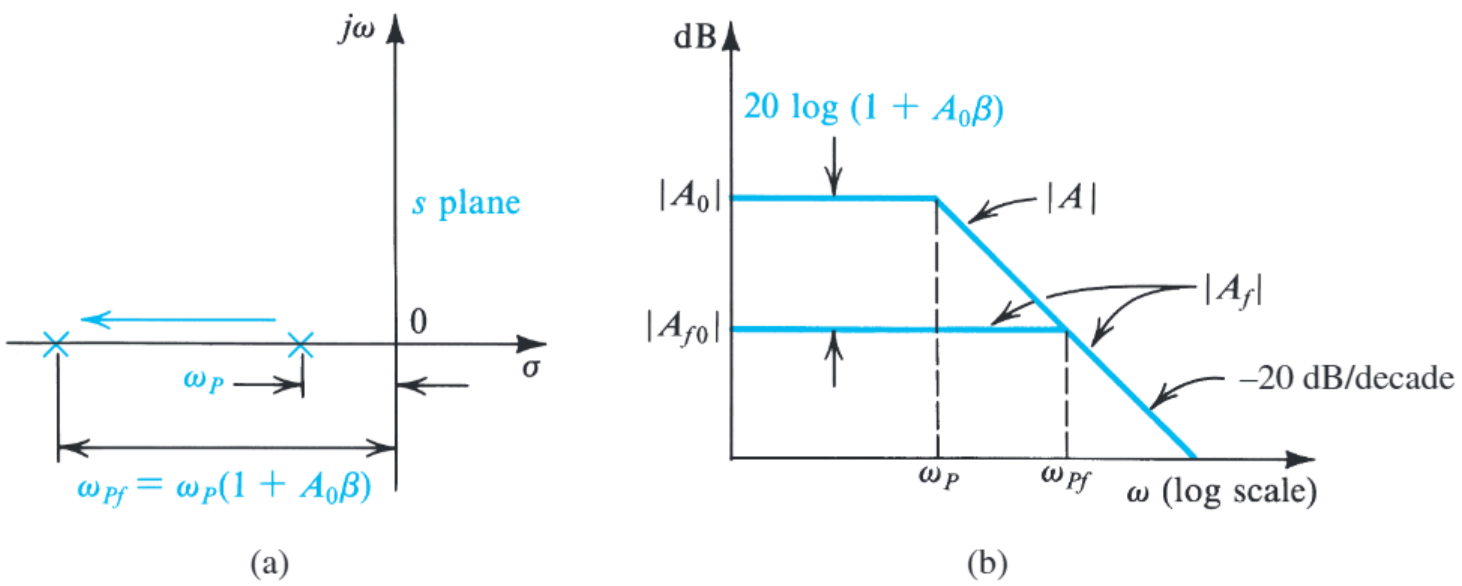


Figure 11.30 Effect of feedback on (**a**) the pole location and (**b**) the frequency response of an amplifier having a single-pole, open-loop response.

下面，我们将讨论反馈对放大器极点的影响。为此，我们将假定开环放大器具有实极点，没有有限零点（即所有零点都位于 s = ∞）。这将简化分析，使我们能够将注意力集中在相关的基本概念上。我们还将假设反馈系数 β 与频率无关。

11.8.3 单极响应放大器

首先考虑放大器的情况，其开环传递函数以单极为特征：

$$A(s) = \frac{A}{1 + s/\omega} \tag{11.62}$$

闭环传递函数如下

$$A(s) = \frac{A/(1 + A\beta)}{1 + s/\omega(1 + A\beta)} \tag{11.63}$$

因此，反馈将极点沿负实轴移动到频率 ω、

$$\omega=\omega(1 + A\beta) \tag{11.64}$$

图 11.30(a) 展示了这一过程。图 11.30(b) 显示了 |A| 和 |A| 的博德图。 |A|.
请注意，在低频时，两条曲线的差值为 20 log(1+Aβ)，而在高频时，两条曲线重合。通过对频率 ω ω(1 + Aβ)的公式 (11.63) 进行近似，可以证明情况确实如此：

$$A(s) \quad \frac{A\omega}{s} \quad A(s) \tag{11.65}$$

从物理上讲，在如此高的频率下，环路增益远小于 1，反馈也就无效了。
图 11.30(b)清楚地说明了一个事实，即对放大器应用负反馈会以降低增益为代价延长其带宽。由于

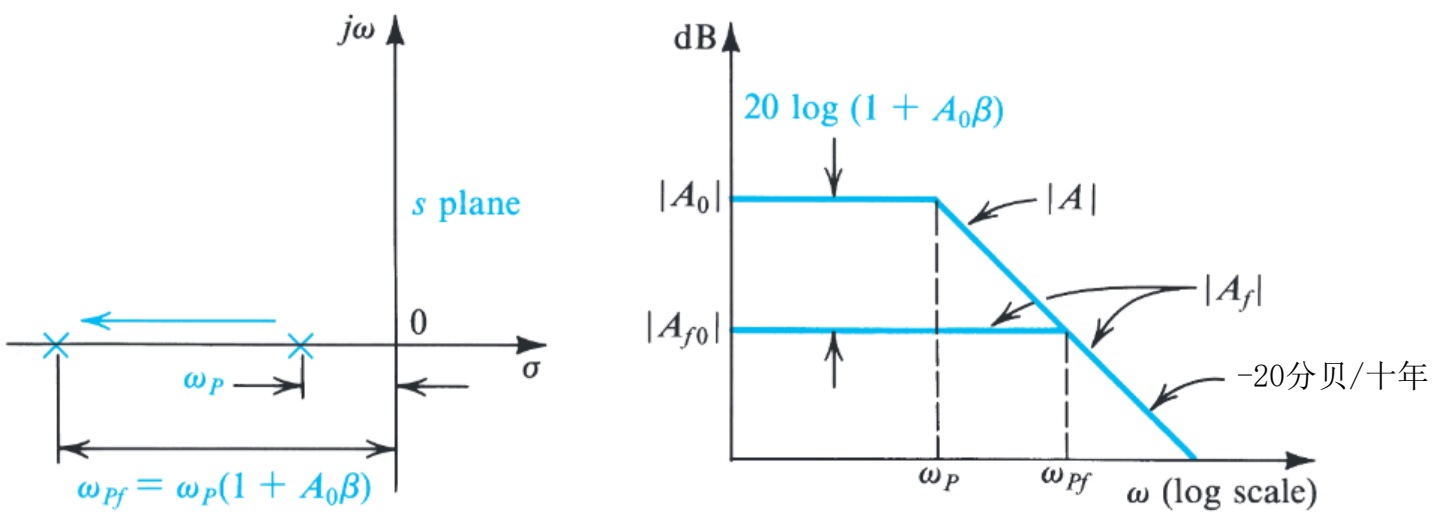


图 11.30 反馈对具有单极开环响应的放大器 (a) 极点位置和 (b) 频率响应的影响。

closed-loop amplifier never enters the right half of the s plane, the single-pole amplifier is stable for any value of β. Thus this amplifier is said to be **unconditionally stable**. This result, however, is hardly surprising, since the phase lag associated with a single-pole response can never be greater than 90°. Thus the loop gain never achieves the 180° phase shift required for the feedback to become positive.

EXERCISE

11.23 An op amp having a single-pole rolloff at 100 Hz and a low-frequency gain of 10^5 is operated in a feedback loop with $\beta = 0.01$. What is the factor by which feedback shifts the pole? To what frequency? If β is changed to a value that results in a nominal closed-loop gain of +1, to what frequency does the pole shift?

Ans. 1001; 100.1 kHz; 10 MHz

11.8.4 Amplifier with a Two-Pole Response

Consider next an amplifier whose open-loop transfer function is characterized by two real-axis poles:

$$A(s) = \frac{A_0}{(1 + s/\omega_{P1})(1 + s/\omega_{P2})} \tag{11.66}$$

In this case, the closed-loop poles are obtained from $1 + A(s)\beta = 0$, which leads to

$$s^2 + s(\omega_{P1} + \omega_{P2}) + (1 + A_0\beta)\omega_{P1}\omega_{P2} = 0 \tag{11.67}$$

Thus the closed-loop poles are given by

$$s = -\frac{1}{2}(\omega_{P1} + \omega_{P2}) \pm \frac{1}{2}\sqrt{(\omega_{P1} + \omega_{P2})^2 - 4(1 + A_0\beta)\omega_{P1}\omega_{P2}} \tag{11.68}$$

From Eq. (11.68) we see that as the loop gain $A_0\beta$ is increased from zero, the poles are brought closer together. Then a value of loop gain is reached at which the poles become coincident. If the loop gain is further increased, the poles become complex conjugate and move along a vertical line. Figure 11.31 shows the locus of the poles for increasing loop gain. This plot is called a **root-locus diagram**, where "root" refers to the fact that the poles are the roots of the characteristic equation.

From the root-locus diagram of Fig. 11.31 we see that this feedback amplifier also is unconditionally stable. Again, this result should come as no surprise; the maximum phase shift of $A(s)$ in this case is 180° (90° per pole), but this value is reached at $\omega = \infty$. Thus there is no finite frequency at which the phase shift reaches 180°.

Another observation to make on the root-locus diagram of Fig. 11.31 is that the open-loop amplifier might have a dominant pole, but this is not necessarily the case for the closed-loop amplifier. The response of the closed-loop amplifier can, of course, always be plotted once the poles have been found from Eq. (11.68). As is the case with second-order responses generally, the closed-loop response can show a peak (see Chapter 17). To be more specific,

因此，这种放大器可以说是无条件稳定的。然而，这一结果并不令人惊讶，因为与单极响应相关的相位滞后永远不会大于 90° 。因此，环路增益永远无法达到反馈变为正值所需的 180° 相移。

运动

11.23 一个运算放大器的单极滚降频率为 100 Hz，低频增益为 10，它在一个 β = 0.01 的反馈回路中工作。反馈使极点移动的系数是多少？移动到什么频率？如果将 β 改为一个导致额定闭环增益为 +1 的值，极点会移动到什么频率？答案：1001；100.1 kHz；10 MHz

11.8.4 双极响应放大器

再看一个放大器，其开环传递函数的特点是有两个实轴极点：

$$A(s) = \frac{A}{(1 + s/\omega)(1 + s/\omega)} \tag{11.66}$$

在这种情况下，闭环极点由 1 + A(s)β = 0 得出，从而得出

$$s + s(\omega + \omega) + (1 + A\beta)\omega\omega = 0 \tag{11.67}$$

因此，闭环极点由以下公式给出

$$s = -\frac{1}{2}(\omega+\omega) \pm \frac{1}{2}\sqrt{(\omega+\omega) - 4(1 + A\beta)\omega\omega} \tag{11.68}$$

从公式 (11.68) 中我们可以看出，随着环路增益 Aβ 从零开始增大，极点会逐渐靠近。然后达到一个环路增益值，此时极点变得重合。如果进一步增大环路增益，两极就会变成复共轭，并沿垂直线移动。图 11.31 显示了环路增益增加时的极点位置。该图称为根-焦点图，其中 "根 "指的是极点是特性方程的根。

从图 11.31 的根焦点图中我们可以看出，这个反馈放大器也是无条件稳定的。在这种情况下，A(s) 的最大相移为 180° （每极 90° ），但这个值在 ω = ∞ 时达到。因此，不存在相移达到 180° 的有限频率。

从图 11.31 的根焦点图中还可以看出，开环放大器可能有一个主要极点，但闭环放大器不一定如此。当然，从公式 (11.68) 中找到极点后，就可以绘制出闭环放大器的响应图。与一般的二阶响应一样，闭环响应也会出现峰值（见第 17 章）。具体来说

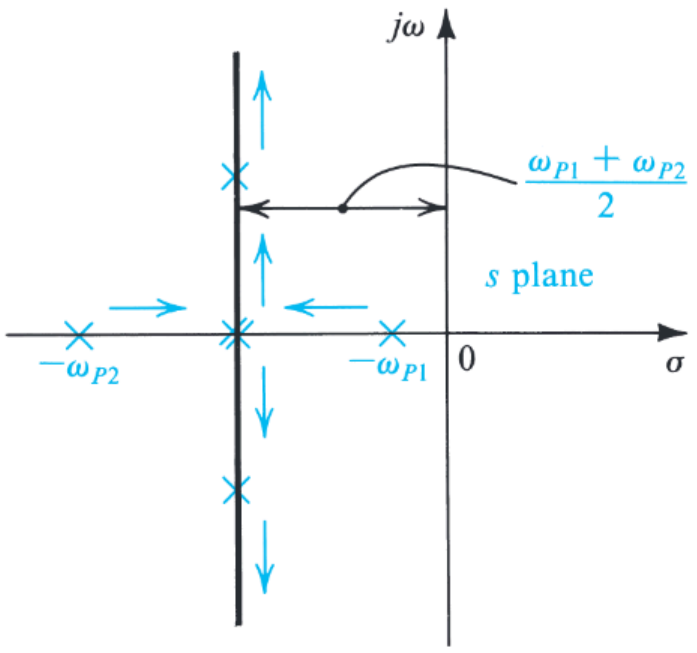


Figure 11.31 Root-locus diagram for a feedback amplifier whose open-loop transfer function has two real poles.

the characteristic equation of a second-order network can be written in the standard form

$$s^2 + s\frac{\omega_0}{Q} + \omega_0^2 = 0 \tag{11.69}$$

where ω_0 is called the **pole frequency** and Q is called **pole Q factor**. The poles are complex if Q is greater than 0.5. A geometric interpretation for ω_0 and Q of a pair of complex-conjugate poles is given in Fig. 11.32, from which we note that ω_0 is the radial distance of the poles from the origin and that Q indicates the distance of the poles from the $j\omega$ axis. Poles on the $j\omega$ axis have $Q = \infty$.

By comparing Eqs. (11.67) and (11.69), we obtain the Q factor for the poles of the feedback amplifier as

$$Q = \frac{\sqrt{(1+A_0\beta)\omega_{P1}\omega_{P2}}}{\omega_{P1}+\omega_{P2}} \tag{11.70}$$

From the study of second-order network responses in Chapter 17, it will be seen that the response of the feedback amplifier under consideration shows no peaking for $Q \leq 0.707$. The boundary case corresponding to $Q = 0.707$ (poles at 45° angles) results in the **maximally flat** response. Figure 11.33 shows a number of possible responses obtained for various values of Q (or, correspondingly, various values of $A_0\beta$).

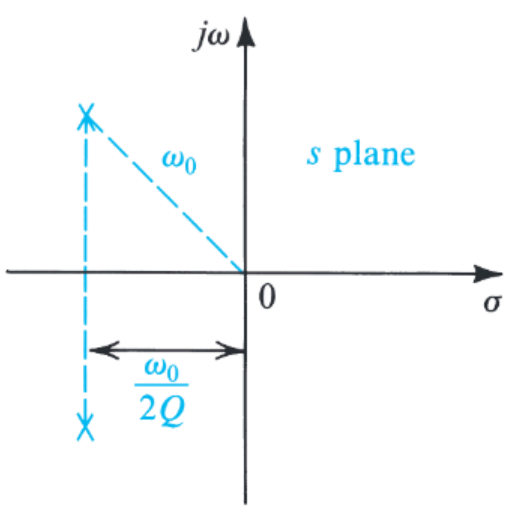


Figure 11.32 Definition of ω_0 and Q of a pair of complex-conjugate poles.

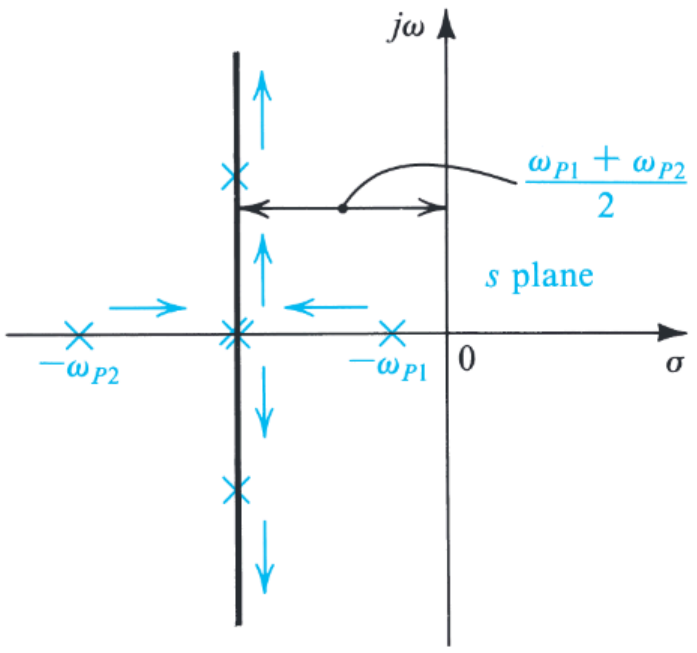


图 11.31 开环传递函数有两个实极点的反馈放大器的根焦点图。

二阶网络的特征方程可以用标准形式写出

$$s^2 + s\frac{\omega}{Q} + \omega = 0 \tag{11.69}$$

其中，ω 称为极点频率，Q 称为极点 Q 因子。如果 Q 大于 0.5，则极点为复极点。图 11.32 给出了一对复共轭极点的 ω 和 Q 的几何解释，从中我们可以看出，ω 是极点离原点的径向距离，Q 表示极点离 jω 轴的距离。在 jω 轴上的极点的 Q = ∞。

比较公式（11.67）和（11.69），我们可以得出反馈放大器极点的 Q 因子为

$$Q = \frac{\sqrt{(1 + A\beta)\omega\omega}}{\omega + \omega} \tag{11.70}$$

从第 17 章对二阶网络响应的研究中可以看出，当 Q ≤ 0.707 时，反馈放大器的响应不会出现峰值。与 Q = 0.707 相对应的边界情况（极点成 45° 角）会产生最大平坦响应。图 11.33 显示了不同 Q 值（或相应的不同 Aβ 值）下可能产生的响应。

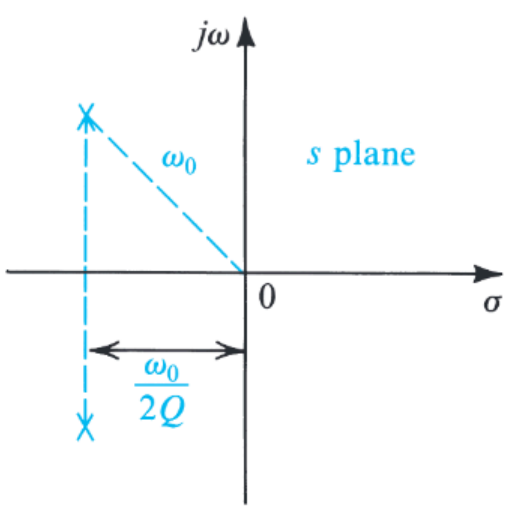


图 11.32 一对复共轭极点的 ω 和 Q 的定义。

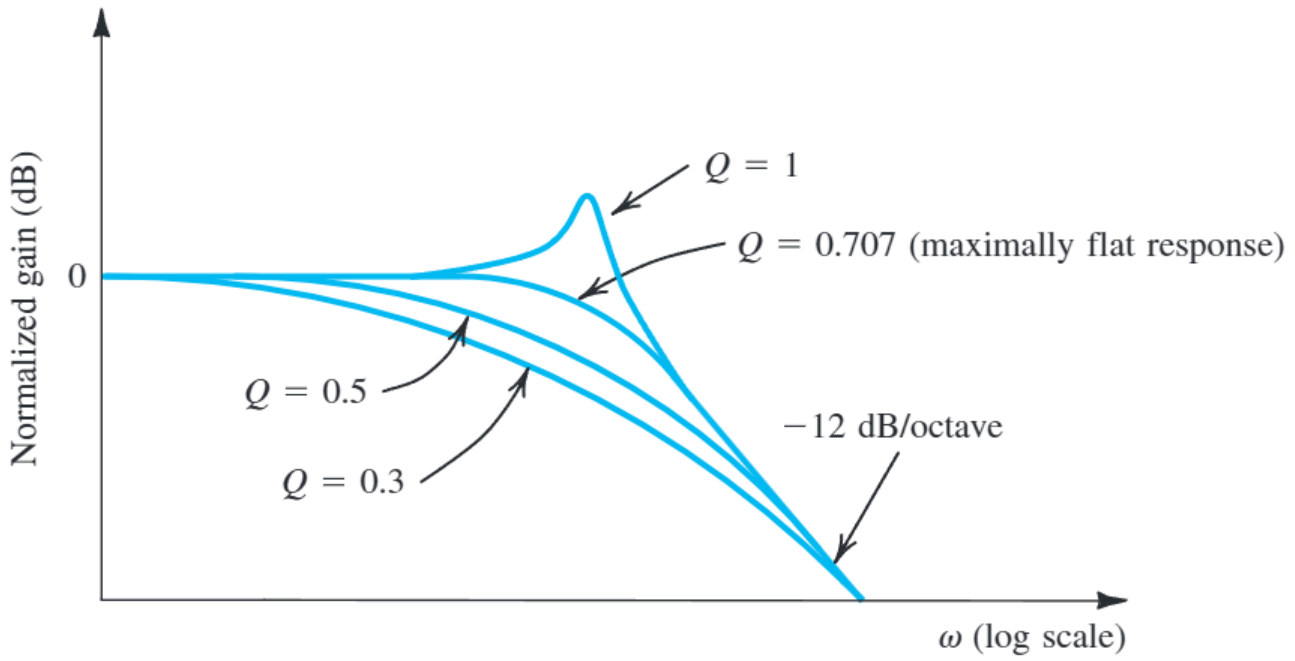


Figure 11.33 Normalized gain of a two-pole feedback amplifier for various values of Q. Note that Q is determined by the loop gain according to Eq. (11.70).

EXERCISE

11.24 An amplifier with a low-frequency gain of 100 and poles at 10^4 rad/s and 10^6 rad/s is incorporated in a negative-feedback loop with feedback factor β. For what value of β do the poles of the closed-loop amplifier coincide? What is the corresponding Q of the resulting second-order system? For what value of β is a maximally flat response achieved? What is the low-frequency closed-loop gain in the maximally flat case?

Ans. 0.245; 0.5; 0.5; 1.96 V/V

Example 11.11

As an illustration of some of the ideas just discussed, we consider the positive-feedback circuit shown in Fig. 11.34(a). Find the loop transmission $L(s)$ and the characteristic equation. Sketch a root-locus diagram for varying K, and find the value of K that results in a maximally flat response and the value of K that makes the circuit oscillate. Assume that the amplifier has frequency-independent gain, infinite input impedance, and zero output impedance.

Solution

To obtain the loop transmission, we short-circuit the signal source and break the loop at the amplifier input. We then apply a test voltage V_t and find the returned voltage V_r, as indicated in Fig. 11.34(b). The

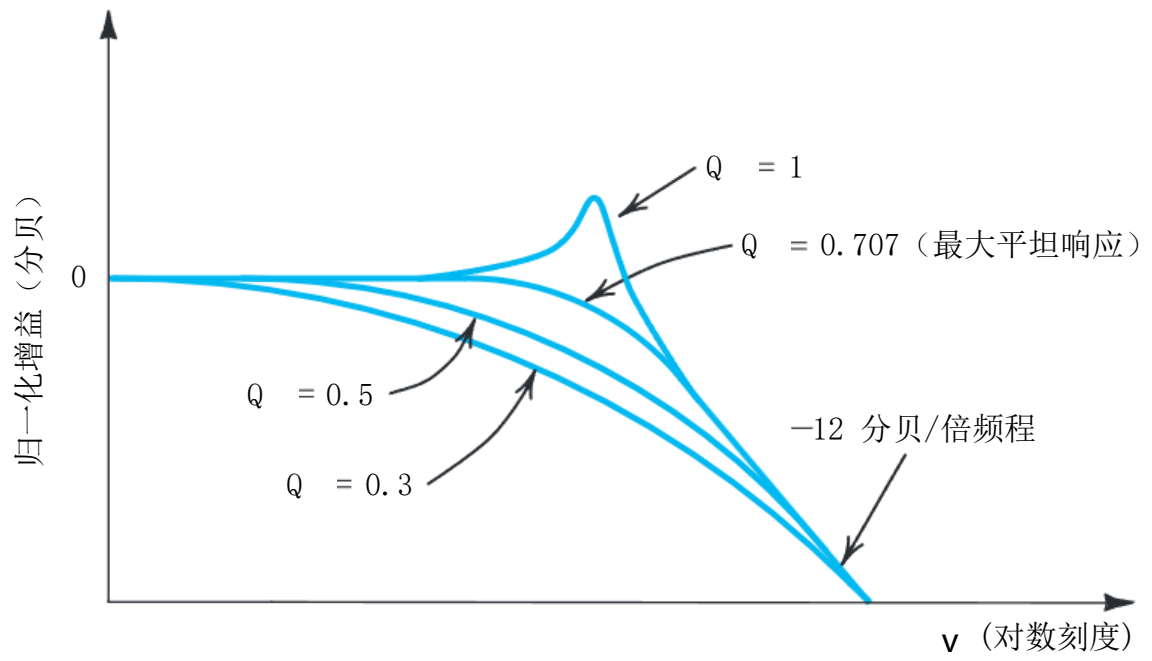


图 11.33 不同 Q 值下两极反馈放大器的归一化增益。请注意，Q 值由公式（11.70）中的环路增益决定。

运动

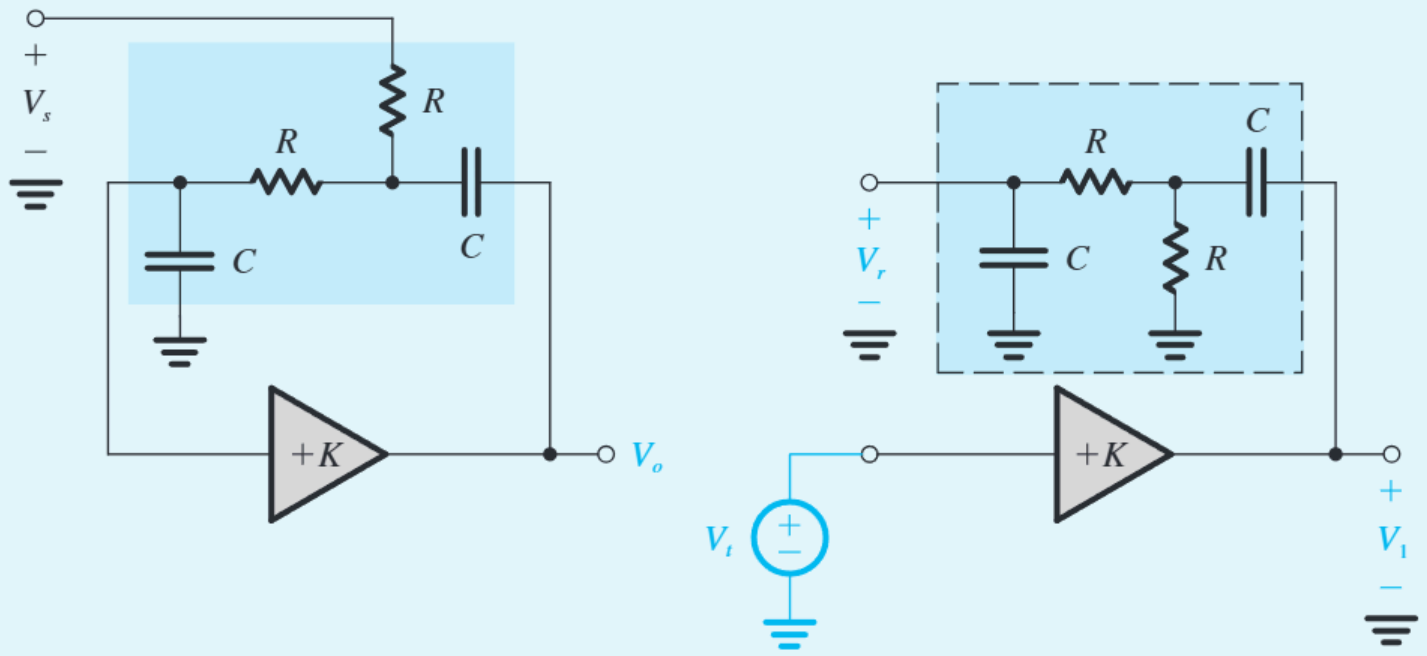
11.24 一个低频增益为 100、极点分别为 10 rad/s 和 10 rad/s 的放大器被纳入一个反馈系数为 β 的负反馈环路中。由此产生的二阶系统的相应 Q 值是多少？在 β 为何值时，可获得最大平坦响应？最大平坦情况下的低频闭环增益是多少？答案：0.245；0.5；0.5；1.96 V/V

例 11.11

为了说明刚才讨论的一些观点，我们来看看图 11.34(a)所示的正反馈电路。求环路传输 L(s) 和特性方程。绘制 K 变化时的根焦点图，并找出导致最大平坦响应的 K 值和使电路振荡的 K 值。假设放大器的增益与频率无关，输入阻抗无限大，输出阻抗为零。

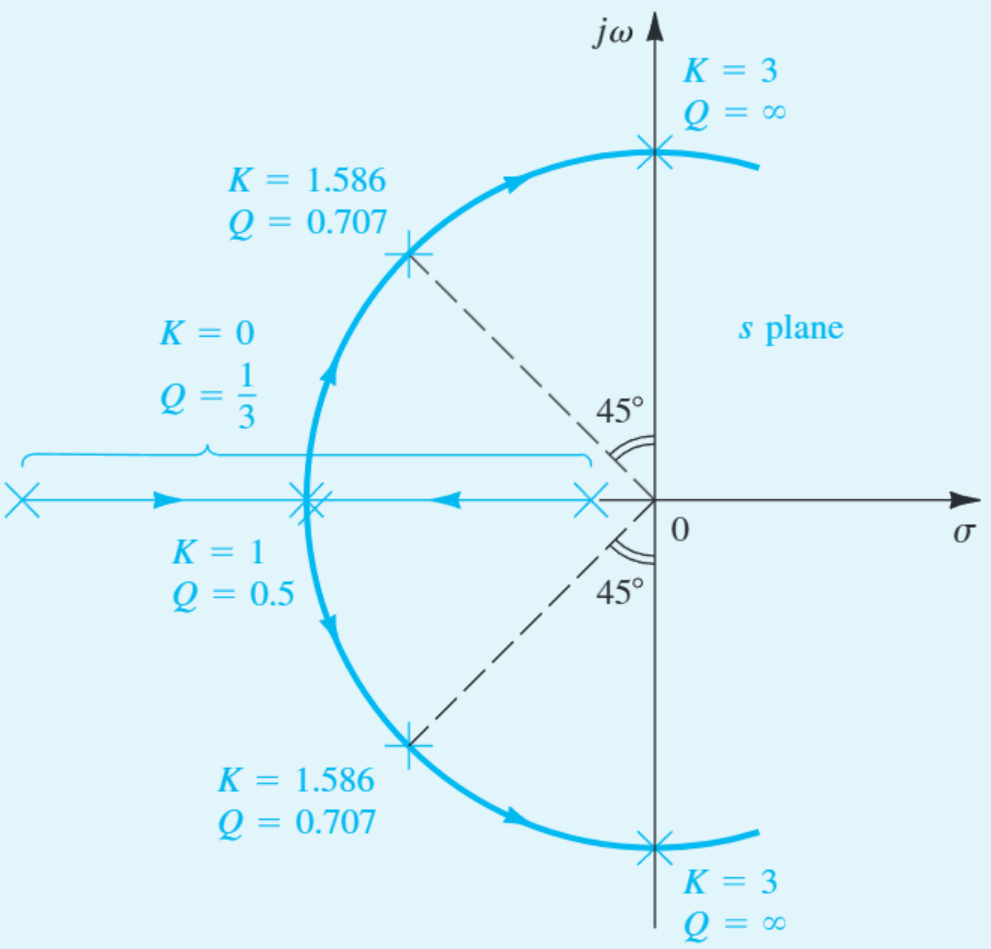
解决方案

要获得环路传输，我们需要短路信号源，并在放大器输入端断开环路。然后施加测试电压 V，并求得返回电压 V，如图 11.34(b)所示。图 11.34(b)



(a)

(b)

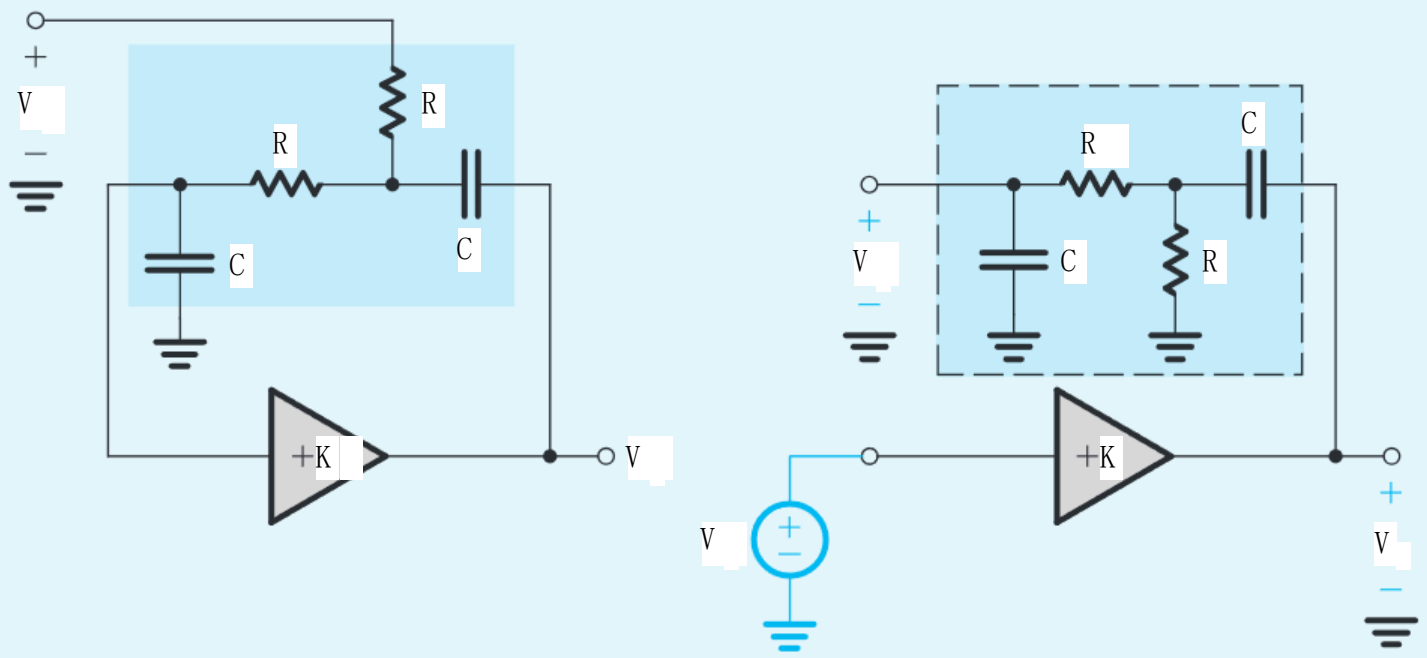


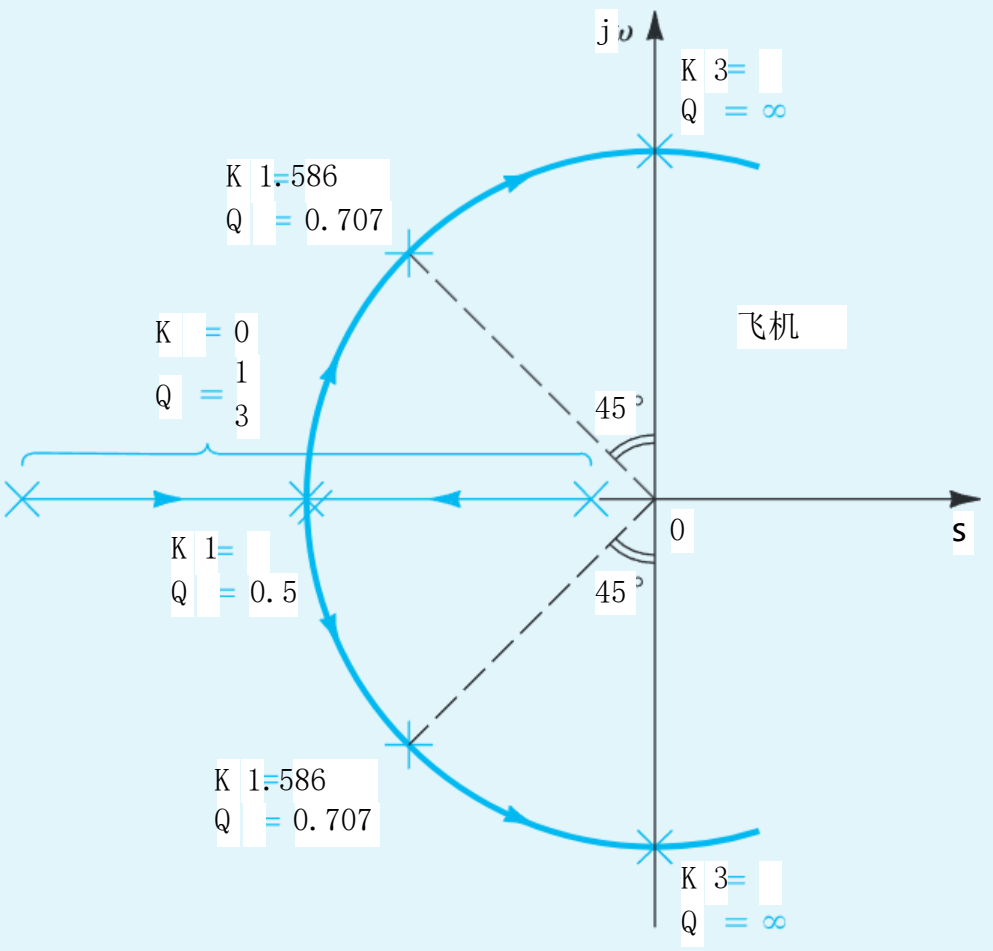
(c)

Figure 11.34 Circuits and plot for Example 11.11.

loop transmission $L(s) \equiv A(s)\beta(s)$ is given by

$$L(s) = -\frac{V_r}{V_t} = -KT(s) \tag{11.71}$$





(c)

图 11.34 例 11.11 的电路和绘图。

环路传输 $L(s) \equiv A(s)\beta(s)$ 的计算公式为

$$L(s) = -\frac{V}{V} = -KT(s) \tag{11.71}$$

Example 11.11 *continued*

where $T(s)$ is the transfer function of the two-port RC network shown inside the broken-line box in Fig. 11.34(b):

$$T(s) \equiv \frac{V_r}{V_1} = \frac{s(1/CR)}{s^2 + s(3/CR) + (1/CR)^2} \tag{11.72}$$

Thus,

$$L(s) = \frac{-s(K/CR)}{s^2 + s(3/CR) + (1/CR)^2} \tag{11.73}$$

The characteristic equation is

$$1 + L(s) = 0 \tag{11.74}$$

that is,

$$s^2 + s\frac{3}{CR} + \left(\frac{1}{CR}\right)^2 - s\frac{K}{CR} = 0$$

$$s^2 + s\frac{3-K}{CR} + \left(\frac{1}{CR}\right)^2 = 0 \tag{11.75}$$

By comparing this equation to the standard form of the second-order characteristic equation (Eq. 11.69), we see that the pole frequency ω_0 is given by

$$\omega_0 = \frac{1}{CR} \tag{11.76}$$

and the Q factor is

$$Q = \frac{1}{3-K} \tag{11.77}$$

Thus for $K = 0$, the poles have $Q = \frac{1}{3}$ and are therefore located on the negative real axis. As K is increased, the poles are brought closer together and eventually coincide ($Q = 0.5$, $K = 1$). Further increasing K results in the poles becoming complex and conjugate. The root locus is then a circle because the radial distance ω_0 remains constant (Eq. 11.76) independent of the value of K.

The maximally flat response is obtained when $Q = 0.707$, which results when $K = 1.586$. In this case the poles are at 45° angles, as indicated in Fig. 11.34(c). The poles cross the $j\omega$ axis into the right half of the s plane at the value of K that results in $Q = \infty$, that is, $K = 3$. Thus for $K \geq 3$ this circuit becomes unstable. This might appear to contradict our earlier conclusion that the feedback amplifier with a second-order response is unconditionally stable. Note, however, that the circuit in this example is quite different from the negative-feedback amplifier that we have been studying. Here we have an amplifier with a positive gain K and a feedback network whose transfer function $T(s)$ is frequency dependent. This feedback is in fact *positive,* and the circuit will oscillate at the frequency for which the phase of $T(j\omega)$ is zero (which is $1/CR$).

Example 11.11 illustrates the use of feedback (positive feedback in this case) to move the poles of an RC network from their negative real-axis locations to complex-conjugate locations. One can accomplish the same task using negative feedback, as the root-locus diagram of

例 11.11 续

71） 其中，T(s) 是图 11.34(b)断线框内所示双端口 RC 网络的传递函数：

$$T(s) \equiv \frac{V}{V} = \frac{s(1/CR)}{s + s(3/CR) + (1/CR)} \tag{11.72}$$

因此、

$$L(s) = \frac{-s\ (K/CR)}{s + s(3/CR) + (1/CR)} \tag{11.73}$$

特征方程为

$$1 + L(s) = 0 \tag{11.74}$$

也就是说

$$s + s\left(\frac{3}{CR} + \frac{1}{CR}\right) - s\frac{K}{CR} = 0$$

$$s + s\left(\frac{3 - K}{CR}\right) + \frac{1}{CR} = 0 \tag{11.75}$$

通过将该方程与二阶特征方程的标准形式（式 11.69）进行比较，我们可以看出极点频率 ω 的计算公式为

$$\omega = \frac{1}{CR} \tag{11.76}$$

Q 因子为

$$Q = \frac{1}{3 - K} \tag{11.77}$$

因此，当 K = 0 时，两极的 Q = 并位于负实数轴上。随着 K 的增大，两极逐渐靠近并最终重合（Q = 0.5，K = 1）。进一步增大 K 会导致极点变得复杂和共轭。这时根点是一个圆，因为径向距离 ω 保持不变（公式 11.76），与 K 值无关。

当 Q = 0.707 时可获得最大平坦响应，当 K = 1.586 时可获得最大平坦响应。在这种情况下，如图 11.34(c)所示，极点成 45° 角。极点穿过 jω 轴进入 s 平面右半部分的 K 值为 Q = ∞，即 K = 3。因此，当 K ≥ 3 时，该电路变得不稳定。这似乎与我们之前的结论相矛盾，即具有二阶响应的反馈放大器是无条件稳定的。但请注意，本例中的电路与我们一直在研究的负反馈放大器截然不同。这里我们有一个正增益 K 的放大器和一个传递函数 T(s) 与频率有关的反馈网络。这种反馈实际上是正反馈，电路将在 T(jω) 相位为零的频率（即 1/CR）上振荡。

例 11.11 展示了如何利用反馈（本例中为正反馈）将 RC 网络的极点从负实轴位置移动到复数共轭位置。使用负反馈也可以完成同样的任务，如

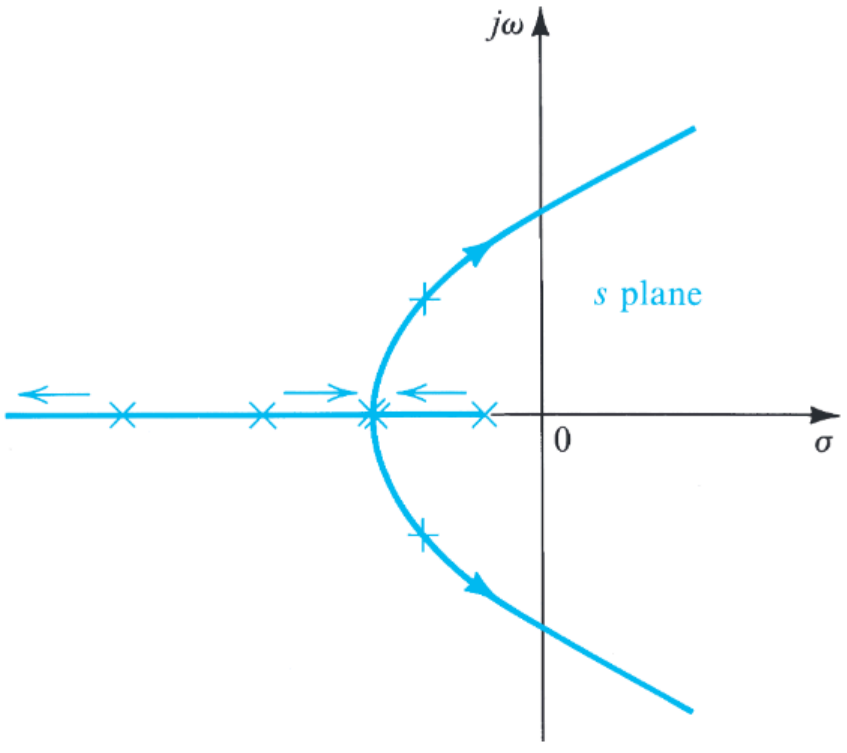


Figure 11.35 Root-locus diagram for an amplifier with three poles. The arrows indicate the pole movement as $A_0\beta$ is increased.

Fig. 11.31 demonstrates. The process of pole control is the essence of *active-filter design,* as will be discussed in Chapter 17.

11.8.5 Amplifiers with Three or More Poles

Figure 11.35 shows the root-locus diagram for a feedback amplifier whose open-loop response is characterized by three poles. As indicated, increasing the loop gain from zero moves the highest-frequency pole outward while the two other poles are brought closer together. As $A_0\beta$ is increased further, the two poles become coincident and then become complex and conjugate. A value of $A_0\beta$ exists at which this pair of complex-conjugate poles enters the right half of the s plane, thus causing the amplifier to become unstable.

This result is not entirely unexpected, since an amplifier with three poles has a phase shift that reaches –270° as ω approaches ∞. Thus there exists a finite frequency, ω_{180}, at which the loop gain has 180° phase shift.

From the root-locus diagram of Fig. 11.35, we observe that one can always maintain amplifier stability by keeping the loop gain $A_0\beta$ smaller than the value corresponding to the poles entering the right half-plane. In terms of the Nyquist diagram, the critical value of $A_0\beta$ is that for which the diagram passes through the (–1, 0) point. Reducing $A_0\beta$ below this value causes the Nyquist plot to shrink and thus intersect the negative real axis to the right of the (–1, 0) point, indicating stable amplifier performance. On the other hand, increasing $A_0\beta$ above the critical value causes the Nyquist plot to expand, thus encircling the (–1, 0) point and indicating unstable performance.

For a given open-loop gain A_0 the conclusions above can be stated in terms of the feedback factor β. That is, there exists a *maximum value* for β above which the feedback amplifier becomes unstable. Alternatively, we can state that there exists a *minimum value* for the closed-loop gain A_{f0} below which the amplifier becomes unstable. To obtain lower values of closed-loop gain, one needs therefore to alter the loop transfer function $L(s)$. This is the process known as *frequency compensation*. We shall study the theory and techniques of frequency compensation in Section 11.10.

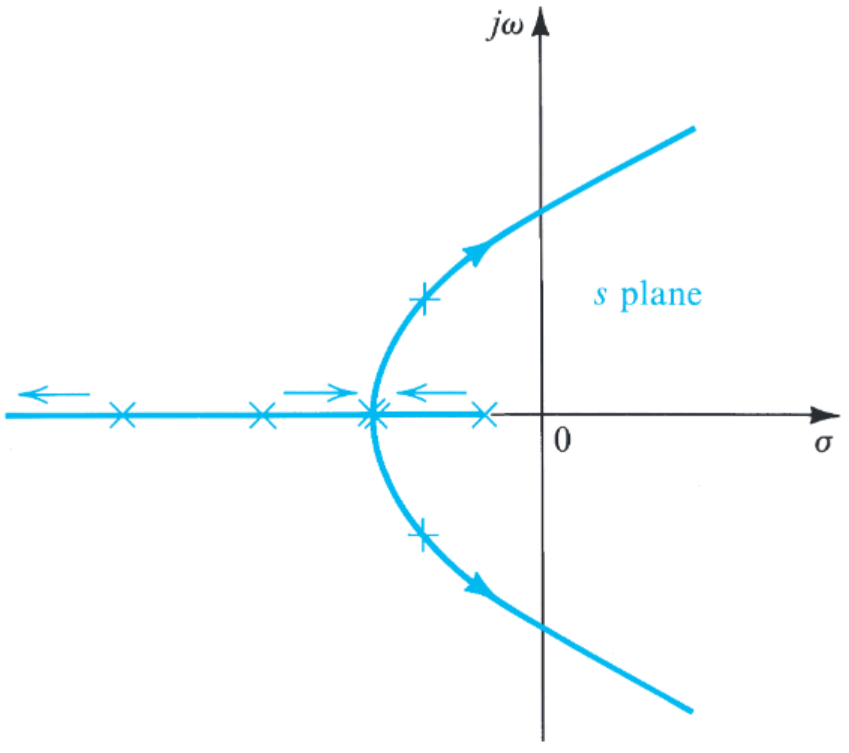


图 11.35 具有三个极点的放大器的根焦点图。箭头表示 Aβ 增加时的极点移动。

如图 11.31 的根焦点图所示。极点控制过程是有源滤波器设计的精髓所在，这将在第 17 章中讨论。

11.8.5 三极或多极放大器

图 11.35 显示了反馈放大器的根焦点图，该放大器的开环响应有三个极点。如图所示，环路增益从零开始增加时，最高频率极点向外移动，而另外两个极点则靠拢。随着 Aβ 进一步增大，两个极点变得重合，然后变成复共轭极点。当 Aβ 值达到一定程度时，这对复共轭极点会进入 s 平面的右半部分，从而导致放大器变得不稳定。

这一结果并不完全出乎意料，因为当 ω 接近 ∞ 时，具有三个极点的放大器的相移会达到 -270°。因此，存在一个有限频率 ω，在该频率下，环路增益的相移为 180°。

从图 11.35 的根焦点图中我们可以发现，只要保持环路增益 Aβ 小于进入右半平面的极点对应的值，就能始终保持放大器的稳定性。就奈奎斯特图而言，Aβ 的临界值就是图中经过（-1，0）点的值。将 Aβ 降低到该值以下会导致奈奎斯特图缩小，从而与负实数轴相交于（-1，0）点的右侧，表明放大器性能稳定。另一方面，将 Aβ 增大到临界值以上会导致奈奎斯特图扩大，从而环绕 (-1, 0) 点，表明放大器性能不稳定。

对于给定的开环增益，上述结论可以用反馈系数 β 来表示。也就是说，β 存在一个最大值，超过该值，反馈放大器将变得不稳定。或者，我们也可以说，存在一个闭环增益 Abel 的最小值，低于该值，放大器就会变得不稳定。因此，为了获得更低的闭环增益值，我们需要改变环路传递函数 L(s)。这就是所谓的频率补偿过程。我们将在第 11.10 节研究频率补偿的理论和技术。

Before leaving this section, we point out that construction of the root-locus diagram for amplifiers having three or more poles as well as finite zeros is an involved process for which a systematic procedure exists. However, such a procedure will not be presented here, and the interested reader should consult Haykin (1970). Although the root-locus diagram provides the amplifier designer with considerable insight, other, simpler techniques based on Bode plots can be effectively employed, as will be explained in Section 11.9.

EXERCISE

11.25 Consider a feedback amplifier for which the open-loop transfer function $A(s)$ is given by

$$A(s) = \left(\frac{10}{1 + s/10^4}\right)^3$$

Let the feedback factor β be frequency independent. Find the closed-loop poles as functions of β, and show that the root locus is that of Fig. E11.26. Also find the value of β at which the amplifier becomes unstable. (*Note:* This is the same amplifier that was considered in Exercise 11.22.)

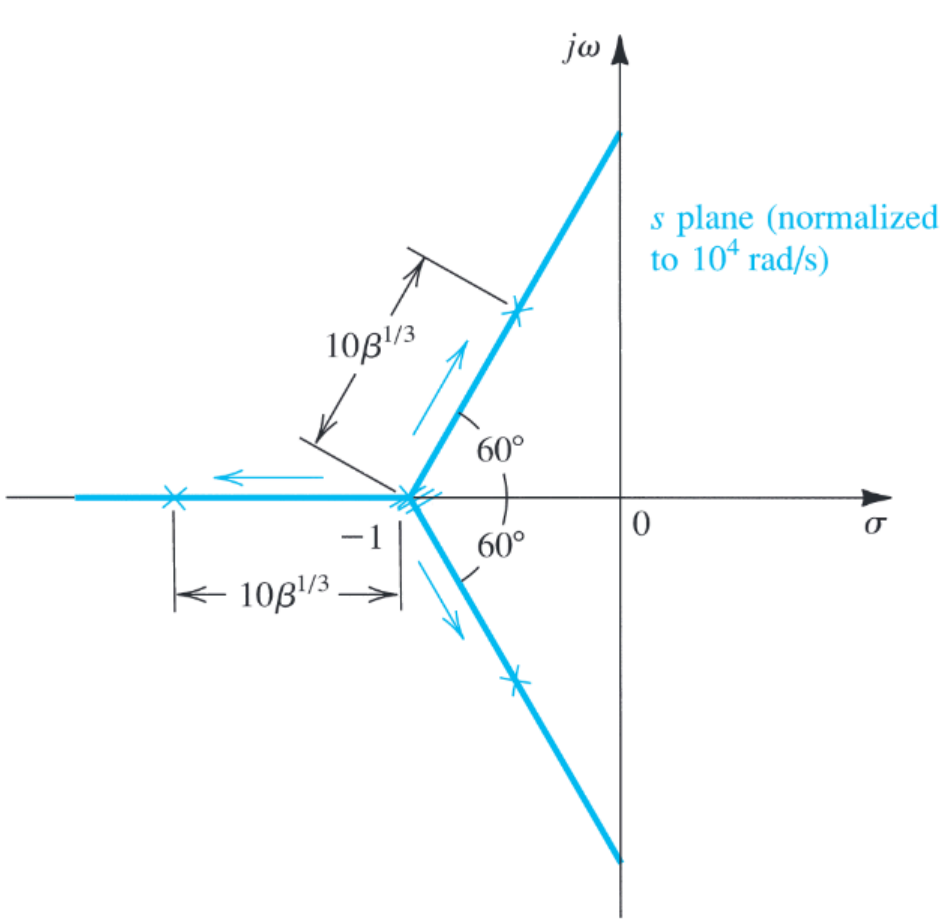


Figure E11.26

Ans. See Fig. E11.26; $\beta_{\text{critical}} = 0.008$

在结束本节之前，我们要指出的是，为具有三个或更多极点以及有限零点的放大器构建根焦点图是一个复杂的过程，其中存在一个系统的程序。不过，本文将不介绍这一过程，感兴趣的读者可参阅 Haykin (1970)。尽管根焦点图为放大器设计者提供了相当深入的见解，但正如第 11.9 节所解释的，基于 Bode 图的其他更简单的技术也可有效使用。

运动

11.25 考虑一个反馈放大器，其开环传递函数 A（s）由以下公式给出

$$A(s) = \left(\frac{10}{1 + s/10} \right)$$

假设反馈系数 β 与频率无关。找出闭环极点与 β 的函数关系，并证明其根点与图 E11.26 一致。同时求出放大器变得不稳定的 β 值。(注：这与练习 11.22 中的放大器相同）。

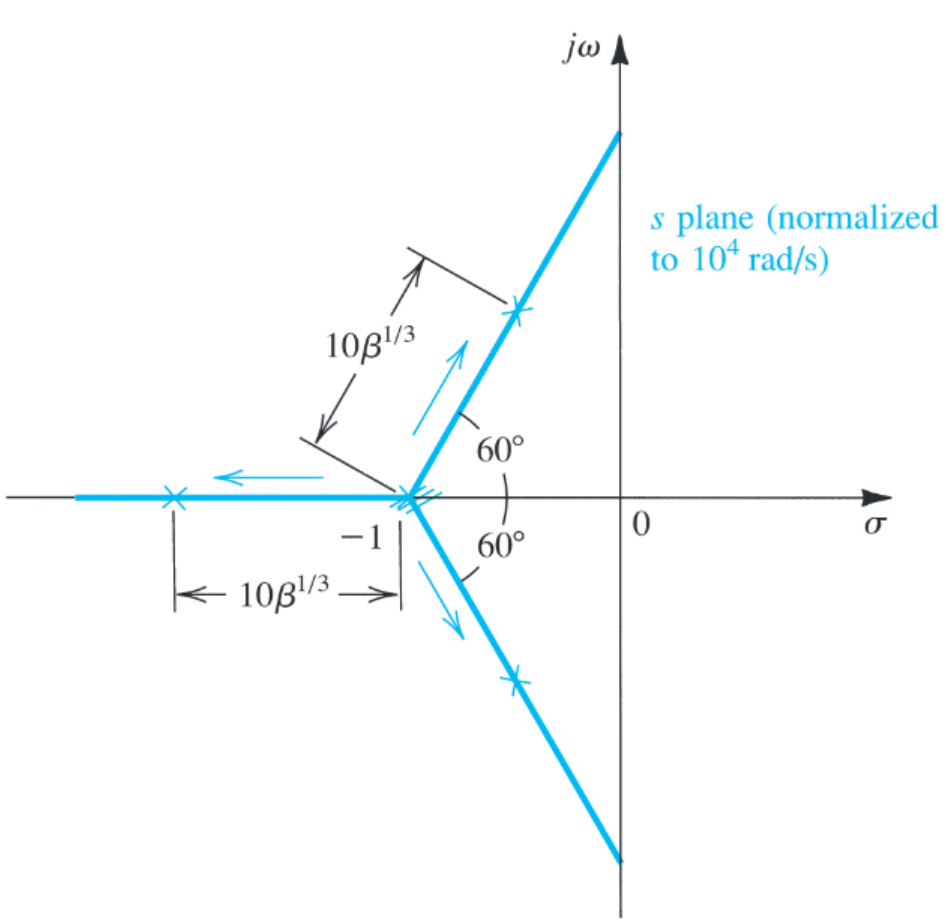


图 E11.26

答案：见图 E11.26; βcritical = 0.008

11.9 Stability Study Using Bode Plots

11.9.1 Gain and Phase Margins

From Sections 11.17 and 11.18 we know that whether a feedback amplifier is or is not stable can be determined by examining its loop gain $A\beta$ as a function of frequency. One of the simplest and most effective means for doing this is through the use of a Bode plot for $A\beta$, such as the one shown in Fig. 11.36. (Note that because the phase approaches –360°, the circuit examined is a fourth-order one.) The feedback amplifier whose loop gain is plotted in Fig. 11.36 will be stable, since at the frequency of 180° phase shift, ω_{180}, the magnitude of the loop gain is less than unity (negative dB). The difference between the value of $|A\beta|$ at ω_{180} and unity, called the **gain margin**, is usually expressed in decibels. The gain margin represents the amount by which the loop gain can be increased while stability is maintained. Feedback amplifiers are usually designed to have sufficient gain margin to allow for the inevitable changes in loop gain with temperature, time, and so on.

Another way to investigate the stability and to express its degree is to examine the Bode plot at the frequency for which $|A\beta| = 1$, which is the point at which the magnitude plot crosses the 0-dB line. If at this frequency the phase angle is less (in magnitude) than 180°, then the amplifier is stable. This is the situation illustrated in Fig. 11.36. The difference between the phase angle at this frequency and 180° is termed the **phase margin**. On the other hand, if at the frequency of unity loop-gain magnitude, the phase lag is in excess of 180°, the amplifier will be unstable.

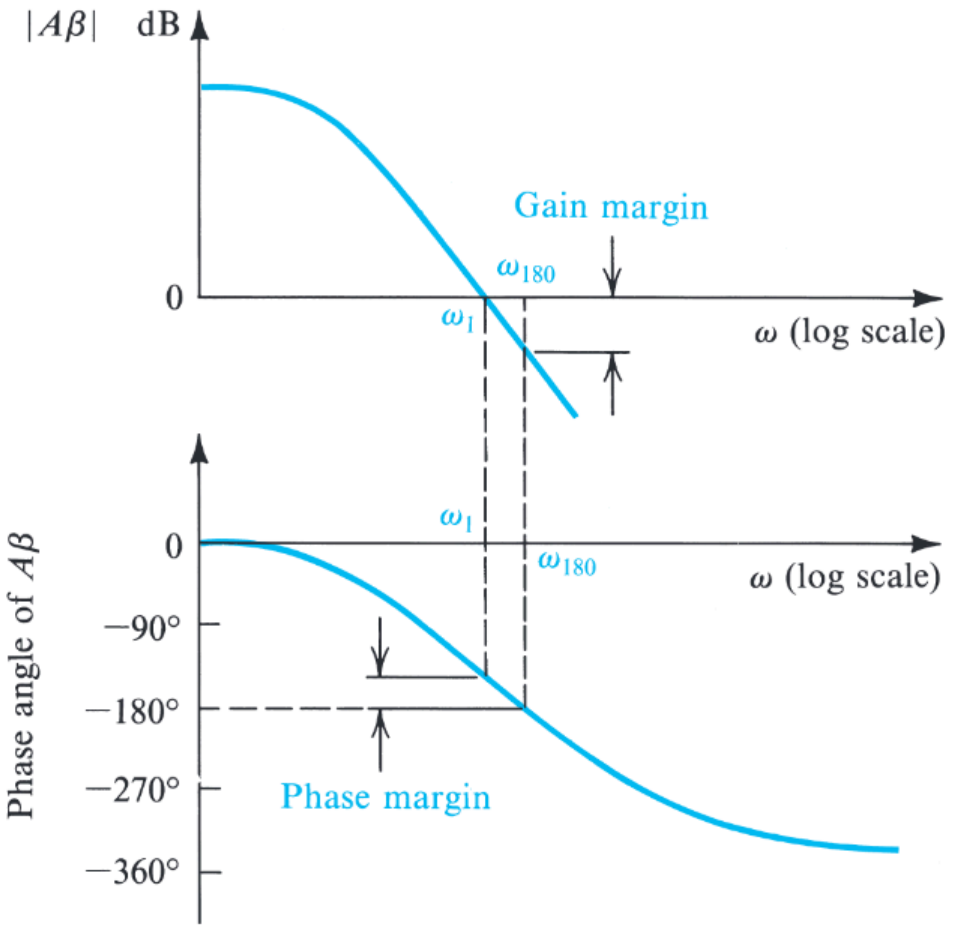


Figure 11.36 Bode plot for the loop gain $A\beta$ illustrating the definitions of the gain and phase margins.

11.9 利用博德图进行稳定性研究

11.9.1 增益和相位差

从第 11.17 节和第 11.18 节中我们知道，反馈放大器是否稳定，可以通过检查其环路增益 Aβ 与频率的函数关系来确定。最简单有效的方法之一是使用 Aβ 的 Bode 图，如图 11.36 所示。(注意，由于相位接近 -360°，因此所研究的电路是四阶电路）。环路增益如图 11.36 所示的反馈放大器将是稳定的，因为在相移 180° 的频率 ω 下，环路增益的幅度小于 1（负 dB）。ω处的|Aβ|值与合一值之间的差值称为增益余量，通常用分贝表示。增益裕量表示在保持稳定的情况下，环路增益可以增加的幅度。反馈放大器通常设计有足够的增益裕量，以适应环路增益随温度、时间等不可避免的变化。

另一种研究稳定性并表示其程度的方法是，在 |Aβ|=1 的频率下检查 Bode 图，即幅度图与 0-dB 线交叉的点。如果在该频率下相位角（幅度）小于 180°，则放大器是稳定的。这就是图 11.36 所示的情况。该频率下的相位角与 180° 之间的差值称为相位裕量。另一方面，如果在环路增益幅度为 1 的频率上，相位滞后超过 180°，则放大器将不稳定。

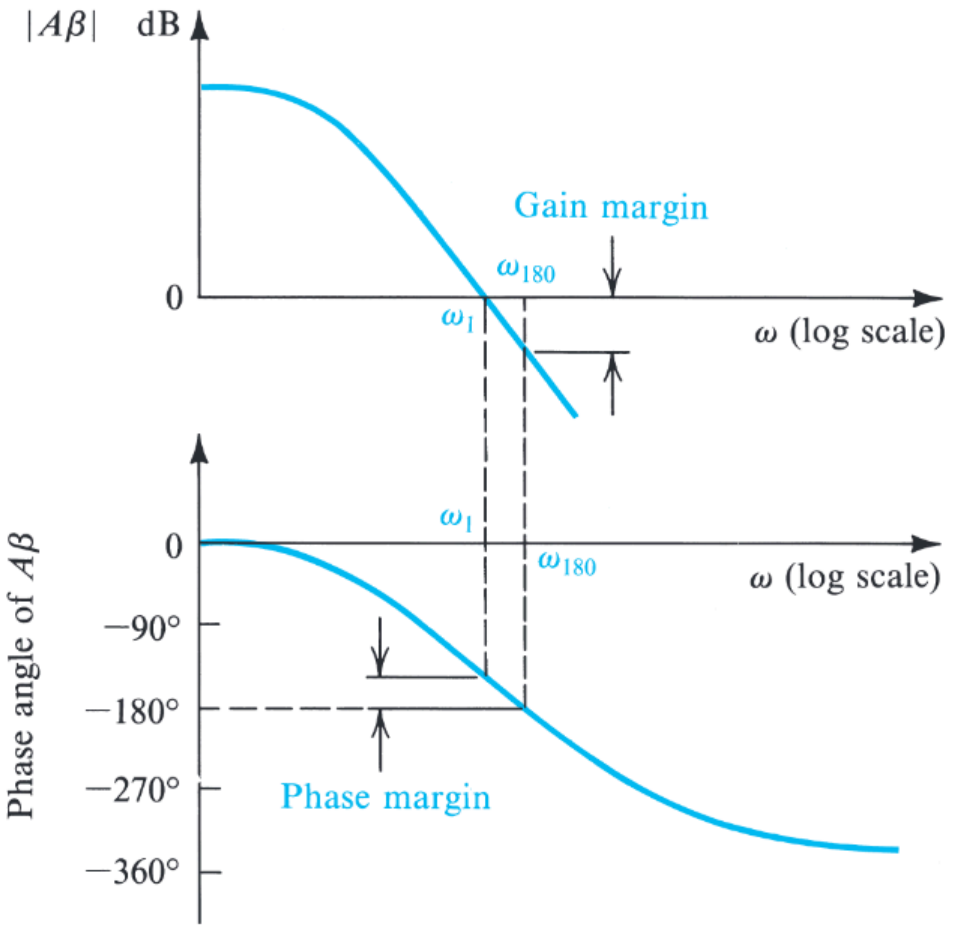


图 11.36 环路增益 Aβ 的 Bode 图，说明增益和相位余量的定义。

EXERCISE

11.26 Consider an op amp having a single-pole, open-loop response with $A_0 = 10^5$ and $f_P = 10$ Hz. Let the op amp be ideal otherwise (infinite input impedance, zero output impedance, etc.). If this amplifier is connected in the noninverting configuration with a nominal low-frequency, closed-loop gain of 100, find the frequency at which $|A\beta| = 1$. Also, find the phase margin.
Ans. 10^4 Hz; 90°

11.9.2 Effect of Phase Margin on Closed-Loop Response

Feedback amplifiers are normally designed with a phase margin of at least 45°. The amount of phase margin has a profound effect on the shape of the closed-loop gain response. To see this relationship, consider a feedback amplifier with a large low-frequency loop gain, $A_0\beta \gg 1$. It follows that the closed-loop gain at low frequencies is approximately $1/\beta$. Denoting the frequency at which the magnitude of loop gain is unity by ω_1, we have (refer to Fig. 11.36)

$$A(j\omega_1)\beta = 1 \times e^{-j\theta} \tag{11.78}$$

where

$$\theta = 180° - \text{phase margin} \tag{11.79}$$

At ω_1 the closed-loop gain is

$$A_f(j\omega_1) = \frac{A(j\omega_1)}{1 + A(j\omega_1)\beta} \tag{11.80}$$

Substituting from Eq. (11.78) gives

$$A_f(j\omega_1) = \frac{(1/\beta)e^{-j\theta}}{1 + e^{-j\theta}} \tag{11.81}$$

Thus the magnitude of the gain at ω_1 is

$$\left|A_f(j\omega_1)\right| = \frac{1/\beta}{|1 + e^{-j\theta}|} \tag{11.82}$$

For a phase margin of 45°, $\theta = 135°$; and we obtain

$$\left|A_f(j\omega_1)\right| = 1.3\frac{1}{\beta} \tag{11.83}$$

That is, the gain peaks by a factor of 1.3 above the low-frequency value of $1/\beta$. This peaking increases as the phase margin is reduced, eventually reaching ∞ when the phase margin is zero. Zero phase margin, of course, implies that the amplifier can sustain oscillations [poles on the $j\omega$ axis; Nyquist plot passing through (–1, 0)].

运动

11.26 考虑一个具有单极开环响应的运算放大器，A= 10，f= 10 Hz。让运算放大器保持理想状态（输入阻抗无限大、输出阻抗为零等）。如果该放大器以非反相配置连接，标称低频闭环增益为 100，请求得 |Aβ| = 1 时的频率。同时，求相位裕度。答案：10 赫兹；90

11.9.2 相位差对闭环响应的影响

反馈放大器的相位裕度通常至少为 45° 。相位裕量对闭环增益响应的形状有很大影响。要了解这种关系，请考虑一个具有较大低频环路增益（Aβ 1）的反馈放大器。由此可见，低频时的闭环增益约为 1/β 。用 ω 表示环路增益大小为 1 的频率，我们可以得出（参见图 11.36）

$$A(j\omega)\beta = 1 \times e \tag{11.78}$$

其中

$$\theta = 180° - \text{相位差} \tag{11.79}$$

在 ω 时，闭环增益为

$$A(j\omega) = \frac{A(j\omega)}{1 + A(j\omega)\beta} \tag{11.80}$$

代入公式（11.78）即可得出

$$A(j\omega) = \frac{(1/\beta)e}{1 + e} \tag{11.81}$$

因此，ω 处的增益大小为

$$\left|A(j\omega)\right| = \frac{1/\beta}{|1 + e|} \tag{11.82}$$

相位差为 45° 时， θ = 135° ；由此得出

$$\left|A(j\omega)\right| = 1.3\ \frac{1}{\beta} \tag{11.83}$$

也就是说，增益峰值比低频值 1/β 高出 1.3 倍。随着相位裕度的减小，峰值也随之增大，最终在相位裕度为零时达到 ∞。当然，零相位差意味着放大器可以维持振荡[jω 轴上的极点；通过（-1，0）的奈奎斯特图]。

EXERCISE

11.27 Find the closed-loop gain at ω_1 relative to the low-frequency gain when the phase margin is 30°, 60°, and 90°.
Ans. 1.93; 1; 0.707

11.9.3 An Alternative Approach for Investigating Stability

Investigating stability by constructing Bode plots for the loop gain $A\beta$ can be a tedious and time-consuming process, especially if we have to investigate the stability of a given amplifier for a variety of feedback networks. An alternative approach, which is much simpler, is to construct a Bode plot for the open-loop gain $A(j\omega)$ only. Assuming for the time being that β is independent of frequency, we can plot $20\log(1/\beta)$ as a horizontal straight line on the same plane used for $20\log|A|$. The difference between the two curves will be

$$20\log|A(j\omega)| - 20\log\frac{1}{\beta} = 20\log|A\beta| \tag{11.84}$$

which is the loop gain (in dB). We may therefore study stability by examining the difference between the two plots. If we wish to evaluate stability for a different feedback factor, we simply draw another horizontal straight line at the level $20\log(1/\beta)$.

To illustrate, consider an amplifier whose open-loop transfer function is characterized by three poles. For simplicity let the three poles be widely separated—say, at 0.1 MHz, 1 MHz, and 10 MHz, as shown in Fig. 11.37. Note that because the poles are widely separated, the phase is approximately –45° at the first pole frequency, –135° at the second, and –225° at the third. The frequency at which the phase of $A(j\omega)$ is –180° lies on the –40-dB/decade segment, as indicated in Fig. 11.37.

The open-loop gain of this amplifier can be expressed as

$$A = \frac{10^5}{(1+jf/10^5)(1+jf/10^6)(1+jf/10^7)} \tag{11.85}$$

from which $|A|$ can be easily determined for any frequency f (in Hz), and the phase can be obtained as

$$\phi = -\left[\tan^{-1}(f/10^5) + \tan^{-1}(f/10^6) + \tan^{-1}(f/10^7)\right] \tag{11.86}$$

The magnitude and phase graphs shown in Fig. 11.37 are obtained using the method for constructing Bode plots (Appendix F). These graphs provide approximate values for important amplifier parameters, with more exact values obtainable from Eqs. (11.85) and (11.86). For example, the frequency f_{180} at which the phase angle is 180° can be found from Fig. 11.37 to be approximately 3.2×10^6 Hz. Using this value as a starting point, a more exact value can be found by trial and error using Eq. (11.86). The result is $f_{180} = 3.34 \times 10^6$ Hz. At this frequency, Eq. (11.85) gives a gain magnitude of 58.2 dB, which is reasonably close to the approximate value of 60 dB given by Fig. 11.37.

运动

11.27 当相位差为 30°、60° 和 90°时，求相对于低频增益，ω 处的闭环增益。答案：1.93；1；0.707

11.9.3 研究稳定性的另一种方法

通过绘制环路增益 Aβ 的 Bode 图来研究稳定性是一个繁琐耗时的过程，尤其是当我们需要研究特定放大器在各种反馈网络下的稳定性时。另一种简单得多的方法是只绘制开环增益 A(jω) 的 Bode 图。暂时假定 β 与频率无关，我们可以在 20 log|A| 的同一平面上绘制 20 log(1/β) 的水平直线。两条曲线的差值为

$$20\log|A(j\omega)| - 20\log\frac{1}{\beta} = 20\log|A\beta| \tag{11.84}$$

即环路增益（dB）。因此，我们可以通过观察两幅图之间的差异来研究稳定性。如果我们想评估不同反馈系数的稳定性，只需在 20 log(1/β) 处画出另一条水平直线即可。

举例说明：考虑一个放大器，其开环传递函数有三个极点。为简单起见，让这三个极点相距较远，例如 0.1 MHz、1 MHz 和 10 MHz，如图 11.37 所示。请注意，由于极点相距较远，在第一个极点频率时相位约为 -45°，在第二个极点频率时相位约为 -135°，在第三个极点频率时相位约为 -225°。如图 11.37 所示，A(jω) 相位为-180° 的频率位于-40 分贝/分年代段。

该放大器的开环增益可表示为

$$A = \frac{10\ (1 + jf/10)(1 + }{jf/10)(1 + jf/10\ )} \tag{11.85}$$

由此可轻松确定任何频率 f（单位 Hz）下的 |A|，相位的计算公式为

$$\varphi = -[\tan(f/10) + \text{棕褐色}(f/10) + \text{棕褐色}(f/10)] \tag{11.86}$$

图 11.37 所示的幅值和相位图是使用构建博德图的方法（附录 F）获得的。这些图提供了重要放大器参数的近似值，更精确的值可通过公式 (11.85) 和 (11.86) 获得。例如，从图 11.37 中可以发现相位角为 180° 的频率约为 3.2 × 10 Hz。以此值为起点，可通过公式 (11.86) 尝试找出更精确的值。结果是 f= 3.34 × 10 Hz。在此频率下，公式 (11.85) 得出的增益幅度为 58.2 dB，与图 11.37 得出的近似值 60 dB 非常接近。

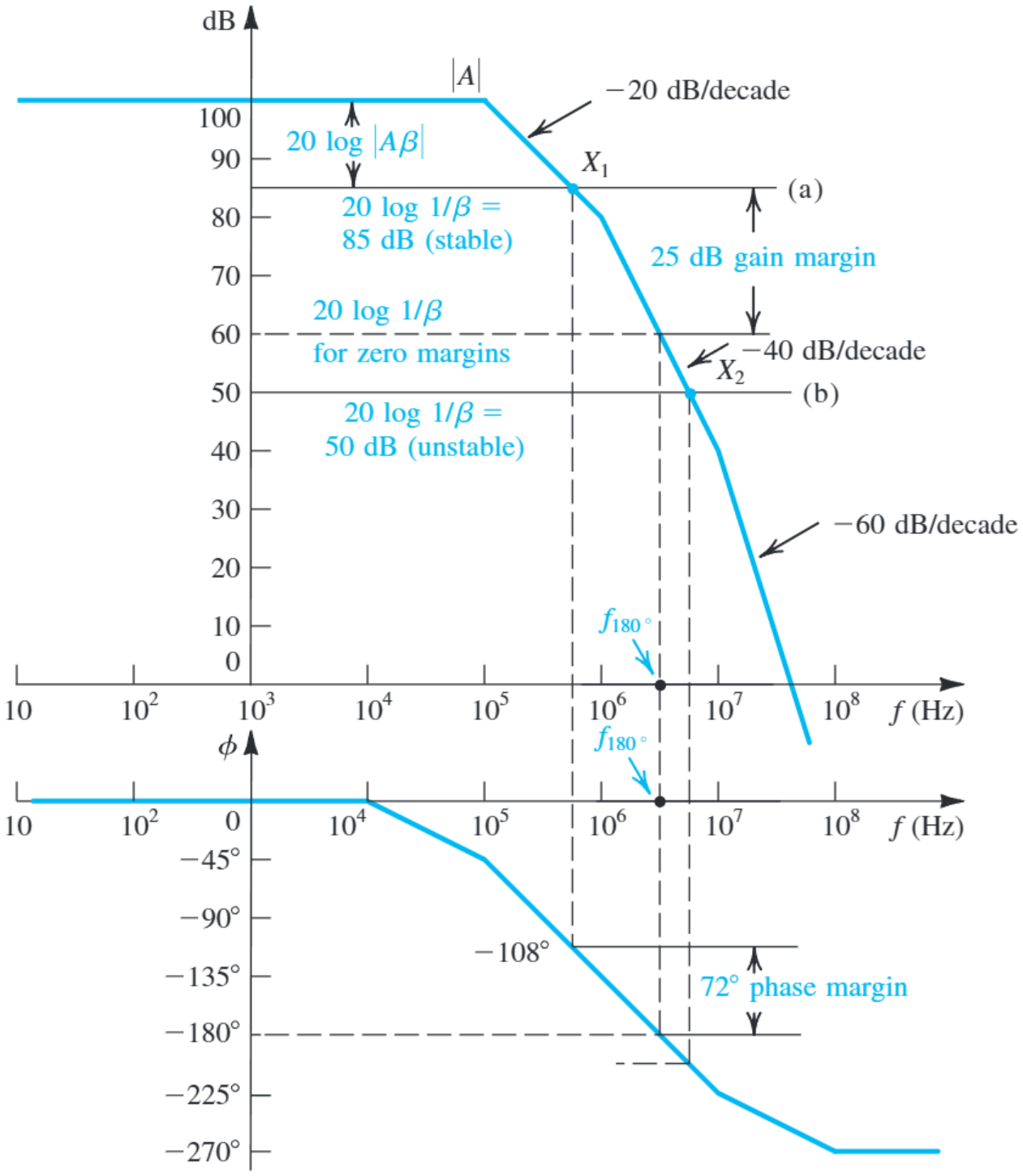


Figure 11.37 Stability analysis using Bode plot of |A|.

Consider next the straight line labeled (a) in Fig. 11.37. This line represents a feedback factor for which $20\log(1/\beta) = 85$ dB, which corresponds to $\beta = 5.623 \times 10^{-5}$ and a closed-loop gain of 83.6 dB. Since the loop gain is the difference between the $|A|$ curve and the $1/\beta$ line, the point of intersection X_1 corresponds to the frequency at which $|A\beta| = 1$. Using the graphs of Fig. 11.37, this frequency can be found to be approximately 5.6×10^5 Hz. A more exact value of 4.936×10^5 can be obtained using the transfer function equations. At this frequency the phase angle is approximately –108°. Thus the closed-loop amplifier, for which $20\log(1/\beta) = 85$ dB, will be stable with a phase margin of 72°. The gain margin can be easily obtained from Fig. 11.37; it is 25 dB.

Next, suppose that we wish to use this amplifier to obtain a closed-loop gain of 50-dB nominal value. Since $A_0 = 100$ dB, we see that $A_0\beta \gg 1$ and $20\log(A_0\beta) \simeq 50$ dB, resulting in $20\log(1/\beta) \simeq 50$ dB. To see whether this closed-loop amplifier is or is not stable, we draw line (b) in Fig. 11.37 with a height of 50 dB. This line intersects the open-loop gain curve at point X_2, where the corresponding phase is greater than 180°. Thus the closed-loop amplifier with 50-dB gain will be unstable.

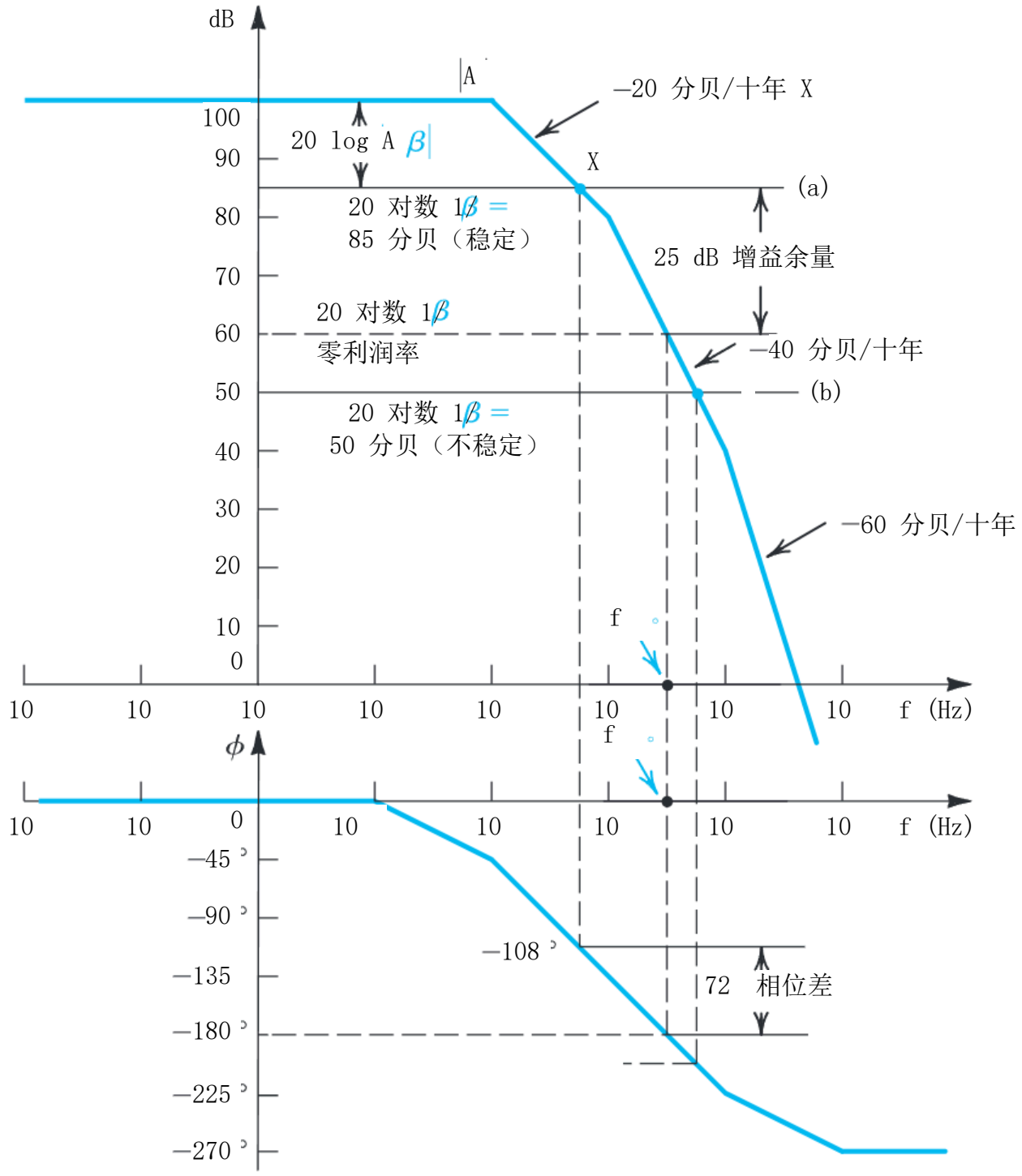


图 11.37 利用 |A| 的 Bode 图进行稳定性分析。

再看图 11.37 中标有 (a) 的直线。这条直线代表的反馈系数为 20 log(1/β) = 85 dB，相当于 β = 5.623×10 和 83.6 dB 的闭环增益。由于环路增益是 |A| 曲线与 1/β 线之间的差值，因此交点 X 相当于 |Aβ| = 1 时的频率。利用图 11.37 的图形，可以发现该频率约为 5.6 × 10 Hz。利用传递函数方程可以得到更精确的值 4.936 × 10。在此频率下，相位角约为 -108° 。因此，对于 20 log(1/β) = 85 dB 的闭环放大器来说，相位裕度为 72° ，是稳定的。增益裕量可从图 11.37 中轻松获得，即 25 dB。

接下来，假设我们希望使用该放大器获得额定值为 50 dB 的闭环增益。由于 A= 100 dB，我们可以看到 Aβ 1 和 20 log(Aβ) 50 dB，从而得到 20 log(1/β) 50 dB。为了了解这个闭环放大器是否稳定，我们在图 11.37 中画出高度为 50 dB 的线 (b)。这条线与开环增益曲线相交于 X 点，此时相应的相位大于 180° 。因此，增益为 50 dB 的闭环放大器将是不稳定的。

In fact, it can easily be seen from Fig. 11.37 that the *minimum* value of 20 log(1/β) that can be used, with the resulting amplifier being stable, is 60 dB. In other words, the minimum value of stable closed-loop gain obtained with this amplifier is approximately 60 dB. At this value of gain, however, a manufactured version of this amplifier may still oscillate, since no margin is left to allow for possible changes in gain.

Since the 180°-phase point always occurs on the –40-dB/decade segment of the Bode plot for |A|, a rule of thumb to guarantee stability is as follows: *The closed-loop amplifier will be stable if the* 20 log(1/β) *line intersects the* 20 log|A| *curve at a point on the* –20-dB/decade *segment*. Following this rule ensures that a phase margin of at least 45° is obtained. For the example of Fig. 11.37, the rule implies that the maximum value of β is 10^{-4}, which corresponds to a closed-loop gain of approximately 80 dB.

The rule of thumb above can be generalized for the case in which β is a function of frequency. The general rule states that *at the intersection of* 20 log[1/|$\beta(j\omega)$|] *and* 20 log|$A(j\omega)$| *the difference of slopes* (called the **rate of closure**) *should not exceed* 20 dB/decade.

EXERCISE

11.28 Consider an op amp whose open-loop gain is identical to that of Fig. 11.37. Assume that the op amp is ideal otherwise. Let the op amp be connected as a differentiator. Use the rule of thumb above to show that for stable performance the differentiator time constant should be greater than 159 ms. [*Hint:* Recall that for a differentiator, the Bode plot for 1/|$\beta(j\omega)$| has a slope of +20 dB/decade and intersects the 0-dB line at $1/\tau$, where τ is the differentiator time constant.]

11.10 Frequency Compensation

In this section, we shall discuss methods for modifying the open-loop transfer function $A(s)$ of an amplifier having three or more poles so that the closed-loop amplifier is stable for a given desired value of closed-loop gain. This process is referred to as frequency compensation.

11.10.1 Theory

The simplest method of frequency compensation consists of introducing a new pole in the function $A(s)$ at a frequency, f_D, sufficiently low that the modified open-loop gain, $A'(s)$, intersects the 20 log(1/|β|) curve with a slope difference of 20 dB/decade. As an example, let it be required to compensate the amplifier whose $A(s)$ is shown in Fig. 11.38 such that closed-loop amplifiers with β as high as 10^{-2} (i.e., closed-loop gains as low as approximately 40 dB) will be stable. First, we draw a horizontal straight line at the 40-dB level to represent 20 log(1/β), as shown in Fig. 11.38. We then locate point Y on this line at the frequency of the first pole, f_{P1}. From Y we draw a line with –20-dB/decade slope and determine the point at which this line intersects the dc gain line, point Y'. This latter point

事实上，从图 11.37 中不难看出，在使用 20 log(1/β)时，放大器稳定的最小值为 60 dB。换句话说，使用该放大器获得的稳定闭环增益的最小值约为 60 dB。然而，在此增益值下，该放大器的制造版本仍可能会发生振荡，因为没有留出余量来考虑增益可能发生的变化。

由于 180° 相位点总是出现在 Bode 图中 |A| 的-40-分贝/衰减段上，因此保证稳定性的经验法则如下：如果 20 log(1/β) 线与 20 log|A| 曲线相交于 -20 分贝/衰减段上的某一点，则闭环放大器将保持稳定。遵循这一规则可确保获得至少 45° 的相位余量。在图 11.37 的示例中，该规则意味着 β 的最大值为 10，相当于约 80 dB 的闭环增益。

上述经验法则可以推广到 β 是频率函数的情况。一般规则规定，在 20 log[1/|β(jω)|] 和 20 log|A(jω)| 的交点处，斜率差（称为闭合率）不应超过 20 dB/10。

运动

11.28 考虑一个开环增益与图 11.37 相同的运算放大器。假定运算放大器在其他方面都是理想的。将运算放大器连接为微分器。根据上述经验法则，微分器的时间常数应大于 159 ms，以获得稳定的性能。[提示：回想一下，对于微分器，1/|β(jω)| 的 Bode 图的斜率为 +20 dB/decade，与 0 分贝线相交于 1/τ，其中 τ 为微分器时间常数。]

11.10 频率补偿

在本节中，我们将讨论如何修改具有三个或更多极点的放大器的开环传递函数 A(s)，从而使闭环放大器在给定的闭环增益理想值下保持稳定。这一过程称为频率补偿。

11.10.1 理论

最简单的频率补偿方法是在函数 A(s)中引入一个新的极点，该极点的频率 f 要足够低，以便修改后的开环增益 A(s) 与 20 log(1/|β|) 曲线相交，斜率差为 20 dB/decade。举例来说，我们需要对图 11.38 所示 A(s) 的放大器进行补偿，以使 β 高达 10（即闭环增益低至约 40 dB）的闭环放大器保持稳定。首先，我们在 40 分贝处画一条水平直线，表示 20 log(1/β)，如图 11.38 所示。然后，我们将这条直线上的 Y 点定位在第一个极点的频率 f 上。从 Y 点出发，我们绘制一条斜率为 -20 分贝/分贝的直线，并确定该直线与直流增益线（Y 点）的交点。后一点

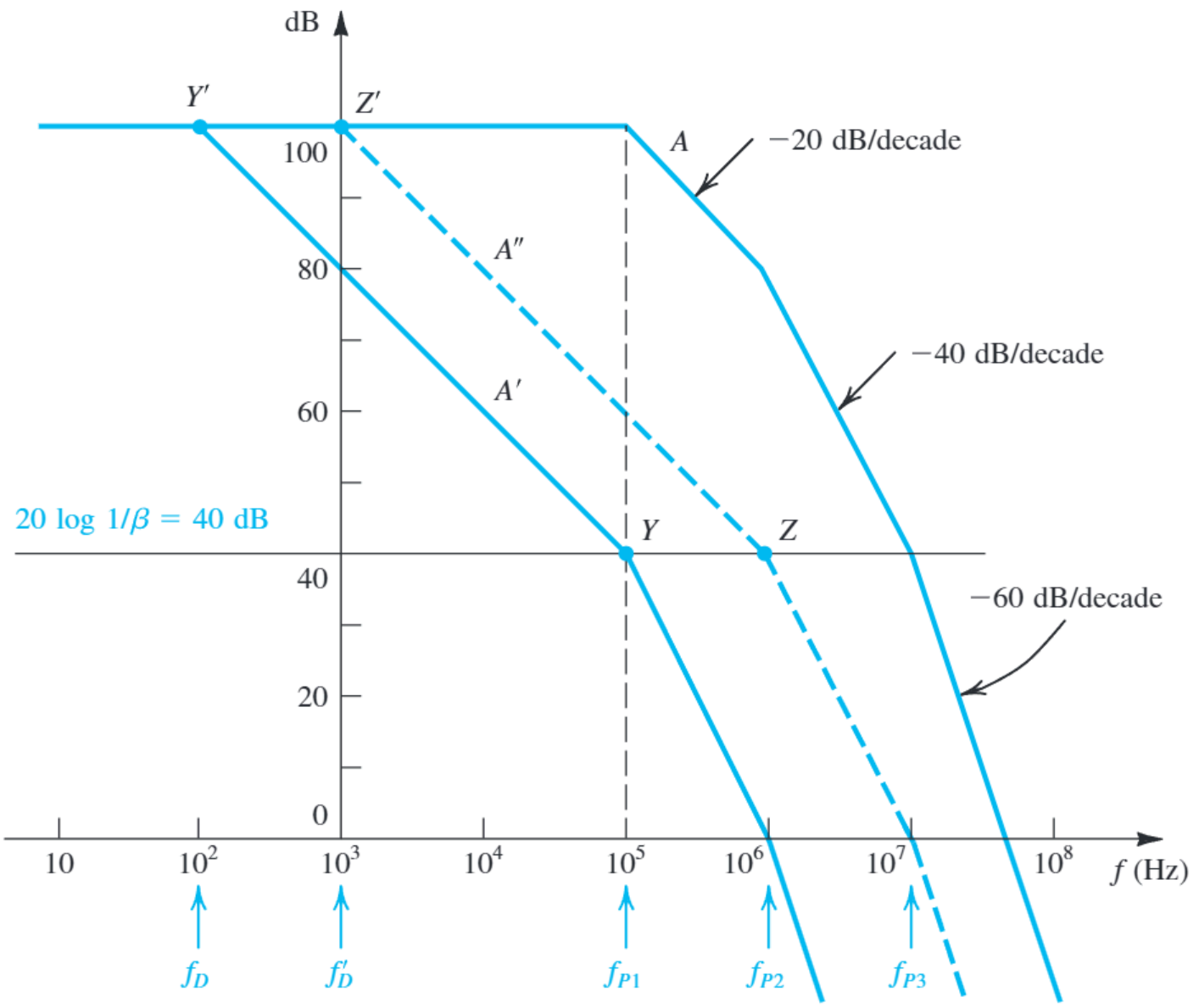


Figure 11.38 Frequency compensation for $\beta = 10^{-2}$. The response labeled A' is obtained by introducing an additional pole at f_D. The A'' response is obtained by moving the original low-frequency pole to f_D'.

gives the frequency f_D of the new pole that has to be introduced in the open-loop transfer function.

The compensated open-loop response $A'(s)$ is indicated in Fig. 11.38. It has four poles: at f_D, f_{P1}, f_{P2}, and f_{P3}. Thus $|A'|$ begins to roll off with a slope of –20 dB/decade at f_D. At f_{P1} the slope changes to – 40 dB/decade, at f_{P2} it changes to – 60 dB/decade, and so on. Since the $20 \log(1/\beta)$ line intersects the $20 \log|A'|$ curve at point Y on the –20-dB/decade segment, the closed-loop amplifier with this β value (or lower values) will be stable.

A serious disadvantage of this compensation method is that at most frequencies the open-loop gain has been drastically reduced. This means that at most frequencies the amount of feedback available will be small. Since all the advantages of negative feedback are directly proportional to the amount of feedback, the performance of the compensated amplifier will be impaired.

Careful examination of Fig. 11.38 shows that the gain $A'(s)$ is low because of the pole at f_{P1}. If we can somehow eliminate this pole, then—rather than locating point Y, drawing YY', and so on—we can start from point Z (at the frequency of the second pole) and draw the line ZZ'. This would result in the open-loop curve $A''(s)$, which shows considerably higher gain than $A'(s)$.

Although it is not possible to eliminate the pole at f_{P1}, it is usually possible to shift that pole from $f = f_{P1}$ to $f = f_D'$. This makes the pole dominant and eliminates the need for introducing an additional lower-frequency pole, as will be explained next.

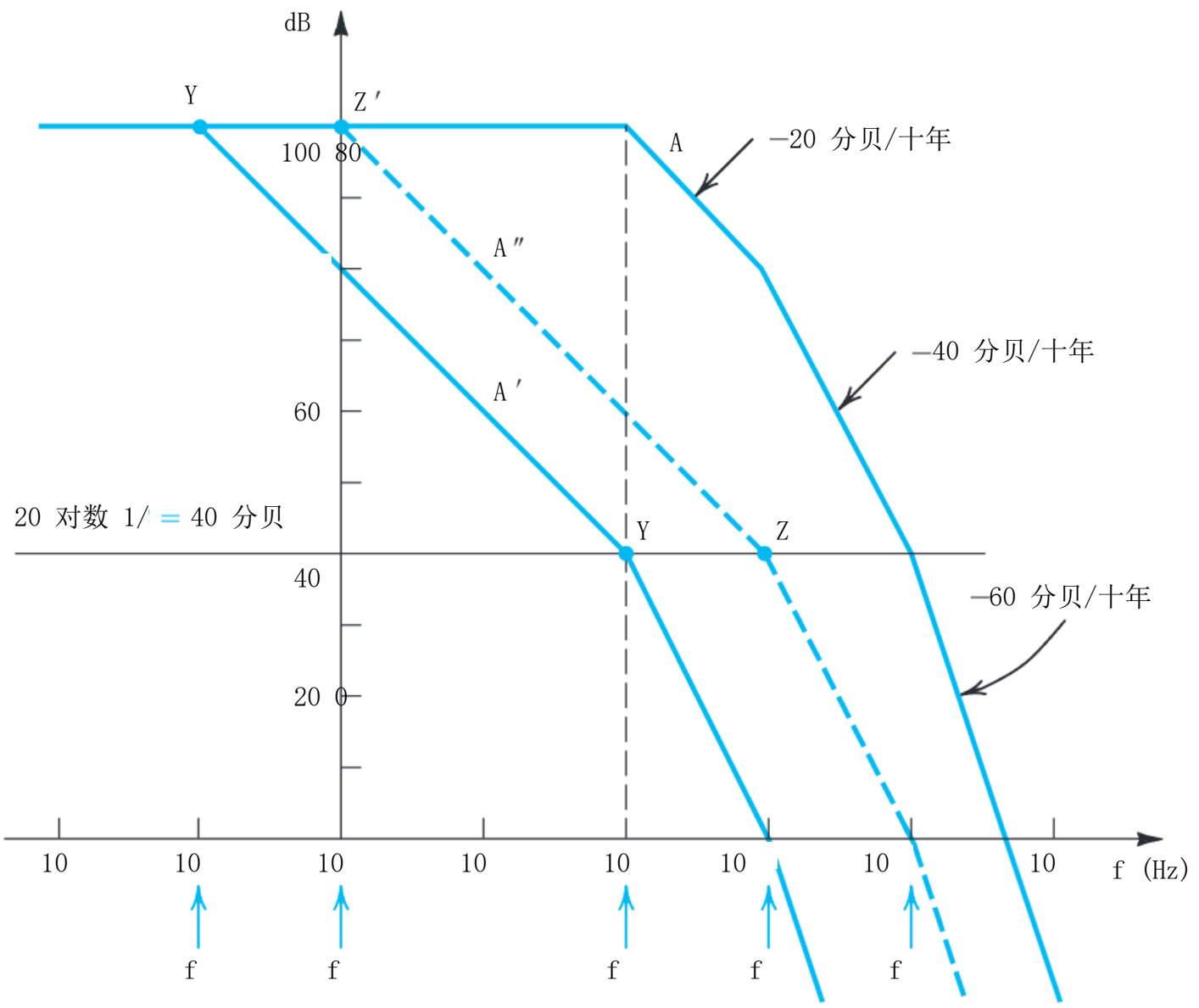


图 11.38 β = 10 时的频率补偿。标为 A 的响应是通过在 f 处引入一个附加极点得到的。

给出了必须引入开环传递函数的新极点的频率 f。

补偿后的开环响应 A(s) 如图 11.38 所示。它有四个极点：f、f、f 和 f。因此在 f 时，| | 开始以-20 dB/decade 的斜率下降。在 f 处，斜率变为-40 dB/decade，在 f 处，斜率变为-60 dB/decade，依此类推。由于 20 log（1/β）线与 20 log | | 曲线上的 Y 点为 -20 分贝/十进制段，具有此 β 值（或更低值）的闭环放大器将保持稳定。

这种补偿方法的一个严重缺点是，在大多数频率下，开环增益大幅降低。这意味着在大多数频率下，可用的反馈量很小。由于负反馈的所有优点都与反馈量成正比，因此补偿放大器的性能将受到影响。

仔细观察图 11.38 可以发现，由于 f 处有一个极点，因此增益 A(s) 很低。如果我们能以某种方式消除这个极点，那么，我们就可以从 Z 点（第二个极点的频率处）开始，画出 ZZ 线，而不是找到 Y 点，画出 YY 线，然后依次类推。这样就可以得到开环曲线 A(s)，它的增益比 A(s)要高得多。

虽然不可能消除 f 处的极点，但通常可以将该极点从 f = f 移至 f = f 处。这将使该极点成为主要极点，从而无需引入额外的低频极点，下文将对此进行解释。

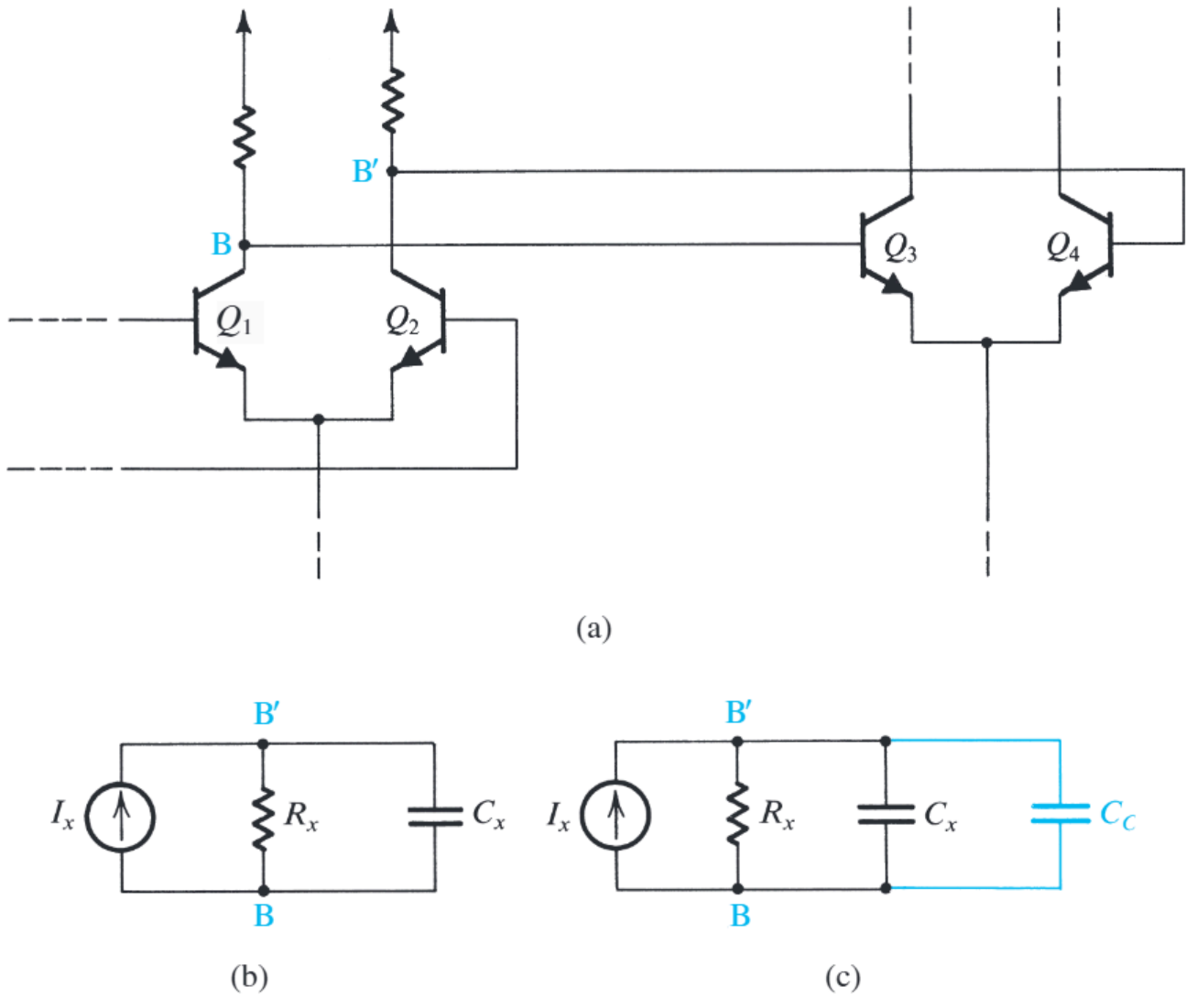


Figure 11.39 (**a**) Two cascaded gain stages of a multistage amplifier. (**b**) Equivalent circuit for the interface between the two stages in (**a**). (**c**) Same circuit as in (**b**), but with a compensating capacitor C_C added. Note that the analysis here applies equally well to MOS amplifiers.

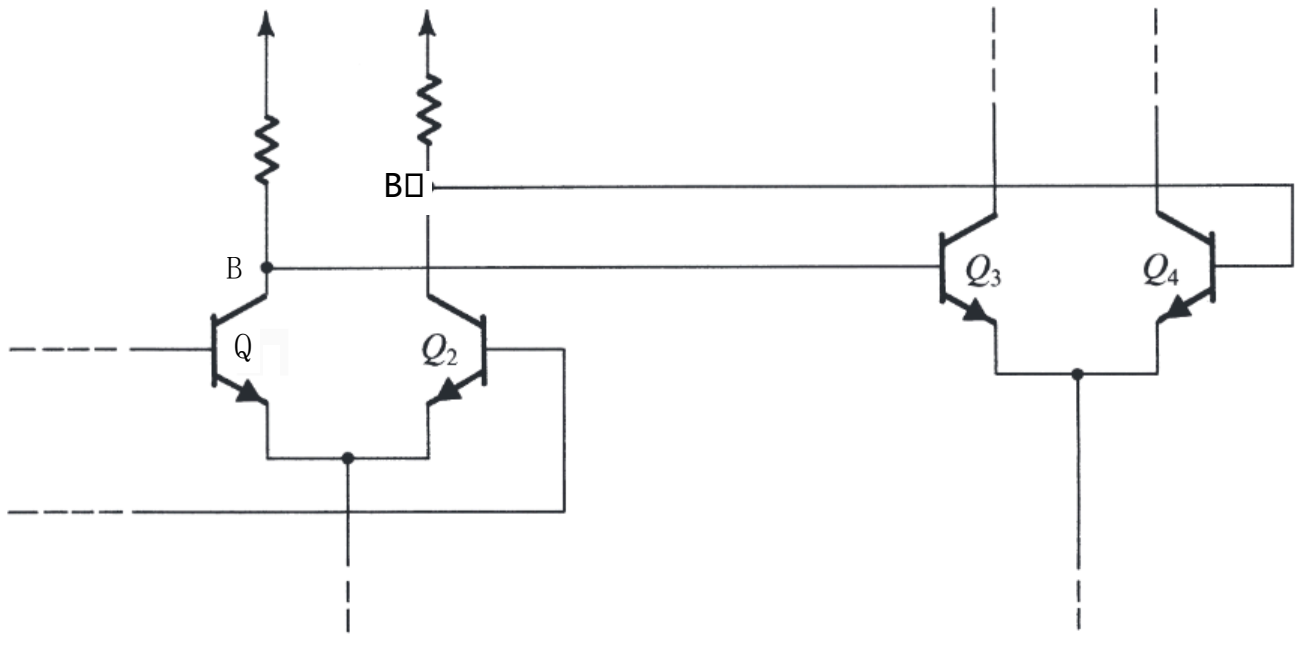
11.10.2 Implementation

We shall now address the question of implementing the frequency-compensation scheme discussed above. The amplifier circuit normally consists of a number of cascaded gain stages, with each stage responsible for one or more of the transfer function poles. Through manual and/or computer analysis of the circuit, one identifies the stage that introduces each of the important poles f_{P1}, f_{P2}, and so on. For the purpose of our discussion, assume that the first pole f_{P1} is introduced at the interface between the two cascaded differential stages shown in Fig. 11.39(a). In Fig. 11.39(b) we show a simple small-signal model of the circuit at this interface. Current source I_x represents the output-signal current of the Q_1–Q_2 stage. Resistance R_x and capacitance C_x represent the total resistance and capacitance between the two nodes B and B′. It follows that the pole f_{P1} is given by

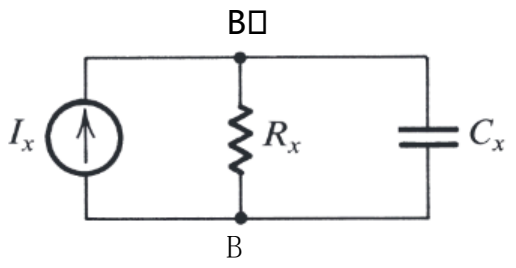
$$f_{P1} = \frac{1}{2\pi C_x R_x} \tag{11.87}$$

Let us now connect the compensating capacitor C_C between nodes B and B′. This will result in the modified equivalent circuit shown in Fig. 11.39(c), from which we see that the pole introduced will no longer be at f_{P1}; rather, the pole can be at any desired lower frequency f'_D :

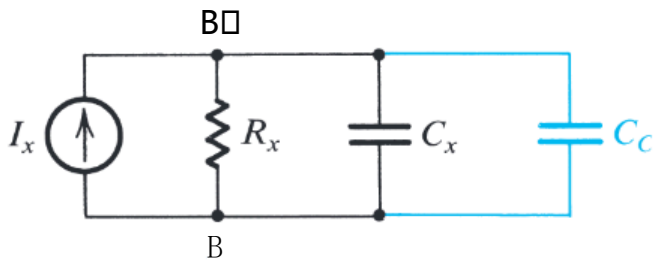
$$f'_D = \frac{1}{2\pi (C_x + C_C) R_x} \tag{11.88}$$



(a)



(b)



(c)

图 11.39 (a) 多级放大器的两个级联增益级。(b) (a) 中两级之间接口的等效电路。(c) 与 (b) 中的电路相同，但增加了补偿电容 C。请注意，此处的分析同样适用于 MOS 放大器。

11.10.2 实施

现在，我们将讨论如何实施上述频率补偿方案的问题。放大器电路通常由多个级联增益级组成，每一级负责一个或多个传递函数极点。通过人工和/或计算机对电路的分析，我们可以确定引入每个重要极点 f 、f 等的级。为便于讨论，假设第一个极点出现在图 11.39(a)所示的两个级联差分级之间的接口处。图 11.39(b) 显示了该接口处电路的简单小信号模型。电流源 I 代表 Q-Q 级的输出信号电流。电阻和电容 C 表示两个节点 B 和 B 之间的总电阻和总电容。

$$f = \frac{1}{2\pi C\ R} \tag{11.87}$$

现在，让我们将补偿电容器 C 连接到节点 B 和 B 之间。这将产生图 11.39(c)所示的修改后等效电路，从中我们可以看到，引入的极点将不再位于 f 处；相反，极点可以位于任何所需的较低频率 f 处：

$$f = \frac{1}{2\pi (C + C)R} \tag{11.88}$$

We thus conclude that one can select an appropriate value for C_C to shift the pole frequency from f_{P1} to the value f_D' determined by point Z' in Fig. 11.38.

At this juncture it should be pointed out that adding the capacitor C_C will usually result in changes in the location of the other poles (those at f_{P2} and f_{P3}). One might therefore need to calculate the new location of f_{P2} and perform a few iterations to arrive at the required value for C_C.

A disadvantage of this implementation method is that the required value of C_C is usually quite large. Thus if the amplifier to be compensated is an IC op amp, it will be difficult, and probably impossible, to include this compensating capacitor on the IC chip. (As pointed out in Chapter 8 and in Appendix A, the maximum practical size of a monolithic capacitor is about 100 pF.) An elegant solution to this problem is to connect the compensating capacitor in the feedback path of an inverting amplifier stage. Because of the Miller effect (Section 10.3), the compensating capacitance will be multiplied by the stage gain, resulting in a much larger effective capacitance. Furthermore, as explained later, another unexpected benefit accrues.

11.10.3 Miller Compensation and Pole Splitting

Figure 11.40(a) shows one gain stage in a multistage amplifier. For simplicity, the stage is shown as a common-emitter amplifier, but in practice it can be a more elaborate circuit. In the feedback path of this common-emitter stage we have placed a compensating capacitor C_f.

Figure 11.40(b) shows a simplified equivalent circuit of the gain stage of Fig. 11.40(a). Here R_1 and C_1 represent the total resistance and total capacitance between node B and ground. Similarly, R_2 and C_2 represent the total resistance and total capacitance between node C and ground. Furthermore, it is assumed that C_1 includes the Miller component due to capacitance C_μ, and C_2 includes the input capacitance of the succeeding amplifier stage. Finally, I_i represents the output signal current of the preceding stage.

In the absence of the compensating capacitor C_f, we can see from Fig. 11.40(b) that there are two poles—one at the input and one at the output. Let us assume that these two poles are f_{P1} and f_{P2} of Fig. 11.38; thus,

$$f_{P1} = \frac{1}{2\pi C_1 R_1} \qquad f_{P2} = \frac{1}{2\pi C_2 R_2} \tag{11.89}$$

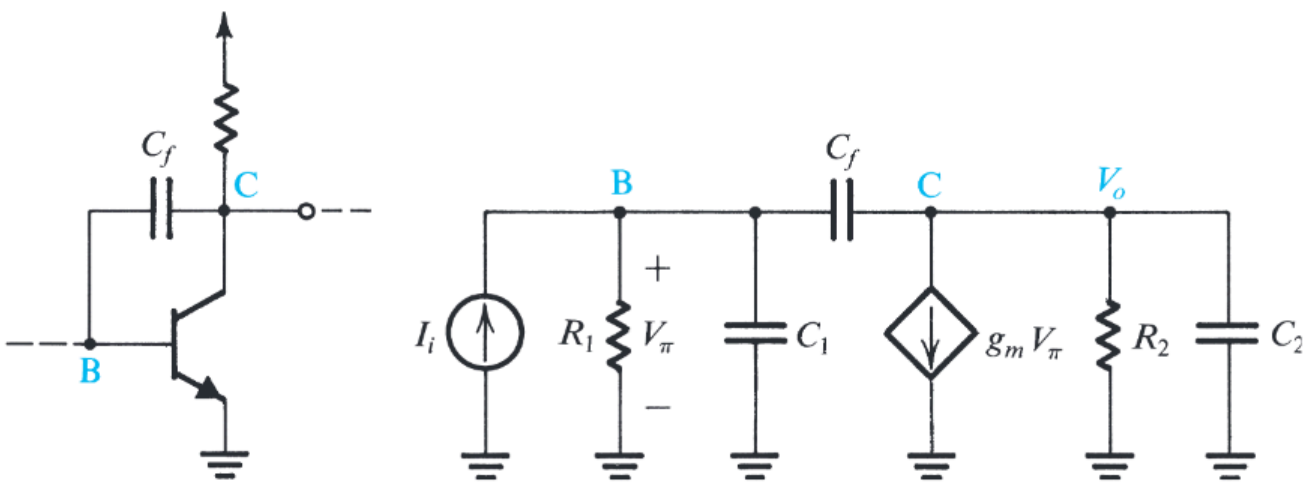


Figure 11.40 **(a)** A gain stage in a multistage amplifier with a compensating capacitor connected in the feedback path, and **(b)** equivalent circuit. Note that although a BJT is shown, the analysis applies equally well to the MOSFET case.

因此，我们可以得出结论：选择一个合适的 C 值，就可以将极点频率从 f 移至图 11.38 中 Z 点确定的 f 值。

在此需要指出的是，添加电容 C 通常会导致其他极点（f 和 f 处的极点）的位置发生变化。因此，我们可能需要计算 f 的新位置，并进行几次迭代，以得出所需的 C 值。

这种实现方法的缺点是所需的 C 值通常很大。因此，如果需要补偿的放大器是集成电路运算放大器，则很难甚至不可能在集成电路芯片上安装补偿电容。(正如第 8 章和附录 A 所指出的，单片电容器的最大实际尺寸约为 100 pF）。解决这一问题的有效方法是在反相放大器级的反馈路径中连接补偿电容。由于米勒效应（第 10.3 节），补偿电容将乘以级增益，从而产生一个大得多的有效电容。此外，正如后面所解释的，还能带来另一个意想不到的好处。

11.10.3 米勒补偿和极点分裂

图 11.40(a) 显示了多级放大器中的一个增益级。为简单起见，该级显示为共射极放大器，但实际上它可以是一个更复杂的电路。在共发射级的反馈路径中，我们放置了一个补偿电容 C 。

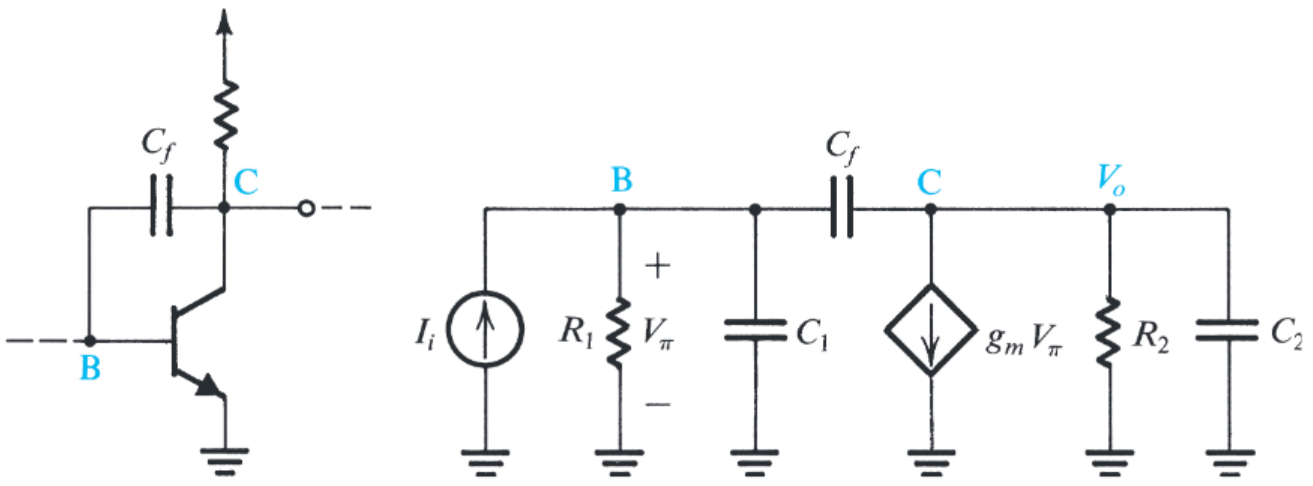
图 11.40(b) 是图 11.40(a) 中增益级的简化等效电路。

这里，Rand Crep 表示节点 B 和地之间的总电阻和总电容。同样，Rand Crep 表示节点 C 和地之间的总电阻和总电容。此外，假定 Cin 不包括电容 C 产生的米勒分量，也不包括后续放大器级的输入电容。

最后，I 代表前级的输出信号电流。

在没有补偿电容 C 的情况下，从图 11.40(b)可以看出有两个极点--一个在输入端，一个在输出端。假定这两个极点分别为图 11.38 中的 f 和 f，则

$$f = \frac{1}{2\pi CR} \qquad f = \frac{1}{2\pi CR} \tag{11.89}$$



(a) (b)

图 11.40 (a) 多级放大器中的增益级，反馈路径中连接有补偿电容；(b) 等效电路。请注意，虽然图中显示的是 BJT，但该分析同样适用于 MOSFET 的情况。

With C_f present, analysis of the circuit yields the transfer function

$$\frac{V_o}{I_i} = \frac{(sC_f - g_m)R_1R_2}{1 + s[C_1R_1 + C_2R_2 + C_f(g_mR_1R_2 + R_1 + R_2)] + s^2[C_1C_2 + C_f(C_1 + C_2)]R_1R_2} \tag{11.90}$$

The zero is usually at a much higher frequency than the dominant pole, and we shall neglect its effect. The denominator polynomial $D(s)$ can be written in the form

$$D(s) = \left(1 + \frac{s}{\omega'_{P1}}\right)\left(1 + \frac{s}{\omega'_{P2}}\right) = 1 + s\left(\frac{1}{\omega'_{P1}} + \frac{1}{\omega'_{P2}}\right) + \frac{s^2}{\omega'_{P1}\omega'_{P2}} \tag{11.91}$$

where ω'_{P1} and ω'_{P2} are the new frequencies of the two poles. Normally one of the poles will be dominant; $\omega'_{P1} \ll \omega'_{P2}$. Thus,

$$D(s) \simeq 1 + \frac{s}{\omega'_{P1}} + \frac{s^2}{\omega'_{P1}\omega'_{P2}} \tag{11.92}$$

Equating the coefficients of s in the denominator of Eq. (11.90) and in Eq. (11.92) results in

$$\omega'_{P1} = \frac{1}{C_1R_1 + C_2R_2 + C_f(g_mR_1R_2 + R_1 + R_2)}$$

which can be approximated by

$$\omega'_{P1} \simeq \frac{1}{g_mR_2C_fR_1} \tag{11.93}$$

To obtain ω'_{P2} we equate the coefficients of s^2 in the denominator of Eq. (11.90) and in Eq. (11.92) and use Eq. (11.93):

$$\omega'_{P2} \simeq \frac{g_mC_f}{C_1C_2 + C_f(C_1 + C_2)} \tag{11.94}$$

From Eqs. (11.93) and (11.94), we see that as C_f is increased, ω'_{P1} is reduced and ω'_{P2} is increased. This action is referred to as **pole splitting**. Note that the increase in ω'_{P2} is highly beneficial; it allows us to move point Z (see Fig. 11.38) further to the right, thus resulting in higher compensated open-loop gain. Finally, note from Eq. (11.93) that C_f is multiplied by the Miller effect factor g_mR_2, thus resulting in a much larger effective capacitance, $g_mR_2C_f$. In other words, the required value of C_f will be much smaller than that of C_C in Fig. 11.39.

Example 11.12

Consider an op amp whose open-loop transfer function is identical to that shown in Fig. 11.37. We wish to compensate this op amp so that the closed-loop amplifier with resistive feedback is stable for any gain (i.e., for β up to unity). Assume that the op-amp circuit includes a stage such as that of Fig. 11.40 with $C_1 = 100$ pF, $C_2 = 5$ pF, and $g_m = 40$ mA/V, that the pole at f_{P1} is caused by the input circuit of that stage, and that the pole at f_{P2} is introduced by the output circuit. Find the value of the compensating capacitor

在 C 存在的情况下，对电路进行分析可得到传递函数

$$\frac{V}{I} = \frac{[\text{CR+ CR+ C(g RR+ R+ R)} \quad (sC - g) RR]}{1 + s} + s[\text{CC+ C(C+ C)} \quad]\text{RR} \tag{11.90}$$

零点的频率通常远高于主极点，我们将忽略其影响。分母多项式 D(s) 可以写成以下形式

$$D(s) = \left(1 + \frac{s}{\omega}\right)\left(1 + \frac{s}{\omega}\right) = 1 + s\left(\frac{1}{\omega} + \frac{1}{\omega}\right) + \frac{s}{\omega\omega} \tag{11.91}$$

其中，ω 和 ω 是两个极点的新频率。通常，其中一个极点会占主导地位；ω ω。因此

$$D(s)\ 1 + \frac{s}{\omega} + \frac{s}{\omega\omega} \tag{11.92}$$

将式（11.90）分母中的 s 系数与式（11.92）中的 s 系数相等，结果是

$$\omega = \frac{1\ \text{CR+ CR+ C（g RR+}}{\text{R+ R）}}$$

可近似为

$$\omega \approx \frac{1}{g\ RC\ R} \tag{11.93}$$

为了得到 ω，我们将式（11.90）和式（11.92）中 sin 的分母相等，并使用式（11.93）：

$$\omega \approx \frac{gC\ \text{CC+ C(C+ C)}}{} \tag{11.94}$$

从式（11.93）和（11.94）可以看出，随着 C 的增大，ω 会减小，ω 会增大。这种作用称为极分裂。请注意，ω 的增大是非常有益的；它允许我们将 Z 点（见图 11.38）进一步向右移动，从而获得更高的补偿开环增益。最后，请注意公式（11.93）中的 C 乘以米勒效应系数 gR，从而得到更大的有效电容 g RC。

换句话说，所需的 C 值将比图 11.39 中的 C 值小得多。

例 11.12

考虑一个运算放大器，其开环传递函数与图 11.37 所示相同。我们希望对该运算放大器进行补偿，从而使带有电阻反馈的闭环放大器在任何增益下（即在 β 达到 1 时）都能保持稳定。假设运算放大器电路包括图 11.40 所示的级，C= 100 pF，C= 5 pF，g= 40 mA/V，f 处的极点由该级的输入电路引起，fis 处的极点由输出电路引入。求补偿电容的值

Example 11.12 *continued*

for two cases: either if it is connected between the input node B and ground or in the feedback path of the transistor.

Solution

First we determine R_1 and R_2 from

$$f_{P1} = 0.1\ \text{MHz} = \frac{1}{2\pi C_1 R_1}$$

Thus,

$$R_1 = \frac{10^5}{2\pi}\ \Omega$$

$$f_{P2} = 1\ \text{MHz} = \frac{1}{2\pi C_2 R_2}$$

Thus,

$$R_2 = \frac{10^5}{\pi}\ \Omega$$

If a compensating capacitor C_C is connected across the input terminals of the transistor stage, then the frequency of the first pole changes from f_{P1} to f_D':

$$f_D' = \frac{1}{2\pi(C_1 + C_C)R_1}$$

The second pole remains unchanged at 1 MHz. The required value for f_D' is determined by drawing a –20-dB/decade line from the 1-MHz frequency point on the $20\log(1/\beta) = 20\log 1 = 0$ dB line. This line will intersect the 100-dB dc gain line at 10 Hz. Thus,

$$f_D' = 10\,\text{Hz} = \frac{1}{2\pi(C_1 + C_C)R_1}$$

which results in $C_C \simeq 1\ \mu\text{F}$, which is quite large and certainly cannot be included on the IC chip.

Next, if a compensating capacitor C_f is connected in the feedback path of the transistor, then both poles change location to the values given by Eqs. (11.93) and (11.94):

$$f_{P1}' \simeq \frac{1}{2\pi g_m R_2 C_f R_1} \qquad f_{P2}' \simeq \frac{g_m C_f}{2\pi\left[C_1 C_2 + C_f(C_1 + C_2)\right]} \tag{11.95}$$

To determine where we should locate the first pole, we need to know the value of f_{P2}'. As an approximation, let us assume that $C_f \gg C_2$, which enables us to obtain

$$f_{P2}' \simeq \frac{g_m}{2\pi(C_1 + C_2)} = 60.6\ \text{MHz}$$

Thus it appears that this pole will move to a frequency higher than f_{P3} (which is 10 MHz). Let us therefore assume that the second pole will be at f_{P3}. This requires that the modified first pole be located at

$$f_{P1}' = \frac{f_{P3}}{A_0} = \frac{10^7\,\text{Hz}}{10^5} = 100\,\text{Hz}$$

例 11.12 续

找出两种情况下补偿电容的值：要么补偿电容连接在输入节点 B 和地之间，要么补偿电容连接在晶体管的反馈路径上。

解决方案

首先，我们从

$$f = 0.1\ \text{MHz} = \frac{1}{2\pi CR}$$

因此、

$$R = \frac{10}{2\pi}\ \square$$

$$f = 1\ \text{MHz} = \frac{1}{2\pi CR}$$

因此、

$$R = \frac{10\ \pi}{}\ \square$$

如果在晶体管级的输入端子上连接一个补偿电容 C，那么第一极的频率就会从 f 变为 f：

$$f = \frac{}{2\pi}\left(\frac{1}{C+\ C}\right) R$$

第二个极点在 1 MHz 时保持不变。从 20 log(1/β) = 20 log 1 = 0 dB 线的 1 MHz 频率点开始，画一条-20-dB/decade 线，即可确定 f 的所需值。这条线将与 100 分贝直流增益线相交于 10 赫兹处。因此

$$f = 10\ \text{赫兹} = \frac{1}{2\pi (C+\ C)R}$$

这就导致 C 1 μF，这个数值相当大，当然不可能包含在集成电路芯片中。

接下来，如果在晶体管的反馈路径上连接一个补偿电容 C，则两个极点的位置都会改变，变为公式 (11.93) 和 (11.94) 所给出的值：

$$f\square \frac{1}{2\pi g\ RC\ R} \qquad f\square \frac{}{2\pi}\left[\frac{g\ C}{CC+\ C}\left(\frac{}{C+\ C}\right)\right] \tag{11.95}$$

为了确定第一极点的位置，我们需要知道 f 的值。作为近似值，我们假设 C 为 C，这样就可以得到

$$f\square \frac{}{2\pi}\left(\frac{g}{C+\ C}\ \frac{}{}\right) = 60.6\ \text{兆赫}$$

由此看来，这个极点将移动到高于 f（即 10 MHz）的频率。因此，我们假设第二个极点将位于 f 处。这就要求修改后的第一极点位于

$$f = \frac{f}{\dot{A}} = \frac{10\ \text{赫兹}}{10} = 100\ \text{赫兹}$$

Thus,

$$f'_{P1} = 100\,\text{Hz} = \frac{1}{2\pi g_m R_2 C_f R_1}$$

which results in $C_f = 78.5$ pF. Although this value is indeed much greater than C_2, we can determine the location of the pole f'_{P2} from Eq. (11.95), which yields $f'_{P2} = 57.2$ MHz, confirming that this pole has indeed been moved past f_{P3}.

We conclude that using Miller compensation not only results in a much smaller compensating capacitor but, owing to pole splitting, also enables us to place the dominant pole a decade higher in frequency. This results in a wider bandwidth for the compensated op amp.

EXERCISES

11.29 A multipole amplifier having a first pole at 1 MHz and an open-loop gain of 100 dB is to be compensated for closed-loop gains as low as 20 dB by the introduction of a new dominant pole. At what frequency must the new pole be placed?

Ans. 100 Hz

11.30 For the amplifier described in Exercise 11.29, rather than introducing a new dominant pole, we can use additional capacitance at the circuit node at which the first pole is formed to reduce the frequency of the first pole. If the frequency of the second pole is 10 MHz and if it remains unchanged while additional capacitance is introduced as mentioned, find the frequency to which the first pole must be lowered so that the resulting amplifier is stable for closed-loop gains as low as 20 dB. By what factor must the capacitance at the controlling node be increased?

Ans. 1000 Hz; 1000

Summary

- Negative feedback is employed to make the amplifier gain less sensitive to component variations; to control input and output resistances; to extend bandwidth; to reduce nonlinear distortion; and to enhance signal-to-interference ratio.
- The advantages above are obtained at the expense of a reduction in gain and at the risk of the amplifier becoming unstable (that is, oscillating). The latter problem is solved by careful design.
- The structure of an ideal negative-feedback amplifier is shown in Fig. 11.1. Table 11.1 summarizes the parameters and relationships governing the operation of the ideal structure.
- For each of the four basic types of amplifier, there is an appropriate feedback topology. The four topologies, together with their analysis procedure and their effects on input and output impedances, are summarized in Table 11.2 in Section 11.6.
- The key feedback parameters are the loop gain ($A\beta$), which for negative feedback must be a positive dimensionless number, and the amount of feedback ($1 + A\beta$). The latter directly determines gain reduction, gain

因此、

$$f = 100\ \text{Hz} = \frac{1}{2\pi g\ RC\ R}$$

得出 C = 78.5 pF。虽然这个值确实远大于 C，但我们可以根据公式 (11.95) 确定极点 f 的位置，得出 f = 57.2 MHz，从而确认这个极点确实已经移过了 f。

我们得出的结论是，使用米勒补偿不仅能使补偿电容器的体积大大缩小，而且由于极点分裂，还能使我们将主导极点的频率提高 10 倍。这使得补偿运算放大器的带宽更宽。

练习

11.29 一个多极放大器的第一极在 1 MHz 处，开环增益为 100 dB，需要通过引入一个新的主极来补偿低至 20 dB 的闭环增益。新磁极必须置于什么频率？答案：100 赫兹

11.30 对于练习 11.29 中描述的放大器，我们可以在形成第一极点的电路节点上使用附加电容来降低第一极点的频率，而不是引入一个新的主极点。如果第二个磁极的频率为 10 MHz，并且在引入上述附加电容时保持不变，那么请找出必须将第一个磁极降低到什么频率，才能使产生的放大器在闭环增益低至 20 dB 时保持稳定。控制节点处的电容必须增加多少倍？

答案：1000 赫兹；1000

摘要

采用负反馈可降低放大器增益对元件变化的敏感性；控制输入和输出电阻；扩展带宽；降低非线性失真；以及提高信号干扰比。

上述优点是以降低增益和放大器变得不稳定（即振荡）为代价的。后一个问题可以通过精心设计来解决。

理想负反馈放大器的结构如图 11.1 所示。表 11.1 汇总了以下参数以及理想结构的运行关系。

对于四种基本放大器类型中的每一种，都有合适的反馈拓扑结构。第 11.6 节表 11.2 概述了这四种拓扑结构及其分析程序以及对输入和输出阻抗的影响。

关键的反馈参数是环路增益（Aβ）（负反馈时必须是无量纲正数）和反馈量（1 + Aβ）。后者直接决定增益减小、增益

desensitivity, bandwidth extension, and changes in R_i and R_o.

- The loop gain $A\beta$ can be determined by breaking the feedback loop, as illustrated in Figs. 11.2 and 11.9. The value of $A\beta$ can be used together with the feedback factor β to determine A and hence A_f. This method, though simple, is incomplete as it does not enable the determination of the input and output resistances. For these, we utilize the systematic method for feedback analysis (refer to Table 11.2).
- The ideal or upper-bound value of the closed-loop gain A_f is $1/\beta$ and is approached when $A\beta \gg 1$.
- Since A and β are in general frequency dependent, the poles of the feedback amplifier are obtained by solving the characteristic equation $1 + A(s)\beta(s) = 0$.
- For the feedback amplifier to be stable, its poles must all be in the left half of the s plane.
- Stability is guaranteed if at the frequency for which the phase angle of $A\beta$ is 180° (i.e., ω_{180}), $|A\beta|$ is less than unity; the amount by which it is less than unity, expressed in decibels, is the gain margin. Alternatively, the amplifier is stable if, at the frequency at which $|A\beta| = 1$, the phase angle is less than 180°; the difference is the phase margin.
- The stability of a feedback amplifier can be analyzed by constructing a Bode plot for $|A|$ and superimposing on it a plot for $20 \log 1/|\beta|$. Stability is guaranteed if the two plots intersect with a difference in slope no greater than 6 dB/octave.
- To make a given amplifier stable for a given feedback factor β, the open-loop frequency response is suitably modified by a process known as frequency compensation.
- A popular method for frequency compensation involves connecting a feedback capacitor across an inverting stage in the amplifier. This causes the pole formed at the input of the amplifier stage to shift to a lower frequency and thus become dominant, while the pole formed at the output of the amplifier stage is moved to a very high frequency and thus becomes unimportant. This process is known as pole splitting.

PROBLEMS

Computer Simulation Problems

SIM Problems identified by the Multisim/PSpice icon are intended to demonstrate the value of using SPICE simulation to verify hand analysis and design, and to investigate important issues such as allowable signal swing and amplifier nonlinear distortion. Instructions to assist in setting up PSpice and Multisim simulations for all the indicated problems can be found in the corresponding files on the website. Note that if a particular parameter value is not specified in the problem statement, you are to make a reasonable assumption.

Section 11.1: The General Feedback Structure

11.1 A negative-feedback amplifier has a closed-loop gain $A_f = 200$ and an open-loop gain $A = 10^4$. What is the feedback factor β? If a manufacturing error results in a reduction of A to 10^3, what closed-loop gain results? What is the percentage change in A_f corresponding to this factor of 10 reduction in A?

11.2 Consider the op-amp circuit shown in Fig. P11.2, where the op amp has infinite input resistance and zero output resistance but finite open-loop gain A.

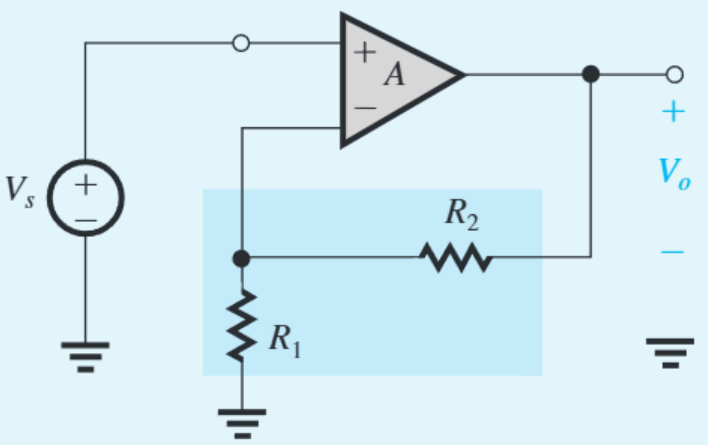


Figure P11.2

(a) Convince yourself that $\beta = R_1/(R_1 + R_2)$.
(b) If $R_1 = 10$ kΩ, find R_2 that results in $A_f = 10$ V/V for the following three cases: (i) $A = 1000$ V/V; (ii) $A = 200$ V/V; (iii) $A = 15$ V/V.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

如图 11.2 和图 11.9 所示，环路增益 Aβ 可通过断开反馈环路来确定。这种方法虽然简单，但并不完整，因为它无法确定输入和输出电阻。为此，我们采用系统法进行反馈分析（参见表 11.2）。

闭环增益 A 的理想值或上限值为 1/β，当 Aβ 为 1 时，闭环增益 A 接近于 1/β。

增益 由于 A 和 β 通常与频率有关，因此反馈放大器的极点可通过求解特性方程 1 + A(s)β(s) = 0 得到。

要使反馈放大器保持稳定，其极点必须全部位于 s 平面的左半部分。

如果在 Aβ 的相位角为 180 °（即 ω）的频率上，|Aβ| 小于|Aβ|，则可保证稳定性。

以分贝表示的小于合一的量即为增益裕量。或者，如果在 |Aβ| = 1 的频率上，相位角小于 180°，放大器就是稳定的；差值就是相位裕量。

反馈放大器的稳定性可通过绘制|A|的 Bode 图并叠加 20 log 1/|β| 的图来分析。如果两幅图的斜率相差不超过 6 dB/倍频程，则可保证稳定性。为了使给定放大器在给定反馈系数 β 下保持稳定，开环频率响应可通过称为频率补偿的过程进行适当修改。一种常用的频率补偿方法是在放大器的反相级上连接一个反馈电容器。这将使放大器输入级形成的极点移至较低频率，从而成为主要极点，而放大器输出级形成的极点则移至非常高的频率，从而变得不重要。这一过程被称为极点分裂。

问题

计算机模拟问题

SIM Multisim/PSpice 图标标识的问题旨在展示使用 SPICE 仿真验证手工分析和设计的价值，以及研究允许信号摆幅和放大器非线性失真等重要问题的价值。网站上的相应文件中提供了针对所有指示问题的 PSpice 和 Multisim 仿真设置说明。请注意，如果问题陈述中未指明特定参数值，则应做出合理假设。

第 11.1 节 一般反馈结构一般反馈结构

11.1 负反馈放大器的闭环增益 A= 200，开环增益 A= 10。反馈系数 β 是多少？如果制造误差导致 A 降为 10，则闭环增益为多少？与 A 减少 10 相对应的 Ac 百分比变化是多少？

11.2 考虑图 P11.2 所示的运算放大器电路，其中运算放大器的输入电阻无穷大，输出电阻为零，但开环增益 A 有限。

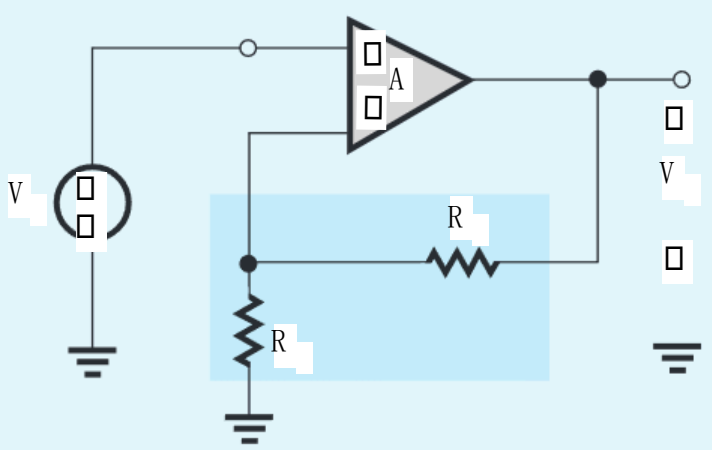


图 P11.2

(a) 说服自己相信 β = R/ (R+ R) .

(b) 如果 R= 10 k，找出在以下三种情况下 A= 10 V/V 的 R：(i) A = 1000 V/V；(ii) A = 200 V/V；(iii) A = 15 V/V。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

(c) For each of the three cases in (b), find the percentage change in A_f that results when A decreases by 20%. Comment on the results.

11.3 The noninverting buffer op-amp configuration shown in Fig. P11.3 provides a direct implementation of the feedback loop of Fig. 11.1. Assuming that the op amp has infinite input resistance and zero output resistance, what is β? If $A = 1000$, what is the closed-loop voltage gain? What is the amount of feedback (in dB)? For $V_s = 1$ V, find V_o and V_i. If A decreases by 10%, what is the corresponding percentage decrease in A_f?

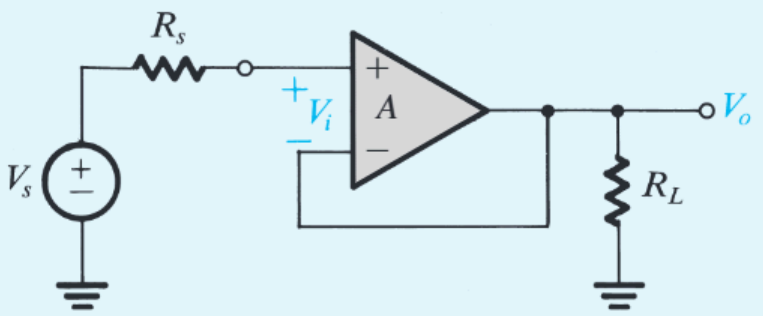


Figure P11.3

11.4 In a particular circuit represented by the block diagram of Fig. 11.1, a signal of 1 V from the source results in a difference signal of 10 mV being provided to the amplifying element A, and 5 V appearing at the output. For this arrangement, identify the values of A and β that apply.

11.5 (a) Show that in a negative-feedback amplifier with loop gain $A\beta \gg 1$, the closed-loop gain A_f is lower than its ideal value of $1/\beta$ by $(100/A\beta)\%$.

(b) What is the minimum loop gain required so that A_f is within (i) 0.1%, (ii) 1%, and (iii) 5% of its ideal value?

11.6 In a particular amplifier design, the β network consists of a linear potentiometer for which β is 0.00 at one end, 1.00 at the other end, and 0.50 in the middle. As the potentiometer is adjusted, find the three values of closed-loop gain that result when the amplifier open-loop gain is (a) 1 V/V, (b) 10 V/V, (c) 100 V/V, (d) 1000 V/V, and (e) 10,000 V/V. Provide your results in a table in which there is a row for each value of A and a column for each value of β.

11.7 A newly constructed feedback amplifier undergoes a performance test with the following results: With the feedback connection removed, a source signal of 2 mV is required to provide a 5-V output; with the feedback connected, a 5-V output requires a 100-mV source signal. For this amplifier, identify values of A, β, $A\beta$, the closed-loop gain, and the amount of feedback (in dB).

D 11.8 An amplifier has an open-loop gain with a nominal value of 1000 but can vary from unit to unit by as much as ±50% of nominal. It is required to apply negative feedback to this amplifier so that the variability of the closed-loop gain of the resulting feedback amplifier is limited to ±1%. What is the largest possible nominal value of closed-loop gain that can be achieved? Now if three of these feedback amplifiers are placed in cascade, what is the nominal value of the gain of the resulting cascade amplifier? What is the expected variability of this gain?

11.9 The op amp in the circuit of Fig. P11.9 has an open-circuit voltage gain μ, a differential input resistance R_{id}, and a negligibly small output resistance. It is connected in the noninverting configuration with a feedback network consisting of a voltage divider (R_1, R_2). While β is still determined by the divider ratio [i.e., $\beta = R_1/(R_1 + R_2)$], the open-loop gain A is no longer simply equal to μ. This is because the feedback network now loads the input of the amplifier (because of the finite R_{id}). To determine the value of A, use the method outlined in Section 11.1.3 to determine the loop gain $A\beta$. Thus show that

$$A = \mu \frac{R_{id}}{R_{id} + (R_1 \parallel R_2)}$$

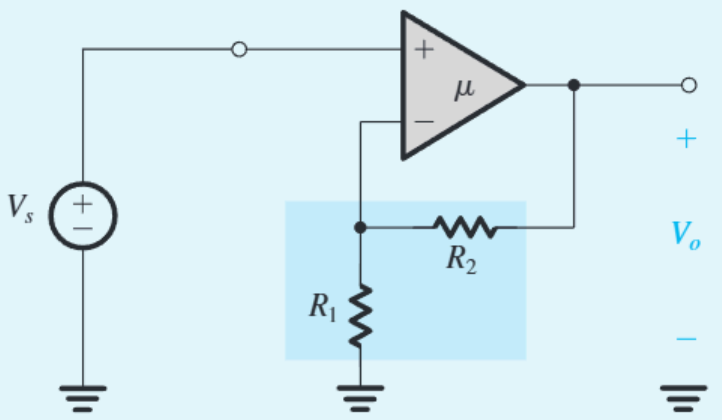


Figure P11.9

Section 11.2: Some Properties of Negative Feedback

11.10 For the negative-feedback loop of Fig. 11.1, find the loop gain $A\beta$ for which the sensitivity of closed-loop gain to open-loop gain [i.e., $(dA_f/A_f)/(dA/A)$] is –40 dB. For what value of $A\beta$ does the sensitivity become 1/5?

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

(c) 对于 (b) 中的三种情况，分别求出 A 减少 20% 时 A 变化的百分比。

对结果发表评论。

11.3 图 P11.3 所示的非反相缓冲运算放大器配置直接实现了图 11.1 的反馈回路。假设运算放大器的输入电阻为无穷大，输出电阻为零，那么 β 是多少？如果 A = 1000，闭环电压增益是多少？反馈量是多少（dB）？如果 A 减小 10%，A 的相应减小百分比是多少？

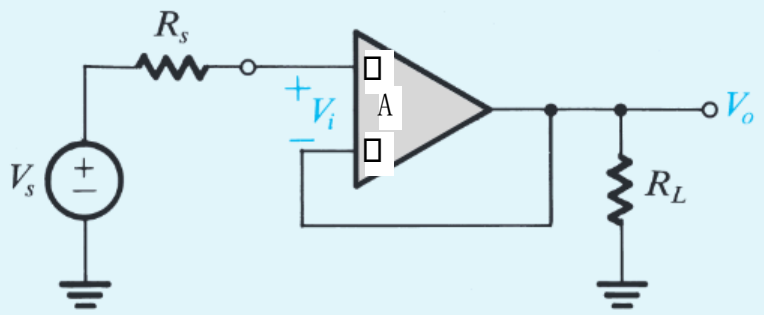


图 P11.3

11.4 在图 11.1 方框图所表示的一个特定电路中，源极发出 1 V 信号后，放大元件 A 得到 10 mV 的差分信号，输出端出现 5 V 电压。对于这种布置，请找出适用的 A 和 β 值。

11.5 (a) 证明在环路增益 Aβ 为 1 的负反馈放大器中，闭环增益 A 比理想值 1/β 低 (100/Aβ)% 。

(b) 要使 A 在理想值的 (i)0.1%、(ii)1% 和 (iii)5%范围内，所需的最小环路增益是多少？

11.6 在一个特定的放大器设计中，β 网络由一个线性电位器组成，β 的一端为 0.00，另一端为 1.00，中间为 0.50。调节电位器时，找出放大器开环增益为 (a) 1 V/V、(b) 10 V/V、(c) 100 V/V、(d) 1000 V/V 和 (e) 10,000 V/V 时产生的三个闭环增益值。请将结果列于表格中，其中 A 值各占一行，β 值各占一列。

11.7 对一个新制造的反馈放大器进行性能测试，结果如下：去掉反馈连接后，需要 2 mV 的源信号才能提供 5 V 输出；连接反馈后，5 V 输出需要 100 mV 的源信号。请确定该放大器的 A、β、Aβ 值、闭环增益和反馈量（dB）。

D 11.8 某放大器的开环增益标称值为 1000，但各单位之间的变化可高达标称值的±50%。需要对该放大器进行负反馈，以便将由此产生的反馈放大器的闭环增益变化限制在 ±1%。可以实现的最大闭环增益标称值是多少？现在，如果将其中三个反馈放大器级联，那么级联放大器的增益标称值是多少？该增益的预期变化率是多少？

11.9 图 P11.9 电路中的运算放大器具有开路电压增益 μ、差分输入电阻 R 和可忽略不计的较小输出电阻。它以非反相配置与一个由分压器 (R, R) 组成的反馈网络连接。虽然 β 仍由分压器比决定 [即 β = R/(R+ R)]，但开环增益 A 不再简单地等于 μ。这是因为反馈网络现在负载放大器的输入（因为 R 有限）。要确定 A 的值，请使用第 11.1.3 节中概述的方法来确定环路增益 Aβ。因此可以得出

$$A = \mu \frac{R}{R + (R \| R)}$$

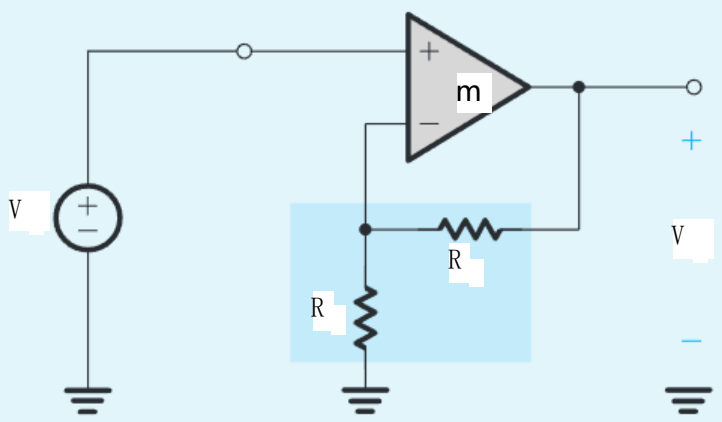


图 P11.9

第 11.2 节：负反馈的一些特性

11.10 对于图 11.1 中的负反馈环路，找出闭环增益对开环增益 [即 (dA/A)/(dA/A)] 的灵敏度为 -40 dB 的环路增益 Aβ。Aβ 的灵敏度变为 1/5 的值是多少？

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

D 11.11 A designer is considering two possible designs of a feedback amplifier. The ultimate goal is $A_f = 10$ V/V. One design employs an amplifier for which $A = 1000$ V/V and the other uses $A = 500$ V/V. Find β and the desensitivity factor in both cases. If the $A = 1000$ amplifier units have a gain uncertainty of $\pm 10\%$, what is the gain uncertainty for the closed-loop amplifiers utilizing this amplifier type? If the same result is to be achieved with the $A = 500$ amplifier, what is the maximum allowable uncertainty in its gain?

D 11.12 A designer is required to achieve a closed-loop gain of $10 \pm 0.1\%$ V/V using a basic amplifier whose gain variation is $\pm 10\%$. What nominal value of A and β (assumed constant) are required?

D 11.13 A circuit designer requires a gain of $25 \pm 1\%$ V/V using an amplifier whose gain varies by a factor of 10 over temperature and time. What is the lowest gain required? The value of β? (*Hint*: Since the change in the open-loop gain is very large, do *not* use differential analysis.)

D 11.14 A power amplifier employs an output stage whose gain varies from 2 to 12 for various reasons. What is the gain of an ideal (nonvarying) amplifier connected to drive it so that an overall gain with feedback of $100 \pm 5\%$ V/V can be achieved? What is the value of β to be used? What are the requirements if A_f must be held within $\pm 0.5\%$? For each of these situations, what preamplifier gain and feedback factor β are required if A_f is to be 10 V/V (with the two possible tolerances)? (*Hint*: Since the change in the open-loop gain is very large, do *not* use differential analysis.)

D 11.15 It is required to design an amplifier with a gain of 100 that is accurate to within $\pm 1\%$. You have available amplifier stages with a gain of 1000 that is accurate to within $\pm 30\%$. Provide a design that uses a number of these gain stages in cascade, with each stage employing negative feedback of an appropriate amount. Obviously, your design should use the lowest possible number of stages while meeting specification.

D *11.16 It is required to design an amplifier to have a nominal closed-loop gain of 10 V/V using a battery-operated amplifier whose gain reduces to half its normal full-battery value over the life of the battery. If only 2% drop in closed-loop gain is desired, what nominal open-loop amplifier gain must be used in the design? (Note that since the change in A is large, it is inaccurate to use differentials.) What value of β should be chosen? If component-value variation in the β network may produce as much as a $\pm 1\%$ variation in β, to what value must A be raised to ensure the required minimum gain?

D 11.17 Design a feedback amplifier that has a closed-loop gain of 100 V/V and is relatively insensitive to change in basic-amplifier gain. In particular, it should provide a reduction in A_f to 99 V/V for a reduction in A to one-tenth its nominal value. What is the required loop gain? What nominal value of A is required? What value of β should be used? What would the closed-loop gain become if A were increased tenfold? If A were made infinite?

11.18 Consider an amplifier having a midband gain A_M and a low-frequency response characterized by a pole at $s = -\omega_L$ and a zero at $s = 0$. Let the amplifier be connected in a negative-feedback loop with a feedback factor β. Find an expression for the midband gain and the lower 3-dB frequency of the closed-loop amplifier. By what factor have both changed?

11.19 A capacitively coupled amplifier has a midband gain of 1000 V/V, a single high-frequency pole at 10 kHz, and a single low-frequency pole at 100 Hz. Negative feedback is employed so that the midband gain is reduced to 10. What are the upper and lower 3-dB frequencies of the closed-loop gain?

D 11.20 Low-cost audio power amplifiers often avoid direct coupling of the loudspeaker to the output stage because any resulting dc bias current in the speaker can use up (and thereby waste) its limited mechanical dynamic range. Unfortunately, the coupling capacitor needed can be large! But feedback helps. For example, for an 8-Ω loudspeaker and $f_L = 100$ Hz, what size capacitor is needed? Now, if feedback is arranged around the amplifier and the speaker so that a closed-loop gain $A_f = 10$ V/V is obtained from an amplifier whose open-loop gain is 1000 V/V, what value of f_{Lf} results? If the ultimate product-design specification requires a 50-Hz cutoff, what capacitor can be used?

D *11.21 It is required to design a dc amplifier with a low-frequency gain of 1000 and a 3-dB frequency of 1 MHz. You have available gain stages with a gain of 1000 but with a dominant high-frequency pole at 20 kHz. Provide a design that employs a number of such stages in cascade, each with negative feedback of an appropriate amount. Use identical stages.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

D 11.11 某设计人员正在考虑两种可能的反馈放大器设计。最终目标是 A= 10 V/V。一种设计采用 A= 1000 V/V 的放大器，另一种设计采用 A= 500 V/V。求这两种情况下的 β 和灵敏度降低系数。如果 A = 1000 放大器单元的增益不确定度为 ±10%，那么使用这种放大器类型的闭环放大器的增益不确定度是多少？如果使用 A = 500 放大器也能获得同样的结果，那么其增益的最大允许不确定度是多少？

D 11.12 设计人员需要使用增益变化为 ± 10% 的基本放大器，实现 10 ± 0.1% V/V 的闭环增益。要求 A 和 β（假定为常数）的标称值是多少？

D 11.13 某电路设计师要求使用增益随温度和时间变化系数为 10 的放大器，其增益为 25 ± 1% V/V。要求的最低增益是多少？β 的值？(提示：由于开环增益的变化非常大，因此不要使用差分分析）。

D 11.14 某功率放大器的输出级由于各种原因增益从 2 到 12 不等。连接一个理想的（无变化的）放大器来驱动它，从而实现 100 ± 5% V/V 的反馈总增益，那么该放大器的增益是多少？使用的 β 值是多少？如果 A 必须保持在 ±0.5% 以内，要求是什么？在上述每种情况下，如果 A 要达到 10 V/V（有两种可能的公差），需要多大的前置放大器增益和反馈系数 β？(提示：由于开环增益的变化非常大，请勿使用差分分析）。

D 11.15 需要设计一个增益为 100 的放大器，其精度应在 ±1%以内。您现有增益为 1000 的放大器级，精度在 ±30% 以内。请提供一个使用若干级联增益级的设计，每个级采用适当数量的负反馈。显然，您的设计应在符合规格的前提下使用尽可能少的级数。

D *11.16 要求使用电池供电的放大器设计一个标称闭环增益为 10 V/V 的放大器，该放大器的增益在电池寿命期间会降低到正常满电池值的一半。如果只希望闭环增益下降 2%，设计中必须使用多大的标称开环放大器增益？(请注意，由于 A 的变化很大，使用差值是不准确的。

）应选择多大的β 值？如果 β 网络中的元件值变化可能会产生高达 ±1% 的 β 变化，那么必须将 A 提高到什么值才能确保所需的最小增益？

D 11.17 设计一个反馈放大器，其闭环增益为 100 V/V，对基本放大器增益的变化相对不敏感。特别是，当 A 减少到额定值的十分之一时，A 应减少到 99 V/V。所需的环路增益是多少？需要多大的 A 标称值？应使用多大的β 值？如果将 A 增大十倍，闭环增益会变成多少？如果将 A 设为无限大？

11.18 考虑一个放大器，其中频增益为 A，低频响应的特点是 s = -ω 时为极点，s = 0 时为零。两者的变化系数是多少？

11.19 某电容耦合放大器的中频增益为 1000 V/V，在 10 kHz 处有一个高频极点，在 100 Hz 处有一个低频极点。采用负反馈后，中频增益降至 10。闭环增益的上限和下限 3 分贝频率分别是多少？

D 11.20 低成本音频功率放大器通常避免扬声器与输出级直接耦合，因为扬声器中产生的任何直流偏置电流都会耗尽（从而浪费）其有限的机械动态范围。遗憾的是，所需的耦合电容器可能很大！不过，反馈也有帮助。例如，对于 8 扬声器和 f= 100 Hz，需要多大的电容器？现在，如果在放大器和扬声器周围布置反馈，以便从开环增益为 1000 V/V 的放大器中获得闭环增益 A= 10 V/V，那么需要多大尺寸的电容器？如果最终的产品设计规格要求 50Hz 截止，可以使用什么电容器？

D *11.21 需要设计一个低频增益为 1000、3 分贝频率为 1 MHz 的直流放大器。现有增益级的增益为 1000，但在 20 kHz 处有一个主要的高频极点。请设计一个采用多级级联的增益级，每级都有适当数量的负反馈。使用相同的级。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

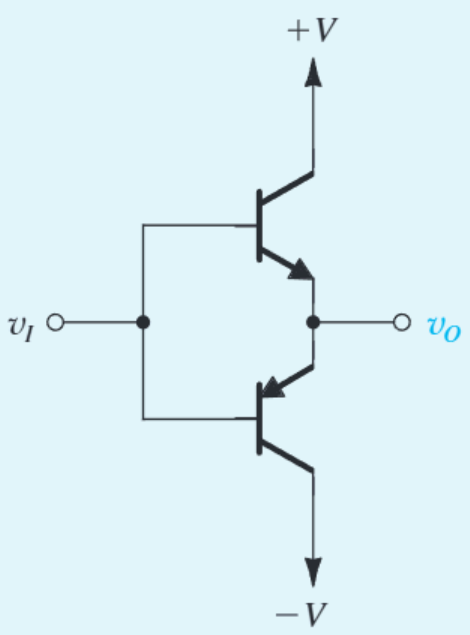
Hint: The 3-dB frequency of a cascade of N identical gain stages, each with a 3-dB frequency $f_{3dB}|_{stage}$ is given by

$$f_{3dB}|_{cascade} = f_{3dB}|_{stage}\sqrt{2^{1/N}-1}$$

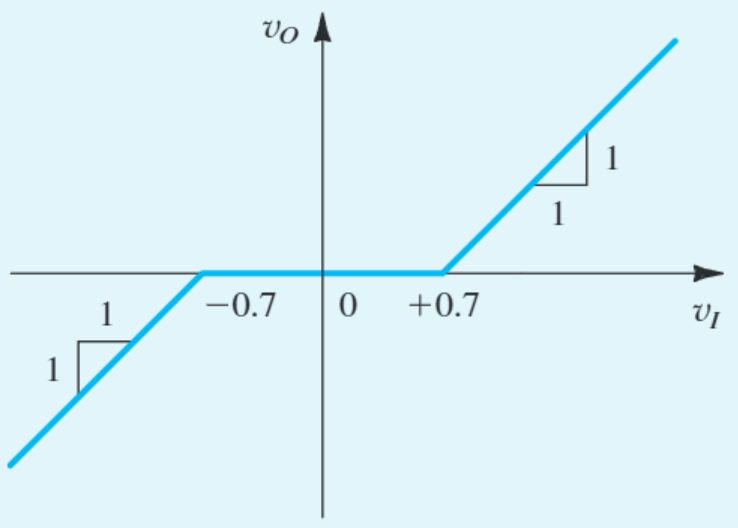
D 11.22 Design a supply-ripple-reduced power amplifier for which the output stage can be modeled by the block diagram of Fig. 11.5, where $A_1 = 0.9$ V/V, and the power-supply ripple $V_N = \pm 1$ V. A closed-loop gain of 10 V/V is desired. What is the gain of the low-ripple preamplifier needed to reduce the output ripple to ±100 mV? To ±10 mV? To ±1 mV? For each case, specify the value required for the feedback factor β.

D 11.23 A feedback amplifier is to be designed using a feedback loop connected around a two-stage amplifier. The first stage is a direct-coupled, small-signal amplifier with a high upper 3-dB frequency. The second stage is a power-output stage with a midband gain of 10 V/V and upper and lower 3-dB frequencies of 8 kHz and 80 Hz, respectively. The feedback amplifier should have a midband gain of 100 V/V and an upper 3-dB frequency of 40 kHz. What is the required gain of the small-signal amplifier? What value of β should be used? What does the lower 3-dB frequency of the overall amplifier become?

***11.24** The complementary BJT follower shown in Fig. P11.24(a) has the approximate transfer characteristic shown in Fig. P11.24(b). Observe that for $-0.7\text{ V} \le v_I \le +0.7\text{ V}$, the output is zero. This "dead band" leads to crossover distortion (see Section 12.3). Consider this follower to be driven by the output of a differential amplifier of gain 100



(a)



(b)

Figure P11.24 *continued*

whose positive-input terminal is connected to the input signal source v_s and whose negative-input terminal is connected to the emitters of the follower. Sketch the transfer characteristic v_o versus v_s of the resulting feedback amplifier. What are the limits of the dead band, and what are the gains outside the dead band?

D 11.25 A particular amplifier has a nonlinear transfer characteristic that can be approximated as follows:

(a) For small input signals, $|v_I| \le 10$ mV, $v_o/v_I = 10^3$.
(b) For intermediate input signals, $10\text{ mV} \le |v_I| \le 60$ mV, $\Delta v_o/\Delta v_I = 10^2$.
(c) For large input signals, $|v_I| \ge 60$ mV, the output saturates.

If the amplifier is connected in a negative-feedback loop, find the feedback factor β that reduces the factor-of-10 change in gain (occurring at $|v_I| = 10$ mV) to only a 10% change. What is the transfer characteristic v_o versus v_s of the amplifier with feedback?

Section 11.3: The Feedback Voltage Amplifier

D 11.26 For the feedback voltage amplifier of Fig. 11.8(a), let the op amp have an infinite input resistance, a zero output resistance, and a finite open-loop gain of 1000 V/V. If $R_1 = 10$ kΩ, what value should R_2 have to obtain an ideal closed-loop gain of 10? Now, calculate the loop gain $A\beta$ and use it to find the actual value of the closed-loop gain A_f. If A_f is to be exactly 10, what must the value of R_2 be?

D 11.27 Consider the series–shunt feedback amplifier in Fig. 11.11(a), which is analyzed in Example 11.3.

(a) If $R_1 = 10$ kΩ, find the value of R_2 that results in an ideal closed-loop gain of 10.

Figure P11.24

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

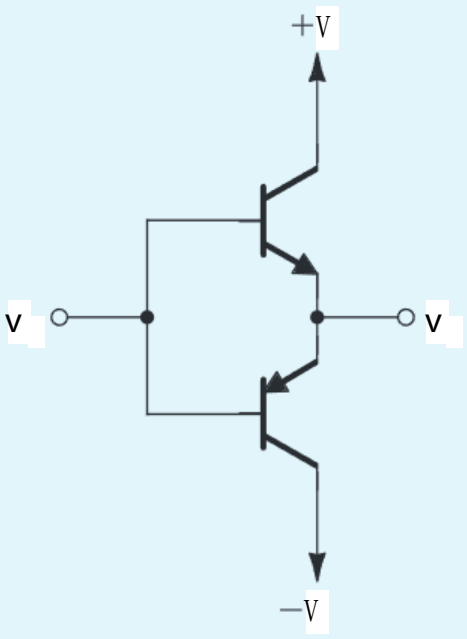
提示：由 N 个相同增益级（每个增益级的 3 分贝频率为 f|）组成的级联的 3 分贝频率为

$$f_{|级联} = f_{|}\sqrt{2^{1/N} - 1}$$

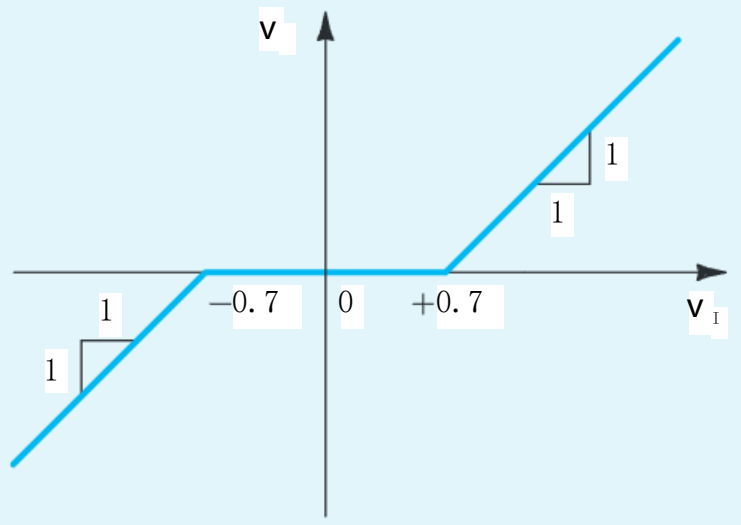
D 11.22 设计一款降低电源纹波的功率放大器，其输出级可用图 11.5 框图建模，其中 A= 0.9 V/V，电源纹波 V= ± 1 V。希望闭环增益为 10 V/V。将输出纹波降至 ±100 mV 所需的低纹波前置放大器增益是多少？降至 ±10 mV？降至 ±1 mV？请说明每种情况下所需的反馈系数 β 值。

D 11.23 在设计反馈放大器时，要使用一个围绕两级放大器连接的反馈回路。第一级是直接耦合的小信号放大器，具有较高的上限 3 分贝频率。第二级是功率输出级，中段增益为 10 V/V，上、下 3-dB 频率分别为 8 kHz 和 80 Hz。反馈放大器的中频增益应为 100 V/V，上 3-dB 频率为 40 kHz。小信号放大器所需的增益是多少？应使用多大的 β 值？整个放大器的较低 3 分贝频率是多少？

*11.24 图 P11.24(a) 所示的互补 BJT 随动器的近似传输特性如图 P11.24(b) 所示。请注意，当 -0.7 V ≤ v≤ +0.7 V 时，输出为零。这个 "死区 "会导致分频失真（见第 12.3 节）。考虑用增益为 100 的差分放大器的输出来驱动该跟随器



(a)



(b)

图 P11.24

图 P11.24 续

的正输入端与输入信号源 v 相连，负输入端与跟随器的发射极相连。请勾画出由此产生的反馈放大器的传递特性 vvers 与 v 的关系。死区的界限是什么？死区以外的增益是多少？

D 11.25 某放大器的非线性传输特性近似如下：

(a) 对于小输入信号、 | | ≤ 10 mV，v/v= 10 。

(b) 对于中间输入信号，10 mV ≤ 10 mV | | ≤ 60 mV、v/v= 10 。

(c) 对于大输入信号、 | | ≥ 60 mV 时，输出达到饱和。

如果放大器连接在负反馈环路中，请找出能降低增益 10 倍变化的反馈系数 β（发生在 | | = 10 mV）的变化仅为 10%。带反馈的放大器的传递特性 vvers 对 v 是多少？

第 11.3 节：反馈电压放大器

D 11.26 对于图 11.8(a)的反馈电压放大器，让运算放大器具有无限大输入电阻、零输出电阻和 1000 V/V 的有限开环增益。如果 R= 10 k，R 的值应为多少才能获得 10 的理想闭环增益？现在，计算环路增益 Aβ，并以此求出闭环增益 A 的实际值。如果 A 恰好为 10，那么 R 的值必须是多少？

D 11.27 考虑图 11.11(a) 中的串并联反馈放大器，例 11.3 对其进行了分析。

(a) 如果 R= 10 k，求理想闭环增益为 10 的 R 值。

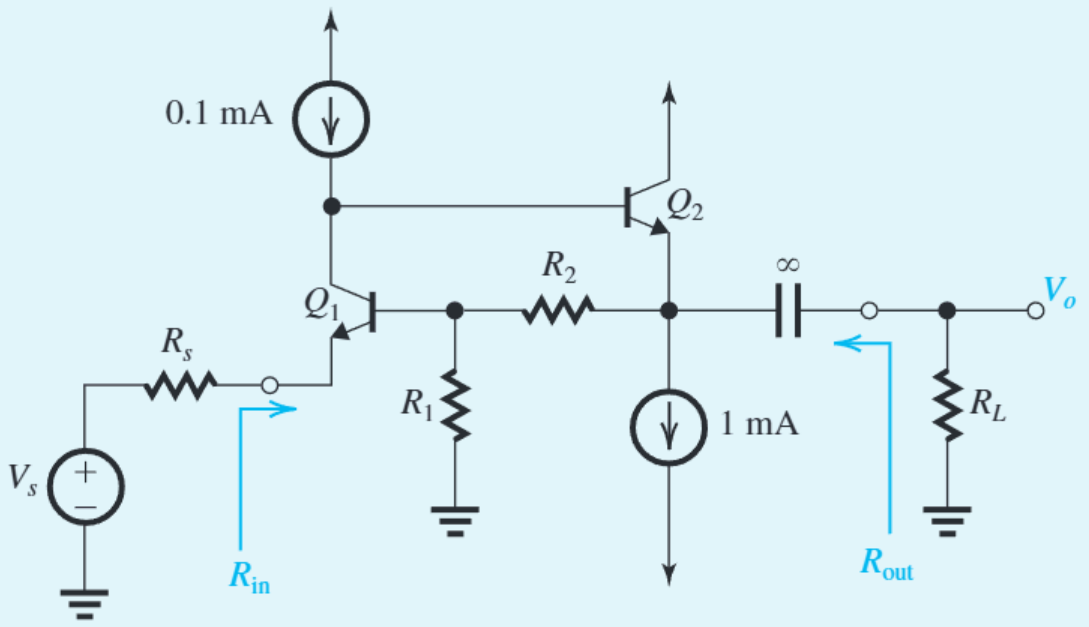


Figure P11.29

(b) Use the expression for $A\beta$ derived in Example 11.3 to find the value of the loop gain for the case $\mu = 1000$, $R_{id} = 100\ \text{k}\Omega$, $r_o = 1\ \text{k}\Omega$, $R_s = 100\ \text{k}\Omega$, and $R_L = 10\ \text{k}\Omega$. Hence determine the value of the closed-loop gain A_f.

(c) By what factor must μ be increased to ensure that A_f is within 1% of the ideal value of 10?

D 11.28 Consider the series–shunt feedback amplifier of Fig. 11.8(b) that is analyzed in Example 11.2.

(a) If $R_1 = 1\ \text{k}\Omega$, what value should R_2 have to obtain a closed-loop gain whose ideal value is 5 V/V?

(b) If $g_{m1} = g_{m2} = 4\ \text{mA/V}$, $R_{D1} = R_{D2} = 10\ \text{k}\Omega$, and the MOSFET's r_o is very large, use the expression for $A\beta$ derived in Example 11.2 to find the value of $A\beta$ and hence determine the closed-loop gain A_f.

***11.29** In the series–shunt feedback amplifier shown in Fig. P11.29, the devices operate with $V_{BE} = 0.7$ V and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100\ \Omega$, $R_1 = 1\ \text{k}\Omega$, $R_2 = 10\ \text{k}\Omega$, and $R_L = 1\ \text{k}\Omega$.

(a) If the loop gain is large, what do you expect the closed-loop gain to be? Give both an expression and its value.

(b) Find the dc emitter current in each of Q_1 and Q_2. Also, find the dc voltage at the emitter of Q_2.

(c) Calculate the value of the loop gain $A\beta$. (*Hint*: Set $V_s = 0$ and break the loop at the base of Q_1. Simplify the circuit by eliminating dc sources.)

(d) Calculate the value of A_f.

D 11.30 Consider the series–shunt feedback amplifier of Fig. 11.8(c), which was the subject of Exercise 11.6. Assume that the voltage divider (R_1, R_2) is implemented with a 1-MΩ potentiometer. Assume that the MOSFET is biased so that $g_m = 4$ mA/V and r_o is large. Also, $R_D = 10\ \text{k}\Omega$. Find the value of R_1 that results in a closed-loop gain of 5 V/V.

D 11.31 Figure P11.31 shows a series–shunt feedback amplifier known as a "feedback triple." All three MOSFETs are biased to operate at $g_m = 4$ mA/V. You may neglect their r_o's.

(a) Select a value for R_F that results in a closed-loop gain that is ideally 10 V/V.

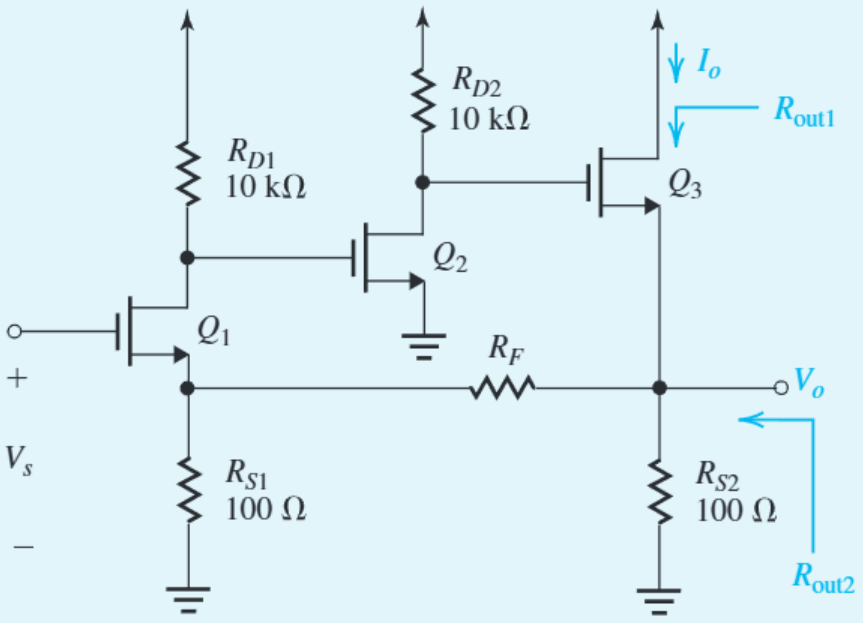


Figure P11.31

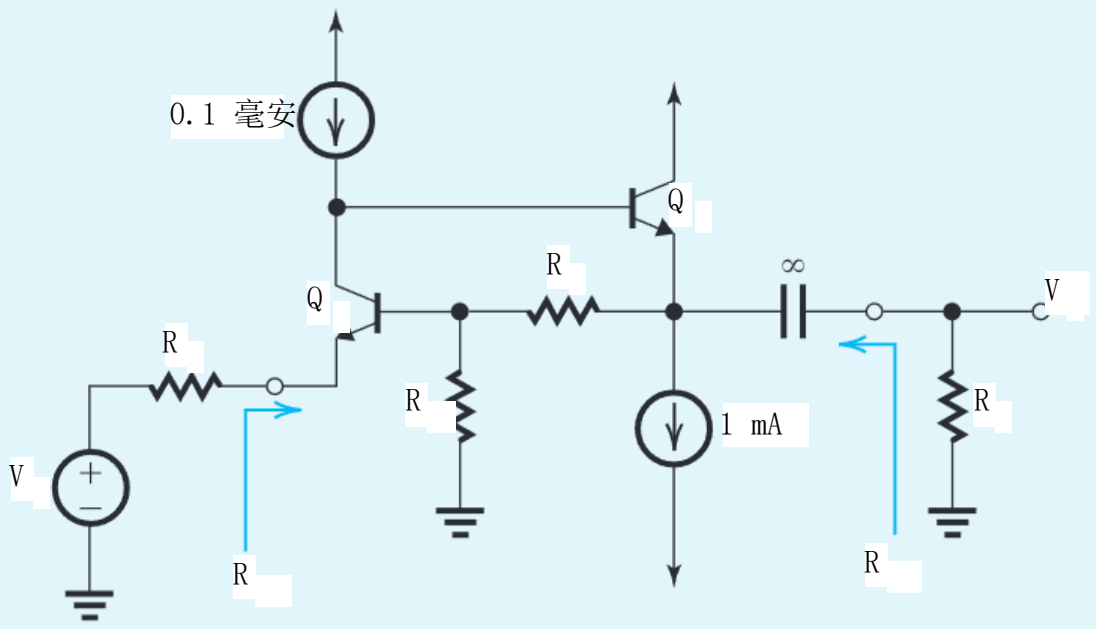


图 P11.29

(b) 利用例 11.3 中得出的 Aβ 表达式，找出 μ = 1000、R= 100 k、r= 1 k、R= 100 k 和 R= 10 k 情况下的环路增益值。

从而确定闭环增益 A 的值。

(c) μ 必须增加多少系数才能确保 A 在理想值 10 的 1%以内？

D 11.28 考虑例 11.2 中分析的图 11.8(b) 的串并联反馈放大器。

(a) 如果 R= 1 k，Rh 取多大值才能获得理想值为 5 V/V 的闭环增益？(b) 如果 g = g= 4 mA/V，R= R= 10 k，且 MOSFET 的 Ris 非常大，请使用例 11.2 中推导的 Aβ 表达式求出 Aβ 值，从而确定闭环增益 A。

*11.29 在图 P11.29 所示的串并联反馈放大器中，器件的工作电压 V= 0.7 V，β= β= 100。输入信号 V 的直流分量为零。电阻 R = 100、R = 1 k、R = 10 k 和 R = 1 k。

(a) 如果环路增益很大，你预计闭环增益会是多少？请给出表达式及其数值。(b) 求 Q 和 Q 的直流发射极电流，同时求 Q 发射极的直流电压。(提示：设置 V= 0 并在 Q 的基极断开回路。）(d) 计算 A 的值。

D 11.30 考虑图 11.8(c) 的串并联反馈放大器，它是练习 11.6 的主题。假设分压器 (R, R) 由 1-M 电位器实现。假设 MOSFET 的偏压为 g= 4 mA/V，且 r 较大。另外，R= 10 k。请找出能使闭环增益达到 5 V/V 的 R 值。

D 11.31 图 P11.31 显示了一个串联并联反馈放大器，称为"反馈三极管"。所有三个 MOSFET 的偏置工作电压均为 g= 4 mA/V。您可以忽略它们的 r 值。

(a) 选择一个 R 值，使闭环增益最好为 10 V/V。

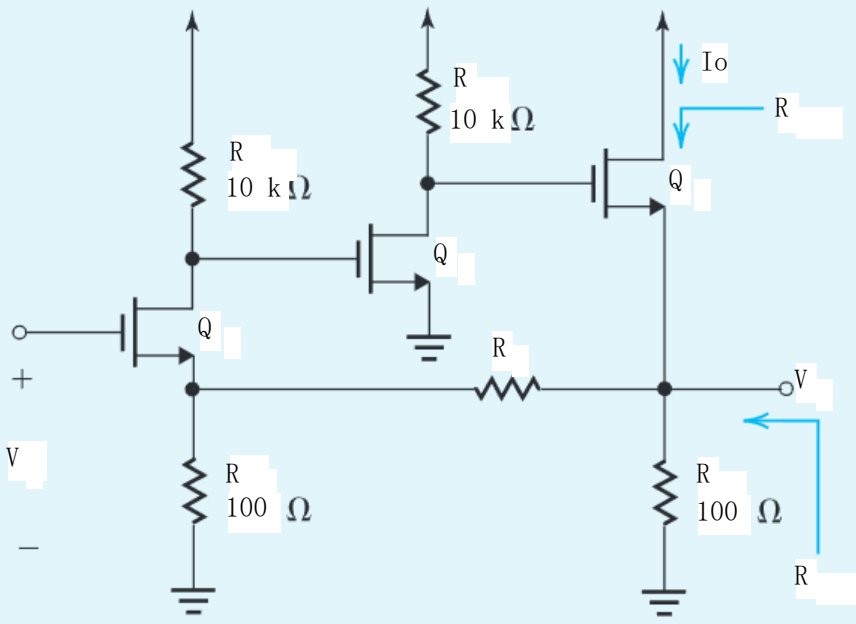


图 P11.31

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

(b) Determine the loop gain $A\beta$ and hence the value of A_f. By what percentage does A_f differ from the ideal value you designed for? How can you adjust the circuit to make A_f equal to 10?

11.32 Figure P11.32 shows a series–shunt feedback amplifier without details of the bias circuit.

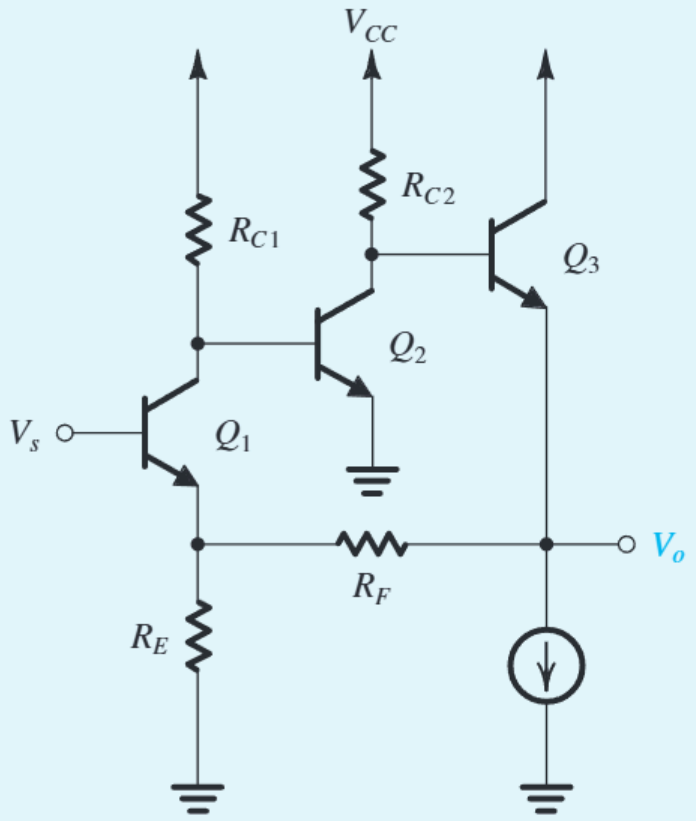


Figure P11.32

(a) If R_E is selected to be 50 Ω, find the value for R_F that results in a closed-loop gain with an ideal value of 25 V/V.

(b) If Q_1 is biased at 1 mA, Q_2 at 2 mA, and Q_3 at 5 mA, and assuming that the transistors have $h_{fe} = 100$ and large r_o, and that $R_{C1} = 2$ kΩ and $R_{C2} = 1$ kΩ, find the value of the loop gain $A\beta$ and hence of the closed-loop gain A_f.

D 11.33 The current-mirror-loaded differential amplifier in Fig. P11.33 has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1$ MΩ. The devices are sized to operate at $|V_{OV}| = 0.2$ V. For all devices, $|V_A| = 10$ V. The input signal source has a zero dc component.

(a) Find the loop gain $A\beta$ and hence the value of A.

(b) Find the values of R_1 and R_2 that result in a closed-loop gain of exactly 5 V/V.

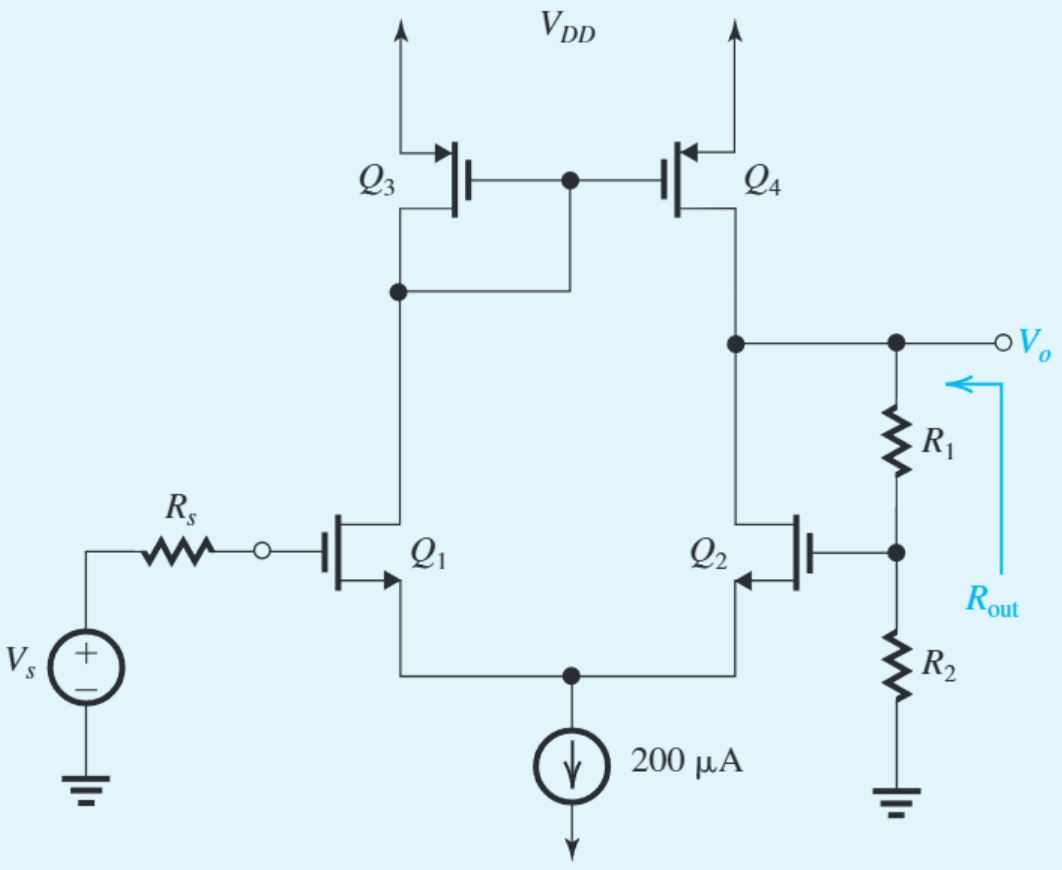


Figure P11.33

Section 11.4: Systematic Analysis of Feedback Voltage Amplifiers (Series–Shunt)

11.34 A series–shunt feedback amplifier employs a basic amplifier with input and output resistances each of 2 kΩ and gain $A = 1000$ V/V. The feedback factor $\beta = 0.1$ V/V. Find

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

(b) 确定环路增益 Aβ，从而确定 A 值。

A 与你设计的理想值相差多少百分比？如何调整电路使 A 等于 10？

11.32 图 P11.32 显示了一个串并联反馈放大器，但没有偏置电路的细节。

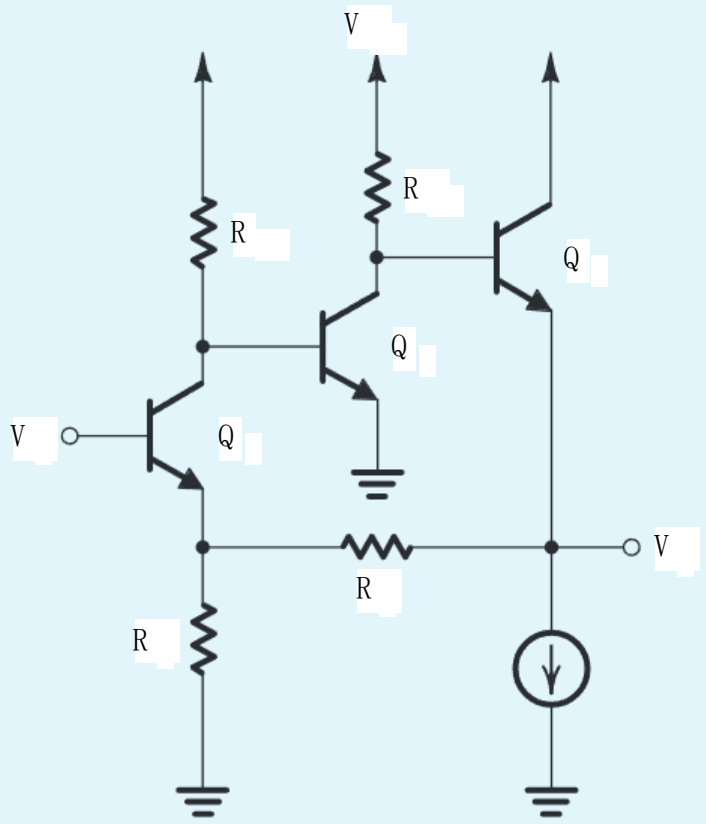


图 P11.32

(a) 如果 Ris 选定为 50，求 R 的值能使闭环增益达到 25 V/V 的理想值。(b) 如果 Q 的偏置电流为 1 mA，Qat 为 2 mA，Qat 为 5 mA，并假设晶体管的 h= 100 和 r 较大，R= 2 k 和 R= 1 k，求环路增益 Aβ，从而求闭环增益 A 的值。

D 11.33 图 P11.33 中的电流镜负载差分放大器的反馈网络由分压器 (R, R) 组成，R+ R= 1 M。器件的工作电压为 |V| = 0.2 V。输入信号源的直流分量为零。

(a) 求出环路增益 Aβ，进而求出 A 的值。 (b) 求出 Rand R 的值，使闭环增益正好为 5 V/V。

第 11.4 节 系统分析反馈电压放大器（串联-并联反馈电压放大器（串联-并联）的系统分析

11.34 串并联反馈放大器采用一个基本放大器，其输入和输出电阻各为 2 k，增益 A = 1000 V/V。反馈系数 β = 0.1 V/V。求

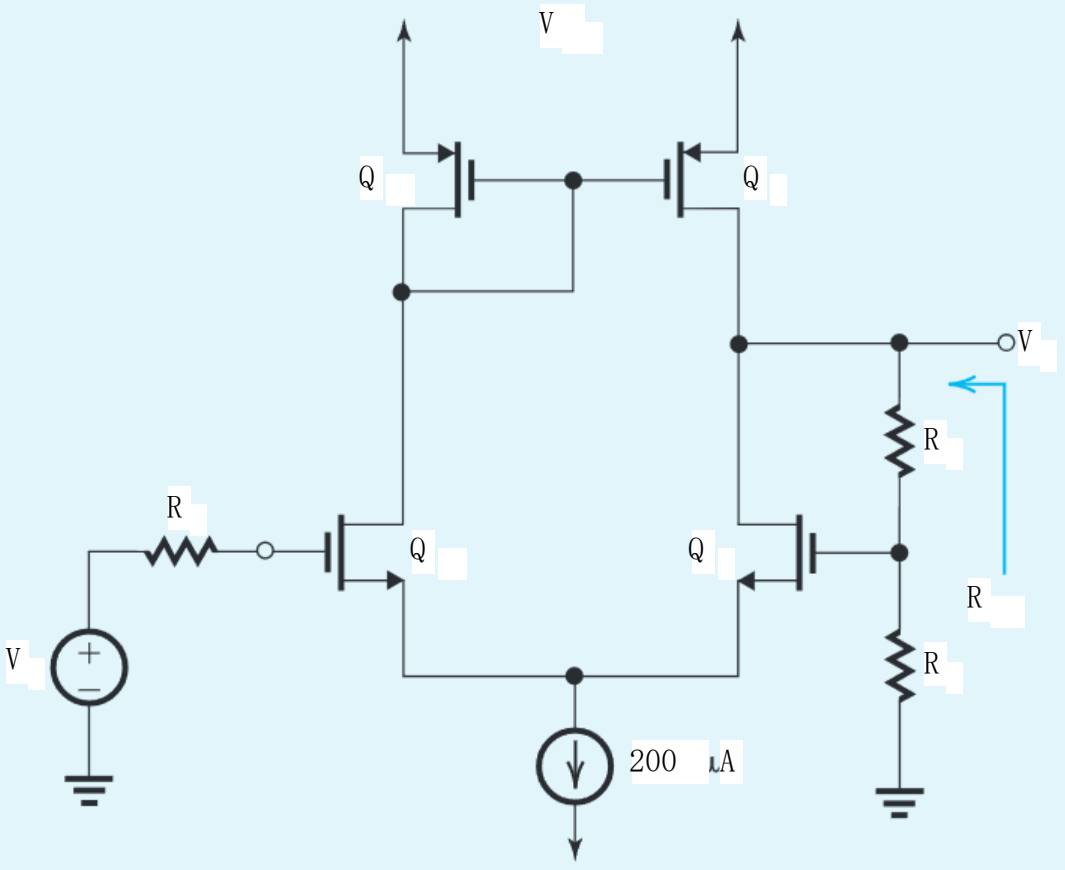


图 P11.33

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

the gain A_f, the input resistance R_{if}, and the output resistance R_{of} of the closed-loop amplifier.

11.35 For a particular amplifier connected in a feedback loop in which the output voltage is sampled, measurement of the output resistance before and after the loop is connected shows a change by a factor of 200. Is the resistance with feedback higher or lower? What is the value of the loop gain $A\beta$? If R_{of} is 100 Ω, what is R_o without feedback?

11.36 The formulas for R_{if} and R_{of} in Eqs. (11.20) and (11.23), respectively, also apply for the case in which A is a function of frequency. In this case, the resulting impedances Z_{if} and Z_{of} will be functions of frequency. Consider the case of a series–shunt amplifier that has an input resistance R_i, an output resistance R_o, and open-loop gain $A = A_0/\left[1 + \left(s/\omega_H\right)\right]$, and a feedback factor β that is independent of frequency. Find Z_{if} and Z_{of} and give an equivalent circuit for each, together with the values of all the elements in the equivalent circuits.

11.37 A feedback amplifier utilizing voltage sampling and employing a basic voltage amplifier with a gain of 1000 V/V and an input resistance of 1000 Ω has a closed-loop input resistance of 10 kΩ. What is the closed-loop gain? If the basic amplifier is used to implement a unity-gain voltage buffer, what input resistance do you expect?

11.38 Consider the noninverting op-amp circuit of Example 11.4 for the case $R_1 = \infty$ and $R_2 = 0$.

(a) What is the value of β, and what is the ideal value of the closed-loop gain?
(b) Adapt the expressions found in Example 11.4 to obtain expressions for A and $A\beta$ for this case.
(c) For $\mu = 10^4$, $R_{id} = 100$ kΩ, $R_s = 10$ kΩ, $r_o = 1$ kΩ, and $R_L = 2$ kΩ, find A, $A\beta$, A_f, R_{in}, and R_{out}.

***11.39** This problem deals with the series–shunt feedback amplifier of Fig. P11.29 and overlaps somewhat with Problem 11.29. Thus, if you have already solved 11.29, you can use some of the results in the solution of this problem. The devices operate with $V_{BE} = 0.7$ V and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100$ Ω, $R_1 = 1$ kΩ, $R_2 = 10$ kΩ, and $R_L = 1$ kΩ.

(a) If the loop gain is large, what do you expect the closed-loop gain V_o/V_s to be? Give both an expression and its approximate value.
(b) Find the dc emitter current in each of Q_1 and Q_2. Also find the dc voltage at the emitter of Q_2.
(c) Sketch the A circuit without the dc sources. Derive expressions for A, R_i, and R_o, and find their values.
(d) Give an expression for β and find its value.
(e) Find the closed-loop gain V_o/V_s, the input resistance R_{in}, and the output resistance R_{out}. By what percentage does the value of A_f differ from the approximate value found in (a)?

SIM **D *11.40** Figure P11.40 shows a series–shunt amplifier with a feedback factor $\beta = 1$. The amplifier is designed so that $v_O = 0$ for $v_S = 0$, with small deviations in v_O from 0 V dc being minimized by the negative-feedback action. The technology utilized has $k_n' = 2k_p' = 120\ \mu\text{A/V}^2$, $|V_t| = 0.7$ V, and $|V_A'| = 24$ V/μm.

(a) Show that the feedback is negative.
(b) With the feedback loop opened at the gate of Q_2, and the gate terminals of Q_1 and Q_2 grounded, find the dc current and the overdrive voltage at which each of Q_1 to Q_5 is operating. Ignore the Early effect. Also find the dc voltage at the output.
(c) Find g_m and r_o of each of the five transistors.
(d) Find expressions and values of A and R_o. Assume that the bias current sources are ideal.
(e) Find the gain with feedback, A_f, and the output resistance R_{out}.
(f) How would you modify the circuit to realize a closed-loop voltage gain of 5 V/V? What is the value of output resistance obtained?

***11.41** Figure P11.41 shows a series–shunt amplifier in which the three MOSFETs are sized to operate at $|V_{OV}| = 0.2$ V. Let $|V_t| = 0.5$ V and $|V_A| = 10$ V. The current sources utilize single transistors and thus have output resistances equal to r_o.

(a) Show that the feedback is negative.
(b) Assuming the loop gain to be large, what do you expect the closed-loop voltage gain V_o/V_s to be approximately?
(c) If V_s has a zero dc component, find the dc voltages at nodes S_1, G_2, S_3, and G_3. Verify that each of the current sources has the minimum required dc voltage across it for proper operation.
(d) Find the A circuit. Calculate the gain of each of the three stages and the overall voltage gain, A.

增益 A、输入电阻 R 和闭环放大器的输出电阻 R。

11.35 某放大器连接了一个反馈回路，在回路中对输出电压进行采样，在连接回路之前和之 后测量输出电阻，结果显示电阻变化了 200 倍。反馈时的电阻是高还是低？回路增益 Aβ 的值是多少？如果是 Ris 100，无反馈时的 R 是多少？

11.36 公式 (11.20) 和 (11.23) 中的 Rand R 公式同样适用于 A 是频率函数的情况。在这种情况下，产生的阻抗 Z 和 Z 将是频率的函数。考虑串并联放大器的情况，该放大器具有输入电阻 R、输出电阻 R 和开环增益 A = A/

[1 +(s/ω)]

以及与频率无关的反馈系数 β 。求 Z 和 Z，并分别给出等效电路，以及等效电路中所有元件的值。

11.37 一个利用电压采样的反馈放大器，采用增益为 1000 V/V、输入电阻为 1000 的基本电压放大器，其闭环输入电阻为 10 k。闭环增益是多少？如果使用基本放大器来实现单增益电压缓冲器，您预计输入电阻是多少？

11.38 在 R= ∞ 和 R= 0 的情况下，考虑例 11.4 中的非反相运算放大器电路。 (a) β 值是多少，闭环增益的理想值是多少？ (b) 根据例 11.4 中的表达式，求出这种情况下 A 和 Aβ 的表达式。(c) 对于 μ = 10、R = 100 k、R = 10 k、r = 1 k 和 R = 2 k，求 A、Aβ、A、R 和 R。

*11.39 本题涉及图 P11.29 中的串并联反馈放大器，与问题 11.29 有一定程度的重叠。因此，如果您已经求解了问题 11.29，则可以在求解本题时使用其中的一些结果。器件的工作电压为 V= 0.7 V， β = β = 100。输入信号 V 的直流分量为零。电阻 R= 100、R= 1 k、R= 10 k 和 R= 1 k。

(a) 如果环路增益很大，你预计闭环增益 V/V 会是多少？请给出表达式及其近似值。

(b) 求 Q 和 Q 的发射极直流电流。推导出 A、R 和 R 的表达式，并求出它们的值。

(d) 给出 β 的表达式并求出其值。

(e) 求闭环增益 V/V、输入电阻 R 和输出电阻 R。

DSIM*11.40 图 P11.40 显示一个反馈系数 β =1 的串并联放大器。放大器的设计使 v= 0 时 v= 0，负反馈作用可将 v 与 0 V 直流的微小偏差降至最低。采用的技术为 k= 2k= 120 μA/V、

| |V | = 0.7 V, and | |V | = 24 V/μm。

(a) 证明反馈是负的。

(b) 在 Q 的栅极打开反馈回路，Q 和 Q 的栅极接地的情况下，求 Q 至 Q 各自工作时的直流电流和过载电压。忽略早期效应。同时求输出端的直流电压。

(c) 求出五个晶体管的 g 和 ro。(d) 假设偏置电流源是理想的，求出 A 和 R 的表达式和数值。(e) 求反馈增益 A 和输出电阻 R。

(f) 如何修改电路以实现 5 V/V 的闭环电压增益？获得的输出电阻值是多少？

*11.41 图 P11.41 显示了一个串联并联放大器，其中三个 MOSFET 的工作频率为 | |V | = 0.2 | |V ，电流源使用单个晶体管和 | |V ，因此输出电阻等于 r 。

(a) 证明反馈是负的。

(b) 假设环路增益很大，你预计闭环电压增益 V/V 大约是多少？(c) 如果 V 的直流分量为零，求节点 S、G、S 和 G 的直流电压。(d) 找出 A 电路。计算三级中每一级的增益和总电压增益 A。

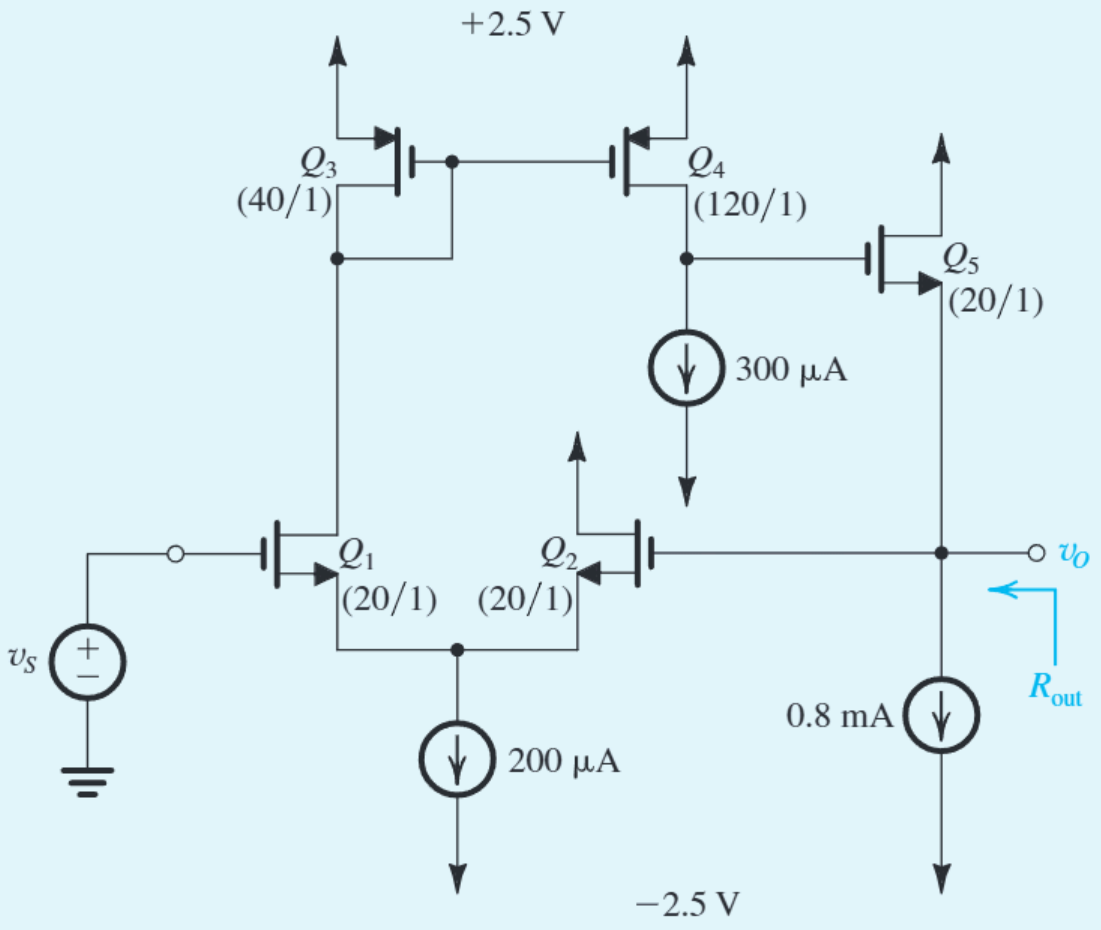


Figure P11.40

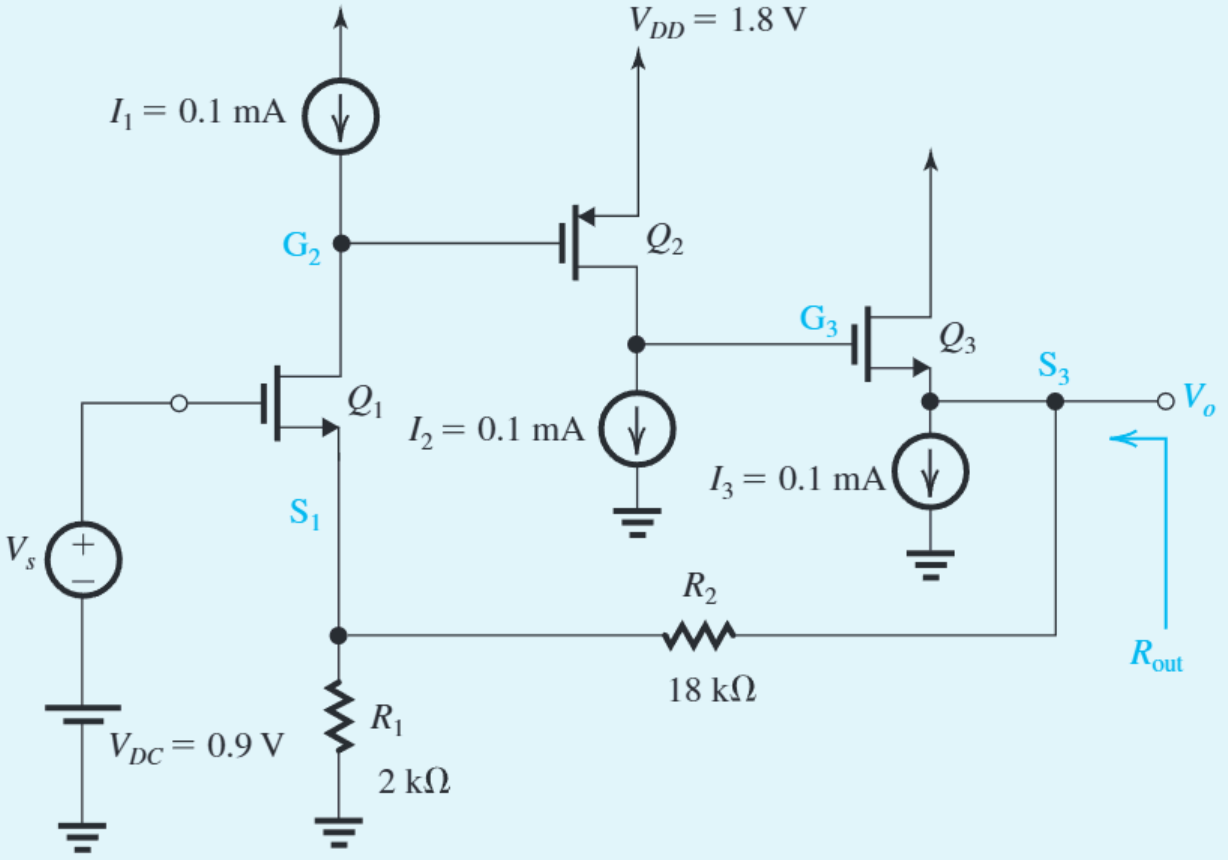


Figure P11.41

[*Hint*: A CS amplifier with a resistance R_s in the source lead has an effective transconductance $g_m/(1+g_mR_s)$ and an output resistance $r_o(1+g_mR_s)$.]

(e) Find β.

(f) Find $A_f = V_o/V_s$. By what percentage does this value differ from the approximate value obtained in (b)?

(g) Find the output resistance R_{out}.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

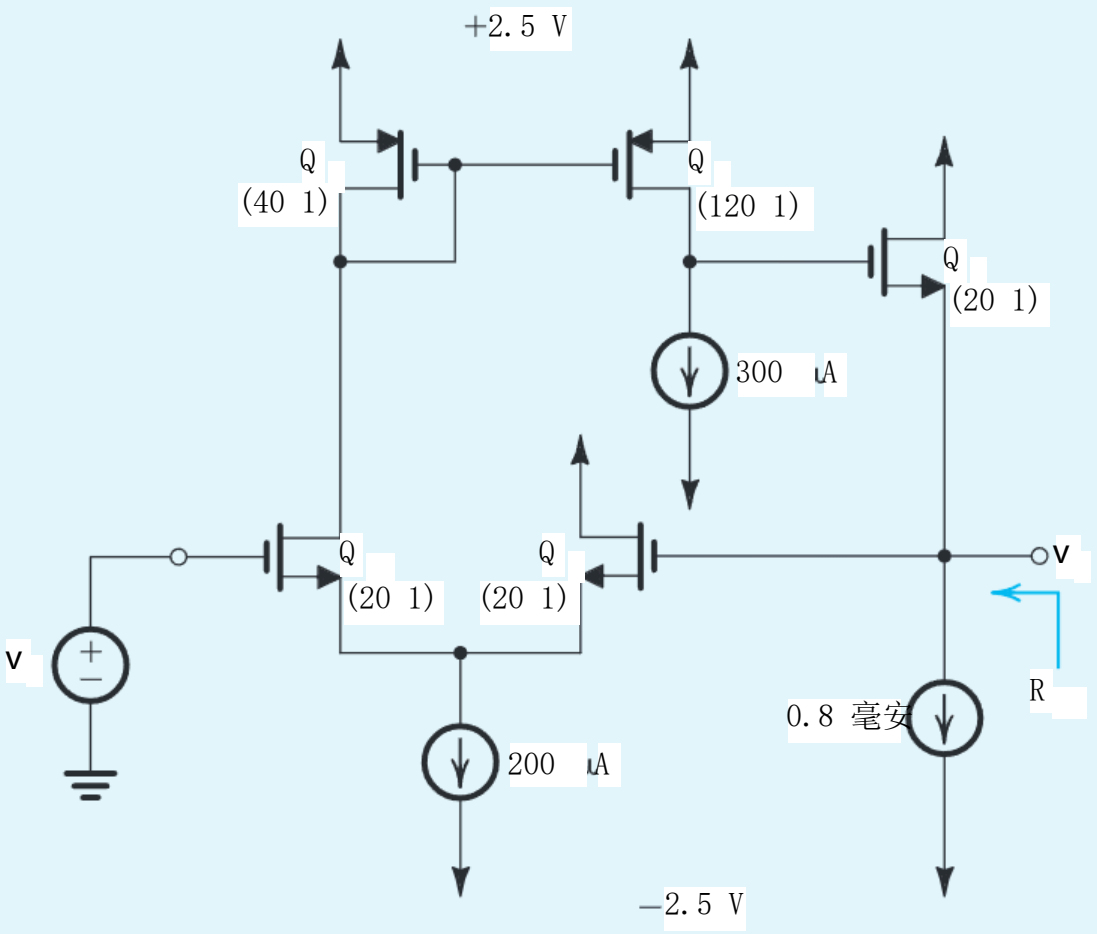


图 P11.40

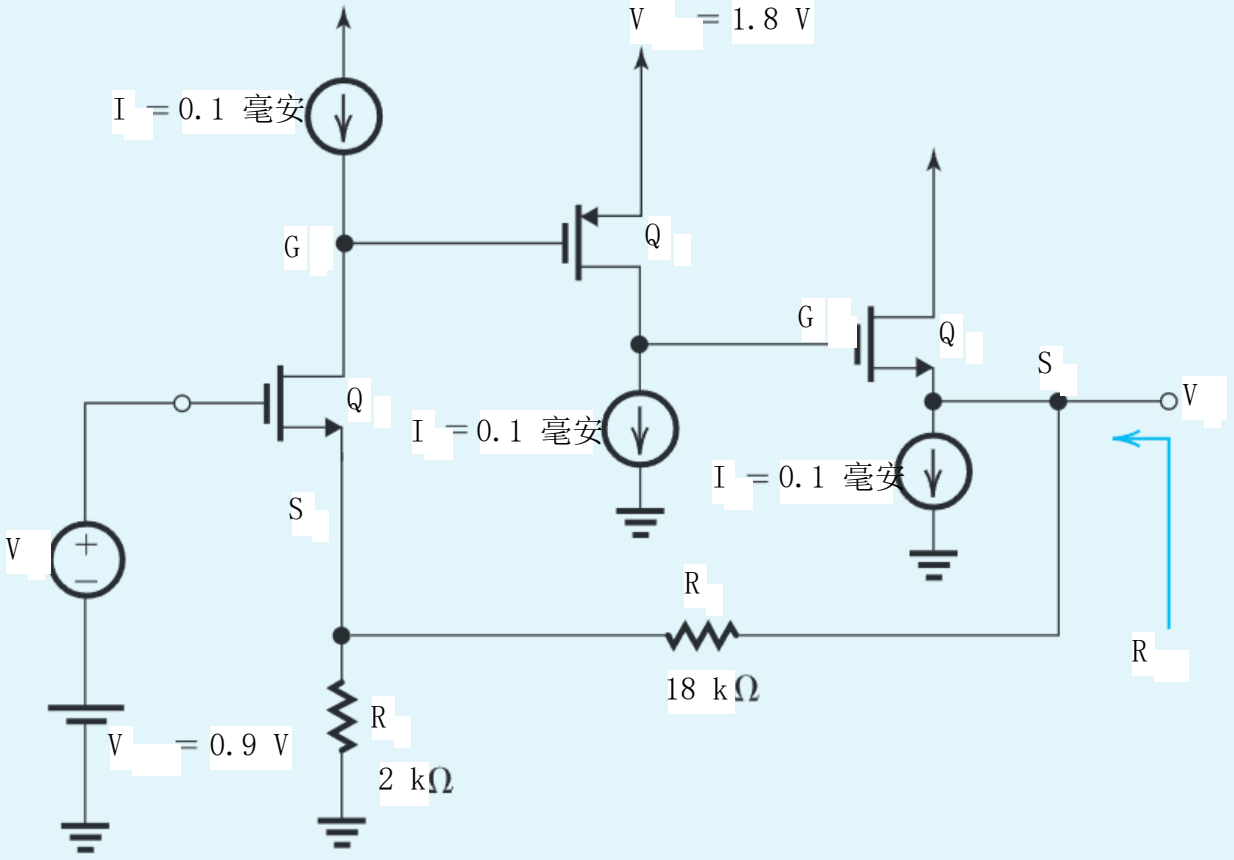


图 P11.41

[提示：带有源极引线电阻 Rin 的 CS 放大器的有效跨导为 $g_m/(1+g_mR)$ 和输出电阻 $r_o(1+g_mR)$。]

(e) 求 β。

(f) 求 A= V/V。该值与 (b) 中得到的近似值相差多少百分比？(g) 求输出电阻 R。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

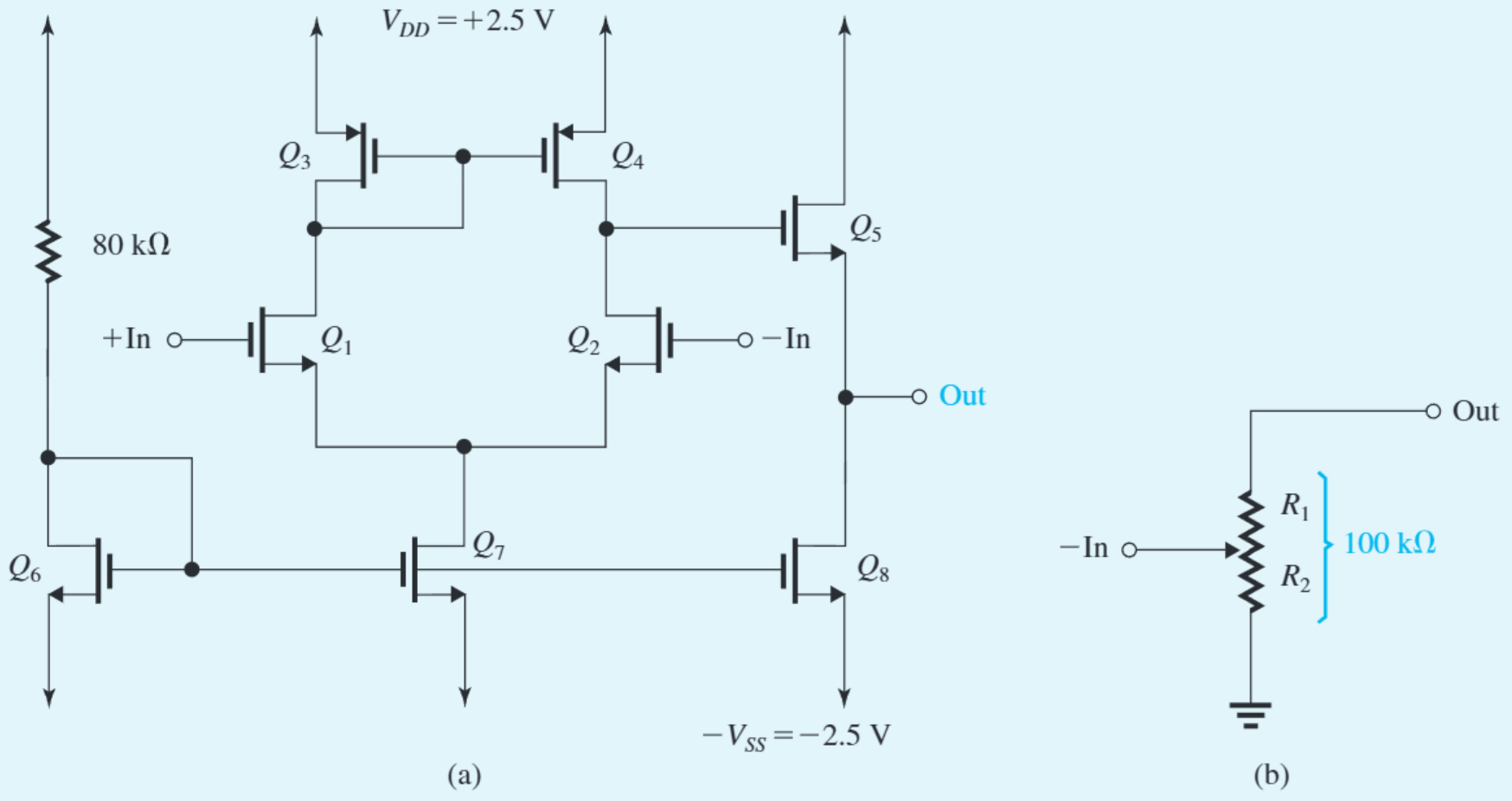


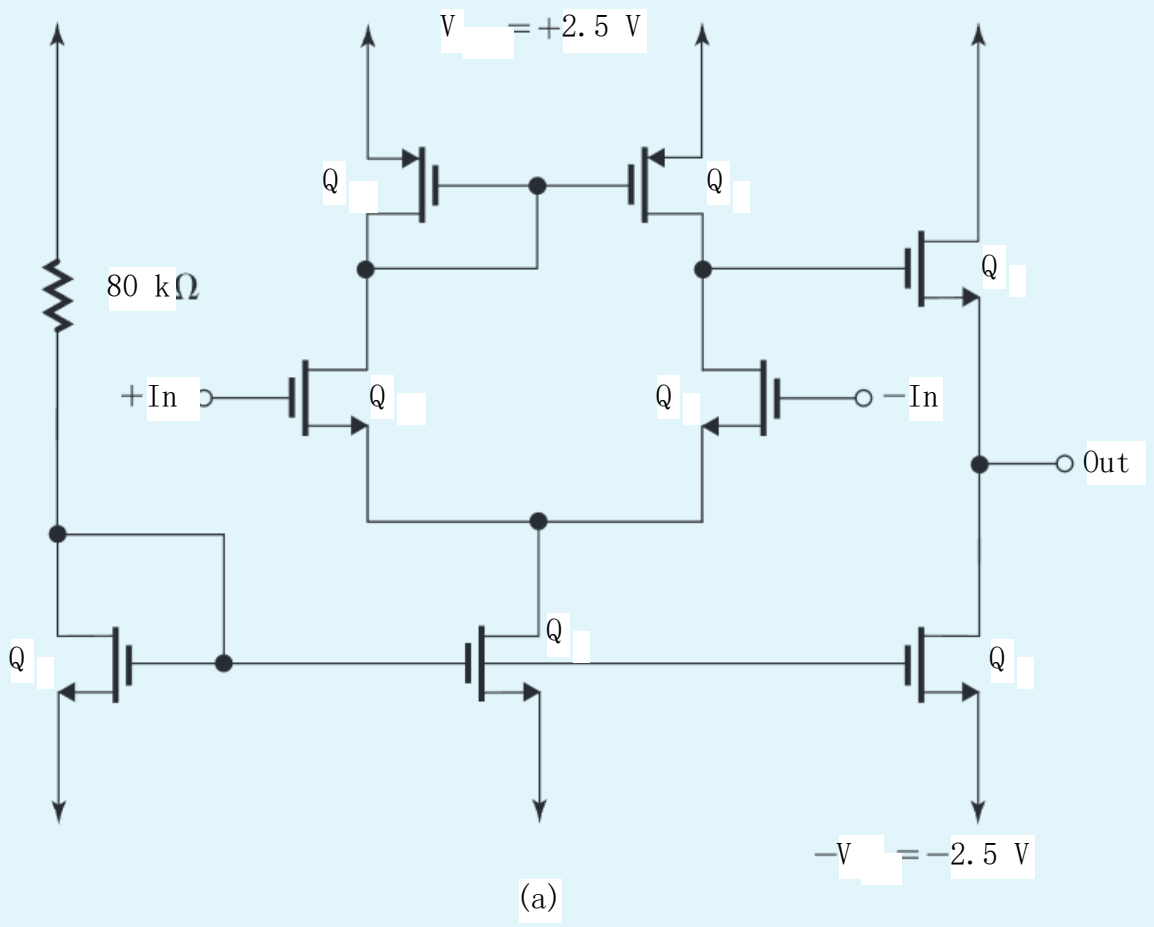
Figure P11.43

D *11.42 This problem deals with the series–shunt feedback amplifier of Fig. P11.33. Certain aspects of this amplifier were considered in Problem 11.33. If you have already solved problem 11.33, you will have the opportunity to compare results. The current-mirror-loaded differential amplifier has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1$ MΩ. The devices are sized to operate at $|V_{OV}| = 0.2$ V. For all devices, $|V_A| = 10$ V. The input signal source has a zero dc component.

(a) Show that the feedback is negative.
(b) What do you expect the dc voltage at the gate of Q_2 to be? At the output? (Neglect the Early effect.)
(c) Find the A circuit. Derive an expression for A and find its value.
(d) Select values for R_1 and R_2 to obtain a closed-loop voltage gain $V_o/V_s = 5$ V/V.
(e) Find the value of R_{out}.
(f) Utilizing the open-circuit, closed-loop gain (5 V/V) and the value of R_{out} found in (e), find the value of gain obtained when a resistance $R_L = 10$ kΩ is connected to the output.
(g) As an alternative approach to (f) above, redo the analysis of the A circuit including R_L. Then utilize the values of R_1 and R_2 found in (d) to determine β and A_f. Compare the value of A_f to that found in (f).

D **11.43 The CMOS op amp in Fig. P11.43(a) is fabricated in a 1-μm technology for which $V_{tn} = -V_{tp} = 0.75$ V, $\mu_n C_{ox} = 2\mu_p C_{ox} = 100$ μA/V^2, and $V_A' = 10$ V/μm. All transistors in the circuit have $L = 1$ μm.

(a) It is required to perform a dc bias design of the circuit. For this purpose, let the two input terminals be at zero volts dc and neglect channel-length modulation (i.e., let $V_A = \infty$). Design to obtain $I_{D1} = I_{D2} = 50$ μA, $I_{D5} = 250$ μA, and $V_O = 0$, and operate all transistors except for the source follower Q_5 at $V_{OV} = 0.25$ V. Assume that Q_1 and Q_2 are perfectly matched, and similarly for Q_3 and Q_4. For each transistor, find I_D and W/L.
(b) What is the allowable range of input common-mode voltage?
(c) Find g_m for each of Q_1, Q_2, and Q_5.
(d) For each transistor, calculate r_o.
(e) The 100-kΩ potentiometer shown in Fig. 11.43(b) is connected between the output terminal (Out) and the inverting input terminal (–In) to provide negative



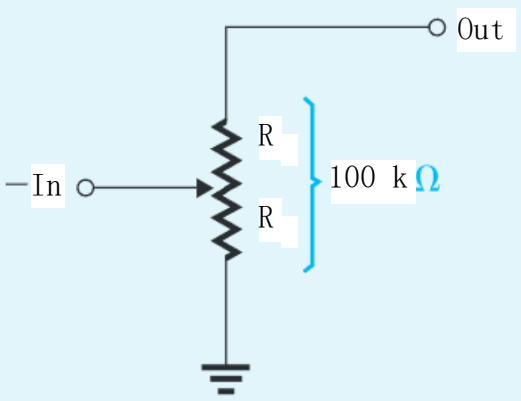


图 P11.43

D *11.42 本问题涉及图 P11.33 中的串并联反馈放大器。问题 11.33 考虑了该放大器的某些方面。如果您已经解决了问题 11.33，您将有机会比较结果。电流镜像加载差分放大器的反馈网络包括分压器

(R, R)

R+ R= 1 M。器件的尺寸为 | | V

| | |

| = 0.2 V.适用于所有设备、V | 输入信号源的直流分量为零。

(a) 证明反馈是负的。

(b) 你预计 Q 的栅极直流电压是多少？输出端呢？(忽略厄尔效应) (c) 找出 A 电路。推导出 A 的表达式并找出其值。(d) 选择 Rand R 的值，以获得闭环电压增益 V/V= 5 V/V。

(e) 求 R 的值。

(f) 利用开路闭环增益（5 V/V）和 (e) 中找到的 R 值，求输出端连接电阻 R = 10 k 时的增益值。(g) 作为上述 (f) 的另一种方法，重新分析包括 R 的 A 电路。然后利用

Rand Rfound in (d) to determine β and A. Compare the value of A to that found in (f).

D **11.43 图 P11.43(a)中的 CMOS 运算放大器采用 1 μm 技术制造，其 V= -V= 0.75 V， μC= 2 μC= 100 μA/V 和 V= 10 V/μm。电路中所有晶体管的 L 均为 1 μm。

(a) 需要对电路进行直流偏置设计。

为此，让两个输入端处于零伏直流，并忽略通道长度调制（即让 V= ∞）。设计得到 I = I = 50 μA、I = 250 μA、V = 0，并在 V= 0.25 V 下操作除源极跟随器 Q 之外的所有晶体管。假设 Q 和 Q 完全匹配，Q 和 Q 也是如此。

求每个晶体管的 I 和 W/L。

(b) 输入共模电压的允许范围是多少？(c) 求 Q、Q 和 Q 的 g 值。

(d) 计算每个晶体管的 r 。

(e) 图 11.43(b)中所示的 100 k 电位器连接在输出端 (Out) 和反相输入端 (-In) 之间，以提供负信号。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

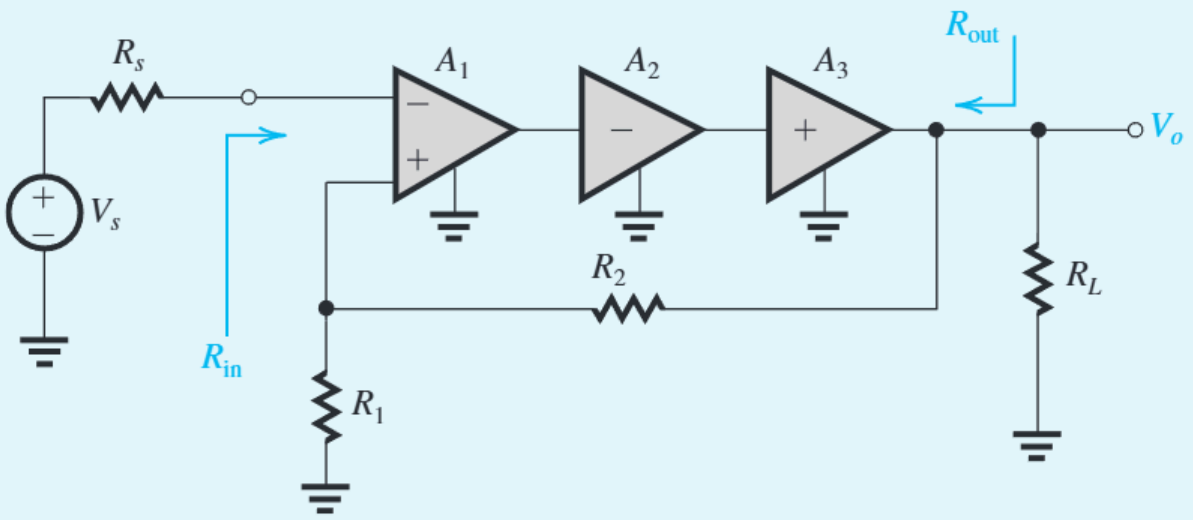


Figure P11.45

feedback whose amount is controlled by the setting of the wiper. A voltage signal V_s is applied between the noninverting input (+In) and ground. A load resistance $R_L = 100$ kΩ is connected between the output terminal and ground. The potentiometer is adjusted to obtain a closed-loop gain $A_f \equiv V_o/V_s \simeq 10$ V/V.

Specify the required setting of the potentiometer by giving the values of R_1 and R_2. Toward this end, find the

(e) For your design, what is the closed-loop voltage gain realized?

(f) Calculate the input and output resistances of the closed-loop amplifier designed.

D ***11.45** Figure P11.45 shows a three-stage feedback amplifier:

A_1 has an 82-kΩ differential input resistance, a 20-V/V open-circuit differential voltage gain, and a 3.2-kΩ output resistance.

A_2 has a 5-kΩ input resistance, a 20-mA/V short-circuit transconductance, and a 20-kΩ output resistance.

A_3 has a 20-kΩ input resistance, unity open-circuit voltage gain, and a 1-kΩ output resistance.

The feedback amplifier feeds a 1-kΩ load resistance and is fed by a signal source with a 9-kΩ resistance.

(a) Show that the feedback is negative.

(b) If $R_1 = 20$ kΩ, find the value of R_2 that results in a closed-loop gain V_o/V_s that is ideally 5 V/V.

(c) Supply the small-signal equivalent circuit.

(d) Sketch the A circuit and determine A.

(e) Find β and the amount of feedback.

(f) Find the closed-loop gain $A_f \equiv V_o/V_s$.

(g) Find the feedback amplifier's input resistance R_{in}.

(h) Find the feedback amplifier's output resistance R_{out}.

(i) If the high-frequency response of the open-loop gain A is dominated by a pole at 100 Hz, what is the upper 3-dB frequency of the closed-loop gain?

(j) If for some reason A_1 drops to half its nominal value, what is the percentage change in A_f?

A circuit (supply a circuit diagram), the value of A, the β circuit (supply a circuit diagram), and the value of β.

(f) What is the output resistance of the feedback amplifier, excluding R_L?

D ***11.44** Figure P11.32 shows a series–shunt feedback amplifier without details of the bias circuit.

(a) Eliminating the dc sources, sketch the A circuit and the circuit for determining β.

(b) Show that if $A\beta$ is large then the closed-loop voltage gain is given approximately by

$$A_f \equiv \frac{V_o}{V_s} \simeq \frac{R_F + R_E}{R_E}$$

(c) If R_E is selected equal to 50 Ω, find R_F that will result in a closed-loop gain of approximately 25 V/V.

(d) If Q_1 is biased at 1 mA, Q_2 at 2 mA, and Q_3 at 5 mA, and assuming that the transistors have $h_{fe} = 100$, find approximate values for R_{C1} and R_{C2} to obtain gains from the stages of the A circuit as follows: a voltage gain of Q_1 of about –10 and a voltage gain of Q_2 of about –50.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

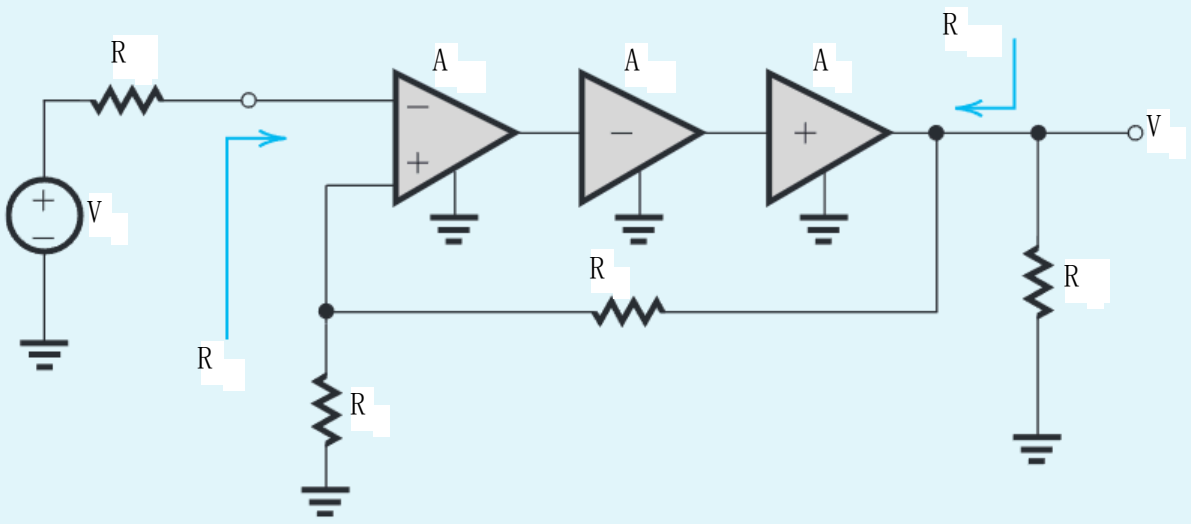


图 P11.45

反馈量由刮片的设置控制。在非反相输入端（+In）和地之间施加电压信号。输出端与地之间连接一个负载电阻 R = 100 k。调节电位器可获得闭环增益 A≡ V/V 10 V/V。为此，请找出 A 电路（提供电路图）、A 值、β 电路（提供电路图）和 β 值。

(f) 除去 R，反馈放大器的输出电阻是多少？

D *11.44 图 P11.32 显示了一个串并联反馈放大器，但没有偏置电路的细节。

(a) 消除直流源，绘制 A 电路和确定 β 的电路草图。 (b) 证明如果 Aβ 较大，则闭环电压增益近似为

$$A \equiv \frac{V}{V} \approx \frac{R+R}{R}$$

(c) 如果选择 R 等于 50，求 R 将导致约 25 V/V 的闭环增益。(d) 如果 Q 的偏置电流为 1 mA，Q 为 2 mA，Q 为 5 mA，并假设晶体管的 h= 100，求 Rand R 的近似值，以便从 A 电路的各级获得如下增益：Q 的电压增益约为 -10，Q 的电压增益约为 -50。

(e) 就你的设计而言，实现的闭环电压增益是多少？(f) 计算所设计的闭环放大器的输入和输出电阻。

D *11.45 图 P11.45 显示了一个三级反馈放大器：A 具有 82 k 差分输入电阻、20 V/V 开路差分电压增益和 3.2 k 输出电阻。

A 具有 5 k 输入电阻、20 mA/V 短路跨导和 20 k 输出电阻。A 具有 20 k 输入电阻、统一开路电压增益和 1 k 输出电阻。反馈放大器馈入 1 k 负载电阻，并由 9 k 电阻的信号源馈入。

(a) 证明反馈是负的。

(b) 如果 R= 20 k，找出 R 的值，使闭环增益 V/V 理想为 5 V/V。

(c) 提供小信号等效电路。

(d) 绘制 A 电路草图并确定 A。

(e) 求出 β 和反馈量。

(f) 求闭环增益 A≡ V/V。

(g) 求反馈放大器的输入电阻 R。

(h) 求反馈放大器的输出电阻 R。 (i) 如果开环增益 A 的高频响应由 100 Hz 处的极点主导，那么闭环增益的上限 3 分贝频率是多少？(j) 如果由于某种原因 Ad 下降到额定值的一半，A 变化的百分比是多少？

Section 11.5: Other Feedback-Amplifier Types

D 11.46 Refer to the circuit in Fig. 11.17(a), which is analyzed in Example 11.6. Select a value for R_F that results in a closed-loop transconductance $A_f \equiv I_o/V_s \simeq 10$ mA/V. Use the formulas derived in Example 11.6 to find the actual value of A_f realized. Let $\mu = 1000$, $R_{id} = 100\ \text{k}\Omega$, $g_m = 2$ mA/V, and $r_{o2} = 20\ \text{k}\Omega$.

D 11.47 Figure P11.47 shows a feedback current amplifier. The feedback network consists of the highlighted two-port network comprising R_M and R_F. It is fed with the output current I_o and delivers a feedback current I_f at its port 1 to the input node. The feedback factor β is the current ratio I_f/I_o measured with port 1 short-circuited (because it is connected in shunt with the amplifier input).

(a) Find an expression for β and hence for the ideal value of $A_f \equiv I_o/I_s$.

(b) Setting $I_s = 0$, break the loop at the gate of Q_2 and thus determine the loop gain $A\beta$. Show that

$$A = -\frac{g_{m2}R_D}{1 + 1/[g_{m1}(R_M + R_F)]}$$

(c) For $g_{m1} = g_{m2} = 4$ mA/V, $R_D = 10\ \text{k}\Omega$, and $(R_M + R_F) = 1\ \text{k}\Omega$, find the value of R_M that results in a closed-loop current gain of 5 A/A.

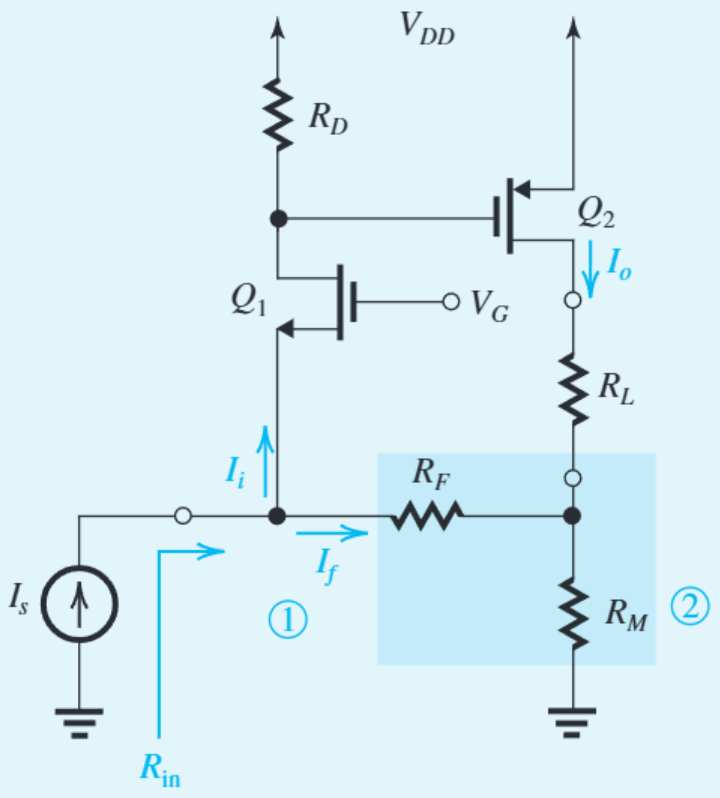


Figure P11.47

D 11.48 Figure P11.48(a) shows a feedback transresistance amplifier formed by an op amp and a feedback resistance R_F. Assume that the op amp is modeled by an input resistance R_{id}, an open-circuit voltage gain μ, and an output resistance r_o.

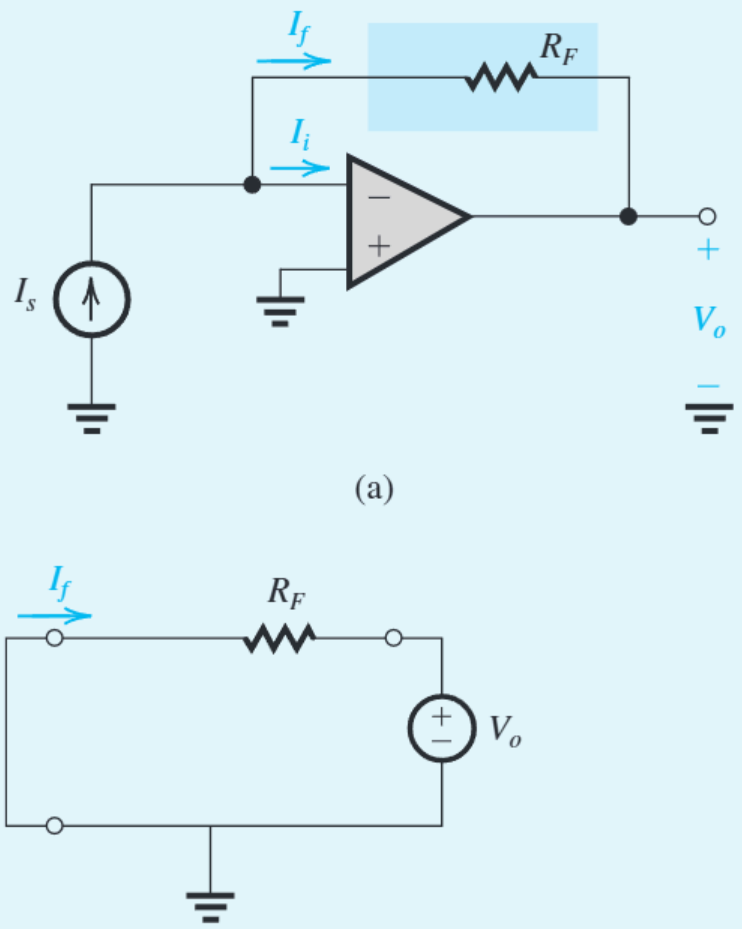


Figure P11.48

(a) Show that the feedback factor β, determined as shown in Fig. P11.48(b), is given by $\beta = -1/R_F$. Hence find the ideal value of the closed-loop gain $A_f \equiv V_o/I_s$. Find R_F that results in A_f of approximately 1 kΩ.

(b) By setting $I_s = 0$ and breaking the loop at the input terminals of the op amp, show that the loop gain is given by

$$A\beta = \mu \frac{R_{id}}{R_{id} + R_F + r_o}$$

(c) For $\mu = 1000$, $R_{id} = 100\ \text{k}\Omega$, $r_o = 1\ \text{k}\Omega$, and R_F having the value found in (a), what is the actual value of A_f realized?

Feedback Transconductance Amplifiers (Series–Series)

11.49 A series–series feedback amplifier employs a transconductance amplifier having a short-circuit transconductance G_m of 0.6 A/V, input resistance of 10 kΩ, and output resistance of 100 kΩ. The feedback network has $\beta = 200\ \Omega$,

第 11.5 节：其他反馈放大器类型

D 11.46 参考图 11.17(a)中的电路，该电路在例 11.6 中进行了分析。选择一个 R 值，使闭环跨导 A ≡ I/V 10 mA/V。使用例 11.6 中得出的公式求出 A 的实际值。假设 μ = 1000，R= 100 k，g= 2 mA/V，r= 20 k。

D 11.47 图 P11.47 显示了一个反馈电流放大器。反馈网络由 R 和 R 组成的双端口网络构成。它接收输出电流 I，并将端口 1 的反馈电流 I 传递到输入节点。反馈系数 β 是端口 1 短路时测量的电流比 I/I（因为端口 1 与放大器输入端并联）。

(a) 求出 β 的表达式，进而求出 A ≡ I /I 的理想值。

(b) 设置 I= 0，在 Q 的栅极处断开环路，从而确定环路增益 Aβ。证明

$$A = -\frac{g\ R}{1 + 1/[g\ (R+ R\ )]}$$

(c) 对于 g = g= 4 mA/V、R= 10 k 和 (R+ R) = 1 k，找出能使闭环电流增益达到 5 A/A 的 R 值。

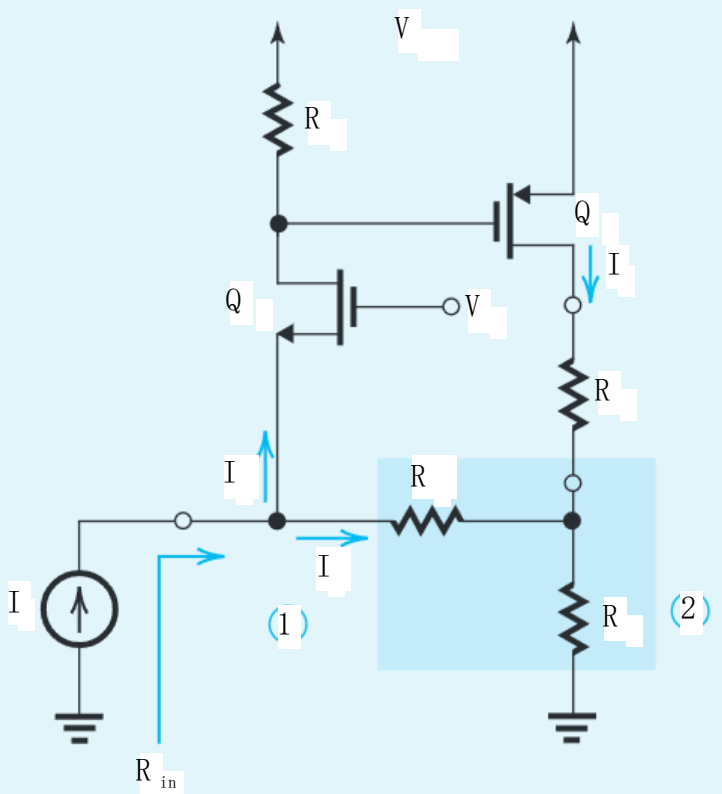
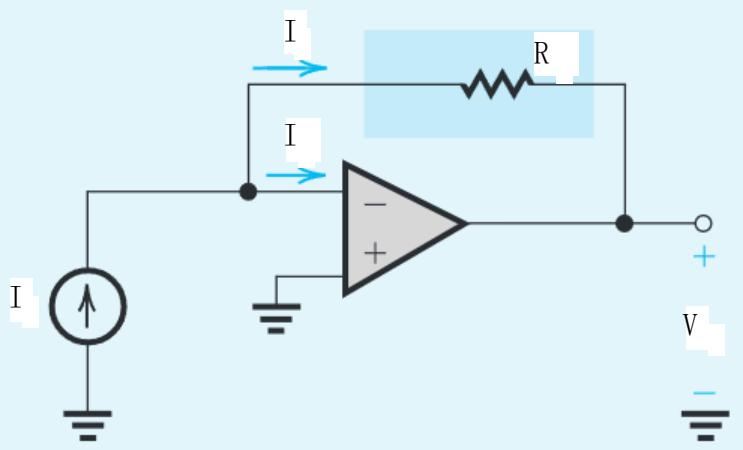


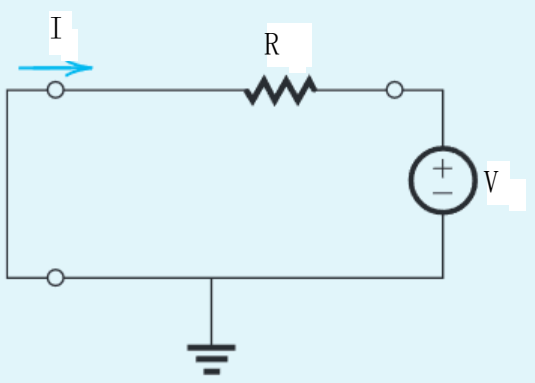
图 P11.47

D 11.48 图 P11.48(a) 显示了由运算放大器和反馈电阻 R 组成的反馈跨阻放大器。

假设运算放大器由输入电阻 R、开路电压增益 μ 和输出电阻 r 组成。



(a)



(b)

图 P11.48

(a) 证明如图 P11.48(b) 所示确定的反馈系数 β 由 β = -1/R 给出。因此，求闭环增益 A ≡ V/I 的理想值。找出 R，使 A 约为 1 k。(b) 设置 I = 0 并在运算放大器的输入端断开环路，说明环路增益为

$$A\beta = \mu\ \frac{R}{R+\ R\ +\ r}$$

(c) μ = 1000，R = 100 k，r = 1 k，R 为 (a) 中的值，A 的实际值是多少？

反馈跨导放大器（串联系列）

11.49 一个串联反馈放大器采用的跨导放大器的短路跨导 Go 为 0.6 A/V，输入电阻为 10 k，输出电阻为 100 k。反馈网络的 β = 200 、

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

an input resistance (with port 1 open-circuited) of 200 Ω, and an input resistance (with port 2 open-circuited) of 10 kΩ. The amplifier operates with a signal source having a resistance of 10 kΩ and with a load resistance of 10 kΩ. Find A_f, R_{in}, and R_{out}.

11.50 Reconsider the circuit in Fig. 11.21(a), analyzed in Example 11.8, this time with the output voltage taken at the emitter of Q_3. In this case the feedback can be considered to be of the series–shunt type. Note that R_{E2} should now be considered part of the basic amplifier and not of the feedback network.

(a) Determine β.
(b) Find an approximate value for $A_f \equiv V_{e3}/V_s$ assuming that the loop gain remains large (a safe assumption, since the loop in fact does not change).
[*Note*: If you continue with the feedback analysis, you'll find that $A\beta$ in fact changes somewhat; this is a result of the different approximations made in the feedback analysis approach.]
(c) If the loop gain remains at the value calculated in Example 11.8 (i.e., 246.3), find the output resistance R_{out} (measured between the emitter of Q_3 and ground). (Neglect the effect of r_{03}.)

D *11.51 Figure P11.31 (page 851) shows a feedback triple utilizing MOSFETs. All three MOSFETs are biased and sized to operate at $g_m = 4$ mA/V. You may neglect their r_o's (except for the calculation of R_{out1} as indicated below).

(a) Considering the feedback amplifier as a transconductance amplifier with output current I_o, find the value of R_F that results in a closed-loop transconductance of approximately 100 mA/V.
(b) Sketch the A circuit and find the value of $A \equiv I_o/V_i$.
(c) Find $1 + A\beta$ and $A_f \equiv I_o/V_s$. Compare to the value of A_f you designed for. What is the percentage difference? What resistance can you change to make A_f exactly 100 mA/V, and in which direction (increase or decrease)?
(d) Assuming that $r_{o3} = 20$ kΩ, find R_o of the A circuit. For this purpose, recall that the resistance looking into the drain of a MOSFET having a resistance R_s in its source is $(r_o + R_s + g_m r_o R_s)$. Hence find the output resistance R_{out1}. Since the current sampled by the feedback network is exactly equal to the output current, you can use the feedback formula.
(e) If the voltage V_o is taken as the output, in which case the amplifier becomes series–shunt feedback, what is the value of the closed-loop voltage gain V_o/V_s? Assume that R_F has the original value you selected in (a). Note that in this case R_{S2} should be considered part of the amplifier and not the feedback network. The feedback analysis will reveal that $A\beta$ changes somewhat, which may be puzzling given that the feedback loop did not change. The change is due to the different approximation used.
(f) What is the closed-loop output resistance R_{out2} of the voltage amplifier in (e) above?

11.52 Consider the circuit in Fig. P11.52 as a transconductance amplifier with input V_s and output I_o. The transistor is specified in terms of its g_m and r_o.

(a) Sketch the small-signal equivalent circuit using the hybrid-π model of the MOSFET and convince yourself that the feedback circuit is comprised of resistor R_F.
(b) Find the A circuit and the β circuit.
(c) Derive expressions for A, β, $(1 + A\beta)$, A_f, R_o, and R_{of}.

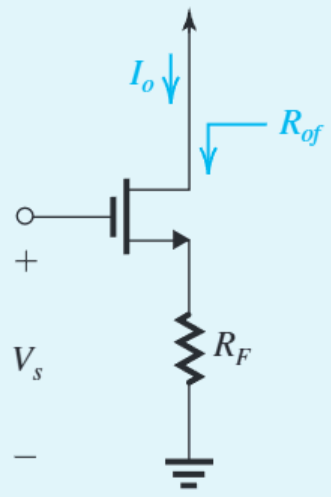


Figure P11.52

D 11.53 The transconductance amplifier in Fig. P11.53 utilizes a differential amplifier with gain μ and a very high input resistance. The differential amplifier drives a transistor Q characterized by its g_m and r_o. A resistor R_F senses the output current I_o.

(a) For $A\beta \gg 1$, find an approximate expression for the closed-loop transconductance $A_f \equiv I_o/V_s$. Hence, select a value for R_F that results in $A_f \simeq 5$ mA/V.
(b) Find the A circuit and derive an expression for A. Evaluate A for the case $\mu = 1000$ V/V, $g_m = 2$ mA/V, $r_o = 20$ kΩ, and the value of R_F you selected in (a).
(c) Give an expression for $A\beta$ and evaluate its value and that of $1 + A\beta$.

输入电阻（端口 1 开路）为 200 k，输入电阻（端口 2 开路）为 10 k。放大器的信号源电阻为 10 k，负载电阻为 10 k。求 A、R 和 R。

11.50 重新考虑例 11.8 中分析的图 11.21(a)中的电路，此时输出电压取自 Q 的发射极。请注意，Rsh 现在应视为基本放大器的一部分，而不是反馈网络的一部分。

(a) 确定 β。

(b) 假设环路增益仍然很大（这是一个安全的假设，因为环路实际上没有变化），求出 A ≡ V/V 的近似值。

[注：如果继续进行反馈分析，您会发现 Aβ 实际上会发生一些变化；这是反馈分析方法中不同近似值的结果。]

(c) 如果环路增益保持在例 11.8 中计算的值（即 246.3），求输出电阻 R（在 Q 发射极和地之间测量）。(忽略 r 的影响）。

D *11.51 图 P11.31（第 851 页）显示了使用 MOSFET 的三路反馈。所有三个 MOSFET 均已偏置，工作电流为 g = 4 mA/V。您可以忽略它们的 r 值（下文所示的 Ras 计算除外）。

(a) 将反馈放大器视为输出电流为 I 的跨导放大器，找出能使闭环跨导达到约 100 mA/V 的 R 值。

(b) 画出 A 电路图，并找出 A ≡ I /V 的值。(c) 求 1 + Aβ 和 A ≡ I/V。与您设计的 A 值进行比较。差异的百分比是多少？

要使 A 精确达到 100 mA/V，您可以改变哪个电阻，以及改变的方向（增大还是减小）？

(d) 假设 r= 20 k，求 A 电路的 R。为此，请回想一下，在 MOSFET 的漏极观察到的电阻为 (r + R+ grR)，而在其源极观察到的电阻为 (r+R+grR)。由于反馈网络采样的电流正好等于输出电流，因此可以使用反馈公式。(e) 如果将电压维斯作为输出，在这种情况下放大器变成串并联反馈，则闭环电压增益 V/V 的值？假设 R 的值与您在 (a) 中选择的值相同。请注意，在这种情况下，R 应被视为放大器的一部分，而不是反馈网络。反馈分析将显示 Aβ 发生了一些变化，这可能令人费解，因为反馈回路并未改变。这种变化是由于使用了不同的近似值。

(f) 上文（e）中电压放大器的闭环输出电阻 R 是多少？

11.52 将图 P11.52 中的电路视为具有输入 V 和输出 I 的跨导放大器。根据晶体管的 g 和 r 指定晶体管。

(a) 利用 MOSFET 的混合-π 模型绘制小信号等效电路草图，并确信反馈电路由电阻 R .

(b) 找出 A 电路和 β 电路。

(c) 推导出 A、β、（1 + Aβ）、A ，R 和 R 的表达式。

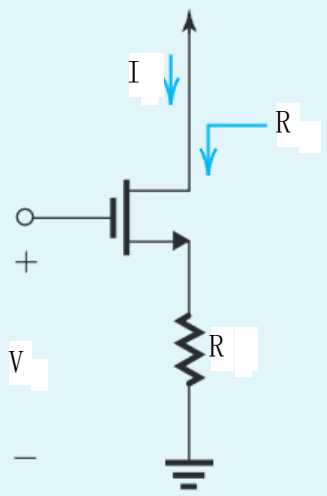


图 P11.52

D 11.53 图 P11.53 中的跨导放大器采用增益为 μ 的差分放大器和极高的输入电阻。差分放大器驱动一个晶体管 Q，晶体管 Q 的特性是 g 和 r。

(a) 对于 Aβ 1，求闭环跨导 A≡ I/V 的近似表达式。因此，请选择一个 R 值，使 A 达到 5 mA/V。

(b) 找出 A 电路并推导出 A 的表达式。

在 μ = 1000 V/V、g = 2 mA/V、r = 20 k 和 (a) 中选定的 Ryou 值的情况下，计算 A。

(c) 给出 Aβ 的表达式，并评估其值和 1 + Aβ 的值。

(d) Find the closed-loop gain A_f and compare to the value you designed for in (a) above.

(e) Find expressions and values for R_o and R_{of}. [*Hint*: The resistance looking into the drain of a MOSFET with a resistance R_s in its source is $(r_o + R_s + g_m r_o R_s)$.]

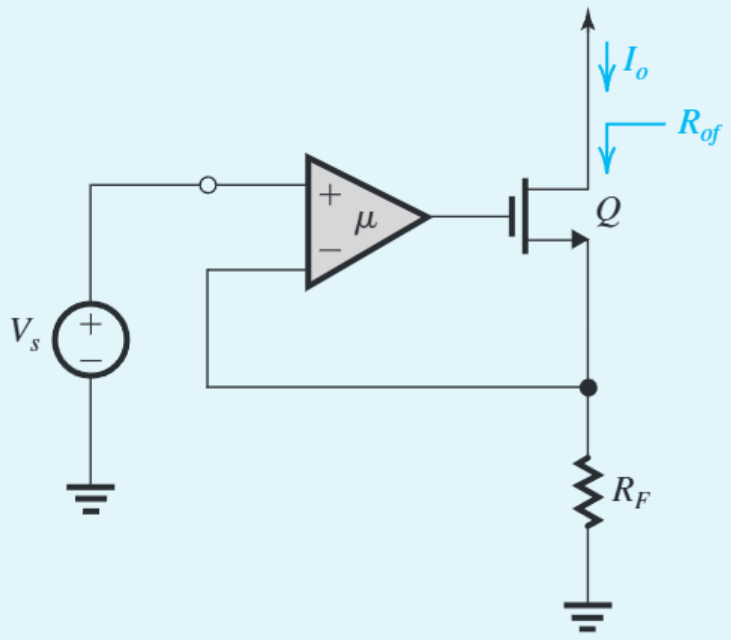


Figure P11.53

***11.54** It is required to show that the output resistance of the BJT circuit in Fig. P11.54 is given by

$$R_o = r_o + [R_e \| (r_\pi + R_b)] \left(1 + g_m r_o \frac{r_\pi}{r_\pi + R_b}\right)$$

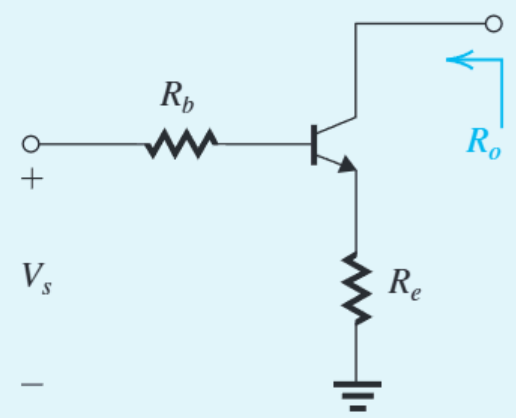
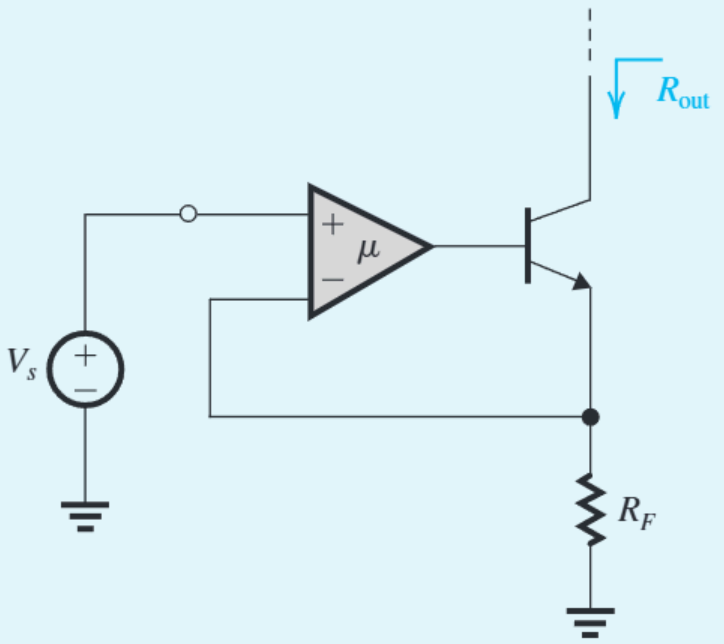


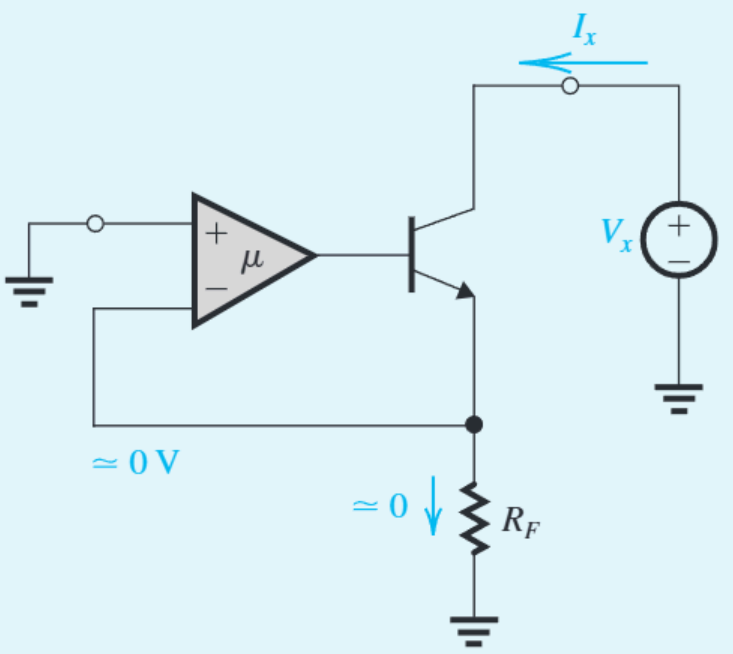
Figure P11.54

To derive this expression, set $V_s = 0$, replace the BJT with its small-signal, hybrid-π model, apply a test voltage V_x to the collector, and find the current I_x drawn from V_x and hence R_o as V_x/I_x. Note that the bias arrangement is not shown. For the case of $R_b = 0$, find the maximum possible value for R_o. Note that this theoretical maximum is obtained when R_e is so large that the signal current in the emitter is nearly zero. In this case, with V_x applied and $V_s = 0$, what is the current in the base, in the $g_m V_\pi$ generator, and in r_o, all in terms of I_x? Show these currents on a sketch of the equivalent circuit with R_e set to ∞.

11.55 As we found out in Example 11.8, whenever the feedback network senses the emitter current of the BJT, the feedback output resistance formula cannot predict the output resistance looking into the collector. To understand this issue more clearly, consider the feedback transconductance amplifier shown in Fig. P11.55(a). To determine the output resistance, we set $V_s = 0$ and apply a test voltage V_x to the collector, as shown in Fig. P11.55(b). Now, let μ be increased to the point where the feedback signal across R_F almost equals the input to the positive terminal of the differential amplifier,



(a)



(b)

Figure P11.55

(d) 找出闭环增益 A，并与上述 (a) 中设计的值进行比较。(e) 找出 Rand R 的表达式和数值。[提示：源极电阻为 R 的 MOSFET 漏极电阻为 (r+ R + g rR)。

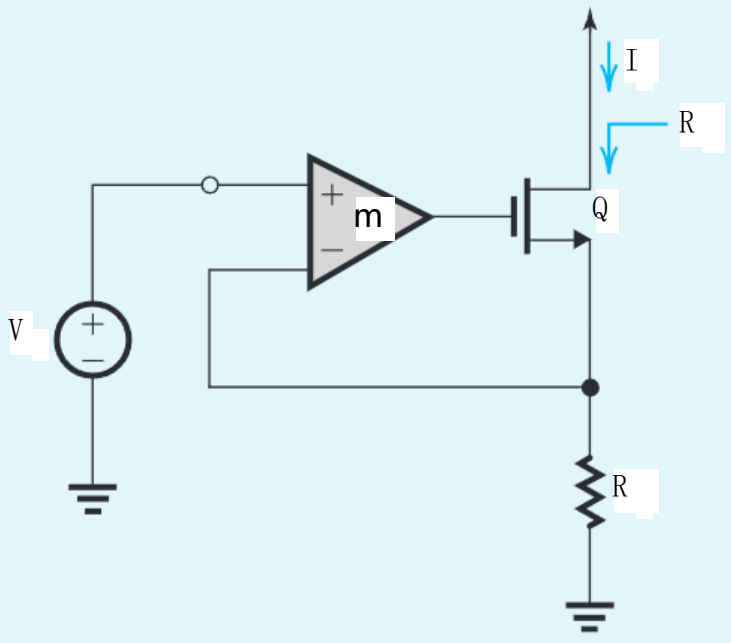


图 P11.53

*11.54 需要说明图 P11.54 中 BJT 电路的输出电阻为

$$R = r + [R \,||\, (r + R)](1 + g_m r \frac{r}{r + R})$$

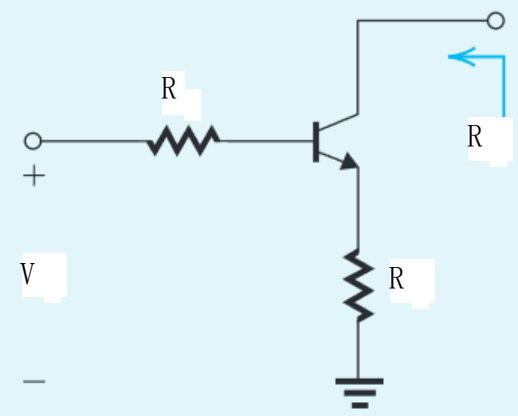
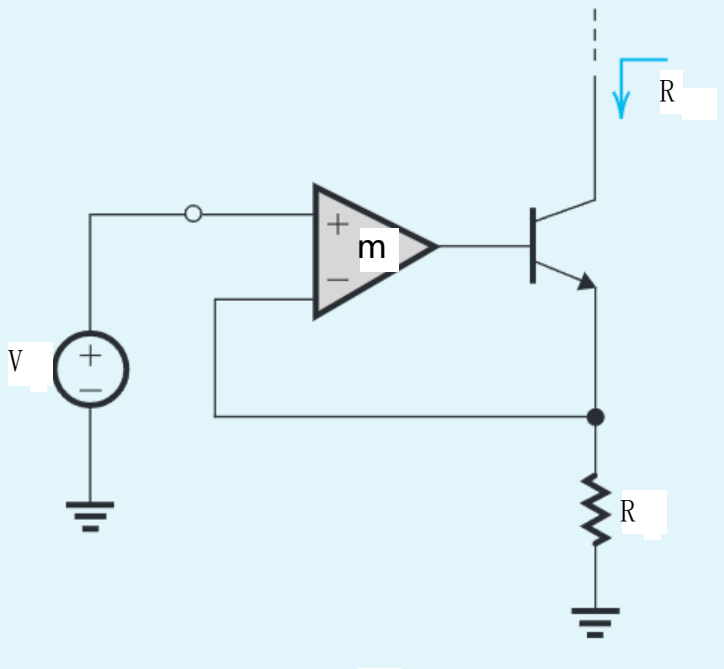


图 P11.54

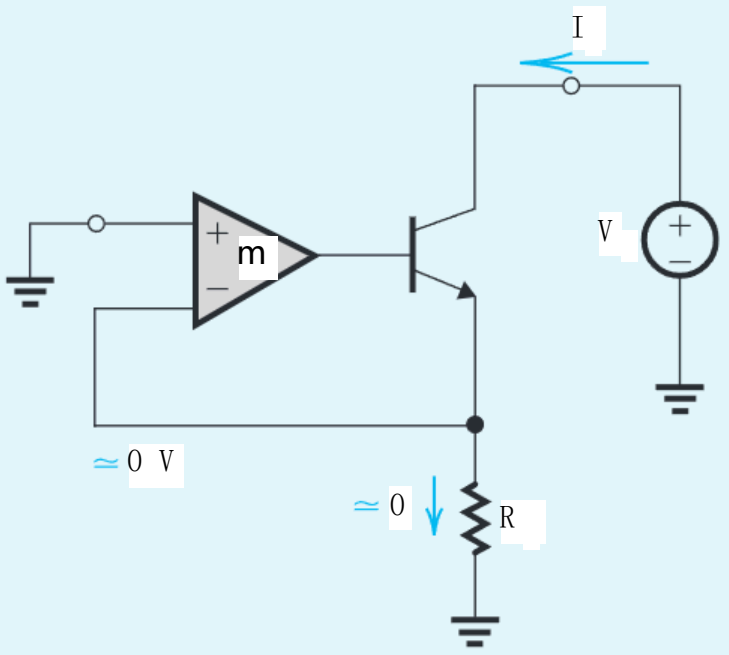
要推导出这一表达式，可设置 V= 0，用小信号混合 π 模型替换 BJT，在集电极上施加测试电压 V，然后求出从 V 和 R 吸取的电流 I，即 V/I。请注意，图中未显示偏压布置。对于 R= 0 的情况，请找出 R 的最大可能值。请注意，当 R 大到发射极中的信号电流几乎为零时，就会得到这个理论上的最大值。在这种情况下，施加 V 且 V= 0

以 I 表示，基极、gV 发电机和 r 中的电流分别是多少？在 R 设为 ∞ 的等效电路草图上显示这些电流。

11.55 正如我们在例 11.8 中发现的，只要反馈网络检测到 BJT 的发射极电流，反馈输出电阻公式就无法预测集电极的输出电阻。为了更清楚地理解这个问题，请看图 P11.55(a) 所示的反馈跨导放大器。如图 P11.55(b) 所示，为确定输出电阻，我们设置 V= 0，并向集电极施加测试电压 V。现在，让 μ 增大到 R 两端的反馈信号几乎等于差分放大器正极的输入、



(a)



(b)

图 P11.55

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

now zero. Thus the signal current through R_F will be almost zero. By replacing the BJT with its hybrid-π model, show that

$$R_{\text{out}} = r_\pi + (h_{fe} + 1)r_o \simeq h_{fe} r_o$$

where h_{fe} is the transistor β. Thus for large amounts of feedback, R_{out} is limited to a maximum of $h_{fe} r_o$ independent of the amount of feedback. This phenomenon does *not* occur in the MOSFET version of this circuit, where the output resistance can be theoretically made infinite.

11.56 For the feedback transconductance amplifier of Fig. P11.56 derive expressions for A, β, $A\beta$, A_f, R_o, and R_{of}. Evaluate A_f and R_{of} for the case of $g_{m1} = g_{m2} = 4$ mA/V, $R_D = 20$ kΩ, $r_{o2} = 20$ kΩ, $R_F = 100$ Ω, and $R_L = 1$ kΩ. For simplicity, neglect r_{o1} and take r_{o2} into account only when calculating output resistances.

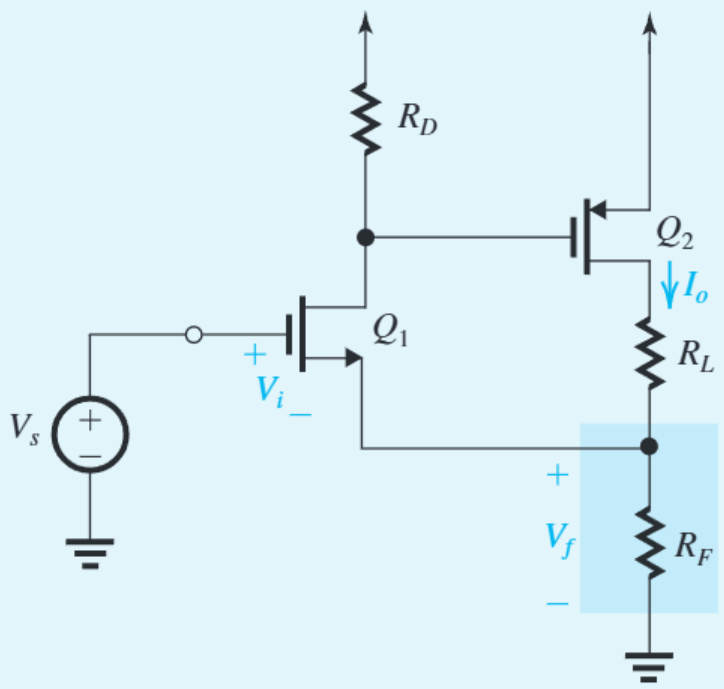


Figure P11.56

D 11.57 For the feedback transconductance amplifier in Fig. P11.57, derive an approximate expression for the closed-loop transconductance $A_f \equiv I_o/V_s$ for the case of $A\beta \gg 1$. Hence select a value for R_2 to obtain $A_f = 100$ mA/V. If Q is biased to obtain $g_m = 1$ mA/V, specify the value of the gain μ of the differential amplifier to obtain an amount of feedback of 60 dB. If Q has $r_o = 50$ kΩ, find the output resistance R_{out}. [*Hint*: Recall that for a MOSFET with a resistance R_s in its source, the resistance looking into the drain is $(r_o + R_s + g_m r_o R_s)$.]

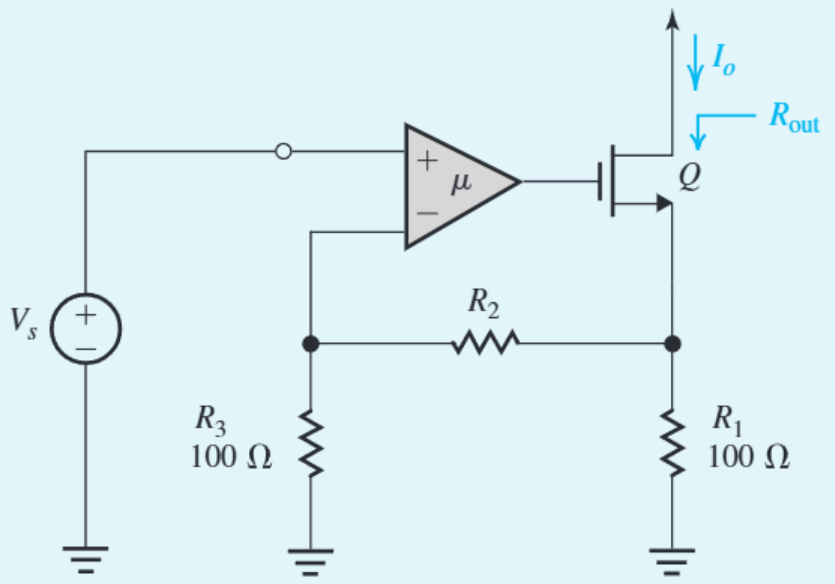


Figure P11.57

SIM 11.58 All the MOS transistors in the feedback transconductance amplifier (series–series) of Fig. P11.58 are sized to operate at $|V_{OV}| = 0.2$ V. For all transistors, $|V_t| = 0.4$ V and $|V_A| = 20$ V.

(a) If V_s has a zero dc component, find the dc voltage at the output, at the drain of Q_1, and at the drain of Q_2.
(b) Find an approximate expression and value for $A_f \equiv I_o/V_s$ for the case $A\beta \gg 1$.
(c) Use feedback analysis to obtain a more precise value for A_f.
(d) Find the value of R_{out}.
(e) If the voltage at the source of Q_5 is taken as the output, find the voltage gain using the value of I_o/V_s obtained in (c). Also find the output resistance of this series–shunt voltage amplifier.

11.59 By setting $V_s = 0$ and breaking the feedback loop, show that the loop gain of the amplifier circuit in Fig. P11.58 is

$$A\beta = g_{m1,2}(r_{o2} \parallel r_{o4}) \frac{R_F \parallel r_{o5}}{(R_F \parallel r_{o5}) + 1/g_{m5}}$$

where $g_{m1,2}$ is the g_m of each of Q_1 and Q_2.

Feedback Transresistance Amplifiers (Shunt–Shunt)

11.60 For the transresistance amplifier analyzed in Example 11.9, use the formulas derived there to evaluate A_f, R_{in}, and R_{out} when μ is one-tenth the value used in the example. That is, evaluate for $\mu = 10^3$ V/V, $R_{id} = \infty$, $r_o = 100\,\Omega$, $R_F = 10\,\text{k}\Omega$, and $R_s = R_L = 1$ kΩ. Compare to the corresponding values obtained in Example 11.9.

现在为零。因此，通过 Rw 的信号电流几乎为零。将 BJT 换成混合-π 模型，说明

$$R = r + \frac{(h+1)}{r\ h\ r}$$

因此，当反馈量较大时，Ris 的最大值限制在 hr，而与反馈量无关。这种现象在 MOSFET 版本的电路中不会出现，因为理论上输出电阻可以做到无限大。

11.56 对于图 P11.56 中的反馈跨导放大器，推导出 A、β、Aβ、A、R 和 R 的表达式。计算 g= g= 4 mA/V、R= 20 k、r= 20 k、R= 100 和 R= 1 k 时的 A 和 R。为简单起见，仅在计算输出电阻时忽略 r 并考虑 rint。

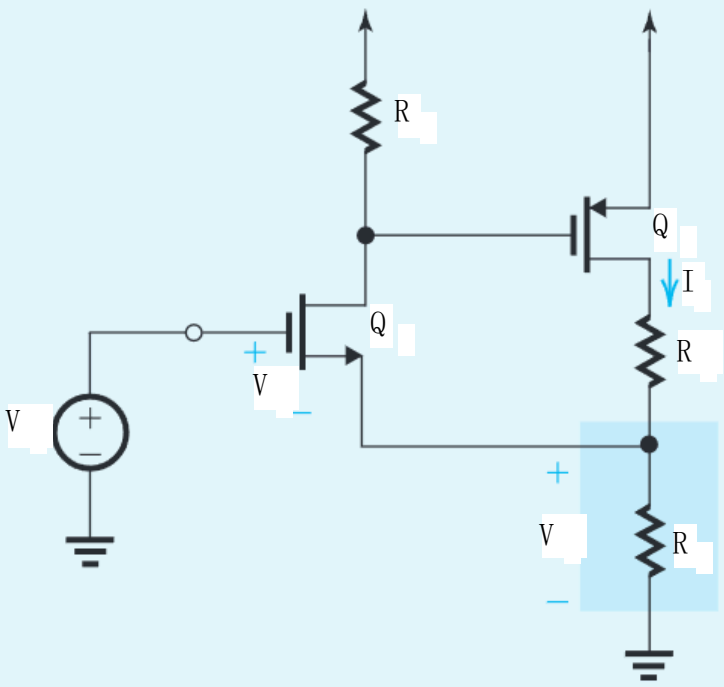


图 P11.56

D 11.57 对于图 P11.57 中的反馈跨导放大器，请推导出 Aβ 为 1 时闭环跨导 A≡ I /V 的近似表达式。因此，请选择一个 R 值，以获得 A= 100 mA/V。如果 Q 的偏压为 g= 1 mA/V，请指定差分放大器的增益 μ 值，以获得 60 dB 的反馈量。如果 Q 的 r = 50 k，求输出电阻 R。[提示：回忆一下，对于源极电阻为 Rin 的 MOSFET，看向漏极的电阻为 (r + R + g rR)。

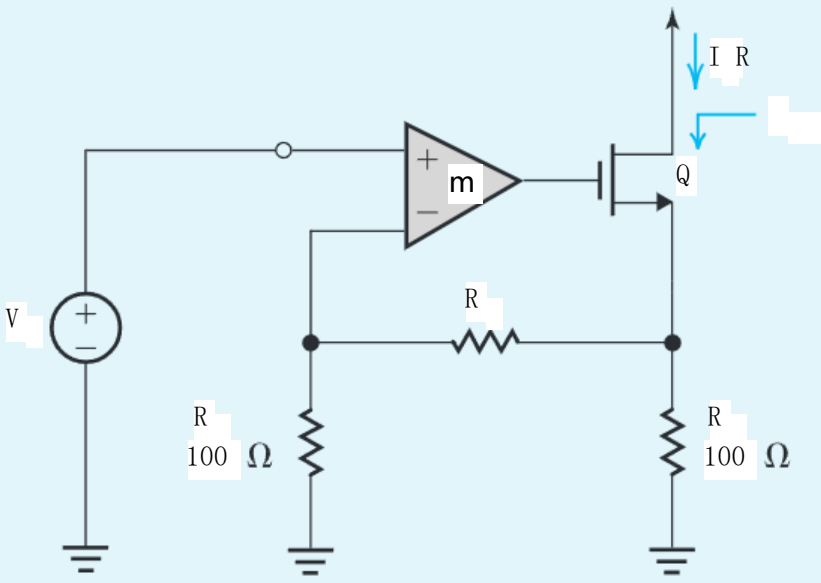


图 P11.57

图 P11.58 中的反馈跨导放大器（串联系列）中的所有 MOS 晶体管的工作电压均为 $|V| = 0.2$ V.适用于所有晶体管、$|V| = 0.4$ V 和 $|V| = 20$ V.

(a) 如果 V 的直流分量为零，求输出端、Q 的漏极和 Q 的漏极的直流电压。 (b) 求 A ≡ I /V 的近似表达式和 A β 1 的值。 (c) 利用反馈分析法求出更精确的 A 值。

(d) 求 R 的值。

(e) 如果将 Q 源的电压作为输出，利用 (c) 中得到的 I/V 值求出电压增益。同时求出该串并联电压放大器的输出电阻。

11.59 通过设置 V = 0 并断开反馈回路，说明图 P11.58 中放大器电路的环路增益为

$$A\beta = g\ (r \parallel r)\left(\frac{R \parallel r}{R \parallel r}\right)\frac{}{+\ 1/g}$$

其中，g 是 Q 和 Q 各自的 go。

反馈跨阻放大器（分流-并流）

11.60 对于例 11.9 中分析的跨阻放大器，当 μ 为例题中所用值的十分之一时，使用该例题中得出的公式求出 A、R 和 Rw。也就是说，在 μ = 10 V/V、R= ∞、r= 100、R= 10 k 和 R= R= 1 k 时进行计算。与例 11.9 中得到的相应值进行比较。

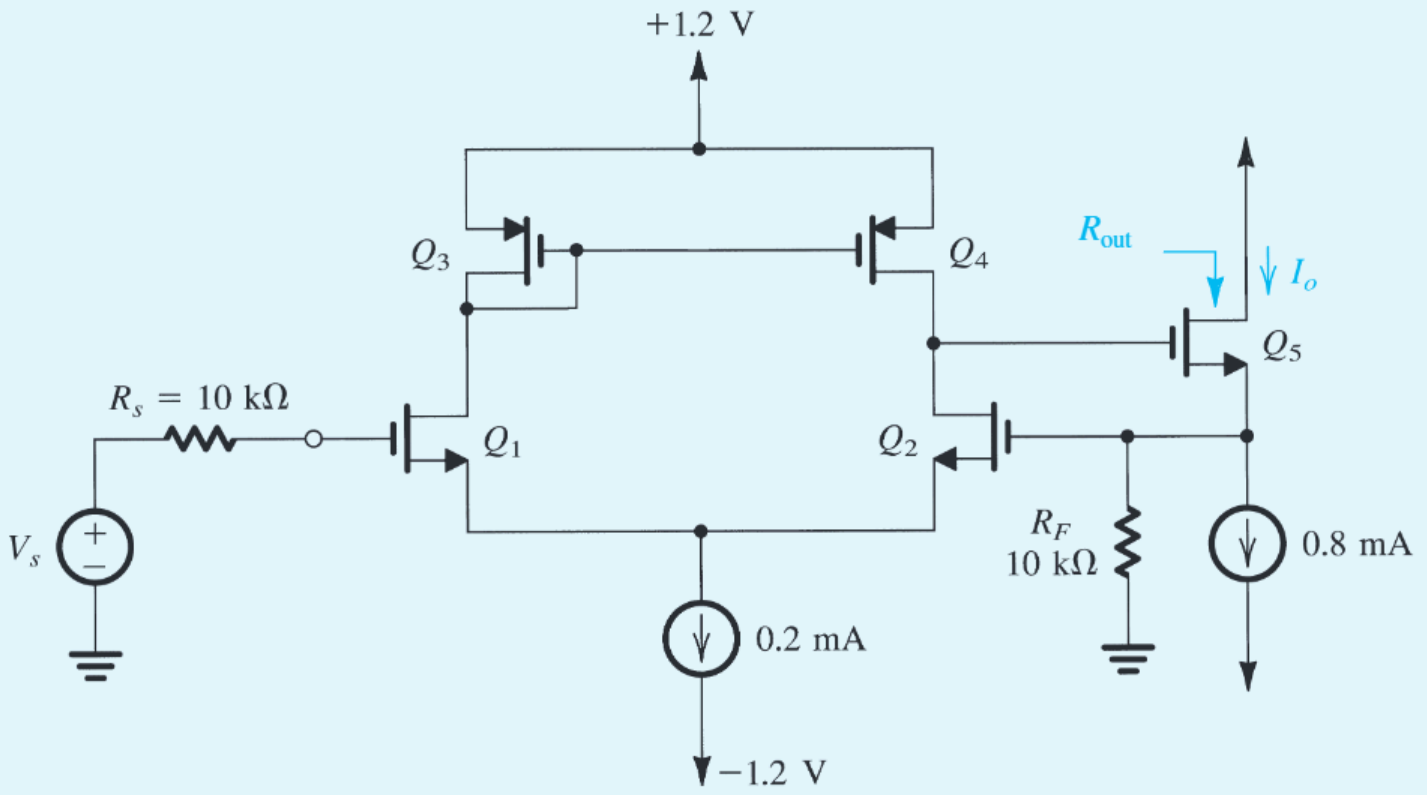


Figure P11.58

11.61 Use the formulas derived in Example 11.9 to solve the problem in Exercise 11.19. Show that the results are identical to those given in the answer to Exercise 11.19.

11.62 By setting $I_s = 0$, replacing the MOSFET with its hybrid-π model, and breaking the feedback loop, determine the loop gain of the feedback amplifier in Fig. E11.19. Hence find the open-loop gain. Evaluate $A\beta$, β, A, and A_f for the numerical values given in Exercise 11.8. Why do the results differ somewhat from those given in the answer to Exercise 11.19?

11.63 The CE BJT amplifier in Fig. P11.63 employs shunt–shunt feedback: Feedback resistor R_F senses the output voltage V_o and provides a feedback current to the base node.

(a) If V_s has a zero dc component, find the dc collector current of the BJT. Assume the transistor $\beta = 100$.
(b) Find the small-signal equivalent circuit of the amplifier with the signal source represented by its Norton equivalent (as we usually do when the feedback connection at the input is shunt).
(c) Find the A circuit and determine the value of A, R_i, and R_o.
(d) Find β and hence $A\beta$ and $1 + A\beta$.
(e) Find A_f, R_{if}, and R_{of} and hence R_{in} and R_{out}.
(f) What voltage gain V_o/V_s is realized? How does this value compare to the ideal value obtained if the loop gain is very large and thus the signal voltage at the base becomes almost zero (like what happens in an inverting op-amp circuit). Note that this single-transistor poor-man's op amp is not that bad!

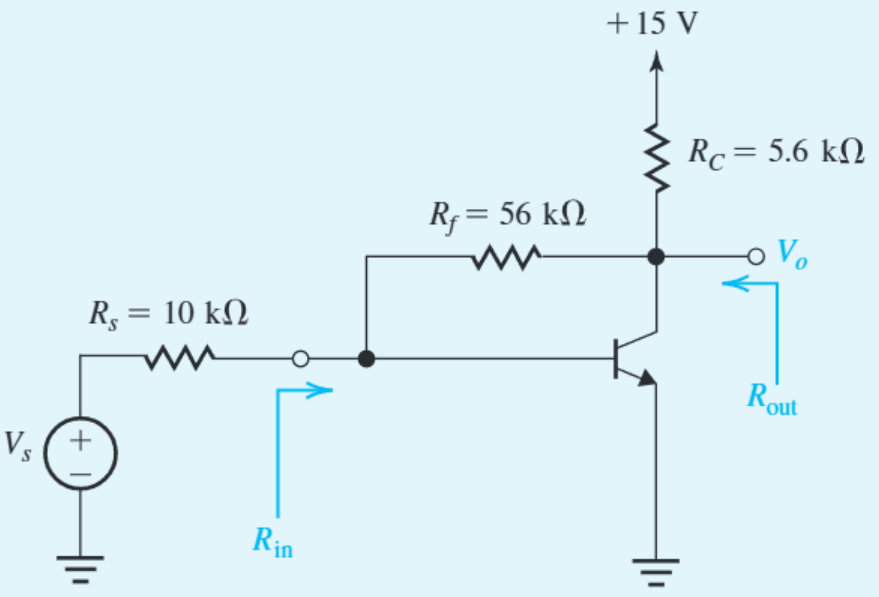


Figure P11.63

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

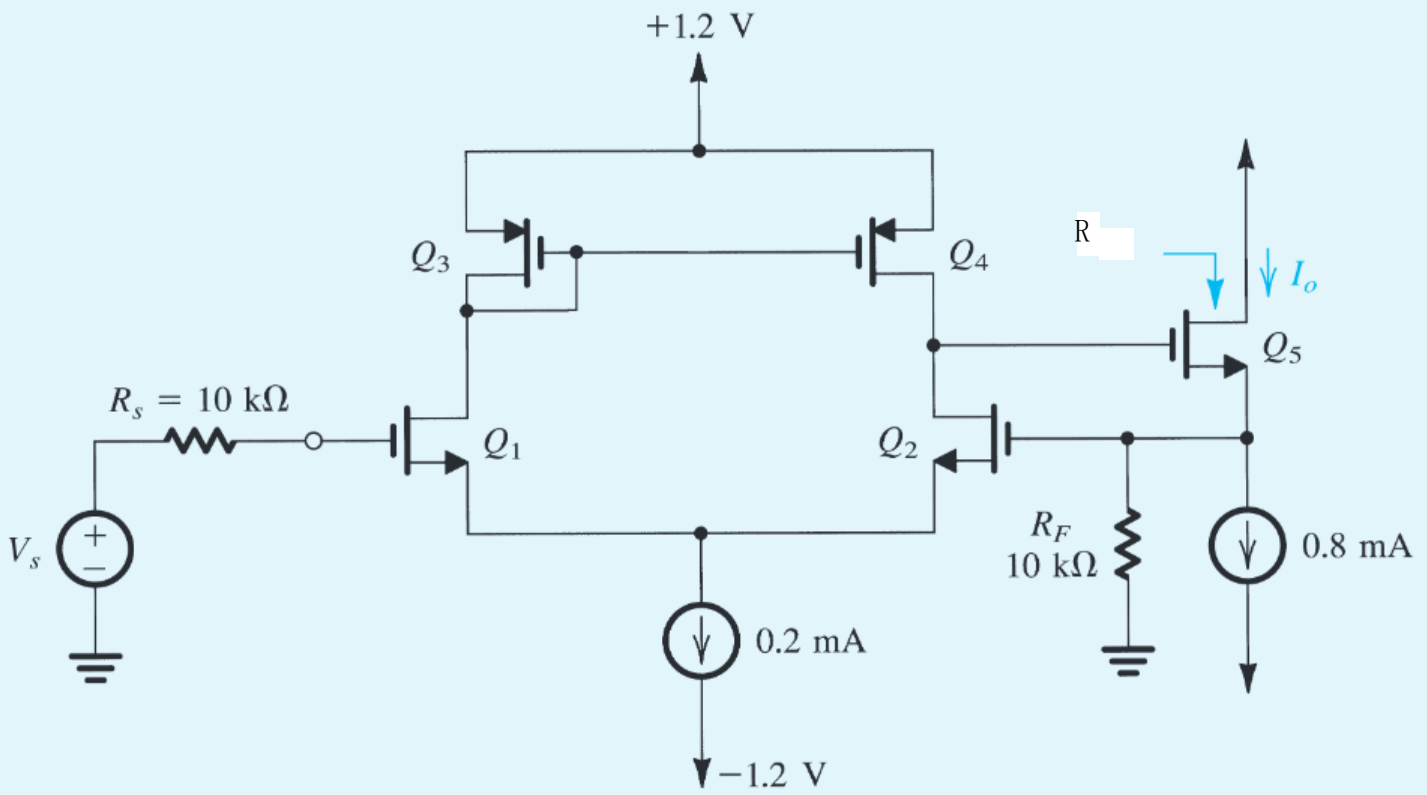


图 P11.58

11.61 使用例 11.9 中推导的公式解决练习 11.19 中的问题。说明结果与练习 11.19 的答案相同。

11.62 设置 I= 0，用混合π模型取代 MOSFET，并断开反馈回路，确定图 E11.19 中反馈放大器的环路增益。从而求出开环增益。根据练习 11.8 中给出的数值，评估 Aβ、β、A 和 Af。为什么结果与练习 11.19 的答案有些不同?

11.63 图 P11.63 中的 CE BJT 放大器采用了并联反馈：反馈电阻 Rsenses 输出电压 V，并为基极节点提供反馈电流。(a) 如果 V 的直流分量为零，求 BJT 的直流集电极电流。假设晶体管 β = 100。(b) 找出放大器的小信号等效电路，信号源用其诺顿等效电路表示（当输入端的反馈连接为并联时，我们通常这样做）。(c) 找出 A 电路，并确定 A、R 和 R 的值。(d) 求出 β，进而求出 Aβ 和 1 + Aβ。(e) 求 A、R 和 Rand，从而求出 Rand R。(f) 实现的电压增益 V/Vis 是多少？如果环路增益非常大，基极的信号电压几乎为零（就像反相运放电路中的情况），这个值与理想值相比如何？请注意，这种单晶体管贫民运算放大器并不差！

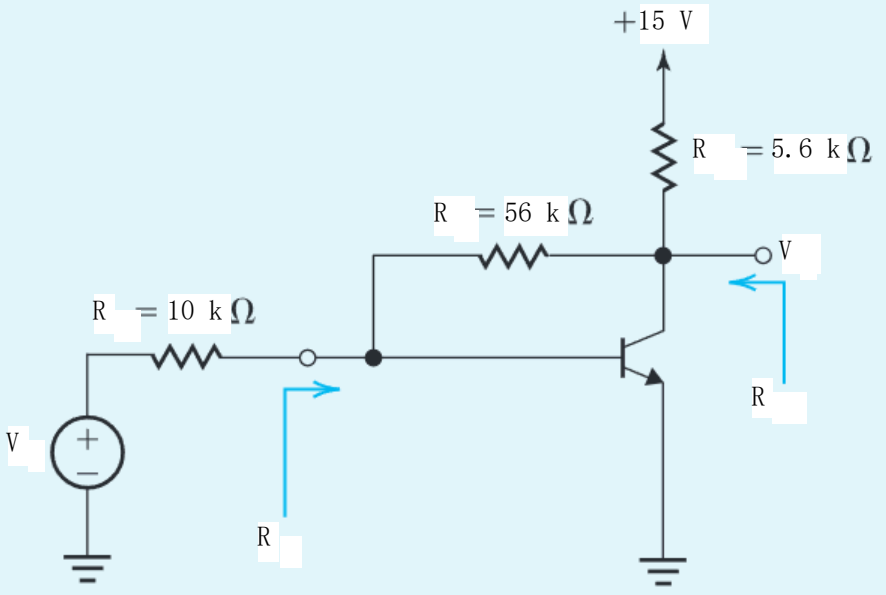


图 P11.63

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

D 11.64 The circuit in Fig. P11.64 utilizes a voltage amplifier with gain μ in a shunt–shunt feedback topology with the feedback network composed of resistor R_F. In order to be able to use the feedback equations, you should first convert the signal source to its Norton representation. You will then see that all the formulas derived in Example 11.9 apply here as well.

(a) If the loop gain is very large, what approximate closed-loop voltage gain V_o/V_s is realized? If $R_s = 2\ \text{k}\Omega$, give the value of R_F that will result in $V_o/V_s \simeq -10$ V/V.

(b) If the amplifier μ has a dc gain of 10^3 V/V, an input resistance $R_{id} = 100\ \text{k}\Omega$, and an output resistance $r_o = 2\ \text{k}\Omega$, find the actual V_o/V_s realized. Also find R_{in} and R_{out} (indicated on the circuit diagram). You may use formulas derived in Example 11.9.

(c) If the amplifier μ has an upper 3-dB frequency of 1 kHz and a uniform −20-dB/decade gain rolloff, what is the 3-dB frequency of the gain $|V_o/V_s|$?

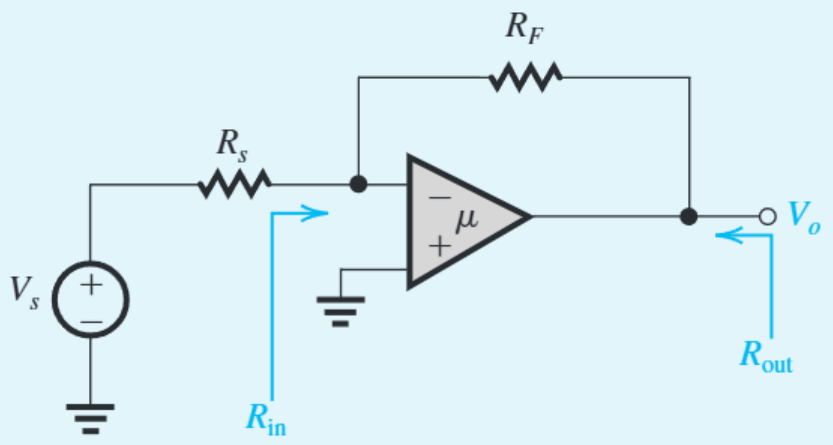


Figure P11.64

11.65 The feedback transresistance amplifier in Fig. P11.65 utilizes two identical MOSFETs biased by ideal current sources $I = 0.4$ mA. The MOSFETs are sized to operate at $V_{OV} = 0.2$ V and have $V_t = 0.5$ V and $V_A = 16$ V. The feedback resistance $R_F = 10\ \text{k}\Omega$.

(a) If I_s has a zero dc component, find the dc voltage at the input, at the drain of Q_1, and at the output.

(b) Find g_m and r_o of Q_1 and Q_2.

(c) Provide the A circuit and derive an expression for A in terms of g_{m1}, r_{o1}, g_{m2}, r_{o2}, and R_F.

(d) What is β? Give an expression for the loop gain $A\beta$ and the amount of feedback $(1 + A\beta)$.

(e) Derive an expression for A_f.

(f) Derive expressions for R_i, R_{in}, R_o, and R_{out}.

(g) Evaluate A, β, $A\beta$, A_f, R_i, R_o, R_{in}, and R_{out} for the component values given.

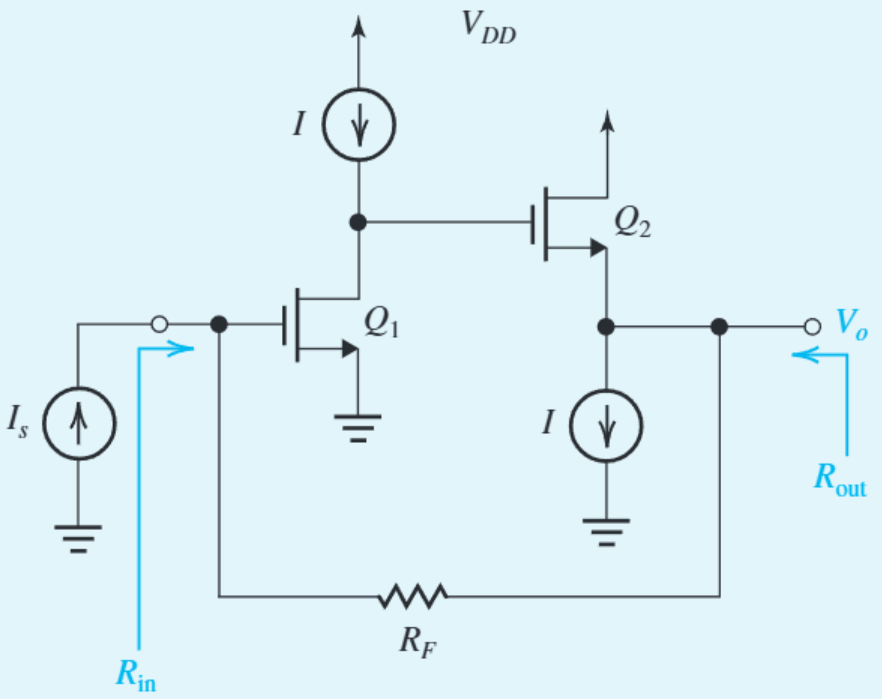


Figure P11.65

11.66 By setting $I_s = 0$ and breaking the feedback loop, find the loop gain of the feedback amplifier in Fig. P11.65. If you have already solved Problem 11.65, compare results. Which result do you think is more accurate, and why? For the numerical values given in Problem 11.65, by how much (in percent) do the two values of loop gain differ?

11.67 Analyze the circuit in Fig. E11.19 from first principles (i.e., do not use the feedback approach) and hence show that

$$A_f \equiv \frac{V_o}{I_s} = -\frac{(R_s \parallel R_F)\left(g_m - \dfrac{1}{R_F}\right)(r_o \parallel R_F)}{1 + (R_s \parallel R_F)\left(g_m - \dfrac{1}{R_F}\right)(r_o \parallel R_F)/R_F}$$

D 11.64 图 P11.64 中的电路采用分流-并联反馈拓扑结构中增益为 μ 的电压放大器，反馈网络由电阻 R 组成。然后您就会发现，例 11.9 中推导的所有公式在这里也同样适用。

(a) 如果环路增益非常大，闭环电压增益 V/V 近似值是多少？如果 R= 2 k，请给出导致 V/V - 10 V/V 的 R 值。(b) 如果放大器 μ 的直流增益为 10 V/V，输入电阻 R= 100 k，输出电阻 r= 2 k，求实际实现的 V/V。同时求出 Rand R（在电路图上标出）。您可以使用例 11.9 中得出的公式。

(c) 如果放大器 μ 的 3 分贝上限频率为 1 kHz，增益滚降为均匀的 -20-分贝/衰减，那么增益的 3 分贝频率是多少？

| V/V | ?

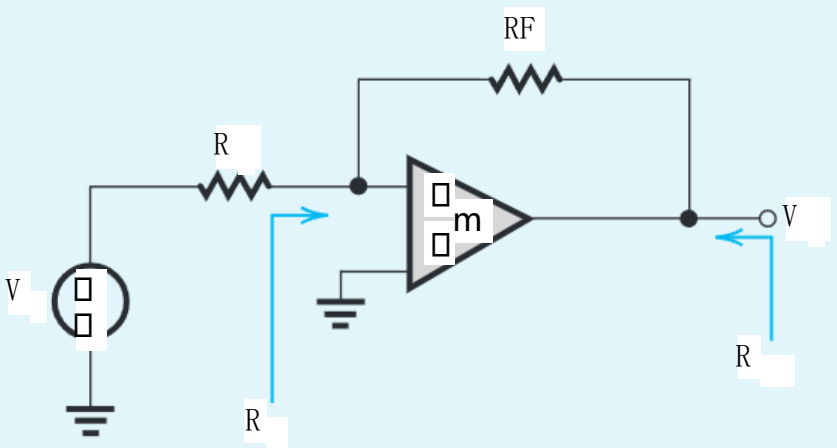


图 P11.64

11.65 图 P11.65 中的反馈跨阻放大器采用两个相同的 MOSFET，由理想电流源 I = 0.4 mA 偏置。MOSFET 的工作电压 V= 0.2 V，V= 0.5 V 和 V= 16 V，反馈电阻 R= 10 k。

(a) 如果 I 的直流分量为零，求输入端、Q 的漏极和输出端的直流电压。

(b) 求 Q 和 Q 的 g 和 rof。

(c) 提供 A 电路，并根据 g、r、g、r 和 R 推导出 A 的表式。给出回路增益 Aβ 和反馈量 (1 + Aβ) 的表达式。

(e) 求出 A 的表达式。

(f) 求出 R 、R、R 和 R 的表达式。

(g) 根据给出的分量值计算 A、β、Aβ、A ， R ， R ， R 和 R。

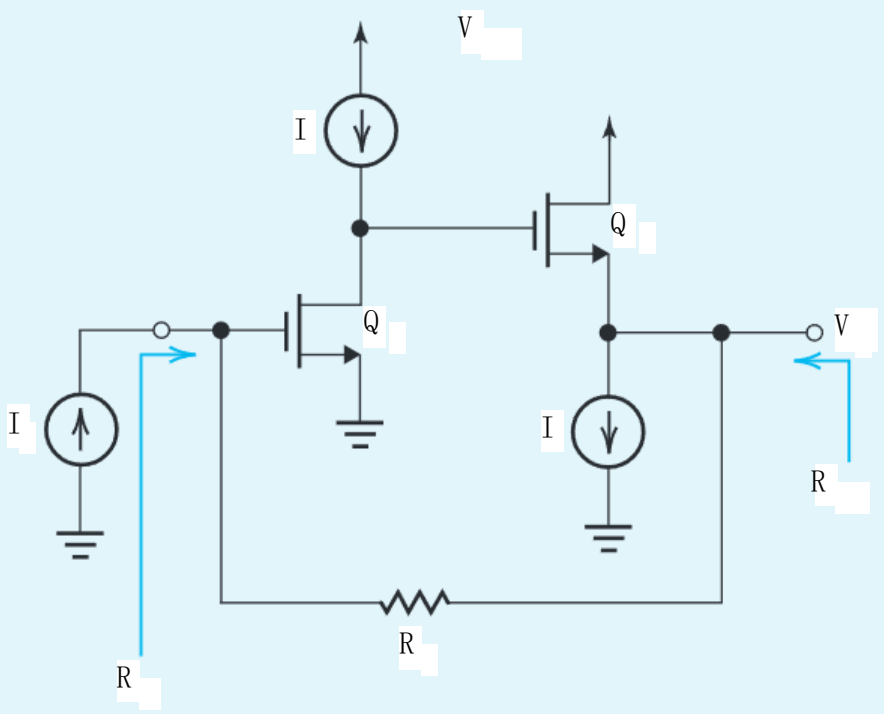


图 P11.65

11.66 设置 I= 0 并断开反馈回路，求图 P11.65 中反馈放大器的回路增益。如果您已经解决了问题 11.65，请比较结果。您认为哪个结果更准确，为什么？对于问题 11.65 中给出的数值，两个环路增益值相差多少（百分比）？

11.67 从第一原理分析图 E11.19 中的电路（即不使用反馈方法），从而说明

$$A \equiv \frac{V}{I} = -\frac{(\quad \| \quad)\left(g - \frac{1}{R}\right)(\quad \| \quad)}{1 + (\quad \| \quad)\left(g - \frac{1}{R}\right)(\quad \| \quad)/R}$$

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

Comparing this expression to the one given in Exercise 11.19, part (b), you will note that the only difference is that g_m has been replaced by $(g_m - 1/R_F)$. Note that $-1/R_F$ represents the forward transmission in the feedback network, which the feedback-analysis method neglects. What is the condition then for the feedback-analysis method to be reasonably accurate for this circuit?

D 11.68 For the feedback amplifier in Fig. P11.68, select a value for R_F that results in a closed-loop gain $A_f \equiv V_o/I_s \simeq -10$ kΩ. Then, analyze the circuit to determine the actual value of A_f realized. As well, determine R_{in} and R_{out}. Transistors Q_1 and Q_2 are operated so that $g_{m1} = g_{m2} = 4$ mA/V and r_{o1} and r_{o2} can be neglected. Also, $R_{D1} = R_{D2} = 10$ kΩ.

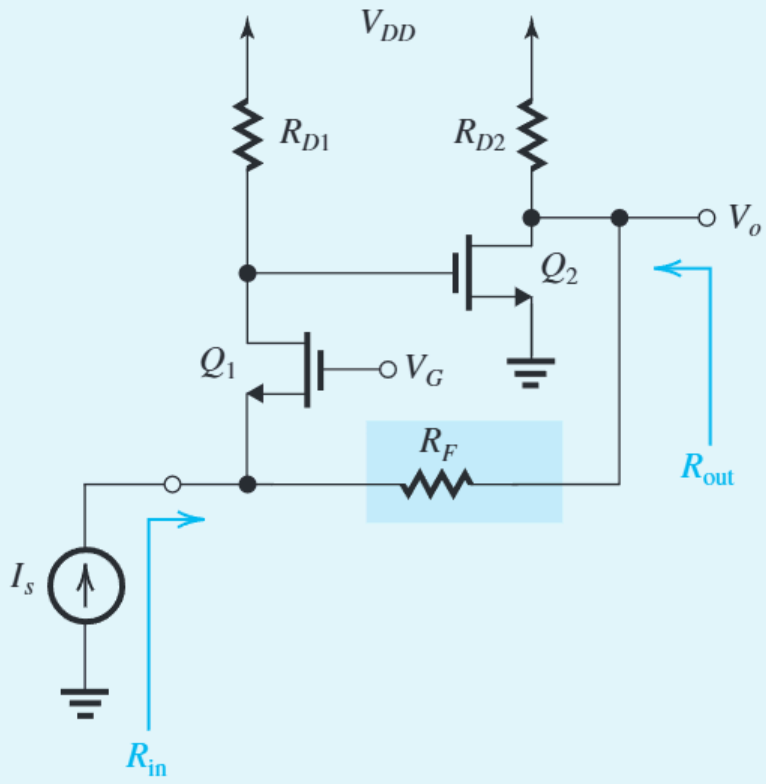


Figure P11.68

11.69 For the feedback transresistance amplifier in Fig. P11.69, let $V_{CC} = -V_{EE} = 5$ V, $R_C = R_E = R_F = 10$ kΩ. The transistors have $V_{BE} = 0.7$ V and $\beta = 100$.

(a) If I_s has a zero dc component, show that Q_1 and Q_2 are operating at dc collector currents of approximately 0.35 mA and 0.58 mA, respectively. What is the dc voltage at the output?

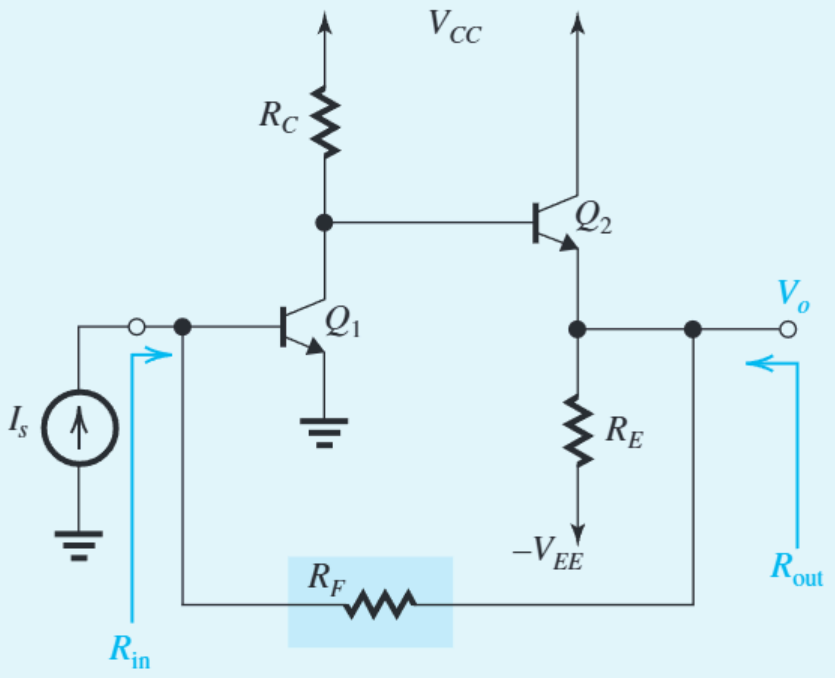


Figure P11.69

(b) Find the A circuit and the value of A, R_i, and R_o. Neglect r_{o1} and r_{o2}.

(c) Find the value of β, the loop gain, and the amount of feedback.

(d) Find $A_f \equiv V_o/I_s$, the input resistance R_{in}, and the output resistance R_{out}.

D **11.70 (a) Show that for the circuit in Fig. P11.70(a), if the loop gain is large, the voltage gain V_o/V_s is given approximately by

$$\frac{V_o}{V_s} \simeq -\frac{R_f}{R_s}$$

(b) Using three cascaded stages of the type shown in Fig. P11.70(b) to implement the amplifier μ, design a feedback amplifier with a voltage gain of approximately −100 V/V. The amplifier is to operate between a source resistance $R_s = 10$ kΩ and a load resistance $R_L = 1$ kΩ. Calculate the actual value of V_o/V_s realized, the input resistance (excluding R_s), and the output resistance (excluding R_L). Assume that the BJTs have h_{fe} of 100. [*Note:* In practice, the three amplifier stages are not made identical, for stability reasons.]

将此表达式与练习 11.19 第（b）部分中的表达式进行比较，你会发现唯一的不同是 g 被替换成了 $(g_m R)$。请注意，-1/R 表示反馈网络中的前向传输，而反馈分析法忽略了这一点。那么，对于该电路，反馈分析法合理准确的条件是什么？

D 11.68 对于图 P11.68 中的反馈放大器，请选择一个能使闭环增益 A≡ V /I -10 k 的 R 值。然后，分析电路以确定 Arealized 的实际值。晶体管 Q 和 Q 工作时，g= g = 4 mA/V，r 和 r 可忽略不计。此外，R = R = 10 k。

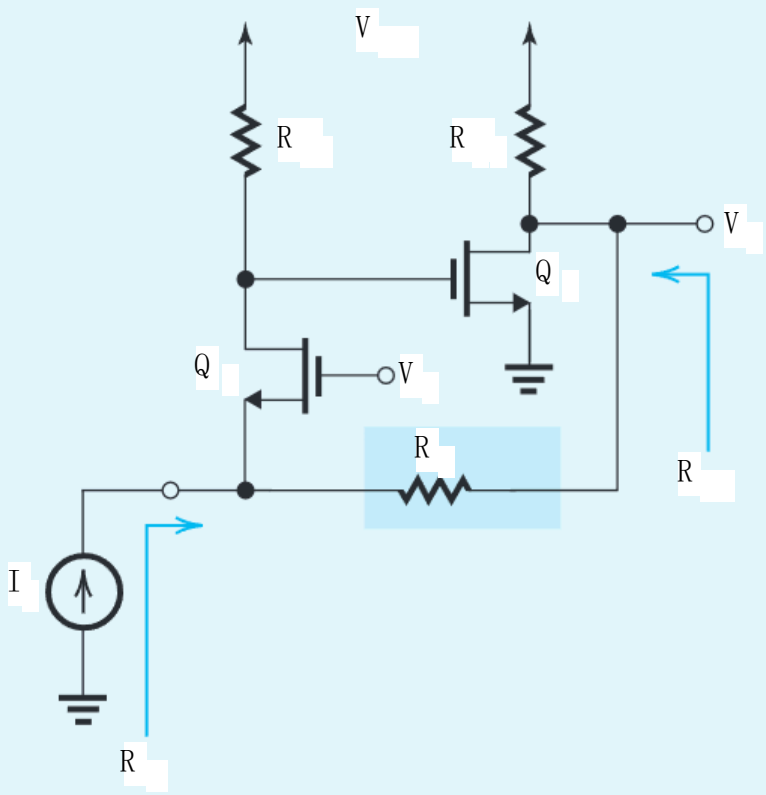


图 P11.68

11.69 对于图 P11.69 中的反馈跨阻放大器，设 V= -V= 5 V，R= R= R= 10 k。

晶体管的 V= 0.7 V，β= 100。

(a) 如果 I 的直流分量为零，说明 Q 和 Q 的工作直流集电极电流分别约为 0.35 mA 和 0.58 mA。输出端的直流电压是多少？

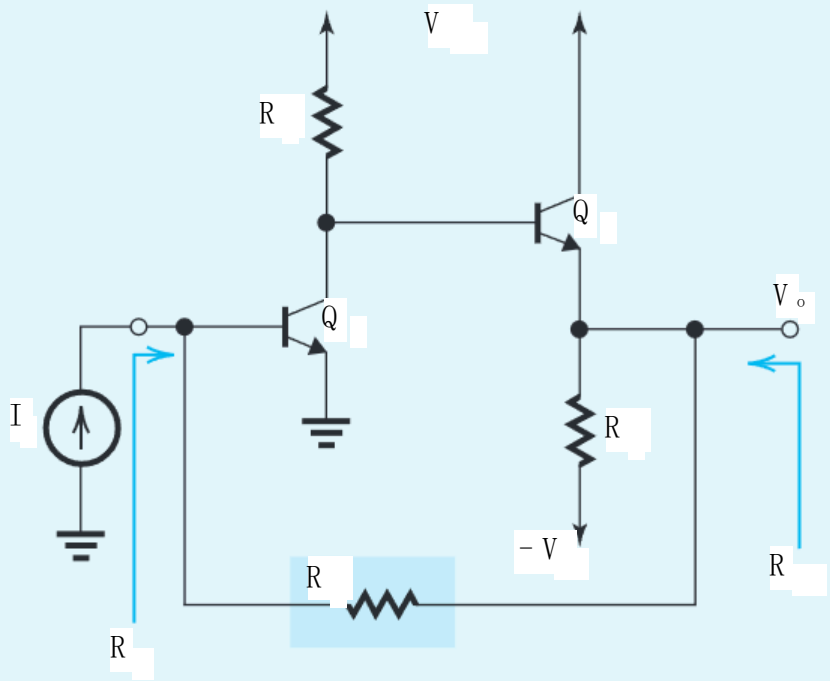


图 P11.69

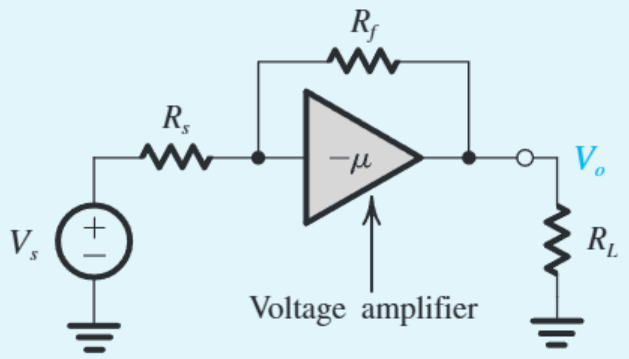
(b) 找出 A 电路以及 A、R 和 R 的值。忽略 r 和 r。 (c) 求 β 值、环路增益和反馈量。(d) 求 A ≡ V/I、输入电阻 R 和输出电阻 R。

D **11.70 (a) 说明对于图 P11.70(a)中的电路，如果环路增益很大，则电压增益 V/V 近似为

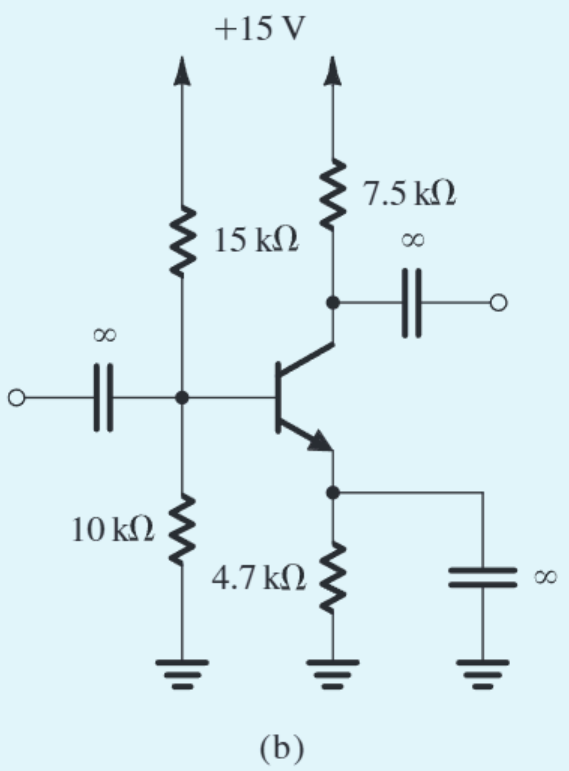
$$\frac{V}{V} \approx -\frac{R}{R}$$

(b) 使用图 P11.70(b)所示的三个级联级来实现放大器 μ，设计一个电压增益约为 -100 V/V 的反馈放大器。放大器将在源电阻 R= 10 k 和负载电阻 R= 1 k 之间工作。计算 V/V 实现值、输入电阻（不包括 R ）和输出电阻（不包括 R ）的实际值。假设 BJT 的 h 为 100。[注：在实际应用中，出于稳定性考虑，三级放大器并不完全相同。]

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题



(a)



(b)

Figure P11.70

is not shown. It is required to derive expressions for $A_f \equiv V_o/I_s$, R_{in}, and R_{out}. Assume that C_1 and C_2 are sufficiently small that their loading effect on the basic amplifier can be neglected. Also neglect r_o. Find the values of A_f, R_{in}, and R_{out} for the case in which $g_{m1} = 5$ mA/V, $R_D = 10$ kΩ, $C_1 = 0.9$ pF, $C_2 = 0.1$ pF, and $g_{mf} = 2$ mA/V.

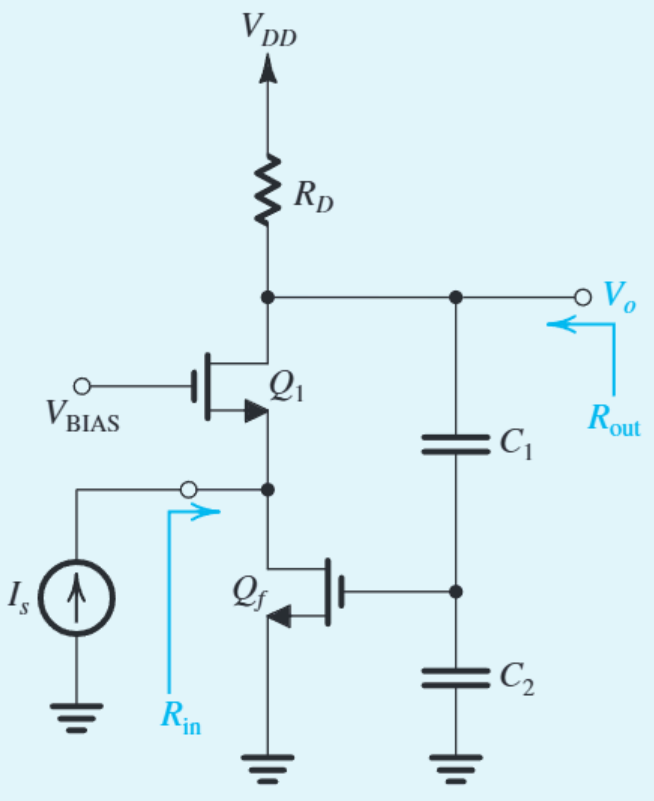


Figure P11.72

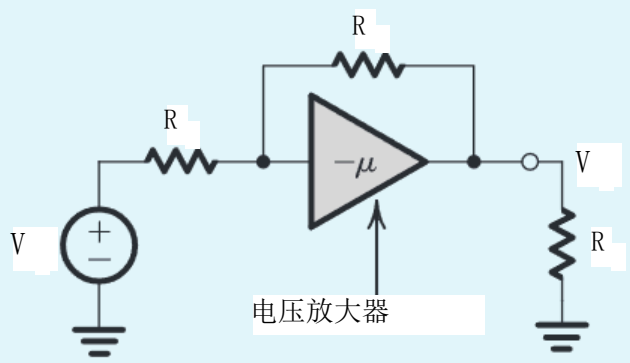
D 11.71 Negative feedback is to be used to modify the characteristics of a particular amplifier for various purposes. Identify the feedback topology to be used if:

(a) input resistance is to be lowered and output resistance raised.
(b) both input and output resistances are to be raised.
(c) both input and output resistances are to be lowered.

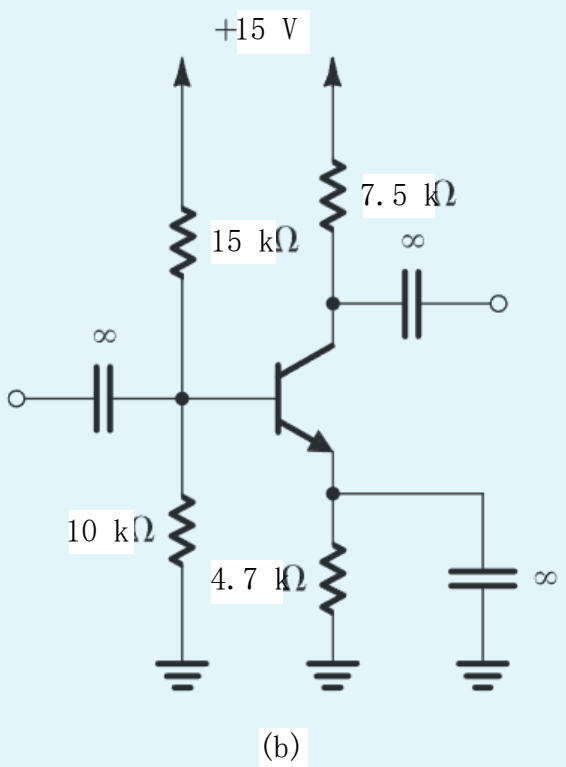
11.72 The feedback amplifier of Fig. P11.72 consists of a common-gate amplifier formed by Q_1 and R_D, and a feedback circuit formed by the capacitive divider (C_1, C_2) and the common-source transistor Q_f. Note that the bias circuit for Q_f

D *11.73 Figure P11.73 shows a shunt–shunt feedback amplifier. The MOSFETs have $V_{tn} = 0.6$ V, $V_A = 20$ V, and $\mu_n C_{ox} = 200\ \mu\text{A/V}^2$. The power supply $V_{DD} = 3.3$ V, and $R_L = 2$ kΩ. The coupling capacitor C_C can be assumed to be very large.

(a) Perform a dc design to meet the following specifications: $I_{D1} = 100\ \mu$A, $I_{D2} = 1$ mA, $I_{R2,R1} = 10\ \mu$A, $V_{OV1} = V_{OV2} = 0.2$ V. Neglect the Early effect. Specify the values required for I_1, R_1, R_2, $(W/L)_1$, and $(W/L)_2$.
(b) Find an expression for β and hence an expression for the ideal value of V_o/V_s.
(c) Find the value of R_s that results in V_o/V_s being ideally −6 V/V.



(a)



(b)

图 P11.70

请注意，图中未显示 Q 的偏置电路。需要推导出 A≡V/I、R 和 R 的表达式。假设 Cand Care 足够小，可以忽略它们对基本放大器的负载影响。同时忽略 r 。在 g= 5 mA/V、R= 10 k、C= 0.9 pF、C= 0.1 pF 和 g= 2 mA/V 的情况下，求 A、R 和 R 的值。

图 P11.72

D 11.71 负反馈用于改变特定放大器的特性，以达到各种目的。请确定在下列情况下使用的反馈拓扑结构：

(a) 输入电阻降低，输出电阻升高。

(b) 输入和输出电阻都要提高。

(c) 输入和输出电阻都要降低。

11.72 图 P11.72 中的反馈放大器包括一个由 Q 和 R 组成的共门放大器，以及一个由电容分压器 (C, C) 和共源晶体管 Q 组成的反馈电路。

D *11.73 图 P11.73 显示了一个并联反馈放大器。MOSFET 的 V= 0.6 V，V = 20 V， μC= 200 μA/V。电源 V= 3.3 V，R= 2 k。耦合电容器 C 可以假定非常大。

(a) 进行直流设计，以满足以下规格：I = 100 μA，I = 1 mA，I = 10 μA，V = 0.2 V。指明 I、R、R、（W/L）和（W/L）所需的值。(b) 求出 β 的表达式，从而求出 V /V 的理想值的表达式。(c) 求 V/V 理想值为 -6 V/V 的 R 值。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

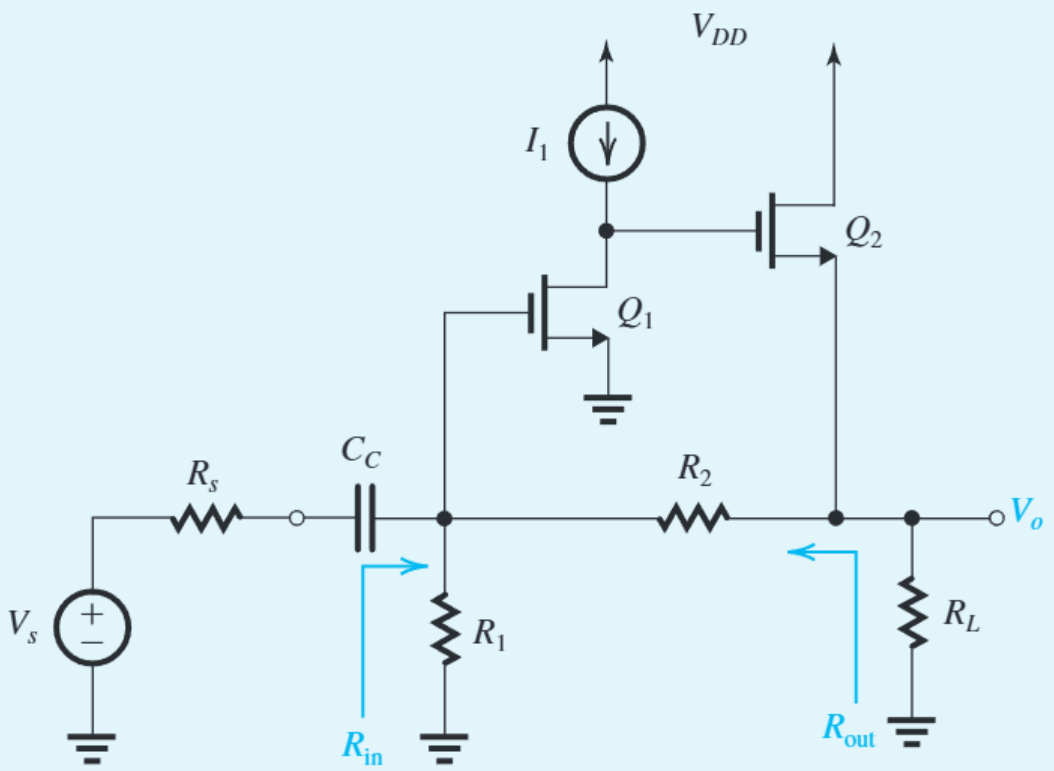


Figure P11.73

(d) Find the A circuit and use it to determine the values of A, R_i, and R_o.
(e) Find the value obtained for V_o/V_s.
(f) Find R_{in} and R_{out}.

Feedback Current Amplifiers (Shunt–Series)

11.74 For the feedback current amplifier in Fig. P11.47:

(a) Provide the A circuit and derive expressions for R_i and A. Neglect r_o of both transistors.
(b) Provide the β circuit and an expression for β.
(c) Find an expression for $A\beta$.
(d) For $g_{m1} = g_{m2} = 5$ mA/V, $R_D = 20$ kΩ, $R_M = 10$ kΩ, and $R_F = 90$ kΩ, find the values of A, β, $A\beta$, A_f, R_i, and R_{if}.
(e) If $r_{o2} = 20$ kΩ and $R_L = 1$ kΩ, find the output resistance as seen by R_L.

D 11.75 Design the feedback current amplifier of Fig. 11.27(a) to meet the following specifications:

(i) $A_f \equiv I_o/I_s = -100$ A/A
(ii) amount of feedback $\simeq 40$ dB
(iii) $R_{in} \simeq 1$ kΩ

Specify the values of R_1, R_2, and μ. Assume that the amplifier μ has infinite input resistance and that $R_s = \infty$. For the MOSFET, $g_m = 5$ mA/V and $r_o = 20$ kΩ. What R_{out} is obtained?

11.76 Consider the feedback current amplifier in Fig. 11.27(a) (which was analyzed in Example 11.10). Let $R_s = R_{id} = \infty$. By setting $I_s = 0$ and breaking the feedback loop at the gate of Q, find an expression for the loop gain $A\beta$. Evaluate $A\beta$ for the component values given in Example 11.10 and hence determine A and A_f. Why do the results differ somewhat from those found in Example 11.10?

11.77 The feedback current amplifier in Fig. P11.77 utilizes two identical NMOS transistors sized so that at $I_D = 0.2$ mA they operate at $V_{OV} = 0.2$ V. Both devices have $V_t = 0.5$ V and $V_A = 10$ V.

(a) If I_s has zero dc component, show that both Q_1 and Q_2 are operating at $I_D = 0.2$ mA. What is the dc voltage at the input?
(b) Find g_m and r_o for each of Q_1 and Q_2.
(c) Find the A circuit and the value of R_i, A, and R_o.
(d) Find the value of β.
(e) Find $A\beta$ and A_f.
(f) Find R_{in} and R_{out}.

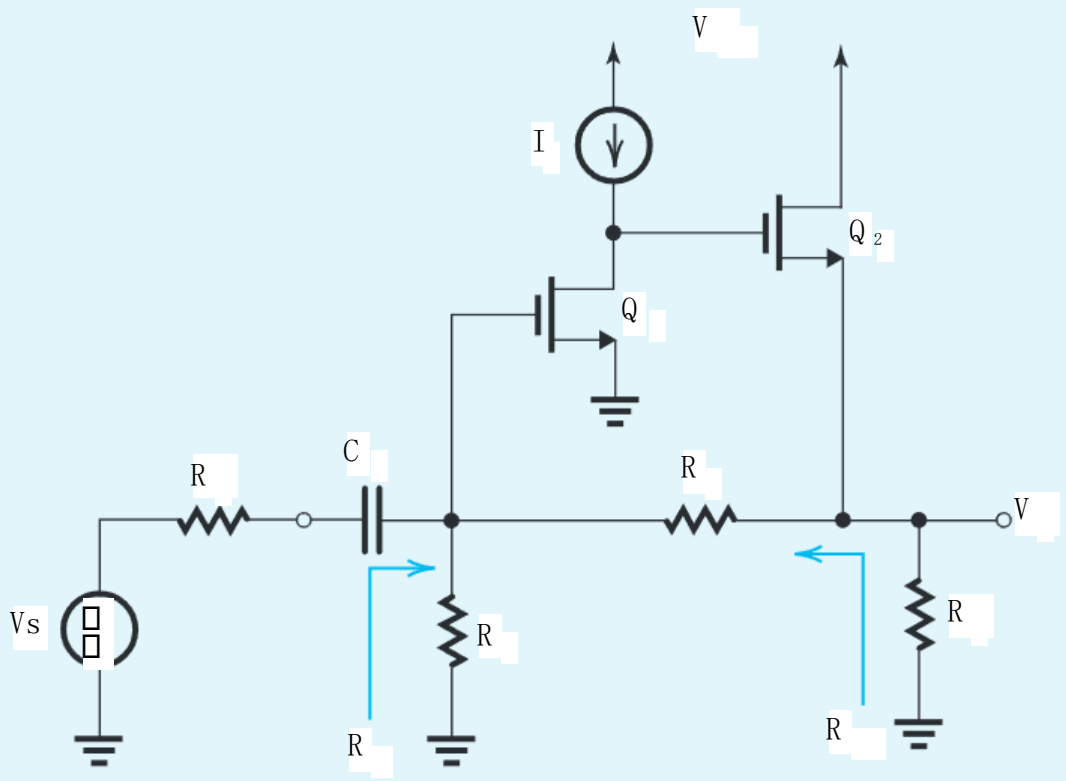


图 P11.73

(d) 找出 A 电路，并用它确定 A、R 和 R 的值。

(e) 求出 V/V 的值。

(f) 求 Rand R。

反馈电流放大器（Shunt 系列）

11.74 对于图 P11.47 中的反馈电流放大器：

(a) 提供 A 电路并推导出 Rand A 的表达式。忽略两个晶体管的 r。

(b) 提供 β 电路和 β 的表达式。

(c) 求 Aβ 的表达式。

(d) 对于 g= g= 5 mA/V、R= 20 k、R= 10 k 和 R= 90 k，求 A、β、Aβ、A、R 和 R 的值。 (e) 如果 r= 20 k 和 R= 1 k，求 R 看到的输出电阻。

D 11.75 设计图 11.27（a）中的反馈电流放大器，以满足下列规格要求：

(i) A≡ I /I= -100 A/A (ii) 反馈量 40 dB (iii) R 1 k

指定 R、R 和 μ 的值。假设放大器 μ 的输入电阻为无穷大，且 R= ∞

对于 MOSFET，g = 5 mA/V，r = 20 k。得到什么结果？

11.76 考虑图 11.27(a) 中的反馈电流放大器（在例 11.10 中分析过）。设 R = R= ∞。设置 I= 0 并在 Q 的栅极处断开反馈回路，求回路增益 Aβ 的表达式。为什么结果与例 11.10 中的结果有些不同？

11.77 图 P11.77 中的反馈电流放大器使用了两个相同的 NMOS 晶体管，其大小使其在 I= 0.2 mA 时的工作电压为 V= 0.2 V，两个器件的工作电压分别为 V= 0.5 V 和 V= 10 V。

(a) 如果 I 的直流分量为零，说明 Q 和 Q 都在 I = 0.2 mA 时工作。输入端的直流电压是多少？(b) 求 Q 和 Q 的 g 和 r。

(c) 找出 A 电路以及 R、A 和 R 的值。

(d) 求 β 的值。

(e) 求 Aβ 和 A。

(f) 求 Rand R。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

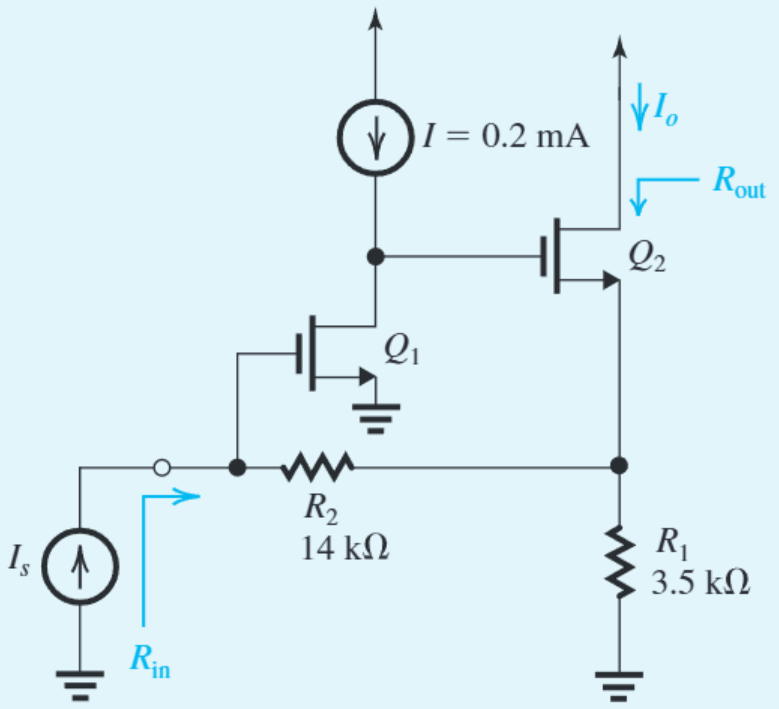


Figure P11.77

***11.78** The feedback current amplifier in Fig. P11.78(a) can be thought of as a "super" CG transistor. Note that rather than connecting the gate of Q_2 to signal ground, an amplifier is placed between source and gate.

(a) If μ is very large, what is the signal voltage at the input terminal? What is the input resistance? What is the current gain I_o/I_s?

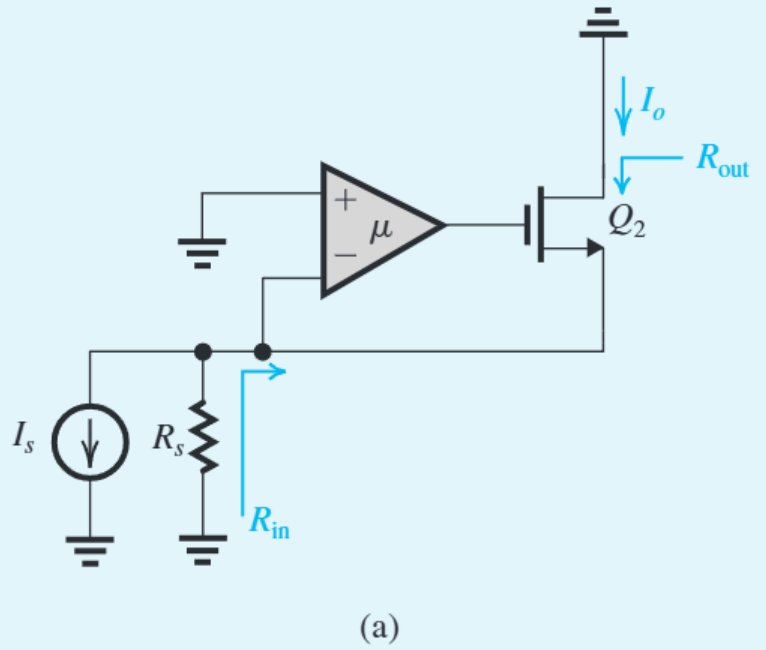
(b) For finite μ but assuming that the input resistance of the amplifier μ is very large, find the A circuit and derive expressions for A, R_i, and R_o.

(c) What is the value of β?

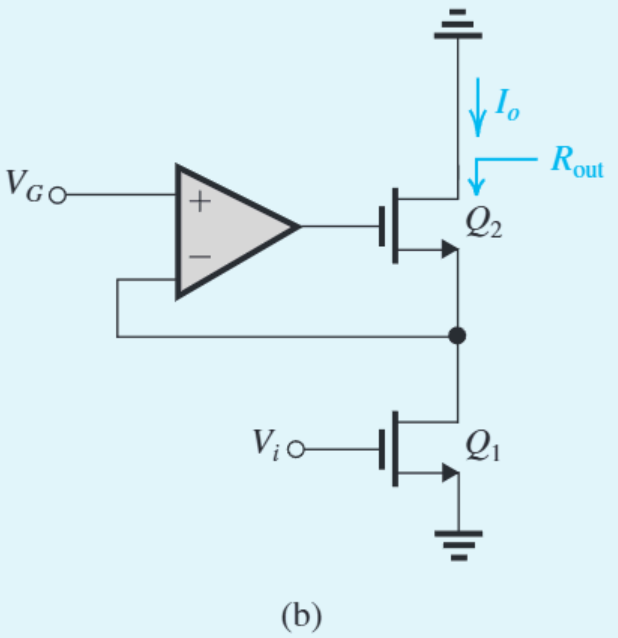
(d) Find $A\beta$ and A_f. If μ is large, what is the value of A_f?

(e) Find R_{in} and R_{out} assuming the loop gain is large.

(f) The "super" CG transistor can be utilized in the cascode configuration shown in Fig. P11.78(b), where V_G is a dc bias voltage. Replacing Q_1 by its small-signal model, use the analogy of the resulting circuit to that in Fig. P11.78(a) to find I_o and R_{out}.



(a)



(b)

Figure P11.78

***11.79** Figure P11.79 shows an interesting and very useful application of feedback to improve the performance of the current mirror formed by Q_1 and Q_2. Rather than connecting the drain of Q_1 to the gate, as is the case in simple current mirrors, an amplifier of gain $+\mu$ is connected between the drain and the gate. Note that the feedback loop does not include transistor Q_2. The feedback loop ensures that the value of the gate-to-source voltage of Q_1 is such that I_{o1} equals I_s. This regulated V_{gs} is also applied to Q_2. Thus, if W/L of Q_2 is n times W/L of Q_1, $I_{o2} = nI_{o1} = nI_s$. This current tracking, however, is *not* regulated by the feedback loop.

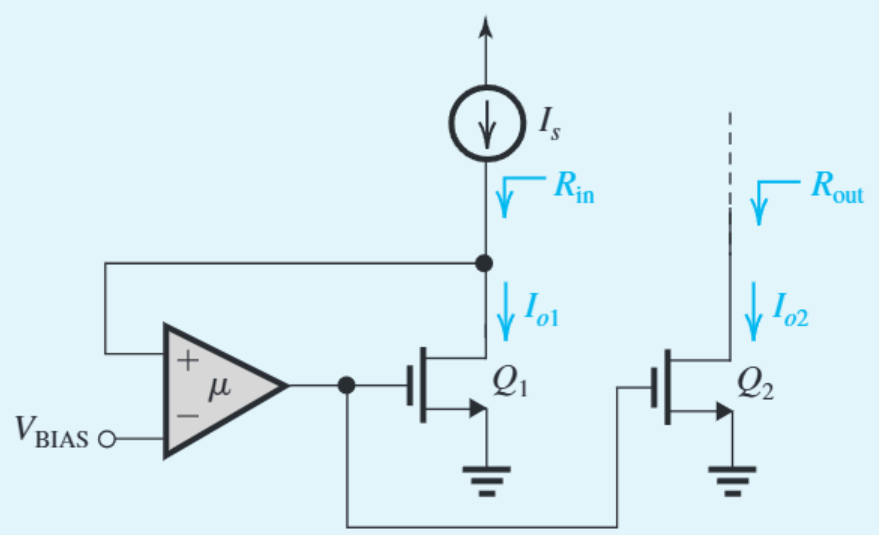


Figure P11.79

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

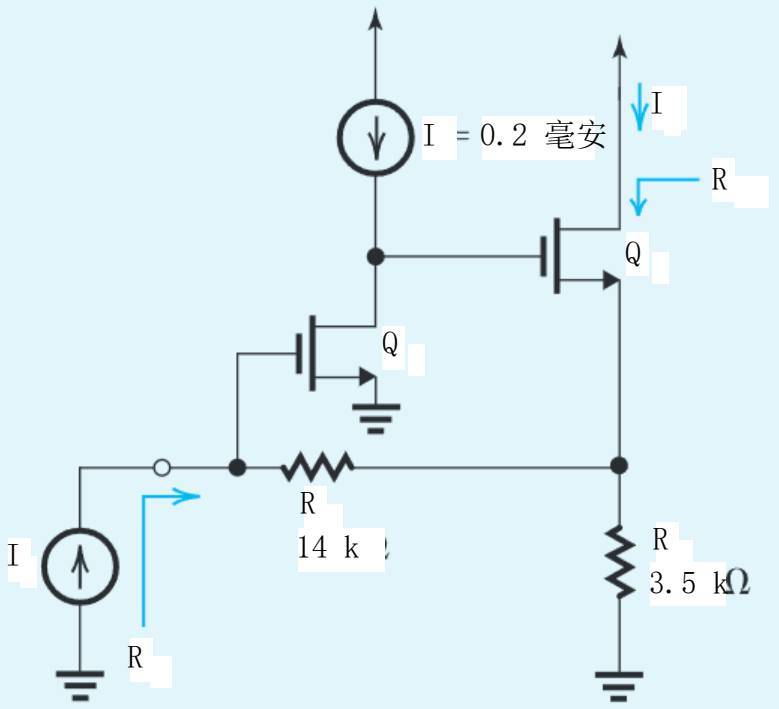
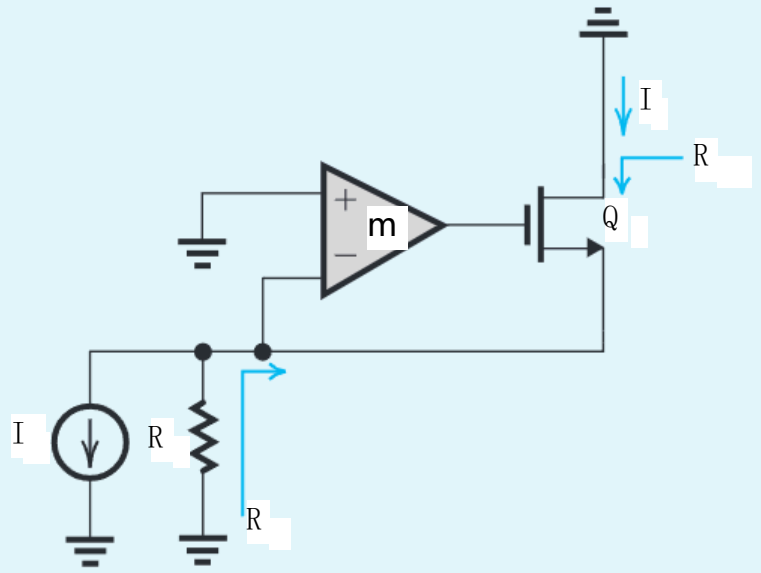


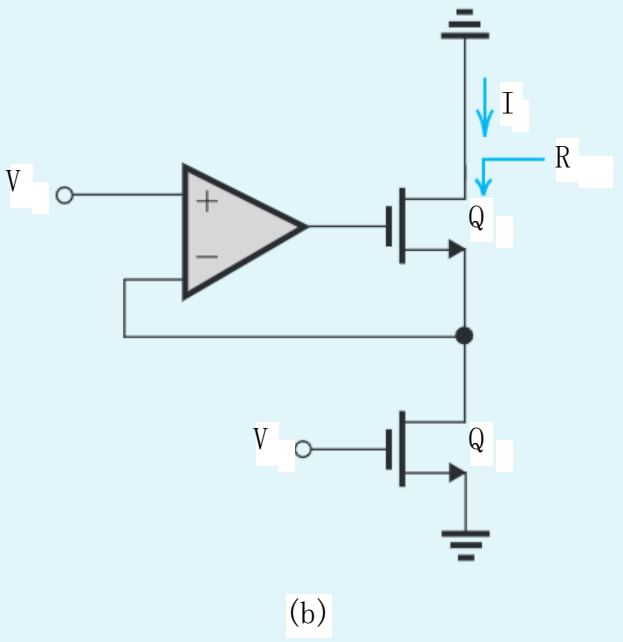
图 P11.77

*11.78 图 P11.78(a)中的反馈电流放大器可视为 "超级 "CG 晶体管。请注意，在源极和栅极之间并没有连接 Q 的栅极和信号地，而是放置了一个放大器。

(a) 如果 μ 非常大，输入端的信号电压是多少？输入电阻是多少？电流增益 I/I 是多少？(b) 对于有限的 μ，但假设放大器的输入电阻 μ 非常大，请找出 A 电路，并推导出 A、R 和 R 的表达式。



(a)



(b)

图 P11.78

(c) β 的值是多少？

(d) 求 Aβ 和 A。如果 μ 较大，A 的值是多少？

(e) 假设环路增益很大，求 Rand R。(f) "超级 "CG 晶体管可用于图 P11.78(b)所示的级联配置，其中 Vis 为直流偏置电压。将 Q 替换为其小信号模型，利用所得电路与图 P11.78(a) 的类比，求出 I 和 R。

*图 P11.79 显示了反馈在改善由 Q 和 Q 形成的电流镜性能方面的一个有趣且非常有用的应用。Q 的漏极并没有像简单电流镜那样连接到栅极，而是在漏极和栅极之间连接了一个增益为 +μ 的放大器。请注意，反馈回路不包括晶体管 Q。反馈回路确保 Q 的栅极至源极电压值与 I 相等。因此，如果 Q 的 W/L 值等于 I，则 Q 的 W/L 值等于 I。

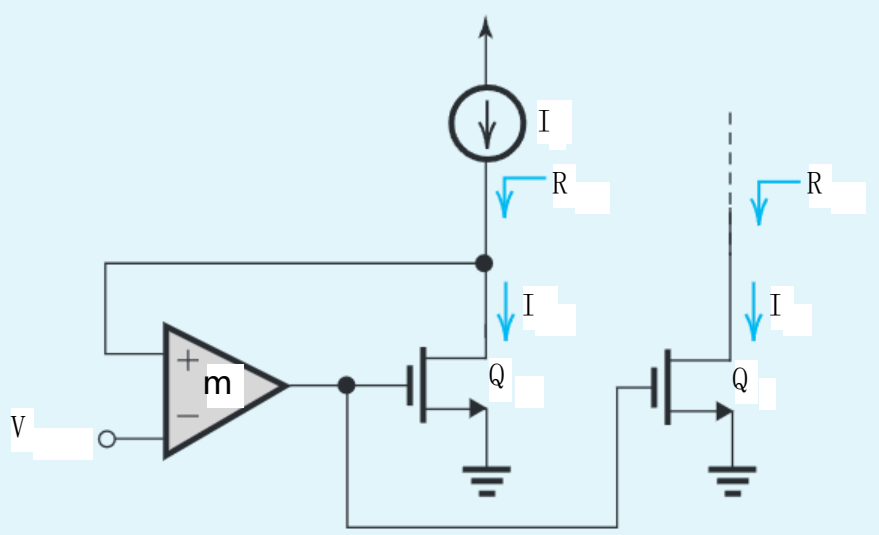


图 P11.79

为 Q 的 n 倍 W/L，即 I= nI= nI。不过，这种电流跟踪不受反馈回路的调节。

(a) Show that the feedback is negative.
(b) If μ is very large and the input resistance of the amplifier μ is infinite, what dc voltage appears at the drain of Q_1? If Q_1 is to operate at an overdrive voltage of 0.2 V, what is the minimum value that V_{BIAS} must have?
(c) Replacing Q_1 by its small-signal model, find an expression for the small-signal input resistance R_{in} assuming finite gain but infinite input resistance for the amplifier μ. Note that here it is much easier to do the analysis directly than to use the feedback-analysis approach. For large μ, what does R_{in} become?
(d) What is the output resistance R_{out}?

***11.80** The circuit in Fig. P11.80 is an implementation of a particular circuit building block known as **second-generation current convoyer** (CCII). It has three terminals besides ground: x, y, and z. The heart of the circuit is the feedback amplifier consisting of the differential amplifier μ and the complementary source follower (Q_N, Q_P). (Note that this feedback circuit is one we have encountered a number of times in this chapter, albeit with only one source-follower transistor.) In the following, assume that the differential amplifier has a very large gain μ and infinite differential input resistance. Also, let the two current mirrors have unity current-transfer ratios.

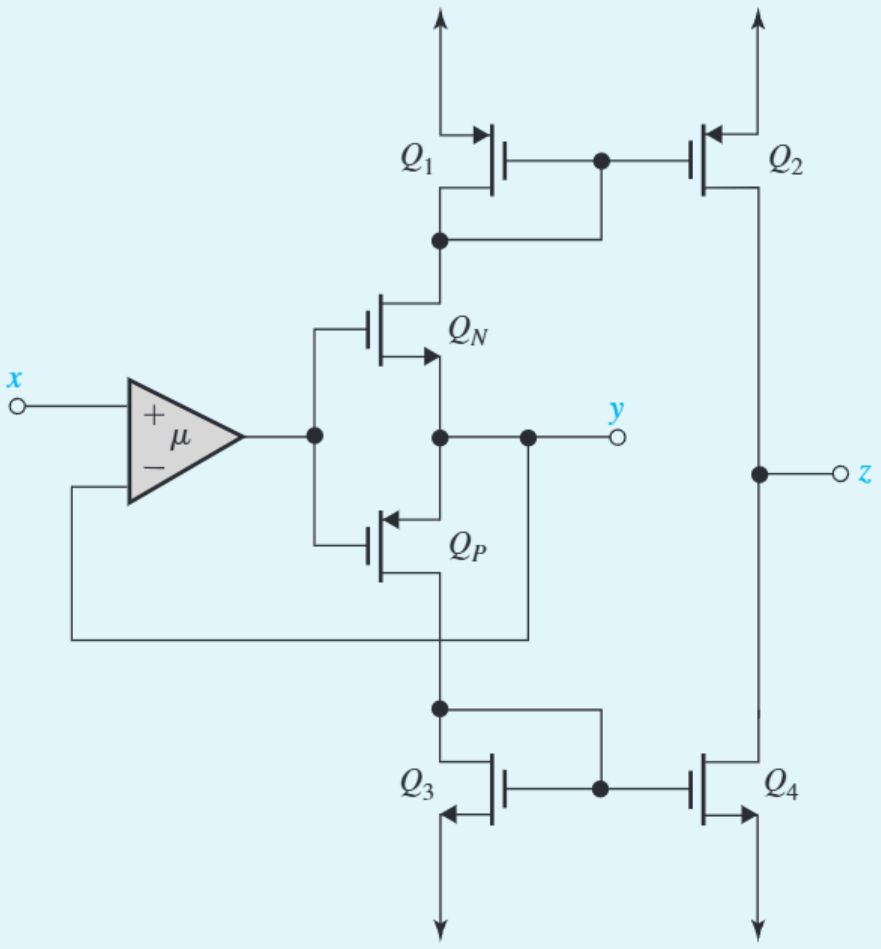


Figure P11.80

(a) If a resistance R is connected between y and ground, a voltage signal V_x is connected between x and ground, and z is short-circuited to ground. Find the current I_z through the short circuit. Show how this current is developed and its path for V_x positive and for V_x negative.
(b) If x is connected to ground, a current source I_y is connected to input terminal y, and z is connected to ground, what voltage appears at y and what is the input resistance seen by I_y? What is the current I_z that flows through the output short circuit? Also, explain the current flow through the circuit for I_y positive and for I_y negative.
(c) What is the output resistance at z?

SIM ***11.81** For the amplifier circuit in Fig. P11.81, assuming that V_s has a zero dc component, find the dc voltages at all nodes and the dc emitter currents of Q_1 and Q_2. Let the BJTs have $\beta = 100$. Use feedback analysis to find V_o/V_s and R_{in}. Let $V_{BE} = 0.7$ V.

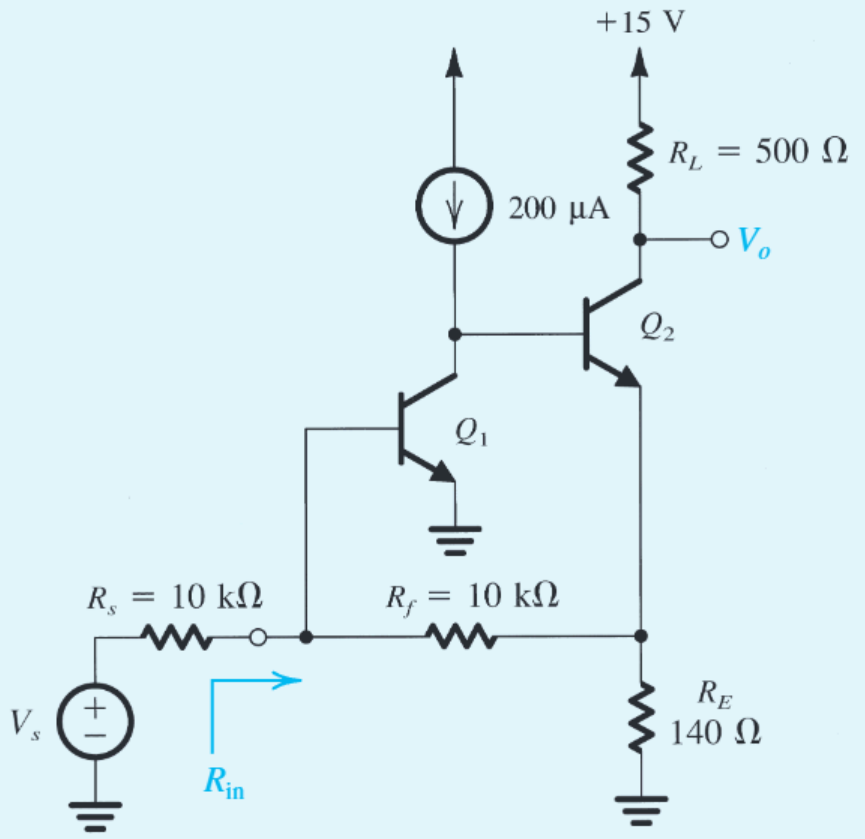


Figure P11.81

****11.82** Figure P11.82 shows a feedback amplifier utilizing the shunt–series topology. All transistors have $\beta = 100$ and $V_{BE} = 0.7$ V. Neglect r_o except in (f).

(a) Perform a dc analysis to find the dc emitter currents in Q_1 and Q_2 and hence determine their small-signal parameters.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

(a) 证明反馈是负的。

(b) 如果 μ 非常大，而放大器的输入电阻 μ 无穷大，Q 的漏极会出现多大的直流电压？

如果 Q 要在 0.2 V 的过驱动电压下工作，那么 Vm 的最小值是多少？

(c) 假设放大器 μ 的增益有限但输入电阻无限，用其小信号模型代替 Q，求出小信号输入电阻 R 的表达式。请注意，在这里直接进行分析要比使用反馈分析方法容易得多。当 μ 较大时，R 是多少？(d) 输出电阻 R 是多少？

*11.80 图 P11.80 中的电路是第二代电流调制器（CCII）这一特殊电路构件的实现。电路的核心是由差分放大器 μ 和互补源跟随器 (Q , Q) 组成的反馈放大器。(请注意，这种反馈电路我们在本章中已多次遇到，尽管只有一个源极跟随器晶体管）。在下文中，假设差分放大器具有非常大的增益 μ 和无限大的差分输入电阻。另外，让两个电流镜具有统一的电流传输比。

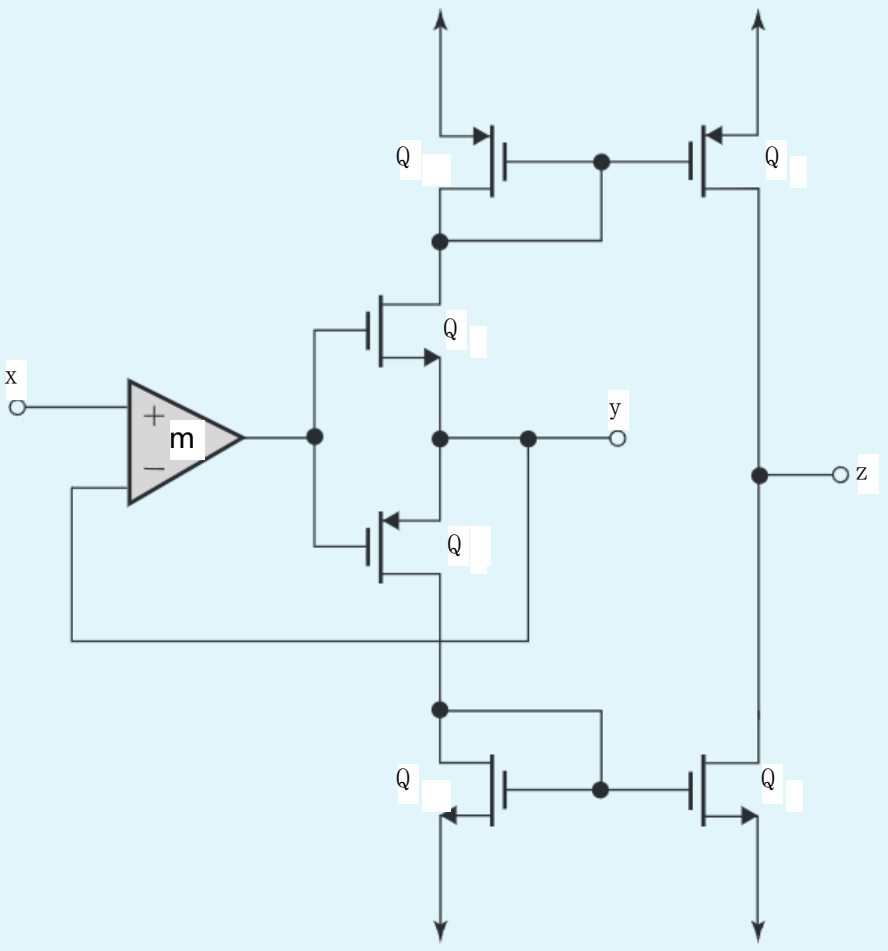


图 P11.80

(a) 如果 y 和地之间连接一个电阻 R，x 和地之间连接一个电压信号 Vis，z 与地短路。求通过短路的电流 I。说明该电流是如何产生的，以及正 V 和负 V 时的路径。(b) 如果 x 接地，输入端 y 接一个电流源 I，z 接地，那么 y 端会出现什么电压，I 看到的输入电阻是多少？流过输出短路的电流 I 是多少？另外，请解释 I 为正和 I 为负时流过电路的电流。

(c) z 处的输出电阻是多少？

*11.81 对于图 P11.81 中的放大器电路，假设 V 的直流分量为零，求所有节点的直流电压以及 Q 和 Q 的直流发射极电流。使用反馈分析法求出 V/V 和 R。

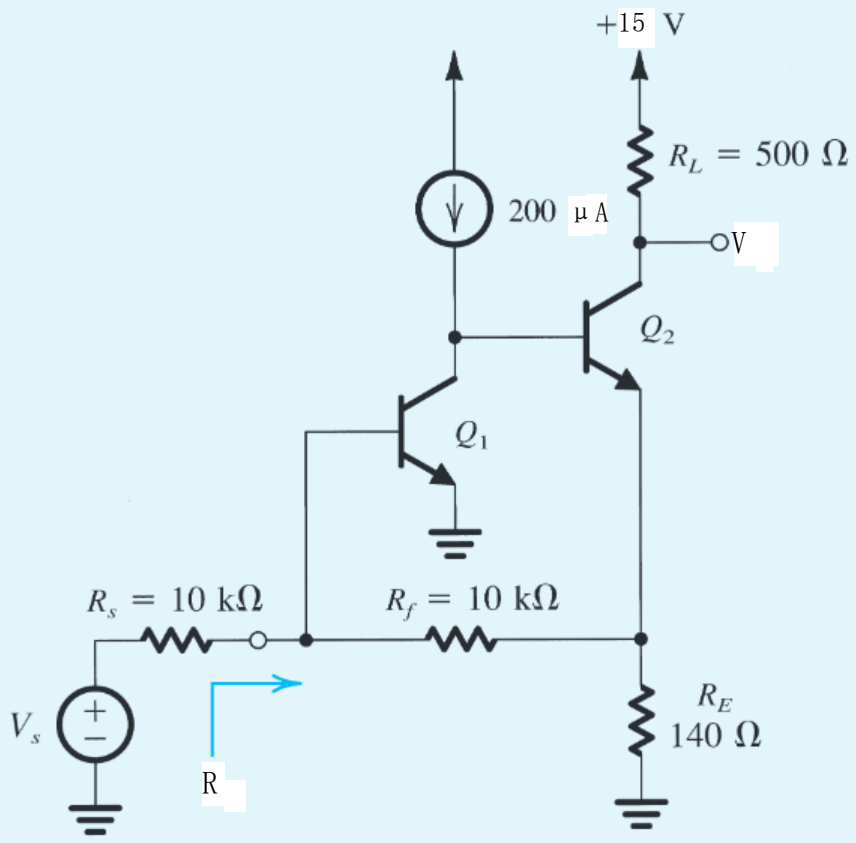


图 P11.81

**11.82 图 P11.82 显示了采用并联串联拓扑结构的反馈放大器。所有晶体管的 β = 100，V = 0.7 V。除 (f) 外，忽略 r。

(a) 进行直流分析，找出 Q 和 Q 的直流发射极电流，从而确定它们的小信号参数。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

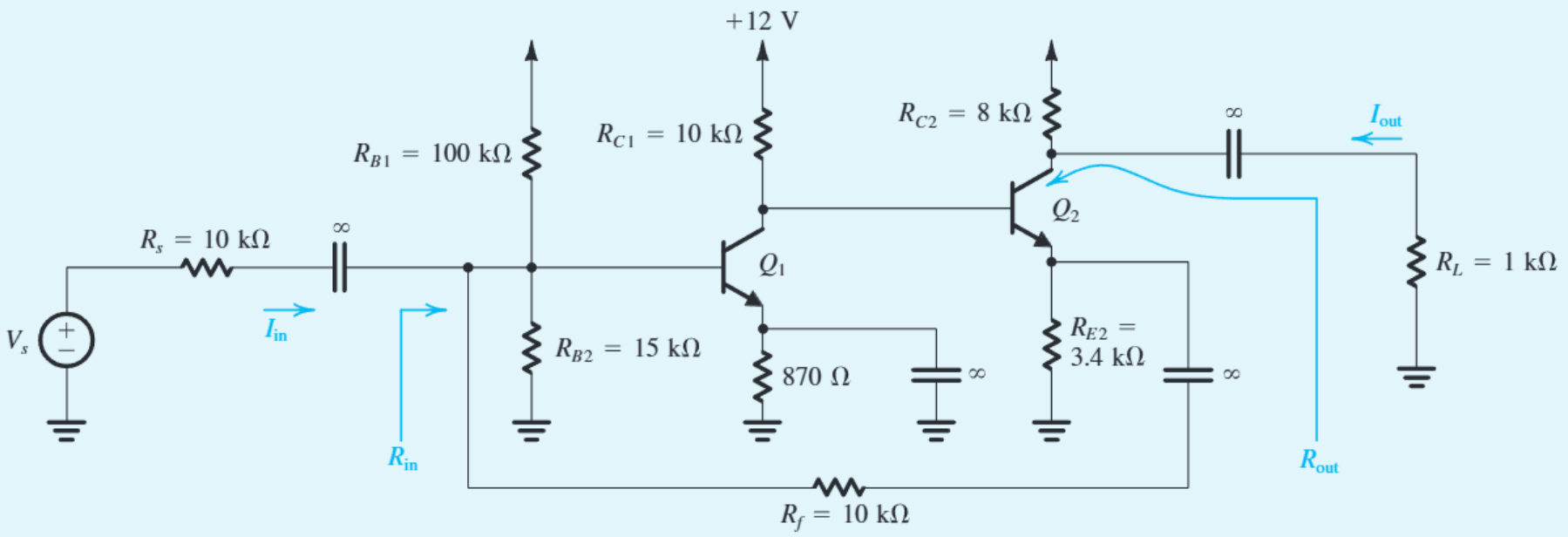


Figure P11.82

(b) Replacing the BJTs with their hybrid-π models, give the equivalent circuit of the feedback amplifier.

(c) Give the A circuit and determine A, R_i, and R_o. Note that R_o is the resistance determined by breaking the emitter loop of Q_2 and measuring the resistance between the terminals thus created.

(d) Find the β circuit and determine the value of β.

(e) Find $A\beta$, $1 + A\beta$, A_f, R_{if}, and R_{of}. Note that R_{of} represents the resistance that in effect appears in the emitter of Q_2 as a result of the feedback.

(f) Determine R_{in}, I_{out}/I_{in}, and R_{out}. To determine R_{out}, use $V_{A2} = 75$ V and recall that the maximum possible output resistance looking into the collector of a BJT is approximately βr_o, where β is the BJT's β (see Problem 11.55).

Section 11.7: The Stability Problem

11.83 An op amp designed to have a low-frequency gain of 10^5 and a high-frequency response dominated by a single pole at 100 rad/s acquires, through a manufacturing error, a pair of additional poles at 20,000 rad/s. At what frequency does the total phase shift reach 180°? At this frequency, for what value of β, assumed to be frequency independent, does the loop gain reach a value of unity? What is the corresponding value of closed-loop gain at low frequencies?

***11.84** For the situation described in Problem 11.83, sketch Nyquist plots for $\beta = 1.0$ and 10^{-3}. (Plot for $\omega = 0$ rad/s, 100 rad/s, 10^3 rad/s, 10^4 rad/s, 2×10^4 rad/s, and ∞ rad/s.)

11.85 An op amp having a low-frequency gain of 10^4 and a single-pole rolloff at 10^3 rad/s is connected in a negative-feedback loop via a feedback network having a transmission k and a two-pole rolloff at 10^3 rad/s. Find the value of k above which the closed-loop amplifier becomes unstable.

11.86 Consider a feedback amplifier for which the open-loop gain $A(s)$ is given by

$$A(s) = \frac{10{,}000}{(1 + s/10^4)(1 + s/10^5)^2}$$

If the feedback factor β is independent of frequency, find the frequency at which the phase shift is 180°, and find the critical value of β at which oscillation will commence.

Section 11.8: Effect of Feedback on the Amplifier Poles

11.87 A dc amplifier having a single-pole response with pole frequency 10 Hz and unity-gain frequency of 1 MHz is operated in a loop whose frequency-independent feedback factor is 0.1. Find the low-frequency gain, the 3-dB frequency, and the unity-gain frequency of the closed-loop amplifier. By what factor does the pole shift?

11.88 An amplifier has dc open-loop gain of 80 dB and a single pole with 100-Hz frequency. It is utilized to design a feedback amplifier with a 3-dB frequency of 10 kHz. What β is needed? What is the dc closed-loop gain realized? Give an expression for $A_f(s)$.

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

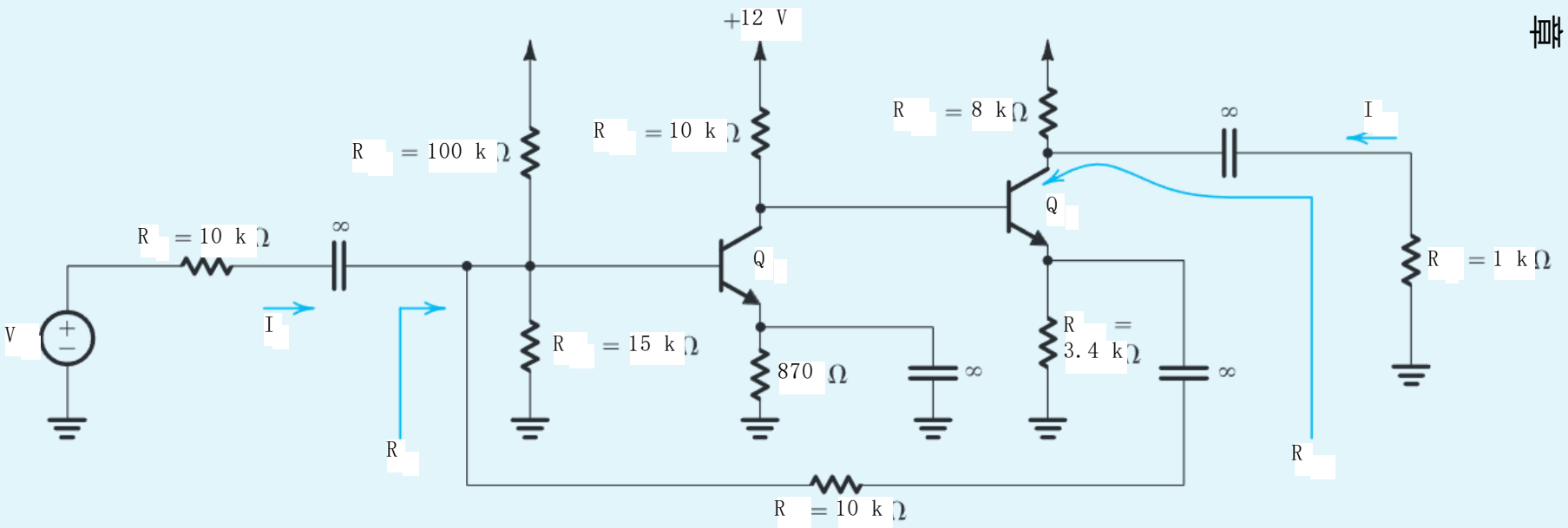


图 P11.82

(b) 将 BJT 换成混合-π 型，给出反馈放大器的等效电路。(c) 给出 A 电路并确定 A、R 和 R。请注意，R 是通过断开 Q 的发射极回路并测量由此产生的两端之间的电阻而确定的电阻。

(d) 找出 β 电路，并确定 β 值。(e) 找出 Aβ、1 + Aβ、A、R 和 R。请注意，R 代表因反馈而实际出现在 Q 发射极的电阻。(f) 确定 R、I/I 和 R。要确定 R，请使用 V= 75 V，并回忆 BJT 集电极可能的最大输出电阻约为βr，其中 β 是 BJT 的 β（见问题 11.55）。

11.85 一个运算放大器的低频增益为 10，在 10 拉德/秒时有单极滚动，它通过一个传输速率为 k 的反馈网络连接在负反馈环路中，在 10 拉德/秒时有双极滚动。求 k 值大于多少时，闭环放大器会变得不稳定。

11.86 考虑一个反馈放大器，其开环增益 A(s) 由以下公式给出

$$A(s) = \left(\frac{10,000}{1 + s/10}\right)\left(\frac{}{1 + s/10}\right)$$

如果反馈系数 β 与频率无关，求相移为 180° 时的频率，并求开始振荡的临界值 β。

第 11.7 节：稳定性问题

11.83 某运放的低频增益为 10，高频响应以单极为主，频率为 100 rad/s，由于制造误差，在 20,000 rad/s 时获得了一对额外的极点。在什么频率下总相移达到 180°？在此频率下，假定 β 与频率无关，则环路增益达到多少？低频时闭环增益的相应值是多少？

*11.84 针对问题 11.83 中描述的情况，绘制 β = 1.0 和 10 的奈奎斯特图（绘制 ω = 0 rad/s、100 rad/s、10rad/s、10rad/s、2 × 10 rad/s 和 ∞ rad/s）。

第 11.8 节：反馈对放大器极点的影响

11.87 某直流放大器具有单极响应，极点频率为 10 Hz，单位增益频率为 1 MHz，在一个与频率无关的反馈系数为 0.1 的环路中运行。求闭环放大器的低频增益、3 分贝频率和单位增益频率。极移系数是多少？

11.88 某放大器的直流开环增益为 80 dB，单极频率为 100 赫兹。利用它设计一个 3 分贝频率为 10 kHz 的反馈放大器。需要多少 β？实现的直流闭环增益是多少？给出 A(s) 的表达式。

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

***11.89** An amplifier having a low-frequency gain of 10^4 and poles at 10^4 Hz and 10^5 Hz is operated in a closed negative-feedback loop with a frequency-independent β.

(a) For what value of β do the closed-loop poles become coincident? At what frequency?
(b) What is the low-frequency, closed-loop gain corresponding to the situation in (a)? What is the value of the closed-loop gain at the frequency of the coincident poles?
(c) What is the value of Q corresponding to the situation in (a)?
(d) If β is increased by a factor of 10, what are the new pole locations? What is the corresponding pole Q?

D 11.90 A dc amplifier has an open-loop gain of 1000 and two poles, a dominant one at 1 kHz and a high-frequency one whose location can be controlled. It is required to connect this amplifier in a negative-feedback loop that provides a dc closed-loop gain of 10 and a maximally flat response. Find the required value of β and the frequency at which the second pole should be placed. What is the 3-dB frequency of the closed-loop amplifier?

11.91 Reconsider Example 11.11 with the circuit in Fig. 11.34, modified to incorporate a so-called tapered network, in which the components immediately adjacent to the amplifier input are raised in impedance to $C/10$ and $10R$. Find expressions for the resulting pole frequency ω_0 and Q factor. For what value of K do the poles coincide? For what value of K does the response become maximally flat? For what value of K does the circuit oscillate?

D 11.92 A feedback amplifier having a dc closed-loop gain of 10 and a maximally flat second-order response with a 3-dB frequency of 1 kHz is required. The open-loop amplifier utilizes a cascade of two identical amplifier stages, each having a single-pole frequency response. Find the values required for β, the 3-dB frequency, and the dc gain of each of the two amplifier stages. Give an expression for $A_f(s)$.

11.93 Three identical inverting amplifier stages, each characterized by a low-frequency gain K and a single-pole response with $f_{3\text{dB}} = 100$ kHz, are connected in a feedback loop with $\beta = 1$. What is the minimum value of K at which the circuit oscillates? What would the frequency of oscillation be?

Section 11.9: Stability Study Using Bode Plots

11.94 Reconsider Exercise 11.26 for the case of the op amp wired as a unity-gain buffer. At what frequency is $|A\beta| = 1$? What is the corresponding phase margin?

11.95 Reconsider Exercise 11.26 for the case of a manufacturing error introducing a second pole at 10^3 Hz. What is now the frequency for which $|A\beta| = 1$? What is the corresponding phase margin? For what values of β is the phase margin 45° or more?

11.96 For what phase margin does the gain peaking have a value of 5%? Of 10%? Of 0.1 dB? Of 1 dB? Of 3 dB? [*Hint:* Use the result in Eq. (11.82).]

11.97 An amplifier has a dc gain of 10^4 and poles at 10^5 Hz, 3.16×10^5 Hz, and 10^6 Hz. Find the value of β, and the corresponding closed-loop gain, for which a phase margin of 45° is obtained.

11.98 A two-pole amplifier for which $A_0 = 10^3$ and having poles at 1 MHz and 10 MHz is to be connected as a differentiator. On the basis of the rate-of-closure rule, what is the smallest differentiator time constant for which operation is stable? What are the corresponding gain and phase margins?

11.99 For the amplifier described by Fig. 11.37 and with frequency-independent feedback, what is the minimum closed-loop voltage gain that can be obtained for phase margins of 90° and 45°?

Section 11.10: Frequency Compensation

D 11.100 A multipole amplifier having a first pole at 1 MHz and a dc open-loop gain of 80 dB is to be compensated for closed-loop gains as low as unity by the introduction of a new dominant pole. At what frequency must the new pole be placed?

D 11.101 For the amplifier described in Problem 11.100, rather than introducing a new dominant pole we can use additional capacitance at the circuit node at which the pole is formed to reduce the frequency of the first pole. If the frequency of the second pole is 20 MHz and if it remains unchanged while additional capacitance is introduced as

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

*11.89 某放大器的低频增益为 10，极点频率分别为 10Hz 和 10Hz，在闭合负反馈环路中运行，β 与频率无关。

(a) 闭环极点重合的 β 值是多少？频率是多少？(b) 与 (a) 中的情况相对应的低频闭环增益是多少？重合极点频率下的闭环增益值是多少？(c) 与（a）中情况相对应的 Q 值是多少？(d) 如果将 β 增大 10 倍，新的极点位置是什么？相应的极点 Q 值是多少？

D 11.90 某直流放大器的开环增益为 1000，有两个极点，一个是 1 kHz 时的主极点，另一个是高频极点，其位置可以控制。需要将该放大器连接到负反馈环路中，以提供 10 的直流闭环增益和最大平坦响应。请找出所需的 β 值和第二极点的频率。闭环放大器的 3 分贝频率是多少？

11.91 用图 11.34 中的电路重新考虑例 11.11，并对其进行修改，加入所谓的锥形网络，其中紧邻放大器输入端的元件的阻抗提高到 C/10 和 10R。求由此产生的极点频率 ω 和 Q 因子的表达式。极点重合的 K 值是多少？K 为何值时，响应达到最大平坦度？在 K 值为多少时，电路出现振荡？

D 11.92 需要一个直流闭环增益为 10、3 分贝频率为 1 kHz 的最大平坦二阶响应的反馈放大器。开环放大器采用两个相同放大器级的级联，每个级都具有单极频率响应。请分别求出两级放大器所需的 β 值、3 分贝频率和直流增益。给出 A(s) 的表达式。

11.93 β = 1 的反馈回路中连接着三个相同的反相放大器级，每个级都具有低频增益 K 和 f= 100 kHz 的单极响应。电路振荡时 K 的最小值是多少？振荡频率是多少？

第 11.9 节：利用博德图进行稳定性研究

11.94 重新考虑练习 11.26，将运算放大器连接成单增益缓冲器。在什么频率下 |Aβ| = 1？

相应的相位差是多少？

11.95 重新考虑习题 11.26，在 10Hz 处引入第二个极点的制造误差。现在 |Aβ| = 1 的频率是多少？相应的相位裕度是多少？相位差为 45° 或以上的 β 值是多少？

11.96 增益峰值为 5%，相位差为多少？10%？0.1 dB？1 分贝？3 dB？[提示：使用公式 (11.82) 中的结果。]

11.97 放大器的直流增益为 10，极点频率为 10Hz、3.16×10 Hz 和 10 Hz。求相位裕度为 45 ° 的 β 值和相应的闭环增益。

11.98 将一个 A= 10、极点分别为 1 MHz 和 10 MHz 的两极放大器连接为微分器。根据闭合率规则，能够稳定工作的最小微分器时间常数是多少？相应的增益和相位余量是多少？

11.99 对于图 11.37 所描述的放大器和与频率无关的反馈，在相位差为 90°和 45°时，可获得的最小闭环电压增益是多少？

第 11.10 节：频率补偿

D 11.100 一个多极放大器的第一极在 1 MHz，直流开环增益为 80 dB，需要通过引入一个新的主极来补偿低至 1 的闭环增益。新极点必须位于哪个频率？

D 11.101 对于问题 11.100 中描述的放大器，我们可以在形成极点的电路节点上使用额外的电容来降低第一个极点的频率，而不是引入一个新的主极点。如果第二个磁极的频率为 20 MHz，并且在引入额外电容时保持不变，则如下图所示

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题

mentioned, find the frequency to which the first pole must be lowered so that the resulting amplifier is stable for closed-loop gains as low as unity. By what factor is the capacitance at the controlling node increased?

11.102 For the amplifier whose $A(s)$ is depicted in Fig. 11.38, to what value must the first pole frequency be lowered to obtain stable performance for (a) $\beta = 0.001$ and (b) $\beta = 0.1$?

11.103 Contemplate the effects of pole splitting by considering Eqs. (11.89), (11.93), and (11.94) under the conditions that $R_1 \simeq R_2 = R$, $C_2 \simeq C_1/10 = C$, $C_f \gg C$, and $g_m = 100/R$, by calculating ω_{P1}, ω_{P2}, and ω'_{P1}, ω'_{P2}. Comment on the results.

D 11.104 An op amp with open-loop voltage gain of 10^5 and poles at 10^6 Hz, 10^7 Hz, and 10^8 Hz is to be compensated by the addition of a fourth dominant pole to operate stably with unity feedback ($\beta = 1$). What is the frequency of the required dominant pole? The compensation network is to consist of an RC low-pass network placed in the negative-feedback path of the op amp. The dc bias conditions are such that a 1-MΩ resistor can be tolerated in series with each of the negative and positive input terminals. What capacitor is required between the negative input and ground to implement the required fourth pole?

D *11.105 An op amp with an open-loop voltage gain of 80 dB and poles at 10^5 Hz, 10^6 Hz, and 2×10^6 Hz is to be compensated to be stable for unity β. Assume that the op amp incorporates an amplifier equivalent to that in Fig. 11.40, with $C_1 = 150$ pF, $C_2 = 5$ pF, and $g_m = 40$ mA/V, and that f_{P1} is caused by the input circuit and f_{P2} by the output circuit of this amplifier. Find the required value of the compensating Miller capacitance and the new frequency of the output pole.

SIM **11.106 The op amp in the circuit of Fig. P11.106 has an open-loop gain of 10^5 and a single-pole rolloff with $\omega_{3dB} = 10$ rad/s.

(a) Sketch a Bode plot for the loop gain.
(b) Find the frequency at which $|A\beta| = 1$, and find the corresponding phase margin.
(c) Find the closed-loop transfer function, including its zero and poles. Sketch a pole-zero plot. Sketch the magnitude of the transfer function versus frequency, and label the important parameters on your sketch.

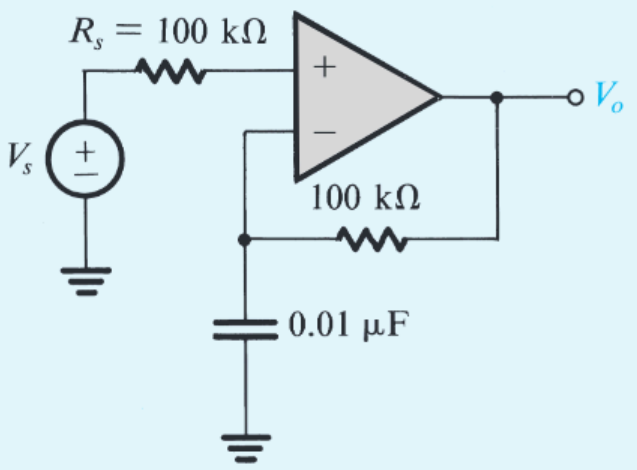


Figure P11.106

SIM = Multisim/PSpice; * = difficult problem; ** = more difficult; *** = very challenging; D = design problem

请找出第一极点必须降低到什么频率，才能使产生的放大器在闭环增益低至 1 时保持稳定。控制节点处的电容要增加多少倍？

11.102 对于 A(s) 如图 11.38 所示的放大器，在 (a) β = 0.001 和 (b) β = 0.1 时，必须将第一极点频率降低到什么值才能获得稳定的性能?

11.103 在 R R= R、C C/10 = C、C C 和 g= 100/R 的条件下，考虑公式 (11.89)、(11.93) 和 (11.94)，通过计算 ω、ω 和 ω、ω，思考极分裂的影响。对结果进行评论。

D 11.104 一个运算放大器的开环电压增益为 10，极点频率分别为 10Hz、10Hz 和 10Hz。所需主极点的频率是多少？补偿网络由一个放置在运算放大器负反馈路径上的 RC 低通网络组成。在直流偏置条件下，负输入端和正输入端之间可以串联一个 1-M 电阻器。要实现所需的第四极，负输入端和地之间需要多大的电容？

D *11.105 某运算放大器的开环电压增益为 80 dB，极点频率为 10 Hz、10 Hz 和 2 × 10 Hz，需要对其进行补偿，使其在统一 β 条件下保持稳定、

C= 150 pF，C= 5 pF，g= 40 mA/V，且 f 由该放大器的输入电路引起，f 由输出电路引起。求补偿米勒电容的所需值和输出极的新频率。

**11.106 图 P11.106 电路中的运算放大器具有 10 的开环增益和 ω = 10 rad/s 的单极滚降。

SIM

(a) 绘制环路增益的 Bode 图。

(b) 找出 |Aβ| = 1 时的频率，并找出相应的相位差。(c) 找出闭环传递函数，包括其零点和极点。绘制极点-零点图。绘制传递函数大小与频率关系草图，并在草图上标注重要参数。

图 P11.106

SIM = Multisim/PSpice； * = 难题； ** = 更难； *** = 极具挑战性； D = 设计问题